CHAPTER 7

Clinical Psychology

IN THIS CHAPTER

7.1 ●● The Profession: History, Models, Ethics and Indian Law: Define clinical psychology and tell it apart from neighbouring professions, trace its history in the West and in India, compare the models of abnormality, apply the APA ethical principles and the Tarasoff duty, and state the main provisions of India's mental health and disability laws.

7.2 ○ The Clinical Interview, Mental Status Examination and Formulation: Conduct and record a clinical assessment: choose an interview format, use the core interviewing microskills, take a case history, carry out and write a mental status examination, assess suicide and violence risk, run a functional analysis, build a 5 Ps formulation, and use DSM-5-TR and ICD-11 with an eye on reliability and labelling.

7.3 ●●● Psychodynamic Therapies and Their Descendants: Explain how Freud's psychoanalysis works as a treatment and how its descendants (ego, object relations and self psychology, brief dynamic therapy, interpersonal psychotherapy, Jungian and Adlerian therapy, transactional analysis and supportive therapy) differ in theory, technique and evidence.

7.4 ●●● Behaviour Therapy: Trace the origins and assumptions of behaviour therapy and describe, attribute and choose among its classical, operant, modelling, relaxation and self-management techniques, from systematic desensitisation, flooding and ERP to token economies, habit reversal and parent training.

7.5 ●●● Cognitive and Cognitive-Behavioural Therapies, and the Third Wave: Explain and compare Ellis's REBT and Beck's cognitive therapy, describe the other cognitive-behavioural and third-wave therapies (DBT, ACT, MBCT and others) with their founders and techniques, and weigh the evidence for CBT.

7.6 ●●● Humanistic, Existential, Experiential and Brief Therapies: Explain person-centred, Gestalt, existential, reality, experiential and brief therapies, motivational interviewing with the stages of change, and the helping-skills models, naming each founder and signature technique.

7.7 ●● Family, Couples and Group Therapies: Explain why families are treated as systems, what the double bind and expressed emotion research found, how the Bowen, structural, strategic, Milan, experiential and later family, couples and sex therapies work, and how group therapy, psychodrama, self-help groups and therapeutic communities help, naming who devised each.

7.8 ○ Does Psychotherapy Work? Evidence, Common Factors and the Alliance: Explain how psychotherapy outcome is evaluated, from Eysenck's challenge and meta-analysis to RCTs, empirically supported treatments and clinical significance, and weigh the evidence for common factors, the alliance and therapist and client variables.

7.9 ○ Health, Community, Crisis and Rehabilitation Psychology: Explain health psychology and the models of health behaviour, the psychology of physical and chronic illness, community mental health and prevention, crisis intervention and suicide prevention, and psychosocial, neuropsychological and addiction rehabilitation.

7.10 ○ Working with Children, Older Adults and Special Settings: Adapt assessment and therapy to children, adolescents, older adults and counselling, forensic, creative and digital settings, naming each method's originator and the Indian laws that apply.

Weightage. 366 of the 3180 questions mapped in this book (11.5%) test this chapter, 191 of them from real past papers. 4 of its 10 sections are High yield.

Paris in the 1790s. On the southern edge of the city stood La Bicêtre, a huge hospice for poor, old, sick and criminal men. In its wards for the insane, men were kept in chains, some of them for years, and treated more like dangerous animals than patients. The story most books tell is that in 1793, at the height of the French Revolution, the hospital's physician, **Philippe Pinel**, ordered the irons struck off, and that men who had not seen the sky for years walked calmly into the courtyard. Nineteenth-century painters made the scene famous.

The real story, pieced together by historians, is less heroic and more useful. The chains at Bicêtre were first removed not by the doctor but by **Jean-Baptiste Pussin**, the lay superintendent of the ward, who had seen what irons did to the men in his care and had begun to treat them as people: he spoke with them, kept them occupied and replaced irons with gentler restraint. Pinel's contribution was to recognise that Pussin's methods worked, to record his patients' histories carefully and count who recovered, and to take the approach, with Pussin himself, to the women's hospital, the Salpêtrière. Their *moral treatment* rested on three things: a relationship, a daily routine and careful observation of results.

Keep both men in mind: the attendant who knew how to be with a frightened, disturbed person, and the physician who wrote down what happened and checked whether it helped. Between them they hold the whole of clinical psychology: understanding and assessing a person, building a relationship that heals, choosing treatments by their evidence, caring for people in their communities and protecting their dignity in law. This chapter follows each of these threads, and at its end we return to Bicêtre.

7.1 The Profession: History, Models, Ethics and Indian Law

MEDIUM YIELD · 39 QS

Before learning how a clinical psychologist interviews, tests or treats, you need to know who this professional is, where the profession came from, which theories of abnormality it uses, and which ethical rules and Indian laws bind it.

What clinical psychology is

The American Psychological Association's Division 12, the Society of Clinical Psychology, describes a field that "integrates science, theory, and practice to understand, predict, and alleviate maladjustment, disability, and discomfort as well as to promote human adaptation, adjustment, and personal development". A **clinical psychologist** therefore researches, assesses, treats and promotes well-being. In India the profession is defined by qualification and registration: under the Rehabilitation Council of India (RCI) Act, a clinical psychologist holds an RCI-recognised qualification and is on the RCI's register, and the Mental Healthcare Act 2017 likewise defines one by an RCI-recognised qualification.

The work spans psychological assessment, diagnosis, psychotherapy, research, teaching and supervision, consultation and administration, in hospitals, rehabilitation centres, schools, universities, courts, prisons, the armed forces, private practice and online. Table 7.1 separates the clinical psychologist from four neighbours.

Table 7.1 The clinical psychologist and neighbouring professions in India.

Profession	Training	Main work	Prescribes?
Clinical psychologist	Psychology degree, then two-year M.Phil (now the RCI MA); RCI registration	Assessment, psychodiagnosis, psychotherapy, research	No
Psychiatrist	MBBS, then MD, DNB or DPM in psychiatry	Medical diagnosis, medication, ECT	Yes
Counselling psychologist	Master's in psychology or counselling	Adjustment, career and relationship problems of relatively healthy people	No
Psychiatric social worker	MSW, then M.Phil in psychiatric social work	Family and social assessment, casework, rehabilitation	No
Psychiatric nurse	Nursing degree, psychiatric specialisation	Ward care, medicines, monitoring, psychoeducation	No

From demons to moral treatment

How people explained strange behaviour decided how they treated it. The oldest explanation, the **supernatural model**, blamed possession, sin or sorcery. Some Stone Age skulls show **trephining**, a hole chipped through the skull, perhaps to let an evil spirit out. The Greek physician **Hippocrates** (460 to 377 BC) made the first break, finding natural causes in the brain and body. His **humoral theory**, developed by the Roman physician **Galen** (about AD 129 to 198), held that health is a balance of blood, yellow bile, black bile and phlegm, and that excess black bile causes melancholia. Medieval Europe swung back to demonology, and outbreaks of mass madness such as St Vitus's dance and tarantism were taken as signs of possession. From the sixteenth century the mentally ill were confined in **asylums** that were more prisons than hospitals; London's Bethlehem Hospital, shortened to **Bedlam**, was so notorious that its name now means uproar.

Reform came at the end of the eighteenth century. In Paris the physician **Philippe Pinel** (1745 to 1826) and the superintendent **Jean-Baptiste Pussin** (1746 to 1811) removed the chains from patients at La Bicêtre (in 1793, by the usual account) and then at the Salpêtrière. In England the Quaker **William Tuke** (1732 to 1822) opened the **York Retreat** in 1796, where patients lived and worked in a calm, family-like house. Both reforms practised **moral treatment** ("moral" meaning psychological): kindness, routine, work and social contact. From the 1840s the American schoolteacher **Dorothea Dix** (1802 to 1887) persuaded legislatures to build state hospitals, but they filled faster than they could be staffed and moral treatment decayed into custodial care. Then a former patient, **Clifford Beers** (1876 to 1943), described his abuse in hospital in *A Mind That Found Itself* (1908) and, through the mental hygiene societies he founded in 1908 and 1909, launched the **mental hygiene movement** for prevention, early treatment and child guidance.

Medicine meanwhile built the **biological** (medical) **model**. **Emil Kraepelin** (1856 to 1926), in a textbook first published in 1883, treated mental disorders as distinct syndromes with their own symptoms, course and outcome, separating dementia praecox (renamed schizophrenia by **Eugen Bleuler** in 1908) from manic-depressive insanity; modern classification descends from him.

> **! EXAM TRAP**
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> Pinel gets the credit, but Pussin had begun the humane regime before Pinel arrived at Bicêtre (1793; some textbooks print 1791), persuaded him to follow, and himself banned the chains there, a step historians date to 1797. "Mental hygiene movement" usually means Beers (1908 to 1909), though some textbooks also give the name to Dix's crusade; check the date.

Witmer to Vail: a profession is born ●●

The profession began in 1896, when **Lightner Witmer** (1867 to 1956), a former student of Wundt, opened the first **psychological clinic** at the University of Pennsylvania for children with learning problems. In 1907 he founded the journal *The Psychological Clinic* and in it first used the term **clinical psychology** in print. For decades the work was mainly testing: **Alfred Binet** and **Theodore Simon** published the first practical intelligence scale in 1905, and in 1917 **Robert Yerkes** (1876 to 1956) led the testing of about 1.75 million American recruits with the **Army Alpha**, a written test for literate men, and the non-verbal **Army Beta** for those who could not read English. After the Second World War, psychological casualties far outnumbered psychiatrists, and from 1946 the **Veterans Administration** (VA) paid to train clinical psychologists as therapists.

How should they be trained? The **Boulder conference** (Colorado, 1949) chose the **scientist-practitioner model**: a PhD training clinicians equally in research and practice, so that they consume research, evaluate their own work and create new knowledge. The **Vail conference** (Colorado, 1973) endorsed the **practitioner-scholar model** and the **PsyD** (Doctor of Psychology), a professional degree centred on practice. **Richard McFall**'s 1991 "Manifesto for a Science of Clinical Psychology" set out the stricter **clinical science model**: training grounded in research, and only scientifically supported methods in practice.

The APA formed a clinical section in 1919, which became **Division 12** when the APA reorganised into divisions in 1945. The newest debate is over **prescription privileges** for psychologists trained in psychopharmacology: supporters cite shortages of psychiatrists, opponents doubt the training and fear a loss of psychological identity. New Mexico (2002) and Louisiana (2004) were the first American states to grant them; in India clinical psychologists do not prescribe.

> **! EXAM TRAP**
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> The first clinic and the first printed "clinical psychology" are Witmer's, not Wundt's (first laboratory, 1879), G. Stanley Hall's (first APA president, 1892), James McKeen Cattell's (coined "mental test", 1890), Freud's, Erikson's, Rogers's or Skinner's. Boulder 1949 = scientist-practitioner, PhD; Vail 1973 = practitioner-scholar, PsyD. Salt Lake City hosted a later conference on graduate education (1987), and Adelphi is a university; neither names a training model.

Clinical psychology in India ●●

Girindrasekhar Bose (1887 to 1953) founded the **Indian Psychoanalytical Society** at Calcutta in 1922 (Chapter 1). Formal training began at the All India Institute of Mental Health, Bangalore (founded 1954), which began training clinical psychologists in 1955 and be-

came **NIMHANS** (National Institute of Mental Health and Neuro Sciences) in 1974; the Hospital for Mental Diseases, Ranchi, now the **Central Institute of Psychiatry** (CIP), had a clinical psychology department from the late 1940s and started its diploma course in 1962. Their post-MA training in medical and social psychology became the two-year **M.Phil in Clinical Psychology**, for decades the only qualifying degree. The **Indian Association of Clinical Psychologists** (IACP) was founded in 1968.

The **Rehabilitation Council of India** (RCI) began in 1986 as a registered society; the **RCI Act 1992** made it a statutory body (from 22 June 1993), and the Act was amended in 2000. The RCI regulates rehabilitation and disability professionals, clinical psychologists among them, so it sits under the Ministry of Social Justice and Empowerment rather than the Health Ministry. It recognises courses, sets syllabi and keeps the **Central Rehabilitation Register** (CRR); practising without registration is an offence under the Act. After the National Education Policy 2020 ended the M.Phil, the RCI replaced it with a two-year professional **MA in Clinical Psychology** from 2025-26.

> **! EXAM TRAP**
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> 1968 is the IACP. NIMHANS dates from 1974 and its community psychiatry unit from 1975; it was made an Institute of National Importance by an Act of Parliament in 2012; the PsyD is the American degree from Vail. RCI: society 1986, statutory body under the 1992 Act; Ministry of Social Justice and Empowerment, not Health. Registration is mandated by the RCI Act, not the Persons with Disabilities Act 1995 or the mental health Acts.

What is abnormal, and why? The models ●●●

No single criterion marks behaviour as abnormal. The **statistical criterion** calls rare behaviour abnormal, but genius is rare too. The **social norm criterion** flags rule-breaking, but norms vary: a devotee in trance at a temple festival is following custom. The **maladaptiveness criterion** asks whether behaviour stops a person meeting daily demands, the **personal distress criterion** whether the person suffers (a manic person may feel wonderful). **Marie Jahoda** (1907 to 2001) in 1958 defined **ideal mental health** instead: positive self-attitude, growth and self-actualisation, integration, autonomy, accurate perception of reality and environmental mastery. Textbooks now condense the criteria into the **four Ds**: **deviance**, **distress**, **dysfunction** and **danger** to self or others.

A **model** then says why a disorder arose and so what will help. Beyond the supernatural and biological models, the **psychodynamic model** of **Sigmund Freud** (1856 to 1939) blames unconscious childhood conflict; the **behavioural model** maladaptive learning; the **cognitive model** faulty beliefs; the **humanistic-existential model** blocked growth and lost meaning, answered with respect, empathy, unconditional positive regard and faith in human potential; the **sociocultural model** poverty, culture and social labels; and the **family systems model** disturbed family interaction, with the patient as symptom bearer. Table 7.2 lines them up.

Table 7.2 Models of psychopathology.

Model	Key names	Cause of disorder	Treatment
Supernatural	Ancient belief	Spirits, sin, sorcery	Exorcism

Model	Key names	Cause of disorder	Treatment
Biological	Hippocrates, Kraepelin	Brain, genes, neurotransmitters	Drugs, ECT
Psychodynamic	Freud	Unconscious conflict	Insight (7.3)
Behavioural	Pavlov, Skinner	Faulty learning, reinforcement	Relearning (7.4)
Cognitive	Ellis, Beck	Irrational, distorted thinking	Change thinking (7.5)
Humanistic-existential	Rogers, Maslow, Frankl	Blocked growth, lost meaning	Growth, choice (7.6)
Sociocultural	Szasz	Poverty, culture, labels	Social change (7.9)
Family systems	Bateson, Minuchin	Family interaction	Family therapy (7.7)

Since each model explains only some cases, two integrative models dominate. In a 1977 paper in *Science*, the psychiatrist **George L. Engel** (1913 to 1999) argued that the biomedical model cannot explain why two people with the same lesion fare so differently, and proposed the **biopsychosocial model**: every illness arises from interacting biological, psychological and social factors, and treatment must address all three.

The **diathesis-stress model** explains who falls ill and when. A **diathesis** is a predisposition, usually genetic but sometimes psychological (a pessimistic style, an early loss); **stress** is the triggering event. Neither alone suffices. **Paul Meehl** (1920 to 2003) applied the idea to schizophrenia in 1962, and **Joseph Zubin** and **Bonnie Spring** formalised it in 1977 as the **vulnerability-stress model**: everyone has a threshold, and the more vulnerable you are, the less stress it takes to cross it (Figure 7.1). Adding **protective factors** (social support, coping skills, continued medication), which raise the threshold, gives the **vulnerability-stress-protective factor model** used in relapse prevention.

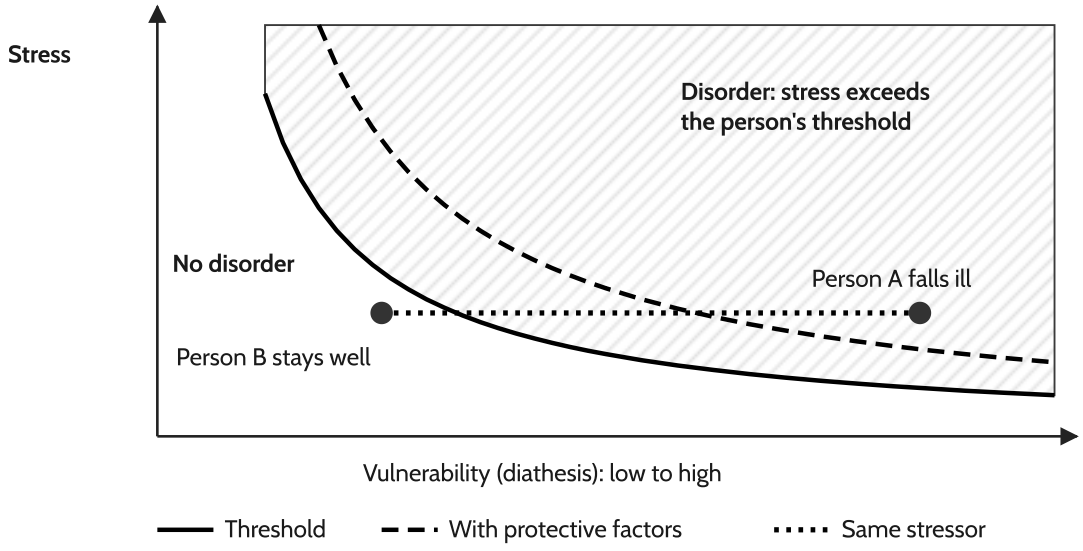


Figure 7.1 The diathesis-stress (vulnerability-stress) model. The same life event pushes highly vulnerable Person A over the threshold but not Person B; protective factors raise the threshold.

> **! EXAM TRAP**
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> "Predisposition plus trigger" is diathesis-stress, not the neurodevelopmental model (faulty early brain development in schizophrenia; Chapter 8); "clinical-reductionist" and "syndrome-sign" are not models. Single-cause options (genes, stress, learning, unconscious conflict or spirits alone) are wrong. The biopsychosocial model is Engel's, not Kraepelin's, Bleuler's, Beck's, Freud's, Walter Cannon's (fight or flight) or Richard Lazarus's (stress appraisal). Reinforcement is behavioural, never humanistic.

Ethical principles and practice ●●●

The APA's **Ethical Principles of Psychologists and Code of Conduct** (2002, amended 2010 and 2016) has two layers: five aspirational **General Principles** (A to E) that guide judgement but cannot be enforced, and ten sections of enforceable **Ethical Standards** (on competence, confidentiality, research, assessment, therapy and more). Table 7.3 pairs the principles with the four principles of biomedical ethics of **Tom Beauchamp** and **James Childress** (1979).

Table 7.3 The APA's five General Principles and Beauchamp and Childress's four.

APA General Principle	What it asks	Biomedical principle
A. Beneficence and Non-maleficence	Benefit clients; do no harm	Beneficence; non-maleficence
B. Fidelity and Responsibility	Trust; professional and social responsibility	None
C. Integrity	Accuracy, honesty, truthfulness	None
D. Justice	Fair access and equal quality of service	Justice
E. Respect for People's Rights and Dignity	Privacy, confidentiality, self-determination	Respect for autonomy

Behind any code lie two ways of reasoning. A **deontological** ethic (**Immanuel Kant**) judges an act by its intrinsic rightness: a duty such as keeping a confidence binds whatever the consequences. A **utilitarian** ethic (**Jeremy Bentham**, **John Stuart Mill**) weighs costs against benefits and chooses the greatest good for the greatest number. **Competence** means working only within one's training and supervised experience and referring beyond it; **ethical competence** means knowing the codes, recognising, analysing and resolving dilemmas, and taking responsibility for one's actions.

Informed consent is free agreement by a capable person who has been told, understandably, the nature and course of treatment, fees, the limits of confidentiality and the right to withdraw. A child cannot consent legally, so a guardian gives permission and the clinician seeks the child's **assent**. Research participants must be told the purpose, procedures, risks and benefits and their right to decline or withdraw (the consent form merely records this). Deception is allowed only when justified and must end in **debriefing**, a full explanation of the study's true purpose after participation. Research ethics go beyond legal compliance: protection from harm, confidential data, ethics committee approval and honest reporting; fabricating or manipulating data is misconduct.

Confidentiality is the clinician's ethical duty not to reveal a client's disclosures; **privileged communication** is the client's legal right to bar the clinician's testimony in court, which the client may waive. Neither is absolute. Disclosure is allowed or required with the client's consent; when there is serious risk of suicide or of harm to an identifiable person; when child abuse is suspected (India's POCSO Act 2012 makes reporting sexual offences against children

mandatory); under a court order, including evidence of guilt or innocence in a criminal case; when clients sue the clinician or put their mental state at issue; and in emergencies. The Mental Healthcare Act 2017 grants patients confidentiality with similar exceptions.

KEY STUDY: TARASOFF V. REGENTS OF THE UNIVERSITY OF CALIFORNIA (1974, 1976)

In 1969 **Prosenjit Poddar**, a graduate student from India at Berkeley, told his university therapist he intended to kill a young woman who had rejected him, readily identifiable as **Tatiana Tarasoff**. Campus police questioned and released him; no one warned her, and two months later he killed her. The California Supreme Court ruled in 1974 that therapists have a **duty to warn** an identifiable victim, and in 1976 widened it to a **duty to protect** (warning, informing police, hospitalisation): "The protective privilege ends where the public peril begins." Critics fear it deters dangerous clients from speaking openly.

! EXAM TRAP

Duty to warn comes from Tarasoff. The M'Naghten (1843) and Durham (1954) rules are insanity tests, and the Daubert standard (1993) governs expert testimony (7.10). "There are never exceptions to confidentiality" is always false.

A **multiple (dual) relationship** exists when a psychologist in a professional role with someone also holds another role with that person or someone close to them, or promises one: treating a friend, employee or relative, or doing business with a client. Under APA Standard 3.05 it is unethical when it risks impaired objectivity, exploitation or harm. Co-therapy by two psychologists, a client seeing two therapists or a psychologist working in two clinics is not a dual relationship. **Sexual intimacy** with a current client is forbidden, and with a former client for at least two years after therapy and even then only in the most unusual circumstances. **Thomas Gutheil** and **Glen Gabbard** (1993) distinguished a **boundary crossing**, a harmless departure from the usual frame (a brief self-disclosure, seeing a client at the workplace for exposure work), from a **boundary violation**, which exploits or harms the client to serve the therapist, such as a romantic relationship or business deal.

Finally, **records** must be accurate, secure and open to clients. In **assessment**, psychologists use valid tests, explain results, release **test data** (raw scores and responses) to the client or whoever the client names, and guard **test security** by never releasing manuals or items to untrained people. **Teletherapy** requires technical competence, secure platforms, consent covering privacy risks, confirming the client's location and an emergency plan. Trainees work under **supervision**, clients are told so, and the supervisor carries responsibility. The RCI Act lets the Council prescribe a code of ethics, and misconduct can remove a professional from its register.

! EXAM TRAP

A romance with a current client is a boundary violation; rapport and convenient appointments are just good practice. The APA has five General Principles; "equity" is not one. Integrity is APA's, not one of Beauchamp and Childress's four (autonomy, non-maleficence, beneficence, justice). Non-maleficence is refraining from harm; doing good is beneficence. Consent comes before a study, debriefing after.

Indian mental health law and policy ●●●

The **Indian Lunacy Act 1912** was custodial, built around detaining "lunatics" in asylums. The **Mental Health Act 1987** (in force 1993) modernised terms and licensing but gave few rights, exempted government hospitals from licensing and left attempted suicide a crime under section 309 of the Indian Penal Code. The **Mental Healthcare Act 2017** (MHCA, in force 2018) replaced it with a rights-based law aligned with the UN Convention on the Rights of Persons with Disabilities.

- **Rights.** A **right to access mental healthcare** (section 18) from government services, free below the poverty line, with rights to community living, dignity, equality (including insurance parity with physical illness, section 21(4)), confidentiality, information, legal aid and medical records.
- **Choices.** Every person, including one with mental illness, is presumed capable (section 4), and treatment needs informed consent. An adult may write an **advance directive** (section 5) on how to be treated, or not treated, in a future illness and appoint a **nominated representative** (NR, section 14). A minor can do neither; a minor's NR is the legal guardian unless the Board orders otherwise (section 15). Under section 87 a minor is admitted only on the NR's application after two independent examinations, with the minor's wishes taken into account, is kept apart from adults with the NR or an attendant always present, and is treated with the NR's consent; the Board is informed within 72 hours and must review any stay beyond 30 days.
- **Prohibitions (section 95).** No **ECT** without anaesthesia and muscle relaxant (unmodified ECT is never lawful, and section 94 bars any ECT as emergency treatment); no ECT for minors unless a psychiatrist judges it necessary, the guardian consents and the Board gives prior permission; no sterilisation as treatment; no chaining. Section 97 also bans seclusion and solitary confinement and allows physical restraint only as a last resort authorised by the psychiatrist.
- **Bodies.** The **Central Mental Health Authority** and **State Mental Health Authorities** register all mental health establishments, government or private, and professionals; **Mental Health Review Boards**, each for a district or group of districts, chaired by a serving or retired District Judge and including a psychiatrist, a medical practitioner and two service-user or carer members (section 74), register advance directives, appoint NRs, review admissions and hear complaints.
- **Section 115.** Anyone who attempts suicide is presumed, unless proved otherwise, to have **severe stress** and is not punished; in effect, decriminalisation, with a government duty to provide care.
- **Professionals.** Section 2 defines a clinical psychologist by an RCI-recognised qualification in clinical psychology, or a postgraduate degree plus a two-year M.Phil in Clinical or Medical and Social Psychology with supervised clinical training; registered with the State Authority, the clinical psychologist is a **mental health professional** under the Act.

The **Rights of Persons with Disabilities Act 2016** (RPwD, in force 2017) replaced the **Persons with Disabilities Act 1995** (7 disabilities) with 21 specified disabilities, including mental illness, intellectual disability, autism spectrum disorder and specific learning disabilities. At least 40 percent of a specified disability is a **benchmark disability**, qualifying for reservation (4 percent of government posts, 5 percent of higher education seats). Table 7.4 adds the other laws, programmes and surveys.

Table 7.4 Indian mental health law, policy and programmes.

Year	Law, programme or survey	Remember
1982	**National Mental Health Programme** (NMHP)	Minimum mental healthcare for all through primary care
1985	**NDPS Act** (Narcotic Drugs and Psychotropic Substances)	Controls drugs of abuse; addicts who seek treatment are spared prosecution
1996	**District Mental Health Programme** (DMHP)	District services on the **Bellary model**, a NIMHANS pilot in Karnataka
1999	**National Trust Act**	Welfare and guardianship for autism, cerebral palsy, mental retardation, multiple disabilities
2014	**National Mental Health Policy**	India's first
2015-16	**National Mental Health Survey** (NIMHANS)	Adults: current prevalence 10.6 percent; **treatment gap** 70 to 92 percent
2022	**Tele-MANAS**	Free 24-hour tele-mental-health helpline (14416)

! EXAM TRAP

Advance directives, NRs, Review Boards, section 115 and the ECT ban belong to the MHCA 2017, never the 1987 Act, the RPwD Act or the RCI Act. RPwD = 2016 and 21; PwD = 1995 and 7. A 2024 paper keyed "a minor has the right to appoint a nominated representative, and their opinion must be considered". Pick it, since the other options are false, but only its second half is exact: a minor's NR is the legal guardian, and admitting a minor needs no prior Board consent. The NR cannot decide "irrespective" of the minor, whose views must be weighed, and a minor can be admitted on the NR's application even when unwilling. ECT is never allowed as emergency treatment.

Practising in India

Trained in an RCI-recognised programme with supervised hospital work, registered with the RCI and working as a scientist-practitioner, the Indian clinical psychologist also needs **cultural competence**, the awareness, knowledge and skills to work across cultures. Families usually share treatment decisions, so the clinician works with them; stigma delays help-seeking, and many people first visit faith healers, temples and dargahs, beliefs to respect while explaining what treatment offers. With a treatment gap above 70 percent, **task-shifting**, training lay counsellors and health workers to deliver simple psychological treatments under supervision, has become central (7.9).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The first psychological clinic was opened in 1896 by _____.
2. The 1949 _____ conference adopted the scientist-practitioner model.
3. The biopsychosocial model was proposed in 1977 by _____.
4. The client's legal right to bar a therapist's testimony in court is _____ communication.
5. The duty to protect an identifiable victim comes from the _____ case.
6. Section _____ of the Mental Healthcare Act 2017 presumes severe stress in a person who attempts suicide.

Answers: (1) Lightner Witmer, (2) Boulder, (3) George L. Engel, (4) privileged, (5) Tarasoff, (6) 115.

7.2 The Clinical Interview, Mental Status Examination and Formulation

LOWER YIELD · 11 QS

Ravi, 24 (a composite case), has failed the bank examination for the third time, in the week his cousin was selected, and has told his mother that everyone would be better off without him. The clinician will interview him, take his history, examine his mental state, judge his safety and build a formulation: the steps this section follows.

Purposes and types of clinical interview ○

The **clinical interview** is a conversation with a professional purpose: it gathers information, tests hypotheses and begins a working relationship at once. Interviews are named by purpose: the **intake interview** (first contact: why now, how urgent?), the case-history interview, the **diagnostic interview** (symptoms checked against criteria), the mental status examination, the **crisis interview** (brief, active, focused on safety; Section 7.9) and the **termination interview** (reviewing progress and planning relapse prevention).

They also differ in structure. In an **unstructured interview** the clinician chooses the questions: flexible and good for rapport, but two clinicians may reach different diagnoses. A **semi-structured interview** fixes topics and core questions but lets a trained clinician rephrase, probe and judge. A **fully structured interview** fixes every word and scoring rule, so a lay interviewer or computer can give it. Structure buys **reliability** (interviewers agree) at the cost of flexibility and rapport, but not **validity**: interviewers can agree and still be wrong, and a schedule misses what it does not ask. Clinicians often open freely, then use a schedule (Table 7.5).

Table 7.5 The main diagnostic interview schedules.

Instrument	Structure	Use
SCID-5, Structured Clinical Interview for DSM-5 (First, Spitzer and colleagues)	Semi-structured, despite its name	DSM diagnosis
Present State Examination (PSE; Wing, 9th edition 1974); its WHO successor SCAN	Semi-structured	ICD diagnosis; cross-national research

Instrument	Structure	Use
Diagnostic Interview Schedule (DIS; Robins, 1981)	Fully structured, lay interviewers	Community surveys
MINI, Mini International Neuropsychiatric Interview (1998)	Structured, about 15 minutes	Rapid screening
K-SADS, Kiddie Schedule for Affective Disorders and Schizophrenia	Semi-structured, parent and child	Ages about 6 to 18
ADIS-5, Anxiety Disorders Interview Schedule (Brown and Barlow)	Semi-structured	Anxiety disorders, with severity

Related interview and observation instruments of developmental assessment are the ADI-R (Autism Diagnostic Interview-Revised, with caregivers), the ADOS-2 (Autism Diagnostic Observation Schedule) and the Vineland-3 (adaptive behaviour, by caregiver interview); see Chapter 2.

> **! EXAM TRAP**
>
> "A structured interview" means the **SCID**. The **MMSE** (Mini-Mental State Examination, 1975) is a 30-point cognitive screen, not an interview and not the MSE; the **WAIS** is an intelligence test; the **TAT** a projective test.

Interviewing skills ●●

Allen Ivey's **microcounselling** (from 1968) teaches the craft one microskill at a time. **Rapport**, the trust that makes disclosure possible, grows from explaining the purpose and the limits of confidentiality and starting with neutral topics. **Attending behaviour** shows that you are listening; **Gerard Egan** (*The Skilled Helper*, 1975) summed it up as **SOLER**: face the client **S**quarely, **O**pen posture, **L**ean forward, **E**ye contact, **R**elaxed, though steady eye contact feels intrusive to many Indian clients.

An **open question** ("Tell me about...") invites a long answer in the client's own words; a **closed question** ("When did the sleep go?") pins down a fact. Move from open to closed. Table 7.6 shows the other skills. **Reflection of feeling** is the signature technique of **Carl Ransom Rogers** (1902 to 1987) in client-centred therapy (Section 7.6); interpretation belongs chiefly to psychodynamic therapy (Section 7.3). **Empathy**, understanding the client's world from inside and showing it, has two levels (Egan): primary empathy reflects what is said; advanced empathy voices what is only hinted, so it waits for trust.

Table 7.6 Microskills. Ravi says: "My father keeps comparing me to my cousin. I don't even go home for meals now."

Skill	What it does	Example
Minimal encourager (furthering)	Keeps him talking, without steering	A nod, leaning forward, "Mm-hmm"
Paraphrase	Restates content in different words	"Being compared makes home hard."
Reflection of feeling	Names the emotion	"You sound hurt."
Clarification	Makes the vague precise	"What does he actually say?"
Checking	Confirms your understanding	"You skip meals to avoid him; is that right?"

Skill	What it does	Example
Summarising	Pulls threads together	"The result, your cousin, your father: all at once."
Confrontation	Gently notes a discrepancy	"You say you don't care, yet you avoid home."
Interpretation	Offers a new meaning	"Avoiding him spares you feeling you failed him."
Silence	Gives room to feel	A pause after pain

> ! EXAM TRAP
>
> Nodding and "umm" are **furthering**, not **prompting** (steering to a topic: "What about work?") or **checking**. Restating **in different words** is **paraphrasing**, not questioning, empathy (the broader capacity) or nonverbal communication. "Tell me about" is an **open-ended question**, even inside an ethnographic interview (learning a culture in the insider's terms) or theory-guided inquiry (questions chosen from a theory).

Common errors are **leading questions** ("You've been depressed, haven't you?"), **double-barrelled questions** (two in one), accusing "why" questions, premature reassurance and premature closure on an early hypothesis. With children, use play, drawing and concrete words and add parents' and teachers' reports; with older adults, check hearing and vision before inferring confusion (Section 7.10); with resistant clients, acknowledge that coming was not their idea; with suicidal clients, ask directly, since asking does not plant the idea. Across cultures, elicit the client's **explanatory model** (his name for the problem, its cause, the cure he expects), as **Arthur Kleinman** (1980) urged, and respect **idioms of distress** such as "tension". Use a trained interpreter, never a relative or child.

The case history

The case history, the story of person and illness, follows a fixed order:

1. **Identifying data** (age, education, occupation, marital status, religion, socioeconomic status), then the **informant** and the account's reliability (relationship, closeness of contact, consistency).
2. **Chief complaints**, in the patient's own words, each with its duration ("Low mood, 3 months").
3. **History of present illness**: onset (abrupt, acute or insidious), course, **precipitating factors** and treatment so far.
4. **Past psychiatric and medical history**, and **family history** drawn as a **genogram** (three generations; squares for males, circles for females).
5. **Personal history**: birth and development, education, occupation, marital and sexual life, substance use and forensic record.
6. **Premorbid personality**: the person before the illness, the baseline for judging change.

The MSE: behaviour, mood, thought and perception ○

The **mental status examination** (MSE) is a snapshot of mental functioning during the interview. It records **appearance and behaviour**, **attitude towards the examiner** (cooperative, guarded, hostile), **psychomotor activity** (retardation, agitation) and **speech** (rate, volume, tone, reaction time, relevance, coherence). **Mood** is the sustained emotion the patient re-

ports: subjective, the climate. **Affect** is the emotion the examiner observes: objective, the weather. Affect is rated for range, reactivity, appropriateness and stability: it may be **blunted** (markedly reduced), **flat** (virtually absent), **labile** (shifting rapidly) or **inappropriate** (laughing about a death).

Thought has four headings: **stream** (speed and flow), **form of thought** (thought process: how ideas connect; Table 7.7), **content of thought** (what is thought: delusions, obsessions, overvalued ideas, phobias, suicidal or homicidal ideas) and **possession** (whose thoughts they feel to be). A **delusion** is a false, unshakeable belief out of keeping with the person's culture: persecutory, of reference (chance events carry a personal message), grandiose, nihilistic (the self, body or world does not exist), of jealousy, erotomanic (a person of higher status is secretly in love with one), of guilt or somatic (the body is diseased or changed). An **overvalued idea** is an understandable belief held with excessive emotion but without delusional fixity. An **obsession** is an intrusive thought recognised as one's own and resisted (Indian case formats file obsessions under possession, as lost control of thinking); a **phobia** is an excessive fear of a specific object, situation or idea. In **thought alienation** thoughts feel inserted, withdrawn or broadcast; these are among the first-rank symptoms of schizophrenia listed by **Kurt Schneider** (1887 to 1967; Chapter 8).

Table 7.7 Disorders of the stream and form of thought.

Term	What you hear	Typical of
Flight of ideas	Rapid jumps by rhyme or chance; links traceable	Mania
Loosening of associations	Slips to unrelated ideas; links lost	Schizophrenia
Tangentiality	Veers off; never reaches the point	Schizophrenia, mania
Circumstantiality	Needless detours, but the point is reached	Epilepsy, obsessional traits
Thought block	Sudden stop; mind goes blank	Schizophrenia
Perseveration	Same answer repeated to new questions	Brain damage
Neologism; word salad	New words; words strung without connection	Schizophrenia
Poverty of speech	Very little speech, brief replies	Schizophrenia, depression

> ! EXAM TRAP
>
> **Circumstantiality** is a disorder of the **form of thought** (thought process), not content, possession or perception (some notes file it under stream). The circumstantial patient arrives late; the tangential one never arrives.

A **hallucination** is a perception without an external stimulus, felt as real: auditory ones are commonest in schizophrenia, visual ones suggest delirium, olfactory and gustatory ones temporal lobe epilepsy, tactile ones cocaine use or alcohol withdrawal; those at sleep onset (hypnagogic) or waking (hypnopompic) are normal. An **illusion** misperceives a real stimulus (a rope seen as a snake). A **pseudohallucination** is perceived in inner space ("inside my head"). In **depersonalisation** the self feels unreal; in **derealisation**, the surroundings.

The MSE: cognition, insight and judgement ○

Cognitive functions are tested in order: **consciousness** (alert to coma); **orientation** to time, place and person (time is lost first); **attention and concentration** (digit span, **serial sevens** from 100, months backwards); **memory**, immediate (repeat three words), recent (recall them after five minutes) and remote (childhood events); **intelligence and general information**; **abstract thinking** (proverbs, similarities: literal answers are concrete); and **judgement**, personal (plans), social (conduct) and test ("What would you do if you saw smoke in a cinema?").

Insight is the patient's awareness and understanding of his own illness, graded on six levels: (1) complete denial; (2) slight awareness of illness while still denying it; (3) awareness, but blamed on others or physical causes; (4) awareness of something unknown within oneself; (5) **intellectual insight**, accepting the illness without changing behaviour; (6) **true emotional insight**, awareness that leads to change.

> **! EXAM TRAP**
>
> Insight is **awareness of one's own illness**, even where a case format files it among the cognitive functions. It is not orientation, consciousness, immediate recall, serial subtraction (attention) or hallucinations (perception).

Write the MSE in the present tense, describing rather than labelling. For Ravi: *Slowed; rapport difficult. Mood "useless"; affect depressed, restricted. Hopelessness, passive death wishes, no plan. Oriented. Insight grade 4.*

Risk assessment: suicide and violence

Suicide risk factors include a previous attempt (the strongest predictor), depression, substance use, schizophrenia, hopelessness, male sex (women attempt more often), living alone, chronic illness, recent loss and access to means such as pesticides. **Protective factors** include family support, responsibility for children, religious objections and reasons for living. The **SAD PERSONS** mnemonic (Patterson and colleagues, 1983) scores one point each for Sex (male), Age (under 19 or over 45), Depression, Previous attempt, Ethanol abuse, Rational thinking loss, Social support lacking, Organised plan, No spouse and Sickness. It aids memory but predicts poorly: suicide is rare, so scales raise many false alarms yet miss many who act. Ask instead about ideation, intent, plan and means, and build a safety plan.

For violence, past violence is the best predictor, with youth, male sex, substance misuse and threats. Structured professional judgement such as the **HCR-20** (Historical, Clinical and Risk management items) beats unaided judgement; dynamic factors, unlike static history, can be treated.

Behavioural assessment ○

Behavioural assessment asks what a behaviour does for the person here and now. Its core, **functional analysis**, rests on the operant conditioning of **Burrhus Frederic Skinner** (1904 to 1990): consequences maintain behaviour. The target behaviour is defined operationally and its **ABC** recorded: **A**ntecedent, **B**ehaviour, **C**onsequence. A boy who throws his glass when told to clear it, and so escapes the chore, is being reinforced. The **SORC** model adds **O**rganism variables (biology, thoughts, learning history) between **S**timulus and **R**esponse.

Data come from **behavioural interviews** (what, when, how often), **self-monitoring** (the client's own diary, essential for private acts such as purging), **role play** (the client enacts a problem situation, such as refusing a request, while the clinician watches), and observation that is **naturalistic** (home, school), **analogue** (a simulated clinic situation) or **participant** (by a parent, teacher or nurse already present). **Event recording** counts occurrences; **interval recording** notes whether the behaviour occurred in each short interval; **time sampling** observes at chosen moments. **Reactivity** is behaviour changing because it is observed; **observer drift** is observers' gradual shift in applying definitions, corrected by retraining.

> ! EXAM TRAP
>
> Functional analysis is based on **operant conditioning**, not classical conditioning (Section 7.4), humanistic or psychodynamic principles. Its ABC is **Antecedent, Behaviour, Consequence**; Albert Ellis's ABC is **Activating event, Belief, Consequence** (Section 7.5).

Case formulation ○

A **diagnosis** places problems in a category: what the person has. A **case formulation** is an individualised, testable hypothesis about why this person has these problems now (causes, triggers, maintaining factors), revised as therapy proceeds. The commonest framework is the **5 Ps**: **presenting** problem, **predisposing** factors (early vulnerabilities), **precipitating** factors (triggers), **perpetuating** factors (what maintains it) and **protective** factors (strengths), as Figure 7.2 shows for Ravi. Filling each P with biological, psychological and social entries gives a **biopsychosocial formulation**, after **George Libman Engel** (1913 to 1999), who proposed the model in *Science* in 1977.

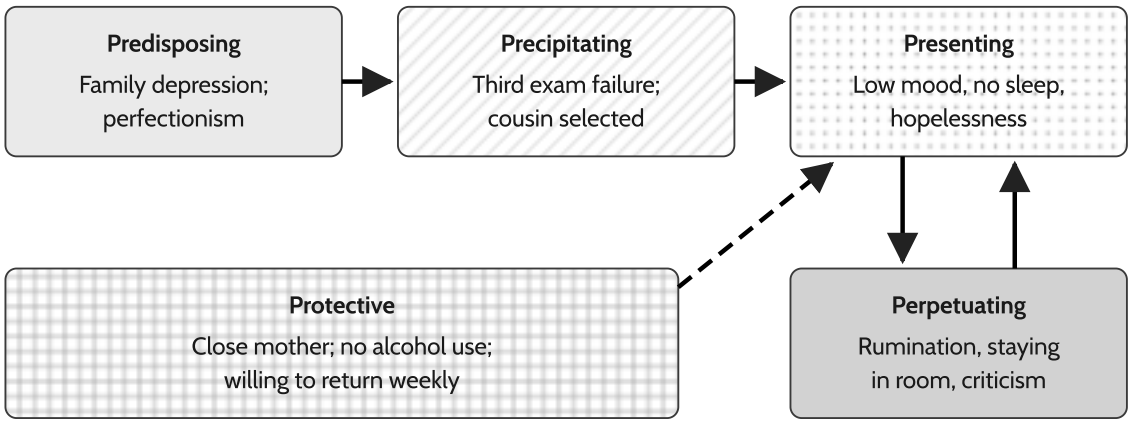


Figure 7.2 The 5 Ps applied to Ravi. Solid arrows run from vulnerability through trigger to the problem, which loops with its perpetuating factors; the dashed arrow marks protection.

In a cognitive-behavioural formulation, the **cognitive conceptualisation diagram** of **Judith S. Beck** (1995) links childhood data to **core beliefs** ("I am a failure"), rules ("If I fail the exam, I am worthless") and compensatory strategies, and traces each situation through automatic thought, meaning, emotion and behaviour. A psychodynamic formulation describes core unconscious conflicts, defences and repeating relationship patterns rooted in early life.

A **differential diagnosis** lists the conditions that could explain the picture and rules them in or out, physical and substance causes first (hypothyroidism can mimic depression). The **clinical report** answers the referral question in plain language (sources, history, MSE, results, formulation, diagnosis, recommendations); **feedback** begins with strengths and ends with an agreed plan.

> **! EXAM TRAP**
>
> A formulation **explains**: it hypothesises causes, precipitants and maintaining factors. Classifying symptoms by DSM-5 is **diagnosis**, predicting course is **prognosis**, and ruling out organic causes is **differential diagnosis**.

Classification as a tool ○

Classification gives clinicians a shared language for communication, treatment, research and statistics (Table 7.8; disorders in Chapter 8).

Table 7.8 DSM-5-TR and ICD-11 compared.

Feature	DSM-5-TR	ICD-11
Publisher	American Psychiatric Association	World Health Organization
Dates	DSM-5 2013; text revision 2022	Adopted 2019; in force 2022
Scope	Mental disorders only	All diseases; mental disorders in Chapter 6
Structure	Three sections: basics; criteria and codes; emerging measures	Codes such as 6A20 (schizophrenia)
Style	Operational criteria; no multiaxial system	Flexible diagnostic requirements

Diagnostic agreement is measured by **Cohen's kappa** (**Jacob Cohen**, 1923 to 1998; 1960), which corrects raw agreement for chance: $\kappa = (P_o - P_e) / (1 - P_e)$. Kappa is 1 for perfect agreement and 0 for chance level; above about 0.6 is good. Even with explicit criteria, the DSM-5 field trials found kappa near 0.28 for major depressive disorder.

Labels can stigmatise, and they reflect values: **pathologisation** means classifying a condition as a mental disorder and **depathologisation** removing it, as when the American Psychiatric Association removed homosexuality from DSM-II in 1973 (the WHO followed in ICD-10, 1990).

> **KEY STUDY: ROSENHAN, "ON BEING SANE IN INSANE PLACES" (1973)**
>
> **David L. Rosenhan** (1929 to 2012) had eight sane volunteers, himself among them, seek admission to 12 US hospitals reporting voices saying "empty", "hollow" and "thud", then behaving normally. All were admitted, all but one diagnosed with schizophrenia, and stayed 7 to 52 days; note-taking was charted as "writing behavior". It became the classic case against labelling. **Robert Leopold Spitzer** (1932 to 2015) replied (1975) that faked symptoms say nothing about diagnosis; later inquiry questioned the study's conduct.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Nodding, leaning forward and saying "umm" are _____ responses.
2. The SCID, though named "structured", is strictly _____.
3. Reaching the point after needless detours is _____, a disorder of the _____ of thought.
4. Mood is _____; affect is _____.
5. Grade 6 insight is _____ insight.
6. Functional analysis records Antecedent, Behaviour and _____, and rests on _____ conditioning.
7. The five Ps are presenting, predisposing, precipitating, _____ and protective.

Answers: (1) furthering, (2) semi-structured, (3) circumstantiality; form, (4) subjective; objective, (5) true emotional, (6) Consequence; operant, (7) perpetuating.

7.3 Psychodynamic Therapies and Their Descendants HIGH YIELD · 39 QS

Every therapy in this section assumes that hidden forces pushing against one another (hence *dynamic*) shape what we feel and do, and that bringing them into awareness relieves suffering. The idea began with one patient in Vienna, became Freud's psychoanalysis, and then branched into a large family.

Freud's psychoanalysis as a treatment ●●●

KEY STUDY: BREUER'S ANNA O. (1880 TO 1882)

The physician **Josef Breuer** (1842 to 1925) treated Bertha Pappenheim, a young woman with hysterical paralyses and weeks in which she could not drink water. When, in a self-induced trance, she talked through the moment a symptom began and released the feeling she had swallowed then, the symptom eased: her "talking cure". Breuer and Freud published her as "Anna O." in *Studies on Hysteria* (1895). Criticism: she was not in fact cured, and spent months in sanatoria afterwards.

Breuer's procedure was the **cathartic method**: hypnotise the patient, take her back to the moment a symptom began and let her relive it. **Abreaction** is the discharge of the emotion attached to a particular repressed memory by reliving it; **catharsis** (Greek, "purging") is the release and relief that follow. **Sigmund Freud** (1856 to 1939), the father of psychoanalysis, then dropped hypnosis: many patients could not be hypnotised, its cures faded, and it went round resistance instead of exposing it. At Nancy in 1889 he had watched **Hippolyte Bernheim** (1840 to 1919) recover "forgotten" memories from waking patients.

The aim of analysis is to make the unconscious conscious, or, in Freud's formula of 1933, "Where id was, there ego shall be". Change comes through **insight**, an emotionally felt understanding of how present feelings repeat past conflicts, and the target is character, not just the symptom. The patient lies on a couch with the analyst out of sight behind, four or five times a

week for years; once- or twice-weekly, face-to-face work on the same principles is psychodynamic psychotherapy. Every analyst first undergoes a training analysis.

The patient's one duty is the **fundamental rule**: say whatever comes to mind, however trivial or embarrassing, without censoring it. Following the rule is **free association**, the defining technique of psychoanalysis. When conscious steering relaxes, Freud argued, thoughts follow paths laid down by unconscious wishes, and the places where association stalls point to what is repressed. The analyst brings evenly suspended attention; **neutrality**, taking no side in the patient's conflicts and giving no advice (the "blank screen"); and **abstinence**, declining to gratify the patient's wishes for love, praise or comfort so that they stay alive to be understood. Abstinence here has nothing to do with alcohol.

Dreams, Freud wrote in *The Interpretation of Dreams* (1900), are "the royal road" to the unconscious. The **manifest content**, the dream as remembered, disguises the **latent content**, the hidden wish, through the dream-work: condensation, displacement, symbolisation and secondary revision. The patient free-associates to each element to recover it.

Sooner or later the patient arrives late, forgets a session or changes the subject just as something painful approaches. This is **resistance**: everything in the patient that opposes the treatment and access to unconscious material. It is repression showing itself in the room, so in **analysis of resistance** the analyst points out the obstruction and explores with the patient what it protects against. **Interpretation** links what the patient says or does to its unconscious meaning; it is prepared by confrontation and clarification and aimed at the defence before the content it hides. Because one insight rarely changes a lifelong pattern, **working through** (Freud, 1914) re-examines the same conflicts and interpretations in context after context until lasting change in behaviour and character follows.

> **REMEMBER**
>
> Origin of disorder by school: psychodynamic, unconscious conflict from childhood emotional experience; behavioural, faulty learning (Section 7.4); cognitive, disturbed thought patterns (Section 7.5); humanistic, failure to fulfil one's potential (Section 7.6).

> **! EXAM TRAP**
>
> Saying whatever comes to mind, uncensored, is free association; systematic desensitisation, flooding and modelling are behaviour therapy (Section 7.4), and paradoxical intention is Frankl's and Adler's. Obstructing the analysis while the therapist explores why is analysis of resistance. Working through is never "giving the interpretation once", advice or ending after one insight.

Transference and countertransference ●●●

In the Dora case (1905) Freud saw that patients come to feel about the analyst as they once felt about their parents. **Transference** is the patient's unconscious redirection onto the therapist of feelings and expectations that belong to significant figures from the past. It may be positive (affection, idealisation, erotic love) or negative (hostility, suspicion). When the patient's neurosis is re-created around the analyst, it becomes a **transference neurosis**, whose resolution is, in classical theory, the cure.

Countertransference, named by Freud in 1910, is the therapist's emotional reaction to the patient arising from the therapist's own unresolved conflicts, a blind spot to be recognised (hence the training analysis). Since **Paula Heimann**'s paper of 1950 the term also covers all the therapist's feelings towards the patient, used as information about the patient. **Negative countertransference** is the therapist's hostility, irritation or boredom towards a patient; positive countertransference is over-involvement or a wish to rescue. Both differ from the working alliance (**Ralph Greenson**, 1965), the patient's rational collaboration with the therapist (Section 7.8).

> **! EXAM TRAP**
>
> Direction decides the term: patient to therapist is transference; therapist to patient is countertransference. "The therapist's emotional reaction to the client" is never transference, "the client's projection onto the therapist" is never countertransference, and a therapist's negative feelings are negative countertransference, not negative transference. Fear of the unknown or low motivation are sources of resistance, not countertransference. Managing both is the hallmark of psychodynamic models, not of behaviour modification, CBT or solution-focused therapy.

Defences and psychic energy ●●

In Freud's language of psychic energy, **libido** is the energy of the sexual (later, life) drive; **cathexis** is its investment in the mental representation of a particular object, person or idea; **fixation** is libido left attached to an earlier psychosexual stage; and **sublimation** redirects a drive into a socially valued aim. **Defence mechanisms** are the ego's unconscious strategies against anxiety. Their theory is in Chapter 4; Table 7.9 shows how a clinician meets them.

Table 7.9 Defences as the clinician meets them.

Defence	What happens	In the room
Repression	Threatening wish or memory kept out of awareness; the primary defence	A painful year cannot be recalled
Projection	Own unacceptable feeling attributed to another	Hostile patient sure the therapist dislikes him
Displacement	Feeling moved from a dangerous target to a safer one	Scolded by the boss, he shouts at his son
Reaction formation	Impulse turned into its exaggerated opposite	Gushing sweetness to a resented sister
Rationalisation	Respectable reasons for behaviour with other motives	"I failed because the paper was unfair"
Isolation of affect	Idea kept, its feeling split off	Flat account of a father's death
Splitting	Self or others seen as all good or all bad	Therapist idealised, then despised
Suppression	Conscious postponement of a worry	"I will think about it after the exam"

Repression is also the usual psychodynamic explanation of dissociative symptoms, and conversion turns a psychological conflict into a bodily symptom, as in Anna O.'s paralysis.

> **! EXAM TRAP**
>
> Energy invested in a particular object, person or idea is cathexis, not libido, fixation or sublimation. Projection gives your feeling to someone else; displacement keeps it yours but moves its target. The primary defence, and the psychodynamic source of dissociation, is repression, not conversion, isolation or "ego state" (Paul Federn's term, later Berne's).

Ego psychology, object relations and self psychology ●●

Freud's successors kept the unconscious but changed what they listened for. **Ego psychology** turned from the id to the ego. **Anna Freud** (1895 to 1982), in *The Ego and the Mechanisms of Defence* (1936), made the analysis of defences central and pioneered child analysis; **Heinz Hartmann** (1894 to 1970) described a conflict-free ego sphere that develops for adaptation (1939); and **Erik Erikson** (1902 to 1994) added psychosocial stages and the adolescent identity crisis (Chapters 4 and 6).

Object relations theory holds that the mind is built from internalised representations of self and others ("objects") and of the relationships between them. **Melanie Klein** (1882 to 1960) analysed children through play. In her paranoid-schizoid position the infant manages terror by **splitting** the world into wholly good and wholly bad objects and by projective identification (placing unwanted parts of the self in another and then controlling that person); in the later depressive position good and bad are seen as one person, bringing guilt and reparation. Paranoia revives the first position, so its defence is splitting. **Donald Winnicott** (1896 to 1971) described the **good-enough mother**, closely attuned at first and then failing tolerably, who lets a true self grow; the **transitional object**, a blanket or soft toy that bridges inner and outer reality; and therapy as a **holding environment**, as reliable as a mother's arms. **Ronald Fairbairn** (1889 to 1964) held that libido is object-seeking, not pleasure-seeking.

Margaret Mahler (1897 to 1985) traced **separation-individuation**: a normal autistic phase, a symbiotic phase of fusion with the mother, then differentiation, practising, rapprochement and object constancy by about three years. Her theory of psychosis (1952) traces schizophrenia to these first phases: the person remains in the autistic phase and fails to develop an adequate, stable symbiosis (Silvano Arieti, Ping-Nie Pao and Edith Jacobson wrote other analytic accounts of psychosis). **Otto Kernberg** (born 1928) described borderline personality organisation (identity diffusion, splitting, intact reality testing) and developed **transference-focused psychotherapy** (TFP), which interprets the split images of self and others as they appear in the transference.

Heinz Kohut (1913 to 1981) founded **self psychology** with *The Analysis of the Self* (1971). The child needs selfobjects who provide **mirroring** (confirming the child's worth), an **idealised** figure to look up to and merge with and, in his later work, **twinship** (feeling like others). Chronic failure of mirroring and idealising leaves a fragile self that fragments into emptiness and rage. In therapy these needs return as mirror, idealising and twinship transferences, met by the analyst's **empathic attunement**.

Relational psychoanalysis (**Stephen Mitchell**, 1946 to 2000, with Jay Greenberg, 1983) treats therapy as a two-person process in which the analyst cannot be neutral. **John Bowlby**'s (1907 to 1990) attachment theory inspired attachment-based therapies in which the therapist is a secure base for revising internal working models. **Mentalisation-based treatment** (MBT), by **Anthony Bateman** and **Peter Fonagy** for borderline personality disorder (first trial 1999),

strengthens mentalisation, understanding one's own and others' behaviour in terms of feelings, wishes and intentions, a capacity that collapses under attachment stress.

> **! EXAM TRAP**
>
> "Object relations" means internalised representations of self and others and their relationships, not the id-ego-superego relation, defences or the genital stage. A client who feels empty, "no one", afraid of abandonment and hungry for admiration is explained by failed parental mirroring and idealising leading to a fragmented self: that is Kohut's self psychology even when the stem says "object relations". Oral fixation, an unresolved Oedipus complex and a harsh superego are classical Freudian explanations of neurotic conflict and guilt (Chapter 4). For paranoia "as given by Klein", answer splitting; projection is Freud's classical explanation of paranoia.

Brief psychodynamic psychotherapies

Brief dynamic therapies keep the psychodynamic ideas but use one agreed focus, a limited number of sessions (usually 12 to 40), an active therapist who questions and confronts instead of waiting for free association, and early work on the transference. **David Malan** of the Tavistock Clinic (*The Frontier of Brief Psychotherapy*, 1976) supplied their map (Figure 7.3): the **triangle of conflict** (hidden feeling, anxiety, defence) and the **triangle of person** (others now, the therapist, parents then). Table 7.10 lists the models of Malan, **Peter Sifneos** (1920 to 2008), **Habib Davanloo** (born 1927), **James Mann** (1973), **Lester Luborsky** (1984), and **Hans Strupp** and Jeffrey Binder (1984).

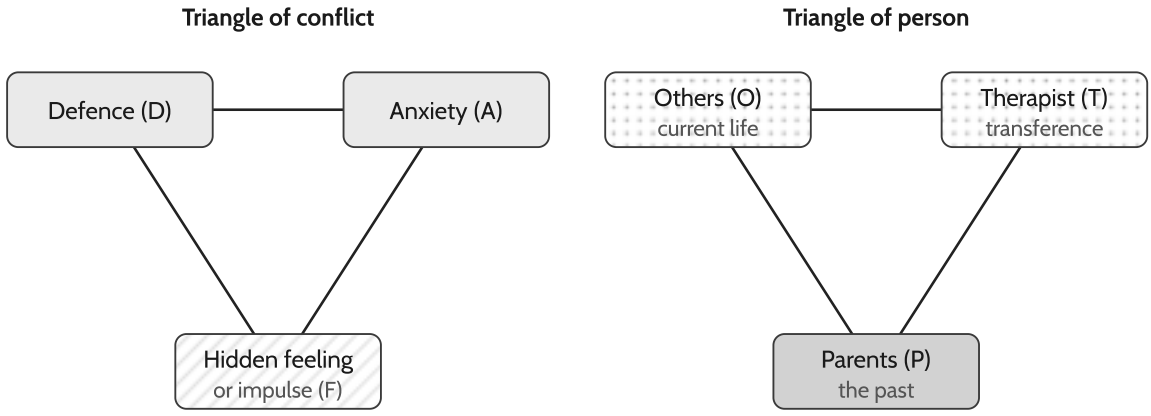


Figure 7.3 Malan's two triangles. The conflict on the left recurs at every corner of the triangle on the right; an interpretation linking therapist, others and parents (T-O-P) is the most powerful.

Table 7.10 The main brief psychodynamic psychotherapies.

Founder	Therapy	Hallmark
Malan	Brief focal psychotherapy	Two triangles; T-O-P links
Sifneos	Short-term anxiety-provoking psychotherapy (STAPP)	Strictly selected, motivated, psychologically minded patients; Oedipal focus
Davanloo	Intensive short-term dynamic psychotherapy (ISTDP)	Relentless pressure on defences

Founder	Therapy	Hallmark
Mann	Time-limited psychotherapy	Exactly 12 sessions; a central issue; time and loss
Luborsky	Supportive-expressive therapy	Core conflictual relationship theme: wish, response of other, response of self
Strupp and Binder	Time-limited dynamic psychotherapy	Cyclical maladaptive pattern

Interpersonal psychotherapy ●●

Interpersonal psychotherapy (IPT) was developed by **Gerald Klerman** (1928 to 1992) and his wife **Myrna Weissman** in the manual *Interpersonal Psychotherapy of Depression* (1984). Rooted in the ideas of Adolf Meyer, Sullivan and Bowlby, it works on current relationships rather than childhood conflict, does not interpret the transference, and treats depression as a medical illness, giving the patient the sick role.

IPT runs 12 to 16 weekly sessions in three phases. The initial phase diagnoses and explains depression, reviews key relationships (the interpersonal inventory) and picks one or two of four problem areas: **grief** (complicated bereavement), **interpersonal role disputes** (with a spouse, parent or employer), **role transitions** (marriage, a new job, retirement, illness) and **interpersonal deficits** (social isolation). The middle phase works on that area through encouragement of affect, clarification, communication analysis (who said what, exactly), role play and decision analysis. Termination treats ending as a small grief and credits the patient's own competence. IPT matched cognitive therapy in the NIMH depression trial (Section 7.8), meta-analyses rate it about equal to CBT, and adaptations treat depressed adolescents (IPT-A), perinatal depression and bulimia.

> **! EXAM TRAP**
>
> Asked what IPT does NOT include beside grief, role conflict and role transition, choose "role playing and role reversal": the other three are its problem areas. The fine print: IPT therapists do use role play after communication analysis; role reversal is psychodrama's (Jacob Moreno, Section 7.7). IPT is Klerman and Weissman's; MBT, Bateman and Fonagy's; TFP, Kernberg's; **Lorna Smith Benjamin** developed interpersonal reconstructive therapy.

Jung, Adler, Horney and Sullivan as therapists ●●●

Freud's early dissenters built their own therapies (their personality theories are in Chapter 4). **Carl Gustav Jung** (1875 to 1961) founded analytical psychology. Beneath the personal unconscious he placed a **collective unconscious** of inherited archetypes: the persona (the social mask), the shadow, the **anima** (the unconscious feminine side of a man), the **animus** (the unconscious masculine side of a woman) and the self. Jung read dreams as direct, creative messages, not disguises. In **amplification** the analyst enlarges a dream image with parallels from myth and religion; in **active imagination** the patient, awake, holds a dialogue with a dream or fantasy figure. The goal is **individuation**, becoming a whole, integrated self.

Alfred Adler (1870 to 1937) left Freud in 1911 and called his school **individual psychology**. From **organ inferiority** (1907: compensation for a weak organ) he moved to universal feelings of inferiority that drive a striving for superiority; when they become overwhelming they form an **inferiority complex**. Therapy is an equal, educational partnership. Lifestyle assess-

ment explores the family constellation, birth order (the dethroned first-born, the competitive second, the pampered youngest) and early recollections, which reveal the present style of life. Its techniques are **encouragement**, the core one; **acting as if** (behaving as though one already had the feared ability); **spitting in the client's soup** (exposing a behaviour's hidden payoff so that it loses its appeal); the **push-button technique** (Harold Mosak: picturing a pleasant and then an unpleasant memory to show that we create our own feelings); and paradoxical intention. Adlerians do not analyse transference.

Karen Horney (1885 to 1952) traced neurosis to basic anxiety, met by moving towards, against or away from people; her therapy dismantled the idealised self-image and its "tyranny of the shoulds". **Harry Stack Sullivan** (1892 to 1949) made the therapist a participant observer of interpersonal patterns, including parataxic distortion (seeing others through the template of earlier relationships), and so influenced IPT.

> **! EXAM TRAP**
>
> Analysis of transference is the technique an Adlerian counsellor would not use; acting as if, the push button and encouragement are all Adlerian. Collective unconscious, anima and animus: Jung. Individual psychology, inferiority complex, organ inferiority: Adler, not Jung, Horney, Frankl (logotherapy, Section 7.6) or Erikson.

Transactional analysis ●●

Transactional analysis (TA) was founded by **Eric Berne** (1910 to 1970), who took the term ego state from his analyst Paul Federn, in *Transactional Analysis in Psychotherapy* (1961) and *Games People Play* (1964). It proceeds through structural, transactional, game and script analysis.

Personality has three **ego states**. The **Parent** holds attitudes and behaviour copied from parents and parental figures, critical or nurturing; the **Adult** processes present reality objectively; the **Child** holds feelings and responses from childhood, free or adapted. A transaction is a stimulus from one ego state and a response from another (Figure 7.4). Complementary transactions run parallel and talk continues. Crossed ones stop it: "Where are my keys?" (Adult to Adult) answered by "Why do you always blame me?" (Child to Parent). Ulterior transactions carry a hidden message beneath the social one.

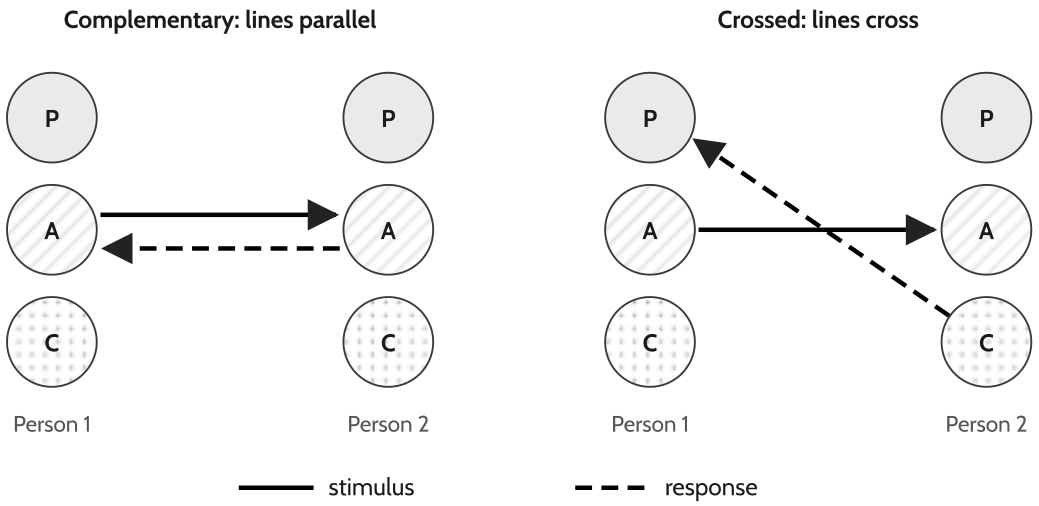


Figure 7.4 Ego states (P, Parent; A, Adult; C, Child). Left: Adult to Adult and back, so talk goes on. Right: the reply comes from Child to Parent, the lines cross and communication stops.

People need **strokes**, units of recognition, and prefer negative strokes to none. A **game** is a series of ulterior transactions with a predictable payoff of bad feelings, such as "Why Don't You, Yes But", in which every suggestion is rejected. A **life script** is an unconscious life plan decided in childhood under parental **injunctions** ("Don't be close", "Don't grow up", sent nonverbally from the parent's Child) and **counterinjunctions** ("Be perfect", "Work hard", spoken from the parent's Parent); Bob and Mary Goulding's redecision therapy helps clients revise them. Scripts rest on one of four **life positions**: I'm OK, you're OK (the healthy one); I'm OK, you're not OK; I'm not OK, you're OK; and I'm not OK, you're not OK, the first made famous by **Thomas Harris**'s best-seller *I'm OK, You're OK* (1969).

> **! EXAM TRAP**
>
> Injunctions, counterinjunctions, games, scripts and strokes all mean transactional analysis, not Gestalt, REBT or person-centred therapy (Sections 7.5 and 7.6). The ego state copied from parents is the Parent; "persona" is Jung's archetype, not an ego state.

Supportive therapy, hypnosis, narcoanalysis and the evidence ○

Psychodynamic work runs along a continuum from expressive (uncovering conflict by interpretation, for patients with ego strength) to supportive. **Supportive psychotherapy** strengthens defences instead of dismantling them, through a warm relationship, reassurance, advice, encouragement, ventilation of feelings, education and practical changes in the environment. It is the mainstay when the ego is overwhelmed: in crisis management, bereavement, acute psychosis and severe physical illness.

Hypnotherapy uses hypnosis, focused attention with heightened suggestibility. It descends from **Franz Anton Mesmer**'s (1734 to 1815) "animal magnetism" and was named by James Braid in 1843. Today it assists the treatment of pain, habits and anxiety, but memories "recovered" under it are unreliable. **Narcoanalysis** gives a slow intravenous barbiturate (sodium amobarbital or thiopentone) so that a mute, stuporous or amnesic patient talks and abreacts,

as in Second World War battle neuroses. It is no "truth serum": drugged people can still lie and fantasise.

> **INDIAN CONTEXT**
>
> In *Selvi v. State of Karnataka* (2010) the Supreme Court held compulsory narcoanalysis, polygraph and brain-mapping unconstitutional (Articles 20(3) and 21); they need consent, and the statements made are not admissible as evidence.

Critics from **Hans Eysenck** (1952; Section 7.8) to the philosopher **Karl Popper**, who called its theory unfalsifiable, attacked psychoanalysis as unproven, long and costly. Yet meta-analyses reviewed by **Jonathan Shedler** (2010) show psychodynamic therapy matching other evidence-based therapies, with gains that grow after treatment ends; short-term dynamic therapy and IPT work for depression, and MBT and TFP for borderline personality disorder. Evidence for the specific mechanism, insight, remains weaker.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The patient's duty to say whatever comes to mind, uncensored, is the _____ rule.
2. Psychic energy invested in a particular person, object or idea is _____.
3. The therapist's emotional reaction to the patient, from the therapist's own unresolved conflicts, is _____.
4. Mann's time-limited psychotherapy lasts exactly _____ sessions.
5. IPT's four problem areas are grief, role disputes, role transitions and interpersonal _____.
6. In TA, a reply from an unexpected ego state that halts communication is a _____ transaction.

Answers: (1) fundamental, (2) cathexis, (3) countertransference, (4) 12, (5) deficits, (6) crossed.

7.4 Behaviour Therapy HIGH YIELD · 88 QS

A child terrified of rabbits, a woman who washes until her hands bleed, a boy with autism who cannot ask for water: behaviour therapy sees in each of them learned behaviour that can be unlearned or replaced. Building on the learning principles of Section 1.6, we trace where the approach came from, then take its techniques family by family.

Origins, assumptions and functional analysis ●●

Behaviour therapy is the application of experimentally established principles of learning to the assessment and treatment of psychological problems. Its spur was a 1952 paper by **Hans Jürgen Eysenck** (1916 to 1997) of the Maudsley Hospital, London, which claimed that about two-thirds of neurotic patients improved within two years whether or not they had psychotherapy (the debate belongs to Section 7.8). If talking did no better than time, Eysenck argued, the clinic needed a treatment derived from learning theory, aimed at the symptom and testable. His creed was blunt: get rid of the symptom and you have eliminated the neurosis.

The new therapy grew from three roots in the 1950s. In Johannesburg, Joseph Wolpe, **Arnold Allan Lazarus** (1932 to 2013) and **Stanley Rachman** worked from classical conditioning. At the Maudsley, Eysenck supplied the theory and **Monte Bernard Shapiro** the method of treating each patient as a single-case experiment. In the United States, **Burrhus Frederic Skinner** (1904 to 1990) and his students carried operant methods onto hospital wards and into schools, a strand later called **applied behaviour analysis** (ABA): operant principles, carefully measured, used to change socially important behaviour. The name *behaviour therapy* first appeared in a 1953 hospital report by Ogden Lindsley, Skinner and Harry Solomon, reached a journal in Lazarus's 1958 paper and was popularised by Eysenck's 1959 paper.

The shared assumptions all contradicted psychoanalysis. Problem behaviour is learned by the same principles as normal behaviour and can be unlearned. The behaviour is the problem, not the symptom of a hidden conflict, so therapy aims to eliminate it and build adaptive behaviour in its place. What maintains a problem now matters more than how it began, and everything is defined and measured. Applied to anxiety, this gives the **learning model of disorder**, best known as the **two-factor theory** of O. Hobart Mowrer (1947): a fear is acquired by classical conditioning and maintained because avoiding the feared object reduces fear and is therefore negatively reinforced. Avoidance stops the person ever learning that the object is harmless, which is why exposure works.

Every case starts with a **functional analysis**: identifying the observable events that trigger and maintain a specific problem behaviour. Its simplest form is the **ABC model** of Antecedents, Behaviour (defined so that two observers would agree) and Consequences; **Frederick Kanfer** and **George Saslow** (1965) expanded it into SORKC (stimulus, organism, response, contingency, consequence). After recording a **baseline**, the frequency, duration or intensity of the behaviour before treatment, the therapist asks what the behaviour achieves. If Arjun's tantrums at homework time get him sent to his room, they are rewarded by escape, and the "punishment" feeds them. Treatment follows the function, not the form.

> **! EXAM TRAP**
>
> Behaviour therapies apply learning principles to **eliminate the unwanted behaviour**; "treat the underlying causes" is the psychodynamic aim, "improve learning and insight" that of insight therapies and "communication and social sensitivity" that of encounter groups (Section 7.7). Behaviour analysis rests on **operant conditioning**, not classical conditioning, modelling, cognitive learning or dream analysis.

Systematic desensitisation ●●●

Can a learned fear be unlearned? In 1924 **Mary Cover Jones** (1897 to 1987), a student of John Watson, treated Peter, a boy of almost three who was terrified of rabbits. He watched other children play happily with a rabbit (modelling), and while he ate a favourite food a caged rabbit was brought a little closer each day, never so close that he stopped eating, until he let it nibble his fingers. This is **counterconditioning**: attaching to a stimulus a new response that is incompatible with the old one, so that the person learns the stimulus is no longer threatening.

KEY STUDY: WOLPE'S CATS (LATE 1940S)

Joseph Wolpe (1915 to 1997), a South African psychiatrist, shocked cats in a small cage until they were too anxious to eat there even when hungry. He then fed them in a room very unlike the laboratory, then in rooms progressively more like it, and finally in the cage, until the anxiety vanished. A response incompatible with anxiety, evoked while the anxiety was weak, had eliminated it. Cats are not neurotic patients, which is why later trials with people mattered.

Wolpe borrowed a term from the physiologist Charles Sherrington. **Reciprocal inhibition** is the principle that two incompatible responses cannot occur together, so a response antagonistic to anxiety, evoked in the presence of anxiety-provoking stimuli, suppresses the anxiety and weakens its bond to them. His main inhibitor was deep relaxation; others were assertion, sexual arousal and, in children, eating. On it he built **systematic desensitisation** (*Psychotherapy by Reciprocal Inhibition*, 1958), which has three steps.

1. **Relaxation training.** The client learns Edmund Jacobson's progressive muscle relaxation over about six sessions, practising 10 to 15 minutes twice a day.
2. **Hierarchy construction.** An **anxiety hierarchy** ranks 10 to 20 feared situations from least to most frightening, each rated on the **subjective units of distress scale** (SUDS), Wolpe's self-rating from 0 (calm) to 100 (the worst imaginable), also called units of disturbance or discomfort.
3. **Desensitisation proper**, the step of gradual exposure. With the client relaxed (SUDS below about 25) the therapist describes the lowest item; at any anxiety the client raises a finger and relaxation is restored. Only when an item evokes no anxiety does the client move up (Figure 7.5).

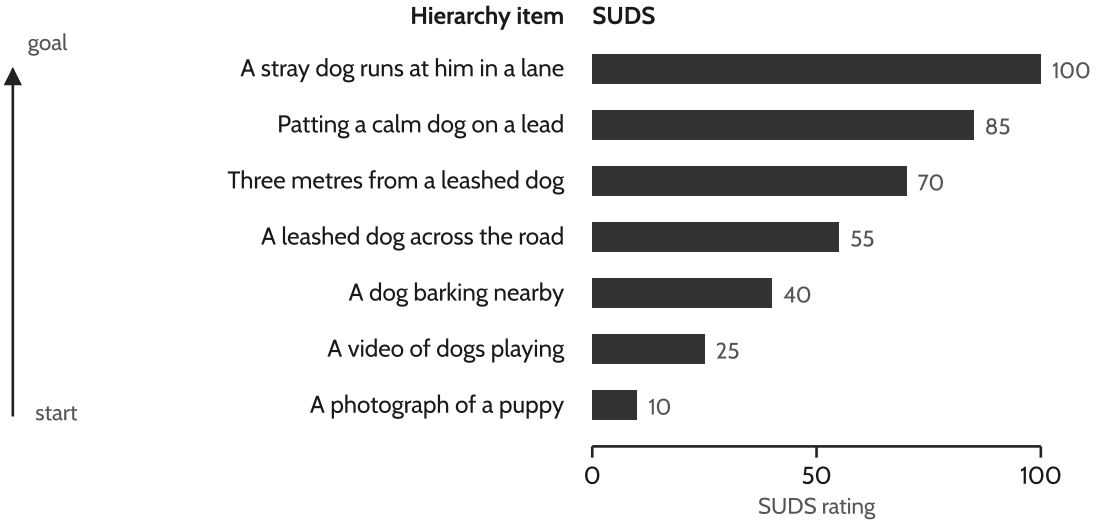


Figure 7.5 An anxiety hierarchy for Ravi, a Mysuru student bitten by a stray dog as a child. Treatment starts at the bottom and climbs only when the current item evokes no anxiety.

Wolpe worked in imagination, called **in vitro desensitisation** (Latin, "in glass", that is, in the consulting room); **in vivo desensitisation** ("in life") climbs the hierarchy in real situations, which generalises better, while imagination suits fears that cannot be staged, such as flying.

Gordon Paul (1966) found desensitisation better than insight therapy for fear of public speaking, but dismantling studies showed that it worked without relaxation or a strict hierarchy, never without exposure. Reciprocal inhibition is Wolpe's theory of why it works, not a proven mechanism.

> **! EXAM TRAP**
>
> Systematic desensitisation is **Wolpe's** (not Skinner's, Pavlov's, Watson's, Rogers's, Perls's, Beck's, Freud's or Bandura's) and rests on **reciprocal inhibition**, not operant extinction, shaping, the law of effect, habituation alone, generalisation, vicarious reinforcement, punishment, adaptation to stress or catharsis. Its relaxation is **Jacobson's**; cognitive restructuring is not one of its steps; it treats **phobia**, not hallucination, delusion or depression. SUDS is keyed to desensitisation (IHBAS 2020), though all exposure therapies now use it; multimodal therapy, acceptance and commitment therapy and mindfulness-based stress reduction belong to Section 7.5. The Fear Survey Schedule (Wolpe and Peter Lang, 1964), State-Trait Anxiety Inventory and Hamilton Anxiety Rating Scale are questionnaires (Chapter 2), not the hierarchy scale.

Exposure, flooding and response prevention ●●●

If exposure is the active ingredient, why not go straight to it? **Exposure therapy** is repeated or prolonged confrontation with a feared stimulus, without escape, until the fear subsides. It works through **habituation**, the reduction in anxiety through repeated or prolonged encounter with a stimulus, and through extinction, as the stimulus keeps appearing without the feared outcome. Exposure may be graded or intense, in vivo or imaginal. **Interoceptive exposure** deliberately provokes feared bodily sensations (spinning in a chair, overbreathing) in panic disorder, and **virtual reality exposure** uses computer-generated scenes for fears of flying, heights or combat.

Flooding exposes the client at once, in real life or in imagination, to the most feared stimulus at full intensity, with no hierarchy and no relaxation, preventing escape until the anxiety peaks and subsides: the person afraid of heights stands on the balcony for as long as it takes. **Implosive therapy** (**Thomas Stampfl**, 1966; set out with **Donald Levis** in 1967) is imaginal flooding with exaggerated, horrific scenes enriched with presumed psychodynamic themes; it is now rare. Both work by extinction, and both are so unpleasant that clients must choose them knowingly.

Exposure and response prevention (ERP) is the first-line behavioural treatment for obsessive-compulsive disorder (OCD). **Victor Meyer** described it in London in 1966, and **Edna Foa** and colleagues refined and tested it. The client touches the "contaminated" object and is prevented from washing. The ritual brings quick relief, so it is negatively reinforced, and it stops the person learning that the anxiety would pass anyway (Figure 7.6). Blocking it allows habituation and extinction, and ERP helps more than 70 percent of patients. Foa also built **prolonged exposure** for post-traumatic stress disorder (PTSD): imaginal reliving of the trauma memory plus in vivo exposure to safe but avoided reminders.

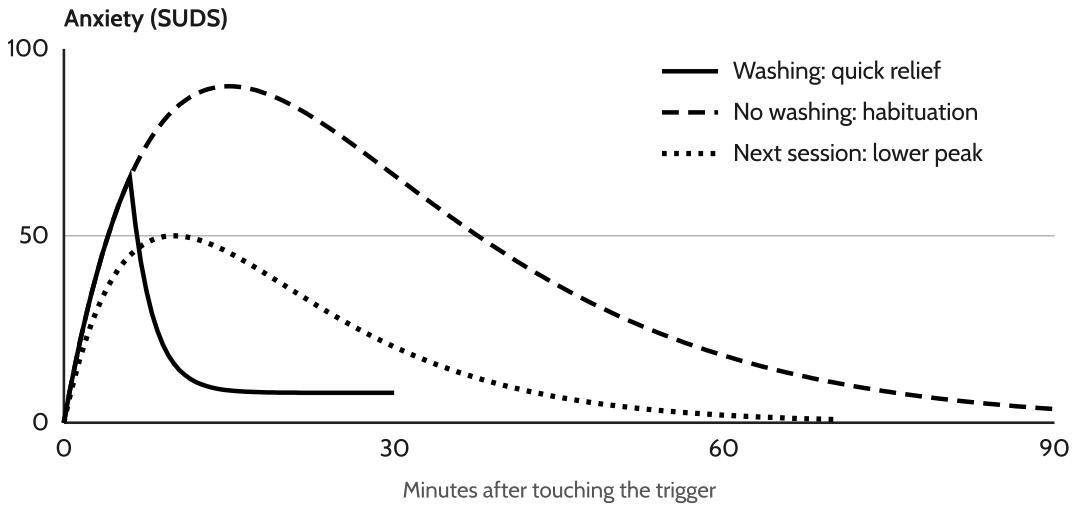


Figure 7.6 Why response prevention works. Washing ends anxiety within minutes and so negatively reinforces the ritual; without it, anxiety peaks and falls by itself, and the next session peaks lower (habituation within and between sessions).

A debated relative of exposure is **eye movement desensitisation and reprocessing** (EMDR), which **Francine Shapiro** (1948 to 2019) developed from a 1987 observation and published in 1989, initially for PTSD. The client holds a traumatic image, its negative belief and its body sensation in mind while following the therapist's fingers back and forth; this **bilateral stimulation** (eye movements, taps or tones) comes in sets of about 15 to 30. Shapiro's standard protocol has **eight phases**, always in this order: (1) history taking, (2) preparation, (3) assessment of the target image, negative belief and distress rating, (4) desensitisation with the eye movements, (5) installation of a positive belief, (6) body scan for leftover tension, (7) closure and (8) re-evaluation at the next session. EMDR works about as well as trauma-focused cognitive behaviour therapy, but the eye movements seem to add little, since holding the memory in mind is itself imaginal exposure. As critics put it, what is effective in EMDR is not new, and what is new is not effective.

> **! EXAM TRAP**
>
> Flooding means full intensity with **no hierarchy and no relaxation**, in vivo or imaginal; pairing the feared stimulus with relaxation step by step is desensitisation, pairing it with shock is aversion therapy, and replacing irrational beliefs (Ellis) or cognitive restructuring is cognitive therapy (Section 7.5). ERP rests on **habituation and extinction**, not counterconditioning or reciprocal inhibition, and is first-line for **OCD**, not phobia, panic, depression, schizophrenia or personality disorder. EMDR is **Shapiro's**, for **PTSD**; prolonged exposure is Foa's, cognitive processing therapy Patricia Resick's and dialectical behaviour therapy Marsha Linehan's (Section 7.5). Bessel van der Kolk researches trauma, Pat Ogden founded sensorimotor psychotherapy and Joseph LeDoux maps the brain's fear circuits.

Aversion therapy, covert sensitisation and the bell and pad ○

Counterconditioning can also run in reverse. **Aversion therapy** pairs an unwanted behaviour, or the stimulus that tempts it, with an unpleasant stimulus, so that by classical conditioning it loses its appeal: a nausea-inducing drug is given just before the patient tastes al-

cohol (chemical aversion), or a brief shock accompanies a cigarette or a deviant sexual image (electrical aversion). It has been used for alcohol use, smoking and paraphilias. It inflicts pain, loses many clients, often fades outside the clinic and was misused against homosexuality, so today it needs informed consent and is tried only after milder methods fail.

Joseph Cautela (1967) moved the procedure into the imagination. In **covert sensitisation** the client imagines lifting a glass of whisky, then vividly imagines vomiting over himself in front of his family, then the relief of turning away; no real aversive stimulus is used.

> **! EXAM TRAP**
>
> An undesirable behaviour imagined together with an imagined unpleasant consequence is **covert sensitisation** (Cautela), not flooding, implosion or desensitisation. Aversion therapy works by pairing (**classical conditioning**), not by intermittent reward, ignoring the behaviour (extinction) or restructuring beliefs.

The oldest classical technique is the **bell-and-pad method** for bed-wetting (nocturnal enuresis), built in 1938 by **Orval Hobart Mowrer** and **Willie Mae Mowrer**. Urine reaching a pad under the sheet completes a circuit and rings a bell; after repeated pairings, bladder tension alone wakes the child. Treatment takes 6 to 12 weeks, and the alarm remains first-line.

Operant techniques that build behaviour ●●●

Operant techniques work on consequences (the grid of reinforcement and punishment, and the schedules, are in Section 1.6). In therapy, **positive reinforcement** means giving something the person values immediately after a target behaviour, every time while the behaviour is new and then intermittently so that it lasts.

A behaviour that never occurs must be built. **Shaping** is reinforcing successive approximations to a target behaviour: for a silent child with autism, first eye contact, then any sound, then "wa", then "water". **Chaining** links the steps of a task, such as dressing, into a sequence; in backward chaining the last step is taught first, so every practice ends in success. A **prompt** is an extra cue (a hint, a gesture, a guiding hand) that makes the right response likely, and **fading** withdraws it gradually until the natural cue alone controls the behaviour.

The **Premack principle** of **David Premack** (1959) states that a more probable activity can reinforce a less probable one: video games after homework, or grandma's rule, vegetables before dessert. A **token economy** rewards target behaviours with tokens (points, stars, chips) that are later exchanged for backup reinforcers such as snacks or privileges. Tokens are **secondary reinforcers**, valued only for what they buy, and they bridge the delay between behaviour and reward.

> **KEY STUDY: AYLLON AND AZRIN'S TOKEN ECONOMY (1968)**
>
> **Teodoro Ayllon** and **Nathan Azrin** ran a ward of chronic psychotic women at Anna State Hospital, Illinois. Tokens earned for self-care and ward jobs bought privileges such as privacy or a walk in the grounds. When tokens were handed out regardless of work, the work almost stopped; when the contingency returned, so did the work. Their book *The Token Economy* (1968) launched the method, though gains often fade outside the ward, and basic needs must never become backup reinforcers.

Contingency management systematically arranges reinforcing consequences for target behaviours, as when clients with substance use disorders earn vouchers for drug-free urine samples. A **contingency contract** puts the arrangement in a signed agreement: "Ravi studies two hours each weekday; his parents allow cricket on Sunday."

> **! EXAM TRAP**
>
> The token economy is **operant** (secondary reinforcers), so it is the treatment NOT derived from classical conditioning; flooding, desensitisation and aversion therapy are classical. Its developers were **Ayllon and Azrin**; Amity 2023 keyed Skinner, who supplied only the principles. Premack runs one way: the **high**-probability behaviour reinforces the **low**-probability one, never the reverse; it is a behaviour modification technique for children, not a theory of relationships, language or parenting. "Pleasant consequences make a behaviour likely to recur" is Thorndike's law of effect, and continuous reinforcement is the *least* resistant to extinction (Section 1.6).

Operant techniques that reduce behaviour, and ABA ●●●

To reduce a behaviour, therapists start with the least restrictive method. **Extinction** withholds the reinforcer that maintains a behaviour: parents who rewarded tantrums with attention now ignore them. Expect an **extinction burst**, a temporary rise when reinforcement stops; giving in at its peak teaches that louder tantrums pay. Extinction works best with **differential reinforcement**, which reinforces something else instead (Table 7.11).

Table 7.11 Operant techniques that reduce behaviour.

Technique	What is done	Category	Example
Extinction	Maintaining reinforcer withheld	Non-reinforcement	Ignoring attention-seeking tantrums
DRO (other behaviour)	Reinforce intervals free of the behaviour	Positive reinforcement	A star for every 10 minutes without hitting
DRI (incompatible behaviour)	Reinforce a behaviour that cannot co-occur	Positive reinforcement	Praise for staying seated, not wandering
DRA (alternative behaviour)	Reinforce a desirable alternative	Positive reinforcement	Praise for asking for help, not screaming
DRL (low rates)	Reinforce only a low rate	Positive reinforcement	Reward if questions stay under five
Time-out	Brief removal from all reinforcement	Negative punishment	Two minutes on a chair away from play
Response cost	Loss of a set amount of a reinforcer	Negative punishment	A fine of tokens; lost television time
Overcorrection	Restore the scene, or practise the right behaviour repeatedly	Positive punishment	Cleaning the whole room after throwing food
Satiation	Reinforcer given in excess	Reinforcer loses value	Ayllon's towel hoarder

Time-out removes the person briefly from all reinforcement and works only if "time-in" is rewarding; **response cost** takes away a set amount of a reinforcer already held, such as tokens. Both are **negative punishment**. **Overcorrection**, from **Richard Foxx** and Azrin (1972), re-

quires restitution (restoring the disturbed scene to better than before) or positive practice (repeating the correct behaviour many times); it is used for aggressive, disruptive and self-stimulatory behaviour in autism and intellectual disability. In **satiation** a reinforcer is given so freely that it loses its value: Ayllon (1963) gave a towel-hoarding patient towels until she threw them out. Positive punishment suppresses quickly but breeds fear, aggression and avoidance and teaches nothing new, so it is a last resort, used with consent and monitoring.

Combined into an intensive, data-driven programme, these methods are ABA, best known in autism. Its core is **discrete trial training**: a skill is broken into small units, each trial being an instruction, a prompt if needed, the child's response and an immediate consequence. In **functional communication training** a child whose tantrums win breaks from work learns to ask for a break with a word or a picture card instead.

KEY STUDY: LOVAAS'S EARLY INTERVENTION FOR AUTISM (1987)

Ole Ivar Lovaas (1927 to 2010) at the University of California, Los Angeles, gave 19 young children with autism 40 or more hours a week of one-to-one treatment for at least two years. Nine (47 percent) reached normal intellectual and educational functioning, against 2 percent of controls. Groups were not randomly assigned and aversives such as slaps were used, but the "Lovaas programme" became the best-known behavioural treatment for autism.

! EXAM TRAP

Deliberately ignoring tantrums is **extinction** (not generalisation or discrimination, Section 1.6). Amity 2023 keyed "operant escape", which is wrong: escape ends an unpleasant stimulus already present and avoidance prevents one (Section 1.6). Time-out and response cost are **negative punishment**; negative reinforcement removes something unpleasant and *increases* behaviour. Overcorrection targets repetitive **aggressive** behaviour in autism; bed-wetting needs the bell and pad. Lovaas treated **autism**, not aphasia, depression or alcoholism, and a picture card to request a break is **ABA**, not CBT, play therapy or psychoanalysis.

Modelling, assertiveness and social skills ○

Albert Bandura (1925 to 2021) turned observational learning into therapy. In **live modelling** the client watches a real person perform the feared or missing behaviour; in **symbolic modelling** the model appears on film or in a story; in **participant modelling** the therapist models the behaviour and then physically guides the client through it. For snake phobia (Bandura, Edward Blanchard and Brunhilde Ritter, 1969), participant modelling beat both symbolic modelling and systematic desensitisation. In **behavioural rehearsal** the client practises the new behaviour in role-play, with feedback, before trying it in life.

Assertiveness training, pioneered by **Andrew Salter** (*Conditioned Reflex Therapy*, 1949) and developed by Wolpe and Lazarus, teaches people to express wishes and feelings directly while respecting others. **Passive** behaviour surrenders one's rights (silence when someone jumps the bus queue), **aggressive** behaviour violates other people's (shouting abuse), and **assertive** behaviour defends one's own without attack ("Please join the queue; we were here first"). **Social skills training** teaches eye contact, conversation and refusal through instruction, modelling, rehearsal, feedback and homework, in autism, social anxiety and schizophrenia, where antipsychotic medication remains front-line.

Relaxation, biofeedback and self-management •••

Several methods teach control of one's own body. **Edmund Jacobson**'s **progressive muscle relaxation** (1929, enlarged in 1938) tenses and releases muscle groups in turn; **Johannes Heinrich Schultz**'s **autogenic training** (1932) uses self-suggestions of heaviness and warmth; **breathing retraining** slows the overbreathing of panic; and the **applied relaxation** of **Lars-Göran Öst** (1987) trains rapid relaxation on cue, used at the first sign of anxiety in panic, generalised anxiety and phobias.

Biofeedback makes a bodily process the person cannot normally sense visible or audible, so that he can learn to control it. Its basis was **Neal Miller**'s 1960s report that rats could change heart rate and other autonomic responses through operant conditioning: the autonomic nervous system could come under voluntary control. Electromyographic (EMG) feedback of muscle tension treats tension-type headache and back pain, thermal feedback of hand temperature treats migraine and Raynaud's disease, and EEG neurofeedback of brain waves is used for ADHD and epilepsy. An early device was the perineometer of **Arnold Kegel** (1948), a pressure gauge that let a patient watch her pelvic floor squeeze.

Habit reversal training (HRT), from Azrin and **Gregory Nunn** (1973), is the treatment of choice for tics and habits such as hair pulling and nail biting. **Awareness training** teaches the client to detect each occurrence and the urge before it; **competing response training** teaches a response physically incompatible with the habit, inconspicuous and held for about a minute or until the urge passes (clenching the fists against hair pulling); **motivation** comes from reviewing the habit's costs and from family praise. A variant, regulated breathing, was applied to stuttering. In **negative practice** (massed practice), proposed by **Knight Dunlap** (1932), the habit is performed deliberately and repeatedly until fatigue makes it aversive. **Thought stopping**, popularised by Wolpe, interrupts an unwanted thought with "Stop!", first shouted, later silent; it has faded because suppressed thoughts rebound.

Self-monitoring, the systematic recording of one's own target behaviour, often changes that behaviour before treatment begins; this **reactivity** spoils the baseline but helps therapy. In **self-reinforcement** the client rewards himself on meeting his own goal, completing Kanfer's sequence of self-monitoring, self-evaluation and self-reinforcement.

Behavioural activation (BA) treats depression by re-engaging clients in meaningful, rewarding activities. **Peter Lewinsohn** proposed in the 1970s that depression follows a loss of response-contingent positive reinforcement. **Neil Jacobson** and colleagues (1996) found the activation component of cognitive therapy as effective as the whole package, and Jacobson and **Christopher Martell** developed BA in its own right: activity monitoring, scheduling and graded tasks that replace avoidance.

For insomnia, **stimulus control** (**Richard Bootzin**, 1972) re-links the bed with sleep: bed only when sleepy and only for sleep and sex, leave it if not asleep within about 10 to 20 minutes, a fixed rising time and no naps. **Sleep restriction** (**Arthur Spielman**, 1987) limits time in bed to the time actually slept, extending it as sleep consolidates. **Paradoxical intention**, from the logotherapist **Viktor Frankl** (Section 7.6), asks the person to do or wish for the very thing he fears: the insomniac tries hard to stay awake, which removes the performance anxiety that kept him awake. Frankl's **dereflection** instead turns attention away from the symptom.

> **! EXAM TRAP**
>
> Operant control of autonomic functions is **biofeedback**, not hypnosis (Section 1.5), sublimation (a defence mechanism) or desensitisation; EMG feedback suits **tension-type headache**, and Kegel's perineometer was early biofeedback. Repeating a habit to fatigue is **negative practice**, not the competing response of habit reversal. HRT is for **tics**; OCD needs ERP. Doing or wishing the opposite of the goal is **paradoxical intention**, not stimulus control, sleep restriction, dereflection, negative practice or reciprocal inhibition. Behavioural activation means **engaging in meaningful activity**, not analysing childhood, relaxation or hypnosis.

Parents, classrooms and an evaluation ○

Gerald Patterson of the Oregon Social Learning Center described a **coercive cycle** at home: a parent makes a demand, the child whines or hits, the parent gives in (negatively reinforcing the aggression) and the child stops (negatively reinforcing the surrender). **Parent management training** (PMT) teaches parents to give clear commands, attend to and praise good behaviour, reward it with points or stickers, ignore minor misbehaviour, and use brief time-out or loss of privileges for serious misbehaviour; negative reinforcement is what it removes, not what it teaches. **Classroom management** applies the same principles at school: clear rules, praise, planned ignoring, token systems, daily report cards and group contingencies such as the Good Behaviour Game.

Behaviour therapy is a first choice for specific phobias, OCD, panic, enuresis, tics and habits, insomnia, conduct problems and skills in autism, and behavioural activation rivals cognitive therapy for depression; in psychosis and personality disorder it is an adjunct. Psychoanalysts predicted **symptom substitution**: remove a symptom without resolving its underlying conflict and another will take its place. Follow-ups from Paul's 1966 trial onwards found little sign of it. Aversive methods raise the sharpest ethical questions: they demand informed consent, the least restrictive alternative first, supervision and special protection for children and institutionalised patients.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Wolpe's systematic desensitisation rests on the principle of _____ inhibition.
2. The 0 to 100 scale used to rank hierarchy items is the subjective units of _____ scale.
3. Exposure to the most feared stimulus at full intensity, with no hierarchy and no relaxation, is called _____.
4. Exposure and response prevention for OCD works through habituation and _____.
5. The token economy was developed for psychiatric inpatients by Ayllon and _____.
6. Time-out and response cost are forms of _____ punishment.
7. In habit reversal training, a response physically incompatible with the habit is the _____ response.
8. Telling an insomniac to try to stay awake is Frankl's technique of _____ intention.

Answers: (1) reciprocal, (2) distress (or disturbance), (3) flooding, (4) extinction, (5) Azrin, (6) negative, (7) competing, (8) paradoxical.

7.5 Cognitive and Cognitive-Behavioural Therapies, and the Third Wave

HIGH YIELD · 69 QS

Two students score 48 out of 90 on the same mock test. One is disappointed and studies her mistakes; the other decides he was never good enough and stops opening his books. Same event, opposite reactions: what differed was what each made of the score. That observation underlies every therapy in this section, from the founders, Albert Ellis and Aaron Beck, to the "third wave", which asks clients not to argue with their thoughts but to change their relationship with them.

The cognitive turn and the three waves ●●

The term **cognitive-behavioural therapy** (CBT) covers a family of treatments that share three propositions: thinking affects emotion and behaviour; thinking can be monitored and changed; and changing maladaptive thought patterns relieves distress, even when the events that triggered it cannot be changed. The idea is old. The Stoic philosopher **Epictetus** (about 50 to 135 CE) wrote that people are disturbed not by things but by the views they take of them.

In the 1950s and 1960s two therapists, Albert Ellis and Aaron Beck, independently concluded that faulty thinking drives emotional disturbance, and both are counted as founders of the cognitive tradition. Behaviour therapists (Section 7.4) joined them in the 1970s; Donald Meichenbaum's *Cognitive-Behavior Modification* (1977) gave the hybrid its name. **Albert Bandura** (1925 to 2021) supplied a bridge in 1977 with **self-efficacy**, a person's belief that he or she can carry out the actions a situation demands. He argued that all effective treatments work largely by raising it, which made self-efficacy the classic cognitive-behavioural explanation of how people change.

In 2004 **Steven C. Hayes** (born 1948) described three waves. The first, behaviour therapy (1950s onwards), applied learning principles to behaviour. The second, the cognitive therapies of Ellis and Beck, changed the *content* of thoughts from irrational or distorted to realistic. The **third wave** (from the 1990s: dialectical behaviour therapy, acceptance and commitment therapy, mindfulness-based cognitive therapy, and compassion-focused, metacognitive and functional analytic therapies) changes a person's *relationship* to thoughts and feelings: noticing and accepting them while acting on one's values. All CBTs are time-limited, focused on present problems, structured, collaborative and educational; they rely on homework and measure symptoms throughout.

> ! EXAM TRAP
>
> "Third-wave CBT includes all EXCEPT": **psychoanalysis** (Section 7.3), which belongs to no wave. REBT and Beck's therapy are second wave; token economies are first wave. Self-efficacy is **Bandura**'s concept, not Beck's, Ellis's or Skinner's (Skinner rejected cognitive explanations).

Ellis's rational emotive behaviour therapy ●●●

Albert Ellis (1913 to 2007) treated his own shyness first: as a young man he made himself talk to a hundred women in New York's Bronx Botanical Garden in one month, and learned that rejection was unpleasant, not terrible. After a PhD at Columbia University (1947) he practised psychoanalysis until about 1953, frustrated that insight rarely made patients better. Drawing on Stoic philosophy, he began rational therapy in 1955; it became rational-emotive therapy (RET) and, in 1993, **rational emotive behaviour therapy** (REBT), a name that acknowledged its behavioural homework. His *Reason and Emotion in Psychotherapy* (1962) is the classic text, and REBT is usually called the first cognitive-behavioural therapy.

Its core is the **ABC model**: **A**, the activating event; **B**, the person's beliefs about A; **C**, the emotional and behavioural consequences. Common sense says A causes C; Ellis says A contributes, but B is the main cause. The two students shared an A, but "one mock does not decide my future" produced disappointment and study, while "I must score well, or I'm a failure" produced depression and avoidance (Figure 7.7). A rational belief is a flexible preference that fits reality, follows logically and serves the person's goals; an irrational belief is a rigid, absolute demand. A rational belief about a real loss still produces a healthy negative emotion (sadness, concern), whereas an irrational one produces an unhealthy emotion (depression, anxiety).

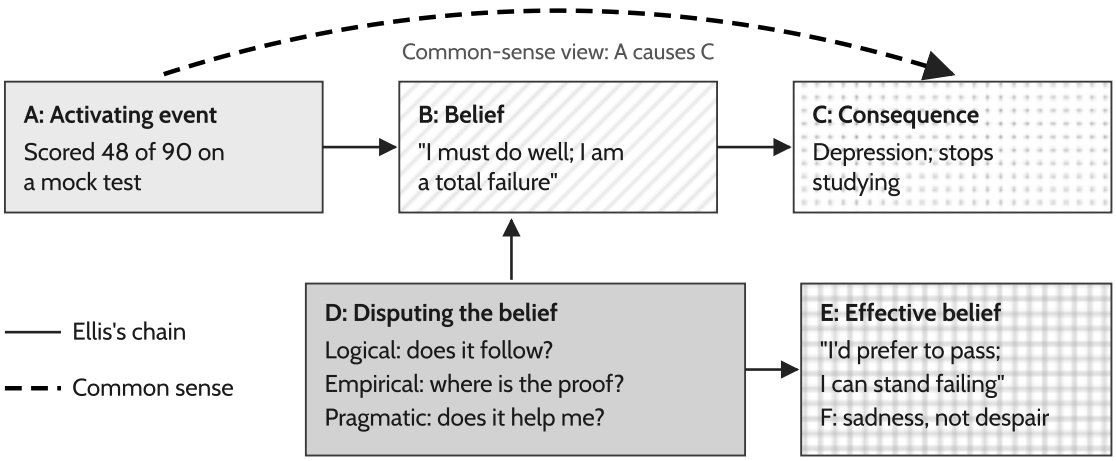


Figure 7.7 Ellis's ABC model extended to D (disputing), E (an effective new philosophy) and F (new feelings and behaviour). The dashed arrow is the common-sense view that events cause feelings directly; Ellis's chain runs through the belief.

Irrational thinking takes four forms. The root is demandingness, turning preferences into musts, shoulds and oughts, which Ellis called **musturbation** (also spelt musterbation). From it follow **awfulising** (rating a setback as more than 100 percent bad, close to Beck's catastrophising), **low frustration tolerance** ("I can't stand it") and global rating of self or others ("I failed, so I am a failure"). Ellis's three major musts concern the self ("I must do well and win approval, or I am no good"), others ("Others must treat me fairly, or they are rotten") and the world ("Life must give me what I want, quickly and easily"). They breed, respectively, anxiety and depression, anger and rage, and self-pity and procrastination.

The active ingredient is D, **disputing** the irrational belief. A **logical** dispute asks whether it follows ("You want to pass; why must you?"), an **empirical** dispute asks for evidence ("Where is the proof that you are a total failure?") and a **pragmatic** dispute asks whether it helps ("Where does this belief get you?"). Success yields E, an effective new philosophy, and F, new feelings and behaviour. The antidote to global rating is **unconditional self-acceptance**: one's acts can be rated, one's whole self cannot. Ellis extended it to others and to life; unlike Carl Rogers (Section 7.6), he thought therapist warmth unnecessary, and his style was famously blunt.

REBT mixes cognitive techniques (disputing, self-help forms, reading), emotive techniques and behavioural homework set at almost every session (in vivo exposure, rewards and penalties). In **rational emotive imagery**, adopted from the psychiatrist **Maxie C. Maultsby Jr.**, the client imagines the activating event, feels the unhealthy emotion, changes it to its healthy counterpart and notes the self-statement that did it. In a **shame-attacking exercise** the client does something harmless but embarrassing in public (Ellis's example was calling out station names in a subway train) to learn that disapproval is bearable.

REBT is clear, brief and teachable, but its evidence is thinner than Beck's: a 1991 meta-analysis of 70 studies found it better than control groups and baseline, and a 2008 trial found it comparable with cognitive therapy and medication for depression. Its chief criticism is that it can be direct and confrontational, so clients may find it intimidating; the therapist's idea of

"rational" may also impose his own values, and its philosophical claims are hard to test. A focus on sexuality is a criticism of Freud, not of Ellis.

> **! EXAM TRAP**
>
> ABC stands for activating event, beliefs, consequences: B is the **belief about the activating event**, not behaviour, baseline, bias or "beliefs about oneself", and C is the consequence of that belief, not "critical analysis". D = **disputing**, not desensitisation, drama or "discuss"; E = effective new philosophy. The behavioural ABC of antecedent, behaviour, consequence is functional analysis (Section 7.4), and affect, behaviour, cognition is the ABC of attitudes (Chapter 3). Musts, musturbation and shame attacking signal Ellis, not Freud, William James, Rogers, Adler, Beck, DBT or Glasser's reality therapy (Section 7.6).

Beck's cognitive model and the distortions •••

Aaron Temkin Beck (1921 to 2021), a psychoanalytically trained psychiatrist at the University of Pennsylvania, noticed in the late 1950s that his patients had a second, barely noticed stream of quick, believable, self-critical thoughts ("He won't like me") that came before their feelings. He called them **automatic thoughts**. Depressed patients' dreams, expected to show anger turned inward, showed loss and defeat instead. Beck set out the model in *Depression* (1967) and the treatment in *Cognitive Therapy of Depression* (1979, with colleagues). He built cognitive therapy for depression and later extended it to anxiety, personality disorders, substance use and psychosis.

At its centre is the **cognitive triad**: negative views of the **self** ("I am worthless"), the **world** or current experience ("everything is against me") and the **future** ("it will never get better"). Remember it as self, world, future: not past, present, future; not self, others, environment; not the thoughts, feelings and behaviours that CBT links in its cognitive model; and not id, ego and superego. Cognition runs on three levels. The deepest are **core beliefs**, or schemas: rigid, global beliefs formed in childhood ("I am unlovable"). From them grow **intermediate beliefs**, also called dysfunctional assumptions: the rules, attitudes and assumptions that follow ("If I am not perfect, I will be rejected"). When a situation activates these, automatic thoughts appear and drive emotion and behaviour. A dormant schema can be woken by a critical incident such as a loss, so the model is a diathesis-stress one. Beck's daughter **Judith S. Beck** laid out these levels in *Cognitive Therapy: Basics and Beyond* (1995).

Negative thinking is kept going by **cognitive distortions**, systematic errors in processing information. Beck named arbitrary inference, selective abstraction, overgeneralisation, magnification and minimisation, personalisation and dichotomous thinking; the psychiatrist **David D. Burns** popularised the rest of Table 7.12 in *Feeling Good* (1980).

Table 7.12 The cognitive distortions, with an example of each.

Distortion	What happens	Example
Arbitrary inference	A conclusion without evidence	A friend does not greet you: "He dislikes me now"
Selective abstraction (mental filter)	One detail taken out of context, the rest ignored	A teacher sees two bored students: "My class dislikes me"
Overgeneralisation	A broad rule from one or a few events	"I failed statistics, so I fail at everything"

Distortion	What happens	Example
Magnification and minimisation	Faults enlarged, strengths shrunk	One slip "ruins" the talk; praise "means nothing"
Personalisation	Blaming oneself for events one did not cause	"The guests left early because of me"
Dichotomous (all-or-nothing) thinking	Only two extremes	"If I don't top the exam, I have failed"
Labelling	A global label for one act	"I'm a loser", not "I made a mistake"
Mind reading	Assuming one knows what others think	"She thinks I'm stupid"
Fortune telling	Predicting a bad outcome	"I'll freeze in the interview"
Emotional reasoning	A feeling taken as proof	"I feel useless, so I am useless"
Should statements	Rigid rules for self and others	"I should never make mistakes"
Catastrophising	Expecting the worst to happen	"If I fail this exam, my life is over"
Disqualifying the positive	Good events "don't count"	"They praised me only to be polite"

! EXAM TRAP

A man whose friend fails to acknowledge him and who concludes "he doesn't like me any more" shows **arbitrary inference**; mind reading and fortune telling are later, narrower names for such jumps, so choose arbitrary inference when they are not offered. **Selective abstraction**, also called selective attention or the mental filter, is fixing on one detail and ignoring the whole, like the teacher with two bored students; overgeneralisation is a broad rule from a few events. **Catastrophising** is worrying that the worst will always happen; all-or-nothing thinking is extreme categories; personalisation is taking things personally.

Cognitive therapy in practice ●●●

Cognitive therapy treats thoughts as hypotheses to be tested, not facts to be argued away. In **collaborative empiricism** therapist and client investigate together, making objective assessments of beliefs against evidence. The main method is **Socratic questioning**, or guided discovery: open questions that help the client discover and attend to previously overlooked information, so that the discovery is the client's own rather than an interpretation from the therapist ("What is the evidence? How else could you see it? If it were true, what would it mean?"). The therapist neither interprets nor tells the client what to think. Collaborative empiricism, Socratic dialogue and guided discovery are the approach's fundamental concepts; free association and dream analysis belong to psychoanalysis (Section 7.3).

- The **dysfunctional thought record** (1979) has columns for the situation, the emotion (rated 0 to 100), the automatic thought, a balanced response and the outcome; with it clients learn to identify, examine and restructure their own automatic thoughts.
- Behavioural experiments test predictions in real life ("If I speak in class, everyone will laugh").
- Activity scheduling plans the week and rates each activity for mastery and pleasure; graded task assignment breaks a daunting task into small steps so that success comes early.

- The **downward arrow technique** asks repeatedly "If that were true, what would it mean to you?" to descend from an automatic thought to the core belief.
- Decatastrophising (the "what if" technique) weighs how likely and how bad the feared outcome is and how the client would cope; reattribution spreads responsibility fairly when the client blames himself for everything.

A session runs in a set order: mood check, bridge from the last session, agenda, homework review, work on the agenda, new homework, then summary and feedback. Progress is measured with Beck's scales: the Beck Depression Inventory (BDI, 1961; BDI-II, 1996), the Beck Anxiety Inventory (BAI, 1988) and the Beck Hopelessness Scale (BHS, 1974), described in Chapter 2.

> **! EXAM TRAP**
>
> The BDI and BAI are **self-report**; the Hamilton depression and anxiety scales of **Max Hamilton** are **clinician-rated**. So "Which is clinician-rated?" is the Hamilton Anxiety Rating Scale, not "both". The downward arrow is not a 0 to 10 distress rating or an exposure hierarchy (Section 7.4).

Ellis and Beck share the central assumption that emotional disturbance arises mainly from how people think about events, not from the events themselves, and both therapies are active, brief and built on homework. Table 7.13 shows how they differ.

Table 7.13 Ellis's REBT and Beck's cognitive therapy compared.

Feature	Ellis (REBT)	Beck (cognitive therapy)
Origin	1955; a psychologist	1960s; a psychiatrist
Target	Irrational beliefs, above all musts	Automatic thoughts, distortions, core beliefs
Theory of disturbance	One core (demandingness) for all problems	Disorder-specific content
Style	Directive, persuasive, confrontational	Collaborative, Socratic
Beliefs are	Disputed as irrational	Tested as hypotheses
Signature methods	Disputing, shame attacking	Thought record, behavioural experiments
Research base	Smaller	Very large

Other cognitive-behavioural therapies ●●

Donald Meichenbaum (born 1940) worked on self-talk. His **self-instructional training**, first used with impulsive children in 1971, moves guidance inward in five steps: the therapist performs the task while talking aloud; the child performs it while the therapist instructs aloud; then the child instructs himself aloud, then in a whisper, then silently. His **stress inoculation training**, named after vaccination, builds resistance by coping with manageable doses of stress, in three phases: conceptualisation (how appraisals and self-talk create stress), skills acquisition and rehearsal (relaxation, cognitive restructuring, problem solving, coping self-statements) and application and follow-through (practice in imagery, role-play and real life, with relapse prevention).

Problem-solving therapy, introduced by **Thomas J. D'Zurilla** and **Marvin R. Goldfried** (1971) and extended to depression by **Arthur M. Nezu**, teaches a positive problem orientation

and a sequence: define the problem, brainstorm alternatives, choose one, act and check the result. **Arnold A. Lazarus** (1932 to 2013) built **multimodal therapy**, which assesses the **BASIC I.D.** (Behaviour, Affect, Sensation, Imagery, Cognition, Interpersonal relationships, Drugs and biology) and borrows techniques from any school (technical eclecticism, Section 7.8). In **fixed-role therapy**, from the personal construct theory (1955) of **George Alexander Kelly** (1905 to 1967), the client lives for about two weeks as a new character sketched by the therapist, to try out new constructs and new relationships.

For clients who failed standard CBT, especially those with personality disorders, **Jeffrey E. Young** developed **schema therapy** (1990). It targets 18 early maladaptive schemas, enduring themes such as abandonment, mistrust, defectiveness and unrelenting standards, formed when childhood needs went unmet, and schema modes, the states a person switches between (vulnerable child, angry child, punitive parent, detached protector, healthy adult). Its signature is **limited reparenting**, meeting some unmet childhood needs within professional limits, with imagery rescripting and chair work. **Anthony Ryle**'s cognitive analytic therapy (CAT, Britain, 1980s) is a brief therapy, often 16 sessions, joining cognitive therapy to psychoanalytic object relations through a reformulation letter and a diagram of the client's reciprocal roles.

> **! EXAM TRAP**
>
> Fixed-role therapy is **George Kelly**'s, not Beck's, Ellis's or Fritz Perls's; BASIC I.D. is **Lazarus**'s; schema therapy, with its schema modes and limited reparenting, is **Jeffrey Young**'s, not DBT's.

Disorder-specific CBT: panic, social anxiety, trauma, psychosis ●●●

Beck's cognitive specificity hypothesis (each disorder has its own theme: loss in depression, danger in anxiety) led to disorder-specific models. **Panic. David M. Clark** (1986) traced panic attacks to the catastrophic misinterpretation of bodily sensations: a racing heart read as a heart attack raises anxiety, which intensifies the sensations, in a vicious circle. **Paul M. Salkovskis** (1991) added **safety behaviours**, such as gripping a shopping trolley or sitting down, which seem to avert disaster but stop the client learning that it would not have happened. Treatment provokes the feared sensations and gradually drops safety behaviours, as experiments that test the catastrophic beliefs. In the model of social anxiety by Clark and **Adrian Wells** (1995), self-focused attention and safety behaviours maintain the fear; effective CBT for social phobia combines exposure, cognitive restructuring and social skills training.

Trauma. Prolonged exposure (PE), developed by **Edna B. Foa**, combines imaginal exposure (repeatedly retelling the trauma) with in vivo exposure to avoided but safe reminders. Cognitive processing therapy (CPT), developed by **Patricia A. Resick** for rape survivors, challenges "stuck points" about blame, safety, trust, power, esteem and intimacy. Trauma-focused CBT (TF-CBT) for children, by **Judith A. Cohen**, **Anthony P. Mannarino** and **Esther Deblinger**, teaches coping skills first; its central aim is to build and process the trauma narrative and so desensitise the traumatic memories. With eye movement desensitisation and reprocessing (EMDR) of **Francine Shapiro** (Section 7.4), these are the first-line trauma treatments.

> **! EXAM TRAP**
>
> PE comes from **Edna Foa and colleagues**, CPT from **Resick**, EMDR from **Shapiro**, DBT from Linehan. For survivors of sexual assault, pick the option naming trauma-focused CBT and EMDR (psychodynamic therapy is also commonly used, with less evidence); DBT, ACT, mindfulness-based therapy and interpersonal psychotherapy (Section 7.3) are not trauma-focused treatments. In panic, safety behaviours are dropped, never prescribed.

Psychosis. CBT for psychosis (CBTp), developed in Britain in the 1990s by **David Kingdon**, **Douglas Turkington** and others, normalises unusual experiences, tests delusional beliefs collaboratively and teaches ways of coping with voices. It is an adjunct, not an alternative: antipsychotic medication remains the front-line treatment for schizophrenia (Chapter 8), with CBTp, social skills training and stress management as psychosocial add-ons.

Dialectical behaviour therapy ●●●

Marsha M. Linehan (born 1943) developed **dialectical behaviour therapy** (DBT; manuals 1993) for chronically suicidal women, most with borderline personality disorder (BPD; Chapter 8). Standard CBT's constant push for change felt invalidating to them and they dropped out, yet acceptance alone left the self-harm untouched. DBT's dialectic synthesises the two: full acceptance of the client together with a firm push for change. The acceptance side, including core mindfulness, came from Zen Buddhism, which Linehan practised; the change side came from behaviour therapy.

Her **biosocial theory** sees BPD as pervasive emotion dysregulation arising when a biological emotional vulnerability (high sensitivity and reactivity, slow return to baseline) meets an invalidating environment that dismisses or punishes the child's emotions. Standard DBT has four modes: individual therapy, a weekly skills group (usually for a year), telephone coaching in crises and a therapist consultation team. Individual work targets life-threatening behaviour first, then therapy-interfering and then quality-of-life-interfering behaviour. The four skills modules are **core mindfulness** (observing without judgement and finding "wise mind" between reasonable mind and emotion mind), **distress tolerance** (surviving crises through distraction, self-soothing and radical acceptance), **emotion regulation** (naming emotions, acting opposite to unhelpful urges) and **interpersonal effectiveness** (asking and refusing while keeping relationships and self-respect).

> **KEY STUDY: LINEHAN'S DBT TRIALS (1991 AND 2006)**
>
> Linehan and colleagues randomly assigned chronically parasuicidal women with BPD to a year of DBT or treatment as usual; the DBT group made fewer and less medically serious parasuicidal acts, spent fewer days in hospital and dropped out less (1991). Critics cited the small sample and weak control, so in 2006 Linehan compared DBT with treatment by community experts in psychotherapy, and DBT still halved suicide attempts.

> **! EXAM TRAP**
>
> DBT is Marsha Linehan's. Its four skills modules are mindfulness, distress tolerance, emotion regulation and interpersonal effectiveness; cognitive restructuring and relaxation are not among them. Family genogram construction is a Bowen technique (Section 7.7), not a DBT mode. DBT's mindfulness is Zen, not Stoic. DBT was built for borderline personality disorder with chronic suicidality, not for schizophrenia, phobias or OCD.

ACT, mindfulness and the rest of the third wave ●●

Hayes, with **Kirk D. Strosahl** and **Kelly G. Wilson**, developed **acceptance and commitment therapy** (ACT, said as the word "act"; their book appeared in 1999). It rests on **relational frame theory**, Hayes's account of language: because humans relate events to one another through words, a thought carries the sting of what it stands for, so remembering a failure hurts like failing. The usual response, experiential avoidance (trying to escape unwanted thoughts and feelings), backfires and narrows life. ACT therefore aims not at symptom elimination but at **psychological flexibility**, acting on one's values with whatever thoughts and feelings are present. It builds flexibility through the six processes of the hexaflex (Figure 7.8): acceptance, **cognitive defusion** (seeing thoughts as words, as in "I'm having the thought that I'm a failure"), contact with the present moment, self-as-context (the observing self), values and committed action.

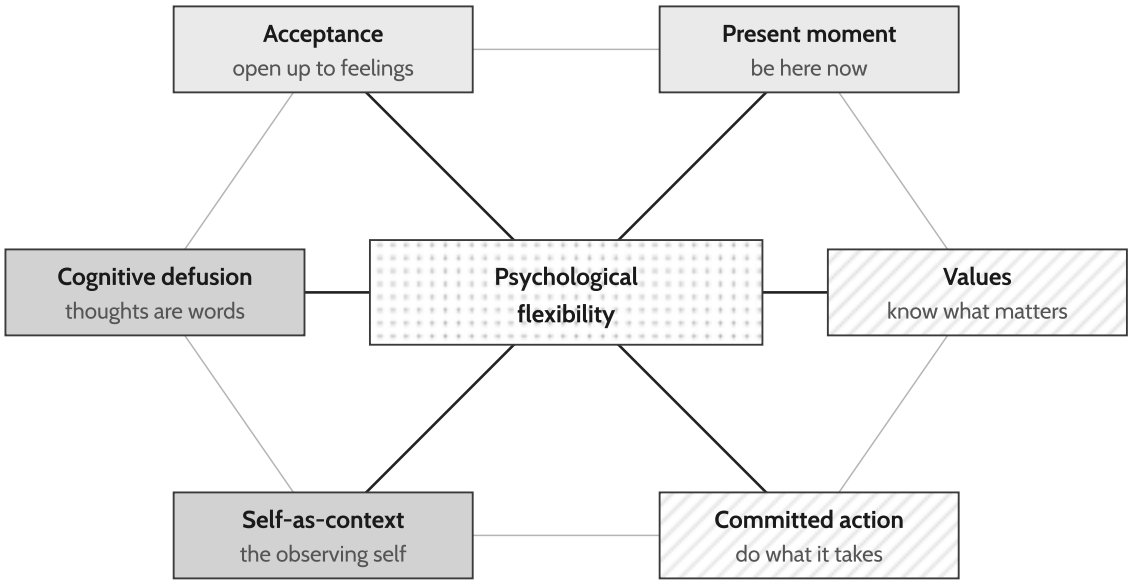


Figure 7.8 The hexaflex of acceptance and commitment therapy. Each process has an opposite that produces inflexibility: experiential avoidance, cognitive fusion, living in the past or future, attachment to a conceptualised self, unclear values, and inaction or impulsivity.

Jon Kabat-Zinn (born 1944) brought mindfulness (attention paid on purpose, in the present moment and non-judgementally) into medicine with **mindfulness-based stress reduction** (MBSR), founded in 1979 at the University of Massachusetts Medical School: an eight-week group course of body scan, meditation and gentle yoga for chronic pain and stress. **Zindel V. Segal**, **J. Mark G. Williams** and **John D. Teasdale** merged it with cognitive therapy as **mind-**

fulness-based cognitive therapy (MBCT; 2002), an eight-week group for recovered patients designed to prevent relapse in recurrent major depression by teaching decentering: seeing thoughts as passing mental events, not facts.

KEY STUDY: MBCT AND DEPRESSIVE RELAPSE (TEASDALE AND COLLEAGUES, 2000)

Recovered patients with recurrent depression were randomly assigned to usual treatment or usual treatment plus MBCT and followed for 60 weeks. For those with three or more past episodes, relapse fell from 66 to 37 percent; for those with two, MBCT made no difference. MBCT is a relapse-prevention treatment, not a treatment for acute illness.

Compassion-focused therapy (CFT) of **Paul Gilbert**, built for shame and self-criticism, describes three emotion-regulation systems: threat (protection, anxiety, anger), drive (seeking resources and achievement) and soothing (contentment, safeness, affiliation); compassionate mind training strengthens the soothing system, which self-critical people lack. **Metacognitive therapy** (MCT) of Adrian Wells targets not what people think but how they respond to thinking. Its central concept, the **cognitive attentional syndrome**, is persistent worry and rumination, threat monitoring and unhelpful coping (thought suppression, avoidance), driven by metacognitive beliefs such as "Worrying helps me cope" and "My worry is uncontrollable"; attention training and detached mindfulness loosen it. **Functional analytic psychotherapy** (FAP) of **Robert J. Kohlenberg** and **Mavis Tsai** (1991) uses the therapy relationship itself: the client's problem behaviours and improvements appear in session, and the therapist naturally reinforces the improvements. **Behavioural activation** grew from **Peter M. Lewinsohn**'s view that depression follows a loss of response-contingent positive reinforcement; when **Neil S. Jacobson** and colleagues (1996) found that the activation component of Beck's therapy worked as well as the whole package, scheduling valued activity and tackling avoidance became a treatment in its own right (Section 7.4).

! EXAM TRAP

ACT comes from Steven Hayes and colleagues, not Linehan, Beck or Ellis. Its hexaflex (acceptance, defusion, present moment, self-as-context, values, committed action) is not DBT's list of skills, and ACT aims at psychological flexibility through six core processes, not symptom elimination, insight into childhood conflicts or disputing beliefs. MBSR is Kabat-Zinn's; MBCT prevents depressive relapse and is not a treatment for acute psychosis or phobias.

How well does CBT work?

CBT is the most researched psychotherapy. A 2006 review of 16 meta-analyses, co-authored by Beck, found large effects for depression, generalised anxiety disorder, panic disorder, social phobia, post-traumatic stress disorder and childhood depression and anxiety. CBT is first-line for most anxiety disorders and for depression, where it matches antidepressants and better protects against relapse after treatment ends; it is also first-line for bulimia nervosa and insomnia, and an adjunct to medication in bipolar disorder and schizophrenia. Its methods travel: in Goa, lay counsellors delivering behavioural activation helped depressed primary-care patients in a 2017 trial led by **Vikram Patel**.

It has limits. Many clients do not fully recover or later relapse; trial samples are more selected than clinic patients; it needs literacy and homework; critics say it neglects social causes such as poverty and assumes a Western, individualist outlook; and against other well-conducted therapies the differences are often small (Section 7.8). Computerised and internet-delivered CBT (iCBT), such as Beating the Blues in Britain and MoodGYM in Australia, delivers modules, thought records and homework on screen. Guided versions approach the results of face-to-face therapy, whereas unguided ones lose many users. It suits stepped care, in which low-intensity help comes first (telepsychology in India, Section 7.10).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Ellis renamed his therapy rational emotive behaviour therapy in _____.
2. In Ellis's ABCDE model, D stands for _____ the irrational belief.
3. Beck's negative views of the self, the world and the future form the cognitive _____.
4. Concluding without evidence that a friend who did not greet you dislikes you is the distortion called _____.
5. Asking "If that were true, what would it mean to you?" to reach a core belief is the _____ technique.
6. The phases of stress inoculation training are conceptualisation, skills acquisition and rehearsal, and _____.
7. Linehan explains borderline personality disorder with her _____ theory.
8. The six processes of the ACT hexaflex build psychological _____.

Answers: (1) 1993, (2) disputing, (3) triad, (4) arbitrary inference, (5) downward arrow, (6) application and follow-through, (7) biosocial, (8) flexibility.

7.6 Humanistic, Existential, Experiential and Brief Therapies

HIGH YIELD · 49 QS

By the 1950s psychotherapy had two systems: psychoanalysis (Section 7.3) and behaviourism (Section 7.4). **Abraham Harold Maslow** (1908 to 1970) called the movement against both the **third force**. Its core, humanistic psychology, studies the whole person from his or her own viewpoint and assumes an inborn drive toward growth; Maslow (Chapter 4) and Carl Rogers, who built the therapy, were its leading proponents. Existential therapists share its respect for freedom but make the person face death, isolation and meaninglessness; experiential therapists work with feeling as it happens in the session. Together these humanistic-experiential psychotherapies aim to promote a more fluid and integrated organisation of the self, not to resolve deep-seated conflicts (the psychodynamic aim), change maladaptive behaviours (the behavioural aim) or train interpersonal skills.

Rogers: the self, conditions of worth and incongruence ○

Carl Ransom Rogers (1902 to 1987) gave his therapy three names: **non-directive therapy** in *Counseling and Psychotherapy* (1942), because the counsellor does not advise, interpret or steer; client-centred therapy in *Client-Centered Therapy* (1951), which added a theory of per-

sonality; and the person-centred approach from the 1970s. A "client", unlike a patient, keeps responsibility for his or her own life.

The theory starts from the **actualising tendency**, the inborn tendency of every organism to develop its capacities in ways that maintain and enhance it (not Maslow's self-actualisation). Infants judge experience by an *organismic valuing process*: food is good when hungry, pain is bad. Part of experience becomes the self-concept, the organised picture of "I" and "me", beside an ideal self, the self one would like to be. With the self comes a need for positive regard, and parents rarely give it without strings ("we love you when you top the class"). The child introjects these rules as **conditions of worth** and starts valuing experience by whether it earns approval.

The result is **incongruence**, a mismatch between self-concept and experience: a split between the organismic valuing process and a self burdened by conditions of worth, felt as tension, anxiety and confusion. Ananya, raised as "the doctor of the family", comes alive only when she draws, so she distorts the experience ("I am just tired") or denies it awareness (Figure 7.9). Rogers's 19 propositions (1951) state the theory: behaviour follows the world as perceived (the phenomenal field), and experience inconsistent with the self is denied or distorted until, in complete safety, it can be admitted. Growth heads toward the **fully functioning person**, a direction rather than a state: open to experience, living fully in each moment, trusting his or her own organism, free and creative.

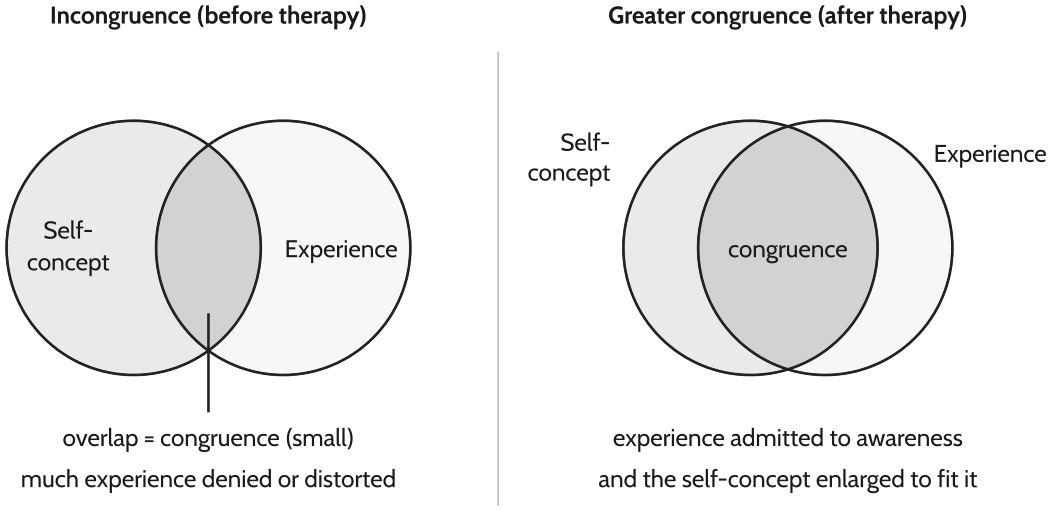


Figure 7.9 Rogers's model of incongruence. The overlap of self-concept and actual experience is the area of congruence; therapy enlarges it by letting denied and distorted experience into awareness.

> **! EXAM TRAP**
>
> "A split between the organismic valuing process and a self-concept burdened by conditions of worth" is **incongruence**. *Introjection* is how conditions of worth get in, *conflict* has no special place in the theory, and the *actualising tendency* is the growth force that incongruence blocks.

The core conditions and person-centred practice ●●●

The cure, then, is a relationship without conditions of worth. In a 1957 paper Rogers listed six **necessary and sufficient conditions** of personality change: (1) two persons are in psychological contact; (2) the client is incongruent, vulnerable or anxious; (3) the therapist is congruent; (4) the therapist feels unconditional positive regard; (5) the therapist empathically understands the client's internal frame of reference and communicates this; (6) the client perceives this empathy and regard, at least minimally. "Necessary and sufficient" means no change without them and nothing more needed. Conditions 3 to 5 are the three core conditions (Table 7.14): **unconditional positive regard**, a term coined in 1954 by Rogers's student **Stanley Standal**; **congruence**, which Rogers later called the most basic; and empathy.

Table 7.14 Rogers's three core conditions.

Core condition	Also called	What it means	What it is not
Congruence	Genuineness, realness	Feelings, awareness and words match; no façade	Disclosing every feeling; playing the expert
Unconditional positive regard	Acceptance, prizing, warmth	Valuing the person without conditions of worth	Approving every act; praise; reassurance
Accurate empathic understanding	Empathy	Sensing the client's world "as if" one's own, never losing the "as if", and saying so	Sympathy; identification; interpretation

The therapist responds from inside the client's frame of reference, above all by **reflection of feeling**: putting into words the feeling the client is expressing and offering it back to be checked. To "I am not sure this course is mine", advice ("Talk to your parents") directs and analysis ("Perhaps you resent your father") imposes theory; a reflection ("You doubt whether this course is really yours") stays with the client's meaning. Interpretation, advice, diagnosis, testing and aversive conditioning (Section 7.4) have no place.

> **! EXAM TRAP**
>
> The core conditions are empathy, unconditional positive regard and congruence ("all of the above" when all three are listed). Interpretation, transference and free association are psychoanalytic; reinforcement and shaping are behavioural; disputation and homework belong to Albert Ellis and Aaron Beck (Section 7.5); confrontation and paradoxical intention belong to Gestalt therapy and logotherapy. Edward Bordin's **working alliance** (1979: goals, tasks and bond; Section 7.8) is not Rogerian; person-centred therapy is the approach least likely to use the term, preferring the "real relationship" and the core conditions.

KEY STUDY: THE Q-SORT AND THE SELF-IDEAL GAP (CHICAGO, 1954)

Question. Does therapy bring the perceived self closer to the ideal self? **Method.** The **Q-sort** was devised by the British psychologist **William Stephenson** (*The Study of Behavior*, 1953) and adapted by Rogers's group. A person sorts about 100 self-statements ("I am a submissive person") from "most like me" to "least like me" into a forced, roughly normal distribution, once for the real and once for the ideal self; correlating the sorts captures within-person variation. **Results.** For John Butler and Gerard Haigh's clients (1954) the mean self-ideal correlation rose from about zero (minus .01) before therapy to .34 after it, and stood at .31 at follow-up. **Criticism.** Falling ideals or learning the "healthy" answer could shrink the gap; samples were small.

Using rating scales of empathy, warmth and genuineness, **Charles Truax** and **Robert Carkhuff** (1967) found that clients of high-scoring therapists improved more. The conditions help in every therapy (Section 7.8) but are rarely sufficient: a severe phobia still needs exposure.

Rogers also championed the **encounter group** (*Carl Rogers on Encounter Groups*, 1970): a small, loosely structured group whose principal role is personal growth and self-change through self-disclosure and honest feedback, not socialising, team building or diagnosis. Kurt Lewin's T-groups and other group formats are in Section 7.7.

Gestalt therapy ●●●

Fritz Perls (1893 to 1970), a German-born psychoanalyst, founded Gestalt therapy with his wife **Laura Perls** and the writer **Paul Goodman**; the founding book, *Gestalt Therapy* (1951), was by Perls, Ralph Hefferline and Goodman, and Perls's workshops at the Esalen Institute (1964 to 1969) made it famous. Therapy stays in the here and now. Its goal is awareness of what one is doing and feeling, and how, not why; with awareness comes responsibility, so clients say "I won't" rather than "I can't". Needs organise experience as *figure* and *ground*: the most pressing need stands out, is met and recedes. A feeling never expressed and completed (an unsaid goodbye to a dead parent) remains as **unfinished business**, an incomplete gestalt that intrudes on the present. Health needs good contact at the boundary between self and environment. Perls, Hefferline and Goodman's five disturbances of contact ended with egotism; the five of Erving and Miriam Polster (1973), the list most textbooks use, drop egotism and add deflection (Table 7.15).

Table 7.15 The contact boundary disturbances of Gestalt therapy.

Disturbance	What happens	Example
Introjection	Swallowing others' views whole, undigested	"Good girls never get angry", held as one's own rule
Projection	Disowning a part of oneself by placing it in others	Blaming the professor for an exam one did not study for
Retroflection	Doing to oneself what one wants to do to others	Biting one's nails instead of expressing anger
Deflection	Turning aside from direct contact	Joking, vagueness, talking about someone instead of to them

Disturbance	What happens	Example
Confluence	Self-other boundary blurred; a high need for acceptance and liking leads to enmeshment	"We feel exactly the same about everything"
Egotism	Excessive self-monitoring; watching oneself instead of engaging	Judging every sentence while talking

The techniques are experiments in awareness. In the **empty chair technique** the client speaks to an absent person, or a part of the self, imagined in an empty chair, often switching seats to reply: the classic way to finish unfinished business. In the two-chair technique each side of an inner split gets a chair, typically the *top dog* (the righteous "you should" voice) and the underdog (the apologetic voice that wins by stalling). On the *hot seat* one workshop member worked with Perls while the group watched. In dream work every person and object in a dream is a projection of a disowned part of the dreamer, so the client speaks as each part, in the present tense, instead of having the dream interpreted. The Gestalt prayer (1969) begins: "I do my thing and you do your thing. I am not in this world to live up to your expectations."

> **! EXAM TRAP**
>
> Gestalt therapy is not Gestalt psychology, the perception school of Max Wertheimer, Wolfgang Köhler and Kurt Koffka (Chapter 1), which rejected Perls's borrowing of its terms. The empty chair always means Gestalt therapy, never person-centred therapy, CBT, REBT, dialectical behaviour therapy, psychoanalysis or Bowen's family systems therapy, though emotion-focused therapy later borrowed it.

Existential therapy and logotherapy ●●●

Existential therapy is more a view of life than a set of techniques. Its chief philosophical source is **Martin Heidegger**, whose *Being and Time* (1927) described human existence as **Dasein** ("being there"), a being-in-the-world: a person is never a thing apart from a world. It begins from the existential givens, which **Irvin Yalom** (born 1931) named in *Existential Psychotherapy* (1980) as four **ultimate concerns**: death, freedom (with its responsibility), isolation and meaninglessness. Anxiety about them is normal; trouble starts when people flee it into inauthentic living, and the therapist helps the client face the givens and choose.

The Swiss psychiatrists **Ludwig Binswanger** (1881 to 1966) and **Medard Boss** (1903 to 1990; *Psychoanalysis and Daseinsanalysis*, 1963) applied Heidegger to psychiatry as Daseinsanalysis. Binswanger described three simultaneous modes of being-in-the-world: the **Umwelt** (world around: body, drives and physical environment), the **Mitwelt** (with-world: relationships with others) and the **Eigenwelt** (own-world: one's relation to oneself); **Emmy van Deurzen** later added the Überwelt of spiritual values. **Rollo May** (1909 to 1994; *The Meaning of Anxiety*, 1950; *Existence*, co-edited 1958) brought these ideas to America. For May, existential anxiety is the anxiety of a being who knows it is free and will die: normal anxiety is proportionate, not repressed and usable; neurotic anxiety is disproportionate, repressed and paralysing. Existential-humanistic therapy (May, James Bugental) is the American blend.

Viktor Emil Frankl (1905 to 1997), a Viennese psychiatrist who survived Auschwitz and other Nazi camps, wrote *Man's Search for Meaning* (German original, 1946). His **logotherapy** (Greek *logos*, meaning), the "third Viennese school", holds that the primary human motive is not Freud's will to pleasure or Adler's will to power but the **will to meaning**. Its frustration

brings an existential vacuum of emptiness and boredom and sometimes a **noogenic neurosis** (Greek *noos*, mind or spirit): one rooted in an existential problem rather than in conflict between drives. Meaning comes from creating or doing (creative values), from experiencing or loving (experiential values), and from the stand one takes toward unavoidable suffering (attitudinal values).

In **paradoxical intention**, used for anticipatory anxiety in phobias, obsessions and insomnia, the client deliberately wishes for the feared symptom: a doctor afraid of sweating is told to show people how much he can really sweat, and the fear collapses. In **dereflection**, which counters hyper-reflection (excessive watching of oneself), attention is turned outward, as when a man anxious about his sexual performance attends to his partner instead. Socratic dialogue, the main technique, uses questions to help clients discover their own meanings.

> **! EXAM TRAP**
>
> The will to meaning, noogenic neurosis or anxiety, paradoxical intention and dereflection all belong to Frankl's logotherapy, an existential therapy: not REBT, acceptance and commitment therapy (Section 7.5) or the broader existential-humanistic therapy. Dereflection is not sensate focus, the graded touching exercise of Masters and Johnson (Section 7.7). Yalom's four ultimate concerns are not his group therapeutic factors (Section 7.7).

Reality therapy, focusing, emotion-focused and positive therapies ••

William Glasser (1925 to 2013), an American psychiatrist, refined reality therapy with delinquent girls at the Ventura School and published *Reality Therapy* in 1965. Its base, control theory, later renamed **choice theory** (*Choice Theory*, 1998), holds that we choose nearly all behaviour to meet five basic, genetically built-in needs: survival, love and belonging, power (achievement), freedom and fun. Behaviour is chosen "total behaviour" (acting, thinking, feeling, physiology), so Glasser spoke of "depressing" rather than being depressed; acting is the easiest part to change. The therapist is warm but firm: present behaviour, responsibility, no excuses, no punishment. **Robert Wubbolding** summed up the method as WDEP: Wants, Doing (and direction), Evaluation (is it working?) and Planning.

Eugene Gendlin (1926 to 2017), working with Rogers, found that improving clients attended to their inner bodily experiencing, and taught this skill as **focusing** (1978): attending to the vague bodily *felt sense* of a problem until a fitting word or image brings a "felt shift". **Leslie Greenberg** joined person-centred empathy to Gestalt tasks in process-experiential therapy, now **emotion-focused therapy**. Adaptive emotions are used, maladaptive ones changed by evoking a healthier emotion, and secondary ones (anger covering hurt) explored; two-chair work treats self-critical splits, the empty chair unfinished business. Greenberg and Susan Johnson also built emotionally focused couples therapy (Section 7.7).

"Positive psychotherapy" has two meanings. **Martin Seligman** (born 1942) launched positive psychology as president of the American Psychological Association in 1998; his positive psychotherapy builds strengths and well-being (PERMA: positive emotion, engagement, relationships, meaning, accomplishment) through exercises such as listing three good things daily. The idea of **flow**, total absorption when challenge matches skill, comes from **Mihaly Csikszentmihalyi** (1934 to 2021), not from Rogers, Maslow or Frankl. The Iranian-born German psychiatrist **Nossrat Peseschkian** (1933 to 2010) founded a separate, transcultural Positive Psychotherapy in the 1970s (Latin *positum*, the given): it starts from capacities, uses

stories from many cultures, balances four areas of life (body, achievement, contact, future) and runs in five stages (observation, inventory, situational encouragement, verbalisation, broadening of goals).

Narrative and solution-focused brief therapy ○

These brief therapies grew from family therapy and social constructionism: no account of a life is the single true one, and the client is the expert. Narrative therapy was created by **Michael White** (1948 to 2008) in Adelaide, Australia, and **David Epston** in New Zealand (*Narrative Means to Therapeutic Ends*, 1990). Clients arrive with a thin, problem-saturated story; the therapist practises **externalising the problem** ("the person is not the problem; the problem is the problem"), searches for *unique outcomes* when the problem did not win, and helps the client *re-author* the self story around them. The **solution-focused brief therapy** (SFBT) of **Steve de Shazer** (1940 to 2005) and **Insoo Kim Berg** (died 2007) was developed at the Brief Family Therapy Center in Milwaukee (founded 1978). It ignores causes and builds, through questions (Table 7.16), on what the client wants and what already works; the *miracle question* (de Shazer, 1988) is its signature. Brief strategic therapy (the Mental Research Institute in Palo Alto; Jay Haley), with reframing and paradoxical directives such as prescribing the symptom, is in Section 7.7.

Table 7.16 Techniques of narrative and solution-focused therapy.

Technique	Approach	Example or purpose
Externalising	Narrative	"When did the Worry first move into your life?"
Unique outcome	Narrative	A time the problem did not dominate
Re-authoring	Narrative	Thickening that exception into a new self story
Miracle question	SFBT	"If a miracle solved the problem tonight, what would be the first sign tomorrow?"
Exception question	SFBT	"When is the problem absent or smaller?"
Scaling question	SFBT	"From 0 to 10, where are you now? What would one point higher look like?"
Coping question	SFBT	"With all this happening, how have you kept going?"
Compliments	SFBT	Genuine praise for strengths and what works

Motivational interviewing and the stages of change ●●

Motivational interviewing (MI) was first described by **William R. Miller** in 1983 for problem drinkers and developed with **Stephen Rollnick** in *Motivational Interviewing* (1991; later editions 2002 and 2013). It is a collaborative, person-centred but goal-directed way of strengthening a person's own motivation by resolving ambivalence, wanting and not wanting change at once; arguing for change only makes the client defend the status quo. The spirit (2013 edition) is PACE: Partnership, Acceptance, Compassion and Evocation (drawing out the client's own reasons). The four processes are engaging, focusing, evoking and planning, and the core skills are OARS: Open questions, Affirmations, Reflections and Summaries. The 2002 edition gave four principles: express empathy, develop discrepancy between behaviour and the client's own goals, roll with resistance (now split into *sustain talk* for the status quo and *discord* in the relationship) and support self-efficacy. The counsellor strengthens **change talk**: DARN (Desire, Ability, Reasons, Need), then CAT (Commitment, Activation, Taking steps).

MI's home ground is substance use; it is also used for smoking, treatment adherence and other health behaviours. The meta-analysis of Burke, Arkowitz and Menchola (2003) found adapted MI effective for alcohol, drug, and diet and exercise problems (central in diabetes care), but not for smoking or HIV-risk behaviour; obsessive-compulsive disorder was not reviewed.

> **! EXAM TRAP**
>
> MI was developed by Miller and Rollnick, not by Prochaska and DiClemente (stages of change), Albert Bandura (self-efficacy) or Alan Marlatt (relapse prevention, Section 7.9). Asked what MI is used to manage, choose substance abuse, its original field, over treatment adherence. Asked which problem Burke and colleagues (2003) did not list, choose obsessive-compulsive disorder: smoking was reviewed (with weak results), OCD not at all.

MI is often paired with the **transtheoretical model** of **James Prochaska** and **Carlo DiClemente**, built from studies of how smokers quit on their own (1983). People pass through stages of change (Figure 7.10): precontemplation (no intention to change within six months), contemplation (intending to, but ambivalent), preparation (acting within a month; small steps taken), action (changed under six months ago), maintenance (change kept over six months) and termination (no temptation, full confidence). Relapse is recycling to an earlier stage, not failure, so progress is a spiral. Ten processes of change drive movement: experiential ones (consciousness raising, dramatic relief, environmental and self-re-evaluation, social liberation) help most early, behavioural ones (self-liberation or commitment, counterconditioning, stimulus control, contingency management, helping relationships) in action and maintenance. Decisional balance (pros and cons) and self-efficacy track progress; the health belief model is in Section 7.9.

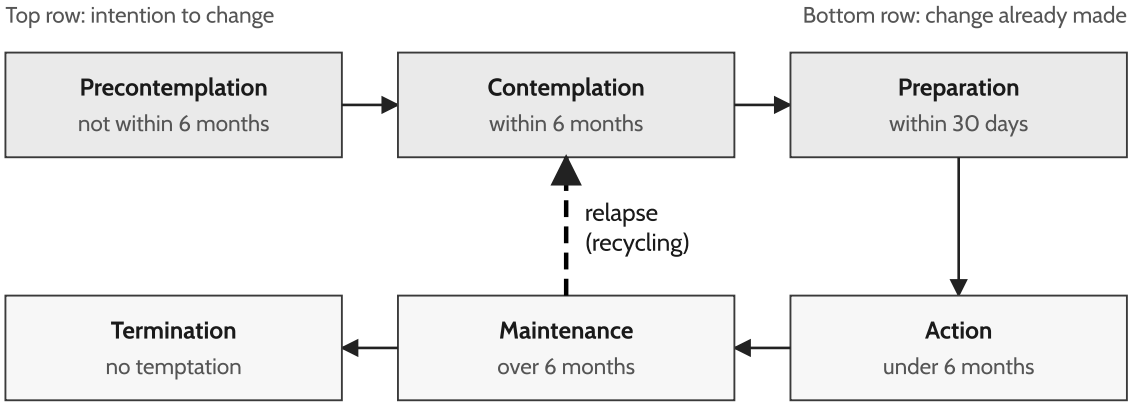


Figure 7.10 The stages of change (Prochaska and DiClemente). Time markers show when change is intended (top row) or how long it has lasted (bottom row); the dashed arrow shows relapse sending a person back to an earlier stage.

Counselling and the helping skills ○

Counselling and psychotherapy overlap so much that Rogers used the words interchangeably. Counselling usually helps fairly well-functioning people with developmental, educational, vocational or situational problems, briefly and in the present; psychotherapy treats

more severe or long-standing disorders, seeking deeper, slower personality change. **Allen Ivey**'s microcounselling (late 1960s) teaches interviewing one skill at a time. His **microskills hierarchy** rises from ethics and multicultural competence, through attending behaviour (culturally appropriate eye contact, vocal quality, verbal tracking, body language) and the basic listening sequence (questions, encouraging, paraphrasing, reflecting feeling, summarising), to focusing, empathic confrontation, reflection of meaning and influencing skills, and finally integration into a personal style. Carkhuff's model (*Helping and Human Relations*, 1969) has the helper attend, respond, personalise and initiate, moving the client through exploration, understanding and action; on his five-level empathy scale, level 3 is interchangeable with what the client said.

Gerard Egan's *The Skilled Helper* (1975) has three stages. Stage I asks "What is going on?" (the current picture: telling the story, seeing blind spots, choosing the key issue); Stage II asks "What do I want instead?" (the preferred picture: possibilities, goals, commitment); Stage III asks "How do I get there?" (strategies, a plan and action). The helper attends with SOLER: face the client Squarely, Open posture, Lean forward, Eye contact, Relaxed. Egan separates primary empathy, understanding what the client has said and felt, from **advanced empathy**, voicing what the client only hints at or half sees, which is a form of challenge.

> **! EXAM TRAP**
>
> Advanced empathy is inappropriate in an initial session: without trust and enough information it feels like an intrusive interpretation. Primary empathy is right from the first minute, and culturally sensitive empathy is always appropriate.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Rogers's term for the mismatch between self-concept and actual experience is _____.
2. Rogers's three core conditions are empathy, congruence and _____.
3. In Gestalt therapy, a high need for acceptance that blurs the self-other boundary is _____.
4. Yalom's four ultimate concerns are death, freedom, isolation and _____.
5. Frankl's technique of turning attention away from oneself, used for hyper-reflection, is _____.
6. Glasser's five basic needs are survival, love and belonging, power, freedom and _____.
7. "If a miracle solved the problem tonight…" is the signature question of _____ therapy.

Answers: (1) incongruence, (2) unconditional positive regard, (3) confluence, (4) meaninglessness, (5) dereflection, (6) fun, (7) solution-focused brief.

7.7 Family, Couples and Group Therapies MEDIUM YIELD · 28 QS

For most of its history psychotherapy meant one patient and one therapist. In the 1950s clinicians noticed that patients who recovered in hospital often relapsed soon after going home, and that when one family member improved another sometimes fell ill. Perhaps the problem lay partly in the relationships. This section follows that idea from the family to the couple and the group.

The family as a system ○

The idea came from biology. **Ludwig von Bertalanffy** (1901 to 1972) proposed a **general systems theory**: a system is a set of interacting parts whose whole is more than their sum, so a change in one part changes the whole. **Norbert Wiener**'s cybernetics (1948) added self-regulation by feedback. A family tends toward **homeostasis**, a balance it defends as a thermostat defends a room temperature. Negative feedback restores the old balance and positive feedback amplifies change; neither word means good or bad.

Systems thinking replaces linear causality (A causes B, so someone is to blame) with **circular causality**, in which each person's behaviour is both cause and effect of the others' (Figure 7.11). The member brought for treatment is the **identified patient**, the visible sign of a troubled system rather than its cause; the symptom may even serve the system, as when a son's misbehaviour unites quarrelling parents, and a child made to carry the family's problems is its scapegoat. Families also pass through a **family life cycle**: **Betty Carter** and **Monica McGoldrick** described six stages (leaving home, the new couple, families with young children, with adolescents, launching children, and later life), and symptoms cluster at the transitions.

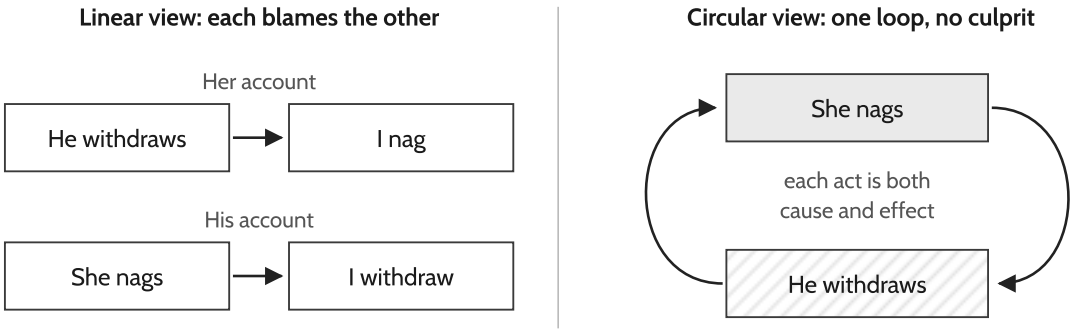


Figure 7.11 Linear and circular causality. Each partner says the other started it; the systems therapist treats the loop and blames no one.

Families and schizophrenia: from blame to expressed emotion ●●●

The first family research asked whether families cause schizophrenia. In 1948 **Frieda Fromm-Reichmann** (1889 to 1957) blamed a dominating, rejecting **schizophrenogenic mother**. In 1956 the anthropologist **Gregory Bateson** (1904 to 1980), with **Don Jackson** (1920 to 1968), Jay Haley and **John Weakland** at Palo Alto, proposed the **double bind**: a person in a vital relationship repeatedly receives two contradictory messages at different levels (typically words contradicted by manner) and can neither comment on the contradiction nor leave, so every response is wrong.

KEY STUDY: BATESON'S DOUBLE BIND (1956)

A young man recovering from an acute schizophrenic episode put his arm round his visiting mother, and she stiffened. He withdrew it; she asked, "Don't you love me any more?" When he blushed she said, "Dear, you must not be so easily embarrassed and afraid of your feelings." After she left he assaulted an aide. Whatever he did was wrong, and he could not say so. **Criticism:** anecdotal and never predictive, it is not accepted as a cause.

Theodore Lidz (1910 to 2001) at Yale described **marital schism**, open conflict in which each parent undermines the other and courts the child, and **marital skew**, in which one disturbed parent dominates while the other pretends all is normal. **Lyman Wynne** (1923 to 2007) described **pseudomutuality** (1958), a facade of harmony that hides distance and leaves no room for a separate identity. None of these survived as a cause of schizophrenia (Chapter 8), and the schizophrenogenic mother is discredited.

What survived concerned *relapse*. In London **George Brown** found that discharged patients who spent long hours with their relatives relapsed more often. The emotional climate of the home, the attitudes relatives express about the patient, is called **expressed emotion** (EE) and is rated from the Camberwell Family Interview. High EE has three components: **critical comments**, **hostility** and **emotional over-involvement** (intrusive, self-sacrificing concern). Brown, **James Birley** and **John Wing** (1972) showed that high EE predicts relapse.

KEY STUDY: VAUGHN AND LEFF (1976)

Christine Vaughn and **Julian Leff** followed patients with schizophrenia for nine months after discharge. About half of those returning to high-EE homes relapsed, against about one in eight in low-EE homes. In high-EE homes, regular medication and less than 35 hours a week of face-to-face contact each lowered the risk.

Hence **family psychoeducation**: teaching relatives about the illness, medication and warning signs, training calm communication and problem solving, and cutting contact hours where the climate will not change (Leff's programmes; **Ian Falloon**'s behavioural family management). It lowers relapse (Section 7.9).

! EXAM TRAP

High EE predicts **relapse**, not onset, negative symptoms or violence. Warmth is rated in the interview but is **not** a high-EE dimension. Reducing contact hours manages high EE. One parent sending two conflicting messages is **double bind** communication; first-rank symptoms are Kurt Schneider's (Chapter 8).

INDIAN CONTEXT

In the 1950s Vidya Sagar at Amritsar Mental Hospital had families camp on the grounds and share in care. Indian families show the lowest rates of high EE in cross-cultural comparisons (Jenkins and Karno, 1992), and closeness that looks like over-involvement by Western norms may be ordinary support. A therapist unaware of a family's culture and customs will misread it and make wrong assessments.

Bowen's family systems therapy ○

Murray Bowen (1913 to 1990), who admitted whole families to hospital wards in the 1950s, founded the transgenerational school. Its master concept is **differentiation of self**: separating thinking from feeling, and staying oneself while emotionally connected; the poorly differentiated are fused and reactive. Under stress a two-person relationship draws in a third person, usually a child: **triangulation**. Table 7.17 lists all eight concepts. Bowen worked as a calm coach who stayed out of the family's triangles, and mapped three or more generations in a **genogram** (squares for males, circles for females, lines for marriages, conflicts and cutoffs).

Table 7.17 Bowen's eight concepts.

Concept	Meaning
Differentiation of self	Thinking separate from feeling
Triangles	A stressed pair draws in a third
Nuclear family emotional system	Fusion shows as marital conflict, an impaired spouse or a symptomatic child
Family projection process	Parents project anxiety onto one child
Multigenerational transmission	Low differentiation and anxiety pass down the generations
Emotional cutoff	Escaping fusion by distance from the family
Sibling position	Birth-order roles shape adult roles
Societal regression	Stressed societies lose differentiation too

> **! EXAM TRAP**
>
> Transgenerational, multigenerational transmission, differentiation and the genogram mean **Bowen**. Diagrams differ: a genogram maps a family across generations; an **ecomap** (**Ann Hartman**, a social worker, in the 1970s) maps a family's links with school, work and services; a **sociogram** (Moreno, below) maps relations and communication within a group; a cognitive map is Tolman's mental map of a maze (Chapter 1).

Minuchin's structural family therapy ●●

Salvador Minuchin (1921 to 2017) worked with poor families of delinquent boys (*Families of the Slums*, 1967) and then at the Philadelphia Child Guidance Clinic (*Families and Family Therapy*, 1974). He studied the family's present *structure*: its subsystems (spousal, parental, sibling), a hierarchy with parents in charge, and boundaries that should be clear, firm yet permeable. Diffuse boundaries produce **enmeshment** (over-involvement, intrusion, little autonomy); rigid boundaries produce **disengagement** (isolation, little support), as Figure 7.12 shows. *Psychosomatic Families* (1978) described families of children with anorexia, diabetes and asthma as enmeshed, overprotective, rigid and unable to resolve conflict.

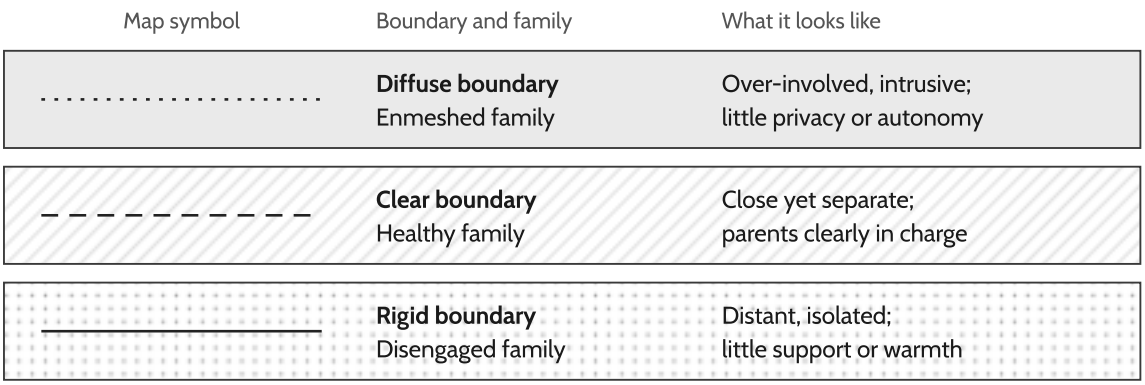


Figure 7.12 Minuchin's boundaries and their structural-map symbols (dotted, dashed, solid). Health is the middle row.

The therapist begins by **joining**, accommodating to the family's style, then maps its structure and asks for an **enactment**: the conflict acted out in the room rather than described. **Unbalancing** (siding with one member, often an overruled parent, to shift power), boundary making and reframing then restructure the family.

> **! EXAM TRAP**
>
> Enmeshment means **diffuse** boundaries; rigid boundaries mean disengagement. Enmeshment is not a complete lack of communication, nor scapegoating. Structural is Minuchin, strategic is Haley, multigenerational is Bowen, experiential is Satir. A therapist's boundary violation with a client is professional ethics (Section 7.1).

Strategic, systemic and other family therapies ○

Jay Haley (1923 to 2007), influenced by the hypnotherapist **Milton Erickson**, made therapy strategic: the therapist plans how to remove the presenting problem through *directives* (tasks) and does not seek insight. With resistant families he used **paradoxical intervention**, such as prescribing the symptom (a quarrelling couple told to quarrel at a set hour); in **ordeal therapy** (1984) the client performs a harmless but tiresome ordeal every time the symptom occurs. **Cloe Madanes** (1981) added pretend techniques: the child pretends to have the symptom and the parent pretends to help.

At the Mental Research Institute in Palo Alto (founded by Jackson, 1959), **Paul Watzlawick**, Weakland and **Richard Fisch** offered brief therapy of about ten sessions. In *Change* (1974) they argued that *attempted solutions* keep problems going: a wife nags her husband to exercise, he resists, she nags more. Trying harder is **first-order change**, within unchanged rules; the goal is **second-order change**, a change in the rules themselves.

The Milan group (**Mara Selvini Palazzoli**, **Luigi Boscolo**, **Gianfranco Cecchin** and **Giuliana Prata**; *Paradox and Counterparadox*, 1978) worked as a team behind a one-way mirror. Their **positive connotation** recasts everyone's behaviour, symptom included, as serving the family's unity, and prescribed rituals interrupt the family's game. Their three guidelines were hypothesising (a provisional systemic explanation, revised each session), circularity (exploring relationships and differences) and neutrality (siding with no one), and **circular questioning**

asks one member about the relationship between others: "Who is most upset when Anu won't eat?"

Among the experiential therapists, **Virginia Satir** (1916 to 1988; *Peoplemaking*, 1972) named four stances that protect low self-esteem under stress: placating (always agreeing), blaming, computing or super-reasonable (cold logic) and distracting or irrelevant, against congruent, levelling communication in which words, feelings and body match. In **family sculpting** she posed members physically to show their relationships. **Carl Whitaker** (1912 to 1995) practised spontaneous, often absurd symbolic-experiential therapy with a co-therapist. Table 7.18 completes the map.

Table 7.18 Other family therapies.

Approach	Founders	Core idea and methods
Contextual	**Ivan Boszormenyi-Nagy** (*Invisible Loyalties*, 1973)	Relational ethics: loyalties and a ledger of fairness across generations
Behavioural family	**Gerald Patterson**; Falloon	Parent training (Section 7.10), contracts, communication and problem-solving skills
Narrative	**Michael White** and **David Epston** (1990)	Social constructionist: lives as stories; externalising the problem, unique outcomes, re-authoring
Solution-focused	**Steve de Shazer** and **Insoo Kim Berg**	Build on what works: miracle, exception and scaling questions (Section 7.6)
Functional family therapy	**James Alexander** and **Bruce Parsons**	A delinquent adolescent's behaviour serves a family function
Multisystemic therapy	**Scott Henggeler**	Intensive home-based work on youth offending across family, peers and school

> **! EXAM TRAP**
>
> Prescribing the symptom is strategic paradox; exception questions are solution-focused; externalisation is narrative. "Individuals, families and cultures hold different power as creators of their realities" describes **narrative therapy**'s social constructionism, not transgenerational, strategic or structural therapy.

Couples and sex therapy ○

When a couple's distress keeps an individual problem going, the pair is treated together. For depression bound up with marital discord, marital and couple therapy is the better choice: behavioural marital therapy relieves the depression about as well as individual cognitive therapy and also mends the marriage.

Behavioural couples therapy (**Neil Jacobson** and **Gayla Margolin**, 1979) increases positive behaviour exchange and teaches communication and problem solving. Jacobson and **Andrew Christensen**'s integrative behavioural couple therapy adds *acceptance* of differences that will not change. **John Gottman**, filming couples in conflict, found that his **four horsemen** predict divorce: criticism (attacking character), contempt (mockery; the strongest predictor), defensiveness and stonewalling (withdrawal). Stable couples keep at least five positive interactions for every negative one, the 5:1 ratio, and his Sound Relationship House builds from love maps and fondness up to shared meaning, on walls of trust and commit-

ment. **Emotionally focused couple therapy** (**Sue Johnson**, 1980s) uses attachment theory: fights are protests at a threatened bond, locked in cycles such as pursue and withdraw.

William Masters and **Virginia Johnson** (*Human Sexual Inadequacy*, 1970) treated the couple as the patient, with a male and female co-therapist team. Their **sensate focus** is graded, non-demand touching with intercourse at first banned, removing performance anxiety; for premature ejaculation they taught the squeeze technique (firm pressure where the head of the penis joins the shaft, just before ejaculation). The stop-start technique, pausing stimulation before the point of no return, is **James Semans**'s (1956). **Helen Singer Kaplan** (*The New Sex Therapy*, 1974) blended these tasks with psychodynamic work in brief sex therapy. **Jack Annon**'s **PLISSIT model** (1976) grades help: Permission, Limited Information, Specific Suggestions, Intensive Therapy.

> **! EXAM TRAP**
>
> Contempt, not criticism, is Gottman's best predictor of divorce. Squeeze is Masters and Johnson's; stop-start is Semans's (Masters and Johnson adapted Semans's method, so some textbooks credit Semans with both). Emotionally focused couple therapy (Johnson) is not Greenberg's individual emotion-focused therapy (Section 7.6).

Group therapy: Yalom's factors and group stages ●●

Why do groups help? **Irvin Yalom**'s *The Theory and Practice of Group Psychotherapy* (first edition 1970) sets out eleven **therapeutic factors** (Table 7.19). Of these, **universality**, discovering that one is not alone in one's suffering, is often a newcomer's first relief.

Table 7.19 Yalom's eleven therapeutic factors.

Factor	What happens
Instillation of hope	Seeing others improve
Universality	Others share my problem
Imparting information	Advice and teaching
Altruism	Helping others raises self-worth
Corrective recapitulation of the primary family group	Family conflicts relived and resolved differently
Development of socialising techniques	Social skills learned through feedback
Imitative behaviour	Modelling the leader and members
Interpersonal learning	Learning how one affects others
Group cohesiveness	Belonging; the group's therapeutic relationship
Catharsis	Expressing strong feeling
Existential factors	Accepting death, aloneness and responsibility

The engine of Yalom's group is the **here-and-now**. The group is a social microcosm, where members soon behave as they do in life, so the leader works with interactions in the session: first experiencing them, then illuminating the process by reflecting on what just happened. The leader creates and maintains the group, builds its norms, and activates and illuminates the here-and-now. **Bruce Tuckman** (1965) traced group development through forming (po-

lite testing), storming (conflict), norming (cohesion) and performing, adding adjourning with **Mary Ann Jensen** in 1977. A closed group keeps the same members for a fixed term and bonds quickly; an open group admits new members as others leave.

> ! EXAM TRAP
>
> "Not alone" is **universality**; seeing others recover is hope; helping is altruism; venting is catharsis. The here-and-now does not ban talk of outside life; it makes the group's own interactions, experienced and then reflected upon, the engine of change, and it is not didactic teaching or disputing irrational beliefs (Section 7.5). Free association, token reinforcement and schema work are not Yalom factors. Yalom's four ultimate concerns belong to existential therapy (Section 7.6).

Psychodrama, encounter and self-help groups, therapeutic communities ●●●

Jacob Levy Moreno (1889 to 1974), who coined the term group psychotherapy, opened a Theatre of Spontaneity in Vienna in 1921. In **psychodrama** the *protagonist* acts out scenes on a *stage*, guided by the therapist as *director*, with members as *auxiliary egos* playing significant others and an *audience* that shares afterwards. In role reversal the protagonist plays the other person; in doubling an auxiliary voices the protagonist's unspoken feelings; in mirroring someone replays the protagonist's behaviour for them to watch. Moreno's sociometry measures attraction and rejection in groups, and its **sociogram** draws members as points and choices as arrows, revealing the group's explicit and implicit communication patterns.

Kurt Lewin (1890 to 1947) inspired the **T-group** (training or sensitivity group, National Training Laboratories, 1947), in which members learn group dynamics and their effect on others. The **encounter group**, championed by **Carl Rogers**, encourages therapy and self-growth through open disclosure and interaction. **Self-help groups** are run by members, not professionals: Alcoholics Anonymous, founded in 1935 by **Bill Wilson** and **Bob Smith**, works through twelve steps, sponsors and anonymity, and Narcotics Anonymous adapted the steps to drugs.

In **milieu therapy** the whole ward environment, staff, routine and fellow patients, is the treatment. Its democratic form, the **therapeutic community**, was pioneered by **Maxwell Jones** (1907 to 1990) at Belmont Hospital in Surrey, whose unit later became the Henderson Hospital: staff and patients share decisions in daily community meetings, hierarchy is flattened, and every interaction becomes feedback. Addiction communities are in Section 7.9.

> ! EXAM TRAP
>
> Maxwell Jones means the **therapeutic community**. Milieu therapy is the broader idea; group analysis is **S. H. Foulkes**'s psychoanalytic group therapy; encounter groups are Rogers's and psychodrama Moreno's; the token economy, aversion therapy and EMDR belong to behaviour therapy (Section 7.4), and logotherapy is Frankl's (Section 7.6). Encounter groups aim at growth, not diagnosis or team building.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Treating the family as a whole with homeostasis and feedback applies the general systems theory of _____.
2. Contradictory messages that can be neither questioned nor escaped form a _____.
3. High expressed emotion comprises critical comments, hostility and _____.
4. Minuchin's diffuse boundaries produce an _____ family.
5. Gottman's strongest predictor of divorce is _____.
6. Yalom's "I am not alone" factor is _____.
7. The therapeutic community was pioneered by _____.

Answers: (1) Ludwig von Bertalanffy, (2) double bind, (3) emotional over-involvement, (4) enmeshed, (5) contempt, (6) universality, (7) Maxwell Jones.

7.8 Does Psychotherapy Work? Evidence, Common Factors and the Alliance LOWER YIELD · 12 QS

Sections 7.3 to 7.7 described a crowded marketplace of therapies. Does any of it help, and if so, why? The story runs from Eysenck's attack, through the statistics that answered him, to a debate that continues: does the technique heal, or the relationship, the client and the therapist?

Eysenck's challenge and the meta-analytic answer

In 1952 **Hans Jürgen Eysenck** (1916 to 1997) reviewed the published outcome reports. Neurotic patients given psychoanalysis improved at about 44 percent and those given eclectic therapy at about 64 percent, but comparable patients without psychotherapy (insurance-disability claimants seen by general practitioners, and state-hospital discharges) improved at about 72 percent. Roughly two-thirds of neurotic patients, he concluded, improve within two years with or without therapy, through **spontaneous remission**: recovery driven by time, life events and ordinary support.

Critics attacked his method: the groups differed in severity, "improved" meant different things in different studies, and the "untreated" patients had received reassurance and sedatives. **Allen E. Bergin** (1971) showed that counting dropouts differently lifted psychoanalysis to about 85 percent improvement among those who completed it; **Michael J. Lambert** (1976) showed that Eysenck had inflated untreated improvement, putting spontaneous remission nearer 40 percent. The decisive answer came from a new statistic.

Gene V. Glass coined the term **meta-analysis** in 1976 for pooling many studies statistically instead of counting how many "worked". Each comparison of a treated with a control group yields an **effect size** (d): the difference between the group means divided by a standard deviation. Effect sizes, all in the same unit, can then be averaged across studies and compared across therapies. A d of 0.85 places the average treated client above about 80 percent of untreated people (Figure 7.13).

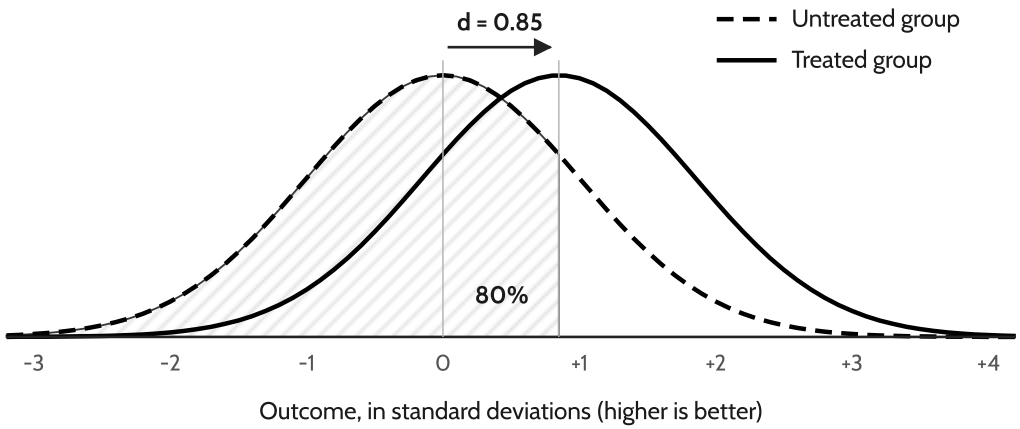


Figure 7.13 What an effect size of 0.85 means. The average treated client (solid curve) scores better than the shaded 80 percent of the untreated group (dashed curve).

KEY STUDY: SMITH AND GLASS (1977)

Mary Lee Smith and Gene Glass pooled 375 controlled studies. The average effect size was 0.68: the typical client was better off than 75 percent of untreated people, and types of therapy differed little. With Thomas Miller (*The Benefits of Psychotherapy*, 1980) they reached 475 studies and 0.85 (80 percent). Eysenck called it "mega-silliness": good and bad studies pooled ("garbage in, garbage out"), unlike outcomes averaged ("apples and oranges").

! EXAM TRAP

Eysenck said therapy adds nothing to spontaneous remission, not that it harms. 0.68 and 75 percent is Smith and Glass (1977); 0.85 and 80 percent is Smith, Glass and Miller (1980).

The dodo bird verdict and common factors ●●

Smith and Glass also found the rival schools roughly equal. **Saul Rosenzweig** (1907 to 2004) had predicted this in 1936: diverse therapies succeed through implicit common factors, he argued, quoting the Dodo in *Alice in Wonderland*: "Everybody has won, and all must have prizes." In 1975 **Lester Luborsky**, Barton Singer and Lise Luborsky reviewed the comparative trials and named the **dodo bird verdict**: bona fide therapies, competently delivered, produce roughly equivalent outcomes, because change comes mainly from the **common factors** they share (the alliance, empathy, hope) rather than from specific techniques.

Specific-effects researchers, holding a medical model in which each therapy has active ingredients like a drug, point to exceptions: exposure beats non-exposure therapies for obsessive-compulsive disorder, phobias and panic. Common-factors researchers reply that most comparisons of bona fide therapies show differences near zero (at most about 0.2 in a 1997 meta-analysis) and that apparent winners often reflect allegiance (7.8.4).

Bruce E. Wampold formalised this view as the **contextual model** (*The Great Psychotherapy Debate*, 2001; in its fullest form in the 2015 edition with Zac Imel): therapy works through a real relationship; through expectations created by a convincing explanation and treatment plan; and through specific ingredients that help because they get the client doing healthy things, not because they correct a specific deficit.

Its root is **Jerome David Frank** (1909 to 2005), whose *Persuasion and Healing* (1961) compared psychotherapy with religious healing and placebo. Clients arrive *demoralised*, and every healing method restores morale through four shared features:

- The client has an emotionally charged, confiding relationship with a helper.
- Help takes place in a healing setting (a clinic, a temple, a consulting room).
- A rationale or "myth" plausibly explains the symptoms and how to overcome them.
- A ritual or procedure, believed in by both, requires the client's active participation.

> **! EXAM TRAP**
>
> "Healing and Persuasion" is Frank's *Persuasion and Healing* with the title reversed. Distractors: **Gerald Klerman** co-developed interpersonal psychotherapy and **Donald Winnicott** was an object-relations analyst (both 7.3); **Alfred Pritz** is the Austrian psychotherapist who was founding president of the World Council for Psychotherapy (1995). The dodo verdict never says therapy is ineffective, that only psychoanalysis lasts or that medication always wins: bona fide therapies tie.

Lambert (1992) estimated, in his famous pie (Figure 7.14), that about 40 percent of improvement comes from client and extratherapeutic factors, 30 percent from relationship factors, 15 percent from expectancy (placebo and hope) and only 15 percent from model-specific techniques: rough judgements from the literature, not statistical results.

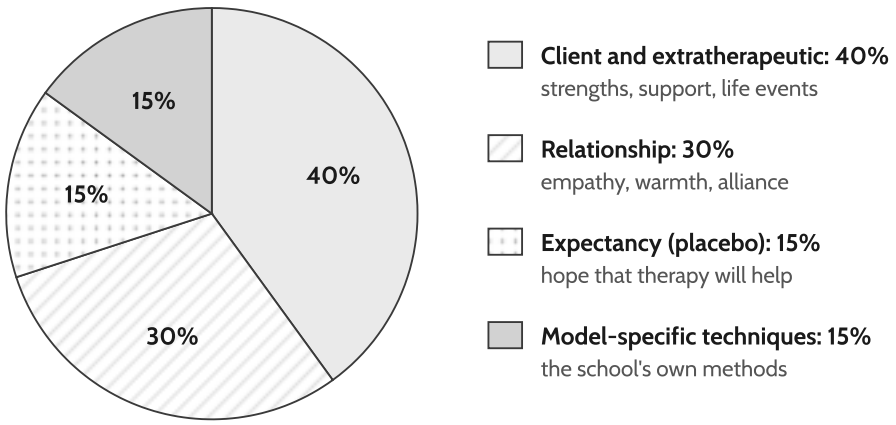


Figure 7.14 Lambert's (1992) rough estimates of what produces improvement. Technique, the slice the schools argue about, is the smallest.

The therapeutic alliance and the core conditions ●●

The alliance idea began in psychoanalysis. **Elizabeth Zetzel** (1956) spoke of the therapeutic alliance and **Ralph Greenson** (1965) of the working alliance, the patient's rational, cooperative tie to the analyst as distinct from transference; Greenson also described the *real relationship*, the genuine person-to-person bond. **Edward S. Bordin** (1979) made the **therapeutic alliance** pantheoretical: a collaborative relationship with three parts, agreement on **goals**, agreement on the **tasks** that reach them, and a **bond** of trust and respect.

A working relationship is thus a contract to work together realistically and collaboratively, visible in how well the two actually work together and resting on a committed belief in the helping relationship; it needs no particular theory. Adam Horvath and Leslie Greenberg's (1989) **Working Alliance Inventory** (36 items, with a 12-item short form) measures goals, tasks and bond. Across meta-analyses the alliance correlates about 0.28 with outcome: modest but robust.

Jeremy D. Safran and J. Christopher Muran described **alliance ruptures**: withdrawal (the client falls silent, goes vague, over-agrees) and confrontation (the client complains or attacks). Repair means noticing the strain, talking openly about what is happening between the two and owning one's part; repaired ruptures predict good outcomes.

Carl Ransom Rogers (1957) held that his core conditions (empathy, unconditional positive regard, congruence; 7.6) were necessary and sufficient. Research supports a weaker claim: empathy correlates about 0.3 with outcome, positive regard and congruence a little less (roughly 0.2 to 0.3): facilitative in every therapy, but not sufficient.

> **! EXAM TRAP**
>
> Bordin's alliance is not a Rogerian concept; Rogers' terms are the actualising tendency, conditions of worth and unconditional positive regard (7.6). Person-centred therapy is the school least likely to say "therapeutic alliance": it treats the relationship and unconditional positive regard as the therapy itself, whereas psychoanalysis coined the term and cognitive-behavioural and interpersonal therapists use Bordin's version. The alliance is not an outcome score, not two clinicians agreeing on a diagnosis (inter-rater reliability, Chapter 2) and not the client's ability to do homework alone. What a working relationship does NOT need: "a psychodynamic theory base".

Therapist, client and allegiance effects ○

Consistent differences between individual therapists' outcomes, whatever their approach, are called **therapist effects**; they account for about 5 percent of outcome variance (roughly 3 in controlled trials, 7 in routine practice), far more than the differences between treatments. An **allegiance effect** appears when researchers find their own favoured therapy superior; Luborsky and colleagues (1999) showed that researcher allegiance predicted most of the outcome differences in comparative trials.

Client factors carry the biggest slice of the pie. **William Schofield** (*Psychotherapy: The Purchase of Friendship*, 1964) noted that therapists prefer, and do best with, the **YAVIS** client: young, attractive, verbal, intelligent and successful. YAVIS describes patient characteristics, not a technique, an illness or a therapist quality; its opposite, HOUND (homely, old, unattractive, nonverbal, dumb), names neglected clients. Readiness to change (stages of change, 7.6) and realistic expectations of outcome (hope) and of role (what a client does) also predict benefit.

In the **hello-goodbye effect**, described by **Starke R. Hathaway**, clients exaggerate distress at intake (to justify seeking help) and well-being at the end (to thank the therapist), so before-and-after self-reports overstate improvement.

Testing outcome: efficacy, effectiveness and ESTs ●●

Research on **efficacy** asks whether a treatment can work under ideal, controlled conditions (internal validity); research on **effectiveness** asks whether it does work in routine practice (external validity), as in **Martin E. P. Seligman**'s (1995) *Consumer Reports* survey, in which most readers reported benefit (Table 7.20).

Table 7.20 Efficacy and effectiveness studies.

Feature	Efficacy study	Effectiveness study
Clients	Homogeneous, one disorder	Mixed, often comorbid
Assignment	Random assignment to treatment or control	Often not random
Therapist	Controlled by a manual	Flexible
Sessions	Fixed number	Set by need

In **analogue research**, clinical questions are studied under conditions that only resemble therapy, usually with healthy, non-clinical participants (students with a mild fear of snakes standing in for phobic patients): control is high, generalisation uncertain. It is neither outcome research on a clinical population nor comparative research pitting two treatments against each other.

In the **randomised controlled trial** (RCT), chance decides who receives which condition, so the groups differ only in treatment. Its safeguard, **allocation concealment**, hides the upcoming allocation sequence from the recruiter until each participant is enrolled, preventing selection bias. Blinding hides the assigned condition after allocation; in psychotherapy only outcome assessors can be blind. The comparison group decides what a trial can show (Table 7.21).

Table 7.21 Control conditions in psychotherapy trials.

Control	Group receives	Controls for
Waiting list	Treatment after a delay	Time, spontaneous remission
Placebo-attention	Credible contact, no active ingredient	Attention and expectancy
Treatment as usual	Routine care	Usual care; shows added benefit

Trials follow treatment manuals (Aaron Beck's 1979 manual for cognitive therapy of depression was an early model) and check **treatment fidelity**: adherence to the manual and competence in delivering it.

In 1995 an APA Division 12 (Clinical Psychology) task force chaired by **Dianne L. Chambless** began listing **empirically supported treatments** (ESTs). A *well-established* treatment needed at least two good between-group experiments, by independent teams, showing it superior to a pill or psychological placebo or another treatment, or equal to an established one (or a large single-case series), with a manual and clearly described clients. A *probably efficacious* treatment needed only two experiments beating a waiting list, or strong trials from one team. Chambless and Steven Hollon (1998) refined the criteria; most listed treatments were cognitive-behavioural.

The APA's 2006 task force defined **evidence-based practice in psychology** more broadly: "the integration of the best available research with clinical expertise in the context of patient characteristics, culture, and preferences."

> **! EXAM TRAP**
>
> Efficacy studies use manuals, homogeneous samples and random assignment; "therapist flexibility is encouraged" marks effectiveness research. Allocation concealment (before enrolment) is not blinding (after allocation).

Landmark trials, dose and clinical significance

KEY STUDY: NIMH TREATMENT OF DEPRESSION COLLABORATIVE RESEARCH PROGRAM (1989)

Irene Elkin's trial randomly assigned about 250 depressed outpatients at three sites to 16 weeks of cognitive-behavioural therapy, interpersonal psychotherapy, imipramine or a pill placebo (the last two with clinical management). All groups improved and the two psychotherapies did not differ; only in severe depression did imipramine clearly beat placebo, with interpersonal therapy showing some advantage. Re-analyses found that the alliance predicted outcome in every arm, and that therapists differed more than treatments.

Project MATCH (1997) assigned about 1,700 people with alcohol problems to twelve-step facilitation, cognitive-behavioural coping-skills therapy or four-session motivational enhancement therapy, to test whether matching clients to treatments helps. All three worked about equally; matching effects were scarcely found.

Kenneth I. Howard and colleagues (1986), pooling about 2,400 outpatients from 15 studies, described the **dose-effect relationship** (Figure 7.15): about 50 percent had improved by 8 sessions, 75 percent by 26 and 85 percent by 52. The curve is negatively accelerated; Howard's later phase model reads it as quick remoralisation, then remediation of symptoms, then slow rehabilitation.

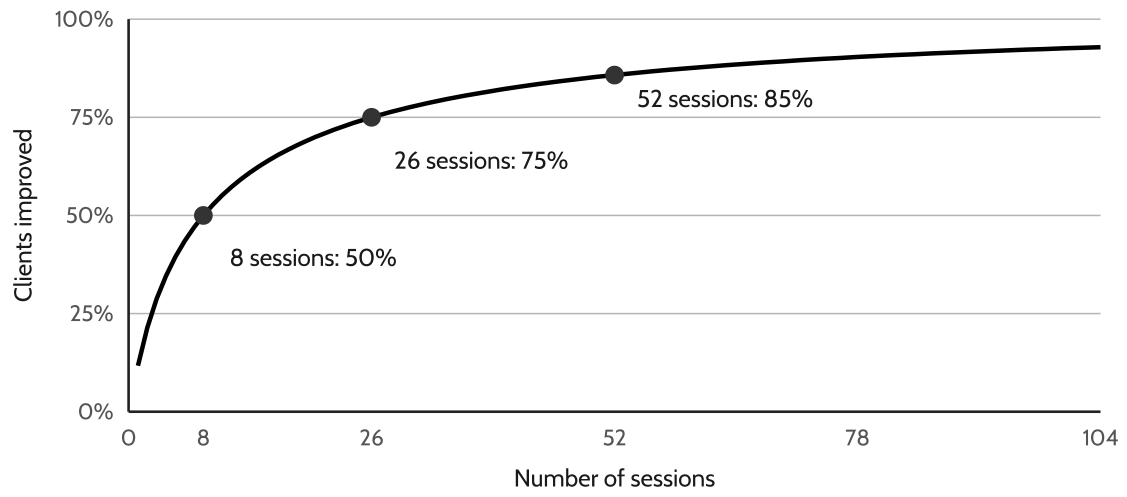


Figure 7.15 The dose-effect relationship (Howard and colleagues, 1986), a smooth curve through the published milestones: most of the benefit comes early.

A large sample can make a small average gain statistically significant while many clients stay unwell. **Neil S. Jacobson** and Paula Truax (1991) therefore judged **clinical significance** client by client, with two tests. The **reliable change index** (RCI), the change divided by the standard error of the difference (derived from the test's reliability), must exceed 1.96, so the change is unlikely to be measurement error; and the client must cross a cut-off from the dysfunctional into the functional range. Both tests passed means recovered; reliable change alone, improved; reliable worsening, deteriorated.

Take Meena, who falls from 32 to 12 on a depression scale with a standard deviation of 8 and reliability of 0.84. The standard error of measurement is $8 \times \sqrt{0.16} = 3.2$, the standard error of the difference $\sqrt{2} \times 3.2 \approx 4.5$, so her RCI is $20 \div 4.5 \approx 4.4$: reliable. With a functional cut-off of 14, she has recovered. A fall from 32 to 27 (RCI about 1.1) is not reliable change.

Therapy can also harm. Bergin (1966) named the **deterioration effect**: roughly 5 to 10 percent of adult clients get worse, which is why Lambert's Outcome Questionnaire-45 feeds each client's progress back to the therapist.

Process research and psychotherapy integration

Where outcome research asks whether therapy works, **process research** studies what happens within sessions and how it produces change. Therapy has a beginning (assessment, alliance, goals, contract, teaching the client's role), a middle or working phase (the main interventions and working through) and a planned termination (consolidating gains, preparing for setbacks, handling the ending).

Many clients end in **premature termination**, leaving before goals are met and against the therapist's advice: near half of outpatients in a 1993 meta-analysis, about one in five in a 2012 one, more often with young and poor clients, weak alliances and mismatched expectations. Classical analysts shunned **therapist self-disclosure** and humanistic therapists value it; it helps when brief, rare, client-focused and about resolved issues. Homework, central to cognitive and behavioural therapy, carries therapy into the client's week; completing it predicts better outcome.

Psychotherapy integration takes four routes (Table 7.22). In **technical eclecticism** techniques are borrowed without their theories; **theoretical integration** fuses theories into a new framework; **assimilative integration** keeps one home theory and absorbs techniques into it.

Table 7.22 Routes to psychotherapy integration.

Route	Example
Technical eclecticism	**Arnold A. Lazarus**: multimodal therapy, BASIC I.D. (7.5)
Theoretical integration	**Paul L. Wachtel**: cyclical psychodynamics; **James O. Prochaska**: transtheoretical approach
Common factors integration	Frank's four features; Wampold's contextual model
Assimilative integration	**Stanley B. Messer**: psychodynamic therapy using borrowed techniques

Wachtel's cycles are vicious circles (inner conflict breeds self-defeating behaviour that deepens the conflict), so he joins psychodynamic exploration to behavioural methods. Prochaska built his approach around the client's stage of readiness (7.6).

Culture-sensitive therapy and therapy in India ○

Western therapy prizes autonomy, insight and an equal, non-directive relationship. A therapist unaware of a family's culture and customs risks wrong assessments and misinterpretations, such as calling a joint family's shared decisions enmeshment. A **culturally adapted therapy** modifies an evidence-based treatment's language, metaphors, content, goals and methods to fit the client's culture (**Guillermo Bernal**'s ecological validity framework), and meta-analyses find adapted versions modestly more effective.

N. C. Surya, writing with S. S. Jayaram (1964), argued that Indian patients are more dependent than Western ones and less inclined to integrate inner processes, so the therapist should use the patient's own language and give direct, situational support rather than intrapsychic interpretation. **Jaswant Singh Neki** (1973) proposed the **guru-chela relationship** as an Indian therapeutic paradigm: a wise, warm, guiding therapist and a client who accepts a phase of dependence, normal in Indian relationships, which the guru gradually turns into self-reliance. Families usually accompany patients and join sessions, so therapists work with them as allies (7.1). Yoga-based programmes have been tested at NIMHANS as add-on treatment for depression and schizophrenia, and mindfulness is taught in familiar forms alongside pranayama and meditation.

Training, supervision and the therapist's self-care

Clinical supervision builds a trainee's skill and protects clients. Developmental models (Cal Stoltenberg's) trace trainees from anxious dependence to autonomy; psychotherapy-based models supervise in the language of the therapy taught; **Janine M. Bernard**'s (1979) discrimination model has the supervisor switch between teacher, counsellor and consultant roles across intervention, conceptualisation and personalisation. In parallel process the trainee re-enacts with the supervisor what is happening with the client.

Carl Gustav Jung's image of the **wounded healer**, after the wounded centaur Chiron, holds that a therapist's understood wounds deepen empathy while unexamined ones become countertransference. The risk of **burnout**, which **Christina Maslach** described as emotional exhaustion, depersonalisation and reduced personal accomplishment, makes self-care (supervision, personal therapy, boundaries, manageable caseloads, rest) an ethical duty.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Eysenck (1952) said two-thirds of neurotic patients improve without therapy, through _____.
2. Smith and Glass (1977) found an average effect size of _____.
3. The _____ verdict holds that bona fide therapies produce equivalent outcomes.
4. Frank's *Persuasion and Healing* (1961) says clients arrive _____.
5. Bordin's alliance has three parts: goals, tasks and _____.
6. Schofield's (1964) preferred client is _____.
7. Ideal-conditions research tests _____; routine-practice research tests _____.
8. Jacobson and Truax's _____ index must exceed 1.96.

Answers: (1) spontaneous remission, (2) 0.68, (3) dodo bird, (4) demoralised, (5) bond, (6) YAVIS, (7) efficacy; effectiveness, (8) reliable change.

7.9 Health, Community, Crisis and Rehabilitation Psychology

LOWER YIELD · 15 QS

Clinical psychology does not end at the consulting-room door. Most deaths in India now come from illnesses driven partly by behaviour, and most people with a mental illness never see a specialist. This section follows the psychologist onto the ward, into the community, into crises and into rehabilitation.

Health psychology and the sick role ○

In 1980 **Joseph D. Matarazzo**, first president of the American Psychological Association's Division of Health Psychology (Division 38, founded 1978), defined **health psychology** as the aggregate of psychology's educational, scientific and professional contributions to promoting and maintaining health, preventing and treating illness, and identifying the causes and correlates of health and illness. It is one discipline's field, leaning towards promotion and prevention. **Behavioural medicine** (**Gary Schwartz** and **Stephen Weiss**, 1978) is an interdisciplinary field that integrates behavioural and biomedical knowledge and techniques for prevention, diagnosis, treatment and rehabilitation. Both apply **George Engel**'s biopsychosocial model (1977; Section 7.1).

What a person does with a symptom, ignoring chest pain for a week or rushing to casualty, is **illness behaviour** (**David Mechanic**, 1962). Once accepted as ill, a person enters the **sick role** of **Talcott Parsons** (*The Social System*, 1951), with two rights (exemption from normal duties; no blame for the illness) and two obligations (to want to get well; to seek competent help and cooperate with it). It fits acute illness better than chronic or stigmatised illness. Seeking the sick role itself is the motive in factitious disorder; malingering seeks external gain (Chapter 8).

> **! EXAM TRAP**
>
> Behavioural medicine, not health psychology, is the interdisciplinary field integrating behavioural and biomedical interventions (JEMAS 2025). Options such as "only uses medication" are nonsense. Health psychology is Division 38; clinical psychology, Division 12.

Models of health behaviour and adherence ●●

Models of health behaviour explain why knowing the risks is not enough (Table 7.23). The **health belief model** grew from a 1950s puzzle, the poor uptake of free tuberculosis X-rays, studied by **Godfrey Hochbaum**, **Irwin Rosenstock** and **Stephen Kegeles** in the United States: people acted when they felt susceptible, judged the disease severe, saw benefits outweighing barriers and met a cue to action. Rosenstock gave it its classic statement (1966), and **Marshall Becker** (1974) extended it to sick-role behaviour and adherence. **Martin Fishbein** and **Icek Ajzen**'s theory of reasoned action (1975) made intention the immediate cause of behaviour; Ajzen's **theory of planned behaviour** added perceived behavioural control (Figure 7.16). **Ronald Rogers**'s **protection motivation theory** (1975) shows that fear appeals work only when coping appraisal is also high, and **Ralf Schwarzer**'s health action process approach (1992) adds planning to turn intention into action.

Table 7.23 Models of health behaviour and their constructs.

Model	Proposed by	Key constructs
Health belief model	Hochbaum, Rosenstock, Kegeles (1950s); Rosenstock 1966	Susceptibility, severity, benefits, barriers, cues to action; self-efficacy added 1988
Theory of planned behaviour	Ajzen 1985, 1991	Attitude, subjective norm, perceived behavioural control
Protection motivation theory	Rogers 1975, revised 1983	Threat appraisal; coping appraisal (response efficacy, self-efficacy, costs)
Social cognitive theory	**Albert Bandura** (self-efficacy 1977)	Self-efficacy (confidence that one can act)
Transtheoretical model	Prochaska and DiClemente 1983	Stages of change (Section 7.6)
Health action process approach	Schwarzer 1992	Motivational, then volitional phase; action and coping plans

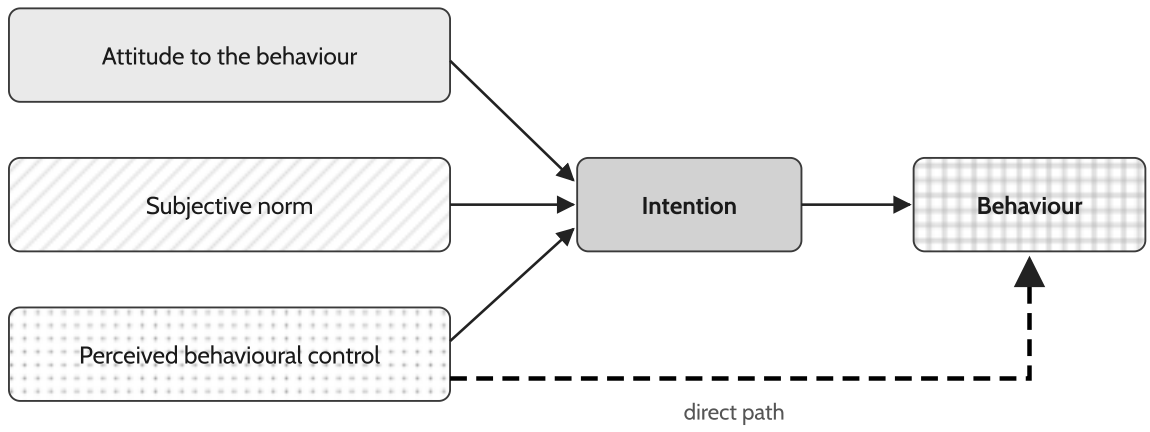


Figure 7.16 Ajzen's theory of planned behaviour; without the lowest box it is the theory of reasoned action.

> **! EXAM TRAP**
>
> "Susceptibility, severity, benefits, barriers" is always the health belief model (JEMAS 2025). RML 2023 keyed the theory of planned behaviour as "attitude, subjective norms and self-efficacy": pick it, though strictly the third factor is perceived behavioural control, which Ajzen (1991) likened to Bandura's self-efficacy; the other options are not its constructs.

Medicine most needs **adherence**, following an agreed treatment; the older word compliance implied obedience (in social psychology it means yielding to a direct request, Chapter 3). Only about half of chronically ill patients in developed countries take treatment as prescribed (World Health Organization, 2003). Clear written instructions, simple regimens, reminders, family involvement and a good relationship all help. **Thomas Szasz** and **Marc Hollender** (1956) ranged patient-practitioner relationships from activity-passivity (the unconscious patient) through guidance-cooperation to mutual participation, best for chronic illness.

Stress and physical illness ●●

Stress itself is taught in Section 1.11: **Hans Selye**'s general adaptation syndrome (alarm, resistance, exhaustion), the life-events scale of **Thomas Holmes** and **Richard Rahe**, and psychoneuroimmunology, begun when **Robert Ader** and **Nicholas Cohen** (1975) conditioned immune suppression in rats. Health psychology adds that an acute stressor is brief with a clear end (a pop quiz), a chronic stressor persists (marital discord, HIV disease, colon cancer), and chronic stress is what suppresses immunity.

Franz Alexander (*Psychosomatic Medicine*, 1950) proposed that a specific unconscious conflict acting on a vulnerable organ caused each of seven **psychosomatic disorders**, the "holy seven": peptic ulcer, essential hypertension, bronchial asthma, rheumatoid arthritis, ulcerative colitis, neurodermatitis and thyrotoxicosis. The specificity idea failed (most ulcers proved bacterial, from *Helicobacter pylori*), but psychological factors are now known to shape the course of every illness (Table 7.24).

Table 7.24 Psychology in five physical illnesses.

Illness	Psychological link	Psychological care
Coronary heart disease	Type A (time urgency, competitiveness, hostility; **Meyer Friedman** and **Ray Rosenman** 1959), not relaxed Type B; hostility its toxic part	Anger and stress management
Cancer	Type C (**Lydia Temoshok**): unassertive, suppresses emotion; weak evidence	Distress, pain, conditioned nausea
HIV and AIDS	Spread by behaviour (unsafe sex, shared needles)	Test counselling, drug adherence, stigma
Type 2 diabetes	Poor diet and sedentary lifestyle	Self-management; lifestyle change cut new cases 58 percent (Diabetes Prevention Program, 2002)
Chronic pain (over three months)	Pain behaviours reinforced by sympathy and rest	Operant approach, CBT, pain clinics

In chronic pain, the gate control theory of **Ronald Melzack** and **Patrick Wall** (1965; Chapter 5) explains why attention and mood change pain. **Wilbert Fordyce** (1976) treated pain behaviours (grimacing, resting, asking for pills) as operants: his operant approach rewards well be-

haviour, raises activity by quotas and gives analgesics on a fixed schedule, not on complaint. Cognitive-behavioural therapy adds coping skills, pacing and challenges to catastrophising, usually in multidisciplinary pain clinics.

> **! EXAM TRAP**
>
> Offered both, RML 2024 keyed **Type A** (the whole pattern) for coronary risk; later research found **hostility** its toxic component, so answer hostility only when the stem asks for the component. For Type 2 diabetes the key is diet and inactivity, the strongest modifiable cause, not genes, stress or toxins.

Behavioural medicine and lifestyle change

Biofeedback displays a bodily signal (muscle tension, finger temperature, skin conductance, brain waves) so that a person can learn to control it, as in tension headache or migraine, though it often does no better than relaxation: **Edmund Jacobson**'s progressive muscle relaxation, **Johannes Schultz**'s autogenic training or **Herbert Benson**'s relaxation response (1975). Stress management adds cognitive restructuring and problem solving (Section 7.5).

Chronic insomnia is treated first with **cognitive behavioural therapy for insomnia**: stimulus control (**Richard Bootzin**, 1972: the bed only for sleep and sex; leave it if awake for about 20 minutes), sleep restriction (**Arthur Spielman**, 1987: time in bed cut to time asleep) and cognitive work on sleep worries. Sleep hygiene (no evening caffeine, a dark quiet room, regular hours) is not enough alone.

Smokers are helped by the five As (ask, advise, assess, assist, arrange follow-up) and nicotine replacement. Drinkers are screened with the World Health Organization's **Alcohol Use Disorders Identification Test** (AUDIT, 1989): 10 items scored 0 to 40, with 8 or more suggesting hazardous or harmful use and 20 or more possible dependence. A positive screen earns a brief intervention, a few short sessions in the style of motivational interviewing (Section 7.6). Weight management relies on self-monitoring, stimulus control and a modest 5 to 10 percent loss; exercise lasts when it is enjoyable, shared and planned.

Chronic illness, dying and grief ○

Chronic illness is self-managed for life; depression is common and worsens self-care, so psychologists teach self-management and involve the family. When cure is impossible, **palliative care** seeks comfort and quality of life for patient and family. It began with **Cicely Saunders**'s St Christopher's Hospice, London (1967), which treated "total pain": physical, emotional, social and spiritual.

Elisabeth Kübler-Ross (*On Death and Dying*, 1969) described five stages in dying patients: denial, anger, bargaining, depression and acceptance. They are neither fixed nor universal, came from the dying rather than the bereaved, and have weak research support. **J. William Worden** (1982) proposed four **tasks of mourning**, in their current wording: accept the reality of the loss, process the pain of grief, adjust to a world without the deceased, and find an enduring connection with the deceased while embarking on a new life. Most grief is normal and runs its course with ordinary support; counselling adds little. **Complicated grief**, intense yearning that persists and disables (prolonged grief disorder, Chapter 8), needs intervention, usually one to one.

> **! EXAM TRAP**
>
> "Crisis-related bereavement should always be considered traumatic grief" is the false statement (RML 2020): grief is usually normal, and "always" gives it away. Worden's are tasks, not stages.

Community mental health and prevention ○

President **John F. Kennedy**'s Community Mental Health Centers Act (1963) funded local centres to replace state hospitals. With antipsychotic drugs it sped **deinstitutionalisation**, but too few centres were built, and many discharged patients became homeless, imprisoned or caught in a "revolving door" of readmissions. Community psychology (Swampscott conference, 1965) turned to prevention and social change.

Gerald Caplan (*Principles of Preventive Psychiatry*, 1964) brought public health's three levels to mental health. **Primary prevention** lowers new cases (incidence) among the healthy: vaccination, health education. **Secondary prevention** lowers prevalence by early detection and prompt treatment. **Tertiary prevention** limits the disability of an established disorder: rehabilitation. **Robert Gordon** (1983) classified prevention by its target instead: universal (everyone), selective (high-risk groups, such as children of depressed parents) and indicated (individuals with early signs). Caplan (1970) also named four types of **mental health consultation**, helping another professional rather than the client: client-centred case, consultee-centred case, programme-centred administrative and consultee-centred administrative. **Julian Rappaport** (1981) preferred **empowerment**, people gaining control over their own lives.

Where specialists are scarce, **task-shifting** trains lay counsellors to deliver structured care under supervision. In Zimbabwe, **Dixon Chibanda**'s Friendship Bench has lay health workers, many of them grandmothers, offer problem-solving therapy on clinic benches. In Goa, **Vikram Patel**'s MANAS trial (2010) showed that lay counsellors in public primary-care clinics improved recovery from common mental disorders, and his group's PRIDE programme took lay-counsellor problem solving to adolescents in Indian schools. Stigma (**Erving Goffman**, 1963) deters help-seeking; contact with people who have recovered reduces it best. These attack the treatment gap, as do India's National Mental Health Programme (1982) and District Mental Health Programme (1996), modelled on a NIMHANS pilot in Bellary, Karnataka (Section 7.1).

> **! EXAM TRAP**
>
> Promoting vaccination is primary prevention (JEMAS 2025). Treating a diagnosed illness, surgery for chronic disease and rehabilitation after injury come later (secondary or tertiary). Caplan's levels say *when* you intervene; Gordon's types say *whom* you target.

Crisis intervention and suicide prevention ●●

KEY STUDY: LINDEMANN AND THE COCOANUT GROVE FIRE (1944)

On 28 November 1942 a fire at Boston's Cocoanut Grove nightclub killed 492 people. **Erich Lindemann** interviewed 101 bereaved people and in 1944 described acute grief as a syndrome (bodily distress, preoccupation with the dead, guilt, hostility) that most work through within weeks by "grief work", while some develop delayed or distorted grief. Timely brief help, he concluded, could prevent lasting disorder: the seed of crisis intervention. Criticism: impressionistic data and too short a time course for grief.

Caplan defined a **crisis** as an obstacle to important life goals that is, for a time, insurmountable by customary problem solving, followed by disorganisation; it usually resolves, for better or worse, within four to six weeks. Crises are developmental (retirement) or situational. Situational events are large-scale (floods, earthquakes, tornadoes) or small-scale, striking one person or family: violence (child abuse, domestic abuse, crime), psychopathology that triggers a crisis (a suicide attempt, an overdose) and personal catastrophe (a car accident, a life-threatening illness).

Crisis intervention is brief, active, directive and focused on the here and now, aiming to restore previous functioning. Its stance is supportive therapy (listening, reassurance, problem solving, mobilising support), not exploratory work, and it is often named the treatment of choice for adjustment disorder (Chapter 8), a short reaction to an identifiable stressor. **Albert Roberts**'s seven stages are: assess lethality, establish rapport, identify the major problems, deal with feelings, explore alternatives, make an action plan, and follow up.

Suicide risk is assessed by asking directly about thoughts, plans and means (Chapter 2). The **safety plan** of **Barbara Stanley** and **Gregory Brown** (2012) is a written list of warning signs, coping strategies, distracting people and places, people to ask for help, professionals and crisis lines, and ways to make the environment safe. Means restriction (pesticide bans cut suicides sharply in Sri Lanka), gatekeeper training for teachers and police, and helplines such as Tele-MANAS (14416) work at population level.

After disasters, **psychological first aid** (World Health Organization, 2011) offers practical support: look (for safety and needs), listen (without forcing talk), link (to services and loved ones). It replaced **critical incident stress debriefing** (**Jeffrey Mitchell**, 1983), a single group session retelling the event, which does not prevent post-traumatic stress disorder and may harm (Cochrane review), so it is not recommended. **Trauma-informed care** (US Substance Abuse and Mental Health Services Administration, 2014) asks services to realise, recognise and respond to trauma and to resist re-traumatising people.

! EXAM TRAP

Crisis management needs **supportive therapy** (IHBAS 2020), not acceptance and commitment, expressive or cognitive behaviour therapy. A tornado or flood is a **large-scale** event; personal events are small-scale.

Rehabilitation ●●

Rehabilitation aims at function and a valued life, not cure. Psychosocial rehabilitation of severe mental illness offers a ladder of services (Table 7.25); its **vocational rehabilitation** rungs, sheltered workshops and supported employment, aim at paid work.

Table 7.25 Services of psychosocial rehabilitation.

Service	What it offers
Day care	Daytime treatment and activity; the person sleeps at home
Halfway house	Supervised transitional home between hospital and independent living
Sheltered workshop	Paid work in a protected setting
Supported employment	A real job first, then support on the job (place, then train)
Assertive community treatment	Mobile team, about 10 clients per worker, round the clock
Social skills training	Modelling, role play, feedback and homework
Family psychoeducation	Teaches families about the illness, lowers expressed emotion and relapse
Clubhouse model	Member-run community with a work-ordered day (Fountain House, New York, 1948)

Leonard Stein and **Mary Ann Test** (1980, Madison, Wisconsin) randomly assigned patients to community treatment instead of hospital; they spent less time in hospital and worked more. This became **assertive community treatment**. **Case management** links a person with long-term, complex needs to services and coordinates them; it seeks skill training, environmental accommodations and empowerment, never cure. Its models are often contrasted as provider-driven, in which the agency sets the client's tasks and the worker controls access to services, and client-driven, in which the worker allies with the client in advocacy and empowerment.

After brain injury, neuropsychological rehabilitation takes two routes. The **restorative approach** retrains the damaged function by graded practice (cognitive retraining), as in the attention process training of **McKay Moore Sohlberg** and **Catherine Mateer** (1987), which drills sustained, selective, alternating and divided attention. The **compensatory approach** works around the deficit with memory aids (diaries, alarms, phone reminders). **Errorless learning** (**Alan Baddeley** and **Barbara Wilson**, 1994) prevents mistakes during learning, because amnesic patients cannot remember, and so cannot correct, their errors.

For substance users, detoxification (medically managed withdrawal) is only the start. **G. Alan Marlatt** and **Judith Gordon**'s **relapse prevention** (1985) sees relapse as failed coping in a high-risk situation (negative emotions, interpersonal conflict, social pressure). Coping raises self-efficacy; without it, self-efficacy falls, a lapse (first use) follows, and the **abstinence violation effect**, the guilt and "I have blown it" thinking after one slip, turns the lapse into relapse. Clients rehearse coping and treat lapses as lessons. **Therapeutic communities**, from Synanon (California, 1958), use the community itself (peer confrontation, graded responsibility, mutual help) as the method of change.

The Rehabilitation Council of India, a statutory body under the Ministry of Social Justice and Empowerment, regulates the training and registration of rehabilitation professionals, including clinical psychologists; disability certification under the Rights of Persons with Disabilities Act 2016 is covered in Section 7.1.

! EXAM TRAP

Case management aims at function, skills, environmental accommodation and empowerment, **not cure** (RML 2020). The same paper's "task or role performance" model (structure-function theory at the organisational level; fiduciary and humanitarian obligations; authority to provide, deny or withdraw services as the client complies with tasks) is, whatever its label, provider-driven, so the exception is the **power alliance** with the client, a client-driven idea. The Rehabilitation Council of India is under Social Justice and Empowerment, not Health, Education or Labour.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The interdisciplinary field defined by Schwartz and Weiss (1978) is behavioural _____.
2. Ajzen turned the theory of reasoned action into the theory of planned behaviour by adding perceived behavioural _____.
3. Alexander's holy seven include peptic ulcer, essential hypertension and bronchial _____.
4. Caplan's _____ prevention lowers the number of new cases.
5. Psychological first aid rests on look, listen and _____.
6. Guilt and "I have blown it" thinking after one slip is the abstinence _____ effect.

Answers: (1) medicine, (2) control, (3) asthma, (4) primary, (5) link, (6) violation.

7.10 Working with Children, Older Adults and Special Settings

LOWER YIELD · 16 QS

Most therapies in this chapter assume an adult who talks and who chose to come. Many clients do not fit: a six-year-old who will not discuss his bed-wetting but will bury a toy tiger in a sand tray; a grandmother who asks each evening when her dead husband is coming home; a youth whose lawyer asks whether he was "of unsound mind". This section adapts therapy to them.

Children as clients, and play therapy ○

Children differ from adult clients. Their developmental level limits what therapy can ask, so clinicians work through play and parents; adults make the referral; and school is a second home. Assessment is therefore **multi-informant assessment** (parents, teachers, the child, observation in clinic, home and class), not because parents over-report or teachers lack information but because *symptoms vary across contexts*: a child with attention-deficit hyperactivity disorder (ADHD) may be calm in a quiet clinic and unmanageable in a class of fifty. Clinic observation alone falls short for the same reason.

Play therapy uses the child's natural language. **Melanie Klein** (1882 to 1960) treated play as the equivalent of free association and interpreted it early and deeply; **Anna Freud** (1895 to 1982) began with a preparatory phase to win trust, involved parents and gave the analyst an educative role. **Virginia Axline** (1911 to 1988) carried Carl Rogers' ideas into the playroom

(*Play Therapy*, 1947; *Dibs: In Search of Self*, 1964). Her **non-directive play therapy** rests on eight principles: build warm rapport; accept the child exactly as he is; create permissiveness so that feelings flow freely; recognise feelings and reflect them back; respect the child's ability to solve his own problems; let the child lead; do not hurry; and set only the limits that anchor therapy to reality. Table 7.26 lists later forms, including **filial therapy** and **sandplay**.

Table 7.26 Later forms of play therapy.

Approach	Founder	Core idea
Relationship play therapy	**Clark Moustakas** (1950s)	The living relationship itself heals
Child-centred play therapy	**Garry Landreth**	The modern form of Axline's model
Filial therapy	**Bernard Guerney** and **Louise Guerney** (1964)	Parents are trained to hold the play sessions at home
Sandplay	**Dora Kalff** (1950s), from **Margaret Lowenfeld**'s World Technique (late 1920s)	The child builds a scene in a sand tray with miniatures; Jungian

> **! EXAM TRAP**
>
> Axline is Rogerian, not Freudian. Klein, not Anna Freud, equated play with free association. Sandplay is Kalff's; the World Technique is Lowenfeld's. In filial therapy parents run the sessions.

Parent training and behaviour change at home and school ●●

Why does defiance worsen in loving homes? **Gerald Patterson** (1926 to 2016) answered with **coercion theory** (*Coercive Family Process*, 1982). A mother orders the television off, her son screams, and she gives in. His screaming is negatively reinforced (the demand goes away), and so is her giving in (the screaming stops; Figure 7.17). Each trains the other, and the exchanges escalate.

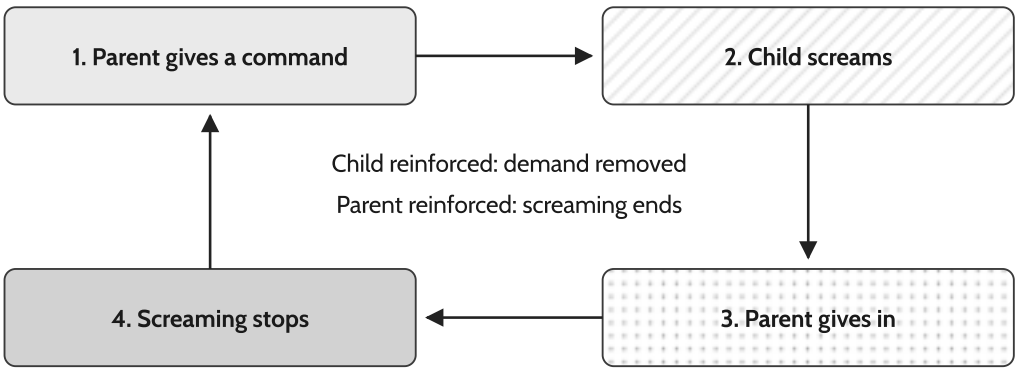


Figure 7.17 Patterson's coercion cycle. Both sides are negatively reinforced, so each round makes the next tantrum likelier.

The treatment, **parent management training** (PMT), reverses the trap. Parents give clear commands, attend to and praise good behaviour and reward it on a token chart, ignore mi-

nor misbehaviour (**extinction**; expect a brief extinction burst), and use time-out and response cost (Section 7.4). Named programmes include **Russell Barkley**'s *Defiant Children* (1987); **Carolyn Webster-Stratton**'s Incredible Years, which teaches parent groups with videotaped scenes; and **Sheila Eyberg**'s **parent-child interaction therapy** (PCIT, ages two to seven), which coaches the parent live through an earpiece, first in child-directed play, then in discipline.

At school, teachers add token economies (Section 7.4) and the daily report card: the teacher rates a few target behaviours daily and the child earns rewards at home. For repetitive aggression, often in autism or intellectual disability, **Richard Foxx** and **Nathan Azrin** devised **overcorrection**: restoring the damaged scene to better than before (restitution) and rehearsing the correct act (positive practice). For anxious children of 7 to 13, **Philip Kendall**'s 16-session Coping Cat (1990) teaches the FEAR plan (Feeling frightened? Expecting bad things? Attitudes and actions that help; Results and rewards), then graded exposure. Family work (Section 7.7) runs through all of these.

> **! EXAM TRAP**
>
> Deliberately ignoring tantrums is **extinction**; some keys wrongly mark "operant escape". Escape (ending an aversive stimulus already present) and avoidance (preventing it) are negative reinforcement, which *strengthens* behaviour. Nor is negative reinforcement a PMT technique; PMT breaks that trap.

Elimination problems, school refusal and ADHD

Enuresis (wetting from age five) is treated best with the **bell-and-pad** of **O. Hobart Mowrer** and **Willie Mae Mowrer** (1938): urine completes a circuit and rings a bell until bladder tension alone wakes the child, and overlearning cuts relapse. Encopresis (soiling from age four) is usually retentive, with constipation and overflow; laxatives are combined with scheduled toilet sitting and rewards.

In **school refusal** the child refuses out of anxiety, with the parents' knowledge; in truancy the child hides the absence, shows no anxiety and often has conduct problems. **Christopher Kearney** and **Wendy Silverman** classified refusal by its function: avoiding distress caused by school, escaping social or evaluative situations, seeking the parents' attention, or gaining tangible rewards outside school. Treatment centres on an early, graded return to school.

> **KEY STUDY: THE MTA STUDY (1999)**
>
> The US National Institute of Mental Health randomly assigned 579 children with ADHD, aged 7 to 9, to 14 months of managed medication, intensive behavioural treatment, both, or community care. Medication and the combination beat the others on core symptoms; the combination added small social and family gains at lower doses. *Criticism:* the advantage faded after the trial. Hence the rule: behavioural treatment plus medication.

Autism, intellectual disability and learning disability ●●●

Autism was first described in 1943 by **Leo Kanner** (1894 to 1981) of Johns Hopkins, in eleven children with extreme aloneness and an anxious insistence on sameness (the disorder is in Chapter 8).

Treatment is early and behavioural, based on **applied behaviour analysis** (ABA), which applies operant principles to socially important behaviour. **Ole Ivar Lovaas** (1927 to 2010), at the University of California, Los Angeles, pioneered early intensive behavioural intervention using **discrete trial training**: a skill is broken into small steps, each taught in repeated trials (instruction, response, prompt, reinforcement), with prompts faded. In functional communication training, a child whose tantrums serve to escape demands learns an acceptable substitute, such as handing over a picture card that says "break". Table 7.27 lists other programmes.

KEY STUDY: LOVAAS (1987)

Nineteen autistic children under four received about 40 hours a week of one-to-one behavioural treatment for two years or more. Nine (47 percent) reached normal intellectual and educational functioning, against 2 percent of controls; a 1993 follow-up found the gains kept. *Criticism:* no random assignment, a narrow outcome measure, and early use of aversives.

Table 7.27 Other named programmes for autism.

Programme	Developer	Core idea
TEACCH	**Eric Schopler** (North Carolina, 1972)	Structured teaching with visual schedules
PECS (picture exchange)	**Andy Bondy** and **Lori Frost** (1985)	The child hands over a picture to get a wanted item
Social stories	**Carol Gray** (early 1990s)	Short stories explaining social situations
Early Start Denver Model	**Sally Rogers** and **Geraldine Dawson**	ABA within play and relationships, from 12 months

Intellectual disability is treated with task analysis, chaining, prompting and reinforcement to build adaptive skills, alongside special or inclusive education. India's **National Trust Act** 1999 covers exactly four conditions (autism, cerebral palsy, mental retardation, multiple disabilities) and provides legal guardians. The Rights of Persons with Disabilities Act 2016 (Section 7.1) lists 21 disabilities, including intellectual disability, specific learning disabilities and autism, but not ADHD. Autism is certified with the 40-item **Indian Scale for Assessment of Autism** (ISAA), developed in 2009 by the National Institute for the Mentally Handicapped (now NIEPID): each item is rated 1 to 5 across six domains, a total of 70 or more indicates autism, and the total grades severity (above 153 is severe) and the disability percentage. Specific learning disability is treated by remedial teaching with multisensory methods, such as Orton-Gillingham phonics and **Grace Fernald**'s tracing method (see, say, hear and trace each word), plus accommodations: extra time, a scribe, exemption from a third language.

! EXAM TRAP

Kanner (1943) first described infantile autism; **Hans Asperger** (1944) described more able children, and **Eugen Bleuler** coined "autism" in 1911 for withdrawal in schizophrenia. The Indian autism tool is the **ISAA**; some keys mark the Childhood Autism Rating Scale (CARS), which is American (Schopler). The CBCL is **Thomas Achenbach**'s Child Behavior Checklist; the CGAS rates global functioning.

Child abuse, POCSO and the adolescent client ○

Child abuse (physical, sexual, emotional, neglect) entered medicine with **C. Henry Kempe**'s "battered child syndrome" (1962). Safety and reporting come first. The best-supported therapy is **trauma-focused cognitive behavioural therapy** (TF-CBT) of **Judith Cohen**, **Anthony Mannarino** and **Esther Deblinger**, for ages 3 to 18, with a non-offending parent. Its PRACTICE components (psychoeducation, relaxation, affect modulation, cognitive coping, trauma narrative, in vivo mastery, conjoint sessions, safety) build coping skills first, but the central goal is the trauma narrative: retelling the trauma desensitises the memories and corrects beliefs such as "it was my fault". (Emotion regulation and mindfulness skills define dialectical behaviour therapy, Section 7.5.)

> **INDIAN CONTEXT**
>
> The **POCSO Act** 2012 (Protection of Children from Sexual Offences) covers everyone under 18 and is gender-neutral. Reporting is mandatory (section 19): anyone, psychologists included, who knows of an offence or fears one is likely must inform the Special Juvenile Police Unit or the local police, and failing to do so is punishable under section 21 (up to six months' imprisonment, a fine or both). It also sets up Special Courts and child-friendly procedures.

Adolescence, Erik Erikson's stage of identity versus role confusion (Chapter 6), is the peak age for risk behaviour such as substance use, unsafe sex and self-harm. Anna Freud and **Peter Blos** held that healthy teenagers must detach from parents; research now shows that continuing interdependence is healthy, so parents remain allies. Engagement comes first: be collaborative and see the teenager alone for part of each session. The limits of **confidentiality with minors** are explained at the start: privacy holds except for risk to life, abuse or legal duties such as POCSO, and the teenager is told before anything is shared; guardians give consent. Schools offer counsellors and WHO life skills education. Adolescent depression is treated with CBT or interpersonal psychotherapy, adding fluoxetine with monitoring for suicidal thinking; self-harm calls for a safety plan, restricted means, family involvement and adolescent dialectical behaviour therapy.

Older adults: dementia care, depression and capacity

The field of **geropsychology** applies clinical psychology to older adults. **Robert Butler** (1927 to 2010) coined "ageism" (1969) and described reminiscence as a healthy life review (1963). In India families, mostly women, give most dementia care; their **caregiver burden**, objective (time, money) and subjective (distress, guilt), is measured with the Zarit Burden Interview and eased by psychoeducation, behaviour-management training, support groups and respite. Methods for the patient, from **reality orientation** to **validation therapy** and **cognitive stimulation therapy**, are in Table 7.28.

Table 7.28 Psychological methods in dementia care.

Method	Originator	What happens
Reality orientation	**James Folsom** (late 1950s)	Repeated orientation to time, place and person; can distress

Method	Originator	What happens
Reminiscence therapy	Butler's tradition	Sharing memories prompted by photographs and music
Life review	Robert Butler (1963)	A structured, evaluative review of the whole life
Validation therapy	**Naomi Feil** (developed 1963 to 1980)	Accept the person's reality and respond to feelings; never correct
Cognitive stimulation therapy	**Aimee Spector** and **Martin Orrell** (2003)	14 themed group sessions over 7 weeks; mild to moderate dementia

> **! EXAM TRAP**
>
> To the grandmother awaiting her dead husband, reality orientation *corrects* ("He died four years ago"); validation *accepts* and explores the feeling ("You miss him"). Life review, unlike plain reminiscence, is evaluative and covers the whole life.

Late-life depression often shows as bodily, sleep and memory complaints rather than sadness. Depressive pseudodementia mimics dementia, but it begins abruptly, the person complains of poor memory and answers "I don't know", and it lifts with treatment. The 30-item Geriatric Depression Scale (**Jerome Yesavage**) omits bodily symptoms. CBT, problem-solving therapy, antidepressants and electroconvulsive therapy all help.

A person's **decision-making capacity** is presumed, and judged for one decision at one time. **Paul Appelbaum** and **Thomas Grisso** (1988) named four abilities: to understand the information, appreciate how it applies to oneself, reason about the options and communicate a choice. India's Mental Healthcare Act 2017 uses a similar test.

Counselling and career guidance ○

Guidance helps people make educational and vocational choices through information, tests and advice. Counselling is a face-to-face relationship that helps a basically healthy person resolve a problem of living and reach his own decision. Psychotherapy treats more disturbed clients, longer and deeper (Section 7.6). Counselling is called the heart of the guidance programme, and its instrument, the interview, was called the heart of the counselling process by **Ruth Strang**. Other tools are the cumulative record (a school-long file of each pupil's background, health, tests and achievement), the case study (an in-depth picture of one pupil drawn from all sources) and tests.

> **! EXAM TRAP**
>
> One key names the aptitude test "the heart of the guidance programme". The sources are not uniform: Indian guidance texts give that title to counselling, Strang gave "heart of the counselling process" to the interview, and a few lesser sources use similar words for the cumulative record, but no standard text found gives it to the aptitude test, which only predicts trainability. If counselling is not offered, choose the interview, and expect some keys to differ.

Frank Parsons (1854 to 1908), father of vocational guidance, set out the **trait-factor approach** in *Choosing a Vocation* (1909): know yourself, know the world of work, and match

them by "true reasoning". The counselling process runs from relationship building through assessment, goals and intervention to termination (microskills are in Section 7.2). Areas include school, career, marital, rehabilitation, de-addiction, grief and HIV counselling (before and after testing, under the five Cs: consent, confidentiality, counselling, correct results, connection to care). Affirmative therapy for LGBTQ+ clients affirms their identities, addresses minority stress and rejects conversion therapy, which India's National Medical Commission called professional misconduct in 2022.

John Holland (1919 to 2008) sorted people and work environments into six **RIASEC** types (Figure 7.18); satisfaction follows from congruence between person and environment. **Donald Super** (1910 to 1994), in his **life-span, life-space theory**, saw a career as lifelong self-concept implementation through five stages (growth to 14, exploration to 24, establishment to 44, maintenance to 64, then decline) and many life roles, drawn as the Life-Career Rainbow; readiness for choice is career maturity.

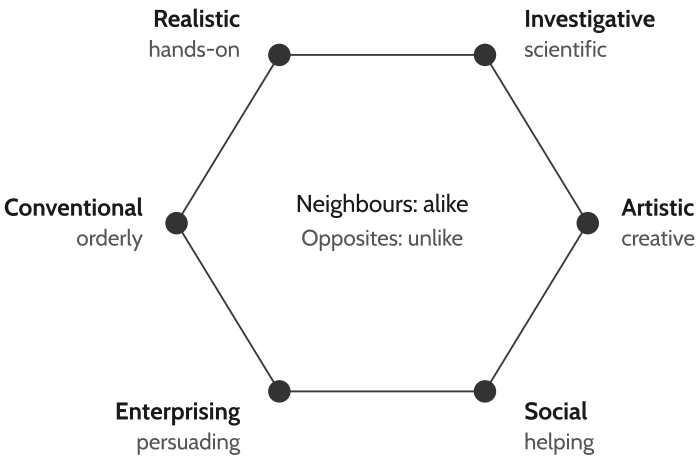


Figure 7.18 Holland's RIASEC hexagon. Neighbouring types are most alike; opposite types (Realistic and Social) are least alike.

Forensic clinical psychology ○

Forensic psychology answers legal questions for a client that is often the court, so confidentiality is limited. An accused must have **competence to stand trial**: the present ability to understand the proceedings and help his lawyer (*Dusky v. United States*, 1960). Criminal responsibility concerns his mind at the offence: the insanity defence (Table 7.29). It began in 1843, when **Daniel M'Naghten**, deluded that he was persecuted, shot Prime Minister Robert Peel's secretary; his acquittal produced the **M'Naghten rules**, which section 84 of the Indian Penal Code (now section 22 of the Bharatiya Nyaya Sanhita 2023) follows. This legal insanity is narrower than medical insanity, and the accused must prove it.

Table 7.29 Tests of insanity.

Test	Origin	Not responsible if he	Kind
M'Naghten rules	England, 1843	did not know the nature of the act, or that it was wrong	Cognitive
Irresistible impulse	US, 1880s	knew it was wrong but could not control himself	Volitional

Test	Origin	Not responsible if he	Kind
Durham rule	US, 1954 (Judge **David Bazelon**)	acted as a product of mental disease or defect (dropped in 1972)	Product
ALI test	American Law Institute, 1962	lacked substantial capacity to appreciate wrongfulness or to conform to the law	Both
IPC 84 / BNS 22	India, 1860 / 2023	could not know the nature of the act, or that it was wrong or contrary to law	Cognitive

> **! EXAM TRAP**
>
> M'Naghten: did he *know*? Irresistible impulse: could he *control*? ALI: both, softened to "substantial capacity". Durham: was the act a *product* of illness? India's section 84 is M'Naghten-based. Competence concerns the mind at trial; insanity, the mind at the offence.

Risk assessment predicts violence or reoffending; unaided clinical prediction was right in at most one case in three (**John Monahan**, 1981), so structured tools such as the HCR-20 and **Robert Hare**'s Psychopathy Checklist-Revised are used. Custody evaluation follows the best interests of the child (in India, the child's welfare, under the Guardians and Wards Act 1890). An expert witness, unlike others, may give opinions (Section 45, Indian Evidence Act 1872; now Section 39, Bharatiya Sakshya Adhiniyam 2023).

The Juvenile Justice (Care and Protection of Children) Act 2015 sends a *child in conflict with law* to a Juvenile Justice Board and a *child in need of care and protection* to a Child Welfare Committee. A 16- to 18-year-old accused of a heinous offence (minimum seven years' imprisonment) may, after the Board's preliminary assessment of his mental and physical capacity to commit the offence and his grasp of its consequences, with psychologists' help, be tried as an adult. In correctional psychology, effective offender programmes follow the **risk-need-responsivity model** of **Donald Andrews** and **James Bonta** (intensity matched to risk, criminogenic needs targeted, cognitive-behavioural methods) and are community-based. The cognitive interview of **Ronald Fisher** and **Edward Geiselman**, which aids eyewitness recall, and offender profiling, which infers an unknown offender's traits from the crime, are investigative tools, not treatments.

Neuropsychology, creative therapies and telepsychology

Clinical neuropsychology, which assesses the effects of brain damage on behaviour, is covered in Chapters 2 and 5; rehabilitation psychology is in Section 7.9.

Art therapy has two founders: **Margaret Naumburg** (1890 to 1983) treated art as symbolic speech from the unconscious (art psychotherapy), and **Edith Kramer** (1916 to 2014) held that the creative process heals through sublimation (art as therapy). Music therapy uses listening and making music; dance and movement therapy (founded by **Marian Chace**, 1896 to 1970) uses movement and mirroring; drama therapy uses role play (psychodrama is in Section 7.7); bibliotherapy is guided reading. Modern **animal-assisted therapy** began in the 1960s with **Boris Levinson** (1907 to 1984), whose dog Jingles drew a withdrawn child into talk. In India yoga and meditation are common adjuncts, and NIMHANS trials support yoga in schizophrenia and depression.

Telepsychology delivers care by phone, video, messaging or apps. Guided internet CBT rivals face-to-face CBT for depression and anxiety, while apps vary widely in quality. India's **Tele-MANAS**, launched on 10 October 2022 with NIMHANS as nodal centre, gives free round-the-clock counselling on 14416. Its ethical issues are consent covering the technology's limits, data security, verifying identity and location, emergencies at a distance, jurisdiction and the digital divide.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. In Axline's _____ play therapy, the child leads.
2. Patterson's _____ theory explains how tantrums and giving in reinforce each other.
3. Deliberately ignoring tantrums is _____.
4. Infantile autism was first described in 1943 by _____.
5. Responding to a disoriented person's feelings without correcting her is _____ therapy.
6. Section 84 of the Indian Penal Code is now section _____ of the Bharatiya Nyaya Sanhita.

Answers: (1) non-directive, (2) coercion, (3) extinction, (4) Leo Kanner, (5) validation, (6) 22.

The Unchaining at Bicêtre, Revisited

Look back at Bicêtre with what this chapter has taught. Pinel, recording his cases and counting recoveries, was a scientist-practitioner a century and a half before Boulder, and his case notes were ancestors of the interview, mental status examination and formulation of section 7.2. Pussin's conversations and routines anticipated what outcome research found in section 7.8: much of what heals in psychodynamic, behavioural, cognitive and humanistic therapy alike is the relationship and the other common factors. Moral treatment collapsed when asylums grew too crowded to give each person attention, the lesson behind community care and rehabilitation in section 7.9. And the chains? India's Mental Healthcare Act 2017 still had to ban chaining in section 95, a reminder that ethics and law protect patients only when clinicians practise them.

Chapter summary

7.1 The Profession: History, Models, Ethics and Indian Law. Clinical psychology joins science and practice to understand, predict and relieve distress; in India it rests on an RCI-recognised qualification and RCI registration. Care moved from supernatural ideas and the humours through the asylums to Pinel and Pussin's **moral treatment**, Dix and Beers. Witmer opened the first clinic (1896); Boulder (1949) set the **scientist-practitioner** model and Vail (1973) the PsyD. In India the IACP dates from 1968 and the RCI Act from 1992. Engel's **biopsychosocial** and the **diathesis-stress** models integrate the older models. Ethics rests on five APA principles, consent, confidentiality

and its limits (Tarasoff) and boundaries; the **Mental Healthcare Act 2017** adds rights, advance directives and section 115.

7.2 The Clinical Interview, Mental Status Examination and Formulation. The **clinical interview** may be intake, diagnostic, crisis or termination, and unstructured, semi-structured (SCID, PSE and SCAN, K-SADS, ADIS) or fully structured (DIS): structure buys reliability, not validity. Microskills include SOLER attending, open questions, furthering, paraphrasing and reflection of feeling. The case history runs from identifying data to premorbid personality. The **MSE** records appearance, speech, mood (subjective) and affect (objective), the stream, form, content and possession of thought, perception, cognition and insight (six grades). SAD PERSONS aids memory but cannot predict suicide. **Functional analysis** (ABC, SORC) is operant. A **formulation** (5 Ps, biopsychosocial) explains, a diagnosis classifies; kappa measures diagnostic reliability, and Rosenhan (1973) exposed labelling.

7.3 Psychodynamic Therapies and Their Descendants. Psychodynamic therapies hold that unconscious conflict drives symptoms. Freud's psychoanalysis grew from Breuer's **cathartic method** with Anna O.; it relies on the **fundamental rule** and **free association**, dream analysis, interpretation, analysis of **resistance** and of **transference** and **countertransference**, and **working through**, under neutrality and abstinence. Ego psychology, object relations (Klein, Winnicott, Mahler, Kernberg), Kohut's self psychology and mentalisation-based treatment changed what analysts listen for. Brief dynamic therapies (Malan, Sifneos, Davanloo, Mann, Luborsky, Strupp and Binder) and Klerman and Weissman's **interpersonal psychotherapy** shortened treatment. Jung, Adler, Horney, Sullivan and Berne's **transactional analysis** offered rival techniques, alongside supportive therapy, hypnosis and narcoanalysis.

7.4 Behaviour Therapy. **Behaviour therapy** applies learning principles to problem behaviour. Spurred by Eysenck's 1952 critique, it grew from Wolpe's Johannesburg, the Maudsley and Skinner's operant work, and every case starts with a functional analysis. Classical techniques include Jones's counterconditioning, Wolpe's **systematic desensitisation** (reciprocal inhibition, a SUDS-rated hierarchy), flooding, implosion, **exposure and response prevention** for OCD, aversion therapy, covert sensitisation, the bell and pad and the debated EMDR. Operant techniques build behaviour (shaping, chaining, the Premack principle, Ayllon and Azrin's token economy) or reduce it (extinction, time-out, response cost, overcorrection), as in ABA for autism. Modelling, assertiveness, relaxation, biofeedback, habit reversal, behavioural activation, insomnia methods and parent training complete the toolkit.

7.5 Cognitive and Cognitive-Behavioural Therapies, and the Third Wave. **Cognitive-behavioural therapies** hold that emotional disturbance follows mainly from how people interpret events. Albert Ellis's **REBT** (1955; renamed 1993) uses the ABC model, disputes irrational musts and aims at unconditional self-acceptance. Aaron Beck's **cognitive therapy** targets the cognitive triad, core and intermediate beliefs, automatic thoughts and distortions through collaborative empiricism, Socratic questioning and the thought record. Meichenbaum's stress inoculation, problem-solving and multimodal therapy, Kelly's fixed-role therapy, Young's schema therapy and disorder-specific models for panic, trauma and psychosis followed. The **third wave** (DBT, ACT, MBSR, MBCT, CFT, MCT, FAP, behavioural activation) changes the relationship to thoughts. CBT is the most researched psychotherapy, now also delivered online.

7.6 Humanistic, Existential, Experiential and Brief Therapies. **Carl Rogers**'s person-centred therapy holds that conditions of worth create **incongruence** between self-concept and experience, and

that six necessary and sufficient conditions, above all **empathy, unconditional positive regard and congruence**, are enough for change. **Fritz Perls**'s Gestalt therapy works on here-and-now awareness, unfinished business and the **empty chair**. Existential therapists (Binswanger, Boss, May, Yalom) confront death, freedom, isolation and meaninglessness, and **Frankl**'s logotherapy rests on the will to meaning. Reality, focusing, emotion-focused, positive, narrative and solution-focused therapies follow. **Motivational interviewing** (Miller and Rollnick) resolves ambivalence alongside Prochaska and DiClemente's stages of change, and Ivey, Carkhuff and Egan model the helping skills.

7.7 Family, Couples and Group Therapies. Family therapy grew from **general systems theory**: families keep homeostasis through feedback, causality is circular, and the identified patient signals a troubled system. Theories that families cause schizophrenia (the schizophrenogenic mother, Bateson's **double bind**, Lidz's schism and skew, Wynne's pseudomutuality) failed, but high **expressed emotion** predicts relapse, which psychoeducation reduces. Bowen stressed differentiation and triangles, Minuchin boundaries, Haley directives and paradox, the MRI second-order change, the Milan team circular questioning. Couples work runs from behavioural therapy to Gottman's four horsemen; Masters and Johnson founded sex therapy. Groups heal through Yalom's eleven factors; Moreno created psychodrama and Maxwell Jones the therapeutic community.

7.8 Does Psychotherapy Work? Evidence, Common Factors and the Alliance. Eysenck (1952) claimed that about two-thirds of neurotic patients improve without therapy (**spontaneous remission**); Smith and Glass's meta-analysis (1977, effect size 0.68; 0.85 in 1980) answered him. Rival therapies tie (the **dodo bird verdict**: Rosenzweig 1936; Luborsky and colleagues 1975), which points to common factors: Frank's *Persuasion and Healing* (1961), Wampold's contextual model, Lambert's pie and Bordin's alliance of goals, tasks and bond. Outcome is tested in efficacy and effectiveness studies, RCTs with waiting-list, placebo-attention and usual-care controls, EST criteria and evidence-based practice (APA 2006), and judged by the dose-effect curve and the reliable change index. Integration, culturally adapted therapy (Neki's guru-chela relationship), supervision and self-care complete the picture.

7.9 Health, Community, Crisis and Rehabilitation Psychology. Health psychology (Matarazzo, 1980) is psychology's contribution to health; **behavioural medicine** (Schwartz and Weiss, 1978) is interdisciplinary. Parsons's **sick role** carries two rights and two duties. The health belief model, the theory of planned behaviour and protection motivation theory explain health behaviour and adherence. Alexander's holy seven, Type A hostility, Fordyce's operant pain treatment, CBT for insomnia and the AUDIT link mind and body; Kübler-Ross and Worden describe dying and grief. Caplan supplied prevention levels, consultation and crisis theory: crisis work is brief and supportive, and debriefing is not recommended. Rehabilitation uses assertive community treatment, case management, errorless learning and relapse prevention.

7.10 Working with Children, Older Adults and Special Settings. Children are assessed through several informants because **symptoms vary across contexts**, and treated with play therapy (Klein, Anna Freud, Axline's non-directive principles, filial therapy, sandplay), parent management training built on Patterson's **coercion theory**, the bell-and-pad for enuresis, behavioural treatment plus medication for ADHD (the MTA study) and applied behaviour analysis for autism (Lovaas 1987). TF-CBT treats abuse, and the POCSO Act 2012 makes reporting mandatory. Dementia care uses reality orientation, reminiscence, life review, validation and cognitive stimulation therapy.

Holland's RIASEC and Super's life-span, life-space theory guide careers. Forensic work covers competence, the M'Naghten-based section 84 (now BNS 22), risk assessment and the Juvenile Justice Act 2015.

Who's who

Table 7.30 The people in this chapter.

Name	When	Examined for
Hippocrates	460 to 377 BC	Natural causes of mental disorder; humoral theory
Galen	About AD 129 to 198	Developed the humoral theory
Philippe Pinel	1745 to 1826	Physician at La Bicetre from 1793; credited with unchaining patients there and at the Salpetriere; moral treatment
Jean-Baptiste Pussin	1746 to 1811	Superintendent at Bicetre who began humane care, persuaded Pinel and banned the chains there (1797)
William Tuke	1732 to 1822	Quaker founder of the York Retreat (1796); moral treatment in England
Dorothea Dix	1802 to 1887	Campaigned from the 1840s for American state hospitals
Clifford Beers	1876 to 1943	A Mind That Found Itself (1908); founded the mental hygiene movement (1908 to 1909)
Emil Kraepelin	1856 to 1926	Classification of mental disorders; separated dementia praecox from manic-depressive insanity
Eugen Bleuler	1857 to 1939	Named schizophrenia (1908)
Lightner Witmer	1867 to 1956	First psychological clinic (1896); first printed 'clinical psychology' (1907)
Wilhelm Wundt	1832 to 1920	First psychology laboratory, Leipzig (1879); Witmer's teacher
Alfred Binet	1857 to 1911	First practical intelligence scale (1905, with Simon)
Theodore Simon	1905	Binet-Simon intelligence scale
Robert Yerkes	1876 to 1956	Army Alpha and Army Beta testing (1917)
Richard McFall	1991	Manifesto for a Science of Clinical Psychology; clinical science model
Granville Stanley Hall	1844 to 1924	First APA president (1892)
James McKeen Cattell	1860 to 1944	Coined 'mental test' (1890)
Sigmund Freud	1856 to 1939	Psychodynamic model
Erik Erikson	1902 to 1994	Psychosocial development (distractor for 'clinical psychology')
Carl Rogers	1902 to 1987	Humanistic model; unconditional positive regard
Burrhus Frederic Skinner	1904 to 1990	Operant conditioning; behavioural model
Ivan Petrovich Pavlov	1849 to 1936	Classical conditioning; behavioural model
Albert Ellis	1913 to 2007	Irrational beliefs; cognitive model
Aaron T. Beck	1921 to 2021	Cognitive model of disorder
Abraham Maslow	1908 to 1970	Humanistic model; self-actualisation
Viktor Frankl	1905 to 1997	Existential model; search for meaning
Thomas Szasz	1920 to 2012	The Myth of Mental Illness (1961); sociocultural critique
Gregory Bateson	1904 to 1980	Family systems model; double bind
Salvador Minuchin	1921 to 2017	Family systems model; structural family therapy
Girindrasekhar Bose	1887 to 1953	Indian Psychoanalytical Society, Calcutta (1922)
Marie Jahoda	1907 to 2001	Ideal mental health (1958): six criteria
George L. Engel	1913 to 1999	Biopsychosocial model (Science, 1977)
Paul Meehl	1920 to 2003	Diathesis-stress account of schizophrenia (1962)
Joseph Zubin	1900 to 1990	Vulnerability-stress model (1977, with Spring)
Bonnie Spring	1977	Vulnerability-stress model (with Zubin)
Walter Cannon	1871 to 1945	Fight or flight (distractor for the biopsychosocial model)
Richard Lazarus	1922 to 2002	Appraisal theory of stress (distractor for the biopsychosocial model)
Tom Beauchamp	1979	Four principles of biomedical ethics (with Childress)

Name	When	Examined for
James Childress	1979	Four principles of biomedical ethics (with Beauchamp)
Immanuel Kant	1724 to 1804	Deontological ethics
Jeremy Bentham	1748 to 1832	Utilitarian ethics
John Stuart Mill	1806 to 1873	Utilitarian ethics
Prosenjit Poddar	1969	Client in the Tarasoff case
Tatiana Tarasoff	1969	Victim in Tarasoff v. Regents of the University of California (1974, 1976)
Thomas Gutheil	1993	Boundary crossings versus boundary violations (with Gabbard)
Glen Gabbard	1993	Boundary crossings versus boundary violations (with Gutheil)
Allen Ivey	1968 (microcoun-selling)	Microcounselling: training interview microskills one at a time
Gerard Egan	1975 (The Skilled Helper)	SOLER attending behaviour; primary versus advanced empathy
Carl Ransom Rogers	1902 to 1987	Client-centred therapy, whose signature technique is reflection of feeling
Arthur Kleinman	1980	The patient's explanatory model of illness in cross-cultural interviewing
Michael B. First	2015 to 2016 (SCID-5)	Lead author of the Structured Clinical Interview for DSM-5
Robert Leopold Spitzer	1932 to 2015	Led DSM-III (1980) and developed the SCID; critic of Rosenhan's study (1975)
John Kenneth Wing	1923 to 2010	Present State Examination (9th edition 1974), core of the WHO's SCAN
Lee Nelken Robins	1922 to 2009	Diagnostic Interview Schedule (1981), a fully structured interview for lay interviewers
Timothy A. Brown	ADIS-5 (DSM-5 edition)	Co-author of the Anxiety Disorders Interview Schedule for DSM-5
David H. Barlow	born 1942	Co-author of the Anxiety Disorders Interview Schedule (ADIS-5)
Kurt Schneider	1887 to 1967	First-rank symptoms of schizophrenia, including thought insertion, with-drawal and broadcasting
Patterson and colleagues	1983	SAD PERSONS mnemonic for suicide risk
George Libman Engel	1913 to 1999	Biopsychosocial model (Science, 1977)
Judith S. Beck	1995	Cognitive conceptualisation diagram for cognitive-behavioural formulation
Jacob Cohen	1923 to 1998	Cohen's kappa (1960), chance-corrected agreement between raters
David L. Rosenhan	1929 to 2012	'On being sane in insane places' (Science, 1973): sane pseudopatients di-agnosed with schizophrenia
Josef Breuer	1842 to 1925	Cathartic method; treated Anna O. (1880 to 1882); Studies on Hysteria (1895) with Freud
Bertha Pappenheim	1859 to 1936	Anna O., the first talking-cure patient; not in fact cured
Hippolyte Bernheim	1840 to 1919	Nancy school of suggestion; Freud visited him in 1889
Paula Heimann	1950	Broad view of countertransference as useful information
Ralph Greenson	1965	Working alliance
Anna Freud	1895 to 1982	Ego psychology; The Ego and the Mechanisms of Defence (1936); child analysis; analyst's neutrality
Heinz Hartmann	1894 to 1970	Ego psychology; conflict-free ego sphere (1939)
Melanie Klein	1882 to 1960	Child analysis through play; paranoid-schizoid and depressive positions; splitting; projective identification
Donald Winnicott	1896 to 1971	Good-enough mother, transitional object, holding environment, true and false self
Ronald Fairbairn	1889 to 1964	Object relations: libido is object-seeking, not pleasure-seeking
Margaret Mahler	1897 to 1985	Separation-individuation; autistic and symbiotic phases; theory of psy-chosis (On Childhood Psychosis and Schizophrenia, 1952)
Silvano Arieti	1955	Psychoanalytic account of schizophrenia (Interpretation of Schizophrenia)
Ping-Nie Pao	1979	Psychoanalytic account of schizophrenic disorders
Edith Jacobson	1964	The Self and the Object World; analytic work on psychosis and depression

Name	When	Examined for
Otto Kernberg	born 1928	Borderline personality organisation; transference-focused psychotherapy (TFP)
Heinz Kohut	1913 to 1981	Self psychology (1971): mirroring, idealising and twinship selfobjects; empathic attunement
Stephen Mitchell	1946 to 2000	Relational psychoanalysis (with Jay Greenberg, 1983)
Jay Greenberg	1983	Object Relations in Psychoanalytic Theory, with Mitchell; relational psychoanalysis
John Bowlby	1907 to 1990	Attachment theory; therapist as a secure base
Anthony Bateman	1999	Mentalisation-based treatment (MBT) for borderline personality disorder, with Fonagy
Peter Fonagy	1999	Mentalisation-based treatment (MBT), with Bateman
David Malan	1976	Brief focal psychotherapy (Tavistock); triangle of conflict and triangle of person
Peter Sifneos	1920 to 2008	Short-term anxiety-provoking psychotherapy (STAPP)
Habib Davanloo	born 1927	Intensive short-term dynamic psychotherapy (ISTDP)
James Mann	1973	Time-limited psychotherapy: exactly 12 sessions
Lester Luborsky	1984	Core conflictual relationship theme (CCRT); supportive-expressive therapy
Hans Strupp	1984	Time-limited dynamic psychotherapy (TLDP), with Binder; cyclical maladaptive pattern
Jeffrey Binder	1984	Time-limited dynamic psychotherapy, with Strupp
Gerald Klerman	1928 to 1992	Interpersonal psychotherapy (IPT) for depression (manual 1984)
Myrna Weissman	1984	Interpersonal psychotherapy, with Klerman
Adolf Meyer	1866 to 1950	Psychobiology; an influence on IPT
Jacob Moreno	1889 to 1974	Psychodrama; role reversal
Lorna Smith Benjamin	2003	Interpersonal reconstructive therapy
Carl Gustav Jung	1875 to 1961	Analytical psychology: collective unconscious, archetypes (persona, anima, animus), amplification, active imagination, individuation
Alfred Adler	1870 to 1937	Individual psychology: organ inferiority (1907), inferiority complex, birth order, early recollections, encouragement, acting as if, spitting in the soup
Harold Mosak	1985	Adlerian push-button technique
Karen Horney	1885 to 1952	Basic anxiety; moving towards, against, away; idealised self; tyranny of the shoulds
Harry Stack Sullivan	1892 to 1949	Interpersonal psychiatry; participant observer; parataxic distortion; influence on IPT
Eric Berne	1910 to 1970	Transactional analysis: Transactional Analysis in Psychotherapy (1961), Games People Play (1964); Parent, Adult, Child ego states
Paul Federn	1871 to 1950	Coined ego states; Berne's analyst
Bob and Mary Goulding	1970s	Redecision therapy; script injunctions
Thomas Harris	1969	I'm OK, You're OK (Harper and Row, 1969): life positions popularised
Franz Anton Mesmer	1734 to 1815	Animal magnetism, forerunner of hypnosis
James Braid	1843	Coined the word hypnosis
Hans Eysenck	1916 to 1997	1952 challenge to the effectiveness of psychotherapy
Karl Popper	1902 to 1994	Called psychoanalytic theory unfalsifiable
Jonathan Shedler	2010	Review showing psychodynamic therapy as effective as other evidence-based therapies
Hans Jürgen Eysenck	1916 to 1997	1952 critique: about two-thirds of neurotics improve without psychotherapy; popularised behaviour therapy (1959)
Joseph Wolpe	1915 to 1997	Systematic desensitisation and reciprocal inhibition (Psychotherapy by Reciprocal Inhibition, 1958); SUDS; experiments on cats; thought stopping
Arnold Allan Lazarus	1932 to 2013	Wolpe's Johannesburg group; printed the term behaviour therapy (1958); assertiveness training; later multimodal therapy
Stanley Rachman	1950s	Member of Wolpe's Johannesburg group of behaviour therapists
Monte Bernard Shapiro	1950s	Maudsley Hospital; treated each patient as a single-case experiment

Name	When	Examined for
Ogden Lindsley	1953	First used the term behaviour therapy, in a hospital report with Skinner and Solomon (1953)
Harry Solomon	1953	First used the term behaviour therapy, in a hospital report with Lindsley and Skinner (1953)
Orval Hobart Mowrer	1907 to 1982	Bell-and-pad method for enuresis (1938, with Willie Mae Mowrer); two-factor theory of fear and avoidance (1947)
Willie Mae Mowrer	1938	Bell-and-pad method for enuresis, with O. H. Mowrer
Frederick Kanfer	1965	SORKC functional analysis (with Saslow, 1965); self-regulation as self-monitoring, self-evaluation and self-reinforcement
George Saslow	1965	SORKC functional analysis, with Kanfer
Mary Cover Jones	1897 to 1987	Removed Peter's fear of rabbits by counterconditioning and modelling (1924)
John Broadus Watson	1878 to 1958	Founder of behaviourism; teacher of Mary Cover Jones
Charles Scott Sherrington	1857 to 1952	Physiologist whose term reciprocal inhibition Wolpe borrowed
Edmund Jacobson	1888 to 1983	Progressive muscle relaxation (Progressive Relaxation, 1929; enlarged 1938), step 1 of desensitisation
Peter Lang	1964	Fear Survey Schedule, with Wolpe (1964)
Gordon Paul	1966	Desensitisation beat insight therapy for public-speaking anxiety, with no symptom substitution at follow-up (1966)
Thomas Stampfl	1966	Implosive therapy
Donald Levis	1967	Implosive therapy, with Stampfl (1967)
Victor Meyer	1966	First described exposure and response prevention for obsessional rituals (1966)
Edna Foa	1980s onwards	Refined and tested ERP for OCD; prolonged exposure for PTSD
Francine Shapiro	1948 to 2019	Eye movement desensitisation and reprocessing (observed 1987, published 1989), for PTSD; eight phases from history taking to re-evaluation
Patricia Resick	1990s	Cognitive processing therapy for PTSD
Marsha Linehan	1990s	Dialectical behaviour therapy
Bessel van der Kolk	1990s onwards	Trauma researcher
Pat Ogden	2000s	Sensorimotor psychotherapy
Joseph LeDoux	1990s	Neuroscience of fear circuits in the brain
Joseph Cautela	1967	Covert sensitisation
David Premack	1959	Premack principle: a more probable activity reinforces a less probable one
Teodoro Ayllon	1968	Token economy at Anna State Hospital (with Azrin, 1968); satiation of a towel hoarder (1963)
Nathan Azrin	1968 to 1973	Token economy (with Ayllon, 1968); overcorrection (with Foxx, 1972); habit reversal training (with Nunn, 1973)
Richard Foxx	1972	Overcorrection: restitution and positive practice, with Azrin
Ole Ivar Lovaas	1927 to 2010	Intensive early behavioural treatment for autism (1987): 47 percent reached normal functioning
Albert Bandura	1925 to 2021	Live, symbolic and participant modelling; participant modelling for snake phobia (1969); self-efficacy
Edward Blanchard	1969	Participant modelling study of snake phobia, with Bandura and Ritter
Brunhilde Ritter	1969	Participant modelling study of snake phobia, with Bandura and Blanchard
Andrew Salter	1949	Pioneer of assertiveness training (Conditioned Reflex Therapy, 1949)
Johannes Heinrich Schultz	1884 to 1970	Autogenic training (1932)
Lars-Göran Öst	1987	Applied relaxation
Neal Miller	1909 to 2002	Operant control of autonomic responses (1960s), the basis of biofeedback
Arnold Kegel	1948	Perineometer for pelvic floor exercises, an early biofeedback device
Gregory Nunn	1973	Habit reversal training, with Azrin
Knight Dunlap	1875 to 1949	Negative (massed) practice for habits (1932)
Peter Lewinsohn	1970s	Depression as a low rate of response-contingent positive reinforcement; roots of behavioural activation

Name	When	Examined for
Neil Jacobson	1949 to 1999	Showed the activation component of cognitive therapy works alone (1996); behavioural activation
Christopher Martell	2001	Developed behavioural activation for depression with Jacobson
Richard Bootzin	1972	Stimulus control treatment for insomnia
Arthur Spielman	1987	Sleep restriction treatment for insomnia
Gerald Patterson	1926 to 2016	Coercive family cycle; parent management training (Oregon Social Learning Center)
Edward Lee Thorndike	1874 to 1949	Law of effect (a wrong option for shaping and desensitisation)
Fritz Perls	1893 to 1970	Gestalt therapy (a wrong option for desensitisation; Section 7.6)
Aaron Beck	1921 to 2021	Cognitive therapy (a wrong option for desensitisation and EMDR; Section 7.5)
Epictetus	about 50 to 135 CE	Stoic philosopher: people are disturbed not by things but by their views of them; a root of REBT
Donald Meichenbaum	born 1940	Cognitive-Behavior Modification (1977); self-instructional training (1971); stress inoculation training
Steven C. Hayes	born 1948	Named the three waves (2004); acceptance and commitment therapy; relational frame theory
Maxie C. Maultsby Jr.	1970s	Psychiatrist; rational emotive imagery, adopted by Ellis
William James	1842 to 1910	Functionalism; a distractor for the founder of RET
William Glasser	1925 to 2013	Reality therapy (a distractor for REBT)
Aaron Temkin Beck	1921 to 2021	Cognitive therapy; automatic thoughts; cognitive triad; Cognitive Therapy of Depression (1979); BDI, BAI, BHS
David D. Burns	1980	Feeling Good; popularised the longer list of cognitive distortions
Max Hamilton	1959 and 1960	Clinician-rated Hamilton anxiety and depression scales
Thomas J. D'Zurilla	1971	Problem-solving therapy, with Goldfried
Marvin R. Goldfried	1971	Problem-solving therapy, with D'Zurilla
Arthur M. Nezu	1980s	Problem-solving therapy for depression
Arnold A. Lazarus	1932 to 2013	Multimodal therapy; BASIC I.D.; technical eclecticism
George Alexander Kelly	1905 to 1967	Personal construct theory (1955); fixed-role therapy
Jeffrey E. Young	1990	Schema therapy; early maladaptive schemas; schema modes; limited reparenting
Anthony Ryle	1980s	Cognitive analytic therapy (CAT)
David M. Clark	1986	Cognitive model of panic: catastrophic misinterpretation of bodily sensations; social anxiety model with Wells (1995)
Paul M. Salkovskis	1991	Safety behaviours in anxiety and panic
Adrian Wells	1995	Social anxiety model with Clark; metacognitive therapy; cognitive attentional syndrome
Edna B. Foa	1990s	Prolonged exposure for post-traumatic stress disorder
Patricia A. Resick	1990s	Cognitive processing therapy for post-traumatic stress disorder, first for rape survivors
Judith A. Cohen	1990s	Trauma-focused CBT for children, with Mannarino and Deblinger
Anthony P. Mannarino	1990s	Trauma-focused CBT for children, with Cohen and Deblinger
Esther Deblinger	1990s	Trauma-focused CBT for children, with Cohen and Mannarino
David Kingdon	1990s	CBT for psychosis, with Turkington
Douglas Turkington	1990s	CBT for psychosis, with Kingdon
Marsha M. Linehan	born 1943	Dialectical behaviour therapy (1993); biosocial theory of borderline personality disorder; trials of 1991 and 2006
Murray Bowen	1913 to 1990	Family systems therapy; the genogram (not a DBT mode)
Kirk D. Strosahl	1999	Acceptance and commitment therapy, with Hayes and Wilson
Kelly G. Wilson	1999	Acceptance and commitment therapy, with Hayes and Strosahl
Jon Kabat-Zinn	born 1944	Mindfulness-based stress reduction (1979, eight weeks)
Zindel V. Segal	2002	Mindfulness-based cognitive therapy, with Williams and Teasdale
J. Mark G. Williams	2002	Mindfulness-based cognitive therapy, with Segal and Teasdale
John D. Teasdale	2000 and 2002	Mindfulness-based cognitive therapy; the 2000 relapse-prevention trial

Name	When	Examined for
Paul Gilbert	2000s	Compassion-focused therapy; threat, drive and soothing systems
Robert J. Kohlenberg	1991	Functional analytic psychotherapy, with Tsai
Mavis Tsai	1991	Functional analytic psychotherapy, with Kohlenberg
Peter M. Lewinsohn	1970s	Depression as loss of response-contingent positive reinforcement; behavioural activation
Neil S. Jacobson	1996	Component study showing behavioural activation alone works as well as full cognitive therapy
Vikram Patel	2017	Goa trial of lay-counsellor behavioural activation for depression
Abraham Harold Maslow	1908 to 1970	Named humanistic psychology the third force; with Rogers its leading proponent
Stanley Standal	1954	Coined the term unconditional positive regard
Edward Bordin	1979	Working alliance: goals, tasks and bond (not a Rogerian concept)
William Stephenson	1953	Devised the Q-sort (The Study of Behavior, 1953)
John Butler and Gerard Haigh	1954	Q-sort study: clients' mean self-ideal correlation rose from about zero (minus .01) before therapy to .34 after it (.31 at follow-up)
Charles Truax	1967	Rating scales of empathy, warmth and genuineness, with Carkhuff
Robert Carkhuff	1967; 1969	Core-condition rating scales with Truax; helping model in Helping and Human Relations (1969)
Kurt Lewin	1890 to 1947	T-groups, forerunners of encounter groups (Section 7.7)
Laura Perls	1905 to 1990	Co-founder of Gestalt therapy
Paul Goodman	1911 to 1972	Co-founder of Gestalt therapy; co-author of Gestalt Therapy (1951)
Ralph Hefferline	1951	Co-author of Gestalt Therapy (1951)
Erving Polster and Miriam Polster	1973	Gestalt Therapy Integrated (1973): five contact boundary disturbances, with deflection in place of egotism
Max Wertheimer	1880 to 1943	Gestalt psychology of perception, not Gestalt therapy (Chapter 1)
Wolfgang Köhler	1887 to 1967	Gestalt psychology of perception (Chapter 1)
Kurt Koffka	1886 to 1941	Gestalt psychology of perception (Chapter 1)
Martin Heidegger	1889 to 1976	Being and Time (1927): Dasein, being-in-the-world
Irvin Yalom	born 1931	Existential Psychotherapy (1980): four ultimate concerns (death, freedom, isolation, meaninglessness)
Ludwig Binswanger	1881 to 1966	Daseinsanalysis; Umwelt, Mitwelt and Eigenwelt
Medard Boss	1903 to 1990	Daseinsanalysis (Psychoanalysis and Daseinsanalysis, 1963)
Emmy van Deurzen	contemporary	Added the Überwelt (spiritual world) to the modes of being
Rollo May	1909 to 1994	The Meaning of Anxiety (1950); Existence (1958); existential anxiety, normal and neurotic anxiety
James Bugental	1915 to 2008	Existential-humanistic therapy
Viktor Emil Frankl	1905 to 1997	Logotherapy; Man's Search for Meaning (1946); will to meaning, noogenic neurosis, paradoxical intention, dereflection
William Masters and Virginia Johnson	1970	Sensate focus in sex therapy, not dereflection (Section 7.7)
Robert Wubbolding	1980s	WDEP summary of reality therapy
Eugene Gendlin	1926 to 2017	Focusing (1978) and the felt sense
Leslie Greenberg	1990s	Process-experiential and emotion-focused therapy
Susan Johnson	1980s	Emotionally focused couples therapy, with Greenberg (Section 7.7)
Martin Seligman	born 1942	Positive psychology (APA president 1998); positive psychotherapy; PERMA
Mihaly Csikszentmihalyi	1934 to 2021	Flow
Nossrat Peseschkian	1933 to 2010	Transcultural Positive Psychotherapy (1970s): balance model, five stages
Michael White	1948 to 2008	Narrative therapy: externalising, unique outcomes, re-authoring (with Epston, 1990)
David Epston	1990	Narrative therapy with Michael White (Narrative Means to Therapeutic Ends, 1990)
Steve de Shazer	1940 to 2005	Solution-focused brief therapy; the miracle question (1988)

Name	When	Examined for
Insoo Kim Berg	died 2007	Solution-focused brief therapy, Brief Family Therapy Center, Milwaukee (1978)
Jay Haley	1923 to 2007	Brief strategic therapy (Section 7.7)
William R. Miller	1983	Motivational interviewing (1983; book with Rollnick 1991, later editions 2002 and 2013)
Stephen Rollnick	1991	Motivational interviewing, with Miller
Burke, Arkowitz and Menchola	2003	Meta-analysis of motivational interviewing: effective for alcohol, drug, and diet and exercise problems; not supported for smoking or HIV-risk behaviour; OCD not reviewed
James Prochaska	1983	Transtheoretical model and stages of change, with DiClemente
Carlo DiClemente	1983	Transtheoretical model and stages of change, with Prochaska
Alan Marlatt	1941 to 2011	Relapse prevention (Section 7.9)
Ludwig von Bertalanffy	1901 to 1972	General systems theory, applied to families
Norbert Wiener	1894 to 1964	Cybernetics: self-regulation by feedback
Betty Carter	1980s	Family life cycle (with Monica McGoldrick)
Monica McGoldrick	1980s	Family life cycle (with Betty Carter); genograms
Frieda Fromm-Reichmann	1889 to 1957	The schizophrenogenic mother (1948), now discredited
Don Jackson	1920 to 1968	Double bind (1956); founded the Mental Research Institute, Palo Alto (1959)
John Weakland	1956	Double bind; MRI brief therapy
Theodore Lidz	1910 to 2001	Marital schism and marital skew
Lyman Wynne	1923 to 2007	Pseudomutuality (1958)
George Brown	late 1950s	Founded expressed emotion research on relapse in schizophrenia
James Birley	1972	Brown, Birley and Wing: high EE predicts relapse
John Wing	1972	Brown, Birley and Wing: high EE predicts relapse
Christine Vaughn	1976	Vaughn and Leff: EE, medication and 35-hour contact and relapse
Julian Leff	1976	Vaughn and Leff; family interventions to lower expressed emotion
Ian Falloon	1980s	Behavioural family management in schizophrenia
Vidya Sagar	1950s	Involved families in inpatient care at Amritsar Mental Hospital
Janis Jenkins	1992	Cross-cultural comparison of expressed emotion (with Marvin Karno)
Marvin Karno	1992	Cross-cultural comparison of expressed emotion (with Janis Jenkins)
Ann Hartman	1970s	The ecomap
Edward Tolman	1886 to 1959	Cognitive map (Chapter 1)
Milton Erickson	1901 to 1980	Hypnotherapist whose methods shaped strategic therapy
Cloe Madanes	1981	Strategic family therapy; pretend techniques
Paul Watzlawick	1974	MRI brief therapy; first-order and second-order change (Change, 1974)
Richard Fisch	1974	MRI brief therapy; attempted solutions
Mara Selvini Palazzoli	1916 to 1999	Milan systemic therapy; Paradox and Counterparadox (1978)
Luigi Boscolo	1978	Milan systemic therapy; circular questioning
Gianfranco Cecchin	1978	Milan systemic therapy; circular questioning
Giuliana Prata	1978	Milan systemic therapy
Virginia Satir	1916 to 1988	Experiential family therapy; communication stances; family sculpting
Carl Whitaker	1912 to 1995	Symbolic-experiential family therapy
Ivan Boszormenyi-Nagy	1920 to 2007	Contextual therapy; Invisible Loyalties (1973)
James Alexander	1970s	Functional family therapy (with Bruce Parsons)
Bruce Parsons	1970s	Functional family therapy (with James Alexander)
Scott Henggeler	1990s	Multisystemic therapy for youth offending
Gayla Margolin	1979	Behavioural couples therapy (with Jacobson)
Andrew Christensen	1996	Integrative behavioural couple therapy: acceptance
John Gottman	1990s	Four horsemen; the 5:1 ratio; Sound Relationship House
Sue Johnson	1980s	Emotionally focused couple therapy, based on attachment

Name	When	Examined for
William Masters	1970	Sex therapy: sensate focus, squeeze technique (Human Sexual Inadequacy, 1970)
Virginia Johnson	1970	Sex therapy with William Masters
James Semans	1956	Stop-start technique for premature ejaculation
Helen Singer Kaplan	1974	Brief sex therapy (The New Sex Therapy, 1974)
Jack Annon	1976	PLISSIT model
Bruce Tuckman	1965	Stages of group development: forming, storming, norming, performing (1965); adjourning added with Mary Ann Jensen (1977)
Mary Ann Jensen	1977	Added the adjourning stage to Tuckman's model
Jacob Levy Moreno	1889 to 1974	Psychodrama; sociometry and the sociogram; coined group psychotherapy
Bill Wilson	1935	Co-founder of Alcoholics Anonymous
Bob Smith	1935	Co-founder of Alcoholics Anonymous
Maxwell Jones	1907 to 1990	The therapeutic community (Belmont Hospital, Surrey, later the Henderson Hospital)
S. H. Foulkes	1898 to 1976	Group analysis
Allen E. Bergin	1966; 1971	Showed (1971) that recounting dropouts lifted psychoanalysis to about 85 percent improvement among completers; named the deterioration effect (1966)
Michael J. Lambert	1976; 1992	Spontaneous remission nearer 40 percent, not two-thirds (1976); the pie of therapeutic factors (1992); Outcome Questionnaire-45
Gene V. Glass	1976; 1977	Coined meta-analysis (1976); Smith and Glass psychotherapy meta-analysis (1977)
Mary Lee Smith	1977; 1980	With Glass: 375 studies, effect size 0.68 (1977); 0.85 in The Benefits of Psychotherapy (1980)
Thomas I. Miller	1980	Co-author of The Benefits of Psychotherapy (effect size 0.85)
Saul Rosenzweig	1907 to 2004	Implicit common factors and the dodo bird quotation (1936)
Barton Singer	1975	Co-author of the dodo bird verdict review
Lise Luborsky	1975	Co-author of the dodo bird verdict review
Bruce E. Wampold	2001; 2015	Contextual model; The Great Psychotherapy Debate (2001; fullest form in the 2015 edition with Imel)
Zac E. Imel	2015	Co-author with Wampold of the 2015 edition of The Great Psychotherapy Debate (contextual model)
Jerome David Frank	1909 to 2005	Persuasion and Healing (1961): demoralisation and four common features of all healing
Alfred Pritz	born 1952	Austrian psychotherapist, founding president of the World Council for Psychotherapy (1995) (distractor)
Elizabeth Zetzel	1956	Coined the therapeutic alliance in psychoanalysis
Edward S. Bordin	1979	Pantheoretical working alliance: goals, tasks and bond
Adam Horvath	1989	Working Alliance Inventory (with Greenberg)
Jeremy D. Safran	2000	Alliance ruptures (withdrawal and confrontation) and their repair
J. Christopher Muran	2000	Alliance ruptures and repair (with Safran)
William Schofield	1964	YAVIS client; Psychotherapy: The Purchase of Friendship
Starke R. Hathaway	1903 to 1984	Described the hello-goodbye effect; co-author of the MMPI
Martin E. P. Seligman	1995	Consumer Reports effectiveness study
Dianne L. Chambless	1995; 1998	Chaired the APA Division 12 task force on empirically supported treatments; criteria with Hollon (1998)
Steven D. Hollon	1998	Refined the EST criteria with Chambless
Irene Elkin	1989	Led the NIMH Treatment of Depression Collaborative Research Program
Kenneth I. Howard	1986	Dose-effect relationship (50 percent improved by 8 sessions, 75 by 26); phase model
Paula Truax	1991	Clinical significance and the reliable change index (with Jacobson)
Paul L. Wachtel	1977	Cyclical psychodynamics; theoretical integration
James O. Prochaska	1979	Transtheoretical approach; stages of change

Name	When	Examined for
Stanley B. Messer	1992	Assimilative integration
Guillermo Bernal	1995	Ecological validity framework for culturally adapted therapy
N. C. Surya	1964	Psychotherapy in the Indian setting (with S. S. Jayaram): Indian patients more dependent, need direct, situational support in their own language
S. S. Jayaram	1964	Co-author with Surya on psychotherapy in the Indian setting
Jaswant Singh Neki	1973	Guru-chela relationship as an Indian therapeutic paradigm
Cal Stoltenberg	1987	Integrated developmental model of supervision
Janine M. Bernard	1979	Discrimination model of supervision (teacher, counsellor, consultant)
Christina Maslach	1981	Burnout: emotional exhaustion, depersonalisation, reduced personal accomplishment
Joseph D. Matarazzo	1980	Defined health psychology; first president of APA Division 38
Gary Schwartz	1978	Defined behavioural medicine with Stephen Weiss
Stephen Weiss	1978	Defined behavioural medicine with Gary Schwartz
George Engel	1913 to 1999	Biopsychosocial model (1977)
David Mechanic	1962	Illness behaviour
Talcott Parsons	1902 to 1979	The sick role, in The Social System (1951)
Godfrey Hochbaum	1950s	Health belief model, from tuberculosis screening studies
Irwin Rosenstock	1966	Health belief model: 1950s origins with Hochbaum and Kegeles; classic statement 1966; self-efficacy added 1988
Marshall Becker	1974	Extended the health belief model to sick-role behaviour and adherence
Stephen Kegeles	1950s	Health belief model
Martin Fishbein	1936 to 2009	Theory of reasoned action with Ajzen (1975)
Icek Ajzen	born 1942	Theory of reasoned action (1975); theory of planned behaviour (1985, 1991)
Ronald Rogers	1975	Protection motivation theory (revised 1983)
Ralf Schwarzer	1992	Health action process approach
Marc Hollender	1956	Activity-passivity, guidance-cooperation and mutual participation, with Szasz
Hans Selye	1907 to 1982	General adaptation syndrome: alarm, resistance, exhaustion
Thomas Holmes	1967	Life-events scale (Social Readjustment Rating Scale) with Rahe
Richard Rahe	1967	Life-events scale (Social Readjustment Rating Scale) with Holmes
Robert Ader	1932 to 2011	Psychoneuroimmunology: conditioned immune suppression in rats (1975)
Nicholas Cohen	1975	Conditioned immune suppression with Ader
Franz Alexander	1891 to 1964	Psychosomatic Medicine (1950); the holy seven; specificity hypothesis
Meyer Friedman	1910 to 2001	Type A behaviour pattern with Rosenman (1959)
Ray Rosenman	1920 to 2013	Type A behaviour pattern with Friedman (1959)
Lydia Temoshok	1980s	Type C (cancer-prone) personality
Ronald Melzack	1929 to 2019	Gate control theory of pain with Wall (1965)
Patrick Wall	1925 to 2001	Gate control theory of pain with Melzack (1965)
Wilbert Fordyce	1976	Operant approach to chronic pain; pain behaviours
Johannes Schultz	1884 to 1970	Autogenic training
Herbert Benson	1975	The relaxation response
Cicely Saunders	1918 to 2005	St Christopher's Hospice (1967); total pain; modern palliative care
Elisabeth Kübler-Ross	1926 to 2004	Five stages of dying: denial, anger, bargaining, depression, acceptance (1969)
J. William Worden	1982	Four tasks of mourning
John F. Kennedy	1917 to 1963	Community Mental Health Centers Act (1963)
Gerald Caplan	1964	Primary, secondary and tertiary prevention; four types of consultation (1970); crisis theory
Robert Gordon	1983	Universal, selective and indicated prevention
Julian Rappaport	1981	Empowerment in community psychology

Name	When	Examined for
Dixon Chibanda	2010s	Friendship Bench: lay health workers deliver problem-solving therapy in Zimbabwe
Erving Goffman	1922 to 1982	Stigma (1963)
Erich Lindemann	1900 to 1974	Acute grief after the Cocoanut Grove fire (1944); founder of crisis intervention
Albert Roberts	1990s	Seven-stage model of crisis intervention
Barbara Stanley	2012	Safety planning intervention with Brown
Gregory Brown	2012	Safety planning intervention with Stanley
Jeffrey Mitchell	1983	Critical incident stress debriefing
Leonard Stein	1980	Assertive community treatment (Training in Community Living) with Test
Mary Ann Test	1980	Assertive community treatment with Stein
McKay Moore Sohlberg	1987	Attention process training with Mateer
Catherine Mateer	1987	Attention process training with Sohlberg
Alan Baddeley	born 1934	Errorless learning with Wilson (1994)
Barbara Wilson	1994	Errorless learning; memory rehabilitation
G. Alan Marlatt	1941 to 2011	Relapse prevention with Gordon (1985); abstinence violation effect
Judith Gordon	1985	Relapse prevention with Marlatt
Virginia Axline	1911 to 1988	Non-directive (client-centred) play therapy and its eight principles; Play Therapy (1947), Dibs: In Search of Self (1964)
Clark Moustakas	1950s	Relationship play therapy
Garry Landreth	1991	Child-centred play therapy, the modern form of Axline's model
Bernard Guerney	1964	Filial therapy, with Louise Guerney: parents trained to hold play sessions
Louise Guerney	1964	Filial therapy, with Bernard Guerney
Margaret Lowenfeld	late 1920s	World Technique (sand tray and miniatures), forerunner of sandplay
Dora Kalff	1950s	Jungian sandplay therapy
Russell Barkley	1987	Defiant Children parent-training programme
Carolyn Webster-Stratton	1980s	Incredible Years video-modelling parent programme
Sheila Eyberg	1970s	Parent-child interaction therapy (PCIT)
Philip Kendall	1990	Coping Cat CBT programme and FEAR plan for anxious children
O. Hobart Mowrer	1938	Bell-and-pad (urine alarm) for enuresis, with Willie Mae Mowrer
Christopher Kearney	1990s	Functional model of school refusal, with Wendy Silverman
Wendy Silverman	1990s	Functional model of school refusal, with Christopher Kearney
Leo Kanner	1894 to 1981	First described early infantile autism (1943)
Eric Schopler	1972	TEACCH programme for autism; Childhood Autism Rating Scale
Andy Bondy	1985	Picture Exchange Communication System (PECS), with Lori Frost
Lori Frost	1985	PECS, with Andy Bondy
Carol Gray	early 1990s	Social stories for autism
Sally Rogers	2010	Early Start Denver Model, with Geraldine Dawson
Geraldine Dawson	2010	Early Start Denver Model, with Sally Rogers
Grace Fernald	1879 to 1950	Multisensory tracing method for reading disability
Hans Asperger	1944	Described more able autistic children (autistic psychopathy)
Thomas Achenbach	1983	Child Behavior Checklist (CBCL)
C. Henry Kempe	1962	Battered child syndrome
Judith Cohen	2006	Trauma-focused CBT (TF-CBT), with Mannarino and Deblinger
Anthony Mannarino	2006	Trauma-focused CBT, with Cohen and Deblinger
Peter Blos	1904 to 1997	Held that healthy adolescents must detach from parents
Robert Butler	1927 to 2010	Life review (1963); coined ageism (1969)
James Folsom	late 1950s	Reality orientation for dementia
Naomi Feil	developed her method 1963 to 1980	Validation therapy for disoriented old people
Aimee Spector	2003	Cognitive stimulation therapy for dementia, with Martin Orrell

Name	When	Examined for
Martin Orrell	2003	Cognitive stimulation therapy, with Aimee Spector
Jerome Yesavage	1982	Geriatric Depression Scale (30 items)
Paul Appelbaum	1988	Four abilities of decision-making capacity, with Thomas Grisso
Thomas Grisso	1988	Four abilities of decision-making capacity, with Paul Appelbaum
Ruth Strang	1895 to 1971	Called the interview the heart of the counselling process
Frank Parsons	1854 to 1908	Father of vocational guidance; trait-factor approach, Choosing a Vocation (1909)
John Holland	1919 to 2008	RIASEC types and the hexagon; congruence
Donald Super	1910 to 1994	Life-span, life-space theory; five career stages; Life-Career Rainbow
Daniel M'Naghten	1843	His acquittal produced the M'Naghten rules of insanity
Robert Peel	1843	Prime Minister whose secretary M'Naghten shot
David Bazelon	1954	Durham rule (product test) of insanity
John Monahan	1981	Showed unaided clinical prediction of violence is right at most one time in three
Robert Hare	1991	Psychopathy Checklist-Revised
Donald Andrews	1990	Risk-need-responsivity model of offender rehabilitation, with James Bonta
James Bonta	1990	Risk-need-responsivity model, with Donald Andrews
Ronald Fisher	1980s	Cognitive interview for eyewitnesses, with Edward Geiselman
Edward Geiselman	1980s	Cognitive interview, with Ronald Fisher
Margaret Naumburg	1890 to 1983	Art psychotherapy: art as symbolic speech from the unconscious
Edith Kramer	1916 to 2014	Art as therapy: healing through sublimation in the creative process
Marian Chace	1896 to 1970	Founder of dance and movement therapy in the United States
Boris Levinson	1907 to 1984	Pioneer of animal-assisted therapy (his dog Jingles, 1960s)

Key terms

clinical psychologist, supernatural model, trephining, humoral theory, moral treatment, mental hygiene movement, psychological clinic, scientist-practitioner model, practitioner-scholar model, PsyD, clinical science model, Rehabilitation Council of India, ideal mental health, four Ds, biopsychosocial model, diathesis-stress model, diathesis, vulnerability-stress-protective factor model, General Principles, deontological, utilitarian, Informed consent, debriefing, Confidentiality, privileged communication, duty to warn, duty to protect, multiple (dual) relationship, boundary violation, advance directive, nominated representative, Mental Health Review Boards, benchmark disability, treatment gap, task-shifting, clinical interview, diagnostic interview, unstructured interview, semi-structured interview, fully structured interview, Rapport, SOLER, open question, Reflection of feeling, Empathy, explanatory model, genogram, Premorbid personality, mental status examination, Mood, Affect, form of thought, delusion, overvalued idea, thought alienation, hallucination, pseudohallucination, Insight, SAD PERSONS, functional analysis, SORC, self-monitoring, Reactivity, observer drift, case formulation, 5 Ps, biopsychosocial formulation, cognitive conceptualisation diagram, differential diagnosis, Cohen's kappa, cathartic method, Abreaction, catharsis, fundamental rule, free association, neutrality, abstinence, manifest content, latent content, resistance, Interpretation, working through, Transference, Countertransference, cathexis, Defence mechanisms, Ego psychology, Object relations, splitting, good-enough mother, transitional object, separation-individuation, self psychology, Mentalisation-based treatment, triangle of conflict, Interpersonal psychotherapy, collective unconscious, active imagination, individual psychology, inferiority complex, Transactional analysis, ego states, life script, Supportive psychotherapy, Narcoanalysis, Behaviour therapy, applied behaviour analysis, two-factor theory, counterconditioning, Reciprocal inhibition, systematic desensitisation, subjective units of distress scale,

Exposure therapy, habituation, Flooding, Implosive therapy, Exposure and response prevention, eye movement desensitisation and reprocessing, Aversion therapy, covert sensitisation, bell-and-pad method, Shaping, Chaining, Premack principle, token economy, Extinction, Time-out, response cost, Overcorrection, participant modelling, Assertiveness training, progressive muscle relaxation, Biofeedback, Habit reversal training, Behavioural activation, stimulus control, Paradoxical intention, Parent management training, symptom substitution, cognitive-behavioural therapy, self-efficacy, third wave, rational emotive behaviour therapy, ABC model, musturbation, awfulising, disputing, unconditional self-acceptance, rational emotive imagery, shame-attacking exercise, automatic thoughts, cognitive triad, core beliefs, intermediate beliefs, cognitive distortions, collaborative empiricism, Socratic questioning, dysfunctional thought record, downward arrow technique, stress inoculation training, multimodal therapy, fixed-role therapy, schema therapy, limited reparenting, safety behaviours, dialectical behaviour therapy, biosocial theory, acceptance and commitment therapy, relational frame theory, psychological flexibility, cognitive defusion, mindfulness-based stress reduction, mindfulness-based cognitive therapy, cognitive attentional syndrome, third force, non-directive therapy, actualising tendency, conditions of worth, incongruence, fully functioning person, necessary and sufficient conditions, unconditional positive regard, congruence, Q-sort, encounter group, unfinished business, empty chair technique, Dasein, ultimate concerns, Umwelt, Mitwelt, Eigenwelt, logotherapy, will to meaning, noogenic neurosis, dereflection, choice theory, focusing, emotion-focused therapy, flow, externalising the problem, solution-focused brief therapy, Motivational interviewing, change talk, transtheoretical model, microskills hierarchy, advanced empathy, general systems theory, homeostasis, circular causality, identified patient, family life cycle, schizophrenogenic mother, double bind, marital schism, marital skew, pseudomutuality, expressed emotion, family psychoeducation, differentiation of self, triangulation, enmeshment, disengagement, joining, enactment, paradoxical intervention, first-order change, second-order change, positive connotation, circular questioning, four horsemen, sensate focus, PLISSIT model, therapeutic factors, universality, here-and-now, psychodrama, sociogram, milieu therapy, therapeutic community, spontaneous remission, meta-analysis, effect size, dodo bird verdict, common factors, contextual model, therapeutic alliance, Working Alliance Inventory, alliance ruptures, therapist effects, allegiance effect, YAVIS, hello-goodbye effect, efficacy, effectiveness, analogue research, randomised controlled trial, allocation concealment, treatment fidelity, empirically supported treatments, evidence-based practice in psychology, dose-effect relationship, clinical significance, reliable change index, deterioration effect, process research, premature termination, therapist self-disclosure, technical eclecticism, theoretical integration, assimilative integration, culturally adapted therapy, guru-chela relationship, wounded healer, burnout, health psychology, Behavioural medicine, illness behaviour, sick role, health belief model, theory of planned behaviour, protection motivation theory, adherence, psychosomatic disorders, cognitive behavioural therapy for insomnia, Alcohol Use Disorders Identification Test, palliative care, tasks of mourning, Complicated grief, deinstitutionalisation, Primary prevention, Secondary prevention, Tertiary prevention, mental health consultation, empowerment, crisis, Crisis intervention, safety plan, psychological first aid, critical incident stress debriefing, Trauma-informed care, vocational rehabilitation, assertive community treatment, Case management, restorative approach, compensatory approach, Errorless learning, relapse prevention, abstinence violation effect, Therapeutic communities, multi-informant assessment, non-directive play therapy, filial therapy, sandplay, coercion theory, parent-child interaction therapy, bell-and-pad, school refusal, discrete trial training, TEACCH, PECS, National Trust Act, Indian Scale for Assessment of Autism, trauma-focused cognitive behavioural therapy, POCSO Act, confidentiality with minors, geropsychology, caregiver burden, reality orientation, validation therapy, cognitive stimulation therapy, decision-making capacity, trait-factor approach, RIASEC, life-span,

life-space theory, competence to stand trial, M'Naghten rules, ALI test, risk-need-responsivity model, animal-assisted therapy, Tele-MANAS

From the past papers

1. To describe a predisposition towards and risk triggers for a mental disorder, modern theorists use a (JEMAS 2025)

   (a) Diathesis-Stress model (b) Neurodevelopmental model (c) Clinical-Reductionist model (d) Syndrome-Sign model

   Answer: (a) Diathesis-stress model. The diathesis is the predisposition and stress is the trigger; disorder appears when stress exceeds the person's threshold. The neurodevelopmental model traces schizophrenia to faulty early brain development, and the other two options are not real models.

2. All of the following statements are TRUE about exceptions to the rule of confidentiality EXCEPT: (RML 2020)

   (a) a court order requires the release of information (b) the information is essential to safeguard the health of others (duty to protect; warn) (c) the information is probative of guilt or innocence in a criminal matter (d) there are never exceptions to the rule of confidentiality

   Answer: (d) There are never exceptions to the rule of confidentiality. Confidentiality is not absolute: court orders, the Tarasoff duty to warn and protect, and evidence a court needs in a criminal case are all recognised exceptions, so (a) to (c) are true.

3. Which of the following is a key legal provision under the Mental Healthcare Act, 2017? (IHBAS 2025)

   (a) Suicide attempts are decriminalized, recognizing mental illness as a contributing factor (b) Electroconvulsive therapy (ECT) can be administered without consent in all cases (c) Only psychiatrists can provide therapy; psychologists are excluded (d) Mental health establishments are exempt from registration if government-run

   Answer: (a). Section 115 presumes severe stress in anyone who attempts suicide, so the person is not punished. Treatment under the Act needs informed consent, and ECT is banned outright without anaesthesia and muscle relaxant; clinical psychologists are mental health professionals under the Act; and every establishment, government or private, must register (it was the 1987 Act that exempted government hospitals from licensing).

4. A structured interview commonly used in clinical psychology is : (IHBAS 2025)

   (a) MMSE (b) SCID (c) WAIS (d) TAT

   Answer: (b) SCID. The Structured Clinical Interview for DSM takes the clinician through the diagnostic criteria with standard questions (strictly it is semi-structured). The MMSE is a 30-point cognitive screening test, the WAIS an intelligence test and the TAT a projective test.

5. Leaning forward, saying umm, nodding, are examples of which one of the following interview techniques? (RML 2020)

   (a) Paraphrasing (b) Prompting (c) Furthering (d) Checking

   Answer: (c) Furthering. Furthering responses (minimal encouragers) keep the client talking without changing the topic. Paraphrasing restates content in new words, prompting steers the client to a particular topic, and checking asks the client to confirm the interviewer's understanding.

6. In the context of psychopathology, a "formulation" differs from a "diagnosis" in that a formulation primarily aims to: (Real PYQ 2024)

   (a) Classify the client's symptoms according to a standardized nosological system like DSM-5 (b) Provide a hypothesis about the cause, precipitants, and maintaining factors of an individual's difficulties (c) Determine the prognosis and likely course of the disorder over time (d) Rule out organic causes for the presenting psychological symptoms

   Answer: (b) A hypothesis about cause, precipitants and maintaining factors. A formulation (for example the 5 Ps) is an individualised explanation. Classifying symptoms is diagnosis, predicting course is prognosis, and ruling out organic causes belongs to differential diagnosis.

7. 'Counter transference' in psychotherapy refers to (JEMAS 2025)

   (a) therapists' emotional reactions to the client based on the therapists' own unresolved conflict (b) client's projection onto the therapist (c) client's resistance to therapy (d) behavioral reinforcement techniques

   Answer: (a) The therapist's emotional reactions to the client, based on the therapist's own unresolved conflict. Countertransference runs from therapist to client. The client's projection onto the therapist is transference, obstruction of therapy is resistance, and reinforcement belongs to behaviour therapy.

8. Which of the following technique or techniques would not typically be included within the repertoire of an Adlerian counsellor? (RML 2022)

   (a) Acting "as if" (b) Push button (c) Encouragement (d) Analysis of transference

   Answer: (d) Analysis of transference. Adlerian therapy is an equal, educational partnership that does not cultivate or analyse a transference. Acting as if, Mosak's push-button technique and encouragement (the core Adlerian technique) are all Adlerian.

9. Interpersonal Therapy (IPT) does NOT typically include which of the following techniques or focus areas? (Real PYQ 2025)

   (a) Role playing and role reversal techniques (b) Grief and bereavement work (c) Role conflict resolution (d) Role transition support

   Answer: (a) Role playing and role reversal techniques. Grief, role disputes and role transitions are three of IPT's four problem areas (the fourth is interpersonal deficits). Strictly, IPT does use role play after communication analysis, but role reversal is a psychodrama technique, so (a) is the best choice.

10. A technique in which a response that inhibits anxiety is paired with an anxiety-provoking stimulus, step by step, to break the conditioned link between that stimulus and anxiety is: (NIMHANS 2026 recall)

    (a) Systematic desensitization (b) Flooding (c) Aversion therapy (d) Exposure and response prevention

    Answer: (a) Systematic desensitization. Wolpe paired relaxation, a response incompatible with anxiety, with each step of a graded hierarchy (reciprocal inhibition). Flooding uses full-intensity exposure with no hierarchy or relaxation, aversion therapy pairs a behaviour with an unpleasant stimulus, and ERP blocks rituals so that anxiety habituates.

11. Exposure & response prevention procedure is based on behaviour therapy principal of : (IHBAS 2024)

    (a) Counter conditioning (b) Habituation and extinction (c) Reciprocal inhibition (d) Graded exposure

    Answer: (b) Habituation and extinction. Kept in contact with the obsessional trigger and prevented from performing the ritual, the patient's anxiety falls by itself (habituation) and the feared outcome fails to occur (extinction). Counterconditioning and reciprocal inhibition underlie systematic desensitisation; graded exposure describes how steps are arranged, not why fear falls.

12. The behavioral technique that involves removing a desirable stimulus following an undesirable behavior in order to reduce the frequency of that behavior is called: (Real PYQ 2025)

    (a) Negative reinforcement (b) Response cost (c) Positive punishment (d) Extinction

    Answer: (b) Response cost. Taking away a reinforcer the person already has (tokens, privileges) after a behaviour is response cost, a form of negative punishment. Negative reinforcement removes something unpleasant and increases behaviour, positive punishment adds something unpleasant, and extinction merely withholds the reinforcer that maintained the behaviour.

13. A patient sees a friend in the street who fails to acknowledge him; he thinks his friend does not like him anymore. He has the following type of cognitive error: (IHBAS 2020)

    (a) Selective abstraction (b) Arbitrary inference (c) Personalization (d) Magnification

    Answer: (b) Arbitrary inference. He draws a specific negative conclusion with no evidence for it, which is Beck's arbitrary inference (mind reading is a narrower name for the same jump). Selective abstraction would need him to fix on one detail while ignoring the rest, personalisation would blame himself for an unrelated event, and magnification would blow up a fault.

14. In the A-B-C-D-E model of cognitive behaviour therapy (RET) 'D' refers to _____ . (IHBAS 2025)

    (a) Drama (b) Desensitization (c) Discuss (d) Disputing

    Answer: (d) Disputing. In Ellis's model D is disputing the irrational belief with logical, empirical and pragmatic arguments, which leads to E, an effective new philosophy. Desensitisation is Wolpe's behavioural technique, and drama and discuss are fillers.

15. Distress tolerance is a major treatment modality in which of the following therapy? (RML 2024)

    (a) Behaviour therapy (b) Dialectical Behaviour therapy (c) Cognitive Behaviour Therapy (d) Psychodynamic therapy

    Answer: (b) Dialectical behaviour therapy. Distress tolerance is one of Marsha Linehan's four DBT skills modules, with mindfulness, emotion regulation and interpersonal effectiveness. Standard behaviour therapy, Beck's CBT and psychodynamic therapy have no such module.

16. A split between the organismic valuing process and a self-concept burdened by conditions of worth which results in feelings of tension and confusion: (Amity 2023)

    (a) Conflict (b) Incongruence (c) Introjection (d) Actualizing tendency

    Answer: (b) Incongruence. Rogers's term for the mismatch between self-concept and experience, felt as tension and anxiety. Introjection is only how conditions of worth get in, conflict has no special place in his theory, and the actualising tendency is the growth force that incongruence blocks.

17. In existential psychology, Dasein encompasses three fundamental psychological dimensions of human existence. Which of the following correctly identifies these three modes of being-in-the-world? (Real PYQ 2025)

    (a) Id, Ego, and Superego (b) Umwelt, Mitwelt, and Eigenwelt (c) Conscious, Preconscious, and Unconscious (d) Sensorimotor, Preoperational, and Concrete Operational

    Answer: (b) Umwelt, Mitwelt and Eigenwelt. Binswanger's three modes: the biological and physical world around, the with-world of relationships and the own-world of one's relation to oneself. The other options are Freud's structural and topographical models and Piaget's stages.

18. Frankl's technique for 'hyper-reflection', in which a person stops watching himself and turns his attention outward, for example from his own sexual performance to his partner, is: (NIMHANS 2026 recall)

    (a) Paradoxical intention (b) Sensate focus (c) Dereflection (d) Socratic dialogue

    Answer: (c) Dereflection. It counters hyper-reflection by turning attention away from the self. Paradoxical intention, also Frankl's, has the client wish for the feared symptom to defuse anticipatory anxiety; Socratic dialogue uncovers meaning; sensate focus is Masters and Johnson's sex-therapy exercise.

19. Which one of the following captures implicit and explicit group communication and interaction patterns? (RML 2020)

    (a) A genogram (b) A cognitive map (c) An ecomap (d) A sociogram

    Answer: (d) A sociogram. Moreno's sociometric diagram draws group members as points and their choices as arrows, exposing stars, isolates, cliques and lines of communication. A genogram maps a family across generations (Bowen), an ecomap maps a family's links with outside systems (Hartman), and a cognitive map is Tolman's mental map of space.

20. Expressed Emotion (EE) is a significant predictor of relapse in schizophrenia. Which of the following is NOT one of the three core dimensions of high Expressed Emotion? (Real PYQ 2024)

    (a) Hostility (b) Critical Comments (c) Emotional Over-involvement (d) Warmth and Positive Regard

    Answer: (d) Warmth and Positive Regard. High EE is defined by critical comments, hostility and emotional over-involvement. Warmth and positive remarks are also rated in the Camberwell Family Interview, but they do not count toward high EE.

21. Which of the following therapeutic modalities was pioneered by Maxwell Jones and is characterized by the democratic involvement of all community members (staff and patients) in decision-making and daily activities to promote rehabilitation? (Real PYQ 2024)

    (a) Encounter Groups (b) Therapeutic Community (c) Group Analysis (d) Milieu Therapy

    Answer: (b) Therapeutic Community. Maxwell Jones built the democratic therapeutic community at Belmont Hospital (later the Henderson Hospital), with shared decisions, daily community meetings and a flattened hierarchy. Milieu therapy is the broader idea of the ward as treatment, group analysis is Foulkes's psychoanalytic group therapy, and encounter groups belong to Rogers.

22. According to the "Dodo Bird Verdict," what is the primary factor responsible for equivalent outcomes across different therapeutic modalities? (Real PYQ 2025)

    (a) The specific techniques used by the therapist (b) Common factors such as the therapeutic alliance and empathy (c) The theoretical orientation of the clinician (d) The specific diagnosis of the patient

    Answer: (b) Common factors such as the therapeutic alliance and empathy. The verdict (Rosenzweig 1936; named by Luborsky, Singer and Luborsky 1975) holds that bona fide therapies produce roughly equal outcomes because they share common factors. Specific techniques and theoretical orientation are exactly what it says do not make the difference, and diagnosis is not the explanation offered.

23. 'Healing and Persuasion' is the work of: (IHBAS 2020)

    (a) Alfred Pritz (b) Jerome Frank (c) Charles Winnicot (d) Jeff Klerman

    Answer: (b) Jerome Frank. The book is *Persuasion and Healing* (1961), its title reversed in the question: all healing restores morale through a confiding relationship, a healing setting, a rationale and a ritual. Gerald Klerman co-developed interpersonal psychotherapy, Donald Winnicott was an object-relations analyst and Alfred Pritz was founding president of the World Council for Psychotherapy.

24. All of the following are the characteristics of efficacy studies EXCEPT (RML 2020)

    (a) Therapist actions are controlled by a protocol or treatment manual (b) Therapist flexibility is encouraged (c) The treatment is applied to a homogenous client population (d) The research design relies on random selection and random assignment to a treatment or control condition

    Answer: (b) Therapist flexibility is encouraged. Efficacy studies test a treatment under ideal, controlled conditions: manualised therapists, homogeneous samples and random assignment. Flexible, client-tailored therapy belongs to effectiveness research in routine practice.

25. Which model explains health behaviour by focussing on perceived susceptibility, severity, benefits and barriers? (JEMAS 2025)

    (a) Health Belief Model (b) Theory of Planned Behaviour (c) Social Cognitive Theory (d) Transtheoretical Model

    Answer: (a) Health Belief Model. Susceptibility, severity, benefits and barriers (with cues to action and, later, self-efficacy) are its signature constructs. The theory of planned behaviour uses attitude, subjective norm and perceived behavioural control; social cognitive theory centres on self-efficacy; the transtheoretical model describes stages of change.

26. Which of the following factor is most directly linked to a higher risk for coronary heart disease? (RML 2024)

    (a) Type A personality (b) Type B personality (c) Hostility (d) Self-efficacy

    Answer: (a) Type A personality (as keyed). Friedman and Rosenman's Type A pattern (time urgency, competitiveness, hostility) predicted coronary heart disease, but later research found hostility to be the toxic component, so choose hostility when a question asks for the component. Type B is relaxed and lower-risk; self-efficacy is not a risk factor.

27. Which therapy is mostly needed in 'Crisis Management'? (IHBAS 2020)
    (a) Cognitive behavior therapy (b) Acceptance & commitment therapy (c) Expressive therapy (d) Supportive therapy

    Answer: (d) Supportive therapy. Crisis intervention is brief, active and here-and-now: it listens, reassures, solves immediate problems and mobilises support to restore the previous level of functioning. CBT, acceptance and commitment therapy and expressive therapy are structured or exploratory treatments, not the first response to a crisis.

28. A parent deliberately ignores a child's temper tantrums in an attempt to discourage them. The parent's strategy is in accordance with: (Amity 2023)
    (a) Negative reinforcement (b) Extinction (c) Operant escape (d) Operant avoidance

    Answer: (b) Extinction. The source key gives (c), operant escape, which is wrong. Ignoring removes the attention that maintained the tantrums, so they weaken: that is extinction. Escape and avoidance are both forms of negative reinforcement, which strengthen behaviour.

29. Which of the following disabilities is explicitly covered under the scope of The National Trust Act (1999) in India for the welfare of persons with the disability? (Real PYQ 2024)
    (a) Autism Spectrum Disorder (b) Specific Learning Disability (c) Intellectual Disability (Mental Retardation) (d) Both (a) and (c)

    Answer: (d) Both (a) and (c). The National Trust Act 1999 covers exactly four conditions: autism, cerebral palsy, mental retardation (intellectual disability) and multiple disabilities. Specific learning disability is a specified disability under the RPwD Act 2016, not under the National Trust Act.

30. Infantile autism was first described by : (IHBAS 2025)
    (a) Hans Asperger (b) Leo Kanner (c) Eugen Bleuler (d) Kurt Schneider

    Answer: (b) Leo Kanner. Kanner described early infantile autism in eleven children in 1943. Asperger described more able children a year later, Bleuler coined the word autism in 1911 for withdrawal in schizophrenia, and Kurt Schneider is known for the first-rank symptoms of schizophrenia.

High-yield sheet

The facts this chapter's questions ask most, most-asked first. Last read before the exam.

1 **Systematic desensitisation** (Joseph Wolpe, *Psychotherapy by Reciprocal Inhibition*, 1958) rests on **reciprocal inhibition**: Jacobson's progressive relaxation is paired, step by step, with a graded anxiety hierarchy rated in SUDS (0 to 100); its main use is phobias. ASKED 21 TIMES

2 **Marsha Linehan**'s dialectical behaviour therapy (1993) was built for borderline personality disorder with chronic suicidality; its four skills modules are mindfulness, distress tolerance, emotion regulation and interpersonal effectiveness. ASKED 10 TIMES

3 **Carl Rogers**'s three core conditions are **empathy, unconditional positive regard and congruence** (genuineness), conditions 3 to 5 of his six necessary and sufficient conditions (1957); interpretation is not one of them. ASKED 10 TIMES

4 In **Albert Ellis**'s ABC model, A is the activating event, B the belief about it and C the emotional and behavioural consequence; B, not A, mainly causes C, and D is disputing the irrational belief. ASKED 9 TIMES

5 The **token economy** (Teodoro Ayllon and Nathan Azrin, 1968) is operant conditioning with tokens as secondary reinforcers, so it is the listed therapy NOT derived from classical conditioning. ASKED 8 TIMES

6 **Aaron Beck**'s cognitive triad in depression is a negative view of the self, the world and the future. ASKED 8 TIMES

7 The **empty chair** technique belongs to **Gestalt therapy** (Fritz Perls), where the client speaks to an absent person or a part of the self to finish unfinished business. ASKED 8 TIMES

8 **Exposure and response prevention** (Victor Meyer, 1966; refined by Edna Foa) is the first-line behavioural treatment for OCD and works through habituation and extinction. ASKED 7 TIMES

9 **EMDR** (Francine Shapiro, published 1989) was developed for PTSD: the client holds the traumatic memory in mind while following bilateral eye movements. ASKED 7 TIMES

10 **Transference** is the patient's unconscious redirection onto the therapist of feelings belonging to significant figures from the past; **countertransference** (named by Freud in 1910) is the therapist's emotional reaction to the patient arising from the therapist's own unresolved conflicts. ASKED 6 TIMES

11 **Viktor Frankl**'s **logotherapy** is an existential therapy built on the **will to meaning**; its frustration causes noogenic neurosis, and dereflection counters hyper-reflection. ASKED 6 TIMES

12 **Motivational interviewing** was developed by **William Miller and Stephen Rollnick** for substance use; its core skills are **OARS** (open questions, affirmations, reflections, summaries) and its spirit **PACE** (partnership, acceptance, compassion, evocation). ASKED 6 TIMES

13 The **diathesis-stress model** explains a disorder as a predisposing vulnerability (the diathesis) triggered by environmental stress; Zubin and Spring formalised it as the vulnerability-stress model in 1977. ASKED 5 TIMES

14 The **Mental Healthcare Act 2017** lets adults make an advance directive and appoint a nominated representative, presumes severe stress in anyone who attempts suicide (section 115) and bans ECT without anaesthesia and muscle relaxant (section 95). ASKED 5 TIMES

15 **Free association**, obeying the **fundamental rule** to say whatever comes to mind without censorship, is the central technique of Freud's psychoanalysis. ASKED 5 TIMES

16 **Interpersonal psychotherapy** (Klerman and Weissman, 1984) treats depression in 12 to 16 sessions through four problem areas: grief, role disputes, role transitions and interpersonal deficits. ASKED 5 TIMES

17 The **Premack principle** (David Premack, 1959): a high-probability behaviour reinforces a low-probability one, as when video games follow homework. ASKED 5 TIMES

18 Socratic questioning (guided discovery) helps the client discover previously overlooked information for himself rather than receive the therapist's interpretation; with collaborative empiricism it is a fundamental concept of Beck's cognitive therapy, whereas free association is not. ASKED 5 TIMES

19 **Salvador Minuchin**'s **structural family therapy** works on subsystems, hierarchy and boundaries: **diffuse** boundaries make an **enmeshed** family (over-involvement, little autonomy) and **rigid** boundaries a **disengaged** one; its techniques are joining, enactment and unbalancing. ASKED 5 TIMES

20 The **Rehabilitation Council of India**, a registered society from 1986, became a statutory body under the RCI Act 1992 (in force 1993); it works under the Ministry of Social Justice and Empowerment, and clinical psychologists must register with it to practise. ASKED 4 TIMES

21 **Informed consent** is agreement after being told a study's or treatment's purpose, procedures, risks and the right to withdraw; **debriefing** is the full explanation of a study's true purpose given after participation. ASKED 4 TIMES

22 **Alfred Adler**'s individual psychology rests on organ inferiority (1907), the inferiority complex and striving for superiority; its techniques (encouragement, acting as if, spitting in the client's soup, the push-button technique) do not include analysis of transference. ASKED 4 TIMES

23 **Object relations** are internalised representations of self and others and the relationships between them; Klein's paranoid-schizoid position rests on **splitting**, and Kohut traced an empty, fragmented self to failed parental mirroring and idealising. ASKED 4 TIMES

24 **Repression** is the primary defence and the psychodynamic explanation of dissociative symptoms; **projection** attributes one's own unacceptable feelings to someone else; **cathexis** is psychic energy invested in a particular object, person or idea. ASKED 4 TIMES

25 **Flooding** is prolonged exposure to the most feared stimulus at full intensity, with no hierarchy, no relaxation and no escape, until the anxiety subsides. ASKED 4 TIMES

26 Rational emotive therapy (1955), renamed rational emotive behaviour therapy in 1993, is **Albert Ellis**'s; challenging absolute musts (musturbation) is its signature. ASKED 4 TIMES

27 **Steven Hayes**'s acceptance and commitment therapy aims at psychological flexibility through the six hexaflex processes: acceptance, cognitive defusion, present moment, self-as-context, values and committed action. ASKED 4 TIMES

28 A person-centred therapist relies on **reflection of feeling**, not advice or interpretation, and speaks of the real relationship rather than Bordin's working alliance. ASKED 4 TIMES

29 **Maxwell Jones** pioneered the **therapeutic community** at Belmont Hospital (later the Henderson Hospital): staff and patients share decisions democratically in daily community meetings. ASKED 4 TIMES

30 The **double bind** (Bateson, Jackson, Haley and Weakland, 1956) is repeated contradictory messages at two levels from a vital figure, with no way to comment or escape; it was proposed as a cause of schizophrenia and is now rejected. ASKED 4 TIMES

CHAPTER 8

Psychiatric Disorders

IN THIS CHAPTER

8.1 ●● Abnormality, Causes and the Classification of Mental Disorders: Define abnormality and the basic terms of psychopathology, explain how biological, psychological and diathesis-stress models account for disorder, and use the DSM-5-TR, ICD-10 and ICD-11 classifications and the cultural concepts of distress.

8.2 ●●● Descriptive Psychopathology: Signs and Symptoms: Define and recognise the signs and symptoms of mental disorder function by function (consciousness, perception, thought, belief, memory, emotion, movement, insight and judgement), including the named delusional syndromes and the disorders each phenomenon points to.

8.3 ●● Neurodevelopmental Disorders: Recognise and tell apart the DSM-5-TR neurodevelopmental disorders (intellectual disability, communication disorders, autism, ADHD, specific learning disorder, and motor and tic disorders) by their criteria, history and causes.

8.4 ●●● Schizophrenia Spectrum and Other Psychotic Disorders: Trace the concept of schizophrenia from Kraepelin, Bleuler and Schneider, distinguish positive, negative and cognitive symptoms, apply the DSM-5-TR, ICD-10 and ICD-11 criteria and the 1 day, 1 month, 6 months duration ladder across the psychotic spectrum, and explain its genetic, dopaminergic, neurodevelopmental and family theories and the side effects of antipsychotic drugs.

8.5 ●●● Depressive and Bipolar Disorders, and Suicide: Define the major depressive, manic and hypomanic episodes, diagnose the depressive and bipolar disorders and their specifiers in DSM-5-TR, ICD-10 and ICD-11, explain the biological, psychological and social causes and first-line drugs, and judge suicide risk.

8.6 ●● Anxiety and Obsessive-Compulsive and Related Disorders: Distinguish fear from anxiety, state the DSM-5-TR features and duration rules of the anxiety disorders, panic attacks, OCD and its related disorders, and explain their biological, learning, cognitive and psychodynamic causes, including the defences of phobia and OCD.

8.7 ●●● Trauma, Dissociative and Somatic Symptom Disorders: Trace hysteria into its modern heirs and diagnose and tell apart PTSD, acute stress, adjustment, grief and attachment disorders, the dissociative disorders, and the somatic symptom disorders, factitious disorder and malingering.

8.8 ○ Feeding, Eating, Elimination and Sleep-Wake Disorders: Diagnose and tell apart the DSM-5-TR feeding and eating, elimination and sleep-wake disorders by their criteria, durations, ages, medical signs and timing in the night.

8.9 ○ Sexual Dysfunctions, Paraphilic Disorders and Gender Dysphoria: Describe the normal sexual response cycle, classify the DSM-5-TR sexual dysfunctions, paraphilic disorders and gender dysphoria, and explain how ICD-11 and the history of homosexuality changed what counts as a sexual or gender disorder.

8.10 ●● Substance-Related, Addictive and Neurocognitive Disorders: Define the core terms of substance use, classify substance-related and addictive disorders in ICD-10, DSM-5-TR and ICD-11, recognise the ten drug classes, and tell apart delirium, the dementias, pseudodementia, Wernicke-Korsakoff syndrome and psychiatric syndromes of medical illness.

8.11 ●● Personality Disorders, Conduct and Impulse-Control Disorders: Define a personality disorder, place the ten DSM-5-TR personality disorders in their clusters with their criteria counts, recognise borderline, antisocial, histrionic, narcissistic and obsessive-compulsive personality and their look-alikes, describe the ICD-11 and AMPD dimensional models, and distinguish oppositional defiant disorder, conduct disorder, intermittent explosive disorder, pyromania and kleptomania.

Weightage. 411 of the 3180 questions mapped in this book (12.9%) test this chapter, 205 of them from real past papers. 4 of its 11 sections are High yield.

Heidelberg, 1891. **Emil Kraepelin**, aged thirty-five, takes up the chair of psychiatry at the university clinic. Years earlier he had worked in Wilhelm Wundt's laboratory in Leipzig, timing how alcohol and other drugs changed simple mental acts, and he brings a laboratory habit with him: record everything, then count. The psychiatry he inherits badly needs it. Diagnostic labels vary from one asylum to the next, and almost all of them describe how a patient looks on the day of admission. None says what will happen to the patient next.

Kraepelin's tool was humble: a card. For every patient admitted, a card recorded the history, the symptoms and a provisional diagnosis. The cards were not filed and forgotten. As months and years passed, the clinic added what became of each patient: recovery, relapse, or a slow slide into apathy and disordered thinking. Sorting the cards again, by outcome rather than by the picture at admission, Kraepelin could see which first impressions had been wrong, and the psychoses fell into two great families. In one, illness began in youth and ran a deteriorating course; he called it *dementia praecox*, premature dementia. In the other, attacks of mania or melancholia came and went, with recovery between them: *manic-depressive insanity*. The sixth edition of his textbook (1899) drew the line.

The idea on those cards is the idea of this chapter. A mental disorder, for Kraepelin, was a recognisable cluster of signs and symptoms with its own onset, course and outcome, and a classification was good if it predicted what would happen to the patient. When the American Psychiatric Association rebuilt its manual around explicit criteria in 1980, the psychiatrists behind the change were nicknamed the neo-Kraepelinians. Every disorder in the pages that follow is, in a sense, one of his piles of cards, redrawn by a century of research.

8.1 Abnormality, Causes and the Classification of Mental Disorders

MEDIUM YIELD · 30 QS

Every disorder in this chapter rests on three prior questions. When does unusual behaviour become a disorder? Why does one person fall ill while another does not? And how do clinicians everywhere agree on what to call what they see? This section answers each in turn, then shows how culture shapes distress itself. (The history of the asylums, and of the models as guides to treatment, is told in Section 7.1.)

What counts as abnormal ●●

No single test separates normal from abnormal, so clinicians weigh the **four Ds** together. **Deviance** takes two forms. The **statistical criterion** calls abnormal whatever deviates from typical patterns; its flaw is that rare talent, a genius IQ, is just as deviant. The **social-norm criterion** calls abnormal whatever violates a society's explicit and implicit norms; its flaw is **cultural relativity**, for a woman in possession trance at a temple festival is following custom, not breaking it. **Distress** is personal unhappiness and suffering, though a person in mania may feel wonderful. **Dysfunction** is maladaptiveness: behaviour that stops you working, studying or keeping relationships. **Danger** to self or others is the least common.

Marie Jahoda (1958) described ideal mental health positively, as self-acceptance, growth, integration, autonomy, accurate perception of reality and environmental mastery. **Jerome Wakefield** (1992) defined disorder as **harmful dysfunction**: a mental mechanism fails at the job evolution designed it for (a fact), and the failure harms the person by the standards of their culture (a value). In DSM-5-TR, a **mental disorder** is a syndrome of clinically significant disturbance in cognition, emotion regulation or behaviour that reflects a dysfunction in the underlying psychological, biological or developmental processes, usually with significant distress or disability. An expectable or culturally approved response to a common stressor or loss, such as the death of a loved one, is excluded, and so is socially deviant behaviour (political, religious or sexual) unless it results from a dysfunction in the person.

Daniel Offer and **Melvin Sabshin** (1966) described four views of normality. **Normality as health**, the medical view, is the absence of manifest psychopathology; judged from the person's own side it becomes the **subjective model**: no distress, no disability, no help-seeking. **Normality as utopia** is ideal, optimal functioning; **normality as average** is the statistical middle of the bell curve; **normality as process** is the product of systems that change across the lifespan.

> ! EXAM TRAP
>
> Statistical abnormality is deviation from typical patterns, not intense anxiety, loss of contact with reality or withdrawal. Optimal functioning is utopia; no distress, disability or help-seeking is the subjective model; absence of any disorder is the medical view; socially acceptable behaviour is the norm criterion.

The language of psychopathology ●●

Psychopathology is the scientific study of mental disorders; **abnormal psychology** is the branch of psychology that studies abnormal behaviour and experience, and the terms largely overlap. A **symptom** is what the patient reports (low mood), a **sign** what the clinician observes (tearfulness), and a **syndrome** a cluster of signs and symptoms that occur together (8.2). **Aetiology** is the study of causes. **Onset** is **acute** (sudden) or **insidious** (gradual). **Course** may be **acute** (short), **chronic** (long-lasting), **episodic** (recurring with recovery between, as in mood disorders) or **time-limited** (improving untreated within a short period). **Prognosis** is the likely course and outcome: a forecast, not the history, the presenting symptoms or the diagnostic criteria.

Prevalence is the proportion of a population that has a disorder, on one day (**point**), over a stated period such as a year (**period**) or ever (**lifetime**); **incidence** counts only new cases arising in a period. **Comorbidity** is the co-occurrence of two or more differently defined disorders in one person, such as depression with an anxiety disorder. In psychiatry it is the rule, not the exception.

> ! EXAM TRAP
>
> Comorbidity is not the frequency of mental illness (that is prevalence), not two or more causes of death, not the degree of suffering, and not dying together or fearing death. "Depression always in winter" garbles the seasonal pattern specifier (8.5).

Causes: from single models to diathesis-stress ●●●

Early theories were **one-dimensional**, tracing a disorder to a single cause. The modern view is **multidimensional** (integrative): biological, psychological, social and cultural influences interact. Table 8.1 gives each classical model's answer.

Table 8.1 Models of psychopathology and the causes they blame.

Model	Cause of disorder
Biological	Genes, brain structure, neurotransmitters
Psychodynamic	Unconscious conflict from emotional trauma in childhood; fixation; defences (Chapter 4)
Behavioural	Maladaptive (inappropriate) learning
Cognitive	Disturbed thought patterns and schemas (Beck and Ellis, 7.5)
Humanistic-existential	Failure to live up to one's potential; incongruence
Sociocultural	Poverty, gender, culture, stigma, labelling
Family systems	Disturbed family interaction

Biological research rests on **family**, **twin** and **adoption studies**. Twin studies compare **concordance**, the rate at which both twins are affected, in identical and fraternal pairs; adoption studies separate genes from upbringing. Together they estimate **heritability**, the share of variation in a population due to genetic differences. Disorders are **polygenic**: many small-effect genes add up to a polygenic risk score. In **gene-environment interaction** a gene matters only in some environments: **Avshalom Caspi** (2003) found that a short serotonin transporter variant raised adult depression mainly after childhood maltreatment. In **gene-environment correlation**, the reciprocal gene-environment model, genes make certain environments likelier, as when an impulsive temperament invites the very stressors that trigger the vulnerability. Neurotransmitter (dopamine) and brain-structure (enlarged ventricles) models complete the strand.

Because there is no single "schizophrenia gene", researchers seek **endophenotypes**: the genetic mechanisms that ultimately produce the problems behind the symptoms, measurable intermediate traits such as the eye-tracking and working-memory deficits of schizophrenia. **Irving Gottesman** and **James Shields** brought the term into psychiatry in the 1970s; Gottesman and **Todd Gould** revived it in 2003. Contrast the **genotype**, a person's genetic make-up, the **phenotype**, the observable disorder, and **molecular genetics**, a method that studies DNA directly.

The **diathesis-stress model** joins the strands: a predisposition, the **diathesis** (genetic, biological or psychological), lies dormant until a stressor sets it off, and neither alone suffices. **Paul Meehl** (1962) proposed an inherited **schizotaxia** that under stress could become schizophrenia; **Joseph Zubin** and **Bonnie Spring** (1977) recast it as the **vulnerability-stress model**: the more vulnerable you are, the less stress tips you into illness (Section 7.1 draws the threshold). Broadest is the **biopsychosocial model** of **George L. Engel** (1977): illness arises from interacting biological, psychological and social factors.

> **! EXAM TRAP**
>
> Any "predisposition plus trigger" or "vulnerability plus stress" wording is diathesis-stress; genes, stress, learning, unconscious conflict or the supernatural alone are wrong. The **neurodevelopmental model** traces schizophrenia to faulty early brain development (8.4); "clinical-reductionist" and "syndrome-sign" are not models. Biopsychosocial is Engel's, not Kraepelin's, Bleuler's (8.4) or Beck's.

> **! EXAM TRAP**
>
> Asked which disorders show strong biological and "little to no" psychosocial evidence, choose schizophrenia and bipolar disorder, the most heritable (bipolar heritability reaches about 90 percent in some twin studies), not bulimia nervosa, in which cultural pressure for thinness weighs heavily (8.8). But the premise is false: identical-twin concordance for bipolar disorder is only 40 to 70 percent and life events trigger relapses, while childhood adversity, migration and urban upbringing raise the risk of schizophrenia. Genes set the diathesis; stress still times the episodes.

Approaches to classification ○

Hippocrates (about 460 to 377 BC) sorted mental illness into **mania**, **melancholia** and **phrenitis** (feverish delirium), and coined the word **hysteria**. Modern classification begins with **Emil Kraepelin** (1856 to 1926), who classified descriptively by symptoms and above all by course, separating deteriorating dementia praecox from episodic manic-depressive insanity (8.4).

The **syndromal approach** classifies disorders by clinical signs and symptoms that cluster together; the **historical approach** (aetiological) classifies them by known cause, as medicine does with tuberculosis. Since most causes are unknown, the DSM and ICD are syndromal. A **categorical approach** says you have a disorder or you do not; a **dimensional approach** rates everyone on continuous scales; a **prototypical approach**, the DSM's, lists the features of a typical case and tolerates variation (five of nine symptoms make a major depressive episode, so two patients may share only one).

Reliability is agreement between clinicians who assess the same patient, measured by Cohen's kappa (agreement corrected for chance); **validity** is whether a category is real and useful, predicting course and treatment response. A diagnosis can be reliable but invalid, never the reverse. A **false positive** diagnoses a normal person as disordered; a **false negative** misses a real disorder. Loose criteria breed false positives, strict ones false negatives. Every diagnosis is also a **label** that can bring **stigma** (Section 7.2 tells the pseudopatient study).

> **! EXAM TRAP**
>
> Syndromal means signs and symptoms; historical means known aetiology; option sets swap them. A high false-positive rate means normal people are diagnosed as disordered (criteria too loose, not too strict); missed cases are false negatives; low reliability is rater disagreement.

The DSM ●●

The **Diagnostic and Statistical Manual of Mental Disorders** (DSM) is published by the **American Psychiatric Association** (APA). Figure 8.1 sets its editions beside the ICD's. **DSM-I** (1952) and **DSM-II** (1968) were short glossaries of descriptions. **DSM-III** (1980), directed by **Robert Spitzer** (1932 to 2015), brought explicit diagnostic criteria, a descriptive stance neutral about causes, and the multiaxial system; **DSM-III-R** (1987), **DSM-IV** (1994) and **DSM-IV-TR** (2000) followed. **DSM-5** (2013) took an Arabic numeral to allow numbered updates. The edition in use, **DSM-5-TR** (2022), added **prolonged grief disorder** and codes for suicidal behaviour and non-suicidal self-injury.

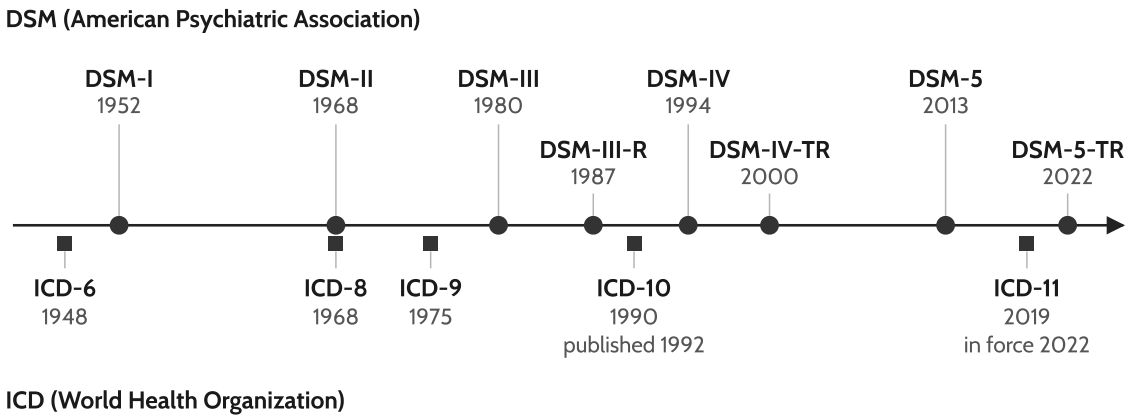


Figure 8.1 DSM editions (circles, above the line) and ICD editions (squares, below). ICD-11, adopted in 2019, took effect in 2022, the year of DSM-5-TR.

From DSM-III to DSM-IV-TR, diagnosis ran on five axes: **Axis I**, clinical disorders; **Axis II**, personality disorders and mental retardation; **Axis III**, general medical conditions; **Axis IV**, psychosocial and environmental problems; **Axis V**, the Global Assessment of Functioning (GAF). **DSM-5 abolished the multiaxial system**: Axes I to III are listed together, psychosocial problems take Z codes, and the GAF gave way to the WHO Disability Assessment Schedule (WHODAS 2.0).

DSM-5-TR has three sections. **Section I** explains how to use it. **Section II** holds the criteria and codes: 20 disorder chapters in rough lifespan order, from neurodevelopmental through paraphilic to other mental disorders, then medication-induced movement disorders and **other conditions that may be a focus of clinical attention**, problems that are not mental disorders, such as relational problems, malingering and borderline intellectual functioning. **Subtypes** are mutually exclusive ("specify whether"), **specifiers** can be combined ("specify if"), and **severity** is rated mild, moderate or severe. A presentation short of full criteria is **other specified** (reason recorded) or **unspecified** (reason not given). **Section III**, "Emerging Measures and Models", holds cross-cutting symptom measures, the WHODAS 2.0, the Cultural Formulation Interview, the Alternative Model for Personality Disorders (8.11) and the **conditions for further study**, proposed criteria sets not approved for clinical use.

REMEMBER: THE SEVEN CONDITIONS FOR FURTHER STUDY

Attenuated psychosis syndrome (milder psychotic symptoms at least weekly for a month); depressive episodes with short-duration hypomania; caffeine use disorder; internet gaming disorder; neurobehavioural disorder associated with prenatal alcohol exposure; suicidal behaviour disorder; and **non-suicidal self-injury disorder** (intentional cutting, burning, stabbing or hitting of one's own body on 5 or more days in a year, without suicidal intent). DSM-5's persistent complex bereavement disorder became DSM-5-TR's prolonged grief disorder (8.7).

! EXAM TRAP

An option "DSM-V" means DSM-5: choose it for the version in use, knowing the current text is DSM-5-TR (2022). DSM-5 abolished the axes; it added no sixth axis and still gives diagnoses, with ICD-10-CM codes beside them. Axis III is general medical conditions, not personality disorders (II) or psychosocial problems (IV).

The ICD ○

The **International Classification of Diseases** (ICD) is the **World Health Organization**'s (WHO) classification of all diseases. **ICD-6** (1948) was the first to include mental disorders. **ICD-10** (endorsed in 1990, published in 1992, used from 1994) put them in Chapter V (F), with the *Clinical Descriptions and Diagnostic Guidelines* (the "blue book", also 1992) and the *Diagnostic Criteria for Research* (1993). Its blocks (Table 8.2) still fill Indian case files.

Table 8.2 ICD-10 Chapter V (F): the blocks.

Block	Disorders
F00–F09	Organic, including symptomatic, mental disorders (delirium, dementia)
F10–F19	Mental and behavioural disorders due to psychoactive substance use
F20–F29	Schizophrenia, schizotypal and delusional disorders
F30–F39	Mood (affective) disorders
F40–F48	Neurotic, stress-related and somatoform disorders
F50–F59	Behavioural syndromes associated with physiological disturbances and physical factors
F60–F69	Disorders of adult personality and behaviour
F70–F79	Mental retardation
F80–F89	Disorders of psychological development
F90–F98	Behavioural and emotional disorders with onset in childhood and adolescence
F99	Unspecified mental disorder

ICD-11, adopted in May 2019, took effect on 1 January 2022. Mental disorders form **Chapter 6**, "Mental, behavioural or neurodevelopmental disorders" (codes 6A00 to 6E8Z). Sleep-wake disorders moved to Chapter 7, and sexual dysfunctions and gender incongruence to Chapter 17, "Conditions related to sexual health" (8.9). New are complex PTSD, prolonged grief disorder, gaming disorder, compulsive sexual behaviour disorder and bodily distress disorder, and a single **personality disorder** graded by severity (8.11). Table 8.3 compares the systems.

Table 8.3 DSM and ICD compared.

Feature	DSM	ICD
Publisher	American Psychiatric Association	World Health Organization
Scope	Mental disorders only	All diseases
Purpose	Explicit criteria for clinicians and researchers	Global health statistics and clinical guidelines
Cost	Sold by the APA	Free
Codes	ICD-10-CM codes (United States)	F00 to F99; ICD-11 6A00 to 6E8Z
Use	United States; research worldwide	Official worldwide; India's health system and Mental Healthcare Act 2017

> **! EXAM TRAP**
> Delirium and dementia are F00-F09 (organic), not F10-F19 (substances); mood is F30-F39. DSM and ICD are classification and diagnostic systems, not tests, therapies or intelligence scales.

Culture and diagnosis ●●

DSM-IV's glossary of **culture-bound syndromes** became, in DSM-5, **cultural concepts of distress**, because all distress is locally shaped. They are of three kinds: **cultural syndromes** (symptom clusters in particular groups), **cultural idioms of distress** (shared ways of voicing suffering, such as "thinking too much") and **cultural explanations** (perceived causes, such as spirits or semen loss). Section III's **Cultural Formulation Interview** (CFI) elicits them in 16 questions. **N. N. Wig** coined the term dhat syndrome in 1960; Table 8.4 lists it with the others.

Table 8.4 Cultural concepts of distress.

Syndrome	Region	Core feature
Dhat	Indian subcontinent, Sri Lanka	Anxiety and weakness blamed on semen loss
Koro	South and East Asia, north-east India	Genitals retracting into the body, feared fatal; epidemics
Suchi-bai	Bengal	Obsessional cleanliness
Possession syndrome	Rural India	Possession by a deity or dead relative's spirit
Bhanmati sorcery	South India	Illness blamed on sorcery
Gilhari syndrome	Rajasthan	Squirrel-like swelling creeping to the neck, feared fatal
Jhinjhinia	West Bengal	Tingling and numbness rising from the legs
Latah	Malaysia, Indonesia	Exaggerated startle to minimal stimuli; echolalia, echopraxia, automatic obedience, obscenities
Amok	Malaysia	Brooding, violent outburst, amnesia
Pibloktoq	Arctic and subarctic Inuit	Dissociative fit: screaming, tearing off clothes, running into snow; amnesia

Syndrome	Region	Core feature
Susto	Latin America	Fright drives the soul from the body
Ataque de nervios	Latin America, Caribbean	Shouting, crying, trembling; loss of control
Brain fag	West Africa (Nigeria)	Students: poor concentration and thinking, head pain
Taijin kyofusho	Japan	Fear that one's body or appearance offends others
Hwa-byung	Korea	Suppressed anger
Shenjing shuairuo	China	Neurasthenia: fatigue, poor concentration
Windigo	Algonquian peoples	Fear of becoming a cannibal
Zar	North Africa, Middle East	Spirit possession
Kufungisisa	Zimbabwe	"Thinking too much"
Ghost sickness	American Indian groups	Preoccupation with death and the dead
Falling out	Southern United States, Caribbean	Sudden collapse, unable to see
Mal de ojo	Mediterranean, Latin America	"Evil eye" illness in children

! EXAM TRAP

Dhat, koro, suchi-bai, possession syndrome, bhanmati, gilhari and jhinjhinia are reported in India; **latah is not**. Startle with obscenities is latah, brooding then violence amok, Inuit running into the snow pibloktoq, soul loss susto. Together they are cultural concepts of distress, not anxiety subtypes, substance-induced or dissociative disorders.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The criterion of abnormality that centres on personal unhappiness is _____.
2. Wakefield defined a mental disorder as a harmful _____.
3. The co-occurrence of two or more differently defined disorders in one person is _____.
4. Measurable intermediate traits between genes and symptoms are _____.
5. Zubin and Spring's 1977 version of the diathesis-stress model is the _____ model.
6. The DSM edition that introduced explicit criteria and five axes, in 1980, was _____.
7. In ICD-10, delirium and dementia fall in block _____.
8. The Malaysian syndrome of exaggerated startle, echolalia and blurted obscenities is _____.

Answers: (1) distress, (2) dysfunction, (3) comorbidity, (4) endophenotypes, (5) vulnerability-stress, (6) DSM-III, (7) F00-F09, (8) latah.

8.2 Descriptive Psychopathology: Signs and Symptoms HIGH YIELD · 45 QS

A psychiatric diagnosis is built from small units: a voice nobody else hears, a belief no argument can shift, a smile at a funeral. This section names and defines those units, one mental function at a time. Section 7.2 shows how the mental status examination draws them out in an interview; the disorders they build begin in Section 8.3.

Jaspers and the descriptive method

Descriptive psychopathology is the precise description and classification of abnormal experiences and behaviour, without claims about their cause. Its founding text is *General Psychopathology* (1913) by the German psychiatrist and philosopher **Karl Jaspers** (1883 to 1969), who brought **phenomenology** into psychiatry: the clinician describes an experience as the patient lives it, setting theory aside. The method is **empathic understanding**: careful questioning until you grasp what the experience is like. Where empathy fails, where an experience cannot be understood from the person's life and mood, Jaspers suspected a disease process.

Three distinctions run through everything that follows. The **form** of a symptom is its type (a hallucination); its **content** is what it is about (a voice calling him a thief). Culture colours the content, but the form carries the diagnosis. A **primary symptom** arises directly from the illness; a **secondary symptom** is an understandable reaction to one, as when a man who hears accusing voices concludes he is being watched. A **symptom** is a subjective complaint the patient reports (low mood); a **sign** is an objective abnormality the clinician observes (flat affect, slowed movement).

Consciousness, attention and orientation

Clouding of consciousness is the mildest lowering of awareness: drowsiness, poor attention and an incomplete grasp of the surroundings. **Confusion** adds muddled thinking and disorientation. **Delirium**, in DSM-5-TR terms, is a disturbance of attention and awareness that develops over hours to days, **fluctuates** during the day (lucid in the morning, worse at night) and brings other cognitive change, often with illusions and visual hallucinations; it signals a medical cause (Section 8.10). In **stupor** the person is mute and motionless and responds only to strong stimuli (catatonia, severe depression, organic states); in **coma** even pain brings no response. A **twilight state** is a narrowed consciousness lasting hours or days, in which complex, seemingly purposeful acts are carried out and later forgotten (after epileptic seizures; alcohol and dissociative states).

Distractibility is attention pulled away by trivial outside stimuli (mania, delirium). **Disorientation** is loss of awareness of time, place or person: **time is lost first**, then place, and one's own identity last.

Disorders of perception ●●

In a **sensory distortion** a real object is perceived but altered in intensity (sounds painfully loud), quality (everything tinged yellow) or size: **micropsia** (objects seen smaller) and **macropsia** (larger). In a **sensory deception** a new perception appears (Figure 8.2).

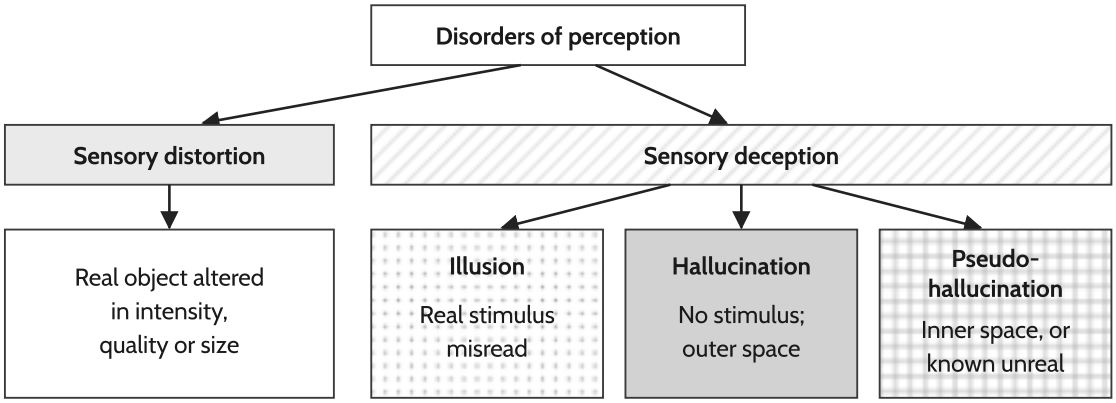


Figure 8.2 Disorders of perception. Only the hallucination needs no stimulus at all.

An **illusion** is a misperception of a real, existing stimulus. **Completion illusions** fill in missing parts (a misprint read as correct), **affect illusions** arise from emotion (a frightened person sees a coat on a hook as an intruder) and **pareidolic illusions** make vivid images from vague material (faces in clouds). A **hallucination** is a perception without an external stimulus, located in outer space and as vivid as a real perception. A **pseudohallucination** is located in inner space ("a voice inside my head") or is recognised as unreal. Hallucinations are most frequent in schizophrenia, and there mostly auditory; voices discussing or arguing about the patient in the third person, a running commentary on his actions and thought echo (hearing one's thoughts aloud) are first-rank symptoms (Section 8.4). Visual hallucinations suggest an organic cause (Table 8.5).

Table 8.5 Hallucinations by modality and special types.

Type	Experience	Typical of
Auditory, second person	A voice addresses the patient ("you are useless")	Psychotic depression, schizophrenia
Auditory, third person	Voices discuss, argue or comment; thought echo	Schizophrenia
Visual	Figures, animals, scenes	Delirium, Lewy body dementia, Charles Bonnet
Olfactory, gustatory	Smells, tastes	Temporal lobe epilepsy
Tactile; formication	Insects crawling under the skin	Cocaine, amphetamine, alcohol withdrawal
Somatic; kinaesthetic	Organs burning; limbs moving	Schizophrenia
Hypnagogic; hypnopompic	Falling asleep; waking	Normal people; narcolepsy
Functional	A real stimulus triggers one in the same sense (voices in a running tap)	Schizophrenia
Reflex	A stimulus in one sense, a hallucination in another	Schizophrenia, drugs
Extracampine	Outside the sensory field (someone behind one's head)	Schizophrenia, epilepsy
Autoscopic	Seeing oneself	Epilepsy, organic states
Lilliputian	Tiny people or animals	Delirium tremens

! EXAM TRAP

A hallucination is a disorder of **perception**, not of thinking, judgement or intelligence. An illusion distorts a stimulus that exists; a hallucination has none. **Hypnagogic** hallucinations occur most commonly in **normal** people (DSM-5-TR counts them as normal experience), not in schizophrenia, delirium or drug dependence; narcolepsy also features them.

KEY STUDY: CHARLES BONNET'S GRANDFATHER (1760)

In 1760 the Swiss naturalist **Charles Bonnet** (1720 to 1793) described his elderly grandfather, almost blind from cataracts, who saw people, birds and buildings that were not there, while knowing they were unreal. **Charles Bonnet syndrome** now names complex visual hallucinations with significant visual loss, intact cognition and preserved insight. It is not a psychosis.

In **depersonalisation** the self feels unreal or changed ("as if I am watching myself in a film"); in **derealisation** the surroundings feel unreal or dreamlike. The person knows it only feels so. The disorder is in Section 8.7.

Disorders of thought ●●

Thought is described by its stream (speed and flow), form (how ideas connect), content (what is thought) and possession (whose the thoughts feel to be). In the stream, **pressure of thought** is a rapid crowding of ideas (mania); **poverty of thought** is a scarcity of slow, empty ones (depression, schizophrenia); **thought block** is a sudden stop in mid-sentence, the mind blank (schizophrenia). Diminished speech output (poverty of speech) is **alogia**.

Formal thought disorder is a disturbance in the links between ideas, inferred from speech (Table 8.6). In **flight of ideas** (mania) thought shifts rapidly from one idea to another before any is completed, but the links can be traced: chance, rhymes, puns and **clang associations** (words chosen for sound). In **loosening of associations**, or derailment (schizophrenia), the links are lost; the oblique jumps are called **knight's move thinking**.

Table 8.6 Disorders of the form of thought.

Term	What you hear	Typical of
Flight of ideas	Rapid jumps; links by chance, rhyme, pun, clang	Mania
Loosening of associations, derailment	Slips to unrelated topics; links lost	Schizophrenia
Tangentiality	Drifts off; never reaches the goal	Schizophrenia, mania
Circumstantiality	Needless detail; finally reaches the goal	Epilepsy, obsessional and organic states
Perseveration	Same idea repeated to new questions	Organic brain disease
Verbigeration	Senseless repetition of words or phrases	Schizophrenia
Neologism	New words, or old words with private meanings	Schizophrenia
Metonym	Approximate word for the exact one ("I ate three menus")	Schizophrenia
Word salad, incoherence	Words without connection; the most severe	Schizophrenia

Term	What you hear	Typical of
Concrete thinking	Literal; cannot abstract (proverbs)	Schizophrenia, organic states
Overinclusive thinking	Concept boundaries lost; irrelevant ideas included	Schizophrenia

Overinclusive thinking was described by **Norman Cameron** (1896 to 1975). Thought content includes preoccupations, **obsessions** (intrusive, unwanted thoughts recognised as one's own and resisted; Section 8.6), **phobias** (excessive fear of a specific object or situation), delusions and **overvalued ideas**, a term of **Carl Wernicke** (1848 to 1905): a single, understandable belief that dominates the person's life and is held with strong feeling but without delusional fixity, such as the anorexic patient's conviction that she is too fat.

In thought alienation, thoughts no longer feel one's own: in **thought insertion** they are put in from outside, in **thought withdrawal** taken out, and in **thought broadcasting** they pass out of the mind so that others know them. All three are first-rank symptoms of **Kurt Schneider** (1887 to 1967); the rest of his list is thought echo, voices arguing, running commentary, delusional perception, somatic passivity and made feelings, made impulses and made actions (Section 8.4). Many Indian texts also file **obsessions and compulsions** under possession: the thought is known to be one's own yet cannot be controlled.

> ! EXAM TRAP
>
> Flight of ideas, tangentiality, derailment, loosening and circumstantiality are disorders of **form** (process), not content (delusions, obsessions) or possession (insertion, broadcasting). Many Indian notes file flight of ideas, circumstantiality and tangentiality under stream; if form is offered, choose it. Word salad is the most severe formal thought disorder; repeating one idea is perseveration.

Delusions ●●●

A **delusion** is a false, firmly held belief, not shaken by reason or contrary evidence, and out of keeping with the person's cultural and educational background (a community's shared belief in the evil eye is not one). Jaspers named its marks: **certainty**, **incorrigibility** and **impossibility** of content. Delusions, hallucinations and grossly disorganised speech or behaviour mark **psychosis**, a loss of contact with reality. Table 8.7 separates the delusion from its neighbours.

A **primary delusion** (autochthonous) cannot be understood from any other experience. It appears as **delusional mood** (something sinister is going on), **delusional perception** (a normal perception suddenly carries a private meaning: the traffic light turns red, so he is chosen to save the world), **delusional memory** or a **sudden delusional idea**. A **secondary delusion** is understandable from another experience: guilt from depressed mood, persecution to explain hostile voices. **Systematised** delusions form a logically connected system around one theme. **Mood-congruent** delusions fit the mood (grandeur in mania); **mood-incongruent** ones do not. DSM-5-TR calls a delusion **bizarre** when it is clearly implausible, not understandable to same-culture peers and not drawn from ordinary life (his organs removed without a scar); a **non-bizarre** delusion could happen (being followed by the police).

A **delusion of persecution**, that one is being deliberately interfered with, plotted against, threatened or mistreated, is the **most common** type, mostly in schizophrenia and delusional disorder; **paranoid** means persecutory or self-referential, and **paranoia** was the old name for chronic, systematised delusional states. In a **delusion of reference** neutral events carry a personal message; an **idea of reference** is held without full conviction. In **delusions of control** (passivity) one's acts or feelings are made by an outside force. **Grandiose** delusions belong to mania; **nihilistic delusions** (oneself, one's organs or the world do not exist or are dead; DSM-5-TR words it as the conviction that a major catastrophe will occur) and delusions of **guilt** and **poverty** to severe depression. **Somatic** (hypochondriacal) delusions concern disease; jealousy, erotomania, infestation and misidentification have eponyms (Table 8.8).

Table 8.7 Delusion, overvalued idea and obsession compared.

Feature	Delusion	Overvalued idea	Obsession
Conviction	Absolute	Strong, emotional, not fixed	Recognised as senseless
Resisted	No	No	Yes
Understandable	No	Yes	Content often absurd
Example	Schizophrenia	Anorexia nervosa	Obsessive-compulsive disorder

> **! EXAM TRAP**
>
> "Others are out to get me" is **persecution** (paranoid), the commonest delusion, most often in **schizophrenia**. Nihilistic delusions point to **depression**, not obsessive-compulsive or somatoform disorder.

Named delusional syndromes ●●●

Four delusional misidentification syndromes lead Table 8.8. **Joseph Capgras** (1873 to 1950) described the impostor delusion, **Capgras syndrome**, in 1923 with Jean Reboul-Lachaux as *l'illusion des sosies* ("the illusion of doubles"). **Jules Cotard** (1840 to 1889) described the nihilistic syndrome in 1880 and named it *délire des négations* in 1882; **Arnold Pick** (1851 to 1924) named reduplicative paramnesia in 1903. Erotomania carries the name of **Gaëtan Gatian de Clérambault** (1872 to 1934), whose classic account of it appeared in 1921. Paul Courbon and Gabriel Fail described Fregoli syndrome in 1927, naming it after the Italian quick-change actor Leopoldo Fregoli. John Todd and Kenneth Dewhurst named the Othello syndrome in 1955, after Shakespeare's jealous husband, and the Swedish neurologist **Karl-Axel Ekbom** (1907 to 1977) described delusional parasitosis in 1938.

Table 8.8 Named delusional and eponymous syndromes.

Syndrome	Core feature	Usual setting
Capgras	A familiar person, usually spouse or relative, replaced by an identical impostor	Schizophrenia; dementia, brain injury
Fregoli	Strangers are one familiar person (often a persecutor) in disguise	Schizophrenia, brain injury
Intermetamorphosis	People exchange identities, in appearance and personality	Psychosis, organic states
Subjective doubles	A double of oneself exists	Psychosis

Syndrome	Core feature	Usual setting
Reduplicative paramnesia	A familiar place or person exists in duplicate	Brain injury, dementia
Cotard	One is dead, does not exist, or has lost organs or blood	Severe psychotic depression
Othello	Morbid jealousy: the partner is unfaithful	Alcohol dependence
de Clérambault	Erotomania: a person of higher status secretly loves the patient	Delusional disorder
Ekbom	Delusional parasitosis: infested by insects or parasites	Delusional disorder, elderly
Couvade	A man develops pregnancy symptoms during his partner's pregnancy	Expectant fathers
Diogenes	Gross self-neglect, squalor and hoarding	Elderly people living alone
Ganser; folie à deux; Munchausen; Kleine-Levin	Approximate answers (Sigbert Ganser, 1898); shared delusion (Charles Lasègue and Jules Falret, 1877); factitious illness; recurrent hypersomnia	Sections 8.7, 8.4, 8.7, 8.8

! EXAM TRAP

Capgras: a familiar person becomes an impostor. **Fregoli**: strangers become one familiar person. **Reduplicative paramnesia** duplicates a place. **Othello** is jealousy, **de Clérambault** love, **Cotard** being dead. Ekbom's name also labels **restless legs syndrome** (Section 8.8); in a delusion question, Ekbom means parasitosis.

Disorders of memory •••

Amnesia is a pathological loss of memory: **anterograde** (new memories cannot be stored after the onset) or **retrograde** (memories from before it are lost). Organic amnesia is in Section 8.10, **psychogenic** (dissociative) amnesia in Section 8.7, and normal memory and false memories in Section 1.7.

Among the paramnesias, **confabulation** fills memory gaps with false, often detailed and believable memories, without intent to deceive; it is the hallmark of Korsakoff syndrome (alcohol-related thiamine deficiency). **Pseudologia fantastica**, described by **Anton Delbrück** in 1891, is fantastic lying that the teller half believes (factitious disorder). **Retrospective falsification** unconsciously rewrites the past to fit present mood; in **cryptomnesia** a forgotten memory returns unrecognised, so it feels original; **hypermnesia** is exaggerated recall (mania, hypnosis).

Déjà vu ("already seen") is the illusion of recognition: a new situation feels already experienced. **Déjà entendu** is the illusion of having heard something before; **déjà vécu**, of having lived the whole moment before. **Jamais vu** is the reverse: the familiar feels unfamiliar. All occur in tired, healthy people and in temporal lobe epilepsy.

! EXAM TRAP

Confabulation is **not lying**: there is no intent to deceive. **Malingering** is deliberate faking for external gain (Section 8.7); **confusion** and **delirium** are disturbances of consciousness, not invented memories.

Disorders of emotion ●●●

Mood is a sustained emotional state that the person reports; **affect** is the immediate, observed expression of emotion. Mood is the climate; affect is the weather. (Emotion is the broader, brief reaction, Chapter 1; feeling is its experienced side.) Affect is described by quality, range, reactivity, intensity, appropriateness and stability; persistence describes mood. Affect may be **restricted** (reduced range), **blunted** (severely reduced intensity), **flat** (almost no expression), **labile** (shifting rapidly without adequate cause) or **inappropriate** (incongruous with thought content: giggling about a death).

Anhedonia is the inability to experience pleasure or interest: a core symptom of depression and a negative symptom of schizophrenia. **Alexithymia**, a term of **Peter Sifneos** (1920 to 2008) in 1973, is difficulty identifying and describing one's own feelings. **Apathy** is lack of feeling and interest. **Ambivalence**, a term of **Eugen Bleuler** (1857 to 1939), is the coexistence of opposite feelings towards the same object. **Dysphoria** is an unpleasant mood; **euthymia** is normal mood. Elevated mood is graded by severity (Figure 8.3).

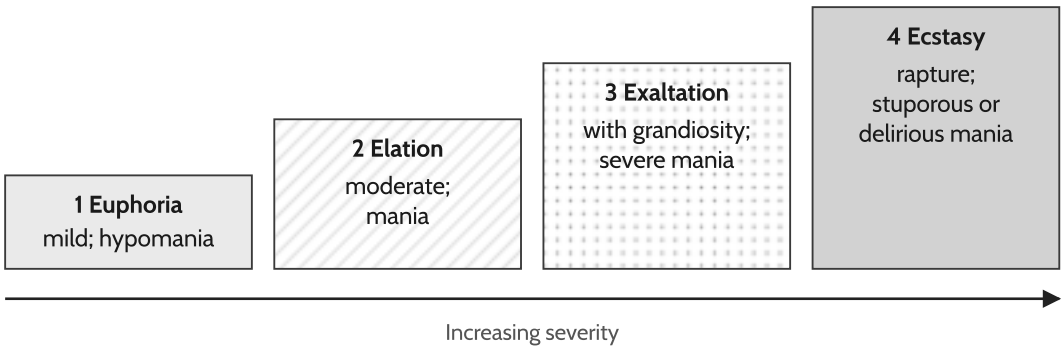


Figure 8.3 The four grades of elevated mood, mildest first.

Emotional incontinence is extreme lability that escapes voluntary control (vascular dementia). **Pseudobulbar affect** is sudden, brief, involuntary laughing or crying, out of proportion to or at odds with the underlying mood, in stroke, multiple sclerosis, amyotrophic lateral sclerosis, brain injury and dementia. *La belle indifférence* is in Section 8.7. A symptom is **egosyntonic** when experienced as consistent with one's self-image, and **ego-dystonic** when intrusive, unwanted and inconsistent with it (obsessions).

> **! EXAM TRAP**
> "Our current mood and feelings", the emotion seen at this moment, is **affect**, not mood. Quality, range and reactivity describe affect; **persistence** describes mood. Severity: **euphoria, elation, exaltation, ecstasy**. Pseudobulbar affect is neurological, not the restricted affect of schizophrenia or histrionic emotional shifts.

Motor behaviour, insight and judgement ○

Psychomotor retardation slows movement, speech and thought (depression); **psychomotor agitation** is purposeless overactivity with inner tension (agitated depression, mania). **Catatonia** has twelve DSM-5-TR signs: stupor; **catalepsy** (a passively induced posture held

against gravity); **waxy flexibility** (slight, even resistance to positioning); mutism; **negativism** (opposing or ignoring instructions); posturing; **mannerisms** (odd caricatures of normal actions); **stereotypies** (repetitive, non-goal-directed movements); agitation; grimacing; **echolalia** (repeating others' words); and **echopraxia** (imitating others' movements). Older signs are **automatic obedience**, **Mitgehen** (moving with the lightest touch despite being told to resist) and **ambitendency** (offering, then withdrawing, a hand).

Compulsions are repetitive acts performed in response to an obsession, to reduce distress (Section 8.6); tics are in Section 8.3. **Avolition** is reduced self-initiated, purposeful activity (a negative symptom: the loss of a normal function; Section 8.4); **abulia** is lack of will, with difficulty initiating action or deciding, typically after frontal lobe damage.

Insight is awareness and understanding of one's own illness and need for treatment, graded on six levels: (1) complete denial; (2) slight awareness while still denying; (3) awareness, but blamed on others, external or physical causes; (4) awareness of something unknown in oneself; (5) **intellectual insight**, accepting the illness without changing; (6) **true emotional insight**, awareness that changes behaviour. **Judgement**, the ability to assess a situation and act appropriately, is personal (one's own plans), social (conduct with others) or test (a hypothetical problem, such as finding a stamped letter in the street). Section 7.2 shows how both are assessed.

> **! EXAM TRAP**
>
> Insight means **awareness of one's own illness**, not orientation, serial subtraction (attention) or hallucinations (perception). Echolalia and echopraxia point to **catatonia**, not depression or panic.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Hallucinations while falling asleep, called _____, occur most often in normal people.
2. Needless detail that finally reaches the goal is _____, a disorder of the form of thought.
3. The belief that a spouse has been replaced by an identical impostor is _____ syndrome.
4. Filling memory gaps with false memories, without intent to deceive, is _____.
5. The observed, moment-to-moment expression of emotion is _____.
6. Elevated mood by severity: euphoria, elation, _____, ecstasy.

Answers: (1) hypnagogic, (2) circumstantiality, (3) Capgras, (4) confabulation, (5) affect, (6) exaltation.

8.3 Neurodevelopmental Disorders MEDIUM YIELD · 29 QS

The disorders in this section are differences in how the mind is built, not illnesses that strike a mind already formed. A child who cannot read a word he can say, or cannot sit still in class, has not lost a skill: his brain developed differently from the start.

The neurodevelopmental group ●●

DSM-5-TR (2022) defines **neurodevelopmental disorders** as conditions with onset in the developmental period, often before school entry, in which differences in brain development impair personal, social, academic or occupational functioning. They frequently co-occur: many autistic children have intellectual disability, and many children with ADHD have a specific learning disorder. ADHD belongs here; obsessive-compulsive and generalised anxiety disorder (Section 8.6) and borderline personality disorder (Section 8.11) do not.

DSM-IV-TR (2000) grouped them as "disorders usually first diagnosed in infancy, childhood or adolescence", which included the **pervasive developmental disorders**: autistic, Rett's, childhood disintegrative and Asperger's disorders and PDD not otherwise specified (PDD-NOS). ICD-10 (1992) spreads them over F70 to F98: mental retardation (F70 to F79), disorders of psychological development (F80 to F89) and childhood behavioural and emotional disorders such as hyperkinetic and tic disorders (F90 to F98).

Children behave differently at home, at school and in the clinic, so diagnosis uses **multi-informant assessment**: parents, teachers, clinician and child, across settings. ADHD even requires symptoms in two or more settings. The Vineland, CARS, ISAA and ADOS are described in Chapter 2.

> **! EXAM TRAP**
>
> Several informants are needed because *symptoms vary across contexts*, not because parents over-report or teachers add nothing. "Observation alone is not enough" is true but is not the reason.

Intellectual disability: criteria and severity ●●●

Intellectual disability (DSM-5-TR: **intellectual developmental disorder**; DSM-IV-TR: mental retardation) has three criteria. (A) Deficits in intellectual functions such as reasoning, problem solving, abstract thinking and learning, confirmed by clinical assessment *and* individual standardised intelligence testing. (B) Deficits in **adaptive functioning**, meeting age and cultural standards of independence and social responsibility, in the conceptual (language, reading, money), social (communication, friendship, judgement) or practical (self-care, work) domain, severe enough to need ongoing support. (C) Onset in the developmental period.

Criterion A means an IQ about two standard deviations (SD) below the mean: 70 on tests with mean 100 and SD 15, a score only about 2.3 percent of people fall below (Figure 8.4). Allowing about 5 points of measurement error, the zone is 65 to 75. Prevalence is about 1 percent, higher in males.

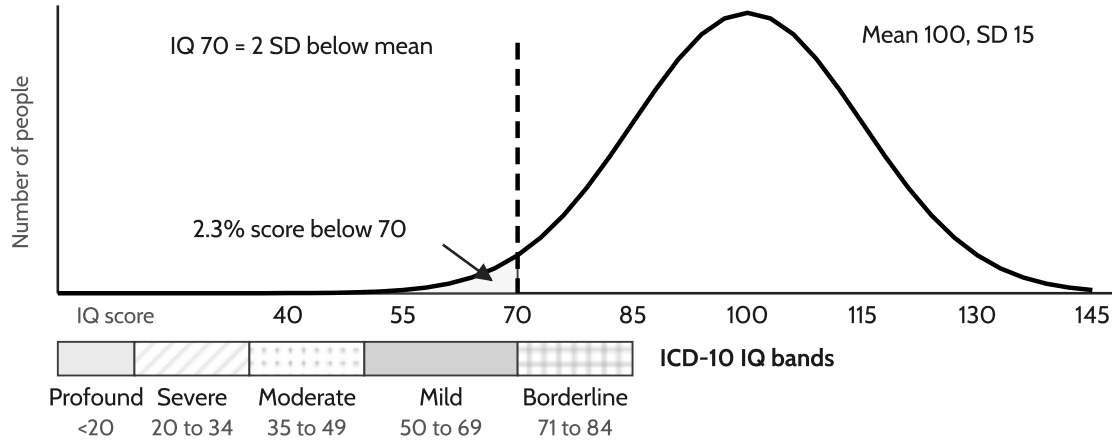


Figure 8.4 IQ is normally distributed; the intellectual criterion sits two SD below the mean. The bands are ICD-10's, with borderline intellectual functioning just above.

ICD-10 and older DSMs graded severity by IQ (Table 8.9; DSM-IV-TR used overlapping limits such as 50-55 to about 70 for mild), and about 85 percent of cases are mild. DSM-5 grades severity by *adaptive functioning*, which decides the support needed, because IQ is least valid at the low end. Severe levels show as delayed milestones in the first two years, mild ones often only at school; most people at every level now live in the community, not institutions.

Table 8.9 Severity levels with ICD-10 IQ bands and adult mental age (years).

Level	IQ	Mental age	Old label and outlook
Mild	50 to 69	9 to under 12	"Educable"; self-supporting adult with training
Moderate	35 to 49	6 to under 9	"Trainable"; partial independence in daily living, supervised work
Severe	20 to 34	3 to under 6	Basic self-care; close supervision
Profound	below 20	below 3	Constant care

An IQ near 70 is not enough: without adaptive deficits there is no diagnosis. **Borderline intellectual functioning** (traditionally IQ 71 to 84) is a condition that may be a focus of clinical attention, not a disorder. **Global developmental delay** is a provisional label for a child under 5 who misses milestones in several areas but whose severity cannot yet be reliably assessed or tested; it requires reassessment.

> **! EXAM TRAP**
>
> Global developmental delay differs from intellectual disability by *age and testability*, not by social, language or motor deficits: a 7-year-old with full-scale IQ 65 and severe adaptive impairment has intellectual disability. On the WAIS-IV, 70 to 79 is labelled "borderline"; without adaptive deficits, FSIQ 70 is not intellectual disability.

Causes of intellectual disability ●●

Causes are grouped by timing, beginning with genes (Table 8.10). **Down syndrome**, described by **John Langdon Down** (1828 to 1896) in 1866, is the commonest chromosomal cause:

trisomy 21, usually from nondisjunction and more likely with older mothers. **Fragile X syndrome**, an expanded CGG repeat silencing the FMR1 gene on the X chromosome, is the commonest inherited cause. In **phenylketonuria** (PKU), found in 1934 by **Asbjørn Følling** (1888 to 1973), recessive deficiency of phenylalanine hydroxylase stops phenylalanine being converted to tyrosine, and the excess damages the brain; newborn screening and a low-phenylalanine diet prevent the disability.

Table 8.10 Genetic and chromosomal causes of intellectual disability.

Condition	Cause	Hallmarks
Down syndrome	Trisomy 21	Flat face, heart defects, early Alzheimer's
Fragile X	FMR1 CGG repeat	Long face, large ears, autistic features
Prader-Willi	Paternal 15q deletion	Overeating, obesity
Angelman	Maternal 15q deletion	No speech, jerky gait, laughter
Williams	7q11.23 deletion	Elfin face, very sociable, poor visuospatial skill
Cri-du-chat	5p deletion	Cat-like cry
Lesch-Nyhan	HPRT deficiency (X-linked)	Self-biting, excess uric acid
Tay-Sachs	Hexosaminidase A deficiency	Cherry-red spot, regression
Klinefelter	47,XXY	Tall male; language and learning problems
Turner	45,X	Short female; IQ usually normal

Prenatally, **fetal alcohol syndrome** from drinking in pregnancy gives a small head, smooth (flat) philtrum, thin upper lip, short palpebral fissures (eye openings), poor growth and intellectual disability; it is among the leading preventable causes, and caffeine, cannabis and mercury do not produce this face. Others are maternal **TORCH infections** (toxoplasmosis, other agents such as syphilis, rubella, cytomegalovirus, herpes) and iodine deficiency causing **cretinism** (congenital hypothyroidism), worldwide a leading preventable cause of brain damage; teratogens are in Chapter 6. Perinatal causes include asphyxia and extreme prematurity; postnatal ones, meningitis, encephalitis, lead, head injury and severe deprivation.

Communication disorders ○

Language disorder is persistent difficulty acquiring and using language: small vocabulary, simple sentences, poor discourse. **Speech sound disorder** is difficulty producing speech sounds, making speech hard to understand, without a physical cause such as cleft palate or deafness. **Childhood-onset fluency disorder** (stuttering) breaks the flow of speech with repetitions, prolongations, broken words and blocks; 80 to 90 percent of cases begin by age 6. **Social (pragmatic) communication disorder** is difficulty using language socially (adjusting to the listener, taking turns, grasping hints and idioms) without the repetitive behaviour of autism.

Ask why a child does not communicate. In **mutism**, as in an inherently mute person, there is no capacity for speech: language itself is the barrier. In **selective mutism**, an anxiety disorder (Section 8.6), the child speaks at home but not in feared settings, so anxiety is the barrier. In autism the barrier is the social use of language; in specific learning disorder, reading and writing.

Autism spectrum disorder ●●●

Eugen Bleuler (1857 to 1939) coined autism (Greek *autos*, self) in 1911 for the withdrawal seen in schizophrenia. In 1943 **Leo Kanner** (1894 to 1981) published "Autistic disturbances of affective contact", on eleven children with extreme aloneness and insistence on sameness, and soon called the condition *early infantile autism*. In 1944 **Hans Asperger** (1906 to 1980) described boys with fluent speech but odd social behaviour and narrow interests. **Bruno Bettelheim**'s (1903 to 1990) "refrigerator mother" theory (1967) blamed cold parenting; it was discredited when autism proved largely genetic. **Lorna Wing** (1928 to 2014) and **Judith Gould** (1979) described a **triad of impairments**: social interaction, communication and imagination.

DSM-5 (2013) merged autistic disorder, Asperger's disorder, PDD-NOS and **childhood disintegrative disorder** (two or more years of normal development, then loss of skills; **Theodor Heller**, 1908) into **autism spectrum disorder** (ASD). **Rett syndrome** (**Andreas Rett**, 1966) was left out as a genetic condition (MECP2): girls regress after early normal development, losing speech and hand use, with hand-wringing.

Criterion A: persistent deficits in social communication and social interaction across multiple contexts, in all three of social-emotional reciprocity, nonverbal communication (eye contact, gesture) and relationships. **Criterion B**: **restricted, repetitive patterns of behaviour**, interests or activities, at least two of stereotyped movements or speech (hand-flapping, lining up toys, echolalia or echoing others' words), insistence on sameness, intense restricted interests, and sensory hyper- or hyporeactivity. Symptoms start early (though they may show only as social demands rise) and impair functioning. Severity is rated per domain: level 1 "requiring support", level 2 "requiring substantial support", level 3 "requiring very substantial support". Specifiers note accompanying intellectual impairment or language impairment, a known genetic or medical condition, and catatonia; many former Asperger's cases are now ASD without either impairment.

> **! EXAM TRAP**
>
> ASD needs social-communication deficits plus repetitive behaviour; delayed speech alone, IQ below 70 and hyperactivity are not required. No eye contact, delayed language and repetitive movements spell ASD, not ADHD, conduct disorder or SLD. Kanner first described infantile autism; Bleuler only coined the word, and Kurt Schneider (Section 8.4) and Cameron, Goldstein and Jaspers (Section 8.2) are distractors. The research deficit is in *encoding social cues*; trouble deceiving follows from mindblindness, and anger control and memory retrieval are not core deficits.

ASD affects about 1 to 2 percent of people, three to four males per female (DSM-5-TR says 3:1; girls are under-recognised). Outside the communication disorders themselves, language development is compromised more often in ASD than in any other neurodevelopmental disorder. The commonest co-occurring conditions are intellectual disability and ADHD (diagnosable alongside ASD only since DSM-5), then anxiety, depression and epilepsy.

Heritability is about 80 percent (estimates 37 to over 90); some cases carry *de novo* (new, uninherited) mutations, and older parents, extreme prematurity and prenatal valproate add risk. Autistic people show deficits in *encoding social cues* such as faces, gaze and tone of voice, and find deliberate deception hard, while rote memory is often a strength. **Simon Baron-Cohen**

(born 1958), **Alan Leslie** and **Uta Frith** (born 1941) proposed a deficit in **theory of mind** (Chapter 6), or "mindblindness". Frith's **weak central coherence** (1989) is a bias to detail over the whole; executive dysfunction explains rigidity; Baron-Cohen's *extreme male brain* theory casts autism as strong systemising with weak empathising.

KEY STUDY: THE SALLY-ANNE TEST (1985)

Baron-Cohen, Leslie and Frith tested 20 autistic, 14 Down syndrome and 27 typical children. Sally puts a marble in her basket and leaves; Anne moves it to her box. Where will Sally look? About 85 percent of typical and Down syndrome children said the basket; 80 percent of autistic children, despite higher verbal mental age than the Down group, said the box, failing to represent Sally's false belief. Criticism: a fifth passed, the deficit is not unique to autism, and it cannot explain repetitive behaviour.

In 1998 **Andrew Wakefield** linked the MMR vaccine to autism in a 12-child *Lancet* paper; it was retracted in 2010, he was struck off the medical register, and large studies find no link. Treatment is behavioural and educational: **Ivar Lovaas**'s applied behaviour analysis (1987), TEACCH and PECS (Sections 7.4 and 7.10).

Attention-deficit/hyperactivity disorder ●●

In 1902 **George Frederic Still** (1868 to 1941) described restless, inattentive children with a "defect of moral control". Later labels were minimal brain dysfunction, hyperkinetic reaction of childhood (DSM-II, 1968), attention deficit disorder (DSM-III, 1980) and ADHD (DSM-III-R, 1987). ICD-10's **hyperkinetic disorder** (F90) is overactivity, distractibility, short attention span and impulsivity; social withdrawal is not a feature.

DSM-5-TR **attention-deficit/hyperactivity disorder** (ADHD) needs six of nine inattention and/or six of nine hyperactivity-impulsivity symptoms (Table 8.11; five from age 17) for at least 6 months; several present before age 12 (DSM-IV: 7) and in two or more settings; and impairment. The *presentation* is combined, predominantly inattentive or predominantly hyperactive-impulsive.

Table 8.11 The 18 DSM-5-TR symptoms of ADHD (each "often").

Inattention (A1)	Hyperactivity-impulsivity (A2)
Careless mistakes	Fidgets or squirms
Cannot sustain attention	Leaves seat
Does not seem to listen when spoken to directly	Runs about or climbs
Does not follow through	Cannot play quietly
Disorganised	"On the go", "driven by a motor"
Avoids sustained mental effort	Talks excessively
Loses things	Blurts out answers
Easily distracted	Cannot wait turn
Forgetful	Interrupts or intrudes

> **! EXAM TRAP**
>
> "Does not seem to listen when spoken to" is an *inattention* item; squirming, not playing quietly and talking excessively are hyperactivity items. The onset age is now 12, not 7. ADHD is not a learning disability (dyslexia, dysgraphia and auditory processing disorder are), though the two often co-occur.

About 7 percent of children and 2.5 percent of adults have ADHD, boys about twice as often as girls; hyperactivity fades, but inattention and impulsivity often persist into adulthood. Heritability is about 74 percent. Dopamine and noradrenaline in frontostriatal circuits are implicated; risks include prenatal smoking and alcohol and low birth weight. **Russell Barkley**'s (born 1949) **behavioural inhibition model** (1997) makes poor response inhibition the core deficit. Stimulants such as methylphenidate are first-line drugs, with the non-stimulant atomoxetine as the main alternative (Section 7.10).

Specific learning disorder ●●

DSM-5 merged reading, mathematics and written expression disorders into one **specific learning disorder** (SLD): difficulty in reading, spelling, writing or mathematics persisting at least 6 months *despite targeted intervention*, achievement well below age expectation on standardised tests, onset in the school years, and no better explanation in intellectual disability, sensory loss or poor teaching. Intelligence is usually average (IQ above about 70 plus or minus 5), hence "unexpected underachievement". It affects 5 to 15 percent of schoolchildren.

The specifiers are impairment in reading (word-reading accuracy, fluency and comprehension), whose core pattern of poor word recognition, decoding and spelling is called **dyslexia**; in written expression, often called **dysgraphia**; and in mathematics, **dyscalculia**.

The Glasgow eye surgeon **James Hinshelwood** (1859 to 1919) described "congenital word-blindness" (a book of that title in 1917), and **Samuel Torrey Orton** (1879 to 1948) in the 1920s blamed incomplete cerebral dominance. The modern **phonological deficit theory** locates dyslexia in linking speech sounds (phonemes) to letters. The old **IQ-achievement discrepancy** model waited for a child to fail; **response to intervention** (RTI) identifies children who do not respond to good teaching. **Auditory processing disorder**, difficulty interpreting heard sounds despite normal hearing, is grouped with the learning difficulties though DSM-5-TR omits it. India's RPwD Act 2016 covers specific learning disabilities (Section 7.1).

> **! EXAM TRAP**
>
> SLD means roughly average intelligence with persistent academic difficulty; IQ below 50 points to intellectual disability, "only attention deficit" to ADHD and "motor coordination deficit only" to dyspraxia. Dyslexia is difficulty reading and decoding words, not understanding speech (language disorder) or concentrating in lectures (ADHD).

Motor disorders and tics ●●

Developmental coordination disorder (**dyspraxia**) is impaired acquisition and execution of coordinated motor skills: clumsiness and slowness in catching, using scissors, handwriting or cycling, without a neurological cause. It differs from syndromes acquired through brain damage (Chapter 5): **apraxia**, loss of learned purposeful movement despite intact

strength; **agnosia**, failure to recognise despite intact sensation; **anomia**, inability to name. **Stereotypic movement disorder** is repetitive, driven, purposeless movement (rocking, head-banging, self-biting), with or without self-injury.

A **tic** is a sudden, rapid, recurrent, non-rhythmic motor movement or vocalisation (blinking, shrugging, sniffing, throat-clearing). Seizures are abnormal electrical discharges that cannot be held back, dissociation disrupts consciousness or identity (Section 8.7), and Parkinson's tremor is rhythmic. Tics are often preceded by a **premonitory urge** and can be briefly suppressed; they begin at 4 to 6 years and peak at 10 to 12. All tic disorders need onset before 18 (Table 8.12). **Tourette's disorder** was described in 1885 by **Georges Gilles de la Tourette** (1857 to 1904); **coprolalia** (involuntary obscenities) occurs in a minority and is not required. Tics often accompany OCD and ADHD; habit reversal training treats them (Section 7.4).

Table 8.12 The DSM-5-TR tic disorders.

Disorder	Tics	Duration
Tourette's disorder	Multiple motor and one or more vocal, not necessarily together	More than 1 year
Persistent (chronic) motor or vocal tic disorder	Motor or vocal, not both	More than 1 year
Provisional tic disorder	Motor and/or vocal	Less than 1 year

! EXAM TRAP

Tourette's needs motor plus vocal tics for more than 1 year, onset before 18; coprolalia is not required. Developmental clumsiness is dyspraxia; apraxia, agnosia and anomia are acquired.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The IQ criterion for intellectual disability is about two _____ below the mean.
2. DSM-5 grades intellectual disability by _____ functioning.
3. Global developmental delay is for children under age _____.
4. In PKU, phenylalanine is not converted to _____.
5. Infantile autism was first described in 1943 by _____.
6. The Sally-Anne task tests understanding of a _____ belief.
7. Several ADHD symptoms must be present before age _____.
8. Tourette's disorder requires tics for more than _____.

Answers: (1) standard deviations, (2) adaptive, (3) 5, (4) tyrosine, (5) Leo Kanner, (6) false, (7) 12, (8) 1 year.

8.4 Schizophrenia Spectrum and Other Psychotic Disorders

HIGH YIELD · 79 QS

Psychosis means losing contact with shared reality: hearing voices no one else hears, or holding a belief no evidence can shake. Schizophrenia is the central psychotic disorder; its neighbours differ from it mainly in duration. We follow the concept's history, symptoms, criteria, causes and drug treatment.

From dementia praecox to schizophrenia •••

In the 1850s the French physician **Benedict Morel** (1809 to 1873) used *démence précoce*, early loss of mind, for young patients who deteriorated. **Ewald Hecker** described **hebephrenia** (silly, disorganised behaviour beginning in youth) in 1871, and **Karl Ludwig Kahlbaum** described **catatonia** (stupor, posturing and excitement) in 1874. **Emil Kraepelin** (1856 to 1926) united these pictures, with paranoid forms, under the Latin **dementia praecox**, 'premature dementia', in 1893. In the sixth edition of his textbook (1899) he separated it from **manic-depressive insanity** by course: dementia praecox began early and deteriorated, while manic-depressive insanity came in episodes with recovery between them. These two clinical types still divide Sections 8.4 and 8.5. Despite the name, dementia praecox is the ancestor of schizophrenia, not of Alzheimer's disease or frontotemporal dementia (Section 8.10).

The Swiss psychiatrist **Eugen Bleuler** (1857 to 1939) saw that not every patient deteriorated or fell ill young. In a 1908 lecture and his 1911 monograph, *Dementia Praecox or the Group of Schizophrenias*, he coined **schizophrenia** (Greek *schizein*, to split, and *phren*, mind) for a splitting between thought, emotion and behaviour, not a split personality. His fundamental symptoms, the **four As**, were present in every case:

- Loosening of **Associations**: thoughts lose their logical links.
- Blunted or incongruous **Affect**, such as smiling at sad news.
- **Ambivalence**: opposite feelings or wishes held at once towards the same object, a term Bleuler coined.
- **Autism**: withdrawal into an inner world (not the developmental disorder of Section 8.3).

Hallucinations and delusions were only **accessory symptoms**, because they occur in other illnesses too.

> ! EXAM TRAP
>
> Auditory hallucinations are **not** one of the four As. Ambivalence was coined by Bleuler, not Kraepelin, Freud or Fromm-Reichmann. Alogia, avolition and anhedonia are later negative-symptom As (8.4.2); apathy, anxiety, aphasia, amnesia, attention and anosognosia are not Bleuler's.

Bleuler's symptoms were hard to rate reliably. The German psychiatrist **Kurt Schneider** (1887 to 1967) listed instead, in *Clinical Psychopathology* (English edition 1959), **first-rank symptoms** (FRS) that, in the absence of brain disease, strongly suggest schizophrenia. Their common theme is a failure of the boundary between self and world:

- **Three kinds of voices**: audible thoughts (*Gedankenlautwerden*, thought echo), voices arguing or discussing the patient in the third person, and voices giving a running commen-

tary on the patient's actions.

- **Thought alienation**: thought insertion, thought withdrawal and thought broadcasting.
- **Passivity phenomena** (delusions of control): made feelings, made impulses, made volitional acts, and somatic passivity (bodily sensations imposed from outside).
- **Delusional perception**: a normal perception suddenly given a private delusional meaning (a traffic light turns red, and he 'knows' he has been chosen).

His **second-rank symptoms** were other hallucinations, perplexity, sudden delusional ideas, mood changes and emotional blunting. FRS are typical of schizophrenia but not specific: they also occur in mania, so DSM-5 gives them no special weight.

> ! EXAM TRAP
>
> Not first-rank: **depersonalisation**, **loose associations** (Bleulerian), **delusions of self-reference**, catatonic signs such as echolalia, negative symptoms, olfactory hallucinations and memory loss. Delusions of control **are** first-rank. FRS are Schneider's: **Karl Jaspers** (*General Psychopathology*, 1913) described primary delusions (Section 8.2), and Bateson and Wynne are family theorists (8.4.7).

Positive, negative and cognitive symptoms ●●●

Positive symptoms are the presence of something abnormal, an excess or distortion of normal function: delusions, hallucinations (commonest in schizophrenia, mostly voices, though mania and psychotic depression can bring them too), disorganised speech and grossly disorganised behaviour. **Negative symptoms** are the absence or reduction of something normal: **affective flattening** (less emotional expression in face, voice and gesture), **alogia** (poverty of speech), **avolition** (little self-started purposeful activity), **anhedonia** (reduced pleasure) and **asociality**. ICD-10 lists its negative symptoms as marked apathy, paucity of speech, and blunting or incongruity of emotional responses. A third, **cognitive** dimension (poor attention, working memory and executive function) best predicts work and social functioning.

Timothy Crow (1980) proposed **Type I schizophrenia** (positive symptoms, acute onset, good response to neuroleptics, normal brain structure, dopamine D2 overactivity) and **Type II** (negative symptoms, chronic course, poor drug response, enlarged ventricles, intellectual decline). **Peter Liddle** (1987) found three syndromes: **reality distortion** (delusions, hallucinations), **disorganisation** (thought disorder, inappropriate affect) and **psychomotor poverty** (poverty of speech, flat affect, slowed movement). Symptoms are rated with **Nancy Andreasen**'s SANS and SAPS (Scales for the Assessment of Negative and of Positive Symptoms) and the 30-item PANSS (Positive and Negative Syndrome Scale, 1987); see Chapter 2.

> ! EXAM TRAP
>
> Delusions, hallucinations and disorganised speech or behaviour are **positive**; flat affect is **negative**, never positive. Ambivalence is Bleulerian, not a negative symptom. Positive means the presence of something unusual, negative the absence of something normal, never the reverse.

Diagnostic criteria: DSM-5-TR, ICD-10 and ICD-11 ●●●

DSM-5-TR (2022) diagnoses **schizophrenia** when these are all met:

- **A.** Two or more of five symptoms, each present for a significant part of a 1-month period (less if successfully treated): delusions, hallucinations, disorganised speech, grossly disorganised or catatonic behaviour, and negative symptoms. At least one must be one of the first three.
- **B.** Functioning in work, relationships or self-care falls markedly below the earlier level.
- **C.** Continuous signs last at least 6 months, including at least 1 month of active-phase symptoms; the rest may be **prodromal** or **residual** phases, showing only negative or attenuated symptoms (odd beliefs, unusual perceptions).
- **D to F.** Schizoaffective and mood disorders, substances and medical conditions are excluded; after childhood autism, prominent delusions or hallucinations must last 1 month.

DSM-5 (2013) dropped the DSM-IV subtypes (paranoid, disorganised, catatonic, undifferentiated, residual) and the rule that one bizarre delusion or one Schneiderian voice was enough. ICD-10 (F20) needs one very clear symptom from groups (a) to (d), or symptoms from two of groups (e) to (h), present most of the time for **1 month** or more: (a) thought echo, insertion, withdrawal or broadcasting; (b) delusions of control or passivity, and delusional perception; (c) voices commenting on or discussing the patient; (d) persistent, culturally impossible delusions; (e) persistent hallucinations with delusions; (f) breaks in the train of thought, incoherence or neologisms; (g) catatonic behaviour; (h) negative symptoms. Its subtypes were paranoid (the commonest), hebephrenic, catatonic, undifferentiated, post-schizophrenic depression, residual and **simple schizophrenia** (insidious negative symptoms without overt psychosis). Table 8.13 compares the three systems.

Table 8.13 Schizophrenia in the three classification systems.

Feature	DSM-5-TR	ICD-10 (F20)	ICD-11 (6A20)
Minimum duration	6 months, including 1 month active phase	1 month	1 month
Symptoms needed	2 of 5, one of the first three	1 clear from (a) to (d), or 2 from (e) to (h)	2, one a core positive symptom
Functional decline	Required	Not required	Not required
Subtypes	Dropped	Seven, from paranoid to simple	Replaced by symptom and course specifiers

> **! EXAM TRAP**
>
> ICD-10 or ICD-11: **1 month**. DSM-5 or DSM-5-TR: **6 months**, including 1 month of active-phase symptoms. Catastrophic thinking is a cognitive distortion (Section 7.5), never a criterion of schizophrenia.

The spectrum: duration, delusions, mood and catatonia ●●●

Symptoms identical to those of schizophrenia can be short-lived, so DSM-5-TR separates the disorders largely by duration (Figure 8.5). **Brief psychotic disorder** lasts at least 1 day but less than 1 month, with full return to premorbid functioning; one symptom suffices if it is

delusions, hallucinations or disorganised speech. Specifiers: with marked stressors (brief reactive psychosis), without marked stressors, or with peripartum onset (in pregnancy or within 4 weeks after delivery). **Schizophreniform disorder** has the full Criterion A picture for at least 1 month but less than 6 months and needs no decline in functioning. Six months or more is schizophrenia.

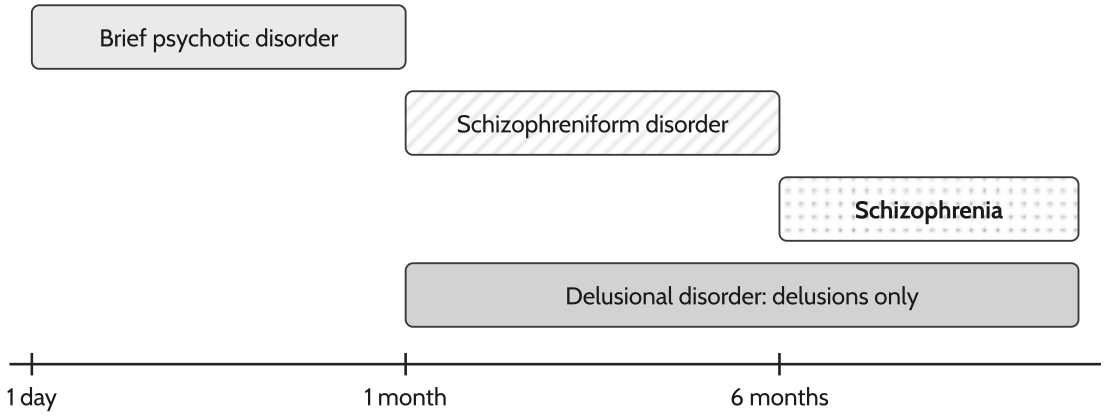


Figure 8.5 The duration ladder: 1 day, 1 month, 6 months. The same psychotic symptoms are named by how long they last; delusional disorder also starts at 1 month.

> **! EXAM TRAP**
>
> One keyed answer (IHBAS 2025) calls psychosis lasting 1 to 6 months 'brief psychotic disorder'. It is **schizophreniform disorder**: brief psychotic disorder ends within a month.

Delusional disorder, Kraepelin's *paranoia*, is one or more delusions lasting 1 month or more (ICD-10 and ICD-11: about 3 months) in someone who has never met Criterion A for schizophrenia; hallucinations are not prominent, and apart from the delusion functioning is not markedly impaired. Types follow the theme: **erotomanic** (another person, usually of higher status such as a film star, is secretly in love with the patient), **grandiose** (an unrecognised great talent or discovery), **jealous** (the partner is unfaithful), **persecutory** (being plotted against, spied on or poisoned; the commonest), **somatic** (bodily functions, such as emitting a foul odour), mixed and unspecified.

Schizoaffective disorder needs an uninterrupted period in which a major mood episode coincides with Criterion A of schizophrenia, plus delusions or hallucinations for **2 weeks or more** without a major mood episode, with mood episodes present for most of the illness; it has bipolar and depressive types. Psychotic depression and mania belong to Section 8.5. **Schizotypal disorder**, eccentricity below the psychotic threshold, is listed in this DSM-5-TR chapter but described in Section 8.11; ICD-10 placed it beside schizophrenia as F21, and ICD-11 keeps it there as 6A22. Psychosis caused directly by a drug, medication or toxin is **substance/medication-induced psychotic disorder**; psychosis caused by a medical illness is **psychotic disorder due to another medical condition**.

Catatonia requires three or more of twelve signs: stupor, catalepsy, waxy flexibility, mutism, negativism, posturing, mannerism, stereotypy, agitation, grimacing, echolalia and echopraxia. DSM-5 makes it not an independent disorder but a **specifier** across several disor-

ders (catatonia associated with schizophrenia, depression and others), plus catatonic disorder due to another medical condition and unspecified catatonia. **Attenuated psychosis syndrome** (Section III, for further study) means weaker psychotic symptoms with insight retained, at least weekly for a month. ICD-10 **acute and transient psychotic disorders** (F23) begin within 2 weeks and usually resolve within 2 to 3 months; ICD-11 6A23 keeps the 2-week onset, with rapidly changing symptoms lasting at most 3 months. They are common in India and recover well.

Folie à deux (**Charles Lasègue** and **Jules Falret**, 1877) is a delusion transmitted from a dominant **primary** case with an established psychotic disorder to a more passive, often socially isolated **secondary** partner in a close relationship, usually a relative. Separation often lets the secondary partner's belief fade. In *folie imposée* the delusion is imposed on a healthy partner; in *folie simultanée* two predisposed people fall ill together; *folie à trois* involves three. ICD-10 coded it **F24 induced delusional disorder**, DSM-IV **shared psychotic disorder**. DSM-5 dropped the category: the partner receives delusional disorder if its criteria are met, otherwise other specified schizophrenia spectrum disorder ('delusional symptoms in the context of relationship with an individual with prominent delusions'). ICD-11 has no separate code either: it describes 'shared or induced delusional disorder' (folie à deux) as a rare form of **6A24 delusional disorder**, in its block 'schizophrenia or other primary psychotic disorders'.

> **! EXAM TRAP**
>
> A belief shared by a whole cultural or religious group is not folie à deux; nor are two independent psychotic episodes, or a stress psychosis in two relatives. ICD-10 F24 is induced delusional disorder; F21 is schizotypal disorder. ICD-11 puts it under neither schizoaffective nor schizotypal disorder: it is a form of delusional disorder among the other primary psychotic disorders.

Epidemiology, course and prognosis ○

The lifetime risk is usually quoted as about **1 percent**; DSM-5-TR gives 0.3 to 0.7 percent. Schizophrenia is above all a disorder of **young adults**: psychosis emerges between the late teens and the mid-30s, peaking in the early to mid-20s in men and the late 20s in women, with a second peak in women in midlife, around the menopause. Prevalence is about equal in the two sexes, but men have more negative symptoms and worse outcomes. A prodrome of withdrawal and odd ideas precedes the active phase. In the WHO **International Pilot Study of Schizophrenia** (1973, nine countries including India) and the later Determinants of Outcome study, patients in developing countries had a better course, often linked to family support. Schizophrenia is commoner in the lowest social classes: **social causation** says poverty and stress produce it; **social drift** says the illness pushes people down the social ladder. Urban birth, migration and heavy adolescent cannabis use raise the risk. Table 8.14 lists the classic prognostic factors.

Table 8.14 Prognostic factors in schizophrenia.

Good prognosis	Poor prognosis
Acute onset	Insidious onset
Late onset	Early onset
Clear precipitating stressor	No precipitant

Good prognosis	Poor prognosis
Mood symptoms; mainly positive symptoms	Prominent negative symptoms
Good premorbid functioning	Poor premorbid adjustment
Female, married, good support	Male, single, withdrawn
Family history of mood disorder	Family history of schizophrenia

Biological causes: genes, dopamine and development ●●●

Irving Gottesman (1991) pooled about 40 family and twin studies (Figure 8.6): risk climbs with shared genes, to about 48 percent, roughly half, for an identical twin. Because it is not 100 percent, environment matters too. The **Genain quadruplets**, identical sisters who all developed schizophrenia but with different onset, severity and outcome, were studied by **David Rosenthal** (1963). Adoption studies separate genes from upbringing: **Leonard Heston** (1966) found schizophrenia in children of mothers with schizophrenia adopted away at birth but in none of the control adoptees, and the Danish adoption studies of **Seymour Kety**, Rosenthal and **Paul Wender** (from 1968) found it concentrated in biological, not adoptive, relatives. Risk is **polygenic**: many common variants of small effect, partly shared with bipolar disorder and autism.

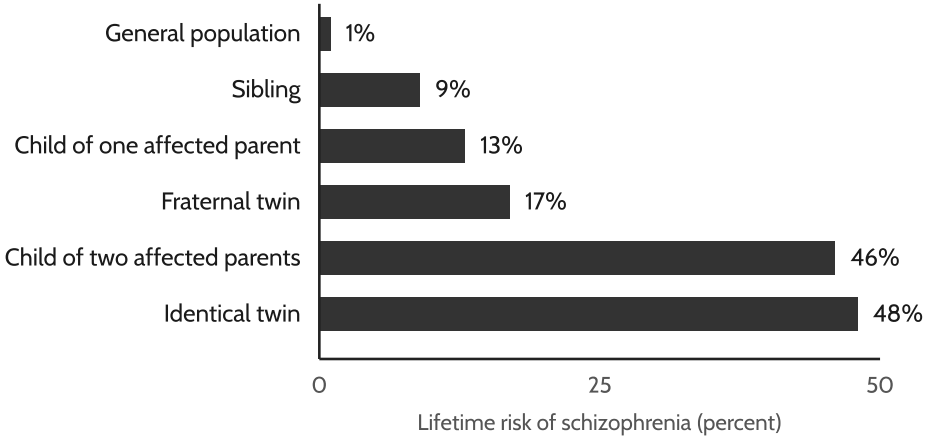


Figure 8.6 Gottesman's (1991) risk figures. Risk rises with the genes shared with an affected relative, but even an identical co-twin falls ill only about half the time.

The **dopamine hypothesis** holds that positive symptoms reflect excess dopamine activity at D2 receptors in the **mesolimbic** pathway. **Arvid Carlsson** showed that dopamine is a neurotransmitter and that antipsychotics act on its system (Nobel Prize, 2000); amphetamine, which releases dopamine, can cause a paranoid psychosis, and L-dopa for Parkinson's disease can cause hallucinations (it does not relieve negative symptoms). The original support came in 1976, when **Philip Seeman** and colleagues, and Ian Creese, David Burt and **Solomon Snyder**, showed that the clinical potency of typical antipsychotics correlates with their affinity for D2 receptors. Dopamine runs in four pathways, so a drug that blocks D2 everywhere both helps and harms (Table 8.15).

Table 8.15 The four dopamine pathways.

Pathway	Route	Role in schizophrenia and treatment

Pathway	Route	Role in schizophrenia and treatment
Mesolimbic	Ventral tegmental area (VTA) to nucleus accumbens (ventral striatum) and limbic areas; reward	Overactive: positive symptoms; D2 blockade treats them
Mesocortical	VTA to prefrontal cortex	Underactive: negative and cognitive symptoms
Nigrostriatal	Substantia nigra to caudate and putamen; voluntary movement	D2 blockade: extrapyramidal symptoms, parkinsonian rigidity
Tuberoinfundibular	Hypothalamus to median eminence and pituitary	D2 blockade: raised prolactin

Kenneth Davis and colleagues (1991) revised the hypothesis: too much subcortical dopamine for positive symptoms, too little prefrontal dopamine for negative ones. **Oliver Howes** and **Shitij Kapur** (2009) located the core fault in increased presynaptic dopamine synthesis in the striatum, a final common pathway to psychosis. Phencyclidine and ketamine block **NMDA glutamate receptors** and produce positive and negative symptoms, suggesting NMDA receptor hypofunction; atypical antipsychotics add blockade of **5-HT2A serotonin** receptors.

The **neurodevelopmental hypothesis** (**Daniel Weinberger**, 1987; **Robin Murray** and Shon Lewis, 1987) proposes that an early brain lesion stays silent until adolescent maturation unmasks it. The evidence: enlarged **lateral ventricles**, first shown on CT by **Eve Johnstone** and colleagues (1976); reduced grey matter; **hypofrontality**, low prefrontal activity during tasks; obstetric complications, prenatal influenza and winter-spring birth; and, later, adolescent cannabis use.

Family and psychological theories ●●●

Early family theories blamed parents. **Frieda Fromm-Reichmann** (1948) described the cold, dominating **schizophrenogenic mother**; the idea is discredited and caused families needless guilt. **Gregory Bateson**, with Don Jackson, Jay Haley and John Weakland (1956), proposed the **double bind**: a person repeatedly receives contradictory messages at two levels from a significant other, and can neither comment on the contradiction nor escape.

KEY STUDY: BATESON'S DOUBLE BIND (1956)

A young man, fairly well recovered from an acute schizophrenic episode, put his arm round his visiting mother; she stiffened. He withdrew, and she asked, 'Don't you love me any more?' He blushed, and she said, 'Dear, you must not be so easily embarrassed and afraid of your feelings.' After she left he assaulted an aide. Bateson called this the **symptomatic double bind**: every response was wrong. No controlled study has shown that double binds cause schizophrenia. The group's cybernetic and homeostatic views of the family belong to family therapy (Section 7.7).

Theodore Lidz (1957) described **marital schism** (parents in open conflict, undermining each other) and **marital skew** (one dominant, disturbed parent whose distortions the other accepts). **Lyman Wynne** (1958) described **pseudomutuality**, a façade of harmony that forbids individuality, and later **communication deviance**, vague and fragmented parental speech. **Ronald David Laing** led **anti-psychiatry**, which saw schizophrenia as an understandable response to an unliveable family situation.

Where these theories sought causes and failed, **expressed emotion** (EE) predicts relapse. EE is the emotional climate of a key relative towards the patient, rated from the **Camberwell Family Interview** on three dimensions: **critical comments**, **hostility** and **emotional over-involvement**. Warmth and positive remarks are also rated, but they are not high-EE dimensions.

KEY STUDY: EXPRESSED EMOTION AND RELAPSE (1972, 1976)

George Brown, James Birley and **John Wing** (1972) followed patients discharged to their families: relapse was about four times commoner in high-EE homes. **Christine Vaughn** and **Julian Leff** (1976) replicated this and found relapse highest when a patient spent more than about 35 hours a week face to face with a high-EE relative and took no medication. Criticism: the data are correlational; a more disturbed patient may provoke more criticism. In WHO studies in Chandigarh, Indian relatives were far less often high in EE than London relatives, one suggested reason for better outcome in India.

High EE is managed by family psychoeducation and by reducing face-to-face contact hours. Table 8.16 summarises the psychodynamic theories; the vulnerability-stress model that integrates these causes is in Section 8.1.

! EXAM TRAP

High EE predicts **relapse** of psychotic illness, not its onset, negative symptoms or violence; relapse is more precise than a vague 'poor treatment outcome'. It means critical, hostile or over-involved relatives, not overwhelmingly positive comments, a family without emotion, or a patient who cannot talk about feelings.

Table 8.16 Psychodynamic theories of schizophrenia.

Theorist	Core idea
Sigmund Freud	Libido withdrawn from the world into the self, with regression; symptoms try to restore contact
Paul Federn	Weak ego boundaries: inner and outer reality blur
Harry Stack Sullivan	Interpersonal: overwhelming anxiety in early relationships
Silvano Arieti	Palaeologic thinking under the von Domarus principle: identity from shared predicates ('the Virgin Mary was a virgin; I am a virgin; so I am the Virgin Mary')
Margaret Mahler	Fixation at the autistic phase, with failure to form an adequate and stable symbiosis with the mother (symbiotic psychosis)
Ping-Nie Pao	Symptoms as the best possible adaptation to overwhelming 'organismic panic'
Edith Jacobson	Psychotic regression fuses self and object representations

Antipsychotic drugs and their side effects ●●●

Antipsychotic medication (the **neuroleptics**) is the front-line treatment for schizophrenia; CBT, social skills training, family work and stress management are added to it, not used instead (Sections 7.4, 7.5, 7.7 and 7.9). Neuroleptics are used chiefly for schizophrenia (and in mania), not for depression or anxiety. The era began in 1952, when **Jean Delay** and **Pierre Deniker** in Paris gave **chlorpromazine** to psychotic patients. **Typical antipsychotics** (first-

generation), such as chlorpromazine and haloperidol, act mainly by blocking dopamine **D2** receptors (not D1); chlorpromazine also blocks muscarinic (M1) receptors, causing dry mouth and constipation. **Atypical antipsychotics** (second-generation), such as risperidone, olanzapine and **clozapine**, block 5-HT2A as well as D2 and cause fewer movement effects; clozapine is reserved for treatment-resistant schizophrenia.

Because a typical drug blocks D2 in the nigrostriatal pathway too, it produces **extrapyramidal symptoms** (EPS), in a fixed order:

- **Acute dystonia**, within hours to days: sustained muscle spasms, a twisted neck, eyes forced upwards.
- **Akathisia**, within days to weeks: a subjective inner restlessness with a compelling need to pace or move.
- **Drug-induced parkinsonism**, within weeks: tremor, rigidity and slowed movement.
- **Tardive dyskinesia**, after months to years: involuntary orofacial movements such as lip-smacking, tongue movements and grimacing; the last to appear and the most likely to be irreversible.

Other dangers are **neuroleptic malignant syndrome**, a rare emergency of high fever, 'lead-pipe' rigidity, unstable pulse and blood pressure and clouded consciousness; **hyperprolactinaemia** from tuberoinfundibular blockade (breast enlargement, milk secretion, missed periods); **metabolic syndrome** (weight gain, diabetes, raised lipids) with atypicals, especially clozapine and olanzapine; and **agranulocytosis** with clozapine, which demands regular blood counts. DSM-5-TR lists these reactions as medication-induced movement disorders and other adverse effects of medication.

> **! EXAM TRAP**
>
> Akathisia and tardive dyskinesia are **secondary** (drug-induced) movement disorders; Huntington's chorea (inherited) and Parkinson's disease (nigral degeneration) are **primary**. Tardive dyskinesia follows neuroleptics, not antidepressants, anxiolytics or barbiturates. Typicals cause EPS by blocking nigrostriatal D2 receptors, not by blocking serotonin reuptake (SSRIs), stimulating GABA-A receptors (benzodiazepines) or blocking mesolimbic receptors selectively.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Kraepelin separated dementia praecox from _____ insanity by its deteriorating course.
2. Bleuler's four As are associations, affect, ambivalence and _____.
3. Thought echo, voices giving a running commentary and delusional perception are Kurt Schneider's _____ symptoms.
4. Psychosis lasting 3 months with no decline in functioning is _____ disorder.
5. ICD-10 and ICD-11 need symptoms for _____; DSM-5-TR needs 6 months.
6. High expressed emotion consists of critical comments, hostility and emotional _____.
7. Involuntary lip-smacking after years on haloperidol is _____.

Answers: (1) manic-depressive, (2) autism, (3) first-rank, (4) schizophreniform, (5) 1 month, (6) over-involvement, (7) tardive dyskinesia.

8.5 Depressive and Bipolar Disorders, and Suicide HIGH YIELD · 49 QS

Sadness after a loss and elation after a win are normal. A **mood disorder** (affective disorder) is a low or high mood, with changed sleep, energy, thinking and behaviour, that takes over a life for weeks or years. Such disorders are built from three kinds of episode, so the episodes come first.

Mood episodes: the building blocks ●●●

A disorder with depressive episodes only is **unipolar**; one in which mood swings between the extremes, mania or hypomania as well as depression, is **bipolar**. DSM-5 (2013) split DSM-IV's "mood disorders" into two chapters, **bipolar and related disorders** and **depressive disorders**, placing bipolar disorders between the psychotic and depressive chapters as a bridge. Table 8.17 defines the three episodes.

Table 8.17 The three mood episodes (DSM-5-TR).

Feature	Major depressive episode	Manic episode	Hypomanic episode
Core mood	Depressed mood or loss of interest or pleasure (one required)	Elevated (often euphoric), expansive or irritable mood, plus raised energy	As in mania
Duration	2 weeks, most of the day, nearly every day	1 week, or any duration if hospitalised	4 consecutive days
Symptoms	5 or more of 9	3 or more of 7; 4 if mood is only irritable	As in mania
Severity	Distress or impairment	Marked impairment, hospitalisation or psychosis	Observable change; no marked impairment, hospitalisation or psychosis

The nine depressive symptoms are depressed mood plus SIG E CAPS: **S**leep change, loss of **I**nterest or pleasure (**anhedonia**), **G**uilt or worthlessness, low **E**nergy, poor **C**oncentration,

Appetite or weight change, **P**sychomotor agitation or retardation, and **S**uicidal thoughts. The seven manic symptoms are DIG FAST: **D**istractibility, **I**ndiscretion (spending sprees and other risky pleasures), **G**randiosity, **F**light of ideas, increased goal-directed **A**ctivity, decreased need for **S**leep and **T**alkativeness or pressured speech. Extremely elevated mood and grandiose ideas are the two hallmarks of mania.

Mania and hypomania differ in degree, not kind: hypomania is shorter, and though others notice it, it never causes marked impairment or hospitalisation; psychotic features make an episode manic by definition. DSM-5 also dropped the **bereavement exclusion**, which had barred a diagnosis of depression in the 2 months after a death, because grief-triggered depression proved to be like any other.

> ! EXAM TRAP
>
> Specific phobias and euphoric mood are not symptoms of a major depressive episode; anhedonia and appetite change are. Social phobia and fear of weight gain (anorexia, Section 8.8) are not manic symptoms, and hallucinations, though possible in severe mania, are not among its seven. Mania lasts longer than hypomania and need not follow depression; hypomania must be observable by others.

Depressive disorders ●●●

Major depressive disorder (MDD), one or more major depressive episodes with no mania or hypomania, is marked by persistent sadness and loss of interest; it may be a single episode or recurrent. Called the **common cold of psychiatry**, it affects about 16 percent of people in a lifetime (estimates run 15 to 20 percent), and rates are rising, fastest among the young. Onset grows likelier at puberty, after which women are affected about twice as often as men (in children the ratio is about equal). Episodes are often triggered by a stressful event, and the more severe the depression, the higher the suicide risk.

> ! EXAM TRAP
>
> Asked which statement least accurately describes depression, pick "men are more vulnerable than women": the reverse is true. Widespread and rising incidence, stress triggers and the severity-suicide link are all accurate.

Persistent depressive disorder (PDD), formerly dysthymia, merges dysthymic disorder and chronic major depression. Mood is depressed most of the day, more days than not, for at least **2 years in adults or 1 year in children and adolescents**, with two or more of six symptoms: appetite change, sleep change, low energy, low self-esteem, poor concentration and hopelessness. The person is never symptom-free for more than 2 months and has never had mania or hypomania. Dysthymia is persistent mild depression, **less severe than MDD but longer**. A major depressive episode on top of it is **double depression**.

Disruptive mood dysregulation disorder (DMDD), new in DSM-5, means severe temper outbursts grossly out of proportion to the provocation, three or more times a week, with persistently irritable mood between them, for 12 months, in two of three settings (home, school, peers). Onset is before age 10 and diagnosis between 6 and 18. It was added to curb the overdiagnosis of bipolar disorder in children, who in fact grow up to have depression and anxiety.

> **! EXAM TRAP**
>
> A 12-year-old whose temper outbursts have just begun cannot have DMDD, because onset must come before age 10. Bipolar disorder, conduct disorder and ADHD (Sections 8.3 and 8.11) all remain possible.

Premenstrual dysphoric disorder (PMDD) needs five or more symptoms, at least one of them mood swings, irritability, depressed mood or anxiety, in the week before menses in most cycles, easing within a few days after menses begin. Depression caused by a drug, or by an illness such as hypothyroidism, is substance-induced or due to another medical condition (Section 8.10).

The "hybrid" disorder is ICD-10's **mixed anxiety and depressive disorder** (F41.2): anxiety and depression both present, neither enough for its own diagnosis. ICD-11 keeps it among the depressive disorders; DSM-5 does not. (Multiple personality, borderline and schizoid disorders belong to Sections 8.7 and 8.11.) **René Spitz** (1887 to 1974) described **anaclitic depression** in 1946: infants separated from their mothers in the second half of the first year grew weepy, then withdrawn, lost weight and stopped developing, recovering if the mother soon returned. His term **hospitalism** named the graver, sometimes fatal, decline of infants reared in impersonal institutions. (Piaget, Skinner and Eysenck are known for cognitive stages, operant conditioning and personality dimensions.) **Masked depression** presents as bodily complaints while sadness is denied, common in Indian primary care; depressive pseudodementia is covered in Section 8.10.

Specifiers ●●

A specifier adds detail to a diagnosis without making a new disorder (Table 8.18).

Table 8.18 Specifiers for depressive and bipolar disorders.

Specifier	Defining features	Applies to
With anxious distress	2 or more of: keyed up, restless, worried, fearing disaster or loss of control	Depressive and bipolar
With mixed features	3 or more symptoms of the opposite pole	Manic, hypomanic or depressive episode
With rapid cycling	4 or more mood episodes in 12 months	Bipolar I and II only
With melancholic features	No pleasure or mood reactivity; worse in the morning, early waking, psychomotor change, weight loss, guilt	Depressive episode
With atypical features	Mood reactivity; weight gain, hypersomnia, leaden paralysis, rejection sensitivity	Depressive episode
With psychotic features	Delusions or hallucinations, mood-congruent or incongruent	Depressive or manic episode
With catatonia	Catatonic features (Section 8.4)	Depressive or manic episode
With peripartum onset	Onset in pregnancy or within 4 weeks of delivery	Any episode
With seasonal pattern	Onset and remission at a set time of year, for 2 years	Recurrent MDD; bipolar I and II

A manic episode **with mixed features** has concurrent depressive symptoms, and a depressive one concurrent manic symptoms; the specifier replaced DSM-IV's mixed episode. In mood disorders, psychosis occurs only during episodes and is usually **mood-congruent**: guilt, disease or deserved punishment in depression, grandiosity in mania. Mood congruence, with this timing, best separates a mood disorder with psychotic features from a primary psychotic disorder (Section 8.4); delusions as such, episode length and anxiety comorbidity do not.

After childbirth, postpartum blues (tearful, labile mood in the first days, gone within a week) is common and not a disorder; postpartum depression is a major depressive episode with peripartum onset; postpartum psychosis (1 to 2 per 1,000 births, often bipolar) risks infanticide and is an emergency. **Norman Rosenthal** and colleagues described **seasonal affective disorder** in 1984: depression each autumn or winter, lifting in spring, with oversleeping and carbohydrate craving, commoner at high latitudes. It responds to bright light therapy.

> **! EXAM TRAP**
>
> Seasonal affective disorder is the "with seasonal pattern" specifier of recurrent major depressive or bipolar disorder: not a separate, anxiety or personality disorder, and not confined to women. "With limited prosocial emotions" belongs to conduct disorder (Section 8.11), not mania.

Bipolar and related disorders ●●●

The two-pole idea came from 1850s France: **Jean-Pierre Falret** (1794 to 1870) described *folie circulaire* and **Jules Baillarger** (1809 to 1890) *folie à double forme*, mania and melancholia alternating in one patient (Falret's son Jules gave us folie à deux). **Emil Kraepelin** (1856 to 1926) united recurrent mood illness as **manic-depressive insanity** in 1899, apart from dementia praecox, and **Karl Leonhard** split unipolar from bipolar forms in 1957.

Bipolar I disorder needs at least one manic episode; major depressive and hypomanic episodes are common but not required, and subthreshold depressive periods are allowed. **Bipolar II disorder** needs at least one hypomanic and one major depressive episode and never a full mania, which would make it bipolar I. Bipolar II is not milder, since its depressions are frequent and disabling, and it needs no psychotic features. **Cyclothymic disorder** is the chronic low-grade form: for at least 2 years (1 in the young), numerous hypomanic and depressive symptoms that never meet episode criteria, never absent for more than 2 months. Swinging between the poles is what makes a disorder bipolar (Figure 8.7). Bipolar disorder affects about 1 percent of people, men and women equally, usually beginning around age 20.

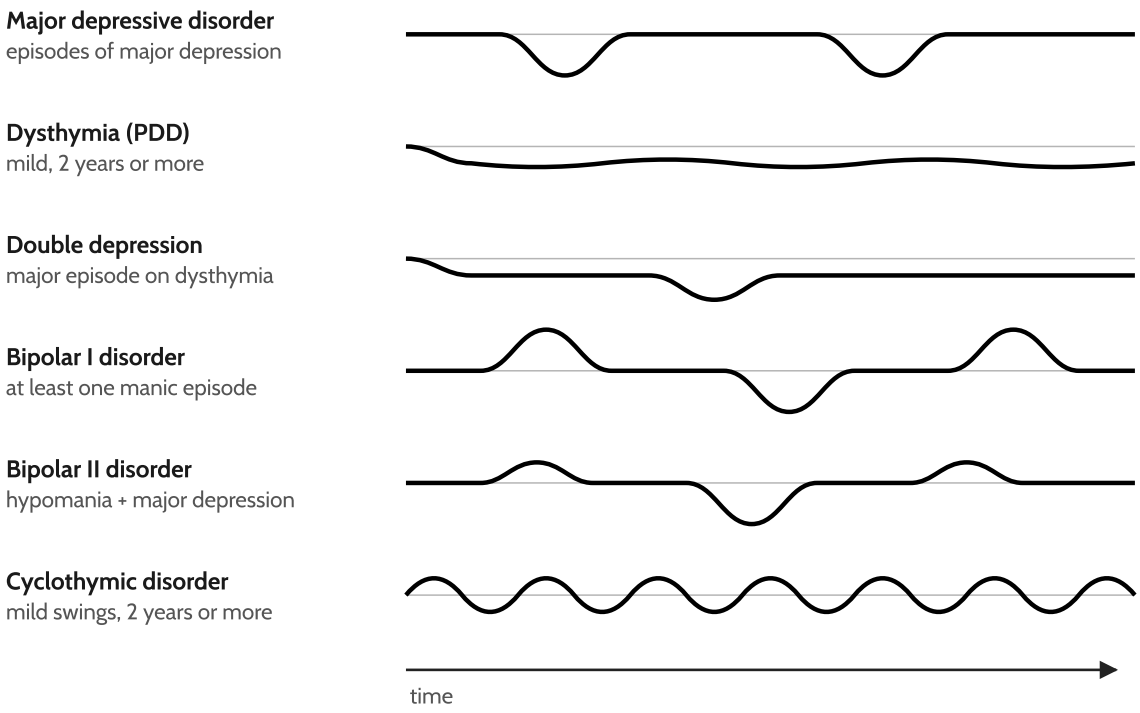


Figure 8.7 Mood over time. The grey line is normal mood: tall peaks are mania, short peaks hypomania, deep troughs major depression and shallow troughs milder depression.

> **! EXAM TRAP**
>
> Bipolar I is defined by mania, so "bipolar I includes hypomania" is the incorrect statement: hypomania defines bipolar II and is optional in bipolar I, which may also include depressive periods, even subthreshold ones. Rapid cycling and psychotic depressions occur in both, and bipolar II has no manic episodes at all, only shorter hypomanias; a full manic episode is what separates I from II.

ICD-10 and ICD-11 compared

ICD-10 codes mood (affective) disorders at F30 to F39 (Table 8.19). Its depressive episode lasts 2 weeks and is graded mild, moderate or severe by how many symptoms are present, three of them typical (depressed mood, loss of interest, reduced energy). Mild and moderate episodes are coded with or without the **somatic syndrome**, ICD-10's counterpart of melancholic features. ICD-11 (in use from 2022) keeps "mood disorders" as an umbrella over bipolar and depressive groupings, retains the mixed episode that DSM-5 dropped, and lets one manic or mixed episode establish bipolar type I.

Table 8.19 Mood disorders in ICD-10, ICD-11 and DSM-5-TR.

ICD-10	ICD-11	DSM-5-TR
F30 Manic episode (including hypomania)	Manic, hypomanic, mixed and depressive episodes	Manic and hypomanic episodes
F31 Bipolar affective disorder: at least two episodes, one elevated	Bipolar type I, type II	Bipolar I, bipolar II
F32 Depressive episode	Single episode depressive disorder	MDD, single episode

ICD-10	ICD-11	DSM-5-TR
F33 Recurrent depressive disorder	Recurrent depressive disorder	MDD, recurrent
F34 Persistent: F34.0 cyclothymia, F34.1 dysthymia	Cyclothymic disorder; dysthymic disorder	Cyclothymic disorder; PDD
F41.2 Mixed anxiety and depressive disorder	Mixed depressive and anxiety disorder	None

Causes ●●●

Mood disorders run in families: heritability is about 40 percent for MDD and up to about 90 percent for bipolar disorder in twin studies, among the highest in psychiatry, though identical-twin concordance of 40 to 70 percent leaves room for the environment. **Joseph Schildkraut**'s **catecholamine hypothesis** (1965) linked depression to a deficit of noradrenaline at brain synapses and mania to an excess; the serotonin **permissive hypothesis** adds that low serotonin lets other transmitters swing towards either pole. Both are now thought too simple. Depression also brings overactivity of the hypothalamic-pituitary-adrenal (HPA) axis with raised cortisol: about half of patients fail to suppress cortisol on the **dexamethasone suppression test**, though the finding is too unspecific for diagnosis. Sleep shows **shortened REM latency**, dream sleep arriving sooner. **Robert Post**'s **kindling hypothesis** (1992) holds that each episode sensitises the brain and lowers the threshold for the next, so early episodes follow major stress and later ones need little or none.

Table 8.20 summarises the psychological and social theories.

Table 8.20 Psychological and social theories of depression.

Approach	Theorist and year	Key idea
Psychodynamic	**Sigmund Freud**, *Mourning and Melancholia* (1917); **Karl Abraham** (1911)	Loss of a loved object; anger at it turned inward as self-reproach
Ego psychology	**Edward Bibring** (1953)	The ego's helplessness to meet its aspirations; loss of self-esteem
Behavioural	**Peter Lewinsohn**	Low rate of response-contingent positive reinforcement
Learned helplessness	**Martin Seligman** (1975)	Uncontrollable bad events teach passivity (dogs given inescapable shock later stopped trying to escape)
Attributional	**Lyn Abramson**, Seligman and **John Teasdale** (1978)	Bad events blamed on internal, stable, global causes ("I am stupid")
Hopelessness	Abramson, **Gerald Metalsky** and **Lauren Alloy** (1989)	Expecting bad outcomes that nothing can change
Cognitive	Aaron Beck (1967)	Negative cognitive triad, schemas, automatic thoughts
Response styles	**Susan Nolen-Hoeksema** (1991)	Rumination (dwelling on one's distress) prolongs depression; women ruminate more, part of the gender gap
Social	George Brown and Tirril Harris (1978)	Vulnerability factors plus provoking life events

The most examined theory is that of **Aaron Temple Beck** (1921 to 2021), who argued in 1967 that depression is maintained by the **negative cognitive triad**: negative views of the **self** ("I am worthless"), the **world** or present experience ("nothing goes right") and the **future** ("it will never improve"). Underlying negative schemas, early core beliefs such as "I am unlovable", are activated by stress and generate quick, unbidden **automatic thoughts**, kept

believable by cognitive distortions such as overgeneralisation and personalisation; the conditional rules linking them ("If I am not liked by all, I am worthless") are dysfunctional assumptions. Cognitive therapy targets all of these (Section 7.5).

> **! EXAM TRAP**
>
> Beck's triad is self, world and future, not id-ego-superego, past-present-future, mother-father-self, self-others-environment, thoughts-feelings-behaviours, body-mind-spirit or mood-behaviour-somatic state. Automatic thoughts, distortions and dysfunctional assumptions are other parts of the model, not the triad. Repressed sexual conflict, too little reinforcement (Lewinsohn) and insecure attachment belong to other theories.

KEY STUDY: SOCIAL ORIGINS OF DEPRESSION (BROWN AND HARRIS, 1978)

George Brown and **Tirril Harris** interviewed women in Camberwell, south London. Severe losses and disappointments provoked depression mainly in women with **vulnerability factors**: no confiding relationship, three or more children under 14 at home, loss of the mother before age 11, and no job outside the home. Criticism: one city, women only, and life events recalled after the fact.

First-line drug treatment ●●

For depression, selective serotonin reuptake inhibitors (SSRIs) such as fluoxetine are first line, safer in overdose than **tricyclics** (imipramine) and **monoamine oxidase inhibitors** (MAOIs, which need a low-tyramine diet). For bipolar disorder the gold standard is **lithium** (**John Cade**, 1949), used for acute mania and for **prophylaxis** (preventing further episodes), and it lowers suicide risk. Its therapeutic range is narrow, so blood levels are monitored; toxicity brings coarse tremor, vomiting, unsteadiness and confusion. Valproate, carbamazepine and lamotrigine are alternatives, and atypical antipsychotics (olanzapine, risperidone) are first line for acute mania; haloperidol and lorazepam are adjuncts for agitation only. Electroconvulsive therapy and the psychotherapies are in Chapter 7.

> **! EXAM TRAP**
>
> Fluoxetine is not a first-line treatment for acute mania: an antidepressant given alone in bipolar disorder can trigger a switch into mania or rapid cycling. Lithium, valproate and atypical antipsychotics are first line.

Suicide ●●

Suicidal ideation is thinking of killing oneself; a plan fixes method, place and time; intent is how much the person wants to die; an attempt is a self-injurious act with some intent to die that is survived; completed suicide ends in death. **Lethality** is the medical danger of the method, so the mode of an attempt (hanging or pesticide against a few tablets) shows its lethality, while intent is inferred from planning and precautions against rescue. **Norman Kreitman** coined **parasuicide** (1969) for non-fatal deliberate self-injury or overdose whatever its intent, later called deliberate self-harm. **Non-suicidal self-injury** (cutting or burning without intent to die) is a DSM-5 condition for further study.

The World Health Organization counts over 700,000 suicides a year. India's National Crime Records Bureau recorded about 1.71 lakh in 2022, about seven in ten of them men and most aged 18 to 45, with family problems (about a third) and illness the leading recorded causes. Women attempt suicide more often, but men die by it more often, using more lethal methods: the gender paradox.

Émile Durkheim (1858 to 1917), in *Suicide* (1897), traced suicide to social ties: **egoistic** (too little integration, the isolated person), **altruistic** (too much, dying for the group as in harakiri), **anomic** (too little regulation, sudden upheaval such as losing a fortune) and **fatalistic** (too much regulation, as for slaves or prisoners). **Edwin Shneidman** (1918 to 2009), founder of suicidology, saw suicide as escape from **psychache**, unbearable psychological pain. In **Thomas Joiner**'s interpersonal theory (2005) the wish to die comes from **perceived burdensomeness** and **thwarted belongingness**, but lethal attempts also need an **acquired capability**, fearlessness of pain and death built by exposure such as self-harm. **Marie Åsberg** (1976) found that depressed patients with low cerebrospinal fluid **5-HIAA**, the serotonin metabolite, made more frequent and more violent suicide attempts.

The strongest single predictor of suicide is a **previous attempt**. Other risk factors are depression and other mental disorders, alcohol, **hopelessness** (Beck's Hopelessness Scale predicts better than depression severity), male sex, older age, living alone, chronic illness, a family history of suicide and access to means such as pesticides. Social support, children at home, religious objection and restricted means protect. Risk is highest when high lethality, a specific plan, frequent ideation and family history converge. SAD PERSONS (Patterson and colleagues, 1983) is in Section 7.2, and the decriminalisation of attempted suicide by the Mental Healthcare Act 2017 in Section 7.1.

> **! EXAM TRAP**
>
> The mode or method of an attempt indicates **lethality**; intentionality is the wish to die, and conviction and survival are not standard dimensions. A recent attempt of low lethality with no family history carries less risk than high lethality, a specific plan, frequent ideation and family history together.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A major depressive episode needs five of nine symptoms in the same _____ period.
2. Persistent depressive disorder requires depressed mood for at least _____ years in adults.
3. Disruptive mood dysregulation disorder requires onset before age _____.
4. Four or more mood episodes in 12 months earn the specifier "with _____".
5. Beck's cognitive triad is a negative view of the self, the world and the _____.
6. The kindling hypothesis of mood disorders was proposed by _____.
7. The strongest single predictor of suicide is a _____.

Answers: (1) 2-week, (2) 2, (3) 10, (4) rapid cycling, (5) future, (6) Robert Post, (7) previous attempt.

8.6 Anxiety and Obsessive-Compulsive and Related Disorders

MEDIUM YIELD · 32 QS

A racing heart before an exam is anxiety doing its job. A man who has avoided buses for three years since he panicked on one, or a woman who washes for three hours a day to cancel a thought of germs, has crossed into disorder. This section sets out the anxiety disorders and why they develop, then obsessive-compulsive disorder and its relatives.

Fear, anxiety and the old neuroses

Fear is the alarm response to danger that is present now: a surge of autonomic arousal and the urge to escape. **Anxiety** is apprehension about a future threat, felt as muscle tension, vigilance and cautious avoidance. Both are normal, and moderate anxiety even helps performance (the Yerkes-Dodson law, Section 1.10). Anxiety becomes pathological when it is out of proportion to the danger, developmentally inappropriate, persistent (typically six months or more) and impairing.

The Scottish physician **William Cullen** coined the word neurosis in 1769 for disorders of sense and motion arising from the nervous system without fever or local disease. **Sigmund Freud** (1856 to 1939) narrowed **neurosis** to a non-psychotic disorder of unconscious conflict, with insight preserved, and in 1895 described an "anxiety neurosis". DSM-III (1980) dropped the term because it implied Freud's theory. DSM-5 (2013) split the old anxiety group into three chapters: anxiety disorders, obsessive-compulsive and related disorders, and trauma- and stressor-related disorders (Section 8.7). ICD-10 kept the umbrella block F40 to F48, "neurotic, stress-related and somatoform disorders", running from phobic anxiety (F40), other anxiety (F41) and OCD (F42) to somatoform (F45) and other neurotic disorders (F48). ICD-11 follows DSM-5, with separate "anxiety or fear-related" and "obsessive-compulsive or related" groupings.

The anxiety disorders and their time rules ●●●

DSM-5-TR orders the anxiety disorders by typical age of onset and separates them by what is feared (Table 8.21); six months is the usual duration. Anxiety caused by a drug or its withdrawal, or by an illness such as hyperthyroidism, is diagnosed instead as substance/medication-induced anxiety disorder or anxiety disorder due to another medical condition.

Table 8.21 The DSM-5-TR anxiety disorders and their duration rules.

Disorder	Feared or key sign	Duration
Separation anxiety disorder	Separation from attachment figures (3 of 8 signs)	4 weeks in children and adolescents; typically 6 months in adults
Selective mutism	Silent at school, speaks at home	1 month, not only the first month of school
Specific phobia	A specific object or situation	Typically 6 months
Social anxiety disorder	Scrutiny and embarrassment	Typically 6 months
Panic disorder	Recurrent unexpected panic attacks	1 month of worry or behaviour change after an attack
Agoraphobia	2 of 5 situations where escape is hard	Typically 6 months

Disorder	Feared or key sign	Duration
Generalised anxiety disorder	Many events; 3 of 6 symptoms (1 in children)	More days than not, at least 6 months

Separation anxiety disorder (excessive fear of separation from attachment figures) and **selective mutism** (consistent failure to speak in specific social situations while speaking in others) moved here from DSM-IV's childhood disorders, and separation anxiety can now begin in adulthood. The selectively mute child's silence at school comes from social anxiety, not a language problem.

A **specific phobia** is a persistent, irrational fear of a specific object or situation, out of proportion to the danger, provoked almost every time and avoided or endured with dread: an anxiety disorder, not a mild stress reaction, a personality disorder or a passing hallucination. Its five types are animal, natural environment (heights, storms, water), blood-injection-injury, situational (aeroplanes, lifts, enclosed places) and other (choking, vomiting). Most phobics show sympathetic arousal, but the blood-injection-injury type shows a diphasic **vasovagal response**: heart rate and blood pressure rise briefly, then drop, and the person may faint.

Social anxiety disorder is fear of situations of possible scrutiny (conversation, eating in public, a speech) because one may act in a way that is negatively evaluated or embarrassing; in children it must occur with peers. The **performance only** specifier covers fear confined to speaking or performing in public.

Generalised anxiety disorder (GAD) is excessive anxiety and worry about a number of events or activities, more days than not for at least six months, that the person finds hard to control, with three or more of six symptoms (only one in children): restlessness, easy fatigue, poor concentration, irritability, muscle tension and sleep disturbance. Hopping from health to money to a child's journey home, it is continued, **free-floating** apprehension about unspecified things. ICD-10 asks for anxiety on most days for several weeks at a time, usually several months; its research criteria fix six months. ICD-11 asks for at least several months, more days than not, with no fixed number.

> **! EXAM TRAP**
>
> GAD is **continued apprehension about unspecified things**, defined by persistent, excessive worry. "Fear of fear" is panic disorder, flashbacks PTSD and safety checking OCD; mania and mood swings are mood symptoms (Section 8.5), hallucinations psychotic (Section 8.4). GAD needs at least **6 months** in adults and specific phobia typically 6 months, not 1 week or 1, 3 or 12 months.

Panic attacks, panic disorder and agoraphobia ●●

A **panic attack** is an abrupt surge of intense fear or discomfort that peaks **within minutes** (literally a few minutes; not seconds, half an hour or hours), with four or more of thirteen symptoms: palpitations, sweating, trembling, shortness of breath, choking, chest pain, nausea, dizziness, chills or heat sensations, paraesthesias (numbness or tingling), derealisation or depersonalisation, fear of losing control or "going crazy", and fear of dying. **Expected** attacks have an obvious cue (a spider); **unexpected** ones come out of the blue. A panic attack is not itself a disorder; in other disorders it is a specifier ("PTSD with panic attacks").

Panic disorder requires recurrent unexpected attacks, at least one followed by a month or more of persistent worry about further attacks or their consequences (a heart attack, going mad), a maladaptive change in behaviour (avoiding exercise), or both. One attack is not panic disorder.

Agoraphobia, literally "fear of the marketplace", is marked fear of two or more of five situations: public transport, open spaces, enclosed places (shops, cinemas), queues or crowds, and being outside the home alone, because escape might be difficult or help unavailable if panic-like symptoms strike. The situations are avoided, endured with dread or faced only with a companion, typically for six months. Since DSM-5, agoraphobia is diagnosed independently of panic disorder; a person meeting both receives both.

In 1978 **Alan Goldstein** and **Dianne Chambless** argued that after a few attacks people come to fear the bodily sensations of anxiety themselves. This **fear of fear** marks panic disorder and agoraphobia.

> **! EXAM TRAP**
>
> Agoraphobia is the **irrational fear of public places**. Hearing voices is psychotic, nervousness in a lift is claustrophobia, fear of tall trees or buildings a fear of heights, and re-experiencing trauma PTSD. Panic disorder is **recurrent unexpected** attacks, not one attack, worry about many things (GAD) or fear of panicking in enclosed spaces.

Naming the phobias ○

Phobia names join a Greek word for the feared thing to *phobos*, fear: *algos* (pain) gives algophobia (Table 8.22). Clinically, glossophobia is social anxiety disorder (performance only) and trypanophobia the blood-injection-injury type.

Table 8.22 Phobia names and their objects.

Name	Fear of	Name	Fear of
Algophobia	Pain	Acrophobia	Heights
Nosophobia	Disease	Claustrophobia	Enclosed spaces
Cynophobia	Dogs	Agoraphobia	The marketplace (public places)
Haematophobia	Blood	Arachnophobia	Spiders
Ophidiophobia	Snakes	Aerophobia	Flying
Hydrophobia	Water	Nyctophobia	Darkness
Thanatophobia	Death	Xenophobia	Strangers
Mysophobia	Dirt and germs	Astraphobia	Thunder and lightning
Glossophobia	Public speaking	Trypanophobia	Needles

Why anxiety develops ●●

Biology. Anxiety is moderately heritable, though what is inherited is a general anxious tendency. Benzodiazepines calm anxiety by boosting GABA, the main inhibitory transmitter, so the **GABA-benzodiazepine system** is the classic anxiety system; noradrenaline from the locus coeruleus drives the alarm, serotonin modulates it, and the amygdala is the hub of the

fear circuit. **Jerome Kagan** (1929 to 2021) found that **behavioural inhibition**, shyness and withdrawal from novelty visible by four months of age, predicts later anxiety.

Learning. The **two-factor theory** of **O. Hobart Mowrer** (1907 to 1982), set out in 1947, explains why fears last. A neutral stimulus paired with a frightening one becomes a conditioned stimulus for fear (classical conditioning, as with Little Albert, Section 1.6); avoiding it then reduces fear, so avoidance is negatively reinforced and the fear is never unlearned. **Martin Seligman** (born 1942) explained why snakes and heights are feared more than electric sockets with **preparedness** (1971): evolution has prepared us to learn ancestral fears quickly and keep them. **Stanley Rachman** (1977) explained phobias with no frightening memory through three pathways: direct conditioning, vicarious learning and information.

Cognition. David M. Clark (1986) traced panic to the **catastrophic misinterpretation** of bodily sensations: a skipped heartbeat is read as a heart attack, and fear and sensations feed each other (Figure 8.8). **Catastrophic thinking** in general means worrying that the worst will always happen. It differs from other distortions (Section 7.5): dichotomous thinking (everything extremely good or extremely bad), selective abstraction (seeing only the negative) and personalisation (taking things personally). **Steven Reiss**'s **anxiety sensitivity**, the belief that anxiety symptoms are harmful, predicts panic. For GAD, **Thomas Borkovec** held that verbal worry avoids the vivid imagery and arousal of fear and is reinforced by that avoidance; **Adrian Wells**' metacognitive model adds positive and negative beliefs about worry itself; and **Michel Dugas** stressed **intolerance of uncertainty**.

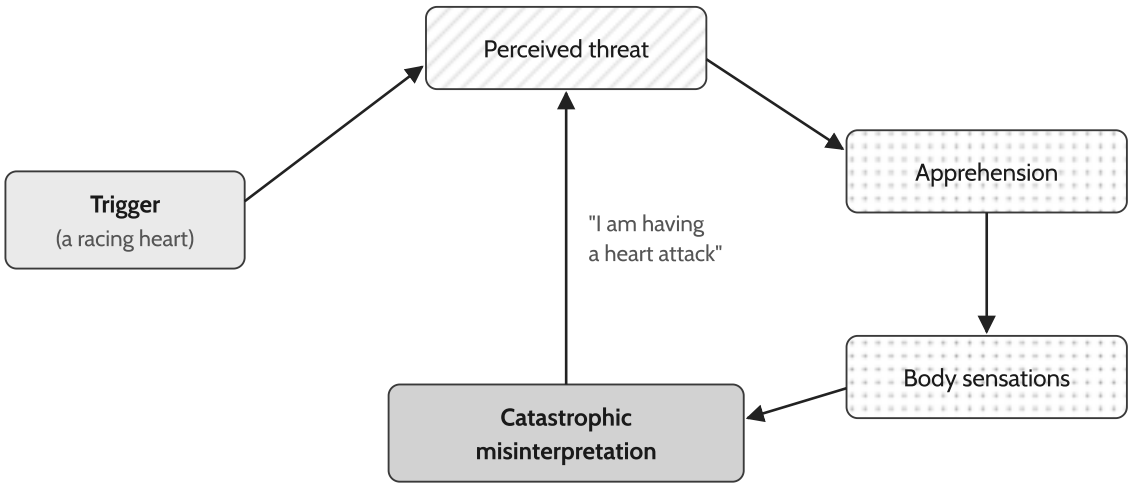


Figure 8.8 Clark's (1986) vicious circle of panic: misread bodily sensations raise the threat, which strengthens the sensations.

The suffocation alarm. Donald F. Klein's **suffocation false alarm theory** (1993) proposes an evolved suffocation monitor that in panic disorder is oversensitive, firing at harmless rises in carbon dioxide; carbon dioxide inhalation and lactate infusion do trigger attacks far more readily in panic patients. It explains panic disorder, not Down syndrome, autism (Section 8.3) or delirium (Section 8.10). **David H. Barlow**'s **triple vulnerability** theory then combines a generalised biological vulnerability (an anxious temperament), a generalised psychological vulnerability (a sense that events are uncontrollable) and a specific psychological vulnerability (learning that certain things are dangerous).

Psychodynamic view. Freud later treated anxiety as **signal anxiety** (1926), a warning to the ego that a forbidden impulse is near, which sets defences going. The defence of phobia is **displacement**, shifting feeling from a threatening object to a safer one. Each classic neurosis had a signature defence: repression in hysteria, displacement in phobia, isolation, undoing and reaction formation in OCD, introjection in depression.

KEY STUDY: LITTLE HANS (FREUD, 1909)

Hans, aged five, would not go out for fear that a horse would bite him. Freud, working mainly through the boy's father, read the fear as an Oedipal fear of the father displaced onto horses, which, unlike a father, could be avoided. Critics noted that the evidence came from a father devoted to Freud, and that Hans had seen a horse fall in the street, which suggests simple conditioning.

! EXAM TRAP

Displacement is the defence of **phobias**, not of hysteria, OCD or depression. Catastrophic thinking is **expecting the worst**, not dichotomous thinking, selective abstraction or personalisation.

Obsessive-compulsive disorder ●●●

Obsessions are recurrent, persistent, intrusive and unwanted thoughts, urges or images that cause anxiety and that the person tries to ignore, suppress or neutralise. **Compulsions** are repetitive behaviours (washing, checking, ordering, tapping) or mental acts (praying, counting) performed in response to an obsession or by rigid rules, to reduce anxiety or prevent a dreaded event, though not realistically connected to it. The cardinal DSM-5 feature is obsessions and/or compulsions that are **time-consuming (more than 1 hour a day) or cause significant distress or impairment**. Merely having obsessions is not enough, and there is no six-hour or two-week rule.

Priya, 22, after touching a door handle, fears she will carry germs that kill her mother and washes three hours a day: this is OCD. GAD worry concerns realistic matters without rituals, illness anxiety disorder is fear of having a disease (Section 8.7), and a specific phobia simply avoids an external object.

Obsessions are **ego-dystonic**: intrusive, unwanted and inconsistent with the self-concept, neither part of one's identity nor pleasurable. Obsessive-compulsive personality disorder (Section 8.11), a separate disorder, is **ego-syntonic**: perfectionism and control felt as right, without true obsessions or compulsions. The **insight specifier** runs from good or fair through poor to absent insight/delusional beliefs (certain the house will burn unless the stove is checked 30 times; 4 percent or fewer). A **tic-related** specifier marks a current or past tic disorder.

Symptoms cluster in four dimensions: contamination and washing, doubt or harm and checking, symmetry with ordering and counting, and taboo (aggressive, sexual, religious) thoughts. Hoarding is no longer an OCD compulsion, since DSM-5 made hoarding disorder separate. Twelve-month prevalence is about 1.2 percent, and onset is in childhood or early adulthood (mean 19.5 years; boys earlier).

> **! EXAM TRAP**
>
> Tapping, washing and checking are compulsions; **hoarding is not**. OCD is **ego-dystonic**, OCPD ego-syntonic; obsessions are not always reality-grounded, OCD is not confined to childhood, and total unawareness that obsessions are unreasonable is absent insight. Grandiosity and need for admiration mark narcissistic personality disorder (Section 8.11).

Biology. OCD responds to clomipramine and the SSRIs, which block serotonin reuptake, implicating the **central serotonin (5-HT) system**; GABA is the anxiety system, and glutamate and dopamine play lesser parts. Imaging implicates the **cortico-striato-thalamo-cortical circuit**: the orbitofrontal cortex and anterior cingulate signal "something is wrong", the caudate nucleus fails to filter the signal, and the thalamus loops it back. **Susan Swedo** (1998) described **PANDAS**, sudden childhood OCD or tics after streptococcal infection, thought to be an autoimmune attack on the basal ganglia. First-degree relatives have about twice the rate, ten times after childhood onset.

Learning. Mowrer's two-factor theory applies directly (Figure 8.9): the obsession acquires anxiety by classical conditioning, the compulsion relieves it, and the relief negatively reinforces the compulsion. Compulsive acts are thus **escape or avoidance responses** (safety behaviours): not conditioned stimuli (the contaminating thought is the stimulus), not unconditioned responses (unlearned reflexes), not positively reinforced (distress is removed, nothing pleasant added) and not involuntary. Conditioning is the general mechanism; escape is the pattern. Exposure and response prevention blocks the ritual (Section 7.4).

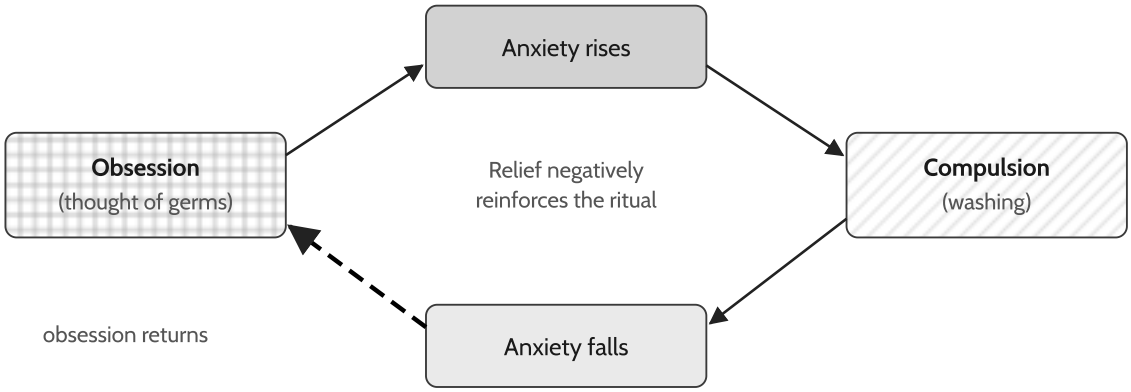


Figure 8.9 The obsessive-compulsive cycle: relief reinforces the ritual, but the obsession returns (dashed line).

Cognition and psychodynamics. Nearly everyone has odd intrusive thoughts; **Paul Salkovskis** (1985) argued that OCD begins when they are appraised as signs of **inflated responsibility** for harm, which must be neutralised. Rachman added **thought-action fusion**, the belief that thinking of an act is as bad as doing it or makes it likelier. Freud's Rat Man (1909), tormented by the thought of rats gnawing into his father, is the classic case of "obsessional neurosis", which Freud traced to regression to anal-stage conflicts over control, dirt and aggression. Its defences are **isolation of affect** (the thought stays, its feeling goes), **undoing** (an act that symbolically cancels a forbidden thought, the most characteristic), **reaction**

formation (scrupulous cleanliness against a wish to soil) and intellectualisation. Repression is the defence of hysteria.

> ! EXAM TRAP
>
> Isolation, undoing and reaction formation are **all** OCD defences. For a single answer choose **undoing**, the defence of obsessive-compulsive neurosis (not depression, mania or schizophrenia). The defence NOT used in OCD is **repression**.

The related disorders ●●

Body dysmorphic disorder (BDD) is preoccupation with perceived defects or flaws in appearance that are not observable or appear slight to others, with repetitive behaviours (mirror checking, grooming, skin picking, reassurance seeking) or mental acts (comparing). Specifiers record insight and **muscle dysmorphia**, the belief that one's body is too small or insufficiently muscular. **Enrico Morselli** called it dysmorphophobia in 1891. DSM-IV listed BDD among the somatoform disorders; DSM-5 placed it in **obsessive-compulsive and related disorders**, not with somatic symptom, anxiety or personality disorders. It concerns appearance, unlike somatic symptom disorder (distressing bodily symptoms), illness anxiety disorder (fear of serious illness) and conversion disorder (neurological symptoms without matching disease), all in Section 8.7.

Hoarding disorder is persistent difficulty discarding possessions regardless of value, from a perceived need to save them, so that clutter blocks the use of living spaces. **Trichotillomania** (hair-pulling disorder), named by **François Hallopeau** in 1889, is recurrent hair pulling causing hair loss, with repeated attempts to stop; **excoriation disorder** is the same with skin picking and lesions. These body-focused repetitive behaviours are not driven by obsessions. Substance/medication-induced OCRD and OCRD due to another medical condition complete the chapter. ICD-11's obsessive-compulsive or related group also includes olfactory reference disorder (a conviction of emitting a foul body odour) and hypochondriasis (health anxiety), and calls trichotillomania and excoriation body-focused repetitive behaviour disorders. Finally, an obsession is resisted as intrusive and usually recognised as excessive; a GAD worry concerns real-life matters and brings no ritual; a delusion is a false belief held as simply true.

> SCAN YOUR BRAIN
>
> Fill in the blanks. The answers are below; cover them first.
>
> 1. Panic disorder needs at least _____ of worry or behaviour change after an attack.
> 2. The phobia type with a diphasic vasovagal fainting response is the _____ type.
> 3. Donald Klein's _____ theory explains panic disorder.
> 4. In Little Hans, fear of the father was shifted onto horses by _____.
> 5. In learning theory, compulsions are _____ responses, negatively reinforced by relief.
> 6. DSM-5 classifies body dysmorphic disorder under _____ disorders.
>
> **Answers:** (1) 1 month, (2) blood-injection-injury, (3) suffocation false alarm, (4) displacement, (5) escape (avoidance), (6) obsessive-compulsive and related.

8.7 Trauma, Dissociative and Somatic Symptom Disorders HIGH YIELD · 50 QS

A soldier who relives a shelling every night, a woman who finds clothes she does not remember buying, a girl whose arm goes limp after a quarrel: all three were once called hysterical. This section follows hysteria as it split into the trauma-related, dissociative and somatic symptom disorders, and ends with illness that is feigned.

Hysteria, from Hippocrates to DSM-III ●●

The word **hysteria** comes from the Greek *hystera*, womb: **Hippocrates** (about 460 to 370 BCE) and his followers blamed its fits and paralyses on a "wandering womb". At the Salpêtrière in Paris, **Jean-Martin Charcot** (1825 to 1893) showed that hysterical phenomena such as loss of feeling in a limb or fainting could be produced and removed by hypnosis and suggestion, so their cause was psychological, not a lesion.

His pupil **Pierre Janet** (1859 to 1947) named the mechanism *désagrégation* (1889), now called **dissociation**: under strain, a cluster of memories and functions splits off from consciousness and runs on its own. The phrase **la belle indifférence**, the calm unconcern some patients show about a dramatic symptom, is usually credited to Janet. In Vienna **Josef Breuer** (1842 to 1925) relieved "Anna O." of her paralyses by letting her talk through their origins under hypnosis, her "talking cure", and with **Sigmund Freud** (1856 to 1939) published *Studies on Hysteria* (1895).

For Freud, hysteria was physical reactions to emotional trauma: a painful memory is pushed out of awareness by **repression** and its energy converted into a bodily symptom, hence **conversion**. Most psychodynamic theorists still see repression behind both conversion and dissociative symptoms. The symptom gives **primary gain**, relief from the anxiety of the conflict, and often **secondary gain**, attention or escape from a duty.

KEY STUDY: HILGARD'S HIDDEN OBSERVER

Ernest Ropiequet Hilgard (1904 to 2001) gave hypnotised volunteers a suggestion that they would feel no pain and put their hands in iced water. They reported little pain, yet a **hidden observer**, addressed by a prearranged cue, reported nearly normal pain. His **neodissociation theory** (*Divided Consciousness*, 1977) holds that awareness can split into parallel streams. Critics such as Nicholas Spanos argued that the instructions create the observer.

In 1980 DSM-III divided hysteria into somatoform and dissociative disorders (Figure 8.10); ICD-10 kept them together as dissociative (conversion) disorders, F44.

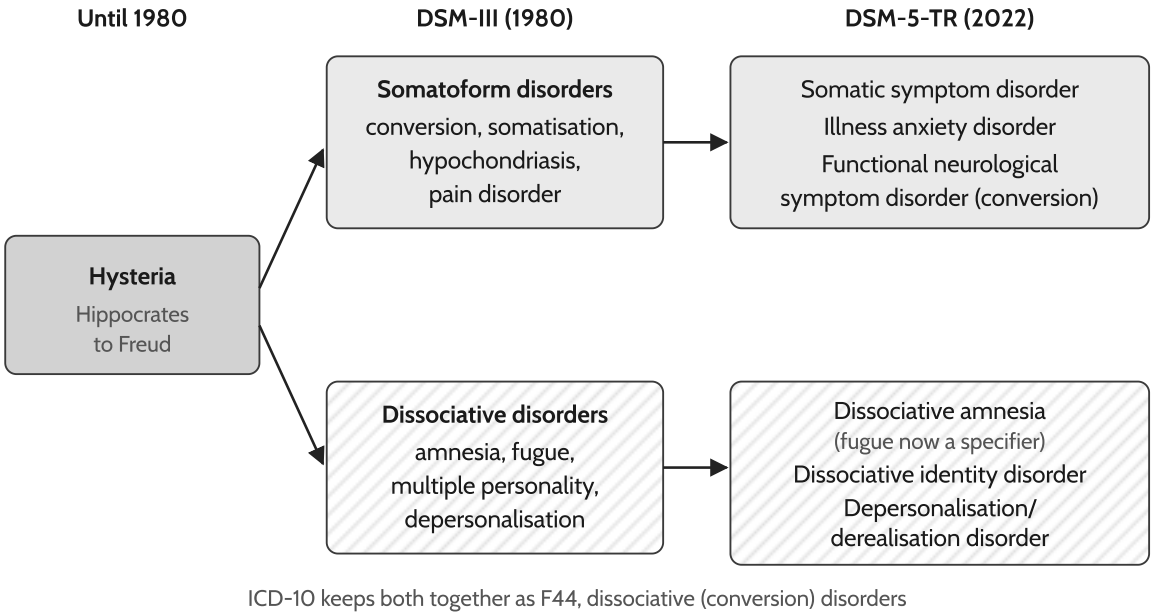


Figure 8.10 How hysteria was divided: DSM-III split it into two classes, which DSM-5 renamed.

> **! EXAM TRAP**
>
> Freud's hysteria is **physical reactions to emotional trauma**; a phobia of horses is Little Hans, and the Oedipus complex and oral traits belong to Chapter 4. The psychodynamic mechanism of dissociative symptoms is **repression**: conflict is what gets repressed. Charcot used **both** hypnosis and suggestion. La belle indifférence is credited to **Janet**, not Freud, Anna Freud (who catalogued the defence mechanisms, Chapter 4) or Hippocrates.

Posttraumatic and acute stress disorders ●●●

First World War soldiers who shook, froze or went mute were said to have shell shock; **Abram Kardiner** (1891 to 1981) described *The Traumatic Neuroses of War* (1941), and DSM-III named **posttraumatic stress disorder** (PTSD) in 1980. DSM-5 moved it from the anxiety disorders into a new trauma- and stressor-related chapter.

The event must involve actual or threatened death, serious injury or sexual violence, experienced directly, witnessed in person, learned of as happening to a close family member or friend, or met through repeated exposure to aversive details, as by police officers hearing about child abuse. Symptoms fall into four clusters:

- **Intrusion** (1 of 5): distressing memories, nightmares, flashbacks (acting or feeling as if the event were recurring), distress at reminders.
- **Avoidance** (1 of 2) of trauma-related thoughts and feelings or of outside reminders.
- **Negative alterations in cognitions and mood** (2 of 7): amnesia for part of the event, self-blame, "no one can be trusted", guilt, detachment, no positive emotions.
- **Alterations in arousal and reactivity** (2 of 6): irritability, recklessness, hypervigilance, exaggerated startle, poor concentration, poor sleep.

Symptoms must last **more than 1 month**. Specifiers are the dissociative subtype (persistent depersonalisation or derealisation) and delayed expression (full criteria met only 6 months or

more after the event); children 6 and younger have separate criteria. **Acute stress disorder** covers **3 days to 1 month** after the trauma and needs nine of fourteen symptoms from five categories. DSM-IV required dissociative symptoms for it; DSM-5 dropped that rule, so timing now separates it from PTSD, and both sit in the trauma chapter.

ICD-11 places PTSD among disorders specifically associated with stress, which require an extremely threatening or horrific event, and keeps three core elements: re-experiencing in the here and now, avoidance, and a persistent sense of current threat. **Complex PTSD** adds disturbances in self-organisation (affect dysregulation, negative self-concept, relationship difficulties), usually after prolonged, inescapable trauma such as torture or childhood abuse. An acute stress reaction is a normal response, not a disorder.

Who develops PTSD depends first on the dose and type of trauma (most prisoners of war do; interpersonal violence is worse than accidents), then on vulnerability: earlier anxiety, family instability and **peritraumatic dissociation** (feeling dazed or unreal during the event) raise risk, while social support protects. Conditioning attaches fear to reminders and avoidance blocks extinction. The amygdala is overactive, the hippocampus smaller and the HPA (hypothalamic-pituitary-adrenal) stress axis dysregulated. Prolonged exposure, cognitive processing therapy and EMDR are covered in Sections 7.4 and 7.5. Trauma can also bring **post-traumatic growth**, which **Richard Tedeschi** and **Lawrence Calhoun** (1996) measured in five domains: appreciation of life, relating to others, personal strength, new possibilities and spiritual change.

> **! EXAM TRAP**
>
> PTSD has **four** clusters (DSM-IV had three). Magical thinking and ideas of reference belong to schizotypal personality and psychosis (Sections 8.4 and 8.11), and "denial, anger, bargaining" are Elisabeth Kübler-Ross's stages of dying (Chapter 6). PTSD lasts more than 1 month; acute stress disorder 3 days to 1 month; delayed expression begins at 6 months.

Adjustment, grief and attachment disorders ○

An **adjustment disorder** is an emotional or behavioural reaction to an identifiable stressor, such as divorce or job loss, with disproportionate distress or impaired functioning; it begins within 3 months of the stressor and ends within 6 months after the stressor or its consequences end. Its subtypes are with depressed mood, with anxiety, with mixed anxiety and depressed mood, with disturbance of conduct, with mixed disturbance of emotions and conduct, and unspecified. Being brief and tied to a present stressor, it is best treated by brief psychotherapy and **crisis intervention** (Section 7.9).

Prolonged grief disorder, added in DSM-5-TR (2022), is intense yearning for or preoccupation with the deceased, with symptoms such as identity disruption, disbelief and meaninglessness, at least 12 months after the death (6 months in children and adolescents; ICD-11 needs 6), beyond cultural norms. Ordinary bereavement is not a disorder.

After severe neglect or institutional rearing, a child with **reactive attachment disorder** is inhibited and emotionally withdrawn, rarely seeking comfort, while one with **disinhibited social engagement disorder** shows indiscriminate sociability, readily going off with strangers. Both need a developmental age of at least 9 months (Table 8.23).

Table 8.23 Trauma- and stressor-related disorders (DSM-5-TR).

Disorder	Trigger	Time rule	Core picture
PTSD	Death, injury or sexual violence	More than 1 month	Four symptom clusters
Acute stress disorder	As for PTSD	3 days to 1 month	9 of 14 symptoms
Adjustment disorder	Any stressor	Starts within 3 months; ends within 6 months of stressor's end	Disproportionate distress
Prolonged grief disorder	Death of someone close	12 months (children 6)	Yearning, preoccupation
Reactive attachment disorder	Severe neglect	Developmental age 9 months; before age 5	Withdrawn, seeks no comfort
Disinhibited social engagement disorder	Severe neglect	Developmental age 9 months	Indiscriminate with strangers

Dissociative identity disorder ●●●

Dissociative disorders disrupt the normal integration of consciousness, memory, identity, emotion, perception and behaviour, and most follow trauma. **Dissociative identity disorder** (DID), called multiple personality disorder until DSM-IV (1994), requires two or more distinct personality states, or an experience of possession, **and** recurrent gaps in the recall of everyday events, personal information or trauma beyond ordinary forgetting, not explained by accepted religious practice.

The identity most often encountered, carrying the person's real name, is the **host**; the others are **alters**, which may differ in age, gender and voice, and changing between them is a **switch**. The amnesia shows in daily life: a woman finds clothes she does not remember buying. About 90 percent of patients report childhood abuse or neglect.

Corbett Thigpen and **Hervey Cleckley**'s *The Three Faces of Eve* (1957) and *Sybil* (1973), a case later criticised as shaped by the therapist, frame the debate on causes. The trauma model sees identities as states split off to escape unbearable abuse. The **sociocognitive model** of **Nicholas Spanos** (1942 to 1994) sees them as roles shaped in suggestible people by therapists' questions, hypnosis and media; in a 1985 experiment, over 80 percent of students playing an accused murderer produced an "alter" when interviewed.

> **! EXAM TRAP**
>
> "Multiple personalities in one body" is **DID, not schizophrenia**, whose "split" is between thought and emotion (Section 8.4). Bipolar disorder alternates moods, and borderline personality disorder an unstable self-image, not separate identities with amnesia. The identity with the real name is the **host**; "ghost" and "mixed" identities do not exist.

Dissociative amnesia, fugue and depersonalisation ●●●

Dissociative amnesia is an inability to recall important autobiographical information, usually traumatic or stressful, too extensive to be ordinary forgetting and not due to a substance or a neurological or medical condition. It is a potentially reversible, mainly retrograde failure of retrieval, in five forms:

- **Localised**: a circumscribed period, such as the hours of an assault (the commonest form).

- **Selective**: some, but not all, events of that period.
- **Generalised amnesia**: one's entire life history, often with identity.
- **Systematised**: one category, such as everything about a violent sibling.
- **Continuous**: each new event, as it occurs.

Fugue, a separate diagnosis in DSM-IV, is now the specifier *with* **dissociative fugue**: apparently purposeful travel or bewildered wandering with amnesia for identity, as in the man found working in another city with no memory of who he is. Organic amnesias differ. **Transient global amnesia** is a sudden, hours-long loss of new learning in an older adult who keeps their identity but repeats the same questions; **Korsakoff syndrome** is thiamine-deficiency amnesia with confabulation in chronic alcohol use; **Alzheimer's disease** begins with gradually failing new learning (Section 8.10); amnesia from head injury or another medical condition is mainly anterograde. In all of these, personal identity is lost last; in dissociative amnesia it can go first.

Depersonalisation/derealisation disorder means persistent or recurrent experiences of either kind, or both. **Depersonalisation** is detachment from oneself: being an outside observer of one's own thoughts, feelings, body or actions, or feeling no longer one's normal self. **Derealisation** is unreality of the surroundings, which seem dreamlike, foggy or distorted. **Reality testing stays intact** ("it is *as if* I were dreaming"), which separates it from psychosis.

> ! EXAM TRAP
>
> Self unreal is **depersonalisation**; world unreal is **derealisation**. Multiple personality, fugue and amnesia are hysterical dissociative states, but **déjà vu** (a new situation feels already lived) is not: it is a disorder of recognition memory, seen in healthy people and temporal lobe epilepsy. A hallucination is perception without a stimulus (Section 8.2).

Other specified dissociative disorder covers possession without amnesia, identity change after brainwashing, acute reactions lasting under a month, and **dissociative trance**, an acute narrowing or loss of awareness of the surroundings with profound unresponsiveness, outside accepted cultural practice. ICD-10 adds **dissociative stupor** and **trance and possession disorders**, in which a spirit or deity seems to take over; possession states are among the commonest dissociative presentations in India, though within an accepted ritual they are not disorders. **Ganser syndrome**, described by **Sigbert Ganser** (1853 to 1931) in 1898 in prisoners awaiting trial, combines **approximate answers** (a cow has "five" legs), clouding of consciousness, pseudohallucinations and conversion symptoms.

Somatic symptom and illness anxiety disorders ●●

Somatoform disorders (DSM-III and DSM-IV) were conditions with symptoms suggesting a medical condition without evidence of physical pathology. They included **somatisation disorder**, earlier **Briquet's syndrome** after **Paul Briquet** (1796 to 1881), who in 1859 described patients returning with endless lists of complaints; **hypochondriasis**; **pain disorder**; and conversion. **Neurasthenia** ("lack of nerve strength"), named by **George Miller Beard** (1839 to 1883) in 1869 for unexplained exhaustion, survives in ICD-10.

DSM-5 renamed the class somatic symptom and related disorders and defined it by positive features. **Somatic symptom disorder** needs one or more distressing somatic symptoms with

excessive thoughts, feelings or behaviours: disproportionate thoughts about their seriousness, high health anxiety, or excessive time and energy spent on them, typically for more than 6 months. The symptoms may have a medical cause: a woman with real arthritis who spends hours a day researching it qualifies.

> ! EXAM TRAP
>
> "Physical symptoms without a medical cause" is the old **somatoform** idea, which DSM-5-TR calls "problematic". Against hallucinations, delusions and obsessions it is still the keyed answer for somatic symptom disorder, but the DSM-5-TR feature is **excessive response** to distressing symptoms.

Illness anxiety disorder (formerly hypochondriasis) is preoccupation with having or acquiring a serious illness when somatic symptoms are absent or mild, with high health anxiety and excessive checking or avoidance, for at least 6 months. It has **care-seeking** and **care-avoidant** types. ICD-11 keeps the name hypochondriasis but places it among obsessive-compulsive and related disorders, and replaces somatoform disorders with **bodily distress disorder**, distressing bodily symptoms with excessive attention to them. **Psychological factors affecting other medical conditions** covers stress or behaviour that worsens a real disease (Section 7.9).

> ! EXAM TRAP
>
> Chronic fatigue syndrome, or **systemic exertion intolerance disease**, has **no DSM-5-TR criteria**. The US Institute of Medicine (2015) requires fatigue for 6 months or more, not relieved by rest; post-exertional malaise; unrefreshing sleep; and cognitive impairment or orthostatic intolerance. **Weight change is not a criterion.**

Conversion: functional neurological symptom disorder ●●●

Hysteria's direct heir is **functional neurological symptom disorder** (conversion disorder): one or more symptoms of altered voluntary motor or sensory function, with clinical findings incompatible with recognised neurological or medical conditions. The symptoms are not intentionally produced. They include paralysis; abnormal movements such as **astasia-abasia** (cannot stand or walk, yet moves the legs normally in bed); **globus**, a lump in the throat; **aphonia**; **psychogenic non-epileptic seizures**; **glove anaesthesia**, ending sharply at the wrist where no nerve territory ends; and tunnel vision or **functional blindness**, explained by psychological factors with no damage to eye, optic nerve or cortex.

Diagnosis rests on positive signs. In **Hoover's sign** a "paralysed" leg cannot press down on request, yet presses down when the patient lifts the healthy leg against resistance. A woman whose arm becomes paralysed after an argument, with a normal examination and the remark "it's not really bothering me", is the classic picture: sudden onset after a stressor, motor or sensory loss, and la belle indifférence. DSM-5-TR no longer requires a stressor and calls la belle indifférence non-specific: in one review 21 percent of conversion patients showed it, against 29 percent with organic disease. ICD-11 calls the condition **dissociative neurological symptom disorder**. Conversion differs from hypochondriasis: neurological-like symptoms without organic basis, versus fear of having a serious illness.

> **! EXAM TRAP**
>
> La belle indifférence points to **conversion disorder**, not malingering, factitious disorder, somatic symptom disorder or illness anxiety. Conversion symptoms "affect motor or sensory function, begin suddenly and follow a stressor" (the classic picture, though DSM-5-TR no longer requires the stressor); "suggest a medical disorder" fits any somatic disorder, and normal investigations alone never make the diagnosis.

Factitious disorder and malingering ●●●

In **factitious disorder** the person falsifies signs or symptoms, or induces injury or disease, with deception and presents as ill even without obvious external rewards; the motive is to assume the sick role. The severe form **imposed on self**, with dramatic stories and many admissions, was named **Munchausen syndrome** by **Richard Asher** (1912 to 1969) in 1951, after the tall tales of **Baron Munchausen**. In factitious disorder **imposed on another** (Munchausen by proxy, **Roy Meadow**, 1977) a carer induces illness in a dependent person, like the nursing assistant who secretly injects insulin into elderly patients; the perpetrator receives the diagnosis. DSM-5 lists it in the somatic symptom chapter, but it was never a somatoform disorder: DSM-IV gave it a chapter of its own.

Malingering is intentionally producing false or grossly exaggerated symptoms for external incentives such as compensation, drugs, or escape from work, military duty or prosecution. It is not a mental disorder but a condition that may be a focus of clinical attention (Table 8.24).

Table 8.24 Telling the conditions apart by intention and motive.

Condition	Intentional?	Motive	Mental disorder?
Conversion disorder	No	Unconscious conflict	Yes
Somatic symptom disorder	No	Excessive response to symptoms	Yes
Illness anxiety disorder	No	Fear of serious illness	Yes
Factitious disorder	Yes	Sick role, no external reward	Yes
Malingering	Yes	External reward	No

> **! EXAM TRAP**
>
> Factitious disorder and malingering differ in the **presence or absence of external rewards**, not severity, age of onset or belief in the symptoms. Confabulation (filling memory gaps without intent to deceive), false identification (misrecognising a person or place) and altered identities are not faking, and the insulin-injecting carer has factitious disorder imposed on another, not antisocial personality disorder.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Charcot produced hysterical symptoms by hypnosis and _____.
2. PTSD symptoms must last more than _____.
3. In DID, the identity carrying the real name is the _____.
4. Loss of memory for one's entire life is _____ amnesia.
5. Surroundings feeling unreal or dreamlike is _____.
6. A "paralysed" leg pressing down when the healthy leg lifts shows _____ sign.
7. Faking illness for an external incentive is _____.

Answers: (1) suggestion, (2) 1 month, (3) host, (4) generalised, (5) derealisation, (6) Hoover's, (7) malingering.

8.8 Feeding, Eating, Elimination and Sleep-Wake Disorders

LOWER YIELD · 20 QS

Eating, bladder and bowel control, and sleep are among the first things a child learns to regulate, and DSM-5-TR gives each its own family of disorders. The eating disorders, above all anorexia nervosa, threaten life most and come first; the elimination and sleep-wake disorders follow.

Pica, rumination and ARFID

DSM-5-TR's **feeding and eating disorders** are persistent disturbances of eating that impair health or psychosocial functioning. Three of the six begin mostly in childhood and involve no body-image concern.

Pica is the persistent eating of non-nutritive, non-food substances (chalk, soil, paper, hair, paint) for **at least 1 month**, inappropriate to the developmental level and not culturally sanctioned, and is diagnosed from about age 2. It is linked with iron or zinc deficiency, pregnancy, intellectual disability and autism. In **rumination disorder** food is repeatedly regurgitated for at least 1 month, then re-chewed, re-swallowed or spat out, without a medical cause such as reflux; it occurs in infants and in intellectual disability. **Avoidant/restrictive food intake disorder** (ARFID) is food avoidance (for texture or smell, lack of interest, or fear of choking) causing weight loss, nutritional deficiency, dependence on supplements or tube feeding, or psychosocial interference; mere picky eating does not qualify. ARFID has **no disturbance in how body weight or shape is experienced**, which separates it from anorexia.

Anorexia nervosa: history and criteria ●●

In 1873 two physicians described young women who starved themselves while insisting they were well. In London **Sir William Withey Gull** (1816 to 1890) named the illness *anorexia nervosa* (in a paper read in 1873 and printed in 1874); in Paris **Ernest-Charles Lasègue** (1816 to 1883) called it *anorexie hystérique*. The name misleads: *anorexia* means loss of appetite, yet most patients are hungry and driven by fear of fatness. **Hilde Bruch** (1904 to 1984), in *The Golden Cage* (1978), described a compliant, high-achieving girl who feels ineffective and finds in control of her body the one thing truly hers.

Anorexia nervosa (AN) has three DSM-5-TR criteria:

1. Restriction of energy intake leading to a **significantly low body weight** for age, sex, development and physical health.
2. **Intense fear of gaining weight** or becoming fat, or persistent behaviour that interferes with weight gain, even at a low weight.
3. Disturbance in how weight or shape is experienced, undue influence of weight or shape on self-evaluation, or persistent failure to recognise the seriousness of the low weight.

Subtypes describe the last 3 months. The **restricting type** has had no recurrent binge eating or purging; weight is lost by dieting, fasting or excessive exercise. The **binge-eating/purging type** has had recurrent binges or purging (vomiting, or misuse of laxatives, diuretics or enemas). Adult severity uses body mass index (BMI, weight in kilograms divided by height in metres squared): **mild** 17 or more, **moderate** 16 to 16.99, **severe** 15 to 15.99, **extreme** below 15.

> ! EXAM TRAP
>
> DSM-5 **dropped amenorrhoea** as a criterion and sets **no BMI cut-off**: a BMI of 17 marks mild severity, not eligibility. ICD-10 used BMI 17.5 or less; ICD-11 uses below 18.5. Purging does not exclude anorexia; it defines a subtype. The core is fear of weight gain with restriction, not obsession with cleanliness (obsessive-compulsive disorder, Section 8.6).

The starving body and the anorexic personality ●●

Most physical signs are the body's defences against starvation, so think "slow and low". Metabolism slows, giving **bradycardia** (slow heart rate), **hypotension** and **hypothermia**. Fine downy hair called **lanugo** grows on the body. Falling hormones produce **amenorrhoea**, the absence of menstrual periods, and in time **osteoporosis**. Vomiting adds **hypokalaemia** (low potassium), which can trigger fatal arrhythmias. Hyperthyroidism, tachycardia, hyperkalaemia and polyphagia (overeating) all point the wrong way.

Anorexia nervosa is the **most lethal eating disorder** and has one of the highest mortality rates of any psychiatric disorder. DSM-5-TR puts its crude mortality at about 5 per cent per decade, from medical complications (cardiac arrhythmia among them) and from suicide. It usually begins in adolescence, far more often in women.

The personality behind it is strikingly consistent: **perfectionism**, obsessionality, rigidity and anxiety; the patient is over-controlled. Impulsivity belongs to bulimia, while avoidance and dependence define the avoidant and dependent personality disorders (Section 8.11).

> ! EXAM TRAP
>
> Amenorrhoea is no longer *required* but is still *common*, so it remains a correct "finding in anorexia". It means absence of menstrual periods, not fear of weight gain or excessive exercise.

Bulimia nervosa and binge-eating disorder ●●●

The London psychiatrist **Gerald Russell** named **bulimia nervosa** (BN) in 1979, calling it "an ominous variant of anorexia nervosa". DSM-5-TR requires:

1. Recurrent **binge eating**: eating, in a discrete period (about 2 hours), an amount definitely larger than most people would, with a **sense of loss of control**.
2. Recurrent **inappropriate compensatory behaviour** against weight gain: self-induced vomiting, misuse of laxatives, diuretics or other medicines, fasting or excessive exercise.
3. Both on average **at least once a week for 3 months** (DSM-IV asked for twice a week).
4. Self-evaluation unduly influenced by body shape and weight.
5. Not occurring only during episodes of anorexia nervosa.

Secret binges, often set off by low mood, are followed by shame and purging, in repeating cycles. Because purging undoes only part of a binge, people with bulimia are usually of **normal weight or overweight**. Vomiting leaves marks: eroded dental enamel, swollen parotid glands, hypokalaemia and **Russell's sign**, calluses on the back of the hand from the teeth. The typical trait is **impulsivity**.

Binge-eating disorder (BED), a full diagnosis only since DSM-5, is binge eating at least once a week for 3 months, with **marked distress** and three of five features: eating very rapidly, until uncomfortably full, when not hungry, alone from embarrassment, and with disgust or guilt afterwards. There is **no regular compensatory behaviour**. Figure 8.11 shows how the three disorders are told apart.

Anorexia and bulimia share a psychology: a **distorted body image**, an **intense fear of being fat**, and eating **regulated by psychological rather than physiological processes** (rules, fear and mood instead of hunger and fullness).

> ! EXAM TRAP
>
> **Significantly low body weight** separates anorexia from bulimia. Purging, fear of fatness and damage to health occur in both. Binges plus compensation is bulimia; binges with distress and no compensation is binge-eating disorder. Only one of the three is diagnosed within an episode.

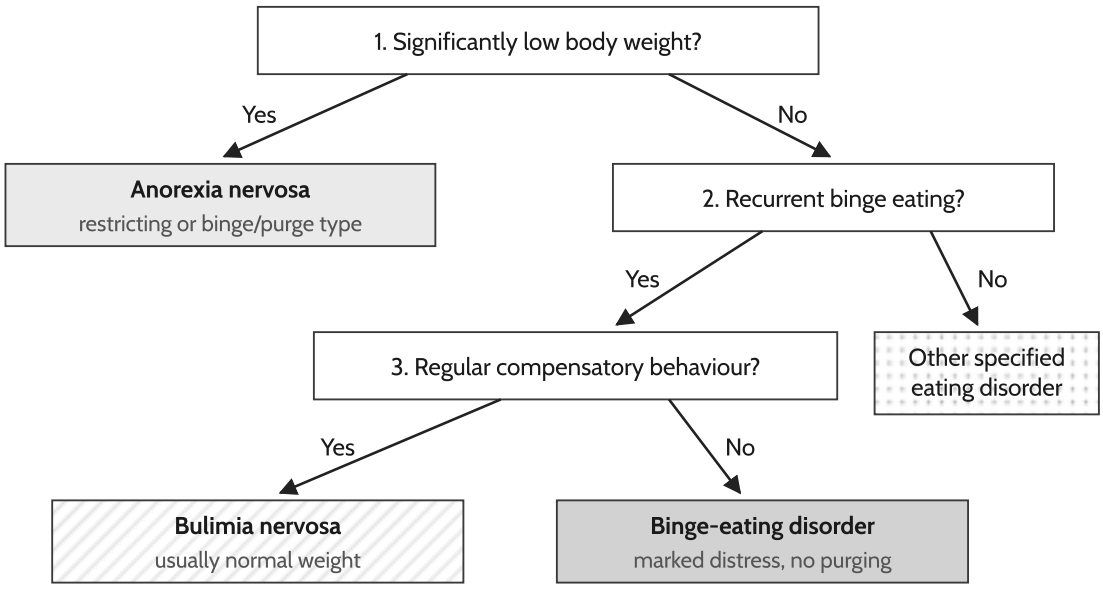


Figure 8.11 Telling the eating disorders apart. Weight is asked first, so a low-weight person who binges and purges has anorexia (binge-eating/purging type), not bulimia.

Causes and other specified eating disorders

Risks pile up. **Sociocultural**: they cluster where a thin ideal is prized, as in modelling and ballet. **Dieting** starts most histories, and restraint provokes bingeing as the body defends its weight (set-point theory, Section 1.10). **Family**: Salvador Minuchin's enmeshed, overprotective, rigid, conflict-avoiding **psychosomatic family** (Section 7.7) features in anorexia. **Personality**: perfectionism in anorexia, impulsivity in bulimia. **Biology**: twin studies put heritability near one half, and low serotonin activity is linked with impulsivity and binge eating.

Presentations fitting no full pattern are **other specified feeding or eating disorder** (OSFED), for example **atypical anorexia nervosa** (every criterion met except low weight: despite significant weight loss, weight is within or above the normal range), bulimia or binge-eating disorder of low frequency or short duration, **purging disorder** (purging without binge eating) and **night eating syndrome** (eating after waking at night, or excessively after the evening meal, with recall).

Enuresis and encopresis ○

The **elimination disorders** are the repeated passage of urine or faeces into inappropriate places by a child developmentally old enough to control them, voluntary or not. **Enuresis** is repeated voiding of urine into bed or clothes, at least **twice a week for 3 consecutive months** or with significant distress or impairment, at an age of at least **5 years**, and not caused by a substance (a diuretic) or a medical condition (diabetes, epilepsy). It is nocturnal only (the commonest), diurnal only, or both. In **primary** enuresis the child has never been dry; **secondary** enuresis returns after a dry period, usually between 5 and 8 years, often after stress. About 5 to 10 per cent of 5-year-olds wet the bed; most outgrow it. The bell-and-pad alarm is described in Section 7.4.

Encopresis is repeated passage of faeces into inappropriate places (clothing, the floor) **at least once a month for 3 months**, at age **4 or older**. It is specified **with constipation and**

overflow incontinence, the commoner form, in which painful defecation leads to withholding and leakage around impacted stool, or **without constipation**, often with oppositional or conduct problems. It too may be primary or secondary.

> **! EXAM TRAP**
>
> Lack of **bowel** control is **encopresis**; lack of bladder control is **enuresis**. Anorexia is an eating disorder, and "analism" is not a recognised elimination disorder. Ages: urine 5, faeces 4.

The sleep-wake disorders and insomnia

The sleep stages, non-rapid eye movement (NREM) stages N1 to N3 and rapid eye movement (REM) sleep, are taught in Section 1.5. DSM-5-TR's **sleep-wake disorders**, ten disorders or groups (Table 8.25), share a complaint about the quality, timing or amount of sleep with daytime distress. ICD-11 moved them out of the mental disorders into their own chapter, Chapter 7. Four follow a rule of threes, at least three times a week for 3 months: insomnia, hypersomnolence, narcolepsy and restless legs syndrome.

Insomnia disorder is difficulty initiating or maintaining sleep, or early waking, on at least 3 nights a week for 3 months **despite adequate opportunity for sleep**. Its treatment, cognitive behaviour therapy with stimulus control, sleep restriction and paradoxical intention, is in Section 7.4. In **hypersomnolence disorder** the person is excessively sleepy despite a main sleep of at least 7 hours, with daytime lapses into sleep, unrefreshing sleep of over 9 hours, or difficulty waking fully.

In the **breathing-related sleep disorders** breathing falters in sleep: the upper airway collapses (obstructive sleep apnoea hypopnoea), the brain stops driving breathing (central sleep apnoea), or shallow breathing lets carbon dioxide rise (sleep-related hypoventilation). In the **circadian rhythm sleep-wake disorders** the body clock is out of step with the required schedule: sleep comes too late or too early (delayed or advanced phase), is scattered across the 24 hours (irregular type), drifts later each day (non-24-hour type, common in blind people) or clashes with working hours (shift work type).

Table 8.25 The sleep-wake disorders of DSM-5-TR.

Disorder	Core feature	Key fact
Insomnia disorder	Poor onset or maintenance, early waking	3 nights a week, 3 months
Hypersomnolence disorder	Sleepy despite 7 hours' sleep	3 times a week, 3 months
Narcolepsy	Irrepressible sleep attacks	Cataplexy, low hypocretin or short REM latency
Obstructive sleep apnoea hypopnoea	Airway collapse: snoring, gasping, sleepiness	5 events an hour with symptoms, or 15
Central sleep apnoea	Breathing effort stops	Cheyne-Stokes breathing; opioids
Sleep-related hypoventilation	Shallow breathing, raised carbon dioxide	Polysomnography
Circadian rhythm sleep-wake disorders	Body clock out of step with schedule	Delayed (adolescents), advanced (elderly), irregular, non-24-hour, shift work
NREM sleep arousal disorders	Sleepwalking, sleep terrors	First third of night; amnesia

Disorder	Core feature	Key fact
Nightmare disorder	Frightening, remembered dreams	Second half of night
REM sleep behaviour disorder	Acting out dreams	Loss of REM atonia
Restless legs syndrome	Urge to move legs at rest, evenings	3 times a week, 3 months
Substance/medication-induced sleep disorder	Intoxication or withdrawal	Insomnia, sleepiness, parasomnia or mixed

Narcolepsy ○

The French physician **Jean-Baptiste-Édouard Gélineau** (1828 to 1906) coined the term narcolepsy in 1880. DSM-5-TR's **narcolepsy** is recurrent periods of irrepressible need to sleep or napping within a day, at least three times a week for 3 months, plus one of three markers: **cataplexy**, brief, sudden, bilateral loss of muscle tone with consciousness preserved, triggered by laughter or joking; **hypocretin (orexin) deficiency** in the cerebrospinal fluid; or a short REM latency (REM within 15 minutes of sleep onset, or a mean sleep latency of 8 minutes or less with two or more sleep-onset REM periods on the multiple sleep latency test). Type 1 has cataplexy or hypocretin deficiency; type 2 has neither.

Hypocretin neurons in the lateral hypothalamus stabilise waking; when they are lost, fragments of REM sleep break into it. Hence the classic **narcoleptic tetrad**: sleep attacks, cataplexy (REM atonia while awake), **sleep paralysis** (awake but unable to move on falling asleep or waking) and vivid **hypnagogic hallucinations** while falling asleep.

> **! EXAM TRAP**
>
> **Cataplexy** is a sudden loss of muscle tone triggered by emotion, in full consciousness, and it marks narcolepsy. **Catalepsy** is the reverse, a posture held for long periods with the waxy flexibility of catatonia (Section 8.4).

Parasomnias and restless legs ○

Parasomnias are abnormal behaviours or experiences arising from sleep, and timing identifies them: deep N3 sleep fills the first third of the night, while REM periods lengthen towards morning. The **NREM sleep arousal disorders** are incomplete awakenings, usually from N3, in the **first third of the night**, with little dream recall and **amnesia**; they are commonest in children. In **sleepwalking** (**somnambulism**) the person sits up, walks about or performs complex acts while asleep, blank-faced and hard to wake. In **sleep terrors** (*pavor nocturnus*) the sleeper wakes abruptly with a panicky scream, intense fear and autonomic arousal (racing heart, rapid breathing, sweating, dilated pupils), and cannot be comforted.

REM parasomnias come later. **Nightmare disorder** is repeated, extended, extremely dysphoric dreams, usually in the second half of the night; the dreamer wakes alert and **remembers vividly**. In **REM sleep behaviour disorder** the normal muscle atonia of REM fails, so the person **acts out dream content**, shouting, punching or kicking. Commonest in men over 50, it often precedes Parkinson's disease, dementia with Lewy bodies or multiple system atrophy, often by more than a decade (about 75 per cent of idiopathic cases within 10 to 15 years).

Restless legs syndrome (Willis-Ekbom disease) is an urge to move the legs, with creeping sensations, that starts at rest, is relieved by movement and worsens in the evening. A **substance/medication-induced sleep disorder** follows intoxication or withdrawal. **Kleine-Levin syndrome**, recurrent hypersomnia with hyperphagia and hypersexuality, mostly in adolescent boys, is a separate diagnosis in the International Classification of Sleep Disorders (ICSD-3); DSM-5-TR names it only as an example of other specified hypersomnolence disorder.

> **! EXAM TRAP**
>
> Sitting, walking or complex behaviour while asleep is **somnambulism**; screaming terror is sleep terrors; acting out dreams is **REM sleep behaviour disorder**, from loss of REM atonia, not absence of REM; uncomfortable leg sensations are restless legs. Bedwetting is enuresis; daytime sleepiness alone suggests hypersomnolence.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Pica requires eating non-food substances for at least _____ month.
2. DSM-5 dropped _____ as a criterion for anorexia nervosa.
3. An adult with anorexia and a BMI of 15.5 is rated _____.
4. In bulimia, binges and compensatory behaviour occur at least once a week for _____ months.
5. Calluses on the back of the hand from induced vomiting are _____ sign.
6. The minimum age is _____ years for enuresis and _____ years for encopresis.
7. Sudden loss of muscle tone triggered by laughter, in narcolepsy, is _____.
8. Sleepwalking usually arises from _____ sleep in the first third of the night.

Answers: (1) 1, (2) amenorrhoea, (3) severe, (4) 3, (5) Russell's, (6) 5; 4, (7) cataplexy, (8) N3 (slow-wave NREM).

8.9 Sexual Dysfunctions, Paraphilic Disorders and Gender Dysphoria

LOWER YIELD · 9 QS

Few corners of psychiatry have been redrawn as often. A man who ejaculates too soon, a person aroused by shoes, a child who insists she is a boy and a contented gay man in 1970 have all been diagnosed; only some would be today. We start with normal sexual response, take the three DSM-5-TR groups in turn, and end with homosexuality's exit from the manuals.

The sexual response cycle and the language of gender

Disorder is judged against the normal picture drawn by the gynaecologist **William Howell Masters** (1915 to 2001) and his research partner **Virginia Eshelman Johnson** (1925 to 2013). In *Human Sexual Response* (1966), from laboratory recordings of nearly 700 volunteers, they described a four-phase **sexual response cycle**. In **excitement**, blood engorges the genitals, producing erection or vaginal lubrication. In **plateau**, arousal levels off just before climax.

Orgasm is a peak of rhythmic muscular contractions, with ejaculation in men. In **resolution** the body returns to rest, and men pass through a **refractory period** in which another orgasm is impossible; many women have none and can have multiple orgasms.

Helen Singer Kaplan (1929 to 1995) saw patients whose bodies worked but who did not want sex, and in *Disorders of Sexual Desire* (1979) put a **desire** phase first (desire, excitement, orgasm), giving the linear sequence of Figure 8.12; dysfunctions are still sorted by the phase they disturb. **Rosemary Basson** argued in 2000 that this fits many women in long relationships poorly. In her **circular model** a woman often starts from sexual neutrality, is receptive for the sake of intimacy, is aroused by sexual stimuli and only then feels **responsive desire**; satisfaction feeds back into intimacy. DSM-5 merged women's desire and arousal disorders partly for this reason.

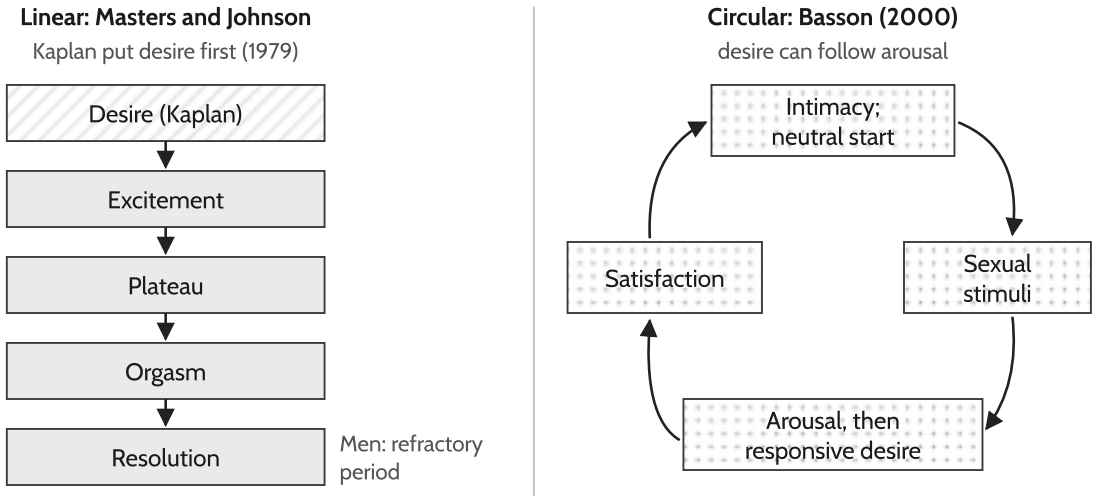


Figure 8.12 Left: Masters and Johnson's four phases with Kaplan's desire phase in front; only men have a refractory period. Right: Basson's circle, in which desire can follow arousal.

Four terms must be kept apart. **Sex** is biological (chromosomes, gonads, hormones, genitals); **gender identity** is the inner sense of being a man, a woman or another gender; **gender role** is the behaviour a culture expects of each gender; **sexual orientation** is the gender of those one is attracted to. **Androgyny**, in the 1974 work of **Sandra Lipsitz Bem** (1944 to 2014) with her Bem Sex Role Inventory, means scoring high on both masculine and feminine traits; it describes a flexible personality, not a disorder.

Sexual dysfunctions ●●

A **sexual dysfunction** is a clinically significant disturbance in the ability to respond sexually or to experience sexual pleasure. Most of DSM-5-TR's eight share one template: the problem occurs on almost all occasions (about 75 to 100 percent), has lasted at least about 6 months, causes clinically significant distress, and is not better explained by another mental disorder, severe relationship distress, a substance or a medical condition. Specifiers give the course: **lifelong** (present since sexual activity began) or **acquired** (after a period of normal function), and **generalised** or **situational** (only with certain partners, stimulation or situations). Table 8.26 gives the eight.

Table 8.26 The DSM-5-TR sexual dysfunctions.

Disorder	Core feature	Note
Delayed ejaculation	Marked delay, infrequency or absence of ejaculation	Older name: inhibited male orgasm
Erectile disorder	Difficulty getting or keeping an erection, or reduced rigidity (1 of 3)	DSM-IV: male erectile disorder
Female orgasmic disorder	Delay, infrequency or absence of orgasm, or weaker orgasm	Specifier: never had an orgasm
Female sexual interest/arousal disorder	At least 3 of 6 signs of low interest or arousal	Merges old desire and arousal disorders
Genito-pelvic pain/penetration disorder	1 or more of 4: penetration difficulty, pain, fear of pain, pelvic-floor tensing	Merges vaginismus and dyspareunia; lifelong or acquired only
Male hypoactive sexual desire disorder	Deficient or absent sexual thoughts, fantasies and desire	Judged against age and context
Premature (early) ejaculation	Within about 1 minute of vaginal penetration and before the person wishes	Severe: under about 15 seconds
Substance/medication-induced	Begins during or soon after intoxication, withdrawal or a medication	No 6-month rule

Vaginismus is involuntary tightening of the vaginal muscles that makes penetration painful or impossible; **dyspareunia** is pain during intercourse. Erectile disorder has the strongest medical ties: diabetes, vascular disease, smoking and age. The substance/medication-induced form follows drugs: selective serotonin reuptake inhibitors (SSRIs) most often delay orgasm or ejaculation, and antipsychotics, antihypertensives and chronic alcohol use are frequent culprits. Masters and Johnson (*Human Sexual Inadequacy*, 1970) described **performance anxiety**, fear of failing, and **spectatoring**, mentally watching and judging one's own performance, so that attention leaves erotic sensation and one failure breeds the next. Relationship conflict, past abuse, depression and religious prohibitions add to the picture. Premature ejaculation is the dysfunction most tied to anxiety, and is commoner in men with anxiety disorders, especially social anxiety disorder.

Treatment is in Section 7.7; here, match tool to problem. The **stop-start technique** of **James Semans** (1956), halting stimulation just before the point of no return, and Masters and Johnson's **squeeze technique**, firm pressure where the head of the penis meets the shaft, teach ejaculatory control. **Sensate focus**, graded non-demanding touch with intercourse at first forbidden, removes performance anxiety in many dysfunctions. Vaginismus is treated with relaxation and graded dilators, erectile disorder often with phosphodiesterase-5 inhibitors such as sildenafil.

> **! EXAM TRAP**
>
> Stop-start and squeeze treat **premature ejaculation**, not erectile disorder, inhibited male orgasm (delayed ejaculation, the opposite problem) or vaginismus. Premature ejaculation is most commonly associated with an **anxiety state**: masturbation does not cause it (a myth that also fuels dhat syndrome, Section 8.1), diabetes is the classic link of erectile disorder, and personality disorder has no special tie.

Paraphilias and paraphilic disorders ●●

DSM-5-TR calls any intense and persistent sexual interest other than in genital stimulation or fondling with physically mature, consenting human partners a **paraphilia**. The key DSM-5 move was to separate interest from disorder. A **paraphilic disorder** is a paraphilia that currently causes distress or impairment, or whose satisfaction has involved harm, or risk of harm, to others. Criterion A gives the interest, recurrent and intense for at least 6 months in fantasies, urges or behaviours; Criterion B the consequences. A shoe fetish that troubles no one is a paraphilia but not a disorder; exposing oneself to strangers meets Criterion B even without distress.

The eight named disorders (Table 8.27) fall into two groups. **Anomalous activity preferences** are the **courtship disorders**, distorted pieces of courtship (looking, showing, touching), and the **algolagnic disorders**, which involve pain. **Anomalous target preferences** are pedophilic, fetishistic and transvestic disorder. Most sets allow the specifiers *in a controlled environment* (such as prison) and *in full remission* (no acting on urges and no distress for 5 years outside such settings).

Table 8.27 The eight DSM-5-TR paraphilic disorders.

Disorder	Recurrent, intense arousal from	Group and notes
Voyeuristic	Observing an unsuspecting person naked, undressing or having sex	Courtship; person at least 18
Exhibitionistic	Exposing one's genitals to an unsuspecting person	Courtship; to children, adults or both
Frotteuristic	Touching or rubbing against a non-consenting person	Courtship
Sexual masochism	Being humiliated, beaten, bound or made to suffer	Algolagnic; with asphyxiophilia
Sexual sadism	Physical or psychological suffering of another	Algolagnic
Pedophilic	Prepubescent children (generally 13 or younger)	Person at least 16 and 5 years older
Fetishistic	Non-living objects or specific non-genital body parts	Not cross-dressing clothes or vibrators
Transvestic	Cross-dressing	With fetishism; with autogynephilia

The fetish object is non-living (shoes, rubber, undergarments) or a specific non-genital body part such as feet, a focus called **partialism**. Other paraphilias go under other specified paraphilic disorder: **necrophilia** (corpses), **zoophilia** (animals), coprophilia (faeces), urophilia (urine), klismaphilia (enemas) and **telephone scatologia** (obscene calls). **Hypoxyphilia**, arousal from restricted breathing, is the 'with asphyxiophilia' specifier of sexual masochism disorder and can kill by accident.

Ray Blanchard coined **autogynephilia** in 1989: a male's recurrent, intense sexual arousal by the thought or image of himself as a woman. It anchors his controversial typology of male-to-female gender dysphoria, and in men with transvestic disorder it raises the likelihood of gender dysphoria.

> **! EXAM TRAP**
>
> Autogynephilia is arousal by the image of **oneself as a woman**; arousal by prepubescent children is pedophilia, by being humiliated is masochism, and mere arousal by oneself is not autogynephilia. An **inanimate object** means fetishism: necrophilia needs a corpse, frotteurism a non-consenting person, voyeurism an unsuspecting one.

Why do paraphilias form? Learning theorists cite **classical conditioning**: **Stanley Rachman** (1966) conditioned mild arousal to a picture of women's boots in the laboratory. **John Money** (1921 to 2006) proposed the **lovemap**, a childhood template of the ideal lover and erotic scene that early experience can distort. **Kurt Freund** (1914 to 1996), in his **courtship disorder hypothesis**, saw voyeurism, exhibitionism, frotteurism and preferential rape as disturbances of successive courtship phases (searching, pre-touch interaction, touching, intercourse), which is why they cluster.

Gender dysphoria and gender incongruence ●●

Gender dysphoria in DSM-5-TR is a marked incongruence between one's experienced or expressed gender and one's assigned gender, lasting at least 6 months and causing clinically significant distress or impairment. A child needs six of eight indicators, one of which must be a strong desire to be, or insistence that one is, the other gender; others include cross-gender clothing, play and playmates and dislike of one's anatomy. An adolescent or adult needs two of six, such as a strong desire to be rid of one's sex characteristics or to be treated as the other gender. Specifiers: 'with a disorder/difference of sex development' and, for adults, 'posttransition'.

DSM-III (1980) introduced transsexualism for adults (a historic term for a person seeking social and often bodily transition to the other sex) and gender identity disorder of childhood. DSM-IV (1994) merged them into **gender identity disorder**, a condition in which one's experienced gender identity does not match one's assigned sex. That made the identity itself the disorder, so DSM-5 (2013) renamed it gender dysphoria, moving the focus from identity to distress. ICD-10 listed transsexualism (F64.0), dual-role transvestism (F64.1) and gender identity disorder of childhood (F64.2) under F64.

ICD-11, in force from 1 January 2022, renamed the condition **gender incongruence**, of adolescence or adulthood and of childhood (about 2 years' persistence in children), and moved it out of the mental disorders chapter into a new Chapter 17, **Conditions related to sexual health**: being transgender is not a mental illness, but a code lets people obtain gender-affirming care. The sexual dysfunctions and sexual pain disorders moved there too. The paraphilic disorders stayed in Chapter 6, now built around non-consent: exhibitionistic, voyeuristic, pedophilic, **coercive sexual sadism disorder** and frotteuristic, plus other paraphilic disorder involving non-consenting individuals and paraphilic disorder involving solitary or consenting individuals with marked distress or risk. Consensual fetishism, fetishistic transvestism and sadomasochism were removed (Table 8.28). DSM-5's gender dysphoria corresponds to ICD-11's gender incongruence.

Table 8.28 Where the sexuality and gender groups went in ICD-11.

Group	ICD-10	ICD-11
Gender identity disorders	F64 (Chapter V)	Gender incongruence, Chapter 17

Group	ICD-10	ICD-11
Sexual dysfunctions	F52 (Chapter V)	Chapter 17
Vaginismus, dyspareunia	F52.5, F52.6	Sexual pain-penetration disorder, Chapter 17
Disorders of sexual preference	F65	Paraphilic disorders, stay in Chapter 6
Fetishism, fetishistic transvestism, sadomasochism	F65.0, F65.1, F65.5	Removed
Egodystonic sexual orientation	F66.1	Removed

> **! EXAM TRAP**
>
> ICD-11 gender incongruence is in **Chapter 17, Conditions related to sexual health**: not Chapter 6, not Chapter 5 (endocrine, nutritional or metabolic diseases), and not deleted. Only the paraphilic disorders stayed in Chapter 6. The same mismatch is gender identity disorder in DSM-IV and ICD-10 and gender dysphoria in DSM-5.

Gender identity is distinct from a paraphilia (what arouses a person), a sexual dysfunction (the response cycle) and androgyny (a blend of traits). Practice today is gender-affirmative, supporting identity rather than trying to change it (Chapter 7).

Homosexuality: pathologised, then depathologised ○

Pathologisation is classifying a behaviour or identity as a mental disorder; **depathologisation** is removing that classification (not changing orientation, the aim of discredited conversion therapies, nor coming out). Homosexuality is the clearest case. DSM-I (1952) placed it under **sociopathic personality disturbance** as a sexual deviation; DSM-II (1968) kept it among the sexual deviations. **Alfred Charles Kinsey** (1894 to 1956), in *Sexual Behavior in the Human Male* (1948), rated behaviour on the **Kinsey scale** from 0 (exclusively heterosexual) to 6 (exclusively homosexual), showing same-sex experience to be common and graded.

> **KEY STUDY: HOOKER'S BLIND JUDGES (1957)**
>
> **Evelyn Hooker** (1907 to 1996) asked whether homosexual men are more maladjusted. She matched 30 homosexual and 30 heterosexual men, none in therapy, for age, IQ and education and gave them projective tests, including the Rorschach. Expert judges blind to group could not tell them apart beyond chance, and rated their adjustment alike. Criticism: a small, non-random sample; but earlier claims had rested on patients in treatment.

With this evidence and activist pressure, the board of the American Psychiatric Association voted in December 1973 to remove homosexuality. The seventh printing of DSM-II (1974) substituted **sexual orientation disturbance** for people distressed by their orientation; DSM-III (1980) renamed it **ego-dystonic homosexuality**, and DSM-III-R (1987) dropped it. ICD-10 (endorsed in 1990) no longer listed homosexuality, stating that orientation by itself is not a disorder, but kept F66, including egodystonic sexual orientation; ICD-11 deleted F66.

INDIAN CONTEXT

In 2018 the Indian Psychiatric Society stated that homosexuality is not a psychiatric disorder. On 6 September 2018 the Supreme Court, in *Navtej Singh Johar v. Union of India*, read down Section 377 of the Indian Penal Code, decriminalising consensual sex between adults of the same sex.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. After orgasm, men pass through a _____ period.
2. In Basson's model, many women feel _____ desire, which follows arousal.
3. Most sexual dysfunctions need the problem on about _____ percent of occasions.
4. Stop-start and squeeze techniques treat _____.
5. Arousal from a specific non-genital body part is _____.
6. Blanchard's term for a male's arousal by the image of himself as a woman is _____.
7. ICD-11 places gender incongruence in Chapter _____.
8. Ego-dystonic homosexuality was dropped in _____ (1987).

Answers: (1) refractory, (2) responsive, (3) 75 to 100, (4) premature ejaculation, (5) partialism, (6) autogynephilia, (7) 17, (8) DSM-III-R.

8.10 Substance-Related, Addictive and Neurocognitive Disorders

MEDIUM YIELD · 36 QS

A man who has drunk heavily for twenty years is admitted for surgery. On his third dry day he is trembling, picking at the sheets and swatting at insects only he can see. Months later, sober, he forgets a nurse within minutes and fills the gap with invented stories. He spans both halves of this section: the disorders drugs produce, and the failures of cognition that drugs, disease and age produce.

The language of substance use ●●

A **psychoactive substance** is a drug that alters mood, cognition, perception or behaviour by acting on the brain; the word has nothing to do with a drug-induced hallucination or an "active psychopath". **Intoxication** is the reversible syndrome of recent use. **Withdrawal** is the syndrome that follows stopping or cutting down after heavy, prolonged use, and it usually mirrors intoxication: a sedative leaves an agitated brain behind, a stimulant a slowed one.

Tolerance is needing markedly more of a drug for the same effect: metabolic tolerance when the liver breaks the drug down faster, pharmacodynamic tolerance when receptors and neurons adapt. **Cross-tolerance** is tolerance that extends to other drugs, especially of the same class, so a heavy drinker needs large doses of benzodiazepines or barbiturates. In **cross-dependence** one drug suppresses another's withdrawal. Together these are why alcohol-dependent patients are detoxified with a long-acting benzodiazepine such as chlordiazepoxide, and

why methadone can replace heroin. **Sensitisation** (reverse tolerance) is the opposite: repeated cocaine or amphetamine use brings paranoia or seizures at ever lower doses.

Physical dependence shows itself in a withdrawal syndrome; psychological dependence is a compelling need to use for pleasure or relief. Addiction, compulsive drug seeking despite harm, is different again: a cancer patient on morphine is physically dependent but not addicted. **Craving** is an intense urge to use, often cued by places and objects linked to the drug, and relapse is a return to use after abstinence (Chapter 7).

ICD-10 codes these disorders F10 (alcohol) to F19 (multiple drugs), and the digit after the point gives the clinical pattern: .0 acute intoxication, .1 **harmful use** (use that is actually damaging health), .2 dependence syndrome, .3 withdrawal state, .4 withdrawal with delirium, .5 psychotic disorder, .6 amnesic syndrome and .7 residual and late-onset psychotic disorder. The **dependence syndrome** was described for alcohol in 1976 by **Griffith Edwards** and **Milton Gross**: a narrowed drinking repertoire, drink given priority over everything else, tolerance, repeated withdrawal, drinking to relieve it, a felt compulsion, and rapid reinstatement after abstinence.

DSM-IV split this ground into substance abuse and substance dependence. DSM-5 (2013) and DSM-5-TR (2022) merge them into one **substance use disorder**: at least 2 of 11 criteria within 12 months, in four groups. Impaired control covers using more or longer than intended, failed efforts to cut down, much time spent, and craving; social impairment covers failed roles, interpersonal problems and activities given up; risky use covers hazardous situations and use despite known harm; the pharmacological criteria are tolerance and withdrawal. Severity is mild with 2 or 3, moderate with 4 or 5, and severe with 6 or more. Craving was added and legal problems dropped. "Addiction" is not a DSM-5-TR diagnosis.

DSM-5-TR also includes **gambling disorder** (four or more of nine criteria in 12 months, such as chasing losses) as its only behavioural addiction; internet gaming disorder is a condition for further study. ICD-11's block "disorders due to substance use or addictive behaviours" codes each drug as an episode of harmful use, a harmful pattern of use, or dependence, and adds gambling disorder and **gaming disorder**: gaming with impaired control, given priority over other interests despite harm.

> **! EXAM TRAP**
>
> "Drug abuse" is not an ICD-10 pattern: abuse was a DSM-IV category, and ICD-10's nearest term is harmful use. Gaming disorder sits in ICD-11's substance use and addictive behaviours block, not under stress-associated, mood or impulse control disorders. Cross-tolerance is tolerance carried over to a related drug, not freedom from addiction.

Why drugs capture the brain ●●

Every drug of abuse, by its own route, ends up raising dopamine in the **mesolimbic dopamine system**, from the ventral tegmental area to the nucleus accumbens in the ventral striatum (Chapter 5). It is the circuit most consistently altered in substance use disorders; the cerebellar vermis (gait), lateral geniculate nucleus (vision) and vestibular nuclei (balance) are not reward structures. Drugs are positive reinforcers because they give pleasure and negative reinforcers because they remove withdrawal, anxiety or pain (Chapter 1).

In the **opponent-process theory** of **Richard Solomon** and **John Corbit** (1974), every drug high (the a-process) triggers an opposing low (the b-process). With repetition the b-process strengthens: the high shrinks (tolerance), the crash deepens (withdrawal), and the user takes the drug to escape misery rather than to feel good.

Using Swedish adoption records in 1981, **C. Robert Cloninger** and colleagues separated type I alcoholism (onset after about 25, guilt, binges with loss of control, both sexes, shaped by environment) from type II (early onset, novelty seeking, antisocial behaviour, inability to abstain, mainly men, highly heritable). **Sigmund Freud** (1856 to 1939) saw alcohol-dependent people as fixated at the oral stage (Chapter 4). **Elvin Morton Jellinek** (1890 to 1963) argued in *The Disease Concept of Alcoholism* (1960) that alcoholism is a disease with five "species": alpha (psychological reliance, no loss of control), beta (physical damage without dependence), gamma (tolerance, withdrawal and loss of control: once started, the drinker cannot stop), delta (inability to abstain) and epsilon (periodic binges).

KEY STUDY: KANDEL'S STAGES OF DRUG USE (1975)

Denise Kandel surveyed New York State high-school students twice and found a regular order: beer or wine, then cigarettes or spirits, then marijuana, then other illicit drugs. Few reached heroin or cocaine without marijuana, so in this **gateway hypothesis** marijuana is the drug most often cited as the gateway, although alcohol and tobacco come first. Criticism: a sequence is not a cause, and shared risk factors could explain it.

Drugs differ in addictive liability, the chance that a user becomes dependent. Nicotine (tobacco) ranks highest: a larger share of people who try it become dependent than of those who try heroin, cocaine, alcohol or marijuana.

The ten substance classes ●●

DSM-5-TR recognises ten classes (Table 8.29).

Table 8.29 The ten DSM-5-TR substance classes.

Class	Examples	Intoxication	Withdrawal
Alcohol	Arrack, toddy, beer	Slurred speech, ataxia, nystagmus	Tremor, sweating, seizures, hallucinations
Caffeine	Coffee, tea	Restlessness, insomnia, tachycardia	Headache, fatigue
Cannabis	Bhang, ganja, charas	Red eyes, hunger, dry mouth	Irritability, poor sleep, low appetite
Hallucinogens; phencyclidine	LSD, psilocybin, mescaline; PCP	Perceptual changes, dilated pupils; PCP: nystagmus, violence	Not defined
Inhalants	Glue, petrol, whitener	Dizziness, nystagmus, euphoria	Not defined
Opioids	Morphine, heroin	Pinpoint pupils, slowed breathing	Yawning, tears, dilated pupils, diarrhoea
Sedatives, hypnotics, anxiolytics	Benzodiazepines, barbiturates	As for alcohol	As for alcohol; seizures
Stimulants	Amphetamines, cocaine	Dilated pupils, agitation, paranoia	Crash: dysphoria, fatigue, hunger

Class	Examples	Intoxication	Withdrawal
Tobacco	Cigarettes, bidis, gutka	No diagnosis	Irritability, restlessness, hunger
Other or unknown	Anabolic steroids, betel nut	Varies	Varies

Alcohol is a positive modulator (functionally an agonist) at GABA-A receptors and an antagonist at NMDA glutamate receptors; it is not a serotonin, acetylcholine or endorphin agonist. DSM-5-TR notes that withdrawal begins within about 4 to 12 hours of the last drink, peaks on the second day and eases markedly by the fourth or fifth. Its complications keep only a rough clock (Figure 8.13): tremor, sweating and insomnia within the first day (roughly 6 to 24 hours); seizures at about 12 to 48 hours; **alcoholic hallucinosis**, hallucinations (usually voices) in clear consciousness, at about 12 to 48 hours; and **delirium tremens** at about 48 to 72 hours, with clouded consciousness, fever, racing pulse, coarse tremor and vivid visual and tactile hallucinations of insects, small animals or tiny (Lilliputian) people. Fewer than 1 in 10 people in withdrawal reach this stage, but it can kill, so benzodiazepines are given to prevent it, with thiamine against Wernicke's encephalopathy.

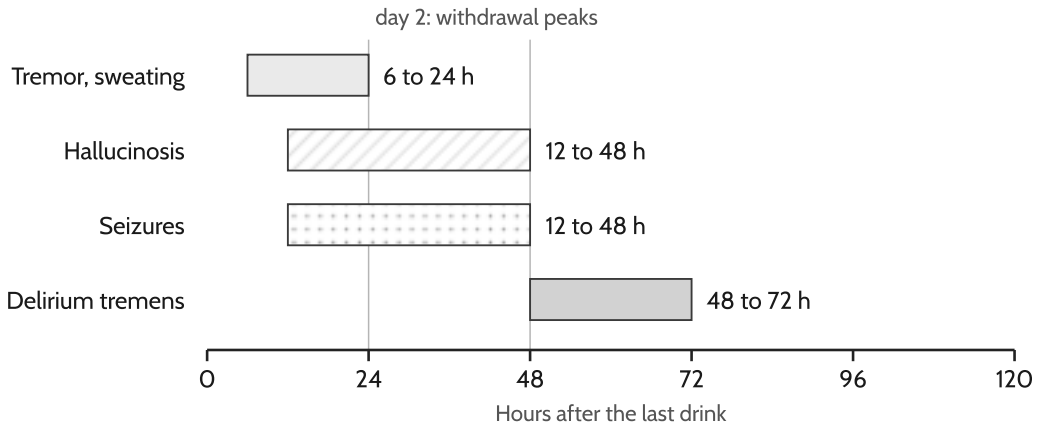


Figure 8.13 Approximate onset windows of alcohol withdrawal features. Tremor comes first and delirium tremens, the most dangerous, last; timing varies between people.

Chronic drinkers may develop alcoholic jealousy, a delusion that the partner is unfaithful; fetal alcohol syndrome is in Section 8.3, and the CAGE and AUDIT questionnaires in Chapter 2. **Disulfiram** inhibits aldehyde dehydrogenase, so acetaldehyde accumulates after a drink and causes flushing, nausea and palpitations. Naltrexone, an opioid antagonist, blunts alcohol's reward and craving, and acamprosate, acting on glutamate, eases protracted withdrawal.

Opium's natural alkaloids are morphine, codeine, thebaine, papaverine and noscapine. Heroin (diacetylmorphine) is made from morphine, so it is semi-synthetic; methadone, pethidine, fentanyl and tramadol are synthetic. Intoxication brings pinpoint pupils and respiratory depression, reversed by naloxone. Withdrawal is miserable but rarely dangerous: yawning, lacrimation, rhinorrhoea, dilated pupils, piloerection ("cold turkey"), diarrhoea and aches. Methadone or buprenorphine substitution stabilises dependent users.

Cocaine blocks dopamine reuptake and amphetamines release dopamine, giving euphoria, energy and, at high doses, paranoia; amphetamine psychosis can resemble paranoid schizophrenia, and a dysphoric crash follows a binge. MDMA (ecstasy) is an amphetamine derivative that releases serotonin. **Formication**, tactile hallucinations of insects crawling on or under the skin ("cocaine bugs"), is most characteristic of stimulant intoxication and of alcohol or sedative withdrawal, not of depression, phobia or generalised anxiety.

Cannabis acts through delta-9-tetrahydrocannabinol (THC); in India it is used as bhang (leaves), ganja (flowering tops) and charas (resin). Heavy use is linked to an **amotivational syndrome** of apathy and to a raised risk of psychosis. LSD (lysergic acid diethylamide), whose effects **Albert Hofmann** (1906 to 2008) discovered in 1943, psilocybin and mescaline alter perception in clear consciousness; some users later re-experience trails and halos (hallucinogen persisting perception disorder, or flashbacks). Inhalants intoxicate like alcohol and damage organs. Nicotine acts on nicotinic acetylcholine receptors; caffeine use disorder is only a condition for further study. Benzodiazepines and barbiturates are cross-tolerant with alcohol, and their withdrawal can also bring seizures and delirium.

> **! EXAM TRAP**
>
> Alcohol is a GABA agonist. Heroin is the odd one out among opium alkaloids because it is semi-synthetic. Opioid intoxication gives pinpoint pupils, withdrawal dilated ones. Disulfiram works by aversion through aldehyde dehydrogenase; reducing reward is naltrexone's action and glutamate modulation acamprosate's.

Neurocognitive disorders: terms and domains ●●

ICD-10 groups disorders of failing cognition as organic mental disorders (F00 to F09). DSM-5 renamed them **neurocognitive disorders** (NCDs), acquired declines from a previous level. It kept delirium and replaced dementia and amnestic disorder with **major neurocognitive disorder**, a significant decline in one or more of six domains (complex attention, executive function, learning and memory, language, perceptual-motor function, social cognition) that costs independence in daily life, and mild NCD, a modest decline with independence kept. One impaired domain is enough, so pure amnesia now counts as major NCD.

Older terms name the function each disorder chiefly affects: dementia, a global decline that includes abstraction and judgement; amnesia, memory; aphasia, alexia and acalculia, language and symbolic function; apraxia, skilled purposeful movement (motor skills); and agnosia, recognition despite intact senses (Chapter 5).

Delirium ●●●

Delirium is a disturbance of attention and awareness (clouded consciousness) that develops over hours to days, fluctuates through the day, and brings disorientation, memory loss or perceptual disturbance, often visual hallucinations. The sleep-wake cycle may reverse, and confusion worsens in the evening (**sundowning**). DSM-5-TR specifies acute or persistent delirium and hyperactive, hypoactive (quiet and easily missed) or mixed types. Motor signs include **asterixis** (a flapping tremor of outstretched hands), multifocal myoclonus, tremor, abnormal tone and reflexes, and **carphologia** or floccillation, picking at the bedclothes: all can occur. The electroencephalogram (EEG) shows diffuse slowing.

Delirium always has a physical cause: infection, metabolic or electrolyte disturbance, hypoxia, drugs, intoxication or withdrawal, or surgery. It affects 11 to 51 percent of older people after operations, and it is the commonest psychiatric syndrome after cardiac surgery, ahead of depression, anxiety and schizophrenia. Table 8.30 compares it with dementia and with depressive pseudodementia.

Table 8.30 Delirium, dementia and depressive pseudodementia.

Feature	Delirium	Dementia	Pseudodementia
Onset	Acute, hours to days	Insidious	Fairly abrupt
Course	Fluctuating, worse at night	Progressive	Follows the mood
Attention, consciousness	Impaired, clouded	Clear until late	Clear
Memory and answers	Impaired; distractible	Impaired; near misses, hides deficits	Complains of poor memory; "don't know"
Hallucinations	Common, visual	Can occur	Uncommon
EEG	Diffuse slowing	Normal or mildly slow	Normal
Outcome	Usually reversible	Usually irreversible	Improves with antidepressants

! EXAM TRAP

The most reliable distinction is acute onset, fluctuating course and disturbed attention or consciousness in delirium versus impaired memory with clear consciousness in dementia. Hallucinations, delusions and memory loss occur in both, and both are commonest in the elderly, so age does not separate them. Delirium, not dementia, is usually reversible. Confabulation points to Korsakoff's syndrome, early aphasia and apraxia to Alzheimer's.

Subtypes of major and mild NCD ●●●

KEY STUDY: AUGUSTE D. (1901 TO 1906)

In 1901 **Alois Alzheimer** (1864 to 1915) examined **Auguste Deter** (1850 to 1906), 51, admitted to the Frankfurt asylum with failing memory, jealousy and disorientation. Asked to write her name, she said "I have lost myself". After her death he described, in 1906, plaques between her neurons and tangles inside them. Her youth made the disease seem presenile; it is now the commonest cause of dementia at any age.

Alzheimer's disease begins insidiously with loss of episodic memory, then aphasia, apraxia and visuospatial deficits. Loss of cholinergic neurons in the nucleus basalis of Meynert causes acetylcholine deficiency, the target of cholinesterase inhibitors (Chapter 5). "Senile dementia" (onset after 65) and cerebral atrophy are older descriptive labels, not specific diseases. Table 8.31 gives the other subtypes.

Table 8.31 Subtypes of major and mild neurocognitive disorder.

Subtype	Hallmarks	Pathology or cause

Subtype	Hallmarks	Pathology or cause
Alzheimer's disease	Insidious; memory, then language and visuospatial loss	Amyloid plaques, tau tangles; apolipoprotein E4 gene; early in Down syndrome
Vascular	Sudden or stepwise; focal signs	Multiple infarcts; Binswanger's small-vessel disease
Frontotemporal (Pick's disease)	Onset often in 50s; personality change, disinhibition or language loss; memory relatively spared	Frontal and temporal degeneration
Lewy bodies	Fluctuating cognition, well-formed visual hallucinations, parkinsonism; REM sleep behaviour disorder; antipsychotic sensitivity	Alpha-synuclein Lewy bodies
Parkinson's disease	Parkinsonism a year or more before cognitive decline	Loss of nigral dopamine neurons
Huntington's disease	Chorea, depression, subcortical dementia	Autosomal dominant CAG repeat, chromosome 4; caudate atrophy
Prion disease	Rapid decline; myoclonus	Creutzfeldt-Jakob disease
Others	HIV, brain injury, substances or medication	Infection, trauma, toxicity

Pick's disease was described by **Arnold Pick** (1851 to 1924) in 1892, and Huntington's disease by **George Huntington** (1850 to 1916) in 1872. Huntington's is autosomal dominant, unlike Down syndrome (trisomy 21), phenylketonuria (recessive) or fragile X (an X-linked repeat), which are in Section 8.3. Cortical dementias (Alzheimer's, frontotemporal) produce aphasia, apraxia, agnosia and amnesia; **subcortical dementia** (Huntington's, Parkinson's, HIV) produces slowing, apathy, depression and motor signs without aphasia. Alzheimer's, Parkinson's, Huntington's, frontotemporal degeneration and amyotrophic lateral sclerosis are neurodegenerative, with progressive neuron death. Epilepsy, a tendency to recurrent seizures, is not; multiple sclerosis is a demyelinating disease.

Pseudodementia, a term popularised by **Leslie Kiloh** in 1961, is cognitive impairment that mimics dementia but has a psychiatric cause, most commonly depression (not schizophrenia, mania or hypochondriasis). Clues: fairly abrupt onset, depressed mood, a patient who complains of poor memory and answers "I don't know", and improvement with antidepressants.

> **! EXAM TRAP**
>
> Lewy body NCD means visual, not auditory, hallucinations; aphasia belongs to Alzheimer's, and loss of smell is only a non-specific early sign. Insidious memory, language and visuospatial decline is Alzheimer's; stepwise decline is vascular.

Wernicke-Korsakoff syndrome ●●●

Return to the opening drinker. Heavy drinkers eat poorly and absorb thiamine (vitamin B1) badly, and neurons need thiamine to use glucose. Acute deficiency causes **Wernicke's encephalopathy**, described by **Carl Wernicke** (1848 to 1905) in 1881: confusion, ataxia, and ophthalmoplegia with nystagmus, an emergency treated with thiamine before glucose. Untreated, it leaves **Korsakoff's syndrome** (Korsakoff's psychosis), described by **Sergei Korsakoff** (1854 to 1900) in 1887: severe anterograde and retrograde amnesia, apathy, and **confabulation**, fluent false accounts filling memory gaps without intent to deceive.

Immediate memory and intelligence are relatively spared. Together these form Wernicke-Korsakoff syndrome, with lesions in the mammillary bodies and dorsomedial thalamus.

> **! EXAM TRAP**
>
> Memory loss, confabulation and nystagmus in a heavy drinker is Wernicke-Korsakoff syndrome. The cause is thiamine deficiency, not vitamin B12, vitamin D, folate, hypoxia or lead. Alzheimer's, Pick's, Parkinson's, Huntington's and Creutzfeldt-Jakob diseases are not caused by drinking.

Disorders due to another medical condition ○

Physical disease can present psychiatrically, so DSM-5-TR has diagnoses such as depressive disorder due to another medical condition (ICD-10 F06 and F07, the "organic" syndromes). Cushing's syndrome (excess cortisol) mainly causes depression, anxiety and cognitive impairment, occasionally mania or psychosis; Addison's disease (cortisol deficiency) causes apathy and depression. Hypothyroidism causes depression and, when severe, a psychosis called "myxoedema madness"; hyperthyroidism causes anxiety. Vitamin B12 and folate deficiency, neurosyphilis (general paresis, with grandiose delusions), HIV and systemic lupus erythematosus can all cause mood change, psychosis or dementia. Epilepsy can bring an interictal psychosis, and head injury lasting personality change. Frontal damage produces the **frontal lobe syndrome**, an organic personality disorder of disinhibition, apathy and poor judgement (Chapter 5).

> **SCAN YOUR BRAIN**
>
> Fill in the blanks. The answers are below; cover them first.
>
> 1. Tolerance that extends to a related drug is called _____.
> 2. A substance use disorder with 6 or more criteria is rated _____.
> 3. The ventral tegmental area to nucleus accumbens pathway is the _____ dopamine system.
> 4. Heroin is not a natural opium alkaloid because it is _____.
> 5. Disulfiram inhibits _____.
> 6. Picking at the bedclothes in delirium is called _____.
> 7. Visual hallucinations, fluctuating cognition and parkinsonism define NCD with _____.
> 8. Korsakoff's syndrome results from deficiency of _____.
>
> **Answers:** (1) cross-tolerance, (2) severe, (3) mesolimbic, (4) semi-synthetic, (5) aldehyde dehydrogenase, (6) carphologia, (7) Lewy bodies, (8) thiamine (vitamin B1).

8.11 Personality Disorders, Conduct and Impulse-Control Disorders

MEDIUM YIELD · 32 QS

Most disorders in this chapter come as episodes. The conditions here describe how a person habitually thinks, feels and relates, year after year. We define personality disorder, sort the

ten types into clusters, study borderline and antisocial personality closely, meet the dimensional models, and end with the disorders of conduct and impulse control.

When personality becomes a disorder ○

Everyone has personality traits, enduring ways of perceiving, relating to and thinking about the world and oneself; they become a **personality disorder** (PD) only when inflexible and maladaptive. DSM-5-TR defines a PD as an enduring pattern of inner experience and behaviour that deviates markedly from the expectations of the person's culture and shows in two or more of four areas: **cognition** (how self, others and events are perceived), **affectivity** (the range, intensity, lability and appropriateness of emotion), **interpersonal functioning** and **impulse control**. The pattern is pervasive and inflexible, stable, traceable to adolescence or early adulthood, causes distress or impairment, and is not better explained by another disorder, a substance or a medical condition. A PD can therefore disturb thinking, mood and relationships alike.

The traits are usually **ego-syntonic**: they feel like "just me", so informants become essential. The median prevalence of any PD is about 10 percent (10.5 percent in DSM-5-TR's review). Diagnosis before 18 needs features present for at least a year, because childhood traits often fade; antisocial PD is never diagnosed before 18. DSM-IV recorded PDs on **Axis II** (Section 8.1). DSM-5-TR keeps the **categorical model**, PDs as distinct syndromes, while noting the **dimensional model**, PDs as extreme variants of normal traits that shade into normality (see the newer models).

The ten disorders in three clusters ●●

DSM-5-TR groups its ten specific PDs into three **clusters** by descriptive similarity (Table 8.32): **Cluster A**, odd or eccentric; **Cluster B**, dramatic, emotional or erratic; **Cluster C**, anxious or fearful (weird, wild, worried). The clusters are poorly validated: one person often shows traits from several.

Table 8.32 The ten DSM-5-TR personality disorders.

Cluster	Disorder	Core pattern	Criteria needed
A: odd, eccentric	Paranoid	Pervasive distrust and suspiciousness; motives read as malevolent	4 of 7
A	Schizoid	Detachment from relationships, restricted emotional expression; prefers solitude	4 of 7
A	Schizotypal	Acute discomfort in close relationships, cognitive and perceptual distortions, eccentric behaviour	5 of 9
B: dramatic, emotional, erratic	Antisocial	Disregard for and violation of others' rights since age 15	3 of 7
B	Borderline	Instability of relationships, self-image and affects; marked impulsivity	5 of 9
B	Histrionic	Excessive emotionality and attention seeking	5 of 8
B	Narcissistic	Grandiosity, need for admiration, lack of empathy	5 of 9
C: anxious, fearful	Avoidant	Social inhibition, inadequacy, hypersensitivity to negative evaluation	4 of 7

Cluster	Disorder	Core pattern	Criteria needed
C	Dependent	Excessive need to be taken care of; submissive, clinging, fears separation	5 of 8
C	Obsessive-compulsive	Orderliness, perfectionism and control, at the expense of flexibility and efficiency	4 of 8

The schizotypal person shows **ideas of reference** (believing that unrelated events, such as strangers laughing, refer to oneself, short of delusion), **magical thinking** (telepathy, a "sixth sense"), odd speech and eccentric dress. It is genetically linked to schizophrenia, so DSM-5-TR also lists it in the schizophrenia spectrum (Section 8.4) and ICD-10 classes it there. DSM-5-TR adds **personality change due to another medical condition** (a lasting change caused by, say, a frontal lobe lesion), and other specified and unspecified PD.

> ! EXAM TRAP
>
> There is **no Cluster D. Avoidant PD is Cluster C**, not A: the schizoid person does not want company; the avoidant person longs for it but fears rejection. Among paranoid, schizoid, schizotypal and borderline, the odd one out is **borderline** (Cluster B).

Borderline personality disorder ●●●

Borderline personality disorder (BPD) is a pervasive pattern of instability of interpersonal relationships, self-image and affects, with marked impulsivity, beginning by early adulthood. Its signature is emotional instability and impulsivity, not detachment, perfectionism or paranoid ideation. Five or more of nine criteria are needed:

1. Frantic efforts to avoid real or imagined abandonment.
2. Unstable, intense relationships alternating between idealisation and devaluation.
3. Identity disturbance: a markedly and persistently unstable self-image.
4. Impulsivity in at least two self-damaging areas (spending, sex, substances, reckless driving, binge eating).
5. Recurrent suicidal behaviour, gestures or threats, or self-mutilation.
6. Affective instability from marked reactivity of mood, episodes lasting hours, rarely days.
7. Chronic feelings of emptiness.
8. Inappropriate, intense anger or difficulty controlling anger.
9. Transient, stress-related paranoid ideation or severe dissociation.

Adolph Stern coined "borderline" in 1938 for patients between neurosis and psychosis. **Otto F. Kernberg** (born 1928) recast it as **borderline personality organisation**: identity diffusion, primitive defences and generally intact reality testing. The primary defence is **splitting**: self and others are seen as all good or all bad, so today's idealised rescuer becomes tomorrow's betrayer. With it come **projective identification** (disowned feelings are placed in another person, who is pressed into feeling them), primitive idealisation and devaluation. Repression (pushing wishes out of awareness), sublimation (channelling them into valued activity), rationalisation (plausible excuses), regression (retreat to earlier behaviour) and undoing (an act that symbolically cancels another) are not the BPD defence (Chapter 4).

Heinz Kohut (1913 to 1981), founder of **self psychology**, held that a child needs **mirroring** (a parent's delighted reflection of its worth) and a calm figure to **idealise**; when both fail, the self stays fragmented, felt as emptiness and being "no one". He built the theory mainly to explain narcissism. **Marsha M. Linehan** (born 1943) proposed the **biosocial theory**: biological emotional vulnerability meets an invalidating environment that dismisses the child's feelings, so regulation is never learned. Treatments (Chapter 7): her dialectical behaviour therapy, mentalisation-based treatment, Kernberg's transference-focused psychotherapy and schema therapy. ICD-10 calls BPD **emotionally unstable personality disorder**, impulsive or borderline type.

Two differentials. BPD mood shifts last hours and follow interpersonal triggers; bipolar episodes last days to weeks, and both may coexist. Conduct disorder is diagnosed under 18 and centres on rule violation and aggression; BPD is an adult pattern of identity and affective instability and fear of abandonment, and childhood conduct disorder leads to ASPD, not BPD.

CLINICAL LINK

BPD is diagnosed more often in women in clinical samples, though not in community samples. Self-harm is common; about 6 percent of hospitalised patients died by suicide over 24 years.

! EXAM TRAP

BPD mood is reactive and unpredictable, never "regular and predictable"; eccentric, odd thinking is schizotypal; cruelty to animals and rule-breaking before 15 are conduct disorder. A 2023 paper asking which is NOT a BPD feature offered both "regular and predictable changes in mood" and "eccentric behaviour with odd thinking"; both are strictly correct, but the key is the mood option, the direct inversion of BPD's affective instability. A 2024 paper keys "failure of parental mirroring and idealising, leading to a fragmented self" under *object relations theory*: strictly it is Kohut's **self psychology**, a cousin of the object relations school of **Melanie Klein** (1882 to 1960) and Kernberg. Choose it; oral fixation, Oedipal conflict and a harsh superego are classical Freudian answers.

Antisocial personality disorder and psychopathy ●●

Antisocial personality disorder (ASPD) is a pervasive pattern of disregard for and violation of the rights of others, occurring since age 15, shown by three or more of seven criteria: failure to conform to laws; deceitfulness and conning; impulsivity; irritability and aggressiveness; reckless disregard for safety; consistent irresponsibility; and lack of remorse. The person must be at least 18, with evidence of **conduct disorder beginning before 15** (Figure 8.14). ICD-10 calls it **dissocial personality disorder**. Antisocial behaviour often remits by about age 40.

Philippe Pinel (1745 to 1826) described **manie sans délire** (mania without delirium) in 1801: impulsive rages with intact reasoning. **James Cowles Prichard** (1786 to 1848) named **moral insanity** in 1835. **Hervey Cleckley** (1903 to 1984), in *The Mask of Sanity* (1941), listed 16 characteristics of the psychopath, among them superficial charm, absence of nervousness, untruthfulness and lack of remorse: a normal mask over an emotional void. **Robert D. Hare** (born 1934) developed the **Psychopathy Checklist-Revised** (PCL-R, 1991), built on Cleckley: 20

items scored 0 to 2 from interview and records (total 40; 30 or more is the usual cut-off for psychopathy), with an interpersonal-affective factor (glibness, grandiosity, callousness) and an antisocial-lifestyle factor (impulsivity, irresponsibility, early behaviour problems).

Psychopathy is defined by traits, ASPD mostly by behaviour: most men in prison meet ASPD criteria, but only a minority are psychopaths. Causes combine genes with adverse rearing (abuse, neglect, erratic discipline). In *Deviant Children Grown Up* (1966), **Lee Nelken Robins** (1922 to 2009) found that adult sociopathy almost never arose without childhood antisocial behaviour: the root of the conduct-disorder rule. The underarousal hypothesis adds that psychopaths have low autonomic arousal (low resting heart rate and skin conductance) and seek stimulation.

KEY STUDY: LYKKEN'S FEARLESS PSYCHOPATHS (1957)

David Thoreson Lykken (1928 to 2006) had primary psychopaths, other prisoners and controls learn a "mental maze" in which some wrong choices brought a shock. Psychopaths learned the maze as well but avoided the shocked errors less, reported less anxiety and showed smaller skin conductance responses to a signal paired with shock. Conclusion, the **fearlessness hypothesis**: punishment teaches psychopaths little. Criticism: small samples, one laboratory task.

! EXAM TRAP

The PCL-R is **Hare's**, built on Cleckley's description; not Cleckley alone, **Aaron Temkin Beck** (1921 to 2021; cognitive therapy) or **Hans Jürgen Eysenck** (1916 to 1997; the PEN model). In ASPD the pattern dates from 15, the diagnosis waits until 18, and conduct disorder must begin before 15, not 18. An option reading "conduct disorder before 15 and disregard for others' rights from 18" is loosely worded but still the answer: 18 is the minimum age for the diagnosis.

Histrionic, narcissistic and the rest of Cluster B ●●

Histrionic personality disorder is excessive emotionality and attention seeking: discomfort when not the centre of attention, seductive or provocative behaviour, shallow and rapidly shifting emotion, impressionistic speech lacking detail, theatricality, suggestibility, and seeing relationships as more intimate than they are. **Narcissistic personality disorder** rests on a core triad of **grandiosity**, **need for admiration** and **lack of empathy**, with fantasies of unlimited success, the belief that one is special and understood only by high-status people, entitlement, exploitation, envy and arrogance. Kohut saw narcissism as arrested development after parents failed to mirror, to be met with empathy; Kernberg saw a pathological grandiose self defending against rage and envy toward cold parents, to be confronted. Table 8.33 separates the look-alikes.

Table 8.33 Telling the attention-drawing disorders apart.

Disorder	Core	Wants	Tell-tale sign
Histrionic	Excessive emotionality	Attention	Seductive, theatrical, shallow, suggestible
Narcissistic	Grandiosity, no empathy	Admiration	Entitlement, envy, feels "special"
Borderline	Instability, impulsivity	Not to be abandoned	Self-harm, emptiness, idealise then devalue

Disorder	Core	Wants	Tell-tale sign
Antisocial	Disregard for rights	Profit or pleasure	Deceit, aggression, no remorse
Dependent (C)	Need to be cared for	Reassurance	Submissive, clinging, cannot decide alone

> ! EXAM TRAP
>
> An admiration seeker who is emotional and seductive rather than grandiose is **histrionic**. Pervasive instability of relationships and self-image is **borderline**, not narcissistic.

Cluster C, and OCPD versus OCD ○

Avoidant personality disorder is social inhibition, inadequacy and hypersensitivity to negative evaluation in someone who wants relationships; it overlaps with social anxiety disorder (Section 8.6). **Dependent personality disorder** is an excessive need to be taken care of: needing reassurance for everyday decisions, never disagreeing for fear of losing support, urgently seeking a new relationship when one ends.

Obsessive-compulsive personality disorder (OCPD) is a preoccupation with orderliness, perfectionism and mental and interpersonal control at the expense of flexibility, openness and efficiency: lists swallow the task, perfectionism stops it being finished, work crowds out friends, morals are rigid, worthless objects are kept and money is hoarded. The motive is **control**, not achievement or autonomy; efficiency is what it sacrifices. **Sigmund Freud** (1856 to 1939) described the same trio in the **anal character** (1908): orderly, parsimonious and obstinate. OCPD is ego-syntonic, with limited insight and no true obsessions or compulsions; OCD is ego-dystonic, its intrusive obsessions and compulsions felt as alien and resisted (Section 8.6). Both can be diagnosed together.

Newer models: ICD-11 and the AMPD

ICD-11 abolished the ten types: one diagnosis of personality disorder is graded **mild, moderate or severe**, with **personality difficulty** below that threshold. Five **trait domain qualifiers** describe its style: negative affectivity, detachment, dissociality, disinhibition and **anankastia** (rigid perfectionism and control), plus a **borderline pattern specifier**. DSM-5's Section III offers the **Alternative Model for Personality Disorders** (AMPD). Criterion A rates the **level of personality functioning** from 0 to 4 on identity and self-direction (self) and empathy and intimacy (interpersonal); at least moderate impairment is needed. Criterion B requires one or more pathological traits, organised in five domains: negative affectivity, detachment, antagonism, disinhibition and psychoticism. Six types survive: antisocial, avoidant, borderline, narcissistic, obsessive-compulsive and schizotypal, plus personality disorder-trait specified.

> ! EXAM TRAP
>
> Dissociality and anankastia are ICD-11 domains; antagonism and psychoticism are AMPD domains.

Oppositional defiant and conduct disorder ○

The disruptive, impulse-control and conduct disorders, failures of emotional and behavioural self-control, mostly begin in childhood and are commoner in boys. **Oppositional defiant disorder** (ODD) is angry or irritable mood, argumentative or defiant behaviour, or vindictiveness for at least 6 months, with four or more of eight symptoms shown with someone other than a sibling. Most children with ODD never develop conduct disorder.

Conduct disorder (CD) is a repetitive, persistent pattern violating the basic rights of others or major age-appropriate norms: three or more of 15 criteria in the past 12 months, one in the past 6 months, from four groups: **aggression to people and animals**, **destruction of property**, **deceitfulness or theft** and **serious violations of rules** (staying out at night or truancy before 13, running away). The **childhood-onset type** shows a symptom before age 10, the **adolescent-onset type** none. The specifier **with limited prosocial emotions**, applied to CD alone, needs two or more of four traits for 12 months: lack of remorse or guilt, callous lack of empathy, unconcern about performance, and shallow or deficient affect. These callous-unemotional traits mark more severe, planned aggression.

Terrie E. Moffitt (1993) explained the two onsets. **Life-course-persistent** antisocial behaviour is rare, starts in early childhood from neuropsychological deficits meeting adverse homes, and continues into adult crime. **Adolescence-limited** antisocial behaviour is common, begins at puberty as teenagers copy older peers, and fades in early adulthood.

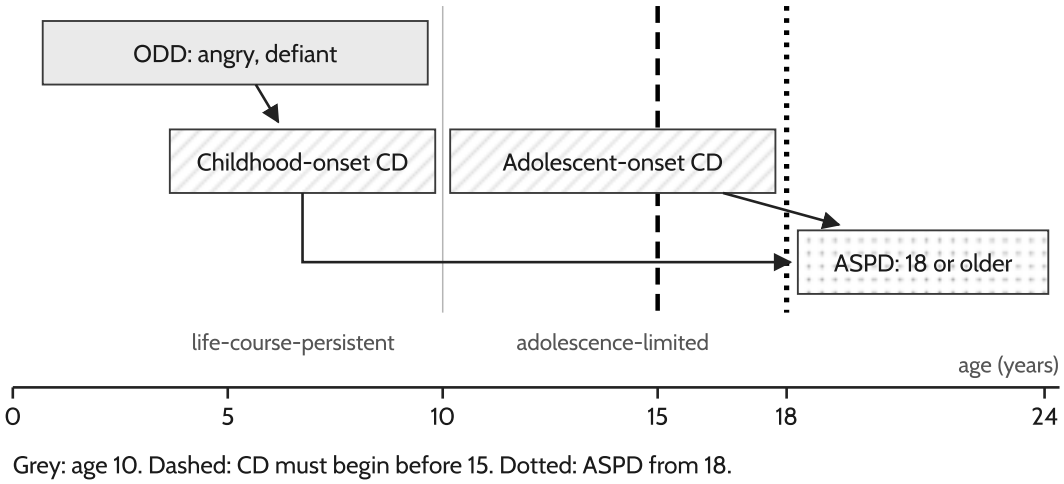


Figure 8.14 From defiance to antisocial personality. ASPD needs conduct disorder before 15 and age 18; the childhood-onset path is the one that usually persists.

Explosive, fire-setting and stealing impulses ○

Intermittent explosive disorder (IED) is recurrent impulsive, verbal or physical, aggressive outbursts grossly out of proportion to the provocation (twice weekly for 3 months, or three destructive or injurious outbursts in a year), unplanned and not aimed at money or power, from age 6. In **pyromania** the person deliberately and purposefully sets fires on more than one occasion, with tension or arousal before, fascination with fire, and pleasure, gratification or relief after; fires set for money, revenge, ideology or to hide a crime do not count. Pyromania is planned fire setting for gratification; IED is impulsive aggression beyond the

trigger. In **kleptomania** the person repeatedly fails to resist impulses to steal objects not needed for personal use or their value, with rising tension before and pleasure or relief during the theft; women outnumber men three to one.

Charles Chrétien Henri Marc and **Jean-Étienne Esquirol** (1772 to 1840) named kleptomania in 1838, renaming the *klopemanie* (stealing madness) of the Swiss physician **André Matthey** (1816). Not the distractors: **Eugen Bleuler** (1857 to 1939) named schizophrenia and described ambivalence and autism; **Enrico Morselli** coined dysmorphophobia (1891); **Alistair Munro** described monosymptomatic hypochondriacal psychosis. ICD-10's habit and impulse disorders (F63) were pathological gambling, fire-setting and stealing, and trichotillomania. ICD-11's impulse control disorders are pyromania, kleptomania, compulsive sexual behaviour disorder and intermittent explosive disorder; gambling moved to addictive behaviours (Section 8.10), trichotillomania to the obsessive-compulsive group (Section 8.6).

> **! EXAM TRAP**
>
> "With limited prosocial emotions" is a specifier of **conduct disorder** only, not ASPD, ODD or IED. IED aggression may be verbal or physical. "Kleptomania" is **Marc's** term (with Esquirol), not Bleuler's, Munro's or Morselli's.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A personality disorder shows in two or more of cognition, affectivity, interpersonal functioning and _____.
2. Avoidant, dependent and obsessive-compulsive PDs form Cluster _____.
3. The primary defence of borderline personality disorder is _____.
4. Borderline PD needs _____ of nine criteria.
5. ASPD requires age 18 and conduct disorder beginning before age _____.
6. The Psychopathy Checklist-Revised was developed by _____.
7. The specifier "with limited prosocial emotions" applies only to _____.
8. ICD-11 grades personality disorder as mild, moderate or _____.

Answers: (1) impulse control, (2) C, (3) splitting, (4) five, (5) 15, (6) Robert Hare, (7) conduct disorder, (8) severe.

Kraepelin's Cards, Revisited

Kraepelin's cards hold this chapter in miniature. Each began with signs and symptoms, the vocabulary of Section 8.2, and ended with onset, course and prognosis, the terms of Section 8.1. The line the cards drew still divides schizophrenia (8.4) from the bipolar and depressive disorders (8.5), and DSM-III turned his method into explicit criteria, which is why every disorder from 8.3 to 8.11 arrives as a criterion set, often with a minimum duration. Yet his deeper hope, that each pile would prove to be a separate brain disease, has not come true. Risk genes for schizophrenia are shared with bipolar disorder and depression, comorbidity is the rule, and the diathesis-stress model explains more than any single cause. So DSM-5 added

severity ratings and spectra to his categories, and ICD-11 grades personality disorder by severity.

Chapter summary

8.1 Abnormality, Causes and the Classification of Mental Disorders. No single criterion defines abnormality: clinicians weigh the **four Ds**, Wakefield's **harmful dysfunction** and the DSM-5-TR definition, which excludes expectable grief and mere social deviance. Psychopathology has its own vocabulary of aetiology, onset, course, prognosis, prevalence, incidence and **comorbidity**. Genes, endophenotypes and stress meet in the **diathesis-stress model** and Engel's **biopsychosocial model**. Classification is syndromal and prototypical, judged by reliability and validity. The **DSM** (APA; DSM-5 dropped the axes; DSM-5-TR, 2022, is current) and the **ICD** (WHO; ICD-10 F codes; ICD-11 Chapter 6) are the two systems, and cultural concepts of distress such as dhat and latah show culture shaping distress.

8.2 Descriptive Psychopathology: Signs and Symptoms. Descriptive psychopathology, founded by Karl Jaspers (*General Psychopathology*, 1913), describes experiences phenomenologically, separating form from content, primary from secondary symptoms and signs from symptoms. An **illusion** misreads a real stimulus; a **hallucination**, a disorder of perception, has no stimulus; hypnagogic hallucinations are normal. Flight of ideas, loosening, tangentiality and circumstantiality are disorders of the **form** of thought. Persecution is the commonest **delusion**; Capgras, Fregoli, Cotard, Othello and de Clérambault name delusional themes. Confabulation fills memory gaps without intent to deceive. Affect is observed and momentary, mood sustained; elevated mood runs euphoria, elation, exaltation, ecstasy. **Insight** has six grades.

8.3 Neurodevelopmental Disorders. Neurodevelopmental disorders begin in the developmental period and often co-occur, so assessment uses several informants. **Intellectual disability** needs intellectual deficits (IQ about two SD below the mean, 65 to 75), adaptive deficits and early onset; DSM-5 grades severity by adaptive functioning, and global developmental delay is used under age 5. Causes include trisomy 21, fragile X, PKU and fetal alcohol syndrome. **Autism spectrum disorder** (Kanner, 1943) needs social-communication deficits plus restricted, repetitive behaviour, not low IQ. **ADHD** needs six of nine symptoms in a domain, several before 12, in two settings. **Specific learning disorder** occurs with average intelligence; Tourette's needs motor and vocal tics for over a year.

8.4 Schizophrenia Spectrum and Other Psychotic Disorders. Kraepelin's **dementia praecox** (1893), separated from manic-depressive insanity by its deteriorating course, became Bleuler's **schizophrenia** (1908, 1911), defined by four As; Schneider's **first-rank symptoms** stress voices, thought alienation, passivity and delusional perception. Symptoms are positive, negative or cognitive. DSM-5-TR needs 6 months including 1 month of active symptoms; ICD-10 and ICD-11 need 1 month. Brief psychotic disorder lasts 1 day to under 1 month, schizophreniform disorder 1 to under 6 months. Risk is about 48 percent for an identical twin. Positive symptoms reflect mesolimbic dopamine excess; high **expressed emotion** predicts relapse. Antipsychotics are first-line; nigrostriatal D2 blockade causes dystonia, akathisia, parkinsonism and, last, tardive dyskinesia.

8.5 Depressive and Bipolar Disorders, and Suicide. Mood disorders are built from three episodes: a **major depressive episode** (5 of 9 symptoms for 2 weeks), a **manic episode** (1 week; marked impair-

ment or psychosis) and a **hypomanic episode** (4 days, no marked impairment or psychosis). MDD affects women twice as often as men; **dysthymia** is milder but lasts 2 years; DMDD begins before age 10. Specifiers include mixed features, rapid cycling, melancholic, atypical, peripartum and seasonal. Bipolar I needs mania; bipolar II needs hypomania plus major depression; cyclothymia is subthreshold. Causes range from monoamines and kindling to **Beck's cognitive triad**. Lithium is the gold-standard mood stabiliser. A previous attempt is the strongest predictor of suicide.

8.6 Anxiety and Obsessive-Compulsive and Related Disorders. **Fear** answers present danger, **anxiety** a future threat. DSM-5 split the old neuroses into anxiety, obsessive-compulsive and trauma chapters. Most anxiety disorders need about **6 months**; panic disorder and selective mutism need 1 month. A **panic attack** peaks within minutes with 4 of 13 symptoms; agoraphobia needs 2 of 5 situations and GAD 3 of 6 symptoms. Causes include the GABA system, Mowrer's two-factor theory, Clark's catastrophic misinterpretation, Klein's suffocation alarm and displacement (Little Hans). **OCD** needs ego-dystonic obsessions or compulsions taking over an hour a day or causing distress; compulsions are negatively reinforced escape responses and undoing its typical defence. BDD, hoarding, trichotillomania and excoriation are its relatives.

8.7 Trauma, Dissociative and Somatic Symptom Disorders. Charcot, Janet, Breuer and Freud explained **hysteria** by hypnosis, **dissociation** and **repression**, and DSM-III (1980) split it into somatoform and dissociative disorders. **PTSD** needs a qualifying trauma and symptoms in four clusters for more than 1 month; **acute stress disorder** lasts 3 days to 1 month, and ICD-11 adds **complex PTSD**. **DID** combines two or more personality states with amnesic gaps; dissociative amnesia, fugue and depersonalisation/derealisation complete the class. **Somatic symptom disorder** is an excessive response to symptoms, **conversion** a neurological-like loss incompatible with disease, and factitious disorder and **malingering** are deliberate, differing in external reward.

8.8 Feeding, Eating, Elimination and Sleep-Wake Disorders. DSM-5-TR's feeding and eating disorders include **pica** (non-food eating for at least 1 month), rumination disorder and ARFID, which restricts food with no body-image concern. **Anorexia nervosa** (Gull and Lasègue, 1873) needs restriction to a significantly low weight, fear of weight gain and disturbed body experience; amenorrhoea was dropped, severity is graded by BMI, and it is the most lethal eating disorder. **Bulimia nervosa** (Russell, 1979) pairs binges with compensation weekly for 3 months at normal weight; binge-eating disorder lacks compensation. Enuresis starts at age 5, encopresis at 4. Among sleep-wake disorders, narcolepsy brings cataplexy, N3 brings sleepwalking and sleep terrors, and REM brings nightmares and REM sleep behaviour disorder.

8.9 Sexual Dysfunctions, Paraphilic Disorders and Gender Dysphoria. Masters and Johnson's **sexual response cycle** (1966: excitement, plateau, orgasm, resolution, with a male refractory period) gained Kaplan's desire phase (1979) and Basson's circular model (2000). Most DSM-5-TR **sexual dysfunctions** need symptoms on 75 to 100 percent of occasions for about 6 months with distress; premature ejaculation is tied to anxiety and treated with stop-start and squeeze. A **paraphilic disorder** is a paraphilia plus distress, impairment or harm to others; DSM-5-TR names eight. **Gender dysphoria** replaced gender identity disorder, and ICD-11 moved **gender incongruence** to Chapter 17 while paraphilic disorders stayed in Chapter 6. The APA removed homosexuality in 1973.

8.10 Substance-Related, Addictive and Neurocognitive Disorders. A **psychoactive substance** alters mood, cognition or behaviour; tolerance, **cross-tolerance**, withdrawal and dependence describe how use becomes disorder. ICD-10 codes F10 to F19 patterns (harmful use, Edwards and Gross's **dependence syndrome**; 'drug abuse' is not one), DSM-5-TR merges abuse and dependence into **substance use disorder** (2 of 11 criteria), and ICD-11 adds gambling and gaming disorder. Drugs converge on the mesolimbic dopamine system. Alcohol is a GABA-A agonist; heroin is semi-synthetic. **Delirium** is acute, fluctuating and clouded, unlike dementia; NCD subtypes differ in course and pathology; depression causes pseudodementia; thiamine deficiency causes **Wernicke-Korsakoff syndrome.**

8.11 Personality Disorders, Conduct and Impulse-Control Disorders. A **personality disorder** is an enduring, pervasive, inflexible pattern deviating from cultural expectations in two or more of cognition, affectivity, interpersonal functioning and impulse control, beginning by adolescence or early adulthood; usually ego-syntonic, it affects about 10 percent. DSM-5-TR's ten types form Clusters A (odd), B (dramatic) and C (anxious). **Borderline PD** (5 of 9 criteria) is instability of relationships, self-image and affect with impulsivity, its defence splitting. **Antisocial PD** needs age 18 and conduct disorder before 15; Hare's PCL-R measures psychopathy. ICD-11 and the AMPD grade severity and traits. Conduct disorder alone takes the specifier **with limited prosocial emotions**; pyromania and kleptomania follow tension with relief.

Who's who

Table 8.34 The people in this chapter.

Name	When	Examined for
Marie Jahoda	1907 to 2001	Ideal (positive) mental health, 1958
Jerome Wakefield	1992	Mental disorder as harmful dysfunction
Daniel Offer	1966	With Sabshin, four perspectives on normality: health, utopia, average, process
Melvin Sabshin	1966	With Offer, four perspectives on normality
Avshalom Caspi	2003	Serotonin transporter gene and childhood maltreatment in adult depression (gene-environment interaction)
Irving Gottesman	1970s; 2003	Brought the endophenotype concept into psychiatry with Shields and revived it with Gould
James Shields	1970s	Endophenotype concept and schizophrenia twin research with Gottesman
Todd Gould	2003	Endophenotype concept paper with Gottesman
Paul Meehl	1920 to 2003	Schizotaxia (1962): an early diathesis-stress account of schizophrenia
Joseph Zubin	1977	Vulnerability-stress model, with Bonnie Spring
Bonnie Spring	1977	Vulnerability-stress model, with Joseph Zubin
George L. Engel	1913 to 1999	Biopsychosocial model (1977)
Hippocrates	about 460 to 377 BC	Classified mania, melancholia and phrenitis; coined the word hysteria
Emil Kraepelin	1856 to 1926	Descriptive classification by course; dementia praecox versus manic-depressive insanity
Robert Spitzer	1932 to 2015	Directed DSM-III (1980): explicit criteria and the multiaxial system
N. N. Wig	1960	Coined the term dhat syndrome
Jacob Cohen	1960	Cohen's kappa, the chance-corrected agreement index used for diagnostic reliability
Aaron T. Beck	1921 to 2021	Cognitive model: disturbed thought patterns and schemas (Section 7.5)
Albert Ellis	1913 to 2007	Irrational beliefs in the cognitive model (Section 7.5)

Name	When	Examined for
Eugen Bleuler	1857 to 1939	Named schizophrenia (Section 8.4); a distractor for the biopsychosocial model
Karl Jaspers	1883 to 1969	General Psychopathology (1913); phenomenology and empathic understanding; primary delusions; the marks of a delusion (certainty, incorrigibility, impossibility)
Charles Bonnet	1720 to 1793	Described (1760) visual hallucinations in his nearly blind grandfather: Charles Bonnet syndrome
Norman Cameron	1896 to 1975	Overinclusive thinking in schizophrenia
Carl Wernicke	1848 to 1905	Introduced the term overvalued idea
Kurt Schneider	1887 to 1967	First-rank symptoms of schizophrenia, including thought insertion, withdrawal and broadcasting
Joseph Capgras	1873 to 1950	Capgras syndrome (1923, 'l'illusion des sosies'): a familiar person replaced by an identical impostor
Jules Cotard	1840 to 1889	Cotard syndrome (described 1880, named 'délire des négations' 1882): nihilistic delusion of being dead or not existing
Arnold Pick	1851 to 1924	Named reduplicative paramnesia (1903)
Gaëtan Gatian de Clérambault	1872 to 1934	de Clérambault syndrome (erotomania; classic account 1921): a person of higher status is secretly in love with the patient
Jean Reboul-Lachaux	1923 (key year)	Co-author with Joseph Capgras of the 1923 description of l'illusion des sosies (Capgras syndrome)
Paul Courbon	1927 (key year)	Described Fregoli syndrome with Gabriel Fail (1927)
Gabriel Fail	1927 (key year)	Described Fregoli syndrome with Paul Courbon (1927)
Leopoldo Fregoli	1867 to 1936	Italian quick-change actor after whom Fregoli syndrome is named
John Todd	1955 (key year)	Named the Othello syndrome (morbid jealousy) with Kenneth Dewhurst (1955)
Kenneth Dewhurst	1955 (key year)	Named the Othello syndrome (morbid jealousy) with John Todd (1955)
Karl-Axel Ekbom	1907 to 1977	Described delusional parasitosis (1938), Ekbom syndrome; his name also labels restless legs syndrome
Sigbert Ganser	1853 to 1931	Ganser syndrome (1898): approximate answers
Charles Lasègue	1816 to 1883	Described folie à deux (shared delusion) with Jules Falret (1877)
Jules Falret	1824 to 1902	Described folie à deux (shared delusion) with Charles Lasègue (1877)
Anton Delbrück	1891 (key year)	Pseudologia fantastica: fantastic lying the teller half believes
Peter Sifneos	1920 to 2008	Introduced the term alexithymia (1973)
Leo Kanner	1894 to 1981	First described early infantile autism, 'Autistic disturbances of affective contact' (1943)
Hans Asperger	1906 to 1980	Described 'autistic psychopathy' in fluent, intelligent boys (1944)
Bruno Bettelheim	1903 to 1990	Discredited 'refrigerator mother' theory of autism (The Empty Fortress, 1967)
Lorna Wing	1928 to 2014	Triad of impairments in autism, with Judith Gould (1979)
Judith Gould	1979	Triad of impairments in autism, with Lorna Wing
Theodor Heller	1908	Dementia infantilis, later called childhood disintegrative disorder
Andreas Rett	1966	Described Rett syndrome (MECP2; girls; regression and hand-wringing)
Simon Baron-Cohen	born 1958	Theory of mind deficit in autism and the Sally-Anne test (1985); mind-blindness; extreme male brain theory
Alan Leslie	1985	Sally-Anne false-belief study with Baron-Cohen and Frith
Uta Frith	born 1941	Sally-Anne study (1985); weak central coherence theory of autism (1989)
Andrew Wakefield	1998	Retracted Lancet paper falsely linking the MMR vaccine to autism
Ivar Lovaas	1927 to 2010	Early intensive applied behaviour analysis for autism (1987)
George Frederic Still	1868 to 1941	First description of ADHD-like children, a 'defect of moral control' (1902)
Russell Barkley	born 1949	Behavioural inhibition model of ADHD (1997)
John Langdon Down	1828 to 1896	Described Down syndrome (1866)
Asbjørn Følling	1888 to 1973	Discovered phenylketonuria (1934)
James Hinshelwood	1859 to 1919	Congenital word-blindness (dyslexia); book of that title (1917)
Samuel Torrey Orton	1879 to 1948	Explained dyslexia by incomplete cerebral dominance (1920s)

Name	When	Examined for
Georges Gilles de la Tourette	1857 to 1904	Described Tourette's disorder (1885)
Kurt Goldstein	1878 to 1965	Concrete versus abstract attitude after brain damage; a distractor in autism questions (Section 8.2)
Benedict Morel	1809 to 1873	Démence précoce (1850s), the first name for the illness
Ewald Hecker	1871	Described hebephrenia
Karl Ludwig Kahlbaum	1828 to 1899	Described catatonia (1874)
Timothy Crow	1980	Type I (positive) and Type II (negative) schizophrenia
Peter Liddle	1987	Three syndromes: reality distortion, disorganisation, psychomotor poverty
Nancy Andreasen	1980s	SANS and SAPS rating scales for negative and positive symptoms
David Rosenthal	1963	Genain quadruplets; Danish adoption studies with Kety and Wender
Leonard Heston	1966	Adoption study: schizophrenia in adopted-away children of mothers with schizophrenia
Seymour Kety	1968	Danish adoption studies
Paul Wender	1968	Danish adoption studies
Arvid Carlsson	1923 to 2018	Dopamine as a neurotransmitter; dopamine hypothesis; Nobel Prize 2000
Philip Seeman	1976	Antipsychotic potency correlates with D2 receptor affinity
Ian Creese	1976	With Burt and Snyder: antipsychotic potency correlates with D2 binding
David Burt	1976	With Creese and Snyder: antipsychotic potency correlates with D2 binding
Solomon Snyder	1976	With Creese and Burt: antipsychotic potency correlates with D2 binding
Kenneth Davis	1991	Revised dopamine hypothesis: subcortical excess, prefrontal deficit
Oliver Howes	2009	With Kapur: presynaptic striatal dopamine as the final common pathway
Shitij Kapur	2009	With Howes: dopamine hypothesis version III
Daniel Weinberger	1987	Neurodevelopmental hypothesis; hypofrontality
Robin Murray	1987	With Shon Lewis: neurodevelopmental hypothesis
Shon Lewis	1987	With Robin Murray: neurodevelopmental hypothesis
Eve Johnstone	1976	First CT evidence of enlarged lateral ventricles in schizophrenia
Frieda Fromm-Reichmann	1889 to 1957	The schizophrenogenic mother (1948), now discredited
Gregory Bateson	1904 to 1980	Double bind theory of schizophrenia (1956); the symptomatic double bind
Don Jackson	1956	Co-author of the double bind theory; family homeostasis
Jay Haley	1956	Co-author of the double bind theory
John Weakland	1956	Co-author of the double bind theory
Theodore Lidz	1957	Marital schism and marital skew
Lyman Wynne	1958	Pseudomutuality; communication deviance
Ronald David Laing	1927 to 1989	Anti-psychiatry; schizophrenia as a response to an unliveable family situation
George Brown	1972	Expressed emotion predicts relapse (with Birley and Wing)
James Birley	1972	Expressed emotion and relapse (with Brown and Wing)
John Wing	1972	Expressed emotion and relapse (with Brown and Birley)
Christine Vaughn	1976	With Leff: expressed emotion, relapse and 35 hours of face-to-face contact
Julian Leff	1976	With Vaughn: expressed emotion and relapse; Chandigarh studies of low EE in India
Sigmund Freud	1856 to 1939	Schizophrenia as withdrawal of libido and regression
Paul Federn	1871 to 1950	Weak ego boundaries in schizophrenia
Harry Stack Sullivan	1892 to 1949	Interpersonal theory of schizophrenia
Silvano Arieti	1914 to 1981	Palaeologic thinking; the von Domarus principle
Eilhard von Domarus	1940s	The von Domarus principle: identity from shared predicates
Margaret Mahler	1897 to 1985	Schizophrenia as fixation at the autistic phase with failure of stable symbiosis; symbiotic psychosis

Name	When	Examined for
Ping-Nie Pao	1979	Symptoms as adaptation to organismic panic
Edith Jacobson	1897 to 1978	Psychotic fusion of self and object representations
Jean Delay	1907 to 1987	With Deniker, introduced chlorpromazine for psychosis (1952)
Pierre Deniker	1917 to 1998	With Delay, introduced chlorpromazine for psychosis (1952)
René Spitz	1887 to 1974	Anaclitic depression in separated infants (1946); hospitalism
Norman Rosenthal	1984	Described seasonal affective disorder and its light therapy
Jean-Pierre Falret	1794 to 1870	Folie circulaire (1850s): alternating mania and melancholia
Jules Baillarger	1809 to 1890	Folie a double forme (1850s), the same alternating illness
Karl Leonhard	1957	Separated unipolar from bipolar mood disorders
Joseph Schildkraut	1965	Catecholamine hypothesis: noradrenaline deficit in depression, excess in mania
Robert Post	1992	Kindling (sensitisation) hypothesis: each episode lowers the threshold for the next
Karl Abraham	1911	Early psychoanalytic account of depression as hostility turned against the self
Edward Bibring	1953	Depression as the ego's awareness of its helplessness; loss of self-esteem
Peter Lewinsohn	1970s	Behavioural theory: low rate of response-contingent positive reinforcement
Martin Seligman	1975	Learned helplessness theory of depression
Lyn Abramson	1978, 1989	Attributional reformulation (internal, stable, global); hopelessness theory
John Teasdale	1978	Co-author of the attributional reformulation of learned helplessness
Gerald Metalsky	1989	Co-author of the hopelessness theory of depression
Lauren Alloy	1989	Co-author of the hopelessness theory of depression
Aaron Temple Beck	1921 to 2021	Cognitive theory of depression (1967): negative cognitive triad of self, world and future
Susan Nolen-Hoeksema	1959 to 2013	Response styles theory (1991): rumination and the gender gap in depression
Tirril Harris	1978	Co-author of Social Origins of Depression
John Cade	1912 to 1980	Introduced lithium for mania (1949)
Norman Kreitman	1969	Coined the term parasuicide
Emile Durkheim	1858 to 1917	Suicide (1897): egoistic, altruistic, anomic and fatalistic suicide
Edwin Shneidman	1918 to 2009	Founder of suicidology; psychache
Thomas Joiner	2005	Interpersonal theory of suicide: burdensomeness, thwarted belongingness, acquired capability
Marie Asberg	1976	Low CSF 5-HIAA in depressed patients who make violent suicide attempts
W. M. Patterson and colleagues	1983	SAD PERSONS suicide risk scale (Section 7.2)
Jean Piaget	1896 to 1980	Cognitive developmental stages (Chapter 6); a distractor to Spitz for anaclitic depression
B. F. Skinner	1904 to 1990	Operant conditioning (Chapter 1); a distractor to Spitz for anaclitic depression
Hans Eysenck	1916 to 1997	Personality dimensions (Chapter 4); a distractor to Spitz for anaclitic depression
William Cullen	1769	Coined the term neurosis (1769)
Alan Goldstein	1978	With Chambless, 'fear of fear' in panic disorder and agoraphobia (1978)
Dianne Chambless	1978	With Goldstein, 'fear of fear' in panic disorder and agoraphobia (1978)
Jerome Kagan	1929 to 2021	Behavioural inhibition, an inherited shy temperament visible by four months that predicts anxiety
O. Hobart Mowrer	1907 to 1982	Two-factor theory (1947): fear acquired by classical conditioning, avoidance kept by negative reinforcement; explains compulsions as escape responses
Stanley Rachman	1977	Three pathways to fear (1977): conditioning, vicarious learning, information; thought-action fusion in OCD
David M. Clark	1986	Catastrophic misinterpretation of bodily sensations in panic (1986)

Name	When	Examined for
Steven Reiss	Contemporary	Anxiety sensitivity, the belief that anxiety symptoms are harmful
Thomas Borkovec	Contemporary	Avoidance theory of worry in GAD
Adrian Wells	Contemporary	Metacognitive model of GAD (beliefs about worry)
Michel Dugas	Contemporary	Intolerance of uncertainty in GAD
Donald F. Klein	1993	Suffocation false alarm theory of panic disorder (1993)
David H. Barlow	Contemporary	Triple vulnerability theory of anxiety
Susan Swedo	1998	PANDAS: childhood OCD or tics after streptococcal infection (1998)
Paul Salkovskis	1985	Inflated responsibility in OCD (1985)
Enrico Morselli	1891	Named dysmorphophobia, now body dysmorphic disorder (1891)
François Hallopeau	1889	Named trichotillomania (1889)
Jean-Martin Charcot	1825 to 1893	Salpêtrière: produced and removed hysterical symptoms by hypnosis and suggestion
Pierre Janet	1859 to 1947	Dissociation (désagrégation, 1889); la belle indifférence usually credited to him
Josef Breuer	1842 to 1925	Treated Anna O.; Studies on Hysteria with Freud (1895)
Anna O. (Bertha Pappenheim)	1859 to 1936	Breuer's hysterical patient; named the 'talking cure'
Anna Freud	1895 to 1982	Defence mechanisms; a distractor for who coined la belle indifférence
Ernest Ropiequet Hilgard	1904 to 2001	Neodissociation theory and the hidden observer (Divided Consciousness, 1977)
Abram Kardiner	1891 to 1981	The Traumatic Neuroses of War (1941)
Elisabeth Kübler-Ross	1969	Stages of dying (denial, anger, bargaining); not PTSD clusters
Richard Tedeschi	1996	Post-traumatic growth, with Calhoun: five domains
Lawrence Calhoun	1996	Post-traumatic growth, with Tedeschi
Corbett Thigpen	1957	The Three Faces of Eve, with Cleckley
Hervey Cleckley	1957	The Three Faces of Eve, with Thigpen
Nicholas Spanos	1942 to 1994	Sociocognitive model of DID; critic of the hidden observer
Paul Briquet	1796 to 1881	Briquet's syndrome (1859), later somatisation disorder
George Miller Beard	1839 to 1883	Named neurasthenia (1869)
Richard Asher	1912 to 1969	Named Munchausen syndrome (1951)
Baron Munchausen	1720 to 1797	Teller of tall tales after whom Munchausen syndrome is named
Roy Meadow	1977	Described Munchausen syndrome by proxy (factitious disorder imposed on another)
Sir William Withey Gull	1816 to 1890	Named anorexia nervosa (paper read 1873, printed 1874)
Ernest-Charles Lasègue	1816 to 1883	Described anorexie hystérique (1873)
Hilde Bruch	1904 to 1984	The Golden Cage (1978): anorexia as a struggle for control by an ineffective-feeling, compliant girl
Gerald Russell	1979	Named bulimia nervosa, 'an ominous variant of anorexia nervosa' (1979); Russell's sign
Salvador Minuchin	1921 to 2017	The psychosomatic family in anorexia (Section 7.7)
Jean-Baptiste-Édouard Gélineau	1828 to 1906	Coined the term narcolepsy (1880)
William Howell Masters	1915 to 2001	Human Sexual Response (1966): four-phase sexual response cycle; Human Sexual Inadequacy (1970): performance anxiety, spectatoring, squeeze technique
Virginia Eshelman Johnson	1925 to 2013	Masters's research partner; co-author of the sexual response cycle (1966) and of sex therapy methods (1970)
Helen Singer Kaplan	1929 to 1995	Added a desire phase to the sexual response cycle (Disorders of Sexual Desire, 1979)
Rosemary Basson	2000	Circular model of female sexual response; responsive desire
Sandra Lipsitz Bem	1944 to 2014	Androgyny; Bem Sex Role Inventory (1974)
James Semans	1956	Stop-start technique for premature ejaculation
Ray Blanchard	1989	Coined autogynephilia: a male's arousal by the image of himself as a woman
John Money	1921 to 2006	Lovemaps: childhood templates of the ideal lover, distorted in paraphilias

Name	When	Examined for
Kurt Freund	1914 to 1996	Courtship disorder hypothesis: voyeurism, exhibitionism and frotteurism as disturbed courtship phases
Alfred Charles Kinsey	1894 to 1956	Kinsey scale (0 to 6); Sexual Behavior in the Human Male (1948)
Evelyn Hooker	1907 to 1996	1957 study: blind judges could not tell homosexual from heterosexual men on projective tests; no difference in adjustment
Griffith Edwards	1976	Described the alcohol dependence syndrome with Milton Gross (1976)
Milton Gross	1976	Co-described the alcohol dependence syndrome with Griffith Edwards (1976)
Richard Solomon	1974	Opponent-process theory of drug use, with John Corbit (1974)
John Corbit	1974	Opponent-process theory, with Richard Solomon (1974)
C. Robert Cloninger	1981	Type I and type II alcoholism, from Swedish adoption studies
Elvin Morton Jellinek	1890 to 1963	The Disease Concept of Alcoholism (1960); alpha to epsilon alcoholism, gamma loss of control, delta inability to abstain
Denise Kandel	1975	Stages of adolescent drug use; the gateway hypothesis (marijuana)
Albert Hofmann	1906 to 2008	Discovered the effects of LSD (1943)
Alois Alzheimer	1864 to 1915	Described Alzheimer's disease (1906): plaques and tangles in Auguste Deter
Auguste Deter	1850 to 1906	Alzheimer's first patient, Auguste D., admitted in 1901
George Huntington	1850 to 1916	Described Huntington's chorea (1872)
Leslie Kiloh	1961	Popularised the term pseudodementia; depression is its common cause
Sergei Korsakoff	1854 to 1900	Korsakoff's syndrome (1887): amnesia with confabulation
Adolph Stern	1938	Coined the term borderline for patients between neurosis and psychosis
Otto F. Kernberg	born 1928	Borderline personality organisation (identity diffusion, primitive defences such as splitting, intact reality testing); transference-focused psychotherapy; narcissism as a pathological grandiose self
Heinz Kohut	1913 to 1981	Self psychology: failed parental mirroring and idealising leave a fragmented self; narcissism as arrested development
Marsha M. Linehan	born 1943	Biosocial theory of borderline personality disorder; dialectical behaviour therapy
Melanie Klein	1882 to 1960	Object relations theory, the school to which splitting belongs
Philippe Pinel	1745 to 1826	Manie sans délire (mania without delirium), 1801, an early description of antisocial personality
James Cowles Prichard	1786 to 1848	Moral insanity, 1835
Robert D. Hare	born 1934	Psychopathy Checklist-Revised (PCL-R, 1991), 20 items scored 0 to 2, cut-off 30 of 40, two factors
Lee Nelken Robins	1922 to 2009	Deviant Children Grown Up (1966): adult sociopathy almost always preceded by childhood antisocial behaviour
David Thoreson Lykken	1928 to 2006	Fearlessness hypothesis of psychopathy (1957): poor avoidance of shock, low anxiety, low skin conductance
Aaron Temkin Beck	1921 to 2021	Cognitive therapy; a distractor for the PCL-R
Hans Jürgen Eysenck	1916 to 1997	PEN model of personality; a distractor for the PCL-R
Terrie E. Moffitt	1993	Life-course-persistent versus adolescence-limited antisocial behaviour
Charles Chrétien Henri Marc	1838	With Esquirol, gave kleptomania its name
Jean-Étienne Esquirol	1772 to 1840	With Marc, named kleptomania (1838)
André Matthey	1816	Swiss physician whose klopemanie (stealing madness) Marc and Esquirol renamed kleptomania
Alistair Munro	Late twentieth century	Monosymptomatic hypochondriacal psychosis (a distractor for kleptomania)

Key terms

four Ds, statistical criterion, cultural relativity, harmful dysfunction, mental disorder, subjective model, Psychopathology, abnormal psychology, syndrome, Aetiology, Prognosis, Prevalence, inci-

dence, Comorbidity, concordance, heritability, gene-environment interaction, gene-environment correlation, endophenotypes, diathesis-stress model, vulnerability-stress model, biopsychosocial model, syndromal approach, historical approach, prototypical approach, Reliability, validity, false positive, false negative, conditions for further study, non-suicidal self-injury disorder, culture-bound syndromes, cultural concepts of distress, Cultural Formulation Interview, Descriptive psychopathology, phenomenology, sign, Delirium, twilight state, illusion, hallucination, pseudohallucination, Charles Bonnet syndrome, depersonalisation, derealisation, thought block, Formal thought disorder, flight of ideas, loosening of associations, overvalued ideas, thought broadcasting, delusion, primary delusion, delusional perception, delusion of persecution, nihilistic delusions, Capgras syndrome, Amnesia, confabulation, Déjà vu, Jamais vu, Mood, affect, Anhedonia, Alexithymia, Pseudobulbar affect, ego-dystonic, Catatonia, Insight, neurodevelopmental disorders, pervasive developmental disorders, multi-informant assessment, Intellectual disability, adaptive functioning, Borderline intellectual functioning, Global developmental delay, Down syndrome, Fragile X syndrome, phenylketonuria, fetal alcohol syndrome, Childhood-onset fluency disorder, Social (pragmatic) communication disorder, selective mutism, triad of impairments, childhood disintegrative disorder, autism spectrum disorder, Rett syndrome, restricted, repetitive patterns of behaviour, theory of mind, weak central coherence, hyperkinetic disorder, attention-deficit/hyperactivity disorder, behavioural inhibition model, specific learning disorder, dyslexia, phonological deficit theory, response to intervention, Auditory processing disorder, Developmental coordination disorder, dyspraxia, apraxia, tic, Tourette's disorder, coprolalia, Psychosis, dementia praecox, schizophrenia, four As, Ambivalence, first-rank symptoms, Positive symptoms, Negative symptoms, affective flattening, alogia, avolition, Brief psychotic disorder, Schizophreniform disorder, Delusional disorder, Schizoaffective disorder, acute and transient psychotic disorders, Folie à deux, social drift, dopamine hypothesis, neurodevelopmental hypothesis, schizophrenogenic mother, double bind, pseudomutuality, expressed emotion, Typical antipsychotics, Atypical antipsychotics, extrapyramidal symptoms, Akathisia, Tardive dyskinesia, neuroleptic malignant syndrome, mood disorder, unipolar, bipolar, bereavement exclusion, Major depressive disorder, Persistent depressive disorder, double depression, Disruptive mood dysregulation disorder, Premenstrual dysphoric disorder, mixed anxiety and depressive disorder, anaclitic depression, hospitalism, Masked depression, mood-congruent, seasonal affective disorder, Bipolar I disorder, Bipolar II disorder, Cyclothymic disorder, somatic syndrome, catecholamine hypothesis, permissive hypothesis, dexamethasone suppression test, shortened REM latency, kindling hypothesis, negative cognitive triad, automatic thoughts, lithium, Suicidal ideation, Lethality, parasuicide, Non-suicidal self-injury, psychache, Fear, Anxiety, neurosis, Separation anxiety disorder, specific phobia, Social anxiety disorder, Generalised anxiety disorder, panic attack, Panic disorder, Agoraphobia, fear of fear, behavioural inhibition, two-factor theory, preparedness, catastrophic misinterpretation, Catastrophic thinking, anxiety sensitivity, suffocation false alarm theory, triple vulnerability, signal anxiety, displacement, Obsessions, Compulsions, insight specifier, PANDAS, thought-action fusion, isolation of affect, undoing, reaction formation, Body dysmorphic disorder, Hoarding disorder, Trichotillomania, excoriation disorder, hysteria, dissociation, la belle indifférence, repression, conversion, primary gain, secondary gain, hidden observer, posttraumatic stress disorder, Acute stress disorder, Complex PTSD, post-traumatic growth, adjustment disorder, Prolonged grief disorder, reactive attachment disorder, disinhibited social engagement disorder, Dissociative identity disorder, host, sociocognitive model, Dissociative amnesia, Generalised amnesia, dissociative fugue, Ganser syndrome, Somatoform disorders, Somatic symptom disorder, Illness anxiety disorder, bodily distress disorder, functional neurological symptom disorder, functional blindness, Hoover's sign, factitious disorder, Munchausen syndrome, Malingering, feeding and eating disorders, Pica, rumination disorder, Avoidant/restrictive food intake disorder,

Anorexia nervosa, restricting type, binge-eating/purging type, lanugo, amenorrhoea, bulimia nervosa, binge eating, inappropriate compensatory behaviour, Russell's sign, Binge-eating disorder, other specified feeding or eating disorder, purging disorder, night eating syndrome, elimination disorders, Enuresis, Encopresis, sleep-wake disorders, Insomnia disorder, hypersomnolence disorder, narcolepsy, cataplexy, sleep paralysis, hypnagogic hallucinations, Parasomnias, NREM sleep arousal disorders, sleepwalking, sleep terrors, Nightmare disorder, REM sleep behaviour disorder, Restless legs syndrome, Kleine-Levin syndrome, sexual response cycle, refractory period, circular model, Sex, gender identity, sexual orientation, Androgyny, sexual dysfunction, Vaginismus, dyspareunia, performance anxiety, spectatoring, stop-start technique, squeeze technique, Sensate focus, paraphilia, paraphilic disorder, courtship disorders, algolagnic disorders, partialism, necrophilia, zoophilia, telephone scatologia, autogynephilia, lovemap, courtship disorder hypothesis, Gender dysphoria, gender identity disorder, gender incongruence, coercive sexual sadism disorder, Pathologisation, depathologisation, Kinsey scale, sexual orientation disturbance, ego-dystonic homosexuality, psychoactive substance, Intoxication, Withdrawal, Tolerance, Cross-tolerance, cross-dependence, Sensitisation, Physical dependence, Craving, harmful use, dependence syndrome, substance use disorder, gambling disorder, gaming disorder, mesolimbic dopamine system, opponent-process theory, gateway hypothesis, alcoholic hallucinosis, delirium tremens, Disulfiram, Formication, amotivational syndrome, neurocognitive disorders, major neurocognitive disorder, sundowning, asterixis, carphologia, subcortical dementia, Pseudodementia, Wernicke's encephalopathy, Korsakoff's syndrome, frontal lobe syndrome, personality disorder, ego-syntonic, categorical model, dimensional model, clusters, ideas of reference, magical thinking, Borderline personality disorder, borderline personality organisation, splitting, projective identification, self psychology, biosocial theory, Antisocial personality disorder, manie sans délire, moral insanity, Psychopathy Checklist-Revised, Psychopathy, fearlessness hypothesis, Histrionic personality disorder, Narcissistic personality disorder, Avoidant personality disorder, Dependent personality disorder, Obsessive-compulsive personality disorder, anal character, personality difficulty, anankastia, Alternative Model for Personality Disorders, Oppositional defiant disorder, Conduct disorder, with limited prosocial emotions, Intermittent explosive disorder, pyromania, kleptomania

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From the past papers

1. Which of the following culture-bound syndromes is NOT reported in India? (Amity 2023)
   (a) Dhat Syndrome (b) Koro (c) Suchi-bai (d) Latah

   Answer: (d) Latah. Latah, an exaggerated startle with echolalia, echopraxia, automatic obedience and blurted obscenities, belongs to Malaysia and Indonesia. Dhat (anxiety over semen loss), koro (fear that the genitals are retracting, with epidemics in north-east India) and suchi-bai (obsessional cleanliness in Bengal) are all reported in India.

2. The diathesis-stress model of psychopathology suggests that mental disorders result from: (RML 2023)
   (a) Genetic factors only (b) Environmental factors only (c) A combination of genetic predisposition and environmental stressors (d) Neurobiological abnormalities only

   Answer: (c) A combination of genetic predisposition and environmental stressors. The diathesis, a genetic, biological or psychological vulnerability, produces disorder only when a stressor sets it off; Zubin and Spring (1977) called this the vulnerability-stress model. Every option ending in 'only' names a single cause, the one-dimensional thinking the model replaced.

3. A high false positive rate in diagnosis means: (IHBAS 2025)

   (a) Diagnosis is too strict (b) Normal people are diagnosed as disordered (c) Diagnosis has low reliability (d) Disordered people go undetected

   Answer: (b) Normal people are diagnosed as disordered. A false positive wrongly labels a healthy person as ill, which happens when criteria are too loose. Disordered people going undetected are false negatives, the result of over-strict criteria, and low reliability means clinicians disagree with each other, a separate problem.

4. Hypnagogic hallucinations most commonly occur in: (RML 2024)

   (a) Schizophrenia (b) Delirium (c) Drug dependence (d) Normal

   Answer: (d) Normal. Hallucinations while falling asleep (hypnagogic) or waking (hypnopompic) lie within normal experience and are commonest in healthy people; narcolepsy also features them. The hallucinations of schizophrenia are mainly auditory and occur in clear consciousness, while delirium and drugs bring visual and tactile ones.

5. Our current mood and feelings are best described by the term: (RML 2024)

   (a) Emotion (b) Mood (c) Affect (d) Feeling

   Answer: (c) Affect. Affect is the immediate, observed, moment-to-moment expression of emotion (the weather); mood is the sustained state the person reports (the climate). Emotion is the broader term for a brief reaction, and feeling is its experienced side.

6. Choose the correct arrangement of these stages of elevated mood based upon severity: i) Elation, ii) Exaltation, iii) Euphoria, iv) Ecstasy. (Amity 2023)

   (a) i, ii, iii & iv (b) iii, i, ii & iv (c) iv, iii, ii & i (d) None of the above

   Answer: (b) iii, i, ii & iv. Elevated mood rises from euphoria (mild; hypomania) through elation (moderate; mania) and exaltation (intense, with grandiosity; severe mania) to ecstasy (rapture; stuporous or delirious mania). Option (a) puts elation first, and (c) reverses the scale.

7. Which is NOT a core symptom to diagnose as autism in children as per DSM- 5? (IHBAS 2024)

   (a) Communication skills (b) Unusual repetitive behavior (c) Social reciprocity (d) I.Q. is below 70

   Answer: (d) I.Q. is below 70. DSM-5 autism rests on two core domains, social communication and reciprocity, and restricted, repetitive behaviour, so (a) to (c) are core. Intellectual impairment is only a specifier (with or without).

8. In which of the following situations would a diagnosis of "global developmental delay" be inappropriate? (RML 2022)

   (a) The patient is a child who is too young to fully manifest specific symptoms or to complete requisite assessments. (b) The patient, a 7-year-old boy, has a full-scale IQ of 65 and severe impairment in adaptive functioning. (c) The patient's scores on psychometric tests suggest intellectual disability, but there is insufficient information about the patient's adaptive functional skills. (d) The patient's impaired adaptive functioning suggests intellectual developmental disorder, but there is insufficient information about the level of cognitive impairment measured by standardized instruments.

   Answer: (b) The 7-year-old with full-scale IQ 65 and severe adaptive impairment. Global developmental delay is a provisional label for children under 5 whose severity cannot yet be reliably assessed, as in (a), (c) and (d); a testable 7-year-old who meets both criteria has intellectual disability.

9. Within attention deficit/hyperactivity disorder, which of the following is NOT classified as a problem relating to hyperactivity/impulsivity? (RML 2021)

   (a) Experiences difficulties in playing quietly. (b) Rather than sitting still, squirms in seat and fidgets. (c) Talks excessively. (d) Doesn't appear to listen when being told something or spoken to

   Answer: (d) Doesn't appear to listen when spoken to. Not seeming to listen when spoken to directly is one of the nine inattention items; not playing quietly, squirming and fidgeting, and talking excessively are hyperactivity-impulsivity items.

10. "First rank symptoms" of schizophrenia given by Schneider include all except (JEMAS 2025)

    (a) Depersonalization (b) Hallucinatory voice in the form of running commentary (c) Delusional perception (d) Thought broadcasting

    Answer: (a) Depersonalization. Running-commentary voices, delusional perception and thought broadcasting are all on Kurt Schneider's list; depersonalisation, like loose associations and delusions of self-reference, is not a first-rank symptom.

11. The ICD-10 duration criterion for schizophrenia requires symptoms to persist for at least: (IHBAS 2025)

    (a) 6 months (b) 2 weeks (c) 3 months (d) 1 month

    Answer: (d) 1 month. ICD-10 and ICD-11 require 1 month. Six months is the DSM-5-TR total (including 1 month of active-phase symptoms); 2 weeks is the onset window of acute and transient psychotic disorders.

12. High EE (Expressed Emotion) is known to cause: (IHBAS 2020)

    (a) Poor treatment outcome (b) Relapse of psychotic illness (c) Negative symptoms of schizophrenia (d) Violent behavior in schizophrenic patients

    Answer: (b) Relapse of psychotic illness. Brown, Birley and Wing (1972) and Vaughn and Leff (1976) showed that critical, hostile or over-involved relatives predict relapse after discharge; EE does not cause the illness, its negative symptoms or violence.

13. Which of the following statements concerning Bipolar disorder is INCORRECT? (RML 2022)

    (a) Bipolar-I Disorder includes mania (b) Bipolar-II Disorder includes hypomania (c) Bipolar-I Disorder includes hypomania (d) Bipolar-I Disorder includes periods of depression even if these do not reach the threshold for major depressive disorder

    Answer: (c) Bipolar-I Disorder includes hypomania. Bipolar I is defined by at least one manic episode; hypomania is the defining high of bipolar II and is optional in bipolar I. Bipolar I may include depressive periods, even below the major depressive threshold, so (a), (b) and (d) are correct statements.

14. A 12-year-old boy begins to have new episodes of temper outbursts that are out of proportion to the situation. Which of the following is NOT a diagnostic possibility in this patient? (RML 2022)

    (a) Disruptive Mood Dysregulation Disorder (b) Bipolar Disorder (c) Conduct Disorder (d) Attention Deficit Hyperactivity Disorder

    Answer: (a) Disruptive Mood Dysregulation Disorder. DMDD requires onset before age 10 (diagnosis between 6 and 18), so outbursts that begin at 12 rule it out. Bipolar disorder, conduct disorder and ADHD can all begin or present at this age.

15. Suicidal risk is assessed to be highest when the following criteria are met in comparison to others, the 'time' factor being the same for all (JEMAS 2025)

    (a) High lethality, specific plan, frequent suicidal ideas, family history (b) Disrupted biological functioning, no recent attempt, no suicidal ideation (c) Adequate biological functioning, no previous attempt, moderate intent (d) Frequent suicidal ideation, recent suicide attempt, low lethality, absence of family history

    Answer: (a). Risk is highest when several markers converge: a highly lethal method, a specific plan, frequent ideation and a family history of suicide. Options (b) and (c) lack ideation or attempts, and (d) has low lethality and no family history.

16. The following defense mechanisms are used in obsessive-compulsive disorder: (RML 2024)

    (a) Isolation (b) Undoing (c) Reaction formation (d) Displacement

    Answer: (b) Undoing. Isolation, undoing and reaction formation are all classic OCD defences, but undoing, an act that symbolically cancels a forbidden thought, is the most characteristic and is the key. Displacement is the defence of phobias (Little Hans).

17. Generalized Anxiety Disorder (GAD) is a pervasive condition in which the sufferer experiences: (IHBAS 2024)

    (a) Continued apprehension about unspecified things (b) Fear of fear (c) Continual flashbacks to past events (d) A desire to check that the environment is safe

    Answer: (a) Continued apprehension about unspecified things. GAD is excessive, hard-to-control, free-floating worry about many matters for at least 6 months. Fear of fear marks panic disorder and agoraphobia, flashbacks PTSD and checking OCD.

18. A client with OCD washes their hands excessively to neutralize the anxiety caused by intrusive thoughts of contamination. This hand-washing behavior is BEST conceptualized as a(n): (Real PYQ 2024)

    (a) Conditioned stimulus (b) Unconditioned response (c) Avoidance response / Safety behavior (d) Operant behavior maintained by positive reinforcement

    Answer: (c) Avoidance response / Safety behavior. In Mowrer's two-factor account the washing removes anxiety, so it is negatively reinforced. The contamination thought is the conditioned stimulus, an unconditioned response is an unlearned reflex, and nothing pleasant is added, so it is not positive reinforcement.

19. The term 'la Bella indifference' seen in hysteria was first used by: (RML 2024)

    (a) Sigmund Freud (b) Anna Freud (c) Pierre Janet (d) Hippocrates

    Answer: (c) Pierre Janet. The phrase for the calm unconcern of some conversion patients is usually credited to Janet, who explained hysteria as dissociation. Freud explained hysteria by repression and conversion, Anna Freud catalogued the defences, and Hippocrates blamed a wandering womb.

20. When an individual has dissociative amnesia it is usually associated with several types of memory disturbances. In contrast, in generalized amnesia: (RML 2022)

    (a) There is a failure of recall of the entire past week (b) There is a failure of recall that encompasses the person's entire life (c) There is a failure of recall that encompasses the person's childhood (d) There is a failure of recall for traumatic events

    Answer: (b) Failure of recall that encompasses the person's entire life. Generalised amnesia covers the whole life history, often with identity. Loss of a set period such as a week is localised amnesia, loss of one category of information is systematised, and loss for traumatic events describes dissociative amnesia in general.

21. The difference between malingering and factitious disorder lies in: (IHBAS 2025)

    (a) The presence or absence of external rewards (b) Whether the individual believes the symptoms are real (c) The severity of symptoms (d) Age of onset

    Answer: (a) The presence or absence of external rewards. Both deliberately falsify symptoms and know they are false; the malingerer seeks an external incentive such as money or escape from duty, whereas factitious disorder persists without obvious external reward, to occupy the sick role.

22. Unlike people with anorexia, individuals with bulimia (RML 2022)

    (a) Often purge themselves to stay thin (b) Are obsessed with being too fat (c) Can do damage to their health (d) May appear normal in weight

    Answer: (d) May appear normal in weight. Significantly low body weight is a criterion of anorexia nervosa; people with bulimia are usually of normal weight or overweight. Purging (the binge-eating/purging type of anorexia), fear of fatness and damage to health occur in both disorders, so they do not distinguish them.

23. In narcolepsy, which of the following is most characteristically present? (IHBAS 2025)

    (a) Both cataplexy and catalepsy equally (b) Cataplexy, sudden loss of muscle tone triggered by emotions (c) Neither cataplexy nor catalepsy (d) Catalepsy, maintenance of imposed posture with waxy flexibility

    Answer: (b) Cataplexy. Narcolepsy type 1 shows brief, sudden, bilateral loss of muscle tone with preserved consciousness, triggered by laughter or joking, from loss of hypocretin (orexin) neurons. Catalepsy, a held posture with waxy flexibility, belongs to catatonia.

24. Lack of bowel control in children is known as: (RML 2024)

    (a) Encopresis (b) Analism (c) Enuresis (d) Anorexia

    Answer: (a) Encopresis. Encopresis is repeated passage of faeces into inappropriate places at least once a month for 3 months from age 4. Enuresis is the urinary counterpart (from age 5), anorexia is an eating disorder, and analism is not a recognised elimination disorder.

25. Premature ejaculation is most commonly associated with: (RML 2024)

    (a) Diabetes mellitus (b) Masturbation (c) Personality disorder (d) Anxiety state

    Answer: (d) Anxiety state. Premature ejaculation is mostly psychogenic and most tied to performance anxiety; it is commoner in men with anxiety disorders. Masturbation does not cause it (a myth), diabetes is the classic link of erectile disorder, and personality disorder has no special association.

26. Autogynephilia is defined by: (RML 2022)

    (a) Recurrent, and intense sexual attraction to prepubescent children (b) Recurrent, and intense sexual attraction to thoughts, images, or thoughts about oneself as a woman (c) Recurrent, and intense sexual attraction to oneself (d) Recurrent, and intense sexual attraction to being humiliated

    Answer: (b) Attraction to thoughts or images of oneself as a woman. Ray Blanchard coined the term in 1989, and DSM-5-TR uses it as a specifier of transvestic disorder. Attraction to prepubescent children is pedophilia and to being humiliated is masochism; mere attraction to oneself is not autogynephilia.

27. When gender incongruence moved out of the mental disorders chapter in ICD-11, which of these stayed in Chapter 6 (Mental, behavioural or neurodevelopmental disorders)? (NIMHANS 2026 recall)

    (a) Sexual dysfunctions (b) Paraphilic disorders (c) Sexual pain disorders (d) Gender incongruence of childhood

    Answer: (b) Paraphilic disorders. ICD-11 moved gender incongruence (of childhood as well as of adolescence or adulthood), the sexual dysfunctions and the sexual pain disorders into Chapter 17, Conditions related to sexual health; only the paraphilic disorders, now centred on non-consent, stayed in Chapter 6.

28. A patient presents with severe memory deficits, confabulation, and nystagmus, with a history of chronic alcohol use. This clinical picture is classic for: (Real PYQ 2024)

    (a) Alzheimer's Disease (b) Wernicke-Korsakoff Syndrome (c) Vascular Dementia (d) Frontotemporal Dementia

    Answer: (b) Wernicke-Korsakoff Syndrome. Thiamine deficiency in chronic drinkers damages the mammillary bodies and dorsomedial thalamus: nystagmus belongs to Wernicke's encephalopathy, amnesia with confabulation to Korsakoff's syndrome. Alzheimer's, vascular and frontotemporal dementia are not caused by alcohol and do not produce this triad.

29. Which of the following substances is NOT a natural alkaloid compound? (Amity 2023)

    (a) Morphine (b) Heroin (c) Codeine (d) Papaverine

    Answer: (b) Heroin. Heroin (diacetylmorphine) is made by chemically modifying morphine, so it is semi-synthetic. Morphine, codeine and papaverine all occur naturally in opium.

30. A key of distinguishing feature of dementia vs. delirium is : (IHBAS 2025)

    (a) Delusions in dementia but not delirium (b) Onset gradual in delirium vs. acute in dementia (c) Hallucinations in dementia but not delirium (d) Impaired memory in dementia vs. fluctuating consciousness in delirium

    Answer: (d) Impaired memory in dementia, fluctuating consciousness in delirium. Delirium is an acute, fluctuating disturbance of attention and awareness; dementia is an insidious decline in memory and cognition in clear consciousness. Delusions and hallucinations occur in both, and option (b) reverses the two onsets.

31. Borderline Personality Disorder is most characteristically associated with which primary defense mechanism? (Real PYQ 2025)

    (a) Repression (b) Splitting (c) Sublimation (d) Rationalization

    Answer: (b) Splitting. Splitting, seeing self and others as all good or all bad, is the primitive defence Kernberg placed at the centre of borderline personality organisation. Repression is the classic neurotic defence, sublimation a mature one and rationalisation a plausible excuse; none defines BPD.

32. Identify the 'odd one out' from the groups: (Amity 2023)

    (a) Paranoid personality disorder (b) Schizoid personality disorder (c) Schizotypal personality disorder (d) Borderline personality disorder

    Answer: (d) Borderline personality disorder. Paranoid, schizoid and schizotypal form Cluster A (odd or eccentric); borderline belongs to Cluster B (dramatic, emotional or erratic).

33. A client reports a persistent pattern of seeking admiration, but their behavior is characterized by excessive emotionality, attention-seeking, and seductive behavior, rather than grandiosity. This presentation is more characteristic of: (Real PYQ 2024)

    (a) Narcissistic Personality Disorder (b) Histrionic Personality Disorder (c) Borderline Personality Disorder (d) Dependent Personality Disorder

    Answer: (b) Histrionic Personality Disorder. Excessive emotionality, attention seeking and seductive behaviour define histrionic PD. Narcissistic PD also seeks admiration but rests on grandiosity and lack of empathy; borderline centres on instability and fear of abandonment; dependent on a need to be taken care of.

High-yield sheet

The facts this chapter's questions ask most, most-asked first. Last read before the exam.

1 Positive symptoms reflect excess dopamine D2 activity in the **mesolimbic** pathway (ventral tegmental area to nucleus accumbens); typical antipsychotics block D2, and blockade in the **nigrostriatal** pathway causes extrapyramidal symptoms. ASKED 13 TIMES

2 **Kurt Schneider's first-rank symptoms**: audible thoughts, voices arguing or commenting, thought insertion, withdrawal and broadcasting, made feelings, impulses and acts, somatic passivity and delusional perception; depersonalisation, loose associations and delusions of self-reference are not first-rank. ASKED 10 TIMES

3 **DSM-5-TR** schizophrenia needs 6 months of continuous signs including 1 month of active-phase symptoms; **ICD-10 and ICD-11** need 1 month. ASKED 10 TIMES

4 **Borderline personality disorder** is a pervasive pattern of instability of interpersonal relationships, self-image and affects, with marked impulsivity; DSM-5-TR requires five or more of nine criteria, including frantic efforts to avoid abandonment, identity disturbance, recurrent self-harm and chronic emptiness. ASKED 9 TIMES

5 **Positive symptoms** are the presence of something abnormal (delusions, hallucinations, disorganisation); **negative symptoms** are the absence of something normal (affective flattening, alogia, avolition, anhedonia, asociality). ASKED 8 TIMES

6 **Aaron Beck**'s cognitive theory of depression (1967): depression is maintained by the **negative cognitive triad**, negative views of the **self, the world (experience) and the future**, arising from negative schemas, automatic thoughts and cognitive distortions. ASKED 8 TIMES

7 **Korsakoff's syndrome** (anterograde and retrograde amnesia with confabulation) follows Wernicke's encephalopathy (confusion, ataxia, ophthalmoplegia with nystagmus); both result from **thiamine (vitamin B1) deficiency**, classically in chronic alcoholism, with lesions in the mammillary bodies and dorsomedial thalamus. ASKED 8 TIMES

8 **Persistent depressive disorder (dysthymia)**: depressed mood most of the day, more days than not, for at least **2 years in adults or 1 year in children and adolescents**, with 2 or more of 6 symptoms and never symptom-free for more than 2 months; less severe than major depression but longer. ASKED 7 TIMES

9 **Bipolar I** needs at least one **manic episode** (1 week or any duration if hospitalised, with marked impairment, hospitalisation or psychotic features); **bipolar II** needs at least one **hypomanic episode** (4 days, no marked impairment or psychosis) plus at least one major depressive episode and never a full mania. ASKED 7 TIMES

10 DSM-5-TR **PTSD** needs symptoms in all **four clusters** (intrusion, avoidance, negative alterations in cognitions and mood, alterations in arousal and reactivity) lasting **more than 1 month**; **acute stress disorder** covers 3 days to 1 month after the trauma. ASKED 7 TIMES

11 **Dissociative identity disorder** (formerly multiple personality disorder) requires two or more distinct personality states, or possession, plus recurrent amnesic gaps; the identity carrying the real name is the **host**, not schizophrenia's 'split'. ASKED 7 TIMES

12 **Cultural concepts of distress** (DSM-5's replacement for culture-bound syndromes): dhat (Indian subcontinent; anxiety over semen loss; named by N. N. Wig, 1960), koro (fear that the genitals are retracting into the body, causing death), pibloktoq (Inuit dissociative fits) and latah (Malaysia and Indonesia; exaggerated startle, echolalia and obscenities; not reported in India). ASKED 6 TIMES

13 Flight of ideas (mania), loosening of associations or derailment (schizophrenia), tangentiality and circumstantiality are disorders of the **form** of thought, not its content; circumstantial speech reaches the goal, tangential speech never does, and **word salad** is the most severe formal thought disorder. ASKED 6 TIMES

14 **Capgras syndrome** (Joseph Capgras, 1923) is the delusion that a familiar person, usually a spouse or relative, has been replaced by an identical impostor; in **Fregoli syndrome** strangers are a familiar person in disguise. ASKED 6 TIMES

15 Antipsychotic movement effects appear in order: acute dystonia (hours to days), akathisia (days to weeks), parkinsonism (weeks), **tardive dyskinesia** (months to years; orofacial, often irreversible); akathisia and tardive dyskinesia are secondary, drug-induced movement disorders. ASKED 6 TIMES

16 **Folie à deux** (Lasègue and Falret, 1877) is a delusion transmitted from a dominant psychotic primary case to a passive partner in a close relationship; ICD-10 F24 induced delusional disorder, DSM-IV shared psychotic disorder; ICD-11 describes it under 6A24 delusional disorder. ASKED 6 TIMES

17 A **major depressive episode** needs **5 or more of 9 symptoms** in the same **2-week** period, at least one being depressed mood or loss of interest or pleasure; phobias and euphoric mood are not among them. ASKED 6 TIMES

18 **La belle indifférence**, calm unconcern about a dramatic symptom, is the classic but non-diagnostic sign of **conversion disorder** (functional neurological symptom disorder); the term is usually credited to **Pierre Janet**. ASKED 6 TIMES

19 **Factitious disorder** (Munchausen syndrome, Asher 1951; imposed on another, Meadow 1977) and **malingering** both feign illness on purpose; they differ in the **absence or presence of external rewards**, and malingering is not a mental disorder. ASKED 6 TIMES

20 The **diathesis-stress model** explains disorder as a predisposition (a genetic, biological or psychological diathesis) set off by stress; Meehl applied it to schizophrenia in 1962, and Zubin and Spring named it the vulnerability-stress model in 1977. ASKED 5 TIMES

21 **Dissociative amnesia** is inability to recall important, usually traumatic autobiographical information beyond ordinary forgetting; **generalised** amnesia covers the whole life history, and **fugue** (purposeful travel with loss of identity) is now a specifier. ASKED 5 TIMES

22 **Bulimia nervosa** needs recurrent binges with loss of control plus compensatory behaviour (vomiting, laxatives, diuretics, fasting, exercise), both at least **once a week for 3 months**; **binge-eating disorder** needs binges once a week for 3 months with marked distress, three of five features and no regular compensation. ASKED 5 TIMES

23 **Cotard syndrome** (Jules Cotard, 1880) is a nihilistic delusion that one is dead, does not exist, or has lost organs or blood; nihilistic delusions are characteristic of severe psychotic depression. ASKED 4 TIMES

24 Depression, the **common cold of psychiatry**, is characterised by persistent sadness and loss of interest; it is widespread and increasing, often triggered by stress, linked to suicide risk by severity, and about **twice as common in women** as in men. ASKED 4 TIMES

25 OCD obsessions are **ego-dystonic** (intrusive, unwanted, inconsistent with the self-concept); obsessive-compulsive personality disorder is **ego-syntonic** perfectionism without true obsessions or compulsions. ASKED 4 TIMES

26 **Generalised anxiety disorder** is excessive, hard-to-control worry about many events, more days than not for at least **6 months**, with 3 of 6 symptoms (restlessness, fatigue, poor concentration, irritability, muscle tension, sleep disturbance; 1 in children): continued apprehension about unspecified things. ASKED 4 TIMES

27 **Undoing** is the most characteristic OCD defence, alongside isolation of affect and reaction formation; **repression** is the defence of hysteria and **displacement** that of phobias. ASKED 4 TIMES

28 **Depersonalisation** is feeling detached from or unreal in oneself ('not my normal self'); **derealisation** is the surroundings feeling unreal or dreamlike; reality testing stays intact. ASKED 4 TIMES

29 DSM-5-TR **anorexia nervosa**: restriction leading to significantly low body weight, intense fear of weight gain (or behaviour blocking gain) and disturbed experience of weight or shape; **amenorrhoea was dropped** and no BMI cut-off is required; the restricting type has no recurrent bingeing or purging in the last 3 months. ASKED 4 TIMES

30 Anorexia nervosa is the **most lethal eating disorder** (crude mortality about 5 per cent per decade in DSM-5-TR), with lanugo, bradycardia, hypotension, hypothermia, amenorrhoea and hypokalaemia, and the trait of **perfectionism**. ASKED 4 TIMES

CHAPTER 9

Research Methodology

IN THIS CHAPTER

9.1 ●● Science and the Research Process: Problems and Hypotheses: Distinguish science from other ways of knowing, inductive from deductive reasoning and the main types of research, and turn a research problem and literature review into a well-formed hypothesis of the right type.

9.2 ●● Variables, Experimental Control and the Validity of Research: Identify and classify the variables in a study, define them operationally, choose the right technique of control, recognise participant and experimenter artefacts and what blinding controls, and judge a study's statistical conclusion, internal, construct and external validity and the threats to each.

9.3 ●● Sampling: Populations, Probability and Non-Probability Methods: Define population, sample, parameter and statistic, tell probability from non-probability sampling, recognise every major sampling method from a description (simple random, systematic, stratified, cluster and multistage, quota, purposive, convenience, snowball), and distinguish sampling error from bias.

9.4 ●● Research Designs: Experimental, Single-Case and Non-Experimental: Distinguish experimental, quasi-experimental, single-case, non-experimental and developmental designs, and judge which conclusions each one licenses.

9.5 ●● Clinical Epidemiology, Randomised Trials and Evidence Synthesis: Calculate and interpret prevalence, incidence, relative risk, odds ratio, NNT, sensitivity, specificity and predictive values, tell the epidemiological designs and trial analyses apart, and explain meta-analysis and the reporting guidelines.

9.6 ○ Methods of Data Collection and Qualitative Research: Choose and critique the main tools of data collection (observation, interview, questionnaire and schedule, content analysis, sociometry), place a study in its research paradigm, and tell the qualitative traditions, their analysis and their quality criteria apart.

9.7 ○ Research Ethics and Writing the Research Report: Trace research ethics from the abuses that produced it to its codes and core principles (informed consent, protection from harm, confidentiality, deception and debriefing, the 3Rs, honest publication), and describe the structure and style of an APA research report and of a research proposal.

9.8 ●●● Measurement Scales, Organising Data and Descriptive Statistics: Classify any variable by Stevens's four levels of measurement, process, tabulate and graph raw data, choose and compute averages and measures of variability, read skewness and kurtosis, and use the normal curve to convert scores through z into percentiles and standard scores.

9.9 ●● Sampling Distributions and the Logic of Hypothesis Testing: Explain how psychologists reason from samples to populations: the sampling distribution and standard error, the central limit theorem, confidence intervals, the steps of a significance test, the p-value, Type I and Type II errors, power and effect size.

9.10 ●●● Comparing Groups: t-Tests, ANOVA and Non-Parametric Tests: Choose the right test for comparing groups from the design and the level of measurement (t-tests, one-way and factorial ANOVA with post hoc tests, ANCOVA and MANOVA, chi-square and the rank tests) and work out its degrees of freedom.

9.11 ●● Correlation, Regression and Factor Analysis: Choose and interpret the right correlation coefficient, read and use a regression equation, and decide how many factors to retain and how to rotate them in a factor analysis.

Weightage. 320 of the 3180 questions mapped in this book (10.1%) test this chapter, 175 of them from real past papers. 2 of its 11 sections are High yield.

Berlin, 1904. A retired schoolteacher named Wilhelm von Osten puts a question to his horse: what is three plus five? Hans taps his hoof eight times and stops. Von Osten had spent years teaching him, and Hans now seemed able to add, subtract, multiply and divide, to work with fractions, tell the time and answer questions about German words, all by tapping. He answered strangers as well as his owner, so deliberate fraud seemed unlikely. In September 1904 a commission that included the psychologist Carl Stumpf watched him perform and found no trick. The newspapers called him Clever Hans.

Stumpf passed the puzzle to his student Oskar Pfungst, who treated it as a research problem rather than a marvel. Pfungst set out rival explanations (trickery, genuine reasoning, cues from the people present) and tested them by changing one condition at a time. When the questioner knew the answer, Hans was nearly always right; when the questioner did not, Hans was nearly always wrong. With blinkers that hid the questioner, or with the questioner standing far off, Hans failed again. The horse was not counting; he was watching. Pfungst found the signal: once the question was put, the questioner bent forward slightly to watch the hoof, and Hans began to tap; when the right count was reached, the questioner's head lifted a fraction and the tension left his body. Hans stopped at that release. None of the questioners, von Osten included, knew they were doing it.

Pfungst published his study in 1907. He even took the horse's part himself in the laboratory, tapping out answers for human questioners, and found that they gave him the same small, unwitting signals. The lesson has outlived the horse: what an observer expects can leak into the very data meant to test it. This chapter is about the methods science has built to stop that from happening.

9.1 Science and the Research Process: Problems and Hypotheses

MEDIUM YIELD · 21 QS

Why should you believe a psychologist's claim? Not because the psychologist is famous or certain, but to the degree that the claim was reached by a method that could have shown it to be wrong. This section describes that method, the reasoning it uses, the kinds of research it produces, and the first two steps of every study: finding a problem and stating a hypothesis.

Ways of knowing ○

In "The Fixation of Belief" (1877) the American philosopher **Charles Sanders Peirce** (1839 to 1914) described four ways in which people settle what they believe, and **Fred N. Kerlinger** (1910 to 1991) made them standard in *Foundations of Behavioral Research* (1964). By the **method of tenacity** you hold a belief because you always have; by the **method of authority**, because a respected source says so; by the **a priori method**, because it seems self-evidently reasonable (intuition). Only the method of science settles belief by public observation that anyone can repeat, so only science corrects its own mistakes.

Older traditions stand behind the list (Table 9.1). **Socrates** (about 470 to 399 BCE) used the **elenchus**, a question-and-answer cross-examination that draws out a person's beliefs until they contradict one another (the Socratic method). **Aristotle** (384 to 322 BCE) observed and

classified nature; **René Descartes** (1596 to 1650) trusted reason and intuition (rationalism); **Galileo Galilei** (1564 to 1642) experimented; and **Francis Bacon** (1561 to 1626) argued in *Novum Organum* (1620) for building knowledge inductively from observations.

Table 9.1 Ways of knowing. Only the last corrects itself.

Way of knowing	Linked with	Belief is settled by
Elenchus	Socrates	Questioning that exposes contradictions
Observation, classification	Aristotle	Watching and sorting nature
Intuition, reason	Descartes	Thinking alone
Experiment, induction	Galileo; Bacon	Varying conditions; generalising
Tenacity	Peirce	Habit and repetition
Authority	Peirce	An expert or text
A priori	Peirce	What "stands to reason"
Science	Peirce	Public, repeatable observation

What makes research scientific ○

Kerlinger defined scientific research as "systematic, controlled, empirical and critical investigation of hypothetical propositions about the presumed relations among natural phenomena". Unpacked, that gives its characteristics.

- **Empiricism**: claims are settled by observation, not by argument or rank.
- Objectivity through systematic, controlled observation: the result should not depend on who observes it.
- **Replicability** and reproducibility: the study can be repeated, and gives the same results when it is. When the Open Science Collaboration (2015) repeated 100 published psychology studies, about 97 percent of the originals but only about 36 percent of the repetitions were significant.
- **Parsimony**: prefer the simpler of two explanations that fit equally well. This is Occam's razor, after **William of Ockham** (about 1287 to 1347); in animal psychology it is Lloyd Morgan's canon (1894), after **Conwy Lloyd Morgan** (1852 to 1936): never explain an action by a higher mental faculty if a lower one will do.
- **Falsifiability**: **Karl Popper** (1902 to 1994), in *Logik der Forschung* (1934), noted that a thousand white swans cannot prove "all swans are white" but one black swan disproves it. A statement is scientific only if some possible observation could show it false; psychoanalysis, which explains any behaviour after the event, was his example of a theory that risks nothing.
- Determinism and tentativeness: behaviour has natural causes and so is lawful, yet every conclusion is provisional.

A. K. Singh adds three assumptions of science (nature is lawful, behaviour is determined, nature is understandable) and six attitudes of the scientist: open-minded, tolerant of uncertainty, aware of being fallible, cautious (a finding is evidence, never "proof"), sceptical and ethical. The goals of science are to describe, explain, predict and control behaviour.

> **! EXAM TRAP**
>
> "A characteristic of good research" is **reproducibility of results**. The distractors are the opposites of the real criteria: good research has clear objectives, unbiased findings, adequate scope and ethical conduct.

Science builds with the terms in Table 9.2. A hypothesis is a guess about one relation, a law a confirmed regularity, a theory an explanatory system; "just a theory" misuses the word.

Table 9.2 The building blocks of scientific knowledge.

Term	Meaning	Example
Fact	An observation others can verify	Pavlov's dogs salivated before feeding
Concept	An abstraction naming a class of observations	Reward, aggression
Construct	A concept invented for science, inferred and not directly observed	Anxiety, intelligence
Theory	Interrelated constructs and propositions that explain and predict (Kerlinger)	Beck's cognitive theory of depression
Law	A highly confirmed, usually narrow regularity	Weber's law
Model	A simplified representation or analogy	Atkinson and Shiffrin's memory model

Popper said how science ought to work; **Thomas S. Kuhn** (1922 to 1996) described how it does, in *The Structure of Scientific Revolutions* (1962). Scientists usually do normal science inside a **paradigm**, a shared package of theory, methods and assumptions. Anomalies pile up, crisis follows, and a scientific revolution installs a new paradigm (a paradigm shift), as when the cognitive revolution displaced behaviourism.

Induction, deduction and abduction ●●●

Induction runs from specific observations to broad generalisations, bottom-up. You interview twenty hostel students in exam week, notice that the poorest sleepers worry most, and generalise. Induction builds theory, so it belongs to the theory-development stage and is typical of exploratory and qualitative research. Its weakness, noted by **David Hume** (1711 to 1776), is that no number of cases guarantees the rule.

Deduction runs top-down, from theory to hypothesis to the observation that tests it: if the cognitive theory of depression is right, depressed patients should recall more negative words, so you test them. It is typical of quantitative, experimental research. **Abduction**, Peirce's term, is inference to the best explanation: from a surprising fact to the hypothesis that would best account for it, as when Oskar Pfungst guessed that Clever Hans was reading his questioners (see the chapter opening).

The **hypothetico-deductive method** joins them: propose a hypothesis, deduce a prediction, test it, then keep, revise or drop the hypothesis. **Clark L. Hull** (1884 to 1952) built *Principles of Behavior* (1943) this way. Figure 9.1 shows the cycle.

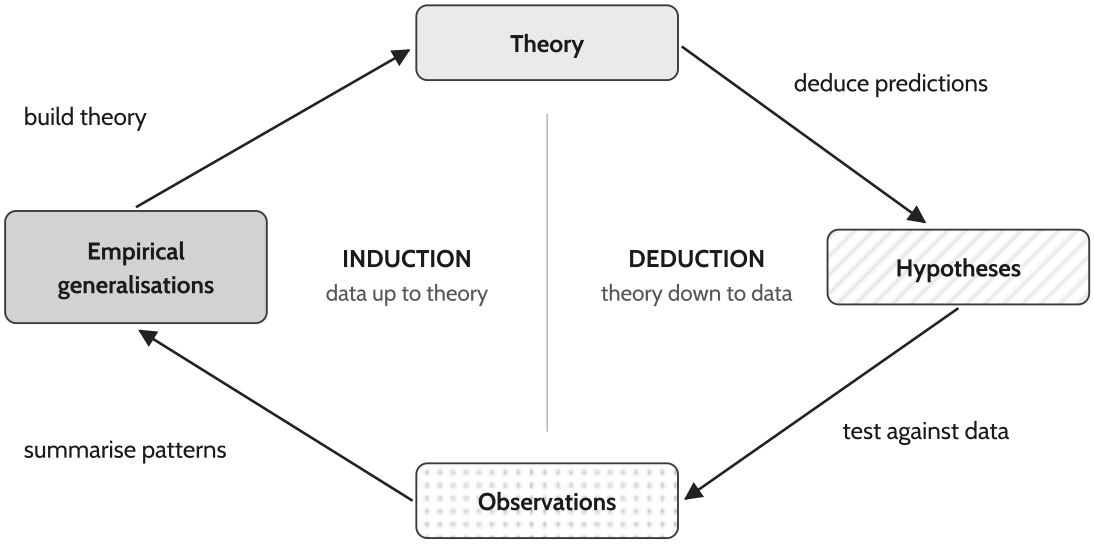


Figure 9.1 The research cycle, after Walter Wallace's "wheel of science" (1971). Deduction carries theory down to data (right); induction carries data back up to theory (left), at the theory-development stage.

> **! EXAM TRAP**
> "Starts with a theory and tests it with observations" is **deductive reasoning**; the hypothetico-deductive method (the whole conjecture-and-test cycle) is the distractor. "Gathers data, finds patterns, then generates a theory", "specific observations to broader generalisations" and "associated with qualitative research" are all **inductive reasoning**, used at the **theory-development** stage, not in formulating hypotheses, collecting data or reviewing literature. Experimental design and quantitative analysis go with deduction; qualitative analysis is a way of handling data, not a direction of reasoning.

Methodology, methods and the steps of research ●●

Research methods are the techniques for collecting and analysing data: an interview, an experiment, a t-test. **Research methodology** is the wider study of the logic behind choosing them: why this method suits this problem, and what its results can show. As **C. R. Kothari** puts it in *Research Methodology: Methods and Techniques*, methods are part of methodology, so methodology has the wider scope.

Research advances theory, informs government policy, guides practice (which therapy works) and, in business, provides insights for decision-making, a role broader than profit and far from merely academic. Kothari's criteria of good research are a clearly defined purpose, a procedure described well enough to be repeated, a design planned for objective results, frankly reported flaws, adequate analysis, and conclusions confined to what the data justify.

Singh's eight steps are: (1) identifying the problem, with the literature review; (2) formulating hypotheses; (3) identifying, manipulating and controlling variables (9.2); (4) the research design (9.4); (5) devices for observation and measurement (9.6); (6) summarising results (9.8); (7) statistical analysis (9.9 to 9.11); and (8) drawing conclusions. Kothari's flow runs: problem, literature survey, hypotheses, research design, sample design (9.3), data collection, execution, analysis, hypothesis testing, generalisation and interpretation, and the report (9.7), which raises the next problem.

Types of research ●●●

One study has a place on each axis of Table 9.3: a randomised trial of yoga for anxiety in a Bengaluru hospital is applied, explanatory, quantitative, experimental, field-based and longitudinal.

Table 9.3 Ways of classifying research.

Basis	Types	What separates them
Purpose (also Kothari)	Basic (pure, fundamental) versus applied	Knowledge for its own sake versus a practical problem
Depth	Exploratory, descriptive, explanatory	Generates hypotheses; describes; tests causes
Kothari	Descriptive versus analytical	Describes things as they are versus critically evaluates available facts
Kothari	Quantitative versus qualitative	Amounts versus qualities and meanings (9.6)
Kothari	Conceptual versus empirical	Abstract ideas versus observation or experiment
Kerlinger	Experimental versus non-experimental	Independent variable manipulated versus measured
Setting	Laboratory versus field	Control versus realism
Time	Cross-sectional versus longitudinal	One occasion versus followed over time (9.4)
Focus	Nomothetic versus idiographic	General laws versus the unique individual

Basic research seeks knowledge for its own sake; **applied research** solves a practical problem. **Action research**, an applied form, was introduced by **Kurt Lewin** (1890 to 1947), often called the father of modern social psychology, in 1946. It is research done by or with the people in a setting to solve a specific local problem there, in cycles now drawn as plan, act, observe and reflect: a ward team studies why patients miss follow-up visits, changes the reminders and measures again (participatory action research is in 9.6). Evaluation research judges whether a programme works; **Michael Scriven** (1967) separated formative evaluation, which improves a running programme, from summative evaluation, which judges it at the end.

Exploratory research probes a poorly understood area without firm hypotheses, so it is hypothesis-generating. **Descriptive research** observes and describes what exists without manipulating variables. **Explanatory research** (causal research) tests hypotheses about why, most strongly by experiment, a variable-manipulating study. A case study is a design for one case (9.4).

Wilhelm Windelband (1848 to 1915) in 1894 separated sciences seeking general laws from those seeking to understand the unique case. The **nomothetic approach** (Greek *nomos*, law) finds laws across many people through experimental and quasi-experimental methods; the **idiographic approach** (*idios*, one's own) studies one person in depth through case studies, ethnography and other qualitative methods. **Gordon W. Allport** (1897 to 1967) brought the pair into personality psychology in 1937 and preferred the idiographic approach (Chapter 4).

> **! EXAM TRAP**
>
> Both experimental and quasi-experimental methods are nomothetic; when both are offered, the expected answer is the **experimental method**. Qualitative methods and ethnography are idiographic. Allport preferred the **idiographic** approach; factorial design (9.4) and chi-square (9.10) are a design and a test, not approaches. Action research is "**research carried out to solve a specific problem**": it is applied but need not be longitudinal, so "all of the above" fails. Lewin founded action research, not psychoanalysis, psychophysics or functionalism (Chapter 1).

The research problem

A **research problem** is a felt difficulty sharpened until data can answer it: "students seem stressed" becomes "Is sleep duration related to test anxiety in first-year students?" Kerlinger's criteria are that it relates two or more variables, is stated clearly as a question, and implies empirical testing. A good research question is clear, answerable, connected to earlier work and worth answering.

Problems come from experience, theory and earlier findings, which reveal a gap, contradictory results, or an isolated unexplained fact, such as the apparent movement that **Max Wertheimer** (1880 to 1943) studied as the phi phenomenon, which led to Gestalt psychology. **Pauline V. Young** (1956) named documentary sources, key informants and library sources; Baker and Keller add programmed activities, research proposals, government policies and social values.

A problem should be novel, significant, feasible (time, money, participants), within the researcher's interest and competence, supported by available data and instruments, and ethical. It must also be solvable by empirical observation. An **unsolvable problem** ("Is there a divinity that shapes our ends?") is metaphysical: unstructured, with terms that cannot be operationally defined and data that cannot be collected. Singh's steps in formulating a problem are: choose a broad area, divide it, select a subarea, raise questions, turn them into objectives, assess their feasibility, and double-check.

Reviewing the literature ●●

The **literature** is the body of prior work related to a research problem; it is not an archive (a store of historical records) or a repository (a storage place for documents and data). The **review of literature** evaluates it. Its primary purpose is to provide a theoretical framework, from which hypotheses are deduced; it also avoids duplication, locates gaps, refines the problem and suggests variables, designs and tools.

Singh's five types are primary (traces a topic's development), content (links a study to related work), methodological (compares methods), integrative or systematic (combines empirical results; see 9.5) and theoretical (compares theories). Sources include peer-reviewed journals, books, review articles, abstracts (*Psychological Abstracts*, begun in 1927, now PsycINFO), indexes, PubMed, dissertations (Shodhganga in India), the internet and **grey literature**, material without an ISBN or ISSN such as theses and government reports. A review is written as critical synthesis organised by theme, with each source annotated, not as a string of summaries.

> **! EXAM TRAP**
>
> The primary objective of a literature review is to **provide a theoretical framework.** Establishing the researcher's credibility is a secondary purpose Singh mentions; summarising your own findings belongs in the results, and ignoring earlier research is the opposite of a review.

The hypothesis: meaning, criteria and functions ○

The problem asks; the **hypothesis** answers. Kerlinger (1973) called it "a conjectural statement of the relation between two or more variables", always declarative: a tentative, testable statement. "Is sleep related to test anxiety?" is a problem, interrogative and not directly testable; "Students who sleep under six hours will report higher test anxiety" is a hypothesis, declarative and testable. Hypotheses can be recast as "if (antecedent), then (consequent)".

A good hypothesis is testable and falsifiable, conceptually clear, specific, consistent with known facts, parsimonious, and related to theory and to available techniques. **William J. Goode** and **Paul K. Hatt** (*Methods in Social Research*, 1952) named three difficulties in formulating one: no clear theoretical framework (or ignorance of it), inability to use that framework logically (Singh renders this as not knowing what theoretical evidence is available), and unfamiliarity with research techniques.

Hypotheses come from theory, observation, earlier research, analogy (learned helplessness in dogs suggested a model of human depression), intuition and culture. A hypothesis guides data collection and analysis (what to collect, what to ignore), delimits the study, suggests the design and links theory to observation. It is never meant to prove the researcher's view or steer the results, and it does not replace the literature review it grows from. A supported hypothesis is retained, never proved.

Types of hypotheses ●●

Table 9.4 lists the types. The **null hypothesis** (H_0) states no difference or relationship in the population; the **alternative hypothesis** (H_1) states that one exists, and is accepted when the null is rejected. An alternative is a **directional hypothesis** if it names the direction (one-tailed test) and a **non-directional hypothesis** if it predicts only a difference (two-tailed); statistical hypotheses and their testing are in 9.9. A **universal hypothesis** holds for all values of the variables at all times and places; an **existential hypothesis** holds for at least one case. Universal hypotheses are preferred because they predict more.

Table 9.4 Types of hypotheses.

Type	States	Example
Universal	All values, all times and places	Adequate light increases reading efficiency
Existential	At least one case	At least one patient improves untreated
Causal	One variable causes another	Boring content causes channel changing
Descriptive	A feature of behaviour, no cause	Viewers alone change channels more
Simple	One independent, one dependent variable	Reward improves learning
Complex	More than two variables	Class and gender jointly affect adjustment

Type	States	Example
Research	Prediction derived from the theory	CBT beats usual care for depression
Working	Provisional guess, revised as data arrive	Dropout may depend on distance
Null (H_0)	No difference or relationship	No difference between male and female incomes in the population
Alternative (H_1)	A difference or relationship	Male and female incomes differ
Directional	Names the direction	Reinforcement will have a positive effect on performance
Non-directional	A difference only	Boys and girls differ in aggression

! EXAM TRAP

"Reinforcement will have a positive effect on performance" predicts an effect (**alternative**, not null) and its direction (**directional**). The statement that "sounds like a null" is the one of **no difference**: "there is a correlation", "the coin is not fair" and "the defendant is guilty" all claim an effect (the courtroom null is "innocent"). For universal hypotheses, options about "two" or "some" values are invented: universal means all values everywhere.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Peirce's four methods of fixing belief are tenacity, authority, a priori and _____.
2. Socrates' cross-examination that exposes contradictions is the _____.
3. For Popper, a scientific statement must be _____.
4. Specific observations to general theory is _____ reasoning.
5. Allport preferred the _____ approach.
6. Action research was founded by _____.
7. A literature review chiefly provides a _____.
8. "Reinforcement will improve performance" is an alternative, _____ hypothesis.

Answers: (1) science, (2) elenchus, (3) falsifiable, (4) inductive, (5) idiographic, (6) Kurt Lewin, (7) theoretical framework, (8) directional

9.2 Variables, Experimental Control and the Validity of Research

MEDIUM YIELD · 22 QS

Suppose a counsellor at a Mysuru college runs a six-week yoga course for forty anxious students, and their anxiety falls. Did the yoga do it, or did the examinations end, or did the students relax because someone was watching? Deciding needs the language of variables, measurement, control, guards against bias and the idea of validity. (Hypotheses are Section 9.1; designs, Section 9.4.)

Variables and their kinds ●●

A **variable** is any characteristic that can take two or more values; one with a single value in a study is a **constant**. A **concept** is an abstraction from observed events (honesty, courage), and

a **construct** is a concept deliberately invented for scientific use, such as intelligence (**Fred N. Kerlinger**, 1910 to 1991, *Foundations of Behavioral Research*, 1964). Measurability is what separates a variable from a concept.

The **independent variable** (IV) is the presumed cause, manipulated by the experimenter or, where that is impossible, selected. The **dependent variable** (DV) is the measured outcome, expected to change with the IV. Find the IV by asking what is presumed to cause what, not by word order: in "old age home residents who have heart disease die earlier than those who do not", the IV is having heart disease, the DV is age at death, and the old age home is a constant.

Heart disease cannot be handed out, which illustrates Kerlinger's split between **active variables**, which the researcher manipulates (dose, instructions), and **attribute variables**, which participants bring and which can only be measured or selected (age, sex, diagnosis). With an attribute IV random assignment is impossible, so causal claims are weaker. Attribute variables are also called organismic, the O in **Robert S. Woodworth**'s (1869 to 1962) stimulus-organism-response scheme. Table 9.5 lists the other contrasts.

Table 9.5 Ways of classifying variables.

Distinction	Meaning	Example
Stimulus, organismic, response	Feature of the task or setting; of the person; the behaviour measured	Noise; age; reaction time
Explanatory and response	IV and DV in observational studies	Smoking; lung disease
Predictor and criterion	IV and DV in correlation and regression	Entrance score; MA marks
Continuous and discrete	Any value in a range; separate, countable values	Reaction time; siblings
Quantitative and qualitative	Differ in amount; differ in kind (categorical)	Anxiety score; diagnosis
True and artificial dichotomy	Two natural categories; a continuum cut in two (biserial correlation, Section 9.11)	Alive or dead; pass or fail

> **! EXAM TRAP**
>
> The IV is the presumed cause (having heart disease), not the first-mentioned variable (dying early) or the setting (the old age home, a constant). "Manipulated by the experimenter" defines the IV; "measured as the outcome" defines the DV; "randomly assigned" describes participants, not a variable.

Extraneous, confounding, moderator and mediator variables ○

An **extraneous variable** is any variable other than the IV that may affect the DV: **subject** or organismic variables (age, intelligence, motivation), **task** variables (difficulty, instructions), **environmental** variables (noise, light) and **situational** variables (time of day, the experimenter's manner). One that varies randomly only adds noise. One that varies systematically with the IV and also affects the DV is a **confounding variable**: it influences both, and its effect cannot be separated from the IV's. If the yoga volunteers were also more motivated, and motivation lowers anxiety, motivation is confounded with treatment.

Reuben M. Baron and **David A. Kenny** (1986) separated two other third variables (Figure 9.2). A **moderator** changes the direction or strength of the IV-DV relationship, saying when or for

whom: if yoga helps mild but not severe anxiety, severity moderates the effect (statistically an interaction, Section 9.10). A **mediator** is the mechanism through which the IV acts: yoga slows breathing, which lowers anxiety. An **intervening variable** is such a link that can only be inferred, never observed, such as learning or fatigue.

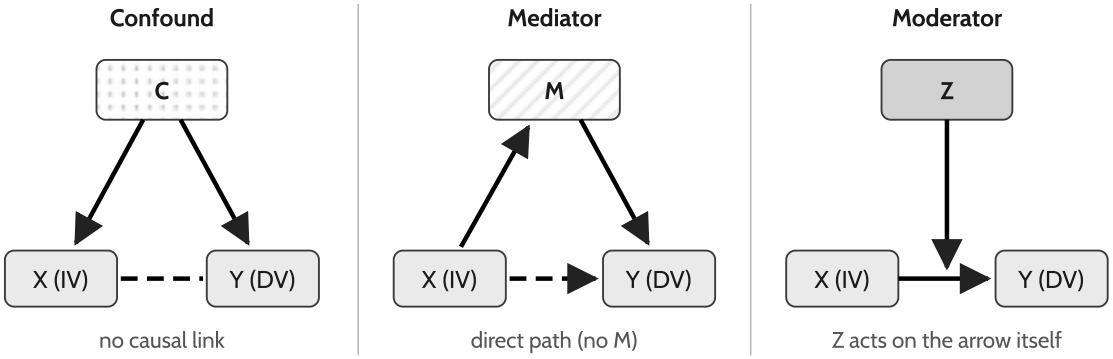


Figure 9.2 A confound (C) drives both X and Y; a mediator (M) carries the effect from X to Y; a moderator (Z) changes how strongly X affects Y.

> **! EXAM TRAP**
> A variable that "only affects the DV" is merely extraneous; a confound must also vary with the IV, which is why it "influences both". Hook: mediator = mechanism (how); moderator = more for some (when, for whom).

Operational definitions and measuring the DV

An **operational definition** defines a variable by the operations used to manipulate or measure it: anxiety is the score on a named inventory; hunger is hours without food. This **operationism** comes from the physicist **Percy Williams Bridgman** (1882 to 1961) in *The Logic of Modern Physics* (1927); **Stanley Smith Stevens** (1906 to 1973) brought it into psychology in 1935. It makes research objective and replicable, though no single operation captures a whole construct. The DV can be measured by direct observation of behaviour, by indirect physiological measures or reaction time, by judgements about a stimulus, or by self-report. A good DV is reliable, valid and sensitive. Scores bunched at the bottom of a scale because the task is too hard show a **floor effect**; scores bunched at the top because it is too easy show a **ceiling effect**. Both hide real differences.

Control and the MAXMINCON principle ••

Control means holding constant, or otherwise neutralising, the influence of extraneous variables on the DV, so that conditions differ only in the IV. Kerlinger summed up the aim of design in the **MAXMINCON principle**: **maximise** systematic (experimental or **primary**) variance, the variance due to the IV, by making its levels as different as possible; **minimise error variance**, the random variance from individual differences and measurement error, by reliable measures and steady conditions; and **control** extraneous (**secondary**) variance, the systematic variance from other variables, by the techniques that follow.

> **! EXAM TRAP**
>
> MAXMINCON reads primary, error, secondary; "true variance" belongs to test theory (Chapter 2). One real paper defines control as holding constant "the possible influence of the *dependent* variable". The stem is careless (extraneous variables are what is held constant), but control is still the answer. Standard deviation is a statistic (Section 9.8); an experiment is the whole procedure (Section 9.4).

Techniques of control ●●

Elimination removes an extraneous variable (a sound-proof room removes noise). **Constancy of conditions** holds it at one value for all (same room, time and instructions), at some cost to generality. **Matching** pairs participants on a relevant variable and splits each pair between groups. **Balancing** spreads a known variable equally across groups, as with equal numbers of men and women in each.

When the same people serve in every condition, **order effects** arise: practice, fatigue and carry-over. **Counterbalancing** varies the order so that each condition appears equally often in each position: **complete** (every order: 2 for two conditions, 6 for three, 24 for four), **partial** (a subset, such as a Latin square, Section 9.4) or **ABBA** (one order, then its reverse). Psychophysics controls **primacy and recency effects** the same way, by counterbalancing the order of presentation.

The most powerful technique is **randomisation**, or random assignment: every participant has an equal chance of being placed in any group. It makes the groups alike at the start on known and unknown variables, so that the main difference between them is the treatment, and is the chief guard against selection bias and confounding. That is why an HIV vaccine trial randomly assigned over 16,000 volunteers to vaccine or placebo: not to reduce non-response or sampling variability (Section 9.3), and not to blind anyone. It does nothing about measurement error (reliable measures reduce that), recall bias (faulty memory of past exposures) or publication bias (selective publishing of positive results; both Section 9.5). Figure 9.3 separates it from random sampling.

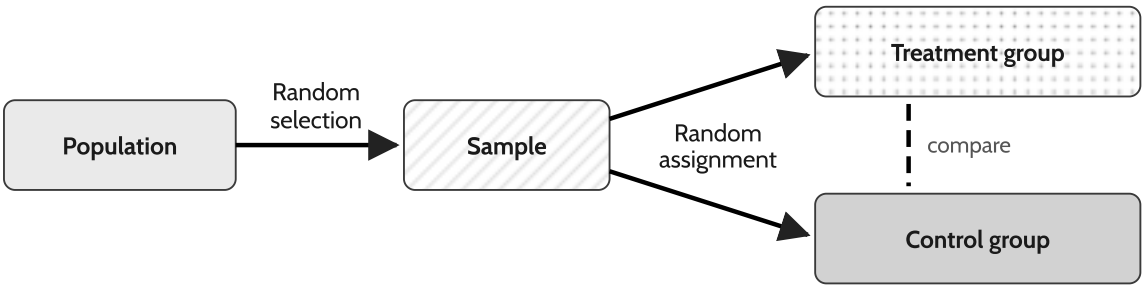


Figure 9.3 Random selection decides who enters the study; random assignment decides who gets which condition.

Finally, a within-subjects design uses **participants as their own control**: stable individual differences drop out, but order effects enter. **Statistical control** removes a measured extra-

neous variable in the analysis (ANCOVA, Section 9.10; partial correlation, Section 9.11). A **control group**, untreated or given a placebo, shows what would have happened anyway.

> **! EXAM TRAP**
>
> Equal probability of being assigned to a treatment group is randomisation. **Random sampling** decides who enters the study, **stratification** splits a population into subgroups before sampling, and **randomness** is mere unpredictability. In the psychophysics item, "pairing" is matching, and randomisation only scatters order effects by chance. "Double-bind technique" is most likely a misprint for double-blind; the double bind proper is a theory of contradictory family messages in schizophrenia (Chapter 8). None of these balances order, so counterbalancing is the answer.

Bias and artefacts ••

Bias is any influence that distorts a study's results away from the truth. (Control is its remedy; sample and population are the people studied, Section 9.3.) Distortions created by the research situation itself are called artefacts; the lesser ones are in Table 9.6. The best known is the **Hawthorne effect**: people change their behaviour because they know they are being observed, whatever the intervention.

> **KEY STUDY: THE HAWTHORNE STUDIES (1924 TO 1932)**
>
> The Western Electric Hawthorne Works near Chicago studied workers' output. In the illumination experiments output rose when lighting was increased, but also in the control group, and sometimes when the light was dimmed. The studies, associated with **Elton Mayo** (1880 to 1949), were reported by **Fritz J. Roethlisberger** (1898 to 1974) and **William J. Dickson** in *Management and the Worker* (1939); **Henry A. Landsberger** coined the term in 1958. Reanalyses find the data weaker than the legend.

Martin T. Orne (1927 to 2000) argued in 1962 that participants are active problem-solvers. **Demand characteristics** are the cues in a setting (rumours, instructions, procedure, the experimenter's manner) that tell participants what is expected and so shape their behaviour. **Stephen J. Weber** and **Thomas D. Cook** (1972) described good (confirming), faithful (literal), negativistic (opposing) and apprehensive subjects, the last showing **evaluation apprehension**, anxiety to be judged favourably (**Milton J. Rosenberg**, 1965). From the other side, **Robert Rosenthal** (born 1933), in *Experimenter Effects in Behavioral Research* (1966), showed that researchers' expectations change how participants are treated and responses scored: the **experimenter expectancy effect**, or experimenter bias.

> **KEY STUDY: PYGMALION IN THE CLASSROOM (1968)**
>
> Rosenthal and **Lenore Jacobson** gave the pupils of Oak School (a pseudonym), a Californian elementary school, an IQ test and told teachers that it had picked out certain pupils, in fact chosen at random (about 20 percent), as about to bloom intellectually. Retested on the same IQ test at the end of the year, these children, especially in grades one and two, had gained more IQ points than their classmates: the **Pygmalion effect**, a self-fulfilling prophecy. Replications find it real but smaller.

The **halo effect**, described by **Edward Lee Thorndike** (1874 to 1949) in 1920, is one salient attribute colouring ratings of others. When a patient's anxiety improves and the rater therefore scores depression as improved too, that is halo: not a ceiling effect (a scale limit), not Hawthorne (a change in the participant) and not a therapeutic effect (a genuine benefit of treatment).

Table 9.6 Further artefacts.

Effect	What happens	Main control
Placebo effect	Improvement from expecting treatment	Placebo control; blinding
Nocebo effect	Symptoms from negative expectations	Placebo control; blinding
John Henry effect	Control group competes harder (compensatory rivalry)	Keep groups apart
Novelty effect	Gain because a treatment is new, fading later	Longer follow-up
Social desirability	Answers chosen to look good (Chapter 2)	Anonymity; lie scales

> **! EXAM TRAP**
>
> Being observed: Hawthorne. Guessing the hypothesis from cues: demand characteristics. The researcher's expectation: experimenter bias. Others' expectations lifting performance: Pygmalion. Expecting treatment: placebo. Demand characteristics are cues, not properties of the hypothesis, statistical demands or the effort an item requires.

Blinding ●●

In a **single-blind** study participants do not know their condition, which controls their expectancy and demand, but the experimenter can still bias the results. In a **double-blind** study neither the participants nor the experimenters who treat and assess them know who is in which group, so it minimises both experimenter bias and expectancy effects. A **triple-blind** study also blinds those who analyse the data. (Allocation concealment is a different safeguard; Section 9.5.)

> **! EXAM TRAP**
>
> Double-blinding is what controls experimenter bias. Random assignment equalises groups, but the experimenter still knows who got what; a control group hides nothing; demand characteristics are a bias, not a control; single-blinding leaves the experimenter free; pretest instructions hide nothing. Double-blind does not mean two control groups or running the study twice.

The four kinds of validity of a study ●●

Here validity belongs to a study, not a test (Chapter 2). **Donald T. Campbell** (1916 to 1996) first set internal against external validity in 1957 and, with **Julian C. Stanley** (1918 to 2005), listed the threats to each in 1963; Cook and Campbell (1979) and **William R. Shadish**, Cook and Campbell (2002) gave four types:

- **Statistical conclusion validity**: is the IV-DV relationship real, or did low power, violated assumptions or many tests mislead?

- **Internal validity**: did the IV cause the change in the DV? Campbell and Stanley called it the basic minimum for interpreting an experiment.
- **Construct validity**: do the manipulation and measures represent the intended constructs? Giving plain versus barbecue-flavoured chips does not manipulate anxiety.
- **External validity**: the extent to which findings generalise to other people (population validity), settings (**ecological validity**) and times (temporal validity). **Glenn H. Bracht** and **Gene V Glass** (1968) split it into population and ecological validity.

The two main kinds trade off: tight laboratory control raises internal validity but makes the setting unlike life, while a busy clinic is realistic but loosely controlled. Researchers establish an effect under control, then replicate it in the field. **Alan Bryman**, in *Social Research Methods* (2001), judges studies by reliability (measures of concepts stable on different occasions), **replicability** (methods stated clearly enough to repeat) and validity (generalising to other social settings is external validity). The researcher's reputation is no criterion.

> **! EXAM TRAP**
>
> In the chips item, internal validity tempts, but no rival variable explains the result; the manipulation simply is not anxiety, so construct validity fails. "Statistical validity" loosely means statistical conclusion validity; "empirical validity" is an old name for a test's criterion-related validity (Chapter 2). Causation is internal validity; internal consistency and reliability belong to measures.

Threats to internal and external validity

A **threat to internal validity** is any rival explanation for a change in the DV. Campbell and Stanley (1963) listed eight, easiest to see in the counsellor's original study, a pretest and posttest with no control group (Table 9.7); Cook and Campbell (1979) added, among others, four that arise when groups influence each other. A control group measured at the same times cancels history, maturation, testing and instrumentation; only random assignment cancels selection. Campbell and Stanley also listed four threats to external validity, to which experimenter effects are usually added (Table 9.8).

Table 9.7 Threats to internal validity, in the yoga study.

Threat	What it is	Example
History	Outside events between pretest and posttest	Examinations end mid-course
Maturation	Change within people over time: growth, fatigue, recovery	Students settle in
Testing	The pretest changes posttest scores	The inventory feels familiar
Instrumentation	The measure or the raters change	Counsellor rates leniently later
Statistical regression	Extreme scorers drift towards the mean on retest	Only the most anxious enrolled
Selection	Groups differ before treatment	Volunteers against non-volunteers
Experimental mortality	Differential drop-out (attrition), not death	The most anxious stop coming
Selection-maturation interaction	Groups would change at different rates anyway	First-years against final-years

Threat	What it is	Example
Diffusion or imitation (1979)	Controls pick up the treatment	Controls copy the breathing
Compensatory equalisation (1979)	Staff give controls something extra	Warden counsels the controls
Compensatory rivalry (1979)	Controls try harder (John Henry effect)	Controls vow to stay calm
Resentful demoralisation (1979)	Controls who miss out give up	Wait-listed students lose heart

Table 9.8 Threats to external validity.

Threat	What it is
Reactive or interaction effect of testing	The pretest sensitises people to the treatment
Interaction of selection and treatment	The effect holds only for the kind of people sampled
Reactive arrangements	The study setting itself changes behaviour (Hawthorne)
Multiple-treatment interference	Earlier treatments carry over into later ones
Experimenter effects	The result depends on who runs the study

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. MAXMINCON says: minimise _____ variance.
2. A variable that varies with the IV and also affects the DV is a _____ variable.
3. The mechanism through which an IV acts is a _____.
4. Bridgman (1927) defined concepts by operations: the _____ definition.
5. Primacy and recency in psychophysics are controlled by _____.
6. Changing behaviour because one is observed is the _____ effect.
7. Blinding both participants and experimenters is a _____ design.
8. Generalisation to other people, settings and times is _____ validity.

Answers: (1) error, (2) confounding, (3) mediator, (4) operational, (5) counterbalancing, (6) Hawthorne, (7) double-blind, (8) external.

9.3 Sampling: Populations, Probability and Non-Probability Methods

MEDIUM YIELD · 24 QS

Suppose you want to know how anxious Indian school students feel before their board examinations. You will test a few hundred of the crores who exist, then speak about all of them. Whether that leap is justified depends almost entirely on how the few hundred were chosen.

Populations, samples and parameters ●●

A **population** (or universe) is the whole, well-defined group about which you want to draw conclusions. The **target population** is the group you would like to generalise to (all school students in India); the **accessible population** is the part you can actually reach (students in the Mysuru schools that admit you). Each member about whom data are collected is an **element**; a **sampling unit** is whatever is selected at a step of sampling, either an element or a

whole group such as a school. The **sampling frame** is the actual list used for selection (a school register, a voters' list), and the **sample** is the smaller set chosen from it by some rule or plan. Studying every element is a **census**; studying a part in order to describe the whole is a sample survey.

A number describing a population is a **parameter**, written with Greek letters (mean μ, standard deviation σ); the same number computed from a sample is a **statistic**, written with Roman letters (M or $\bar{X}$, s). A parameter is fixed but usually unknown; a statistic is known but varies from sample to sample. Sampling exists to estimate the first from the second.

> **! EXAM TRAP**
>
> "A statistical measure based upon a population" is a **parameter**; one based upon a sample is a **statistic**. Pair the initials: Population with Parameter, Sample with Statistic.

Sampling is needed because it saves time and money, often measures more accurately (a few trained investigators beat an army of hasty ones), is the only option for huge or infinite populations, and is unavoidable when testing destroys what is tested. Its principles: a statistic usually differs somewhat from its parameter; parameters are estimated through sampling distributions; and a random sample mirrors its population more closely as it grows. Its aims: a representative sample, unbiased estimates and a known margin of error. A good sample therefore shows **representativeness** (population characteristics in the same proportions), **adequacy** (enough cases), **independence** (choosing one element does not affect choosing another) and **homogeneity** (its units are of the same kind as the population's).

Probability versus non-probability sampling ●●

Following the sociologist **Hubert M. Blalock** (1926 to 1991), whose *Social Statistics* appeared in 1960, sampling methods fall into three families (Figure 9.4). **Probability sampling** methods clearly specify the likelihood of inclusion of each element: every element has a known, non-zero chance of selection, and a random mechanism does the choosing. In **non-probability sampling** that chance cannot be known, because a person (researcher, interviewer, volunteer or friend) does the choosing. **Mixed sampling** combines features of both: Indian texts following Blalock place systematic sampling here, since only its start is random, and the term also covers designs that join a purposive stage to a random one.

Why does the random mechanism matter? It removes the researcher's personal bias from selection, and because selection follows the laws of chance, the size of the chance error can be calculated (Section 9.9), which is what licenses inferential statistics. Probability sampling is therefore the method most likely to yield a representative sample: likely, not certain, since a random draw can still come out lopsided. Non-probability samples allow neither randomisation nor a calculated sampling error.

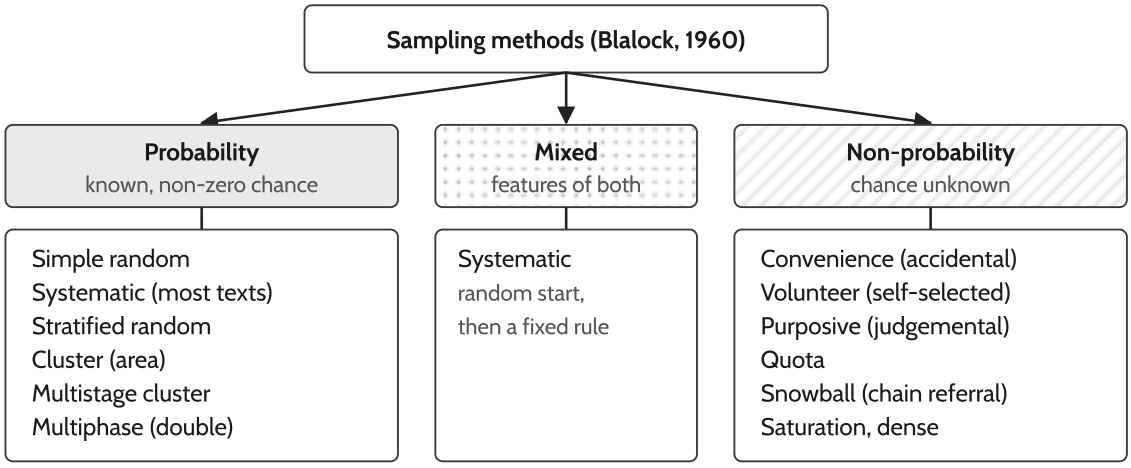


Figure 9.4 The three families of sampling methods. Systematic sampling appears twice: most texts call it probability sampling, Indian texts following Blalock call it mixed.

> **! EXAM TRAP**
>
> A past paper keys "random sampling guarantees a representative sample". Choose it, since the other options (it ensures personal bias, reduces the sample needed, makes data collection harder) are plainly wrong, but know the exact fact: it makes representativeness *likely* and error *calculable*, never guaranteed. "Statistical" and "empirical" sampling are not Blalock's families; methods that specify each element's chance of inclusion are **probability** methods.

Simple random and systematic sampling ○

Simple random sampling (unrestricted random sampling) is the foundation of the others: every element has an equal and independent chance of selection. Equal means none is favoured; independent means one draw does not alter another's chance. Randomness comes from the **lottery or fishbowl method** (names on identical slips, drawn blind), a **table of random numbers** (the most popular: elements are numbered 1 to N, and numbers above N and repeats are skipped) or computer generation for large populations. Returning each drawn element before the next draw is **sampling with replacement**; psychologists usually sample **without replacement**. The method needs no knowledge of the population's make-up and its error is easy to compute, but it requires a complete frame and may, by bad luck, miss small subgroups.

Systematic sampling takes every kth (or nth) element after a random start, with the interval k = N/n. For 100 students from a register of 2,000, k = 20, so a random start of 7 gives the 7th, 27th, 47th student and so on. It is quick and spreads the sample evenly, but beware **periodicity**: if a hostel list runs room by room, four to a room with the prefect always first, k = 4 yields every prefect or none. Chance decides only the start, hence the label mixed in Indian texts following Blalock; it is never a non-probability method.

Stratified random sampling ○

Suppose a university has 1,000 PhD, 2,000 MA and 7,000 BA students, and research anxiety may differ by level. **Stratified random sampling** first divides the population into **strata**,

mutually exclusive and exhaustive subgroups formed on a variable related to the study, then draws a random sample from every stratum. In **proportionate stratified sampling** each stratum gets the same sampling fraction, so a sample of 100 holds 10 PhD, 20 MA and 70 BA students and mirrors the population. In **disproportionate stratified sampling** small strata are deliberately over-sampled (say 40, 30 and 30) so that each can be analysed on its own, and results are weighted back for the whole. **Optimum allocation** takes more cases from strata that are larger, more variable and cheaper to sample, giving the most precise estimate for a fixed cost; **Jerzy Neyman** (1894 to 1981) gave its classic form, weighing stratum size and variability, in 1934.

Stratification guarantees that every important subgroup appears. When strata are internally homogeneous and differ from one another, it is also more precise than a simple random sample of the same size, because chance can no longer hand you too many BA students. Its costs: every element's stratum must be known in advance, and in a homogeneous population it gains nothing.

> **! EXAM TRAP**
>
> Stratified sampling is preferred over simple random sampling when **important subgroups must be represented**. It does not help a homogeneous population, is not chosen to minimise sample size, and needs more knowledge of the frame, not less (with no list of individuals, use cluster sampling). Every nth case is systematic; intact schools are cluster; volunteers are non-probability.

Cluster, multistage and multiphase sampling ●●

No list of every school student in India exists, and a simple random sample would scatter interviewers across thousands of villages. **Cluster sampling** (area sampling) solves both problems: the population is divided into naturally occurring groups (districts, villages, schools, wards), a random sample of these clusters is drawn, and units inside the chosen clusters are studied. Only a list of clusters is needed. It began in agricultural research and now drives large surveys.

In **one-stage** cluster sampling everyone in each chosen cluster is studied; in **two-stage** cluster sampling a random sample is taken within each chosen cluster; in **multistage sampling**, associated with the British statistician **Claus Moser** (1922 to 2015), selection descends through more levels. Randomly select 5 districts from each state, then 10 schools per district, then 30 students per school: districts and schools are intact groups picked at random, so this is multistage cluster sampling, the logic of India's National Family Health Surveys.

The price is precision. Students in one school are alike, so each extra student from a chosen school adds less new information than one drawn from the whole country. For the same n, a cluster sample usually has a larger sampling error than a simple random sample, and a stratified sample a smaller one; cluster sampling buys cheapness, not a larger sample. Ideal clusters are miniatures of the population, the reverse of ideal strata (Table 9.9, Figure 9.5).

Table 9.9 Stratified versus cluster sampling.

Feature	Stratified random	Cluster (area)
Groups are	Strata defined by a characteristic (sex, class)	Natural units (schools, districts)

Feature	Stratified random	Cluster (area)
Groups sampled	Every stratum	Some clusters, chosen at random
Within a group	A random sample	Everyone, or a random subset (two-stage)
Ideal groups	Alike inside, different from each other	Varied inside, alike to each other
List needed	Every element, with its stratum	Only the clusters
Precision versus simple random	Usually higher	Usually lower
Main reason to use	Represent subgroups	Cost; no list of individuals
Family	Probability	Probability

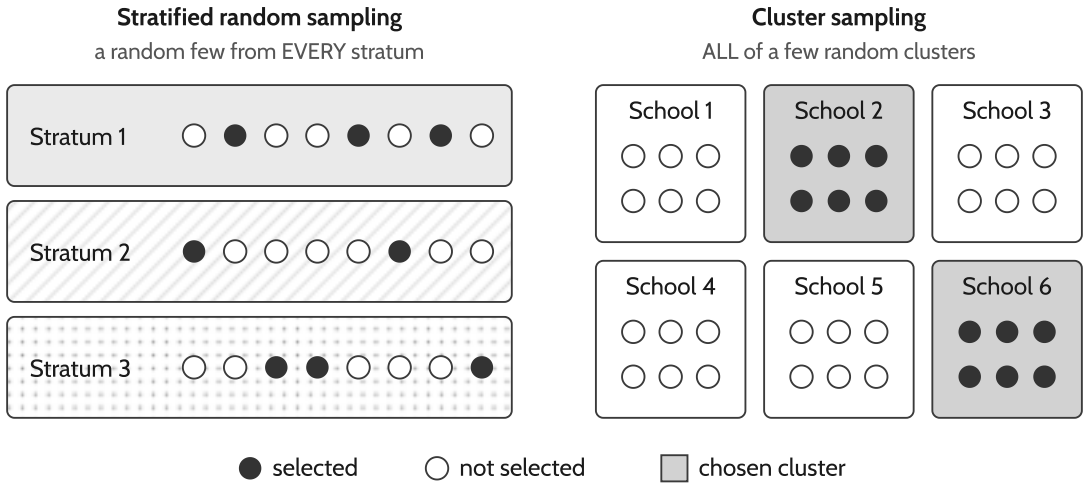


Figure 9.5 Stratified sampling draws a few people at random from every stratum; cluster sampling picks some whole clusters at random (shaded), studies everyone in them and skips the rest.

Two relatives remain. In **multiphase sampling** (with two phases, **double sampling**) the same kind of unit is sampled repeatedly: a large first-phase sample answers a cheap screening question, and a smaller subsample drawn from it gets the detailed study (screen 2,000 students with a brief anxiety scale, then interview 200). Units change across the stages of a multistage design but stay the same across phases. In **panel sampling** one sample is studied repeatedly over time (longitudinal designs are in Section 9.4).

> **! EXAM TRAP**
>
> A stem saying researchers "divide a population into smaller groups, and then randomly select from these groups" is a stock definition of **cluster** sampling and is keyed so: the groups themselves are picked at random. Some groups taken and the rest skipped means cluster; every group contributing a random sample means stratified. In the districts, schools, students stem every state contributes, so states act as strata (strictly, stratified multistage cluster sampling), but the keyed answer is **cluster**. "Two-stage" points to cluster sampling, not to proportionate or disproportionate stratified sampling, and two-stage is not two-phase (double) sampling.

Non-probability methods ●●●

Non-probability methods give up the random mechanism for speed, low cost or access. In all of them the chance of selection is unknown, randomisation is impossible, sampling error cannot be calculated, and generalisation rests on judgement (Table 9.10).

Convenience sampling (accidental or incidental sampling) takes people who are available, volunteer or can easily be recruited: your own classmates, shoppers passing a mall. It is the quickest, cheapest and least controlled method. In a **volunteer** or **self-selected sample** people answer an advertisement or online poll; volunteers are usually more interested and motivated than others. In **purposive sampling** (judgemental sampling) the researcher's knowledge and judgement hand-pick cases thought typical, expert or information-rich, such as senior nurses she considers typical of their wards.

Quota sampling, "a common and poor version of stratified sampling", fixes how many people of each kind are needed to mirror population proportions (60 men and 40 women, say), then lets interviewers fill each quota with whoever they find. The interviewer, not chance, picks people within quotas, so bias enters there. Cheap and needing no frame, it is widely used in market research and opinion polls.

In **snowball sampling** (chain-referral sampling) a few people with the characteristic of interest recruit others like them, so existing participants recruit future participants. It is the route into **hidden populations** that have no list, such as drug users or sex workers. In **saturation sampling** the researcher includes as many available members of a limited group as possible (900 of a school's 1,000 students); in **dense sampling** all or nearly all of the accessible population is taken. Both are rare. Qualitative purposive strategies (theoretical sampling, maximum variation, typical and extreme case, criterion sampling, sampling to saturation) belong to Section 9.6.

Table 9.10 The non-probability methods.

Method	Who is chosen	Main weakness
Convenience (accidental, incidental)	Whoever is available	Least representative
Volunteer (self-selected)	People who offer themselves	Volunteers differ from others
Purposive (judgemental)	Cases the researcher judges typical	Researcher bias
Quota	Proportional quotas filled non-randomly	Interviewer bias within quotas
Snowball (chain referral)	Participants recruit others	Over-represents the well connected
Saturation, dense	Nearly all of a small accessible group	Costly near-census

> **! EXAM TRAP**
>
> A 2024 paper (repeated in mocks) asks which method relies on "the researcher's judgment to select participants who are typical or representative" and allows no calculation of sampling error, and keys **quota**. That description is purposive sampling; quota wins only as the one non-probability option beside simple random, stratified and cluster. Rule: judgement of typical cases, purposive; fixed numbers per category, quota; whoever is available, convenience; participants recruit participants, snowball. Offered both purposive and quota for a judgement stem, choose purposive.

Sampling error, bias and the Literary Digest

Sampling error is the chance difference between a statistic and its parameter; **Everet Franklin Lindquist** (1901 to 1978) defined it in 1940 as "the difference between a parameter and an estimate of that parameter which is derived from a sample". It implies no mistake. It falls as n rises (the standard error of the mean is σ divided by the square root of n, so quadrupling n halves it) and as population variability falls, is zero in a census, and can be calculated only for probability samples.

Non-sampling error afflicts a census too: coverage error (the frame misses part of the population), non-response error (non-repliers differ from repliers), measurement error (poor questions, interviewer effects) and processing error (coding and entry mistakes). Error that pushes results consistently one way is **sampling bias**. Random error shrinks as n grows; bias does not, and a bigger biased sample is only a more confident wrong answer.

KEY STUDY: THE LITERARY DIGEST POLL (1936)

Question. Who would win the 1936 United States presidential election? **Method.** The magazine *Literary Digest* mailed about 10 million ballots to names from its subscriber lists, telephone directories and car registrations; over 2 million came back. **Result.** It predicted a comfortable win for the Republican challenger; the sitting president won by a landslide, carrying all but two states. A young pollster, **George Horace Gallup** (1901 to 1984), with a far smaller sample, called it correctly. **Why.** In the Depression, telephone and car owners were richer than average (coverage bias), and those who returned ballots differed from those who did not (non-response bias). **Lesson.** Size cannot cure bias.

! EXAM TRAP

Random sampling decides who enters the study and supports generalisation (external validity). **Random assignment** decides who gets which condition and supports causal conclusions (internal validity; Section 9.2). A true experiment needs random assignment and can survive a convenience sample.

How large should a sample be?

The size needed rises with population variability, with the precision and confidence level wanted, with smaller effects to detect, with higher power (conventionally .80), with the number of subgroups to analyse separately and with expected non-response; cost, time and the planned analysis set the ceiling. Rules of thumb call 30 or more a large sample and about 100 adequate for most educational research, but the modern answer is a **power analysis** before data collection (Section 9.9). A parameter can be estimated as a single value (**point estimation**) or as a range likely to contain it (**interval estimation**, the confidence interval); both rest on the sampling distribution, taught in Section 9.9.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A statistical measure based upon a population is a _____.
2. In probability sampling every element has a known, _____ chance of selection.
3. Every kth name after a random start is _____ sampling; its danger is _____.
4. Sampling randomly within every mutually exclusive subgroup is _____ sampling.
5. Random districts, then schools, then students is _____ sampling.
6. Interviewers filling proportional quotas non-randomly is _____ sampling.
7. Existing participants recruiting future participants is _____ sampling.
8. Sampling error shrinks as n rises; sampling _____ does not.

Answers: (1) parameter, (2) non-zero, (3) systematic; periodicity, (4) stratified random, (5) multistage cluster, (6) quota, (7) snowball, (8) bias.

9.4 Research Designs: Experimental, Single-Case and Non-Experimental

MEDIUM YIELD · 21 QS

A counsellor in Mysuru believes that slow breathing before class lowers exam anxiety. She could ask 500 students whether they practise it, follow one anxious student as the practice starts and stops, or let chance decide which half of a class practises. Each is a **research design**, a plan fixing who is studied, what is done to them and when they are measured, and each licenses different conclusions. We begin with the only design that can establish cause.

What makes an experiment ●●●

An **experiment** has three features: **manipulation** of the independent variable (IV) by the researcher (variables are defined in Section 9.2), **control** of everything else, and **random assignment**, in which chance (a lottery, random numbers), not the researcher or the participant, decides who enters which condition. The **experimental group** receives the treatment; the **control group** is treated identically but receives no treatment, and shows what would have happened anyway.

Random assignment makes the groups **equivalent** at the start, spreading every characteristic, even unmeasured ones, evenly by chance. If equivalent groups end unequal on the dependent variable (DV), only the IV can explain it. Creating equivalence among participants is therefore vital for experimental research; the issue does not arise in correlational or descriptive research, where nobody is assigned. **John Stuart Mill** (1806 to 1873) gave the logic in *A System of Logic* (1843). By his **method of agreement**, a circumstance shared by every case of an effect is probably its cause; by his **method of difference**, if cases with and without the effect differ in only one circumstance, that circumstance is the cause, which is exactly what an experiment arranges. Causal inference needs **covariation**, **temporal precedence** (cause before effect) and **elimination of alternative explanations**. Only the experiment secures all three, so only it licenses cause-and-effect conclusions.

Ronald Aylmer Fisher (1890 to 1962), in *The Design of Experiments* (1935), gave three principles of experimental design: **replication** (several units per treatment, so that chance variation can be estimated), **randomisation** (every participant has an equal chance of assignment to any group) and **local control** (blocking: grouping similar units so that their differences leave the error term). Homogeneity is not one of the three.

> ! EXAM TRAP
>
> **Random selection** (sampling, Section 9.3) decides who enters a study; **random assignment** decides who gets which condition. A survey of 10,000 randomly selected adults finding regular meditators less stressed has no assignment: people chose to meditate, and calmer people may be the ones who do. Large or random samples never turn an observational study into an experiment, so neither this study nor one linking exercise and depression in 30,000 adults supports a causal conclusion.

Campbell and Stanley's designs ○

Donald Thomas Campbell (1916 to 1996) and **Julian C. Stanley** (1918 to 2005) catalogued designs in 1963 in a notation still used: **R** is random assignment, **X** the treatment and **O** an observation; each row is a group and time runs left to right. They grouped designs by how well they rule out threats to internal validity such as history and maturation (Section 9.2), as Table 9.11 shows. **Pre-experimental designs** have no random assignment and little or no comparison; **true experimental designs** have R and a control group.

A pretest has a hidden cost, **pretest sensitisation**: a questionnaire on mental illness may make students more responsive to an anti-stigma film that follows. **Richard L. Solomon** (1949) answered with the **Solomon four-group design**, which runs the pretest-posttest and posttest-only designs together with four random groups, so the effect of pretesting and its interaction with treatment are measured and controlled. Maturation and selection are already handled by any randomised control group, and researcher bias by blinding (Section 9.2).

A **quasi-experiment** manipulates the IV but uses intact groups (classes, wards) without random assignment. Its main forms are the **nonequivalent control group design**, the **interrupted time-series design** (one group measured many times before and after the intervention) and the **regression discontinuity design** (Donald Thistlethwaite and Campbell, 1960), which assigns treatment by a cut-off score and looks for a jump in the regression line at the cut-off.

Table 9.11 Campbell and Stanley's designs (/ separates the groups).

Family	Design	Notation	Key point
Pre	One-shot case study	X O	Nothing to compare with
Pre	One-group pretest-posttest	O X O	No control group
Pre	Static-group comparison	X O / O	Groups may differ from the start
True	Pretest-posttest control group	R O X O / R O O	Classic trial; pretest may sensitise
True	Posttest-only control group	R X O / R O	Randomisation alone equates
True	Solomon four-group	Both above, four groups	Controls pretest sensitisation
Quasi	Nonequivalent control group	O X O / O O	Pretest shows how unequal the groups are

Family	Design	Notation	Key point
Quasi	Interrupted time series	O O O X O O O	Change judged against the prior trend
Quasi	Regression discontinuity	Assignment by cut-off	Jump in the line at the cut-off

> **! EXAM TRAP**
> Two fourth-grade classes take a test, one with a radio on. Noise was manipulated, so it is not a correlational study or naturalistic observation; no child was randomly assigned, so it is not a true experiment. It is a **quasi-experiment**.

Between, within and mixed designs ●●

In a **between-subjects design** each participant receives one condition: two randomised groups; a **multigroup design** (one IV at three or more levels, which shows the shape of its effect); or the **matched-groups** or **randomised block design**, in which participants are matched on a variable correlated with the DV and the members of each block are randomly assigned to conditions, Fisher's local control in practice.

In a **within-subjects** or **repeated-measures design** everyone receives every condition and serves as his or her own control, so fewer participants are needed and individual differences leave the error term. The cost is **order effects** (practice, fatigue) and **carry-over effects** (a drug still acting in the next condition), handled by **counterbalancing** the order (Section 9.2). A **Latin square** arranges orders so that each condition appears once in every position: four conditions need four orders instead of 24. Latin squares thus control order effects in repeated-measures designs; they are not correlational, and "equal group sizes" or "one IV with several levels" describe balanced and multigroup designs. A **mixed design** has a between factor and a within factor (group × time). In a **crossover design** each participant receives both treatments in sequence, half in each order.

> **! EXAM TRAP**
> Patients take a placebo for four weeks and a viral medication for four more, or the reverse. The researchers assigned the treatments, so this crossover study is an **experiment**, and the **explanatory variable** (the one that explains changes in the response) is the type of treatment; symptom improvement is the response.

Factorial designs ●●

A **factorial design** studies two or more IVs at once, every level of each combined systematically with every level of the others. Each IV is a **factor**, each of its values a **level**, and each combination a **treatment combination** or **cell**. Count the numerals for factors and multiply them for cells: 2 × 3 is two factors and six cells; 2 × 2 × 2 is three factors and eight cells.

A **main effect** is one factor's effect averaged over the levels of the others; a **simple effect** is its effect at one level of another; an **interaction effect** exists when one factor's effect depends on the level of another, shown by non-parallel lines (Figure 9.6). There noise adds 2 errors on an easy task and 8 on a hard one: a main effect of 5 errors that hides an interaction only a factorial design can reveal. Two factors give two main effects and one interaction; three give three main effects, three two-way and one three-way interaction (factorial ANOVA, Section 9.10).

In a **nested design** a factor's levels occur within only one level of another: therapists who each deliver only one therapy are nested within it.

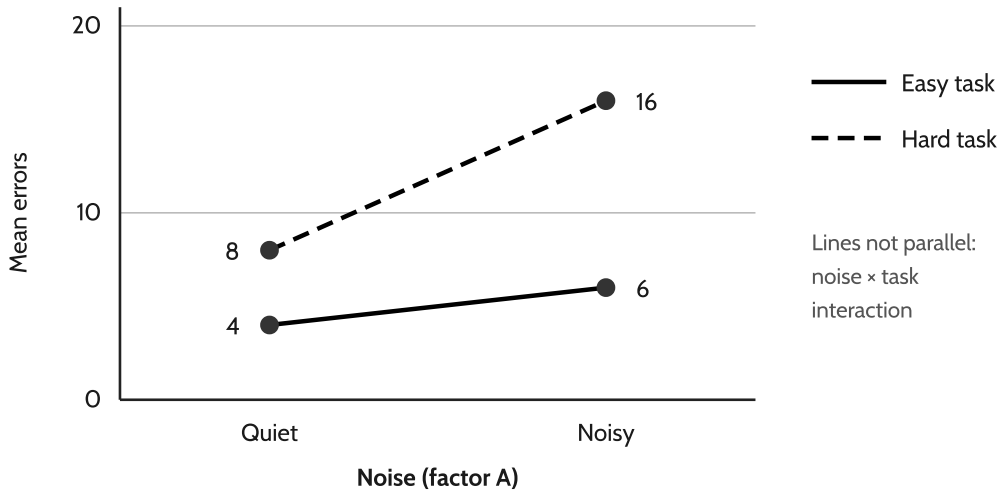


Figure 9.6 A 2 × 2 factorial design (illustrative means). Noise adds 2 errors on the easy task but 8 on the hard one; non-parallel lines signal an interaction.

> **! EXAM TRAP**
> IQ (100 or 120), age (30 or 60) and sex make a **three-factor** (2 × 2 × 2) design: count the IVs, not the levels. "Eight factor" counts cells and "six factor" adds levels.

Laboratory, field and natural experiments ○

A **laboratory experiment**, as defined by **Leon Festinger** and **Daniel Katz** (1953), is one in which the investigator creates exactly the conditions he wants, controlling some variables and manipulating others: high internal validity, but limited external validity, that is, generalisation to real life (Section 9.2). A **field experiment** manipulates an IV in a natural setting, as when Irving Piliavin and colleagues (1969) staged a collapse on New York subway trains to see who helped. Its advantage is higher external validity; the laboratory keeps higher internal validity and easier control, and both can use between-participants designs. A **field study** manipulates nothing but studies relations as they exist in a factory or school; Katz (1953) divided field studies into exploratory (finding the variables) and hypothesis-testing types. In a **natural experiment** nature or policy does the manipulating: people conceived at the height of the Dutch famine of 1944 to 1945 later showed about twice the usual risk of schizophrenia. With no random assignment, it remains quasi-experimental. Analogue studies are covered in Section 9.5.

Single-case and small-N designs

The **single-subject design** (single-case or N = 1 design) asks whether a treatment changes one person's behaviour, by measuring it repeatedly while treatment is switched in planned phases; with a few participants it becomes a **small-N design**. Such designs grew from the intensive study of a few animals by **Burrhus Frederic Skinner** (1904 to 1990). **Murray Sidman**'s *Tactics of Scientific Research* (1960) argued that a group average may fit no individual and that

generality comes from replication, and **Alan E. Kazdin** (1982) systematised the designs for clinical work.

Behaviour is recorded (by frequency, duration or time intervals) through a **baseline**, phase A, until stable, which predicts what would happen untreated; B is the treatment and C a second treatment. An A-B design shows one change and cannot rule out coincidence. The **withdrawal design** (A-B-A) removes the treatment to see behaviour return; the **reversal design** (A-B-A-B) reinstates it, the clearest demonstration (Figure 9.7). Withdrawal is unethical for self-injury and useless for skills, such as reading, that do not reverse; the **multiple-baseline design** avoids it by starting treatment at staggered times across behaviours, participants or settings (Table 9.12).

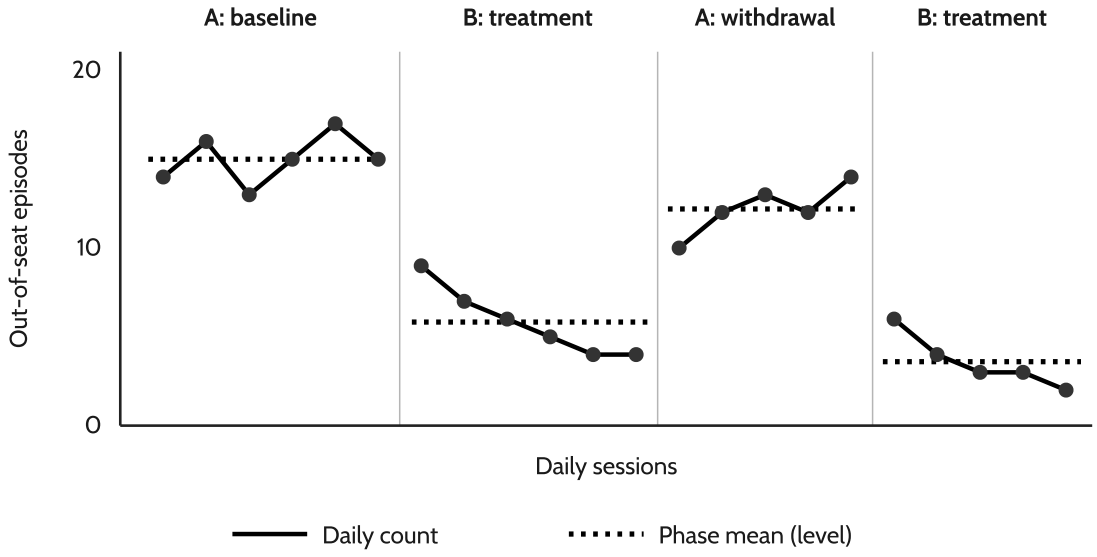


Figure 9.7 An A-B-A-B (reversal) design for one child (illustrative data). Behaviour drops each time treatment starts and climbs when it is withdrawn, so the treatment, not time or chance, controls it.

Table 9.12 The single-case designs (A = baseline; B, C = treatments).

Design	How it works	Strength or limit
A-B	Baseline, then treatment	Cannot rule out coincidence
A-B-A (withdrawal)	Treatment removed	Shows control; ends untreated
A-B-A-B (reversal)	Treatment reinstated	Strongest; ends on treatment
B-A-B	Starts with treatment	Dangerous behaviour; no initial baseline
A-B-A-C-A	Two treatments, each against baseline	Cannot rank B against C
Multiple baseline	Staggered starts across behaviours, participants or settings	No withdrawal; baselines must be independent
Changing criterion	Reward criterion tightened step by step (fewer cigarettes)	Behaviour should track each criterion
Alternating treatments	Two treatments rapidly alternated at random	Quick comparison; treatments may interfere

Data are judged by **visual analysis** of **level** (the phase mean), **trend** (the slope) and **variability**; Kazdin also weighs the **latency** of change after each phase shift. Compared with large-N research, single-case designs show causation within an individual and suit rare disorders and clinical practice; their weakness is external validity, built by replication rather than by sampling.

Non-experimental designs ●●

In **non-experimental** or **observational research** variables are measured as they exist and nothing is assigned. The commonest form, **correlational research**, measures two or more variables in the same people (Section 9.11) and allows prediction but not causation. The **directionality problem** (does poor sleep cause depression, or the reverse?) means it cannot determine the direction of influence; the **third-variable problem** means an unmeasured factor such as loneliness may cause both. It needs no random assignment and is among psychology's most used designs. No sample, however large or random, cures these two problems.

Fred N. Kerlinger (1910 to 1991) defined **survey research** as studying populations by selecting samples, to discover the relative incidence, distribution and interrelations of sociological and psychological variables, but not the causal factors of an event. A survey of an entire population is a **census** (the Census of India is the classic survey); of a part, a **sample survey**. Modes are interview, mail, telephone, online and the panel technique (the same respondents surveyed repeatedly). In Kerlinger's **ex post facto research** ("after the fact"), also called **causal-comparative research**, the IV has already occurred or cannot be manipulated, so pre-existing groups such as smokers and non-smokers are compared; self-selection and the post hoc fallacy are its risks.

Historical research uses **primary sources** (eyewitness records) and **secondary sources** (accounts by people not present). Its evidence faces **external criticism**, which tests authenticity (is the document genuine?), and **internal criticism**, which tests credibility (is it accurate, its author honest and informed?). Table 9.13 gives the other descriptive designs.

Table 9.13 Other descriptive designs.

Design	What it is	Limit
Naturalistic observation	Behaviour recorded in its natural setting without intervening	Reactivity, observer bias
Case study	Intensive study of one person or group (Freud's Little Hans, 1909)	No control; cannot generalise
Archival research	Records kept for other purposes (Durkheim's *Suicide*, 1897)	No control over what was recorded
Sociometry	Jacob Moreno's (1934) mapping of choices within a group (Section 9.6)	Describes, does not explain

> **! EXAM TRAP**
>
> A **census** is a survey of everyone, not ex post facto research, a field study or sociometry. What survey research does NOT do is study the causal factors of an event.

Developmental designs ●●●

A **cohort** is a group born in the same period and sharing its history. The **longitudinal design** studies the same people repeatedly over time (one group across ten years) and is the best design for change over time; its problems are **attrition** (selective drop-out), practice effects, cost and time. The **cross-sectional design** tests different age groups at one time: quick, cheap and covering several ages at once, but limited by the **cohort effect**, generational differences (less schooling in older groups) mistaken for development. The **time-lag design** tests one age in different years. **K. Warner Schaie** (1965) combined these in **sequential designs**, which test several cohorts on several occasions so that age changes can be told apart from cohort differences; in the commonest, the **cross-sequential design**, several age groups are tested and then retested over years (Figure 9.8). A case-control design instead compares people with and without an outcome on past exposures (Section 9.5).

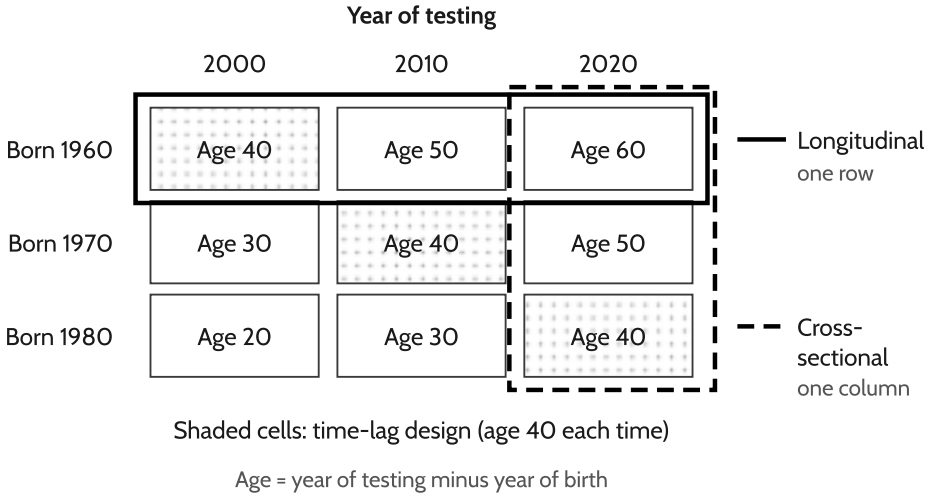


Figure 9.8 Age, cohort and year of testing. A longitudinal study reads one row, a cross-sectional study one column (so age is tangled with cohort) and a time-lag study the diagonal.

KEY STUDY: TERMAN'S GIFTED CHILDREN (FROM 1921)

Question: are gifted children frail and maladjusted? **Method: Lewis Madison Terman** (1877 to 1956) followed about 1,500 California schoolchildren with very high Stanford-Binet IQs from 1921 into old age, the classic longitudinal study. **Result:** they grew up healthier, better adjusted and more successful than average. **Criticism:** the sample was teacher-nominated and largely middle class, and Terman helped his participants' careers.

! EXAM TRAP

Cohort effects limit cross-sectional designs; **attrition** and **practice effects** limit longitudinal ones. Socio-economic factors can confound any comparison but are not the characteristic flaw, age cannot be randomly assigned in either design, and a cross-sectional design studies many ages at once.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Fisher's principle giving every participant an equal chance of assignment to any group is _____.
2. The design that controls pretest sensitisation is the _____ design.
3. Two intact classes, one tested with the radio on, form a _____.
4. Latin squares control _____ effects in repeated-measures designs.
5. IQ, age and sex, each at two levels, make a _____-factor design.
6. The single-case design that staggers treatment across behaviours, participants or settings, with no withdrawal, is the _____ design.
7. Survey research cannot identify the _____ of an event.
8. The main weakness of the cross-sectional design is the _____ effect.

Answers: (1) randomisation, (2) Solomon four-group, (3) quasi-experiment, (4) order, (5) three, (6) multiple-baseline, (7) causal factors, (8) cohort.

9.5 Clinical Epidemiology, Randomised Trials and Evidence Synthesis

MEDIUM YIELD · 27 QS

How common is a disorder, does a test detect it, and does a treatment work? The designs and numbers come from **epidemiology**, the study of how disorders are distributed in populations and what determines them; evidence synthesis pools the answers. In epidemiology the independent variable is called the **exposure** and the dependent variable the **outcome**.

Counting cases: prevalence and incidence ○

A disorder's **prevalence** counts existing cases, old and new together, as a proportion of a population. Its narrowest form, **point prevalence**, is the number of existing cases on a specific date divided by the total population on that date (40 cases among 2,000 Mysuru adults on 1 March: 2 percent). **Period prevalence** counts everyone who was a case at any time during a stated period, such as the past 12 months, and **lifetime prevalence** everyone who has ever met the criteria. By contrast, **incidence** counts only new cases arising during a period among people free of the disorder at its start, either as **cumulative incidence** (new cases divided by the population at risk) or as an **incidence rate** (new cases per person-time at risk, such as per 1,000 person-years).

For a stable condition, **prevalence ≈ incidence × average duration**. Schizophrenia has a low incidence but lasts for years, so its prevalence is relatively high; a brief adjustment disorder can have many new cases yet a low point prevalence. Incidence is used to find causes, prevalence to plan services. A **morbidity rate** expresses illness (its incidence or prevalence) per unit of population over a stated time, and a **mortality rate** gives deaths per unit of population, usually per 1,000 or 100,000 a year.

> **! EXAM TRAP**
>
> "Existing cases on a specific date out of the total population" is **point prevalence**. Plain "prevalence" is less specific, "periodic prevalence" covers a stretch of time, and "incidence" counts new cases only.

Epidemiological designs ●●

The designs differ in where the researcher starts and which way the inquiry runs in time (Table 9.14). An **ecological study** correlates group-level data, such as suicide and unemployment rates across Indian states; inferring individual behaviour from such correlations is the **ecological fallacy**, demonstrated by **William S. Robinson** in 1950. A **cross-sectional study** measures exposure and outcome at one moment, as prevalence surveys do, so it cannot show which came first.

A **case-control study** starts from the outcome: cases who have the disorder and matched controls who do not are compared on past exposure. For a rare dissociative disorder and childhood trauma, a cohort would need thousands of children followed for decades, whereas a case-control study starts from the cases a hospital already has: for rare outcomes it is the most feasible design. It cannot measure incidence, so it yields the odds ratio, and it is prone to **recall bias**, because ill people search their memories harder than healthy controls.

A **cohort study** classifies people who are free of the outcome as exposed or unexposed and follows them to see who develops it, which gives incidence and the relative risk. A **prospective cohort** runs forward from now. In the British Doctors Study, begun by **Richard Doll** (1912 to 2005) and **Austin Bradford Hill** (1897 to 1991), British doctors were questioned about smoking in 1951, again in 1957, 1966 and repeatedly up to 2001, and their deaths were traced: lung cancer mortality rose with the number of cigarettes smoked. Nobody was assigned to smoke, so the study is observational; it ran forward, so it is prospective. The **Framingham Heart Study** of a Massachusetts town (from 1948) is another. A **retrospective cohort** (historical cohort) uses existing records but still reasons from exposure to outcome. Cohorts are slow and costly, and they lose people to attrition.

Table 9.14 Epidemiological designs at a glance.

Design	Starts from	Direction	Yields
Cross-sectional	One sample, one moment	Snapshot	Prevalence
Case-control	Cases and controls	Outcome back to exposure	Odds ratio
Cohort	Exposed and unexposed	Exposure forward to outcome	Incidence, relative risk
Randomised trial	Randomly allocated groups	Forward, exposure assigned	Risk difference, NNT

> **! EXAM TRAP**
>
> **Retrospective is not case-control:** only case-control reasons from outcome back to exposure. The British Doctors Study is a **prospective observational** study, not retrospective, not a sample survey (9.3) and not an experiment. "Rare outcome plus past exposure" means **case-control**: trauma cannot be randomised, and a cross-sectional study cannot show which came first.

Measures of association and effect ●●

Any exposure and yes-or-no outcome fit a 2 × 2 table. In a trial, 200 people recovering from depression receive maintenance cognitive behaviour therapy (CBT) and 200 receive treatment as usual (TAU); 50 and 80 of them relapse. **Risk** is the proportion who develop the outcome: 0.25 with CBT, 0.40 with TAU (the control event rate). The **relative risk** or risk ratio (RR) divides the risks: 0.25/0.40 = 0.625. The **odds ratio** (OR) divides the odds, odds being those with the outcome over those without: (50/150)/(80/120) = 0.50. A ratio of **1.0 means no association**. Above 1 the exposure increases risk (an RR of 2.0 means double the incidence); below 1 it is protective, as CBT is here. A case-control study gives only the OR, there phrased as the odds of exposure among cases over that among controls; it approximates the RR when the outcome is rare.

Absolute measures show size. The **attributable risk** (risk difference) is the risk in the exposed minus the risk in the unexposed; in a trial it becomes the **absolute risk reduction** (ARR): 0.40 − 0.25 = 0.15. The **relative risk reduction** (RRR = ARR/control risk = 1 − RR) is 37.5 percent, a grander-sounding figure. The **number needed to treat** (**NNT = 1/ARR**) is how many patients must receive the treatment instead of the control for one additional patient to benefit: 1/0.15 = 6.7, rounded up to 7. An NNT of 8 for an antidepressant means treating 8 patients for one extra to benefit; the ideal is 1. The **number needed to harm** (NNH) is 1 divided by the absolute risk increase of an adverse effect: a side effect in 30 percent on the drug and 10 percent on placebo gives NNH = 5.

> **! EXAM TRAP**
>
> **NNT = 1/ARR.** The distractors 1/RRR, 1/RR, RR × control event rate (that gives the treated group's event rate), OR − 1, OR × 100, RRR × 100 and "enrolled ÷ events" are all wrong. An NNT of 8 does not mean the drug works in 8 percent of patients or that 8 must drop out, and there is no NNT threshold for drug approval. **RR or OR = 1.0 means no association**: not protective (below 1), not "doubles the risk" (2.0), not "causes the outcome" and not "perfect correlation" (r = 1, 9.11).

Screening and diagnostic accuracy ●●

Against a gold standard such as a clinical interview, a screening test sorts people into four cells (Table 9.15): true positives, **false positives** (healthy people flagged), **false negatives** (cases missed) and true negatives. A test's **sensitivity**, TP/(TP + FN), is the **true positive rate**: the proportion of real cases it catches. Its **specificity**, TN/(TN + FP), is the proportion of non-cases it correctly clears. A depression test with sensitivity 90 percent and specificity 70 percent therefore identifies 90 percent of depressed people as depressed and 70 percent of healthy people as healthy. Both are computed among people whose true status is known, so they do not change with prevalence.

Table 9.15 A screening test against the true diagnosis.

Test result	Disorder present	Disorder absent	Row index
Positive	True positive (TP)	False positive (FP)	PPV = TP/(TP + FP)
Negative	False negative (FN)	True negative (TN)	NPV = TN/(TN + FN)
Column index	Sensitivity = TP/(TP + FN)	Specificity = TN/(TN + FP)	Rows depend on prevalence

The patient asks the reverse: given a positive result, am I ill? The **positive predictive value** (PPV), the proportion of positive scorers who truly have the disorder, and the **negative predictive value** (NPV), the proportion of negative scorers who truly lack it, answer it, and they depend on the prevalence or **base rate**, the pre-test probability. Apply the 90/70 test to 1,000 clinic patients, 20 percent depressed: 180 true and 240 false positives give PPV = 180/420 = 43 percent. Where only 5 percent are depressed, 45 true and 285 false positives give PPV = 14 percent. The test is unchanged; the base rate fell. **Paul Everett Meehl** (1920 to 2003) and **Albert Rosen** warned in 1955 that with rare conditions a test's cutting score can make more errors than predicting from the base rate alone.

Lowering a cut-off score raises sensitivity and lowers specificity. The **receiver operating characteristic (ROC) curve**, from signal detection theory, plots sensitivity (y-axis) against 1 – specificity (x-axis) across all possible cut-offs (Figure 9.9). The **area under the curve** (AUC) runs from 0.5 (chance) to 1.0 (perfect discrimination). Finally, **likelihood ratios** combine both indices: sensitivity/(1 – specificity) shows how much a positive result raises the odds of the disorder (3 for the 90/70 test), and (1 – sensitivity)/specificity how much a negative result lowers them (0.14).

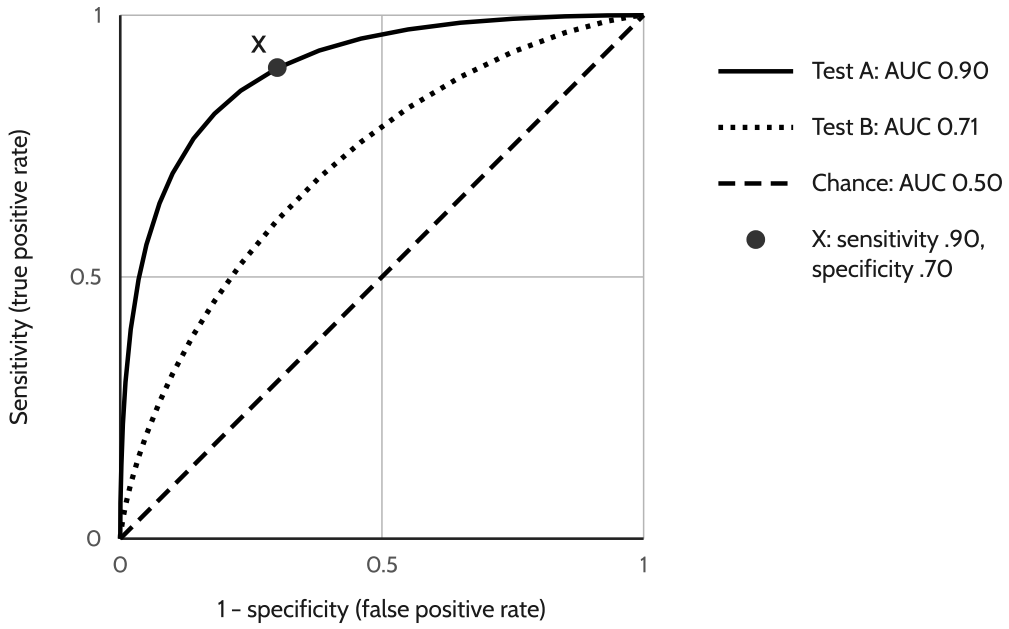


Figure 9.9 ROC curves of two tests (binormal model). Each point is one cut-off; the better test bows towards the top-left corner and has the larger area under the curve.

> **! EXAM TRAP**
>
> **Sensitivity is not PPV.** "A positive test reflects true disease" is PPV, "probability of disease before testing" is prevalence, "true negatives correctly identified" is specificity, and TN/(TN + FN) is NPV. PPV is most affected by **prevalence**, not by sensitivity or specificity alone or by the number of items (Chapter 2). ROC curves do not plot Type I against Type II error (9.9), predictor against outcome (9.11), item difficulty against discrimination, validity against reliability, score against age, or means against standard deviations.

The randomised controlled trial ○

A **randomised controlled trial** (RCT) is a true experiment on patients: participants are randomly allocated to a treatment or a comparison condition and compared on an outcome named in advance. **Randomisation** gives everyone the same chance of each arm, balancing measured and unmeasured characteristics. Next, **allocation concealment** hides the upcoming allocation sequence from those enrolling participants until each is irreversibly enrolled, using central randomisation or sealed opaque envelopes, so that a clinician cannot steer a promising patient into the treatment arm; it prevents selection bias. **Blinding** works after allocation, hiding the arm from the participant (single-blind), also from those treating and assessing (double-blind), and also from the analysts (triple-blind); see 9.2. Psychotherapy trials cannot blind patients or therapists, only assessors and analysts. Controls may receive a **placebo** (an inert, identical pill) or an **active** established treatment; psychotherapy trials also use waiting-list, attention-placebo and treatment-as-usual controls (Chapter 7).

> ! EXAM TRAP
>
> **Allocation concealment is not blinding.** Concealment hides the *upcoming* allocation from recruiters before enrolment; blinding hides the *actual* arm after it. "Blinding the patient to the assigned arm", "hiding outcome assessors' identity", "sealing the analysis dataset", "concealing the primary outcome from the statistician" and "keeping investigators unaware of the hypothesis" are all wrong.

Analysing, designing and registering trials ○

In **intention-to-treat analysis** (ITT) every participant is analysed in the group to which they were randomised, regardless of adherence, drop-out or crossover. A **per-protocol analysis** keeps only those who completed the treatment as planned (completers), and **as-treated analysis** groups people by the treatment they actually received. Drop-outs differ between arms, so discarding them reintroduces selection bias and usually flatters the treatment; ITT preserves randomisation and gives a conservative, real-world estimate. **Attrition**, the loss of participants, reduces power, and when it differs between arms it destroys the balance that randomisation created.

> ! EXAM TRAP
>
> ITT analyses **everyone in their randomised group**. "Only completers" and "excluding protocol deviators" are per-protocol; "by actual treatment received" is as-treated; "only responders" or "only the control group" is no analysis at all.

Most trials are **parallel-group**: each person receives one condition. In a **crossover trial** each person receives both conditions in random order, with a washout between, which fails when effects carry over (9.4). A **cluster randomised trial** randomises whole groups such as schools or villages. A **superiority trial** asks whether the new treatment is better; a **non-inferiority trial** whether it is no worse than the standard by more than a preset margin; an **equivalence trial** whether the difference lies within that margin in both directions.

Drug trials run in phases: **phase I** tests safety and dose in about 20 to 100 mostly healthy volunteers; **phase II** seeks first evidence of efficacy in up to several hundred patients; **phase III** runs confirmatory RCTs in roughly 300 to 3,000 patients, the basis for approval; **phase IV** is

post-marketing surveillance for rare and long-term harms. **Trial registration** records planned outcomes publicly before enrolment; the Clinical Trials Registry-India (CTRI), run by the Indian Council of Medical Research since 2007, has been mandatory since 2009. Finally, **efficacy** is whether a treatment works under ideal conditions (an explanatory trial) and **effectiveness** whether it works in routine care (a pragmatic trial; Chapter 7).

Analogue research ○

In **analogue research**, a clinical phenomenon is modelled under conditions that only resemble the clinical situation, usually with healthy, non-clinical participants (often students) or simulated conditions: snake-fearful undergraduates standing in for phobic patients, or sadness induced in the laboratory. It offers control, easy recruitment and fewer ethical obstacles, but its findings may not generalise to patients (external validity, 9.2). It is not research on clinical populations, nor an efficacy or comparative effectiveness trial.

Systematic reviews, meta-analysis and the evidence hierarchy ○

A **systematic review** uses explicit, pre-specified, reproducible methods to find, appraise and synthesise all the studies on one question, unlike a narrative review whose author picks the studies. A **meta-analysis** is its statistical step: it summarises the results of many studies by pooling their effect sizes (9.9) into one weighted average. **Gene V. Glass** (born 1940) coined the term in 1976.

KEY STUDY: SMITH AND GLASS (1977)

Hans Eysenck (1916 to 1997) had claimed in 1952 that neurotic patients improve about as often without psychotherapy as with it. **Mary Lee Smith** and Glass pooled 375 controlled studies of psychotherapy: the average effect size of 0.68 meant that the average treated client was better off than about 75 percent of untreated controls. Critics, Eysenck foremost, replied that pooling poor studies gives a precise but wrong answer ("garbage in, garbage out").

! EXAM TRAP

One real paper (IHBAS 2024) keys "to conduct a single large experiment" as the purpose of meta-analysis. That is wrong: a single experiment, however large, is primary research. Meta-analysis exists **to summarise findings from multiple studies**; it collects no primary data and does not analyse individual case studies.

A **forest plot** shows each study as a square (its effect, sized by its weight) on a line (its 95 percent confidence interval), with a vertical line of no effect and a diamond for the pooled estimate. **Heterogeneity**, variation between studies beyond chance, is tested with Cochran's Q and expressed as I^2. Pooled effects are inflated by **publication bias**, the preferential publication of significant results; **Robert Rosenthal** (born 1933) named its extreme form the **file-drawer problem** in 1979: null results stay in researchers' drawers. A **funnel plot** of effect size against precision is symmetrical without bias and lopsided when small null studies are missing.

The **hierarchy of evidence** for treatment questions runs from systematic reviews and meta-analyses of RCTs at the top, through single RCTs, cohort studies, case-control studies and case series or reports, down to expert opinion.

Reporting guidelines ●●

Reporting guidelines are design-specific checklists telling authors how to write a study up so that readers can judge it (Table 9.16). **CONSORT** (Consolidated Standards of Reporting Trials) adds a flow diagram accounting for every randomised participant; **PRISMA** (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) governs a systematic review, such as one on CBT for insomnia.

Table 9.16 Reporting guidelines matched to designs.

Guideline	Use it for
CONSORT	Randomised controlled trials
SPIRIT	Trial protocols, before the trial runs
PRISMA	Systematic reviews and meta-analyses
STROBE	Observational studies: cohort, case-control, cross-sectional
STARD	Diagnostic accuracy studies
CARE	Case reports
COREQ	Qualitative research (interviews, focus groups)

! EXAM TRAP

CONSORT is for **randomised controlled trials**, not systematic reviews (PRISMA), observational studies (STROBE), diagnostic accuracy (STARD), qualitative research (COREQ) or case reports (CARE).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Existing cases on one date over the total population: _____.
2. Rare outcome, past exposures: the _____ design.
3. NNT = 1 divided by the _____.
4. Sensitivity = TP/(TP + _____).
5. PPV depends heavily on the _____.
6. Hiding the upcoming allocation from recruiters: _____.
7. Systematic reviews are reported using _____.

Answers: (1) point prevalence, (2) case-control, (3) absolute risk reduction, (4) false negatives, (5) prevalence (base rate), (6) allocation concealment, (7) PRISMA.

9.6 Methods of Data Collection and Qualitative Research LOWER YIELD · 15 QS

A design (Section 9.4) settles when and on whom you measure; a method of data collection settles how the information reaches you: by watching people, questioning them, handing them a form, or studying what they have produced. This section takes those tools and their errors, then the paradigms beneath them, and ends with qualitative research.

Data sources and observation

Primary data are collected first-hand for the present study, by observation, interview, questionnaire or test. **Secondary data** were collected by others for another purpose and are reused: census records, hospital files, archives, published surveys. They are cheap, large and non-reactive, but the researcher controls neither what was recorded nor how accurately.

Observation is the systematic watching and recording of behaviour without manipulating it; it alone does not depend on what people say they do. In **participant observation** the observer joins the group's life; in non-participant observation the observer stays outside. Structured observation codes predefined categories; unstructured observation keeps open field notes. Naturalistic observation records behaviour in its usual setting (play in a school yard); controlled observation arranges standard conditions (mother and child behind a one-way mirror). Overt observers are known to the group; covert observers are not, which raises consent problems (Section 9.7). **James P. Spradley** (1933 to 1982), in *Participant Observation* (1980), described three narrowing phases: **descriptive** observation, broad, to grasp the field; **focused** observation, on the processes found relevant; and **selective** observation, near the end, seeking further evidence for specific practices.

Its two errors are **reactivity**, behaviour changing because people know they are watched (reduced by habituation and one-way mirrors), and **observer bias**, seeing what one expects (reduced by blind observers and inter-observer agreement). The participant observer may also "go native", losing critical distance. Time and event sampling of behaviour are covered in Section 9.4.

Interviews and ethnographic inquiry ●●

An **interview** is a purposeful conversation in which the interviewer asks questions and records the answers. A **structured interview** fixes the wording and order of every question: reliable and comparable, but rigid. A semi-structured interview follows a topic guide but lets the interviewer rephrase and probe; an unstructured interview fixes only the aim, trading reliability for depth. An in-depth interview is a long one-to-one exploration of experience; a **focus group** is a moderated discussion among about 6 to 10 similar people whose interaction is itself the data, though dominant members can sway it.

An **open-ended question** lets respondents answer in their own words ("Tell me about..."); a closed-ended question restricts the answer to yes or no or a fixed choice ("Did you sleep well?"). Probes are neutral follow-ups ("Could you say more?", an expectant pause) that draw out a fuller answer without suggesting one.

Spradley's *The Ethnographic Interview* (1979) treats the respondent as an informant, an expert on their own cultural world, asked to explain its meanings to an outsider through descriptive, structural and contrast questions. "Can you tell me why this band is so popular among your age group?" is an **ethnographic interview** question: the teenager explains the culture of the peer group. Other styles of inquiry differ: **psychodynamic inquiry** (feelings, early relationships, hidden motives: "What does this remind you of?"), **task-focused inquiry** (completing a defined task), **strength-based inquiry** (resources and past successes: "What has helped you cope?") and **theory-guided inquiry** (questions set by one theory, such as cognitive therapy).

The interviewer is part of the instrument, so errors arise from the interviewer's attitudes and expectations, from their sex, age, caste or status, from confusing or overlong questions, coldness and unmotivated respondents, and from interviewers recording similar answers differently. Training and standard wording reduce them.

> **! EXAM TRAP**
>
> "Tell me about..." is an **open-ended question**, not an ethnographic technique. The band question is **ethnographic inquiry** because it asks an insider to explain a culture.

Questionnaires and schedules ○

A **questionnaire** is answered by respondents themselves; a **schedule** is asked and filled in by the investigator or enumerator face to face, as in the census, so it suits respondents who cannot read. Items are open (own words; rich but slow to code) or closed: dichotomous (yes or no), multiple choice, ranking, rating, and the contingency question that routes respondents ("If no, go to question 12"). Attitude scales (Likert, Thurstone, Guttman) are taught in Chapter 2. Table 9.17 lists the wording faults; presuming and hypothetical questions are avoided too.

Table 9.17 Faults in question wording.

Fault	Example
Double-barrelled question: two questions, one answer	"How often and how much time do you exercise?"
Leading: suggests the answer	"Don't you agree therapy should be free?"
Loaded: emotive words	"Should lunatics roam the streets?"
Ambiguous or vague	"Do you drink regularly?"
Double negative	"Do you not agree patients should not be told?"

The **funnel sequence** moves from broad, easy questions to narrow ones, with sensitive items last. A good questionnaire is short and relevant, states its purpose, asks one idea per item in simple, neutral words, follows a logical order, looks neat and has been pilot-tested. Mailed questionnaires are cheap and anonymous but cannot clarify questions, and by one survey estimate about 70 percent are never returned; because non-responders differ from responders, a low response rate biases the sample.

Every tool thus has its own error: reactivity and observer bias, interviewer bias, response sets and non-response, records of unknown accuracy. Careful studies combine tools (triangulation).

Content analysis, sociometry and Verstehen ○

Content analysis studies communications (letters, diaries, case notes, films) rather than people. **Bernard Berelson** (1912 to 1979) defined it in 1952 as "the objective, systematic and quantitative description of the manifest content of communication". **Manifest content** is what is plainly there; latent content, the underlying meaning, is left to qualitative versions. His method, as set out in his 1954 handbook chapter (the date A.K. Singh gives): specify the universe (the content and its categories), choose the unit of analysis (word, theme, item, character, or space and time) and quantify by counting, ranking or rating.

Sociometry measures an individual's social behaviour and relations within a group. **Jacob Levy Moreno** (1889 to 1974), founder of psychodrama, introduced it in *Who Shall Survive?* (1934). Members name whom they would choose for an activity, and the **sociogram** draws the choices as arrows, showing stars (chosen by many), isolates (chosen by none) and mutual choices.

Verstehen ("understanding") is the interpretive grasp of the meaning an action has for the actor. **Wilhelm Dilthey** (1833 to 1911) argued in 1883 that the natural sciences explain from outside while the human sciences must understand from inside; **Max Weber** (1864 to 1920) made Verstehen the method of interpretive sociology. It is an interpretive stance, not a measure of social behaviour; nor is the "sample method", which is simply sampling (Section 9.3). Q-methodology (**William Stephenson**, 1935), which factor-analyses persons through Q-sorts and bridges the two traditions, is covered in Chapter 2.

Research paradigms ○

Every method rests on a paradigm, a set of basic beliefs. **Egon Guba** (1924 to 2008) and **Yvonna Lincoln** (1994) defined paradigms by three questions: **ontology** (what is the nature of reality?), **epistemology** (what is the relation between the knower and the known?) and methodology (how should the inquirer find out?). Table 9.18 gives their four paradigms.

Table 9.18 Guba and Lincoln's (1994) paradigms.

Paradigm	Ontology	Quality criteria
Positivism	Naive realism: a "real" reality that is apprehendable	Internal and external validity, reliability, objectivity
Postpositivism	Critical realism: real, but only imperfectly and probabilistically apprehendable	The same conventional benchmarks
Critical theory	Historical realism: a virtual reality shaped by social, political, cultural, economic, ethnic and gender values	Historical situatedness; erosion of ignorance; stimulus to action
Constructivism	Relativism: local and specific constructed realities	Trustworthiness and authenticity

Knowledge grows in positivism and postpositivism by accretion, building blocks added to an edifice of knowledge; in critical theory by historical revisionism and generalisation by similarity; and in constructivism through more informed reconstructions and vicarious experience. Positivists observe from a detached distance; for critical theorists and constructivists inquirer and participant are interactively linked, so findings are value-mediated (critical theory) or created (constructivism). Quantitative research rests on positivism and postpositivism, qualitative research on constructivism and critical theory, and mixed methods on pragmatism, which lets the question choose the methods.

> **! EXAM TRAP**
>
> Positivism's reality is real and **apprehendable**; "only imperfectly and probabilistically" is postpositivism. Historical situatedness and generalisation by similarity belong to critical theory, vicarious experience to constructivism; none is a positivist answer.

Qualitative versus quantitative research ●●

Qualitative research studies meaning, experience, process and context through words, images and observed action, analysed by interpretation; quantitative research measures variables and tests hypotheses statistically. Qualitative work seeks the insider's view and treats the researcher as the instrument, which calls for reflexivity, an open account of the researcher's influence. It values language and discourse, process and human individuality, and claims neither objective truth nor the researcher's authority over participants' meanings (Table 9.19).

Table 9.19 Quantitative and qualitative research compared.

Feature	Quantitative	Qualitative
Logic	Deductive, top-down	Inductive, bottom-up
Setting and sample	Controlled; probability samples, standardised tools	Natural, context-specific; purposive samples
Aim	Objective truth, prediction	Meaning, process, individuality
Quality	Validity, reliability, objectivity	Trustworthiness
Methods	Experiment, survey	Ethnography, case study, grounded theory

Its roots include Verstehen, phenomenology and **symbolic interactionism**, the name **Herbert Blumer** (1900 to 1987) gave in 1937 to the ideas of **George Herbert Mead** (1863 to 1931), and set out in his 1969 book: people act toward things according to meanings that arise in social interaction and are modified through interpretation. **Norman Denzin** and Lincoln trace its history through eight "moments" (2005), from Malinowski and the Chicago school to the present; **Michael Quinn Patton** (2002) grouped its twelve themes under design (emergent, purposeful, naturalistic), fieldwork and analysis (inductive, holistic, context-sensitive). In India, qualitative research, from ethnography to action research, has been practised since the late nineteenth century.

> **! EXAM TRAP**
>
> The **survey** is quantitative; case study, ethnography and grounded theory are qualitative. The "odd one out" is whatever implies objective truth, standardised tools, probability samples, top-down evidence or the researcher's authority.

The qualitative traditions ●●

Ethnography ("writing about a people") describes a culture or group from the inside through long immersion in its setting, mainly by participant observation and key informants; the researcher's role is to understand from the insider's perspective, not to manipulate variables, test large samples or reanalyse official statistics. The linguist **Kenneth Pike** (1912 to 2000) coined **emic** (insiders' own categories) and etic (outsiders' comparative categories) in 1954: "tension" as an idiom of distress is emic, "generalised anxiety disorder" etic. **Clifford Geertz** (1926 to 2006) made **thick description** the aim (1973): behaviour recorded with the context that gives it meaning, which alone tells a wink from a twitch of the eyelid.

KEY STUDY: MALINOWSKI IN THE TROBRIAND ISLANDS (1915 TO 1918)

Stranded in Australian territory by the First World War, **Bronislaw Malinowski** (1884 to 1942) pitched his tent in a Trobriand village off New Guinea, learned the language and joined daily life for about two years. *Argonauts of the Western Pacific* (1922) set ethnography's goal: "the native's point of view". Criticism: his diary, published in 1967, revealed contempt for the people he studied.

Grounded theory generates theory from data. **Barney Glaser** (1930 to 2022) and **Anselm Strauss** (1916 to 1996) launched it in *The Discovery of Grounded Theory* (1967). Constant comparison of each new incident with earlier ones drives it; theoretical sampling picks whom to study next from the emerging categories; collection stops at theoretical saturation, when new data add nothing new; and memos record the analyst's developing ideas. The coding stages are those of Strauss and **Juliet Corbin** (1990), shown in Figure 9.10: open coding breaks the data apart and labels concepts; **axial coding** weaves the data back together around categories; selective coding integrates everything around one core category.

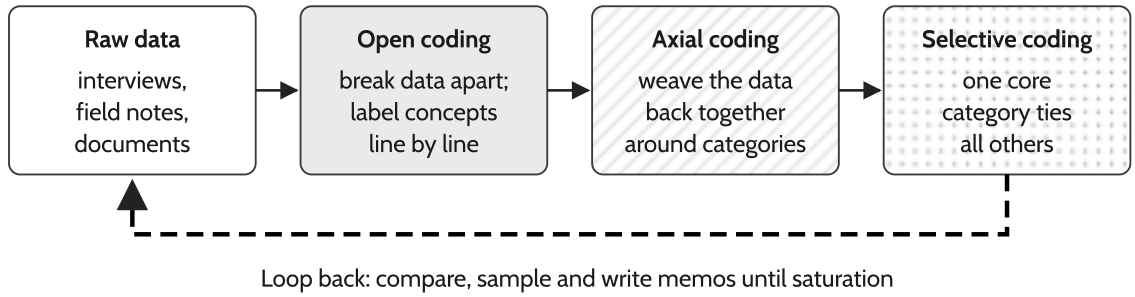


Figure 9.10 Strauss and Corbin's (1990) coding stages: open coding fractures the data, axial coding reassembles it, selective coding integrates it.

! EXAM TRAP

"Weaving the data back together around groups of concepts" is **axial coding**. "Early coding" and "core coding" are not stages; the core category is found in selective coding.

Phenomenology studies lived experience as it appears to the person. **Edmund Husserl** (1859 to 1938) asked researchers to describe it after **bracketing** (epoché), suspending their own assumptions; his former assistant **Martin Heidegger** (1889 to 1976) replied (1927) that we are always immersed in the world, so all understanding is interpretation (hermeneutic phenomenology). **Jonathan Smith**'s interpretative phenomenological analysis (IPA, 1996) follows the hermeneutic line: the researcher makes sense of participants making sense of their experience. Most methods texts call the two lines **descriptive** (Husserl) and **interpretive** or hermeneutic (Heidegger) phenomenology. One source used for this book renames them, contrasting a **direct approach** (Husserlian: a detached researcher lights up the foreground, participants' mental representations) with an **indirect approach** (Heideggerian: an involved observer, immersed in the everyday world, lights up its background).

> **! EXAM TRAP**
>
> "Direct" and "indirect" are not standard labels, and sources disagree on them. The safe fact is that the detached, bracketing approach is Husserl's (descriptive) and the involved one Heidegger's (interpretive). This book's source calls detached **direct** and involved **indirect** ("direct; indirect"); one practice key reverses them ("indirect; direct"). "Semi-direct" is not an approach.

Table 9.20 completes the family. Thematic analysis runs in six phases: familiarising with the data, generating initial codes, searching for themes, reviewing themes, defining and naming themes, and writing the report.

Table 9.20 Other qualitative approaches.

Approach	Key name	Focus
Case study	**Robert Yin** (1984)	One bounded case in depth, from several sources
Narrative and biographical	Several	Life stories kept whole; lives reconstructed retrospectively
Hermeneutics	Dilthey, Heidegger	Interpretation; the circle of parts and whole
Ethnomethodology	**Harold Garfinkel** (1967)	Everyday methods of making social order; breaching experiments
Conversation and discourse analysis	Several	Turn-taking in talk; what talk does and constructs
Thematic analysis	**Virginia Braun** and **Victoria Clarke** (2006)	Patterns of meaning (themes) across a data set
Framework analysis	**Jane Ritchie** and **Liz Spencer** (1994)	Policy research; a matrix of cases by themes
Co-operative inquiry	**John Heron**	Every participant is co-researcher and co-subject
Participatory action research	**Kurt Lewin**'s action research (1946)	Those affected share power; aims at change

Analysis, trustworthiness and triangulation ○

Matthew Miles and **A. Michael Huberman** (1994) saw analysis as three interwoven activities: data reduction (coding and summarising), data display (matrices and networks) and conclusion drawing and verification.

Lincoln and Guba's (1985) **trustworthiness** replaces the experimental criteria with four parallels: credibility (internal validity), transferability (external validity, judged by the reader), dependability (reliability) and confirmability (objectivity). Credibility comes from prolonged engagement, peer debriefing, triangulation and member checking (returning findings to participants); transferability from thick description; dependability and confirmability from an audit trail of decisions and from reflexivity.

Triangulation, a surveyor's term, studies one question from several positions so that each method's errors are offset by the others. Denzin (1970) named four types: data (several sources, times and places), investigator (several researchers), theory (several theoretical lenses) and methodological (several methods). Uniting qualitative with quantitative methods is therefore triangulation; "coalition" is no research term, a "multidimensional approach"

merely studies several aspects of a problem, and co-operative inquiry is Heron's participatory method.

Mixed-methods research integrates both kinds of data on a pragmatist footing. **John Creswell**'s three core designs are convergent (both strands at once, then merged), explanatory sequential (quantitative first, qualitative to explain it) and exploratory sequential (qualitative first, building a tool that a quantitative phase tests).

> **! EXAM TRAP**
>
> Uniting qualitative and quantitative methods is **triangulation**. Explanatory designs start quantitative; exploratory designs start qualitative.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A form asked and filled in by an enumerator is a _____.
2. "How often and how much time do you exercise?" is a _____ question.
3. Moreno's method of measuring social relations in a group is _____.
4. Positivism's ontology is naive realism; constructivism's is _____.
5. Weaving data back together around categories is _____ coding.
6. Uniting qualitative and quantitative methods is called _____.

Answers: (1) schedule, (2) double-barrelled, (3) sociometry, (4) relativism, (5) axial, (6) triangulation.

9.7 Research Ethics and Writing the Research Report LOWER YIELD · 10 QS

A study is finished twice: once when the last participant leaves unharmed, and again when its findings are written up for others to check. This section follows the scandals that produced ethical rules, the rules themselves, and honest reporting.

Why research needs ethics

Modern research ethics was written in response to abuse. After Nazi physicians' lethal experiments on concentration camp prisoners, the verdict of the Doctors' Trial at Nuremberg (1947) set out ten principles, the **Nuremberg Code**. Its first point reads: "The voluntary consent of the human subject is absolutely essential"; another lets the subject stop at any time. In 1964 the World Medical Association adopted the profession's own code, the **Declaration of Helsinki**, which puts the participant's welfare above the interests of science and later required independent committee review.

From 1932 to 1972 the United States Public Health Service ran the Tuskegee syphilis study: about 600 poor Black men, 399 with syphilis, were told they had "bad blood", and penicillin was withheld even after it became the standard cure. Its exposure in 1972 led to the **Belmont Report** (1979) and its three principles, each with one application. *Respect for persons* (treat people as autonomous and protect those with reduced autonomy) is applied through informed consent; *beneficence* (do no harm, maximise benefits and minimise harms) through

risk-benefit assessment; and *justice* (share research's burdens and benefits fairly) through fair selection of participants, the principle Tuskegee violated most plainly.

Psychology's own debated cases matter here for their ethics only (Section 1.6, Chapter 3). **John B. Watson** (1878 to 1958) and **Rosalie Rayner** (1898 to 1935) conditioned fear in an infant, Little Albert (1920), and never removed it. **Philip Zimbardo** (1933 to 2024) cast 24 male students as guards or prisoners in his Stanford prison study (1971); planned for two weeks, it was stopped after six days as guards turned abusive, with Zimbardo himself acting as superintendent.

KEY STUDY: MILGRAM'S OBEDIENCE STUDY (1963) AND BAUMRIND'S CRITIQUE

Stanley Milgram (1933 to 1984) ordered 40 men, told the study concerned punishment and learning, to give rising "shocks" up to 450 volts to a learner who was really an unharmed confederate; 26 (65 per cent) went to the maximum. **Diana Baumrind** (1927 to 2018) argued in 1964 that the deception and stress were unjustified, and critics add that prods such as "you have no other choice, you must go on" undercut the right to withdraw. Milgram replied that all were debriefed and 84 per cent were glad to have taken part.

Codes and ethics committees

Psychologists follow the **APA Ethics Code**, the American Psychological Association's *Ethical Principles of Psychologists and Code of Conduct* (2002, amended 2010 and 2016). Its five aspirational general principles are (A) beneficence and nonmaleficence, (B) fidelity and responsibility, (C) integrity, (D) justice, and (E) respect for people's rights and dignity; beneath them, enforceable standards set specific rules, Standard 8 covering research and publication. In India, the Indian Council of Medical Research issued the **ICMR National Ethical Guidelines** for Biomedical and Health Research Involving Human Participants (2017), which cover behavioural research in health. Under either system no study may begin until an independent **institutional ethics committee** (in the United States, an **institutional review board** or IRB) has approved its protocol. The Rehabilitation Council of India's code for clinical psychologists is in Section 7.1.

! EXAM TRAP

The APA has five general principles. Competence and confidentiality are enforceable standards, the usual intruders in "which is NOT a general principle".

Informed consent and voluntary participation ●●

Informed consent is the principle that participants are told enough about a study (its purpose, procedures, duration, risks, benefits, the protection of their data, and their rights) to decide voluntarily whether to take part, and agree before it begins. It needs understandable *information*, *voluntariness* and *capacity*, the ability to understand, weigh and communicate a choice. The **consent form** is only the signed record of that agreement; a signature extracted with jargon or pressure is not informed consent.

Suppose a Mysuru student reads that your exam-anxiety study takes 20 minutes, may stir up worry (a counsellor's number is given), codes answers by number and can be left at any point

without effect on marks. If she then agrees, she has given informed consent; her signed form merely documents it.

Children cannot give legal consent, so a parent or guardian consents and the child gives **assent**, an age-appropriate agreement of his or her own; a child's refusal is normally respected. An adult lacking capacity needs a legally authorised representative's consent, though a diagnosis alone does not remove capacity. **Voluntary participation** rules out coercion (a threat to grades, treatment or parole) and undue inducement (a payment large enough to distort judgement about risk). It continues as the **right to withdraw** at any time, without a reason or penalty, including withdrawal of one's data.

> ! EXAM TRAP
>
> "Told enough about the research to decide whether to take part" and "permission obtained after informing them of the purpose and procedures" both define **informed consent**, given before participation. The *consent form* is the record, not the principle. *Expectation management* (a business phrase) and *de-concealment* are not research-ethics terms; revealing a deception is dehoaxing, part of debriefing. Random assignment is a design procedure (Section 9.2), not an ethical principle.

Harm, risk and confidentiality ○

Participants should leave no worse off than they came. **Protection from harm** covers physical harm but in psychology chiefly psychological harm (anxiety, guilt, lowered self-esteem) and social harm such as stigma. Above minimal risk (no more than in daily life), the committee makes a **risk-benefit analysis**: are risks as low as possible, outweighed by benefits to participants or society, and covered by safeguards?

The duty most often used to sum up research ethics is ensuring the **confidentiality** of data: the researcher knows who gave the data but does not disclose it, coding names, storing data securely and reporting only group results. **Anonymity** is stricter: no one, not even the researcher, can link data to a person, because no identity is recorded. Confidentiality has limits, stated at consent, such as a serious risk of harm to self or others. Ethics asks more than following the law, and manipulating data to fit a hypothesis (falsification) or ignoring consent is its opposite.

Vulnerable populations, people less able to protect their own interests, need extra safeguards: children, prisoners, people with severe mental illness or intellectual disability, pregnant women, the very poor, and anyone dependent on the researcher, such as students or patients. Justice forbids loading risky research onto them, but also excluding them from research that could benefit them.

> ! EXAM TRAP
>
> Confidential means the researcher could identify you but will not; anonymous means nobody can. A face-to-face interview or a longitudinal study can be confidential but never anonymous.

Deception and debriefing ○

Obedience or conformity cannot be studied if participants know the hypothesis. **Deception** is misleading participants about a study's purpose or procedures, actively (a cover story, a confederate, false feedback) or passively (withholding the hypothesis). APA Standard 8.07 permits it only when the study has significant scientific, educational or applied value; no effective non-deceptive method is feasible; nobody is deceived about anything expected to cause physical pain or severe emotional distress; and the deception is explained as early as feasible, no later than the end of data collection, with the option to withdraw one's data.

That explanation is **debriefing**: after participation, participants receive full information about all aspects of the study, including its true purpose and any deception, and are checked for distress. **David S. Holmes** (1976) named its two parts. **Dehoaxing** convinces participants that the false information was false (the learner received no shocks); **desensitising** helps them deal with distress or unwelcome self-knowledge (that they obeyed). Debriefing is good practice even without deception, but cannot excuse an unjustified deception. It is not description or explanation (goals of science, Section 9.1), nor interpretation, which belongs to a report's Discussion. Figure 9.11 places each safeguard on the participant's path.

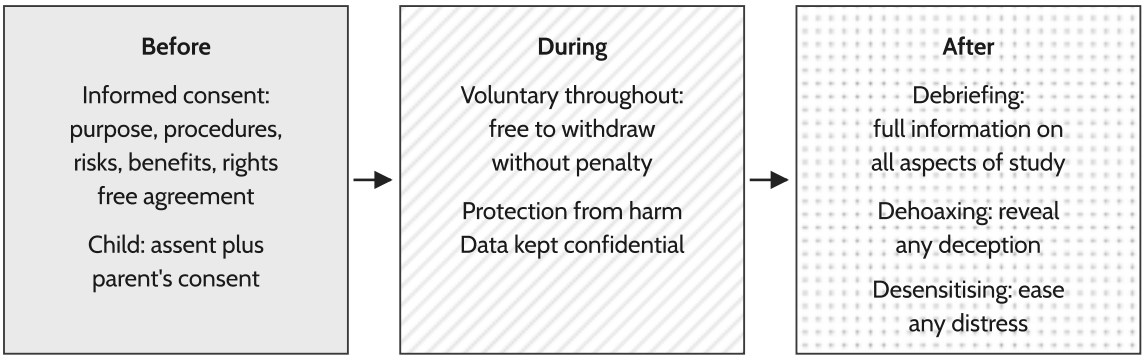


Figure 9.11 The participant's path. Informed consent comes before participation, the right to withdraw and protection last throughout, and debriefing comes after.

Animal research and the 3Rs

Animals cannot consent, so animal research ethics rests on justification and welfare. **William Russell** and **Rex Burch**, in *The Principles of Humane Experimental Technique* (1959), proposed the **3Rs**: *replacement* (non-animal methods, such as cell cultures or computer models, wherever possible), *reduction* (the fewest animals that give a valid answer) and *refinement* (less pain and distress, through anaesthesia and better housing).

Misconduct, plagiarism and giving credit ●●

Research misconduct means **fabrication** (inventing data or results), **falsification** (manipulating materials, procedures or data, or omitting results, so that the record misrepresents the research) and **plagiarism** (presenting another person's ideas, words or results as one's own). Honest error is not misconduct. Table 9.21 gives the forms of plagiarism.

Table 9.21 Forms of plagiarism, and the one practice that is not plagiarism.

Form	What it is

Form	What it is
Copying	Using someone's words verbatim without quotation marks or citation
Patchwriting	Reworking a source's sentences by swapping synonyms and word order
Mosaic plagiarism	Stitching phrases from several sources together without credit
Collusion	Submitting unauthorised joint work as individual work
Ghost-writing	Submitting work someone else wrote
Self-plagiarism	Reusing one's own published text or data as if new
Citation (not plagiarism)	Openly acknowledging the source: the cure for plagiarism

A **citation** is the brief acknowledgement in the text where an idea is used, such as (Milgram, 1963). Its primary purpose is to give credit to the original authors, and it also lets readers trace the idea; avoiding legal trouble is a side effect. **Referencing** is listing the full details of every source used at the end of the report, so that readers can find each one; it is not adding footnotes, presenting raw data or sorting data into categories (Section 9.8). Authorship credit (APA Standard 8.12) goes only to substantial contributors, in order of contribution; adding a head of department who did nothing is gift authorship. Under data sharing (Standard 8.14), published data must not be withheld from competent professionals who wish to verify the findings by reanalysis.

> **! EXAM TRAP**
>
> "Which is NOT a form of plagiarism?" Citation. Collusion, patchwriting and ghost-writing are all plagiarism. Citing gives credit to the original authors; referencing lists the sources used.

Writing the research report ●●

The purpose of a **research report** is to communicate the findings: what was done, why, what was found and what it means, so that others can evaluate and replicate the work. Collecting and analysing data come before it, and it summarises raw data rather than presenting them. In psychology the format is set by the APA Publication Manual, now in its 7th edition (released in 2019, dated 2020). Its sections form an hourglass (Figure 9.12); Table 9.22 lists their contents.

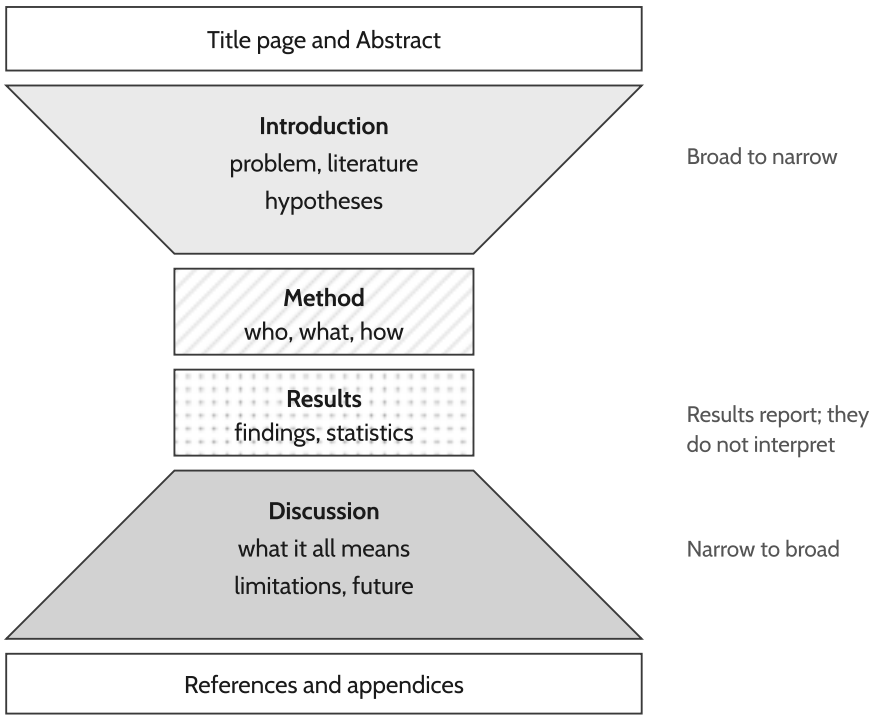


Figure 9.12 The hourglass shape of a research report: the Introduction narrows to the hypotheses, and the Discussion widens again to interpretation.

Table 9.22 The parts of an APA research report.

Part	Contents	Point to remember
Title page	Title, author(s), affiliation, author note; running head (short title) in professional papers	No "A study of"; running head at most 50 characters
Abstract	Problem, method, results, conclusions	About 150 to 250 words (A. K. Singh, following an older edition: 100 to 150); no citations
Introduction	Problem, review of literature, hypotheses	Not headed "Introduction"
Method	Participants (subjects), apparatus or materials, design, procedure	Enough detail to replicate
Results	Findings, statistics, tables and figures, including results against the hypotheses	Reported without interpretation
Discussion	Interpretation, implications, limitations, future research, conclusions	Where findings are interpreted
References	Every work cited in the text, alphabetical	Not a bibliography
Appendices	Tests, stimuli, instructions, long tables	Labelled A, B and so on

A. K. Singh lists the author note (affiliations, funding, acknowledgements) as a ninth part. **References** include only the works cited in the text; a **bibliography** lists everything consulted, cited or not, and is not normally part of an APA report. An APA journal reference runs: Milgram, S. (1963). Behavioral study of obedience. *Journal of Abnormal and Social Psychology*, 67(4), 371-378. https://doi.org/10.1037/h0040525. For three or more authors, the text citation uses the first author and "et al." from the start.

In APA style, numbers below 10 are words and 10 and above numerals, but statistics and measurements always take numerals and a number opening a sentence is spelled out; an abbreviation is spelled out at first use; headings have five levels (Level 1, used for Method, Results and Discussion, is centred and bold; Levels 2 and 3 are flush left, bold and then bold italic; Levels 4 and 5 are indented and run into the paragraph); bias-free language avoids reducing people to labels ("people with schizophrenia", not "schizophrenics"); and procedures take the past tense. Evaluating a report asks whether the hypothesis is testable, the methods sound and the conclusions supported by the data.

> **! EXAM TRAP**
>
> "In which step of a research report are findings interpreted?" One real paper (RML 2023) keys *Results*. The correct answer is **Conclusion**, that is the Discussion: Results report what was found, statistics included, without interpretation, and the Discussion interprets.

The research proposal

A **research proposal** is the detailed plan of a study, an architect's blueprint, written to win approval and often funding. **A. K. Singh** gives nine steps: (1) the problem, in testable terms; (2) definitions (operational definitions, Section 9.2), assumptions, limitations and delimitations; (3) a selective review of related literature; (4) hypotheses; (5) methods: participants, procedure and data analysis; (6) a time schedule; (7) expected results, with alternative outcomes; (8) references; (9) an appendix of instruments and instructions.

Assumptions are statements the researcher believes but cannot prove. **Limitations** are conditions beyond the researcher's control that may weaken the conclusions, such as a low response rate. **Delimitations** are boundaries the researcher deliberately sets, such as studying only first-year students in Mysuru colleges, and they decide to whom the findings apply.

> **SCAN YOUR BRAIN**
>
> Fill in the blanks. The answers are below; cover them first.
>
> 1. The Nuremberg Code of _____ declared voluntary consent "absolutely essential".
> 2. The Belmont Report's principles are respect for persons, beneficence and _____.
> 3. Telling participants enough to decide whether to take part is _____.
> 4. Full information about all aspects of the study, given after it, is _____.
> 5. Holmes's two parts of debriefing are dehoaxing and _____.
> 6. Data the researcher can link to a person but will not disclose are _____.
> 7. The 3Rs are replacement, reduction and _____.
> 8. Findings are interpreted in the _____ section, not the Results.
>
> **Answers:** (1) 1947, (2) justice, (3) informed consent, (4) debriefing, (5) desensitising, (6) confidential, (7) refinement, (8) Discussion.

9.8 Measurement Scales, Organising Data and Descriptive Statistics

HIGH YIELD · 58 QS

Two hundred students in Mysuru have filled in an anxiety questionnaire, and you now own a notebook of numbers that, on their own, say nothing. Before any hypothesis can be tested (Sections 9.9 to 9.11), you must decide what kind of numbers you have, put them in order and describe them. This section does those jobs, ending with the normal curve and the z-score.

Levels of measurement ●●●

Measurement is the assignment of numbers to objects or events according to rules. But numbers mean different things: the 2 on a cricketer's shirt, 2nd rank in class, 2°C and 2 kg are all "2", yet only the last can be doubled meaningfully. In 1946 the Harvard psychophysicist **Stanley Smith Stevens** (1906 to 1973) sorted these meanings into four **levels of measurement** ("On the theory of scales of measurement", *Science*). Each level adds one property: **identity** (a number names a category), **magnitude** (order), **equal intervals** (a unit means the same amount anywhere on the scale) and an **absolute zero** (zero means none of the attribute).

A **nominal scale** has identity only: labels name unordered categories such as marital status, eye or car colour and gender, and coding "married" as 2 does not make it more than "single" at 1. An **ordinal scale** adds magnitude: cases are ranked, but the distances between ranks are unknown. Class ranks, socioeconomic status categories and any task in which respondents place items along a continuum by preference or degree of agreement give ordinal data. A single Likert item (strongly disagree 1 to strongly agree 5) is strictly ordinal, though a total over many items is conventionally treated as interval.

An **interval scale** adds equal intervals (20°C to 30°C is the same rise as 30°C to 40°C), but its zero is *arbitrary*, a reference point rather than the absence of the quantity: 0°C is just the freezing point of water. Differences are meaningful, ratios are not; 40°C is not twice as hot as 20°C. Celsius temperature, calendar years and most test scores, IQ included, are treated as interval. A **ratio scale** adds a *true* zero: height, weight, reaction time and the number of books a person owns (zero books is no books, and 6 is twice 3). The level decides which statistics are permissible (Table 9.23), and each level keeps every statistic of the levels below it (memory hook: NOIR).

Table 9.23 Stevens's four levels of measurement, from least to most informative.

Scale	Properties	Examples	Permissible statistics
Nominal	Identity only	Marital status, eye or car colour, gender	Frequency, percentage, mode, chi-square
Ordinal	Adds magnitude	Class rank, SES category, one Likert item	Adds median, percentiles, rank correlation
Interval	Adds equal intervals; arbitrary zero	Celsius temperature, IQ, most test scores	Adds mean, SD, Pearson r, t and F tests
Ratio	Adds a true zero	Height, weight, reaction time, books owned	All, including coefficient of variation

> **! EXAM TRAP**
>
> One real paper keys "ratio scale" for "Arbitrary zero is the characteristic of which scale?". The key is wrong: an arbitrary zero defines the **interval** scale, and the ratio scale has a **true** zero. Some Indian texts cite "Stevens (1951)", his handbook chapter; the scales date from 1946.

KEY STUDY: STEVENS AND THE SCALES OF MEASUREMENT (1946)

Can sensations be measured at all? A committee of the British Association for the Advancement of Science had debated it for years without agreeing, its physicists doubting it (final report, 1940). Stevens replied by defining measurement broadly, as assigning numbers by rule, and by classing scales according to the transformations that leave their information intact: any one-to-one relabelling (nominal), any order-preserving change (ordinal), a linear change, aX + b (interval), and multiplication by a constant (ratio). The result was the four levels and their permissible statistics. Critics such as **Frederic Lord** (1953, "On the statistical treatment of football numbers") replied that the numbers do not remember where they came from, so scale type should not ban a statistic outright; the debate continues.

Two further classifications cut across the levels. A **categorical variable** (qualitative) sorts cases into classes; a quantitative one records an amount. A **discrete variable** takes separate values, usually counts (books owned, so a count can be discrete yet ratio); a **continuous variable** can take any value in a range (height, reaction time). Nominal variables are always categorical. Ordinal variables are strictly ordered categories, but because ranks are taken to lie on an underlying continuum (anxiety, agreement) they are usually analysed as continuous.

> **! EXAM TRAP**
>
> "Ordered variables are considered ____, nominal variables ____" is keyed **continuous, categorical**. Take the key, knowing that some books call nominal and ordinal variables alike qualitative. "Discontinuous" simply means discrete.

A good measure is **reliable**, that is, consistent (it gives the same result when the same thing is measured again), **valid** (it measures what it claims to), **practical** (cheap, quick, easy to interpret) and **sensitive** to small differences. Objectivity belongs here too; high cost, low reliability and subjectivity mark a poor measure. Test reliability and validity are treated in Chapter 2.

Processing data and statistical software ●●●

Filled-in questionnaires become analysable data through four processing operations, in this order, as set out in **C. R. Kothari**'s *Research Methodology*. **Editing** examines raw data to detect and correct errors, omissions and inconsistencies (an age of 210, a blank item). Field editing is done by the investigator soon after the interview; central editing is done at the office, by one editor or team, once all forms are in. **Coding** assigns numerals or other symbols to answers so that responses fall into a limited number of exhaustive, mutually exclusive categories (male 1, female 2); the key to the codes is the codebook. **Classification** arranges the data into groups or classes sharing a characteristic, by attribute (urban or rural) or by class interval (ages 20 to 29). **Tabulation** arranges the classified data in rows and columns: a simple (one-way) table shows one characteristic, a complex table two or more. Graphs, charts and dia-

grams present the same data visually instead. On computer, data entry is followed by data cleaning: hunting for out-of-range values (7 on a 1-to-5 scale), duplicates and missing-value codes.

> **! EXAM TRAP**
>
> Coding *assigns symbols* (the "assigning numerical values" option); classification *groups* data into classes. One real paper keys "organising data into categories or groups" for both: it is exact for classification and the nearest option for coding. Editing corrects errors; tabulation uses rows and columns.

IBM SPSS (Statistical Package for the Social Sciences), first released in 1968 and owned by IBM since 2009, is psychology's standard package; its primary purpose is data analysis, from descriptive statistics to factor analysis. R, SAS, Stata and the free JASP are alternatives. **Microsoft Excel** is a spreadsheet for data entry, tabulation, charts and calculation: =AVERAGE(A1:A10) gives the mean; SUM, MAX, MIN, MEDIAN and MODE do what they say; STDEV gives the sample SD, CORREL Pearson's r and COUNT the number of numeric cells. Word (report writing), Photoshop (photo editing), Illustrator (drawing) and InDesign (page layout) are not analysis tools.

Frequency distributions and graphs ●●

A list of raw scores, the scores as first recorded, says little. A **frequency distribution** shows how many cases (f) obtained each score (ungrouped) or fell in each class interval (grouped), the intervals being of equal width, usually 10 to 20 of them. An interval's apparent limits (25 to 29) are those written down; its **exact limits** (24.5 to 29.5) extend half a unit beyond, because a score of 29.4 is recorded as 29. The midpoint (27) stands for every score in it. **Cumulative frequency** (cf) is the running total up to an interval's top, and percentages restate f and cf out of 100 (Table 9.24).

Table 9.24 A grouped frequency distribution of 40 test scores.

Class interval	Exact limits	Midpoint	f (%)	cf (cumulative %)
40 to 44	39.5 to 44.5	42	2 (5)	40 (100)
35 to 39	34.5 to 39.5	37	4 (10)	38 (95)
30 to 34	29.5 to 34.5	32	8 (20)	34 (85)
25 to 29	24.5 to 29.5	27	12 (30)	26 (65)
20 to 24	19.5 to 24.5	22	8 (20)	14 (35)
15 to 19	14.5 to 19.5	17	4 (10)	6 (15)
10 to 14	9.5 to 14.5	12	2 (5)	2 (5)

A set of raw scores does not give a unique set of grouped scores: the interval width and starting point are the researcher's choice, and grouping treats every score as lying at its midpoint (grouping error). Pooled scores, by contrast, are simply scores from several groups combined. Raw counts also mislead when groups differ in size: 30 of 200 urban students against 15 of 50 rural ones. **Relative frequencies**, each f divided by N as a proportion (.15 against .30) or a percentage, make such distributions comparable.

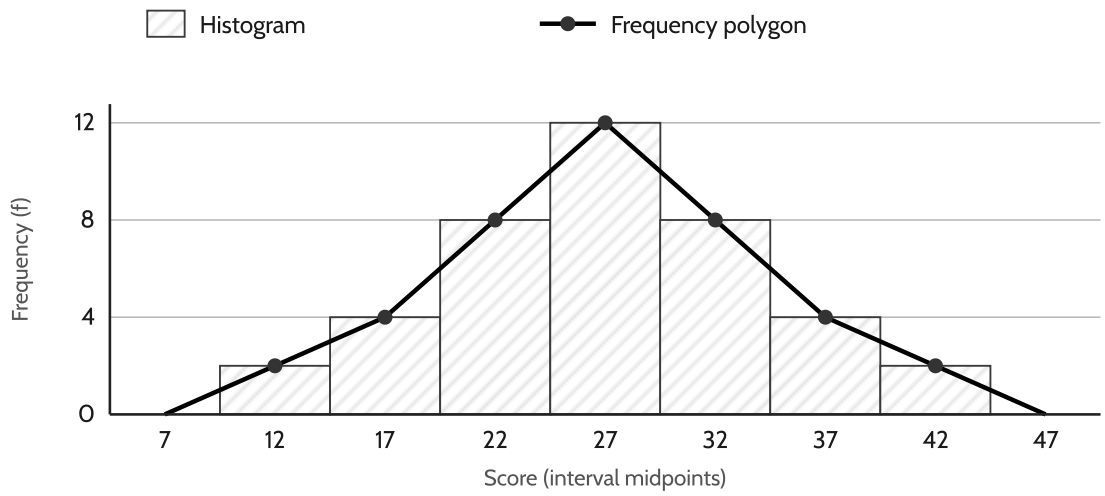


Figure 9.13 The 40 scores of Table 9.24 as a histogram and frequency polygon. The bars touch because the scale is continuous; the polygon joins the interval midpoints and closes on the baseline.

Categorical data take a **bar graph** (separate bars with gaps) or a pie chart (parts of one whole). Grouped continuous data take a **histogram**, adjoining rectangles each standing on a class interval with height equal to its frequency; joining the midpoints of the bar tops gives the **frequency polygon** (Figure 9.13), best for overlaying two distributions. Cumulative frequencies plotted against upper exact limits give the **ogive**, an S-shaped curve from which percentiles are read. **John Wilder Tukey** (1915 to 2000), in *Exploratory Data Analysis* (1977), added the stem-and-leaf display, which keeps every raw score, and the box plot (a box from Q_1 to Q_3, a median line and whiskers), good for comparing groups and spotting outliers. A line graph shows change over time; a **scatter plot** shows the relationship and spread of two continuous variables (Section 9.11).

Measures of central tendency ●●●

A **measure of central tendency** is one score that represents the whole distribution. The arithmetic **mean** ($M = \Sigma X / N$) is the balance point: deviations from it sum to zero, the sum of squared deviations about it is a minimum (least squares), and adding a constant to every score adds it to the mean, while multiplying every score by a constant multiplies the mean by it. Using every score makes it the most stable average across samples, but also lets one extreme score pull it: 4, 7, 7, 9 and 13 have a mean of 8, and adding a 40 lifts it to 13.3. It cannot be computed for an open-ended top class. A combined mean weights each group by its size: 30 boys averaging 60 and 20 girls averaging 70 give (1800 + 1400) / 50 = 64.

The **median** is the middle score of the ordered set, the 50th percentile (for 4, 7, 7, 9, 13, 40 it is 8). Depending only on position, it resists extreme scores, so it is the average for skewed data: community income, where a few rich households inflate the mean, and anxiety or symptom scores, where most people score low and a few very high. It also suits ordinal data.

The **mode** is the most frequent score and the only average for nominal data. One peak makes a distribution unimodal and two bimodal; when two or more scores share the highest frequency it is multimodal. The geometric mean (the Nth root of the product of N scores) suits growth rates; the harmonic mean (N divided by the sum of the reciprocals) suits rates such as speeds.

REMEMBER

Symmetrical interval or ratio data: **mean**. Skewed data, outliers or ordinal data: **median**. Nominal data: **mode**.

! EXAM TRAP

Range, standard deviation, variance and **mean deviation** measure variability, not central tendency. Mean deviation is the favourite decoy because it contains the word "mean".

Measures of variability ●●●

Two classes may share a mean of 50 while one spans 45 to 55 and the other 10 to 90; variability (dispersion) describes this spread. The range (highest minus lowest score) rests on the two most extreme scores. The interquartile range (IQR = $Q_3 - Q_1$) covers the middle 50 per cent; half of it is the **quartile deviation** or semi-interquartile range, $Q = (Q_3 - Q_1) / 2$, the partner of the median. The mean deviation is the average absolute deviation from the mean, $\Sigma|X - M| / N$.

The **variance** squares the deviations instead: for a population $\sigma^2 = \Sigma(X - \mu)^2 / N$, and the **standard deviation** (SD, σ) is its square root, back in the original units: the typical distance of a score from the mean. For 2, 4, 4, 4, 5, 5, 7, 9 the mean is 5; the deviations −3, −1, −1, −1, 0, 0, 2, 4 square to 9, 1, 1, 1, 0, 0, 4, 16, a sum of squares (SS) of 32. So $\sigma^2 = 32 / 8 =$ **4** and $\sigma = 2$ (32 is the SS, 2 the SD, 5 the mean).

To estimate a population's variance from a sample, divide SS by N − 1: $s^2 = \Sigma(X - M)^2 / (N - 1)$. Deviations from a sample's own mean run slightly small, so dividing by N underestimates. The fix is Bessel's correction, after the astronomer **Friedrich Wilhelm Bessel** (1784 to 1846), and N − 1 is the **degrees of freedom**: once the mean is fixed, only N − 1 deviations can vary freely. For the sample 3, 5, 7: SS = 8 and $s^2 = 8 / 2 =$ **4**, so s = 2 (dividing by N gives 2.67). The **coefficient of variation**, CV = (SD / M) × 100, compares spread across different units but needs a ratio scale.

Adding a constant to every score shifts the mean but leaves the SD and variance unchanged; multiplying by c multiplies the SD by c and the variance by c^2. With variance 4 and scores multiplied by 3 and then increased by 10, the new SD is 6 and the new variance $4 \times 9 =$ **36** (not 12, which multiplies the variance by 3 only, nor 46, which also adds the 10).

REMEMBER

Affected by extreme scores: range, mean deviation, variance, SD and the mean. Resistant: IQR, quartile deviation and the median.

Skewness and kurtosis ●●

Skewness is asymmetry, and it is named after the long tail. In positive (right) skew most scores are low and a few high ones trail right (income, a very hard test); in negative (left) skew most are high and a few low ones trail left (a very easy test). The mean, which uses every score's value, is dragged furthest into the tail, the mode stays at the hump and the median lies between. So positive skew gives **mode** < **median** < **mean** and negative skew **mean** < **median** <

mode, the mean lying left of the median (Figure 9.14). **Karl Pearson** (1857 to 1936) measured it as Sk = 3(Mean – Median) / SD, zero for a symmetrical curve.

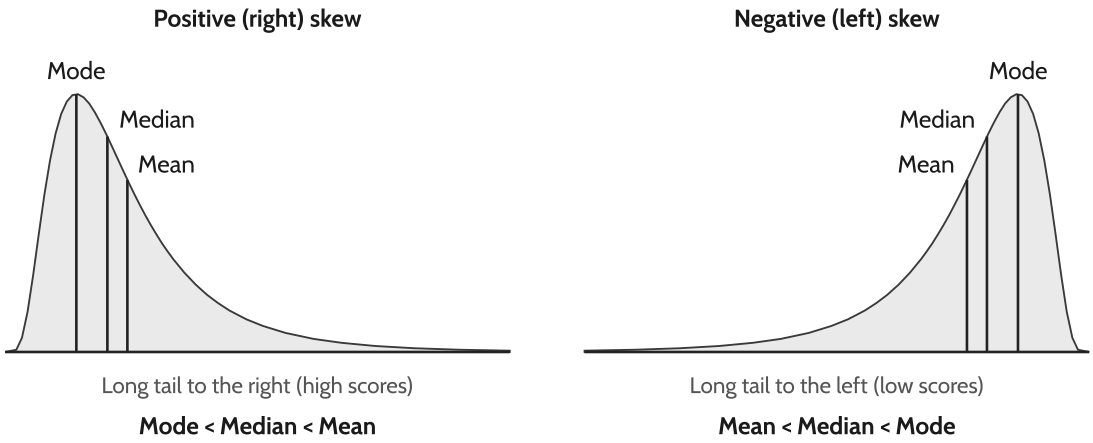


Figure 9.14 Skew is named after the tail. The mean is pulled furthest into the tail, the mode stays at the peak and the median lies between them.

Kurtosis is peakedness relative to the normal curve: leptokurtic (more peaked), mesokurtic (normal) or platykurtic (flatter). The percentile coefficient Ku = Q / (P_{90} – P_{10}) is 0.263 for the normal curve; above it is platykurtic, below it leptokurtic. On the moment measure the normal curve scores 3, and SPSS prints excess kurtosis (minus 3), so 0 means mesokurtic.

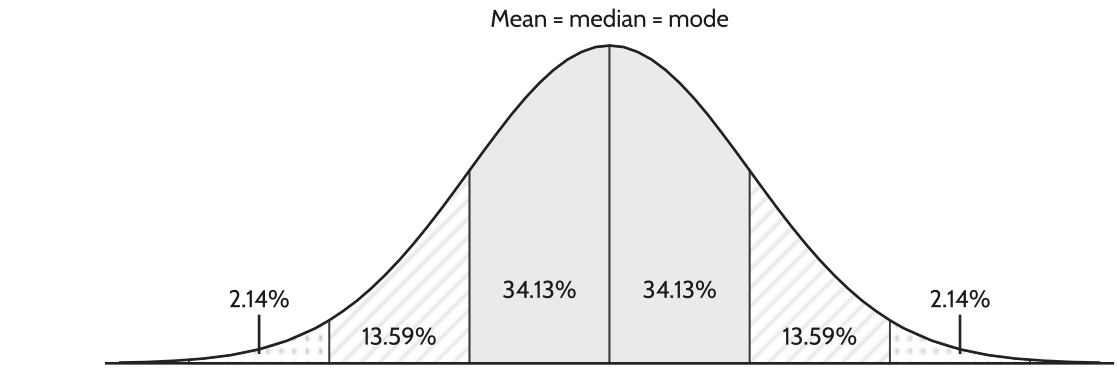
The normal probability curve ●●●

The **normal probability curve** was derived by **Abraham de Moivre** (1667 to 1754) in 1733 as the limit of the binomial distribution. **Pierre-Simon Laplace** (1749 to 1827) developed it, and **Carl Friedrich Gauss** (1777 to 1855) used it in 1809 for errors in astronomical measurement, hence the Gaussian curve. **Adolphe Quetelet** (1796 to 1874) applied it to human characteristics, and **Francis Galton** (1822 to 1911) to human ability.

> **! EXAM TRAP**
> "Who propounded the normal curve?" is keyed **Carl Gauss**, after whom it is named; mark Gauss, but know that de Moivre came first (1733). **Siméon Denis Poisson** (1781 to 1840) gave the distribution of rare events, **Jacob (James) Bernoulli** the law of large numbers (1713), and **Wilhelm Reich** (1897 to 1957) was a psychoanalyst.

The curve is bell-shaped, symmetrical and unimodal, with mean, median and mode coinciding at the centre; a two-peaked curve cannot be normal. Its tails are asymptotic, nearing the baseline without touching it. The total area under it is 1 (100 per cent), so areas read as probabilities. Its points of inflection lie at ±1 SD, and it is defined by its mean and SD alone.

The areas are therefore fixed (Figure 9.15). On each side, 34.13 per cent of cases lie between the mean and 1 SD, 13.59 per cent between 1 and 2 SD and 2.14 per cent between 2 and 3 SD. About **68 per cent** fall within ±1 SD, **95 per cent** within ±2 SD (exactly 95 within ±1.96) and **99.7 per cent** within ±3 SD.



z score	−3	−2	−1	0	+1	+2	+3
T score	20	30	40	50	60	70	80
Deviation IQ	55	70	85	100	115	130	145
Percentile	0.1	2	16	50	84	98	99.9

Figure 9.15 The normal curve with the percentage of cases in each SD band. The rows beneath give each point as z, T, deviation IQ (SD 15) and percentile rank.

z-scores, percentiles and standard scores ●●●

You score 65 in statistics (class mean 50, SD 10) and 70 in English (mean 60, SD 20). Which is better? A **z-score** states how many SDs a score lies from its mean: z = (X − M) / SD. Statistics gives z = +1.5 and English z = +0.5, so the lower raw score is the better performance. That is the purpose of z-scores: they let you compare results from different distributions. A z of +1 lies one SD above the mean, −1 one SD below, and 0 at the mean itself. A set of z-scores has mean 0 and SD 1, and because the conversion is linear, a skewed set stays skewed. A z-score shows relative position. It is not a measure of variability (the SD, variance or quartile deviation) or of central tendency, nor the coefficient of variation, nor the standard error (the SD of a sampling distribution, Section 9.9). The z table, derived from the normal distribution, converts z into percentiles.

A **percentile rank** is the percentage of cases below a score. Quartiles split the distribution into four ($Q_1 = P_{25}$, Q_2 = the median, $Q_3 = P_{75}$), deciles into ten. On the normal curve z = 0 is the 50th percentile, +0.5 about the 69th, +1 the 84th, +1.28 the 90th, +1.5 the 93rd, +1.96 the 97.5th, +2 about the 98th and +3 the 99.9th; z = −1 is the 16th and −2 the 2nd. Percentile units are unequal (crowded in the middle), so they are never averaged.

Deviation IQs have mean 100 and SD 15 (**David Wechsler**'s system, Chapter 2): IQ 100 is the 50th percentile, 85 the 16th, 115 the 84th and 130 (z = +2) about the 98th. Older Binet-type tests used SD 16, so an IQ of 136 is z = 36 / 16 = +2.25; the table puts .9878 below it, so about 98.7 to 98.8 per cent score lower.

> **CLINICAL LINK**
>
> An IQ of 70 lies two SD below the mean, near the 2nd percentile. DSM-5 places the intellectual deficits of intellectual disability at about this point, roughly 65 to 75 once measurement error is allowed for (Chapter 8).

Other **standard scores** avoid the signs and decimals of z: new score = new mean + (new SD × z), so z = +2 is a T of 70, an IQ of 130 and a stanine of 9 (Table 9.25). Such linear standard scores keep the raw distribution's shape; normalised standard scores (normalised T scores, stanines) convert percentile ranks into normal-curve z values, forcing a normal shape. Norms are treated in Chapter 2.

Table 9.25 Common standard-score systems.

System	Mean	SD	Where you meet it
z-score	0	1	The base of all the others
T score	50	10	MMPI and most personality inventories
Deviation IQ	100	15	Wechsler scales; Stanford-Binet 5
Old Stanford-Binet IQ	100	16	Earlier Binet-type tests
Stanine	5	2	Nine bands, 1 to 9
Sten	5.5	2	Ten bands; the 16PF
Scaled score	10	3	Wechsler subtests

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Equal intervals with an arbitrary zero define the _____ scale.
2. The only average usable with nominal data is the _____.
3. The representative income of a community is best given by the _____.
4. In negative skew the mean lies to the _____ of the median.
5. The population variance of 2, 4, 4, 4, 5, 5, 7, 9 is _____.
6. Multiplying every score by 3 multiplies the variance by _____.
7. About _____ per cent of normal-curve cases lie within ±2 SD of the mean.
8. A deviation IQ of 85 lies at about the _____ percentile.

Answers: (1) interval, (2) mode, (3) median, (4) left, (5) 4, (6) 9, (7) 95, (8) 16th.

9.9 Sampling Distributions and the Logic of Hypothesis Testing

MEDIUM YIELD · 37 QS

At the Rothamsted research station in the 1920s, the biologist **Muriel Bristol** claimed she could taste whether the milk or the tea had gone into her cup first. **Ronald Aylmer Fisher** (1890 to 1962) turned the claim into the opening example of *The Design of Experiments* (1935): eight cups, four made each way, in random order. A pure guesser would sort all eight correctly only 1 time in 70 (p = .014). That is the whole logic of this section: assume nothing is going on, work out how probable the result would be by chance, and reject chance when the result would be rare.

From describing to inferring ••

Descriptive statistics summarise the scores you have (section 9.8). **Inferential statistics** reason from a sample to the population it came from, drawing on two things: the sample data, and knowledge of how random processes behave. Their question is how plausible it is that a random process alone, drawing samples from a population in which nothing is going on, could produce results as extreme as those obtained. Inference always runs from the observed sample to the unobserved population, never the reverse.

If a therapy group improves by 6 points and a control group by 4, part of the gap is chance, because two random samples almost never have identical means. What statistics (applied to health data, **biostatistics**) tells the psychologist is whether an observed difference is larger than chance would usually produce. Only a statistical test can show a reliable difference between groups; norms (a reference group's typical scores), reliability (consistency) and validity (measuring what is claimed) are properties of tests (Chapter 2).

> **! EXAM TRAP**
> "Inferential statistics make statements about samples based on observing populations" is the FALSE option: it reverses the logic. Statistics does not choose patients for therapy, design interviews or plan therapeutic exercises; it judges whether differences are due to chance.

Probability: the language of chance

A probability runs from 0 (impossible) to 1 (certain). **Classical probability** is worked out in advance from equally likely outcomes, favourable outcomes divided by all outcomes: a die shows a six with probability 1/6. **Empirical probability** is the observed relative frequency over many trials: if 30 of 200 discharged patients relapse, p(relapse) is about .15. The **addition rule** handles "or": for mutually exclusive events p(A or B) = p(A) + p(B), so a one or a two on a die is 2/6. The **multiplication rule** handles "and": for independent events p(A and B) = p(A) × p(B), so ten heads in a row has probability 1/1024. For dependent events you multiply by a **conditional probability**, p(B given A); remember it when you meet the p-value.

The **binomial distribution** gives the probability of each number of successes in N independent two-outcome trials with fixed probability p (mean Np, SD √Npq): ten coin tosses average 5 heads, yet exactly 5 occurs only about one time in four. As N grows the binomial becomes bell-shaped, as **Abraham de Moivre** (1667 to 1754) showed in 1733. The normal distribution is therefore the probability model of inference: area under it is probability, and since 95 percent lies within ±1.96 SD (section 9.8), a chance value beyond those points has probability .05.

The sampling distribution and the standard error ••

Draw a random sample of 36 students, compute their mean, replace them, and repeat thousands of times. The means vary, and the theoretical distribution they form is the **sampling distribution of the mean**. You never build it, but three facts about it let you judge your one real sample. Its mean equals the population mean, μ. Its standard deviation is the **standard error of the mean**, SE = σ/√N (estimated as s/√N): for IQ (σ = 15), samples of 25 give SE = 3 and samples of 100 give 1.5, so halving the SE takes four times the sample. And its shape is normal once N is large.

That last fact is the **central limit theorem**: with a sufficiently large sample, conventionally 30 or more (A.K. Singh says above 50), the sampling distribution of the mean is approximately normal whatever the shape of the population, centred on μ, with a spread that shrinks as N grows (Figure 9.16). De Moivre's coin result was its first case and **Pierre-Simon Laplace** (1749 to 1827) generalised it. It is why tests on means can use the normal curve even though psychological scores are often skewed.

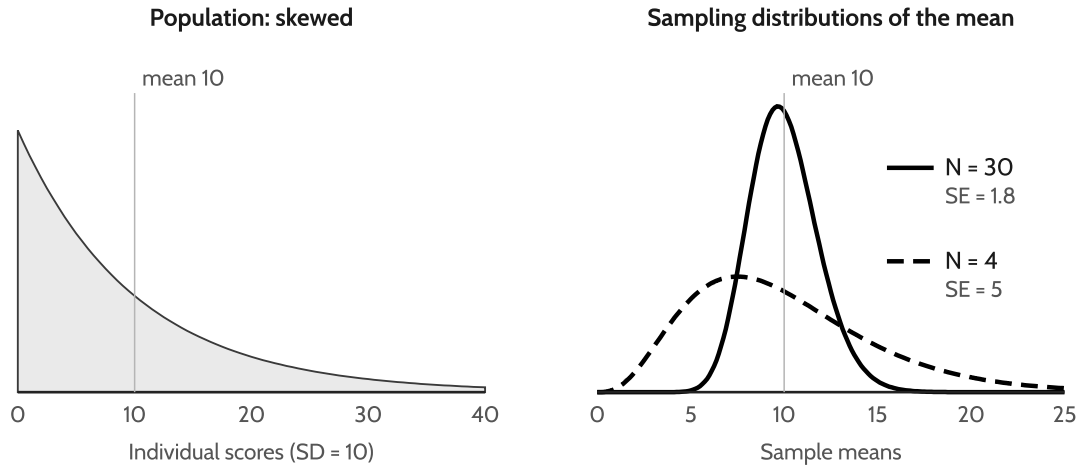


Figure 9.16 The central limit theorem. Means of samples from a skewed population cluster around its mean; with N = 30 they are nearly normal and far narrower (SE about 1.8) than with N = 4.

Sampling error is the gap between one sample's statistic and the parameter (M − μ); it is a random error, unsystematic and averaging to zero. The standard error is the typical size of that gap across all samples. The **law of large numbers** (**Jacob Bernoulli**, 1655 to 1705, published 1713) says only that a sample mean gets closer to the population mean as N grows; it says nothing about the shape of the sampling distribution.

> **! EXAM TRAP**
>
> The standard error is **the SD of the sampling distribution of the mean**: not the SD of the sample (which does not shrink with N) and not M − μ (sampling error). As N rises the sampling distribution approaches normal; it does not grow skewed, does not become identical to the population distribution, and its SE gets smaller. "Normal distribution theorem" and "sampling distribution theorem" are invented names.

Estimation and confidence intervals ○

A **point estimate** is a single value, such as a sample mean of 52 for μ. An interval estimate, the **confidence interval** (CI), adds a margin: M ± 1.96 SE for 95 percent and M ± 2.58 SE for 99 percent. With M = 52, SD = 12 and N = 36, SE = 2, so the 95 percent CI runs from 48.08 to 55.92 and the 99 percent CI, wider, from 46.84 to 57.16. Higher confidence, a larger SD and a smaller N all widen the interval.

The population mean is fixed, so the confidence belongs to the procedure: if the sampling were repeated many times, 95 percent of the intervals so built would contain the true mean. If a 95 percent CI for a difference includes zero, the difference is not significant at α = .05 (two-

sided): a mean difference of 0.4 with a CI of −0.1 to 0.9 is non-significant. That is not proof of no effect, since the interval also allows 0.9; a wide CI signals an imprecise estimate. For ratios the null value is 1 (section 9.5).

> **! EXAM TRAP**
>
> A 95 percent CI is NOT "a 95 percent probability that μ lies in this interval", NOT "95 percent of individual scores fall in it" and NOT "a 95 percent chance of significance". A CI crossing zero means non-significant, not a difference of exactly zero.

The logic and steps of a significance test ○

A significance test assumes the null hypothesis (H_0: no difference or effect), asks how probable the data would be if H_0 were true, and rejects H_0 in favour of the alternative hypothesis (H_1) if they would be rare (wording hypotheses: section 9.1). The steps:

1. State H_0 and H_1; choose a one-tailed or two-tailed test.
2. Set the **significance level** (α), usually .05 or .01, before seeing data.
3. Compute the test statistic (z, t, F, chi-square): the observed effect over its standard error.
4. Compare it with the **critical value** bounding the critical region, or compare p with α.
5. Reject H_0 or fail to reject (retain) it; report the effect size and CI.

A stress scale has a population mean of 50 and SD of 12; after a mindfulness course 36 students average 55. Then $z = (55 - 50)/(12/\sqrt{36}) = 2.5$: beyond 1.96, so significant at .05 two-tailed (p about .012), but short of 2.58, so not at .01. Table 9.26 gives the cut-offs. They explain the rough rule that a score is significant when it lies **more than two standard deviations from the mean**: about 95 percent of a normal distribution lies within two SDs, so a value beyond is in the outer 5 percent, while values within one SD, or between one and two, are ordinary chance results.

Table 9.26 Critical values of z.

α	Two-tailed	One-tailed	Matching CI
.05	±1.96	1.645	95 percent
.01	±2.58	2.33	99 percent

A **two-tailed test** (non-directional H_1) splits α between the tails, .025 each. A **one-tailed test** (directional H_1) puts all of α in one tail, so z = 1.80 is significant one-tailed but not two-tailed. It is more powerful in the predicted direction, blind in the other, and legitimate only if the direction was fixed in advance. Tests other than z also need **degrees of freedom** (df), the number of values free to vary once a constraint is fixed: five scores with a set mean of 10 leave four free, df = N − 1 = 4. Critical values of t, F and chi-square depend on df (section 9.10).

The p-value and what it is not ●●●

The **p-value** is the probability of obtaining the observed result, or one more extreme, if H_0 is true: the probability, revealed by a statistical test, that a random process (random samples of a given size from a given population) could produce that outcome. If $p \leq \alpha$, reject H_0: with α = .05, p = .03 is statistically significant and H_0 is rejected.

Because p assumes H_0 and asks about the data, it is not the probability that H_0 (or H_1) is true, which would need Bayes' theorem. Nor is it the effect size (a huge sample makes a trivial effect give tiny p), the probability of replication, or power. It is continuous: p = .049 and p = .051 are essentially the same evidence, because .05 is a convention, not a natural boundary. Two distractors share a letter: the **q-value**, the false-discovery-rate analogue of p used when thousands of tests are run (brain imaging, genetics), and Tukey's studentised range q (section 9.10).

> **! EXAM TRAP**
>
> The key is always "the probability of the data (or more extreme) **if H_0 is true**"; the favourite wrong option reverses it, "the probability that H_0 is true". Nor is p the magnitude of a correlation or the sample size, and p = .001 does not mean a large, clinically important or certain-to-replicate effect.

Type I and Type II errors ●●●

A decision from a sample can go wrong in two ways (Table 9.27). A **Type I error** (alpha error) is rejecting a true null hypothesis: a false positive. Its probability is α, so the significance level is the probability of a Type I error, in analysis of variance as in every test; it does not certify that F is correct, and the Type II risk is β, not α. A **Type II error** (beta error) is retaining, or loosely "accepting", a false null hypothesis: a false negative, a real effect missed. Its probability, β, is not chosen; it follows from N, the true effect, α and variability.

Table 9.27 The four outcomes of a significance test.

Decision	H_0 true	H_0 false
Reject H_0	Type I error (α), false positive	Correct: power (1 − β)
Retain H_0	Correct (1 − α)	Type II error (β), false negative

In court terms, a Type I error convicts the innocent and a Type II error frees the guilty. With N fixed, a stricter α (.05 to .01) cuts Type I errors but raises β; only a larger sample or a more precise measure reduces both (Figure 9.17). A non-significant result is a failure to reject H_0, never proof of it.

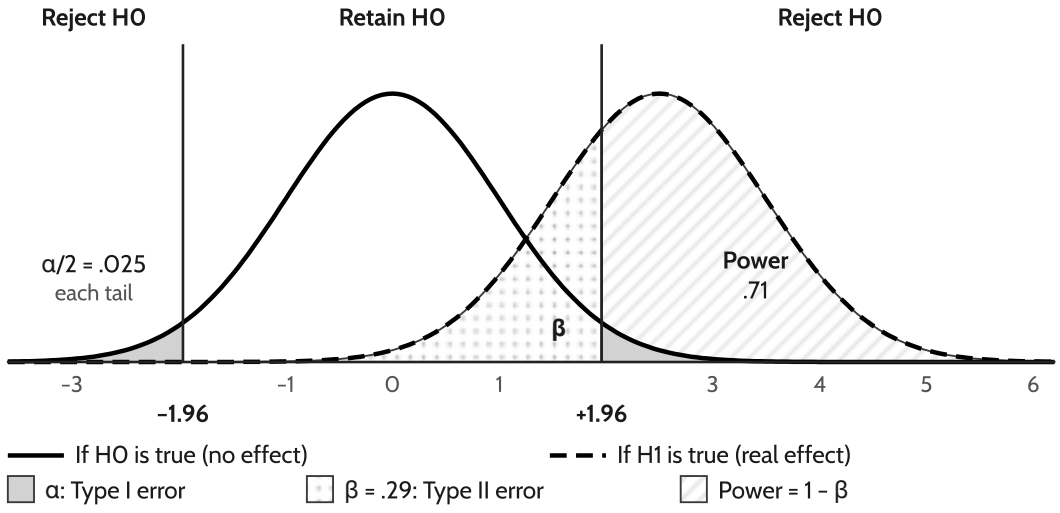


Figure 9.17 Errors and power in a two-tailed z test at α = .05. Under H_0 (solid curve) a result beyond ±1.96 is a Type I error; if the true effect is 2.5 standard errors (dashed curve), the test misses it 29 percent of the time (β) and detects it 71 percent of the time (power).

The procedure fuses two rival traditions. Fisher (*Statistical Methods for Research Workers*, 1925) treated p as graded evidence against a single null hypothesis and offered .05 as a convenient cut-off. **Jerzy Neyman** (1894 to 1981) and **Egon Pearson** (1895 to 1980), in papers of 1928 and 1933, recast testing as a decision between H_0 and H_1 with fixed error rates, giving us Type I and II errors and power.

> **! EXAM TRAP**
>
> Swapped wordings ("accepting a false null" offered as Type I) are the commonest wrong options. **Type III error** is informal: the right answer to the wrong question, or rejecting H_0 but in the wrong direction; "Type IV" is rarer still. Low power, sample size (too small, too large or "large sample bias"), a wide CI, a misused one-tailed test, sampling error, standard error and confounding (section 9.2) are not definitions of these errors, though several change their risk.

Statistical power ○

Power (1 – β) is the probability of correctly rejecting a false null hypothesis, of finding an effect that is really there; the convention is at least 0.80. Power rises with a larger true effect, a larger N (it shrinks the SE, pulling the curves of Figure 9.17 apart), a larger, less strict α (at the price of more Type I errors), lower variability (homogeneous samples, standard procedures, repeated measures, reliable measures) and a correctly predicted one-tailed test. It falls when N shrinks, α becomes stricter or variability grows. **Power analysis**, done before a study, fixes α, the desired power and the smallest effect worth detecting, and solves for N. **Jacob Cohen** (1923 to 1998) made it routine in *Statistical Power Analysis for the Behavioral Sciences* (2nd edition, 1988): detecting a medium effect (d = 0.5) with an independent-samples t-test at α = .05, two-tailed, and power .80 needs about 64 people per group.

Effect size and practical significance ●●

Significance says an effect is probably not zero; **effect size** says how big it is, independent of N. **Cohen's d** is the standardised difference between two means, $(M_1 - M_2)$/pooled SD, with benchmarks 0.2 small, 0.5 medium and 0.8 large. The mindfulness example gives d = 5/12 = 0.42, between small and medium, despite p = .012. The correlation r is also an effect size (.10, .30, .50); its sign gives only direction, so r = −.30 is a medium negative relationship, not "none", not perfect (−1.00) and not causal (section 9.11). Table 9.28 adds the variance-explained indices; the odds ratio belongs to section 9.5.

Table 9.28 Common effect sizes and Cohen's benchmarks.

Index	Expresses	Small	Medium	Large
Cohen's d	Mean difference in SD units	0.2	0.5	0.8
r	Strength of a linear relationship	.10	.30	.50
Eta squared (η^2)	Share of total variance explained (ANOVA)	.01	.06	.14

Eta squared is SS_{effect}/SS_{total}; **partial eta squared**, $SS_{effect}/(SS_{effect} + SS_{error})$, drops the other factors from the denominator, so values across effects can sum past 1; omega squared (ω^2) is a less biased, smaller estimate. Statistical significance (data surprising under H_0, driven by N) differs from practical significance (an effect big enough to matter) and **clinical significance** (patients really function better, often back in the normal range): with 10,000 participants r = .03 is significant and meaningless. **p-hacking**, analysing data many ways (extra measures, dropped cases, stopping once p dips below .05, a one-tailed test chosen afterwards) until something is significant, inflates Type I errors; preregistration, power analysis and reporting effect sizes with CIs are the remedies.

KEY STUDY: THE OPEN SCIENCE COLLABORATION (2015)

About 270 researchers coordinated by **Brian Nosek** repeated 100 studies published in 2008 in three leading psychology journals. Of the originals, 97 percent were significant; only 36 percent of the replications were, with effect sizes about half as large. The project made the **replication crisis** famous: small, underpowered studies plus flexible analysis yield fragile findings, although critics noted that some replications departed from the originals.

! EXAM TRAP

d of 0.2, 0.5 and 0.8 = small, medium, large (not "trivial, weak, moderate"). d measures standardised effect size for a difference between two means, not significance, sample-size adequacy or correlation strength.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The SD of the sampling distribution of the mean is the _____.
2. The _____ theorem says that with a large sample the sampling distribution of the mean is approximately normal, whatever the population's shape.
3. The 99 percent confidence interval is the mean plus or minus _____ standard errors.
4. The p-value is the probability of the observed result, or one more extreme, if the _____ is true.
5. Rejecting a true null hypothesis is a _____ error.
6. Power equals 1 minus _____.
7. A Cohen's d of 0.5 is a _____ effect.

Answers: (1) standard error of the mean, (2) central limit, (3) 2.58, (4) null hypothesis, (5) Type I, (6) β, (7) medium.

9.10 Comparing Groups: t-Tests, ANOVA and Non-Parametric Tests

HIGH YIELD · 57 QS

A counsellor at a Mysuru college tries three ways of revising for examinations: practice testing, flashcards and plain rereading. A month later the groups' recall scores differ a little. Is the difference real, or the wobble any three groups would show by chance? Section 9.9 gave the logic of significance testing; here are the tools. Choosing one always comes down to three questions: what kind of data, how many groups, and are the groups independent or made of the same (or matched) people?

Parametric or non-parametric? ●●

A **parametric test** estimates population parameters (means and variances) and rests on four assumptions: the scores come from a **normally distributed** population; the groups have roughly equal variances (**homogeneity of variance**); observations are **independent** of one another; and the dependent variable (DV) is measured on an **interval or ratio** scale, so that means make sense. The z-test, t-tests, ANOVA's F-test and Pearson's r are parametric. When data are normal with equal variances, a parametric test is the right choice, because it uses the actual distances between scores and so has the most power.

A **non-parametric test**, also called a **distribution-free test**, makes few assumptions about the population's shape and works on ranks, signs or frequency counts. **Sidney Siegel** (1916 to 1961) brought these tests into psychology with *Nonparametric Statistics for the Behavioral Sciences* (1956). They suit ordinal data (ranks), nominal data (counted categories), very small samples and clearly violated assumptions (heavy skew, outliers, very unequal variances). Rank order is thus the classic non-parametric data, whereas height, weight and Celsius temperature are interval or ratio measures (9.8). Chi-square is not a third family: it is the non-parametric test for nominal frequencies.

The price is power, because ranking discards the distances between scores. Siegel measured the cost as **power-efficiency**: the sample a parametric test needs, as a percentage of the sam-

ple a non-parametric test needs for equal power. If a t-test needs 95 cases to match what the Mann-Whitney test does with 100, Mann-Whitney is 95 percent power-efficient. Under normality the Mann-Whitney and Wilcoxon tests reach about 95 percent; the sign test only about 63 percent in large samples.

> **! EXAM TRAP**
>
> "Which is non-parametric data?" Rank order, not height, weight or Celsius temperature (Celsius has an arbitrary zero but equal intervals: interval data). "Distribution-free" is just another name for non-parametric. For normal data with equal variances the answer is parametric tests, never "chi-square tests" or "distribution-free tests".

Checking the assumptions ○

Normality is tested with the **Shapiro-Wilk test** (the usual choice) or the **Kolmogorov-Smirnov test**, which compares the sample's cumulative distribution with a normal one; it is inspected with a **Q-Q plot** (quantile-quantile plot), where normal data hug a straight diagonal, and with skew and kurtosis values near zero (9.8). **Homogeneity of variance**, or **homoscedasticity**, is tested with **Levene's test** (**Howard Levene**, 1960), which stays accurate when data are not normal; **Bartlett's test**, more powerful but upset by non-normality; or **Hartley's F-max**, the largest group variance divided by the smallest (near 1 is fine). These checks precede a t-test, an ANOVA or a regression. **Independence** has no test: it is secured by random sampling and assignment, and its violation is the most damaging.

Repeated-measures ANOVA adds **sphericity**: equal variances of the difference scores between every pair of conditions. It matters only with three or more conditions, is tested by **Mauchly's test**, and when violated the df are reduced by the **Greenhouse-Geisser** correction (more conservative) or the **Huynh-Feldt** correction (less so). ANCOVA adds homogeneity of regression slopes (9.10.6). Read these tests backwards: their null hypothesis is that the assumption holds, so a non-significant result ($p > .05$) is good news.

> **! EXAM TRAP**
>
> Shapiro-Wilk tests normality, NOT homogeneity of variance, so it is the odd one out beside Levene's and Bartlett's tests. Levene's test checks equal variances, not normality, independence or equal group sizes. Equal group sizes are not an assumption, though they make ANOVA more forgiving of unequal variances.

The t-tests ●●●

A **z-test** compares means using the population standard deviation, which you almost never know (older texts allow it above about 30 cases). Estimating the standard deviation from the sample adds a second source of error, and the statistic then follows the t distribution.

KEY STUDY: "STUDENT" AT THE GUINNESS BREWERY (1908)

William Sealy Gosset (1876 to 1937), a chemist at the Guinness brewery in Dublin, had to judge barley and stout from tiny samples. He worked out the exact distribution of a mean divided by its estimated standard error. Guinness barred staff from publishing under their own names, so his 1908 *Biometrika* paper appeared as "Student", hence Student's t. (Some Indian texts print 1915; the paper is from 1908.)

The **t distribution** is a family of curves, one for each number of degrees of freedom (df). Like the normal curve it is symmetrical and bell-shaped around zero; unlike it, it has heavier tails and a lower peak, so its variability and kurtosis are greater. As df rise it closes in on the normal curve (Figure 9.18), which is why critical t values shrink towards 1.96 in large samples.

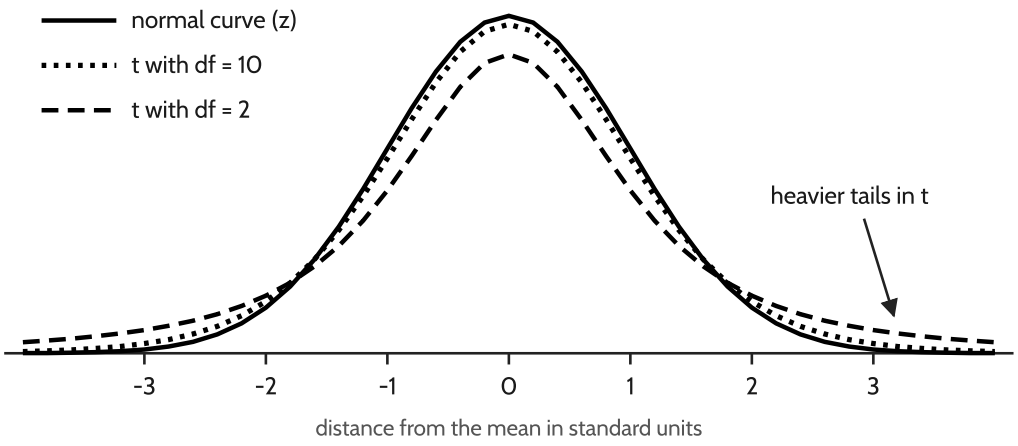


Figure 9.18 The t distribution against the normal curve. All are symmetrical, but with few df t has a lower peak and heavier tails, so a larger t is needed for significance; by df = 10 it is already close to normal.

There are three t-tests, each with its own df.

- The **one-sample t-test** compares one sample mean with a known population value; df = N - 1, since once the mean is fixed the last score cannot vary. N = 20 gives 19; N = 30 gives 29.
- The **independent-samples t-test** (unpaired t) compares the means of two separate groups, using their pooled variance. Its df = N1 + N2 - 2: 25 per group gives 48. If Levene's test shows unequal variances, **Welch's t-test** is used; it does not pool the variances and lowers the df.
- The **paired-samples t-test** (dependent or matched-pairs t) is for the same people measured twice, pre and post, or for matched pairs. It tests whether the mean difference score is zero; df = pairs - 1. Each person acts as their own control, so individual differences leave the error term and power rises.

Why not run three t-tests on three groups? Each carries a 5 percent Type I risk (9.9), and the risks accumulate. With c tests the chance of at least one false positive, the **family-wise error rate**, is $1 - (1 - .05)^c$: about .14 for three tests and .26 for the six that four groups need. ANOVA tests all the means at once at one alpha.

> **! EXAM TRAP**
>
> One-sample df: N - 1. Independent: N1 + N2 - 2 (25 per group gives 48, not 24, 49 or 50). Paired: pairs - 1. The t curve shares only its symmetry with the normal curve; its kurtosis and variability are larger.

One-way ANOVA: partitioning variance ●●●

Analysis of variance (ANOVA) was developed by **Ronald Aylmer Fisher** (1890 to 1962) for agricultural trials at Rothamsted Experimental Station and published in *Statistical Methods for Research Workers* (1925); **George Waddel Snedecor** (1881 to 1974) named its statistic F in Fisher's honour. A **one-way ANOVA** compares the means of independent groups formed by one independent variable (IV, or factor) on one continuous DV. It is used when more than two means are compared, though it works for two as well, where $F = t^2$.

ANOVA compares means by analysing variances. The deviations of all scores from the **grand mean** sum to zero, so they are squared and added to give the **total sum of squares**. Fisher split it into exactly two parts. The **between-groups sum of squares** reflects how far each group mean lies from the grand mean: treatment effect plus chance. The **within-groups sum of squares** reflects how far each person lies from their own group mean: since a group was treated alike, this is pure chance or error. SS total = SS between + SS within, and each part has its own df: between = k - 1 (k groups), within = N - k, total = N - 1. Three groups of 10 give 2, 27 and 29.

Each SS divided by its df is a **mean square** (MS), a variance. MS within is the **pooled error variance**, and it is the denominator of the **F ratio**: F = MS between / MS within. If the null hypothesis is true, both estimate the same chance variance and F is about 1; a real effect swells only the numerator. So an F below 1, such as 0.99, can never be significant whatever the df, and as variability due to chance falls, F rises. Figure 9.19 and Table 9.29 work through the revision study.

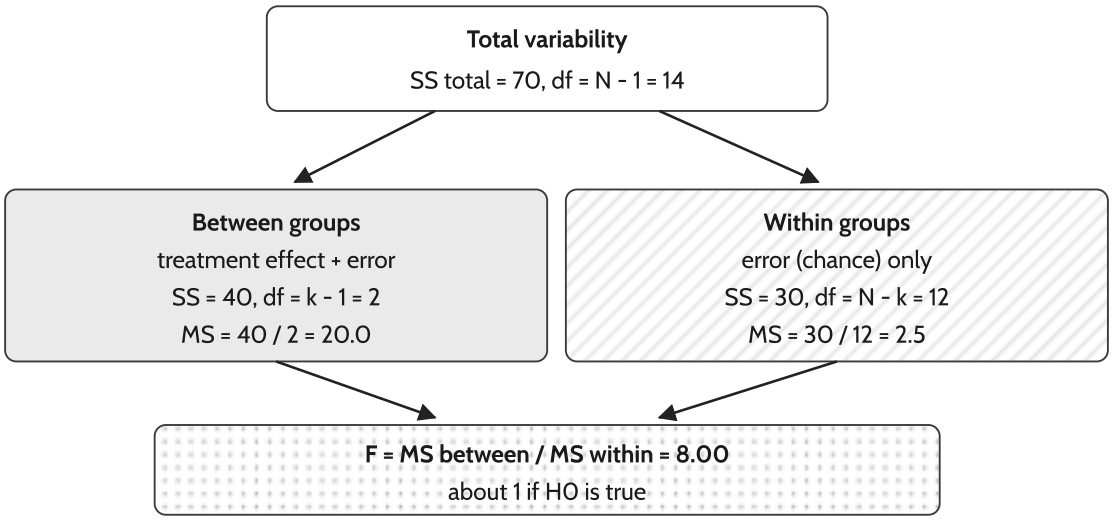


Figure 9.19 Fisher's partition of variance. The two sums of squares add up to the total, each has its own df, and the within-groups (error) mean square is always the denominator of F.

Table 9.29 A worked one-way ANOVA. Practice testing 9, 8, 10, 7, 6 (mean 8); flashcards 6, 5, 7, 4, 8 (mean 6); rereading 4, 3, 5, 2, 6 (mean 4); grand mean 6.

Source	SS	df	MS	F
Between groups	$5(2)^2 + 5(0)^2 + 5(-2)^2 = 40$	3 - 1 = 2	20.0	8.00
Within groups (error)	10 + 10 + 10 = 30	15 - 3 = 12	2.5	
Total	70	15 - 1 = 14		

The critical F for df 2 and 12 is 3.89 at .05 and 6.93 at .01, so the methods differ ($p < .01$), and condition accounts for 40/70, or 57 percent, of the variance (eta squared, 9.9). But F is an **omnibus test**: it says at least one mean differs, not which. Being a ratio of two variances, the **F distribution** is never negative and is positively skewed, its shape set by two df (numerator and denominator); the test is one-tailed, because only a large F counts against the null.

> **! EXAM TRAP**
>
> F's denominator is MS within (the pooled error variance), not MS between. One-way ANOVA has two sources of variance, each with its own df; "error variance" is the within-groups part, not a third source, and "information variance" does not exist. F = 0.99 is never significant; F is never negative. ANOVA tests means, not "variance only", and unequal variances violate its assumption.

After a significant F: post hoc tests ●●

Post hoc tests (a posteriori or multiple comparison tests) compare groups pair by pair while holding the family-wise error rate down. They are used only after a significant F; after a non-significant F you stop. With k groups there are k(k - 1)/2 pairs: 3 for three groups, 6 for four, 10 for five. The tests differ in how conservative they are, that is, how hard they make it to call a difference significant (Table 9.30).

Table 9.30 Common post hoc tests. The first five run from most liberal to most conservative.

Test	What it does	Conservatism
Fisher's LSD (least significant difference)	Plain t-tests, allowed only after a significant F	Least; most power
Duncan's multiple range test	Stepwise comparison of ordered means	Liberal
Newman-Keuls	Stepwise; critical value depends on how far apart the means rank	Between LSD and Tukey
John Wilder Tukey's HSD (honestly significant difference)	All pairs, using the studentised range q; the standard choice with equal n	Moderate
Henry Scheffé's test	Any comparison, including complex ones (group 1 against groups 2 and 3)	Most conservative
Dunnett's test	Each treatment against one control group	Powerful for that job
Games-Howell	Pairs when variances or group sizes are unequal	Moderate

Planned contrasts (a priori comparisons) are instead fixed from theory before data collection; they are few (at most k - 1 independent ones), need no significant F and keep more power. The simplest guard for any set of comparisons is the **Bonferroni correction**, after **Carlo Emilio Bonferroni** (1892 to 1960): divide alpha by the number of comparisons. Six compar-

isons are each tested at .05/6 = .0083, keeping the family-wise Type I error rate at .05 or below. The cost is power: Bonferroni is conservative, raising Type II risk as comparisons multiply. A single planned comparison needs no correction.

> **! EXAM TRAP**
>
> Duncan, Newman-Keuls, Tukey and Scheffé are used only when F is significant. Bonferroni divides alpha, never multiplies it (software may multiply each p-value by the number of tests, the same rule). It controls family-wise Type I error; it does not raise power, fix unequal sample sizes or transform skewed data (skew calls for a log or square-root transformation, or a rank test).

Factorial, repeated-measures and covariance designs ●●

A **factorial ANOVA** (two-way or higher) crosses two or more IVs so that every level of one meets every level of the other; a 2 × 3 design has two factors and six cells. It tests a **main effect** of each IV (averaged over the other) and an **interaction effect**, present when the effect of one IV depends on the level of another: plotted, the lines are not parallel. Of the familiar techniques only factorial ANOVA detects interactions; a t-test has one two-level IV, chi-square tests association between two categorical variables, and correlation measures how two variables move together. A significant interaction is interpreted before the main effects.

Repeated-measures ANOVA tests the same people under three or more conditions, extending the paired t-test. It accounts for the correlation among each person's scores by removing individual differences from the error term, so it needs fewer people; but it is still parametric, assuming normality, plus sphericity. A **randomised block design** uses different people grouped into homogeneous blocks (by pre-test, age or IQ) and randomly assigned to treatments within each block, so block differences leave the error term. With one person per treatment per block, error df = (treatments - 1)(blocks - 1): 4 treatments and 5 blocks give 3 × 4 = 12. A **mixed (split-plot) ANOVA** combines a between-subjects and a within-subjects factor.

> **CLINICAL LINK**
>
> Most therapy trials are mixed designs: group (therapy or waitlist) by time (pre, post, follow-up). Evidence that therapy worked is the group × time interaction, not either main effect.

Analysis of covariance (ANCOVA), also Fisher's, builds a regression into ANOVA. A **covariate** (a continuous variable such as pre-test score or IQ) predicts the DV, its effect is removed, and groups are compared on adjusted means. This raises power and statistically equates groups that started unequal, though it cannot replace random assignment. ANCOVA keeps ANOVA's assumptions and adds a linear covariate-DV relation, a covariate measured reliably before treatment, and above all **homogeneity of regression slopes**: the covariate must predict the DV equally in every group, so the regression lines are parallel. **Multivariate analysis of variance** (MANOVA) tests groups on two or more DVs at once, avoiding the Type I error of many separate ANOVAs; its statistics are Wilks' lambda (most reported; smaller means a bigger effect), Pillai's trace (most robust), Hotelling's trace and Roy's largest root. MANCOVA adds covariates. In all these names, M means more than one DV and C means a covariate.

> **! EXAM TRAP**
>
> The assumption ANCOVA adds to ANOVA is homogeneity of regression slopes. Homogeneity of variance and homoscedasticity (one idea, two names) belong to both; equal sample sizes to neither. Repeated-measures ANOVA is not non-parametric, still assumes normality and is not limited to two conditions.

Chi-square: comparing frequencies ●●●

Nominal data offer no means, only counts of people in categories. For these, **Karl Pearson** (1857 to 1936) introduced the **chi-square test** (χ^2) in 1900. It compares the observed frequency (O) in each category with the frequency expected (E) under the null hypothesis: $\chi^2 = \Sigma (O - E)^2/E$. Being squared, it is never negative and its distribution is positively skewed. It is a non-parametric test for categorical frequencies, not for interval, ratio or parametric data.

The **goodness-of-fit test** asks whether one variable's frequencies match a hypothesised pattern (do students prefer four brands of chai equally?); df = k - 1 for k categories. The **test of independence** asks whether two categorical variables in a **contingency table** are associated (is gender related to treatment preference?). Each expected frequency is row total × column total / N, and df = (rows - 1)(columns - 1), because once the margins are fixed only that many cells can vary: 2 × 2 gives 1, 3 × 4 gives 6, 4 × 5 gives 12.

Chi-square needs raw counts (not percentages), each person in one cell only, and an **expected frequency** of at least 5 per cell. For 2 × 2 tables with small expected values, **Frank Yates** (1902 to 1994) proposed the **Yates' correction for continuity** in 1934, subtracting 0.5 from each |O - E| before squaring, while **Fisher's exact test** computes the exact probability of a small table. Paired yes-or-no data (the same people before and after) need the **McNemar test** (**Quinn McNemar**, 1947), which counts only those who changed. Chi-square shows whether an association exists, not its strength; for that report phi (2 × 2), the contingency coefficient or Cramér's V (9.11).

> **! EXAM TRAP**
>
> Contingency df is a product, (rows - 1)(columns - 1): 3 × 4 gives 6, not 12, 7 or 11; 4 × 5 gives 12, not 20 or 9. It is not N - 1 (one-sample t) or N - 2 (independent t), and it subtracts 1, not 2.

Rank tests and choosing the right test ●●●

Each parametric test has a rank-based partner. The **Mann-Whitney U test** (**Henry Berthold Mann** and **Donald Ransom Whitney**, 1947) pools and ranks the scores of two independent samples and asks whether one group's ranks run higher. It is the non-parametric equivalent of the independent-samples t-test, and identical to the **Wilcoxon rank-sum test**. For two related samples, the **Wilcoxon matched-pairs signed-rank test** (**Frank Wilcoxon**, 1945) ranks the size of each pair's difference and reattaches the signs, using both direction and magnitude; the **sign test** counts only pluses and minuses (direction) and is weaker. For three or more independent groups, the **Kruskal-Wallis H test** (**William Kruskal** and **W. Allen Wallis**, 1952) is the one-way ANOVA by ranks, H being read against chi-square with k - 1 df. For three or more related samples or repeated measures, the **Friedman test** of the economist **Milton Friedman** (1912 to 2006), devised in 1937, ranks conditions within each person: the two-way ANOVA by ranks.

The **median test** splits independent groups at the combined median and applies chi-square. The **Kolmogorov-Smirnov two-sample test** compares two samples' whole cumulative distributions, so it also catches differences in shape and spread. The **Wald-Wolfowitz runs test** counts runs of the same group in the pooled, ordered scores; few runs mean the groups differ. **Cochran's Q** extends McNemar to three or more related yes-or-no samples. Rank correlations (Spearman, Kendall), Pearson's r and regression belong to 9.11. Table 9.31 gathers the choices.

Table 9.31 Which test when, by design and level of measurement.

Design	Interval or ratio	Ordinal	Nominal
One sample against a known value	One-sample t (z if SD known)	Sign test; one-sample Wilcoxon	Chi-square goodness of fit
Two independent groups	Independent t (Welch if variances differ)	Mann-Whitney U; median, Kolmogorov-Smirnov, runs tests	Chi-square; Fisher's exact
Two related samples	Paired t	Wilcoxon signed-rank; sign test	McNemar
Three or more independent groups	One-way ANOVA	Kruskal-Wallis H	Chi-square (r × c)
Three or more related samples	Repeated-measures ANOVA	Friedman	Cochran's Q
Two or more IVs	Factorial ANOVA	None in common use	None in common use
Association of two variables	Pearson r (9.11)	Spearman, Kendall	Chi-square; phi, Cramér's V

! EXAM TRAP

Two names, two independent groups: Mann and Whitney. Wilcoxon signed-rank is for related pairs (but "Wilcoxon rank-sum" is Mann-Whitney in disguise). Kruskal-Wallis is for independent groups, Friedman for related ones. Ordinal differences with magnitude call for Wilcoxon, not the sign test; McNemar is for paired nominal data.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Non-parametric tests are also called _____ tests.
2. Levene's test checks homogeneity of _____; Shapiro-Wilk checks _____.
3. An independent-samples t-test with 25 per group has _____ df.
4. In one-way ANOVA, F = MS between / _____.
5. An IV with 4 levels needs _____ pairwise post hoc comparisons.
6. ANCOVA adds the assumption of homogeneity of _____.
7. Chi-square on a 3 × 4 table has _____ df.
8. The non-parametric equivalent of repeated-measures ANOVA is the _____ test.

Answers: (1) distribution-free, (2) variance; normality, (3) 48, (4) MS within, (5) 6, (6) regression slopes, (7) 6, (8) Friedman.

9.11 Correlation, Regression and Factor Analysis MEDIUM YIELD · 28 QS

Does stress steal sleep? Is a satisfied worker a productive one? Such questions concern quantities that rise and fall together, not the group means of Section 9.10. Correlation measures how strongly two variables go together, regression turns that association into a prediction, and factor analysis asks how many underlying dimensions produced a whole table of correlations.

Correlation: direction, strength and form ●●

A **correlation** is an association between two variables measured on the same people. It has a *direction*, positive when both rise together (study hours and marks) and negative when one rises as the other falls (stress and sleep quality); a *strength*, how closely the scores keep to the pattern; and a *form*, straight or curved. **Sir Francis Galton** (1822 to 1911), studying heredity, described such "co-relations" in 1888, and his protégé **Karl Pearson** (1857 to 1936) gave them an exact formula in 1896: the **Pearson product-moment correlation coefficient**, r. So r measures association; it is not a mean, a variance or a ratio.

Correlational data are graphed as a **scatterplot** (scatter diagram), with one variable on each axis and each person a single dot (Figure 9.20). A rising cloud is a positive correlation, a falling cloud a negative one, and a shapeless cloud no relationship. The last panel is a warning: a clear inverted U gives $r = 0$, because r detects only straight-line relations. Do not confuse the scatterplot with the bell curve or a skewed distribution, which describe a single variable (Section 9.8), or with chi-square, which is a test, not a graph (Section 9.10).

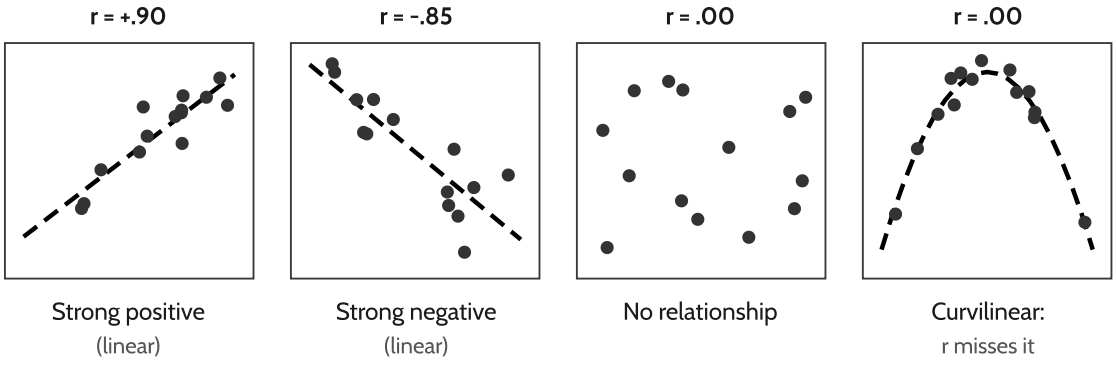


Figure 9.20 Four scatterplots; each dot is one person. The curved relation in the last panel is strong, yet r is zero.

Reading Pearson's r ●●●

Pearson's r runs from -1.00 through 0 to +1.00. Its *sign* gives the direction; its *size*, ignoring the sign, gives the strength. So $r = -0.85$ between stress and sleep quality is a strong negative (inverse) relationship, exactly as strong as +0.85: the more stress, the worse the sleep, and the scores keep closely to that pattern. The range of r is never 0 to 1 (that is the range of r^2 and of multiple R), never -0.04 to +0.04, and never 0 to 100 or -100 to +100. Table 9.32 gives the usual verbal labels; effect-size conventions are in Section 9.9.

Table 9.32 Interpreting the size of r (the sign gives only direction).

Size of r	Usual label	Shared variance (r^2)
.00 to .20	Negligible	0 to 4 per cent
.20 to .40	Low, weak	4 to 16 per cent
.40 to .70	Moderate	16 to 49 per cent
.70 to .90	High, strong	49 to 81 per cent
.90 to 1.00	Very high	81 to 100 per cent

Squaring *r* gives the **coefficient of determination**, the proportion of variance the two variables share. If study hours and marks correlate +0.90, r^2 = 0.81: 81 per cent shared variance, not 90. This is why correlation is related to prediction: the nearer *r* is to plus or minus 1, the better one variable predicts the other, and at 0 it predicts nothing. Correlation does not, however, imply causation. Y may cause X, or a third variable may drive both (ice-cream sales and drownings both follow summer heat). Only a randomised experiment (Section 9.4) licenses a causal claim, so *r* = +0.90 shows a strong positive linear association: it neither proves that X causes Y nor shows that they are causally unrelated.

KEY STUDY: THE HAPPY, PRODUCTIVE WORKER (1985, 2001)

Managers long assumed that satisfied workers perform better. In 1985 **Michelle Iaffaldano** and **Paul Muchinsky** pooled the studies in a meta-analysis (Section 9.5) and found a mean correlation of only about .17. In 2001 **Timothy Judge** and colleagues, pooling more than 300 samples and correcting for measurement error, put it at about .30. Job satisfaction and performance thus show a weak (modest) positive correlation: not zero, not negative, not strong, and silent on which causes which.

! EXAM TRAP

A minus sign never means weak: -0.85 is strong. And *r* = +0.90 is neither "90 per cent of the variance" (it is 81 per cent) nor proof that X causes Y.

Assumptions, limits and testing r ●●

Pearson's *r* is the technique for two continuous (interval or ratio) variables, such as stress level and sleep duration. Chi-square (association between categories), the t-test, analysis of variance (ANOVA) and the Mann-Whitney U test (differences between groups) answer other questions (Section 9.10). Pearson's *r* assumes *linearity*; *homoscedasticity*, an equal scatter of Y at every value of X, whose violation, **heteroscedasticity**, shows as a fan-shaped cloud; *bivariate normality*, for significance tests; and *independence* of cases. Observations that depend on earlier ones, such as daily mood ratings, show **autocorrelation**: a variable correlated with itself over time. Homogeneity of variance is the group version of homoscedasticity, equal variances across groups. *r* is also distorted by **restriction of range** (entrance scores predict grades poorly among admitted students alone, whose scores span a narrow band), by outliers, by curvilinear relations and by pooling unlike groups.

A sample *r* is tested with $t = r\sqrt{(N - 2)} / \sqrt{(1 - r^2)}$ on N - 2 degrees of freedom: *r* = .30 fails with 20 people but passes with 100. To compare two correlations, **Sir Ronald Aylmer Fisher** (1890 to 1962) devised *Fisher's r-to-z transformation*, $z' = 0.5 \ln[(1 + r)/(1 - r)]$, which removes the skew in

the sampling distribution of r. **Fisher's z-test** then divides the difference between the two z' values by its standard error, $\sqrt{[1/(N_1 - 3) + 1/(N_2 - 3)]}$. A t-test compares two means, ANOVA several, and Pearson's r measures a single association; none compares two correlations.

Matching the coefficient to the data ●●

When the data are ranks or categories, another coefficient is used. The key distinction is between a *true dichotomy*, two natural categories (alive or dead), and an *artificial dichotomy*, a continuous variable cut in two (pass or fail from marks). **Charles Spearman** (1863 to 1945) introduced the rank-difference method in 1904. **Spearman's rho**, $\rho = 1 - 6\Sigma D^2 / N(N^2 - 1)$, where D is each person's difference in ranks, correlates ranks; it therefore suits ordinal data and monotonic (steadily rising or falling) but non-linear relations, and it resists outliers. *Kendall's tau*, from **Maurice Kendall** (1938), counts agreeing and disagreeing pairs of ranks and suits small samples with many ties. Table 9.33 gives the rest; agreement between raters uses the intraclass correlation (Chapter 2).

Table 9.33 Choosing a correlation coefficient.

Coefficient	Variable 1	Variable 2	Example or note
Pearson r	Continuous	Continuous	Stress and sleep duration
Spearman rho, Kendall tau	Ranks	Ranks	Two judges' rankings
Point-biserial	True dichotomy	Continuous	Item right or wrong and total score
Biserial	Artificial dichotomy	Continuous	Pass or fail (cut from marks) and IQ
Phi (φ)	True dichotomy	True dichotomy	2 by 2 table
Tetrachoric	Artificial dichotomy	Artificial dichotomy	Both split from continuous scores
Contingency C, Cramér's V	Nominal	Nominal	Built on chi-square
Eta (correlation ratio)	Categories	Continuous	Curvilinear relations

> **! EXAM TRAP**
> "One continuous and one dichotomous variable" means point-biserial unless the dichotomy was made by splitting a continuous variable (biserial). Two continuous variables call for Pearson's r, not Spearman, which is for ranks or monotonic non-linear relations.

Partial, semi-partial and multiple correlation ○

Ordinary r is a zero-order correlation: nothing is held constant. Marital stress may rise with length of marriage partly because long marriages have more children. The **partial correlation**, $r_{12.3}$, is the correlation between two variables after a third variable's influence has been removed statistically from *both*; partialling out the number of children gives the stress and length relation among couples with equally many children. Holding one variable constant gives a first-order partial, two a second-order partial. A **semi-partial correlation** (part correlation) removes the third variable from only *one* of the two, usually the predictor; its square is a predictor's unique contribution in multiple regression, and it is never larger than the partial. The **multiple correlation** (R, from 0 to 1) links one variable with the best weighted combination of several others. Linear (ordinary) correlation, biserial correlation and multiple correlation remove nothing.

> **! EXAM TRAP**
> "Controlling for", "holding constant" or "subtracting out the influence of" a third variable always means partial correlation.

Simple linear regression ●●

Regression also began with Galton. In 1886 he found that children of very tall parents were tall, but less extremely so, and called it "regression towards mediocrity". This **regression to the mean** follows whenever two measures correlate imperfectly: extreme scores on one pair with less extreme scores on the other (a threat to validity, Section 9.2).

Simple linear regression predicts a criterion Y from a predictor X with the line Y = a + bX (Figure 9.21). The *intercept* a is the predicted Y when X is zero; the **regression coefficient** b is the slope, the change in Y for each one-unit rise in X. $b = r(SD_Y/SD_X)$ and $a = M_Y - bM_X$, where SD is the standard deviation and M the mean. If hours and marks correlate .60, the SD of marks is 10 and the SD of hours 2, then b = 3 marks per hour; with means of 5 hours and 50 marks, a = 35, so 8 hours predicts 35 + 24 = 59 marks.

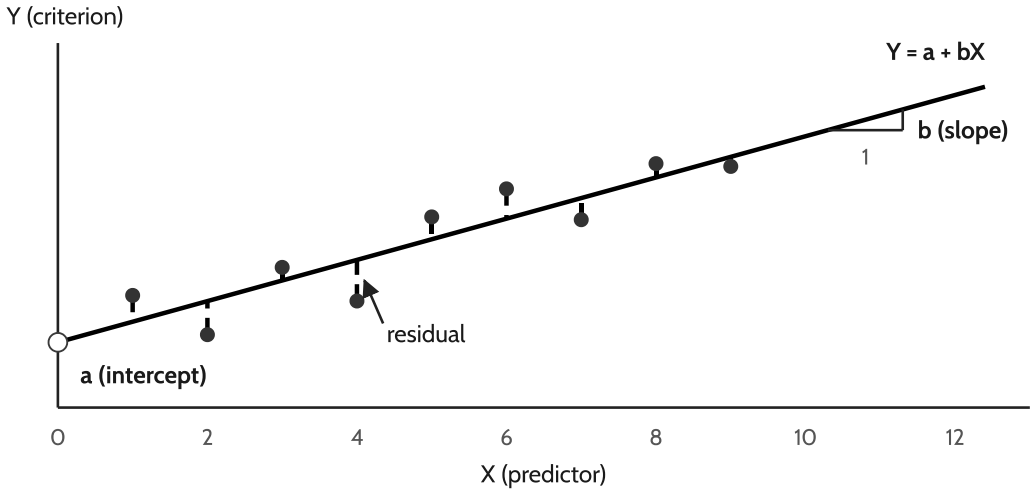


Figure 9.21 A least-squares regression line. The dashed verticals are the residuals whose squared sum the line minimises.

The **method of least squares**, published by **Adrien-Marie Legendre** (1752 to 1833) in 1805 and developed by **Carl Friedrich Gauss** (1777 to 1855), chooses the line that minimises the sum of squared *residuals* (actual minus predicted Y). Their typical size is the **standard error of estimate**, $SD_Y\sqrt{(1 - r^2)}$. Regression assumes linearity, independence of residuals, normality of residuals and equal residual variance (LINE). Correlation is symmetrical and unit-free; regression is directional and gives an equation in real units. A researcher therefore uses regression to predict the value of one variable from another, not to find a mean, count frequencies or estimate sampling error.

> **! EXAM TRAP**
> In Y = a + bX the slope is **b**; a is the intercept, X the predictor and Y the predicted score.

Multiple regression and multicollinearity ●●

Multiple regression uses several predictors, $Y = a + b_1X_1 + b_2X_2 + \ldots$, each b holding the others constant. Standardised **beta weights** (β) let predictors be compared. R^2 is the variance all predictors explain together, and adjusted R^2 corrects its rise with every added predictor. Entry may be simultaneous (all at once), hierarchical (theory-ordered blocks, testing the change in R^2) or stepwise (chosen by the computer, which capitalises on chance).

When predictors correlate highly with each other, the result is **multicollinearity**. Anxiety and neuroticism scores correlating .90 carry the same information, so R^2 stays high but the weights become unstable, their standard errors inflate and signs may flip. A **variance inflation factor** (VIF) above 10 (above 5 on a stricter rule) or a tolerance (1/VIF) below .10 flags it; the cure is to drop or combine predictors.

> **! EXAM TRAP**
>
> Multicollinearity is correlation among predictors, not between a predictor and the outcome. Heteroscedasticity is unequal residual spread, autocorrelation is residuals correlated over time, homogeneity of variance is equal group variances, a categorical outcome calls for logistic regression, and "assumptions are violated" defines nothing.

A dichotomous outcome (relapsed or not) calls for **logistic regression**, which models the log of the odds, estimates weights by maximum likelihood, expresses each as an odds ratio (above 1 raises the odds, below 1 lowers them) and tests it with the Wald test. Regression, the t-test, ANOVA and analysis of covariance (ANCOVA) are all cases of the general linear model: factorial designs are analysed by ANOVA, and ANCOVA blends ANOVA with regression, using the covariate's correlation with the outcome (Section 9.10). Mediation and moderation are tested by regression, following **Reuben Baron** and **David Kenny** (1986), with the Sobel test of **Michael Sobel** (1982) for indirect effects (Section 9.2). *Discriminant analysis* predicts membership of known groups from continuous variables. *Path analysis*, devised by **Sewall Wright** (1889 to 1988) in 1921, chains regressions to test a diagram of directional paths, and *structural equation modelling* (SEM) adds latent variables, joining a confirmatory factor analysis to a path model.

Factor analysis: the dimensions behind correlations

Twenty tests yield 190 correlations, and **factor analysis** asks how few underlying factors could produce them: if verbal tests correlate among themselves and numerical tests among themselves, two factors may explain the whole table. It serves data reduction and construct validation (Chapter 2). Spearman invented it in 1904 to argue for a general factor, g (Chapter 1). **Louis Leon Thurstone** (1887 to 1955) extended it in the 1930s into multiple factor analysis, with the goal of **simple structure**: each variable loading highly on one factor and near zero on the others.

A **factor loading** is a variable's correlation with a factor; about .30 to .40 or more counts. A variable's squared loadings summed across factors give its **communality** (h^2), and $1 - h^2$ is its uniqueness. A factor's squared loadings summed down the column give its **eigenvalue**, the variance it explains in units of one variable. Before factoring, the Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy should be at least .60 and **Bartlett's test of sphericity** significant (the correlation matrix is not an identity matrix), with at least 10 cases per variable and

ideally 300 or more. Of the extraction methods, **principal components analysis** (**Harold Hotelling**, 1933) analyses total variance for data reduction, whereas principal axis factoring and maximum likelihood analyse only common variance, to model latent factors.

Retaining, rotating and confirming factors ●●

How many factors? The **Kaiser-Guttman rule** (**Louis Guttman**, 1916 to 1987, and **Henry Kaiser**) keeps eigenvalues above 1.0, since a factor below 1 explains less than one variable; it tends to keep too many. The **scree test** of **Raymond Cattell** (1905 to 1998), published in 1966, keeps the factors above the elbow of the eigenvalue plot, where the slope flattens into "scree", the rubble below a cliff. **John Horn**'s **parallel analysis** (1965), the most accurate, keeps factors whose eigenvalues beat those of random data. With eigenvalues of 3.5, 1.8, 0.9, 0.7 and 0.5 (Figure 9.22), two exceed 1.0 and two lie above the elbow at factor 3: retain two.

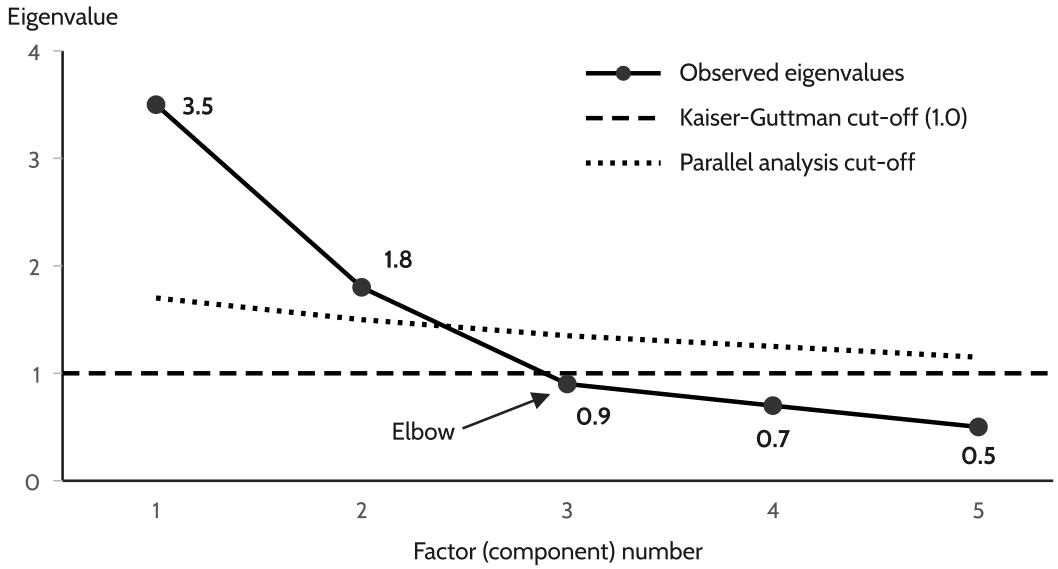


Figure 9.22 Scree plot of the worked example: two eigenvalues lie above both cut-off lines and before the elbow, so every rule retains two factors.

> **! EXAM TRAP**
>
> The elbow factor itself is dropped: an elbow at factor 3 means retain two, not three. Five counts every eigenvalue, and one keeps only the largest.

Factor rotation turns the factor axes towards simple structure without changing the variance explained. An **orthogonal rotation** keeps factors uncorrelated: varimax (Kaiser, 1958) simplifies each factor, quartimax each variable, and equimax compromises. An **oblique rotation** (promax, direct oblimin) lets factors correlate, as psychological constructs usually do. A **second-order factor** is extracted from the correlations among first-order factors (general ability above verbal, numerical and spatial factors, a hierarchical model), so it needs an oblique rotation such as promax; orthogonal factors leave no correlations to factor.

> **! EXAM TRAP**
>
> Second-order factor analysis needs promax (oblique). Varimax, quartimax and equimax are all orthogonal.

All of this is **exploratory factor analysis** (EFA): data-driven, with no prior model. By contrast, **confirmatory factor analysis** (CFA), developed by **Karl Jöreskog** in 1969, tests a prespecified structure, fixing which items load on which factor, and judges fit by the comparative fit index (CFI) and Tucker-Lewis index (TLI), about .95 or above, the root mean square error of approximation (RMSEA), about .06 or below, and the standardised root mean square residual (SRMR), .08 or below, with modification indices suggesting changes. It needs no rotation, takes continuous or categorical items, and is the measurement model of SEM. *Cluster analysis* groups people rather than variables, and *multidimensional scaling* maps judged similarities between objects in as few dimensions as possible.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Correlational data are graphed as a _____.
2. If r = +0.90, the shared variance is _____ per cent.
3. The significance of r is tested on _____ degrees of freedom.
4. A continuous and an artificially dichotomised variable call for the _____ correlation.
5. Removing a third variable's influence from both variables gives the _____ correlation.
6. In Y = a + bX, the slope is _____.
7. Highly intercorrelated predictors show _____.
8. Second-order factors need an _____ rotation such as promax.

Answers: (1) scatterplot, (2) 81, (3) N - 2, (4) biserial, (5) partial, (6) b, (7) multicollinearity, (8) oblique.

The Case of Clever Hans, Revisited

Pfungst's investigation is a compact course in this chapter. He turned a marvel into a research problem with rival, falsifiable hypotheses and preferred the simplest explanation that fitted, as Lloyd Morgan's canon advises (9.1). He varied one condition at a time while holding the others constant, and exposed the questioner's expectation as a hidden variable that undermined every earlier demonstration (9.2). His study of one horse was a single-case experiment (9.4); he observed systematically (9.6) and honestly reported a result that deflated a celebrity (9.7). His verdict rested on counts of right and wrong answers compared across conditions (9.8 to 9.10). And the blinding of modern clinical trials (9.5), in which those who assess outcomes do not know who was treated, exists so that no one plays von Osten again.

Chapter summary

9.1 Science and the Research Process: Problems and Hypotheses. Science differs from tenacity, authority and the a priori method (Peirce, 1877) because it fixes belief by public, repeatable observation. Scientific research is empirical, objective, replicable, parsimonious (Occam's razor, Lloyd Morgan's canon), falsifiable (Popper, 1934) and tentative; Kuhn (1962) described paradigm shifts. **Induction** builds theory from observations (theory development, qualitative research); **deduction** tests theory against data; abduction infers the best explanation. Methodology is wider than methods. Research is basic or applied (including Lewin's action research), exploratory, descriptive or explanatory, nomothetic or idiographic (Allport). A clear problem and a literature review (which provides a theoretical framework) yield hypotheses: null or alternative, directional or non-directional, universal or existential.

9.2 Variables, Experimental Control and the Validity of Research. A **variable** takes two or more values; the **independent variable** is the presumed cause (manipulated or selected) and the **dependent variable** the measured outcome. A **confound** varies with the IV and affects the DV; a **moderator** changes the strength of an effect and a **mediator** carries it (Baron and Kenny, 1986). **Control** follows Kerlinger's MAXMINCON principle (maximise primary, minimise error, control secondary variance), above all through **random assignment.** Hawthorne, demand (Orne), expectancy and Pygmalion (Rosenthal) and halo (Thorndike) effects distort results; **double-blinding** controls expectancy on both sides. Studies are judged by statistical conclusion, internal, construct and external validity (Campbell and Stanley, 1963; Cook and Campbell, 1979).

9.3 Sampling: Populations, Probability and Non-Probability Methods. A **parameter** describes a population and a **statistic** a sample. Following Blalock (1960), sampling methods are probability, non-probability or mixed. In **probability sampling** every element has a known, non-zero chance of selection, so sampling error can be calculated and the sample is likely, though not guaranteed, to be representative: simple random, systematic (every kth after a random start), stratified (random sampling within every stratum) and cluster sampling (random choice of whole natural groups, multistage when repeated down levels). In **non-probability sampling** (convenience, purposive, quota, snowball, saturation, dense) the chance is unknown. Sampling error shrinks as n rises; bias, as in the 1936 *Literary Digest* poll, does not.

9.4 Research Designs: Experimental, Single-Case and Non-Experimental. Only an **experiment**, with manipulation, control and random assignment that makes the groups equivalent, licenses cause-and-effect conclusions; Fisher's (1935) principles are replication, randomisation and local control. Campbell and Stanley (1963) classed designs as pre-experimental, true experimental (including Solomon's four-group design, which controls pretest sensitisation) and quasi-experimental (intact groups, no random assignment). Experiments may be between-subjects, within-subjects (order effects controlled by counterbalancing and Latin squares), mixed or crossover; factorial designs reveal interactions. Single-case designs (A-B-A-B, multiple baseline) test a treatment in one person. Correlational, survey (a census is a survey) and ex post facto designs cannot establish cause; longitudinal designs suffer attrition, cross-sectional designs cohort effects.

9.5 Clinical Epidemiology, Randomised Trials and Evidence Synthesis. Epidemiology counts disorder as **prevalence** (existing cases: point, period, lifetime) or **incidence** (new cases); prevalence is roughly incidence times duration. Cross-sectional studies give prevalence, **case-control** studies

(best for rare outcomes) give odds ratios, and **cohort** studies, such as Doll and Hill's prospective British Doctors Study, give relative risk; a ratio of 1 means no association, and NNT = 1/ARR. Sensitivity and specificity belong to the test, but predictive values depend on the base rate (Meehl and Rosen, 1955). Trials rely on randomisation, allocation concealment, blinding and **intention-to-treat** analysis. Meta-analysis (Glass, 1976) pools many studies; CONSORT reports trials and PRISMA systematic reviews.

9.6 Methods of Data Collection and Qualitative Research. Data are **primary** or **secondary**. Observation is participant or non-participant, structured or unstructured, naturalistic or controlled, overt or covert. Interviews run from structured to unstructured, with open-ended questions, probes and Spradley's **ethnographic inquiry**. A questionnaire is self-answered, a **schedule** filled in by the enumerator; items must not be **double-barrelled**, leading or loaded. Berelson's content analysis, Moreno's **sociometry** and Dilthey's Verstehen follow. Guba and Lincoln's paradigms run from positivism (naive realism) to constructivism (relativism). Qualitative research is naturalistic, inductive and emic; ethnography, grounded theory (open, **axial**, selective coding) and phenomenology lead its traditions, judged by trustworthiness and strengthened by **triangulation**.

9.7 Research Ethics and Writing the Research Report. Research ethics grew out of abuse: the Nazi experiments produced the **Nuremberg Code** (1947), and the **Tuskegee** study (1932 to 1972) led to the **Belmont Report** (1979: respect for persons, beneficence, justice). The **APA Ethics Code** (2002) has five general principles, and in India the **ICMR** guidelines (2017) require ethics committee approval. **Informed consent** tells participants enough to decide voluntarily before taking part; **debriefing** gives full information afterwards (dehoaxing and desensitising). Confidentiality differs from anonymity, deception needs strict justification, and animal research follows the **3Rs**. Misconduct means fabrication, falsification and plagiarism; citation is not plagiarism. A research report communicates findings, and its Discussion, not its Results, interprets them.

9.8 Measurement Scales, Organising Data and Descriptive Statistics. S. S. Stevens (1946) gave four levels of measurement: nominal (unordered categories), ordinal (ranks), interval (equal intervals, arbitrary zero, as in Celsius) and ratio (true zero, as in height). Raw data are edited, coded, classified and tabulated, analysed in SPSS or Excel, and shown as frequency distributions and graphs. The mean suits symmetrical data, the median skewed data and the mode nominal data; the SD, the square root of the variance, measures spread. The normal curve holds about 68, 95 and 99.7 per cent of cases within 1, 2 and 3 SD, and **z = (X − M)/SD** converts any score into a percentile or another standard score.

9.9 Sampling Distributions and the Logic of Hypothesis Testing. Inferential statistics use sample data and the laws of chance to judge whether a result could be due to chance. The **sampling distribution of the mean** centres on the population mean, its SD is the **standard error** ($\sigma/\sqrt{N}$), and by the **central limit theorem** it is near normal once N is about 30. A 95 percent CI is M ± 1.96 SE. A test assumes H0 and rejects it when $p \le \alpha$. A **Type I error** rejects a true H0 (α); a **Type II error** retains a false one (β). **Power** is $1 - \beta$; Cohen's d of 0.2, 0.5 and 0.8 is small, medium and large.

9.10 Comparing Groups: t-Tests, ANOVA and Non-Parametric Tests. Parametric tests (z, t, F) assume normality, equal variances, independence and interval or ratio data; non-parametric or distribution-free tests (Siegel, 1956) handle ranks, categories and small or skewed samples. Gosset's t tests one or two means (df N - 1, N1 + N2 - 2 or pairs - 1). Fisher's ANOVA splits variance into between and within parts, **F = MS between / MS within**, about 1 under H0; a significant F is followed

by post hoc tests or a Bonferroni correction. Factorial ANOVA detects interactions; ANCOVA adds homogeneity of regression slopes. Chi-square handles frequencies, with df (r - 1)(c - 1), and rank tests partner each parametric test.

9.11 Correlation, Regression and Factor Analysis. Correlation (Pearson's *r*, 1896) measures the direction (sign) and strength (size) of a linear association from -1 to +1. It is graphed as a scatterplot; *r* squared is shared variance (+0.90 means 81 per cent), and correlation aids prediction but never proves causation. Spearman's rho, point-biserial, biserial, phi and tetrachoric suit other data; partial correlation removes a third variable from both. **Regression** (Y = a + bX) predicts one variable from another; multiple regression is undermined by multicollinearity among predictors. **Factor analysis** (Spearman, 1904) reduces correlated variables to a few factors, retained by eigenvalues above 1 or the scree elbow and rotated orthogonally (varimax) or obliquely (promax).

Who's who

Table 9.34 The people in this chapter.

Name	When	Examined for
Charles Sanders Peirce	1839 to 1914	Four methods of fixing belief (1877): tenacity, authority, a priori, science; named abduction
Fred N. Kerlinger	1910 to 1991	Foundations of Behavioral Research (1964); definitions of scientific research, problem, hypothesis and theory
Socrates	about 470 to 399 BCE	The elenchus, the Socratic method of question-and-answer cross-examination
Aristotle	384 to 322 BCE	Knowledge through observation and classification
René Descartes	1596 to 1650	Rationalism: knowledge through reason and intuition
Galileo Galilei	1564 to 1642	Knowledge through experimentation
Francis Bacon	1561 to 1626	Inductive method; Novum Organum (1620)
William of Ockham	about 1287 to 1347	Occam's razor (parsimony)
Conwy Lloyd Morgan	1852 to 1936	Lloyd Morgan's canon (1894): never explain behaviour by a higher faculty if a lower one will do
Karl Popper	1902 to 1994	Falsifiability as the mark of science; Logik der Forschung (1934)
A. K. Singh	Indian textbook (several editions)	Tests, Measurements and Research Methods in Behavioural Sciences: eight steps of research, assumptions of science, attitudes of scientists, types of hypotheses and reviews
Thomas S. Kuhn	1922 to 1996	Paradigms, normal science and scientific revolutions (1962)
David Hume	1711 to 1776	The problem of induction
Clark L. Hull	1884 to 1952	Hypothetico-deductive theory of learning; Principles of Behavior (1943)
C. R. Kothari	2004 (2nd revised edition)	Research Methodology: Methods and Techniques; methodology wider than methods; types of research; criteria of good research; research process
Kurt Lewin	1890 to 1947	Father of modern social psychology; founded action research (1946)
Michael Scriven	1967	Formative versus summative evaluation
Wilhelm Windelband	1848 to 1915	Nomothetic versus idiographic sciences (1894)
Gordon W. Allport	1897 to 1967	Brought nomothetic and idiographic into personality psychology (1937); preferred the idiographic approach
Max Wertheimer	1880 to 1943	Phi phenomenon: an isolated unexplained fact that became a research problem and led to Gestalt psychology
Pauline V. Young	1956	Sources of research problems: documentary sources, key informants, library sources
Baker and Keller	cited by A. K. Singh	Sources of research problems: programmed activities, research proposals, government policies, social values

Name	When	Examined for
William J. Goode	1952 (with Hatt)	Methods in Social Research: criteria of good hypotheses and three difficulties in formulating them
Paul K. Hatt	1952 (with Goode)	Methods in Social Research: criteria of good hypotheses and difficulties in formulating them
Walter L. Wallace	1971	The wheel of science: the research cycle of deduction and induction
Ivan Petrovich Pavlov	1849 to 1936	Example of a scientific fact: dogs salivating before feeding (conditioning, Chapter 1)
Aaron T. Beck	1921 to 2021	Cognitive theory of depression, cited as an example of a theory
Ernst Heinrich Weber	1795 to 1878	Weber's law, cited as an example of a scientific law
Richard Atkinson and Richard Shiffrin	1968	Multi-store memory model, cited as an example of a model
Wilhelm von Osten	1904	Owner and trainer of Clever Hans (chapter opening)
Carl Stumpf	1848 to 1936	Sat on the 1904 commission that examined Clever Hans; Pfungst's teacher
Oskar Pfungst	1907	Showed that Clever Hans responded to questioners' unwitting cues; a classic demonstration of experimenter expectancy
Robert S. Woodworth	1869 to 1962	S-O-R scheme: stimulus, organismic and response variables
Reuben M. Baron	1986	Moderator versus mediator distinction (with David Kenny)
David A. Kenny	1986	Moderator versus mediator distinction (with Reuben Baron)
Percy Williams Bridgman	1882 to 1961	Operationism; The Logic of Modern Physics (1927)
Stanley Smith Stevens	1906 to 1973	Brought operational definition into psychology (1935)
Elton Mayo	1880 to 1949	Associated with the Hawthorne studies (1924 to 1932)
Fritz J. Roethlisberger	1898 to 1974	Reported the Hawthorne studies in Management and the Worker (1939, with Dickson)
William J. Dickson	1939	Co-author of Management and the Worker (Hawthorne studies)
Henry A. Landsberger	1958	Coined the term Hawthorne effect
Martin T. Orne	1927 to 2000	Demand characteristics (1962)
Stephen J. Weber	1972	Participant roles: good, faithful, negativistic and apprehensive subjects (with Cook)
Thomas D. Cook	1972; 1979	Participant roles (with Weber, 1972); four types of validity and added threats to internal validity (with Campbell, 1979)
Milton J. Rosenberg	1965	Evaluation apprehension
Robert Rosenthal	born 1933	Experimenter expectancy effect (1966); Pygmalion effect (with Jacobson, 1968)
Lenore Jacobson	1968	Pygmalion in the Classroom (with Rosenthal)
Edward Lee Thorndike	1874 to 1949	Halo effect (1920)
Donald T. Campbell	1916 to 1996	Internal versus external validity (1957); their threats (with Stanley, 1963; with Cook, 1979)
Julian C. Stanley	1918 to 2005	Campbell and Stanley (1963): eight threats to internal validity, four to external validity
Glenn H. Bracht	1968	Split external validity into population and ecological validity (with Gene Glass)
Gene V Glass	1968	Split external validity into population and ecological validity (with Glenn Bracht)
William R. Shadish	2002	Shadish, Cook and Campbell: the four types of validity in current form
Alan Bryman	2001	Social Research Methods: reliability, replicability and validity of a study
Hubert M. Blalock	1926 to 1991	Social Statistics (1960); classification of sampling into probability, non-probability and mixed methods
Jerzy Neyman	1894 to 1981	Neyman (optimum) allocation in stratified random sampling by stratum size and variability (1934)
Claus Moser	1922 to 2015	Associated with multistage sampling in area (cluster) surveys
Everet Franklin Lindquist	1901 to 1978	Definition of sampling error (1940): the difference between a parameter and its estimate from a sample
George Horace Gallup	1901 to 1984	Correctly forecast the 1936 US presidential election with a far smaller sample than the Literary Digest

Name	When	Examined for
John Stuart Mill	1806 to 1873	A System of Logic (1843): methods of agreement and difference, the logic of causal inference
Ronald Aylmer Fisher	1890 to 1962	The Design of Experiments (1935): replication, randomisation and local control
Donald Thomas Campbell	1916 to 1996	With Stanley (1963): R, X, O notation; pre-experimental, true and quasi-experimental designs; regression discontinuity with Thistlethwaite (1960)
Richard L. Solomon	1918 to 1995	Solomon four-group design (1949), which controls pretest sensitisation
Donald Thistlethwaite	1960	Regression discontinuity design, with Campbell
Leon Festinger	1953	Definition of the laboratory experiment (with Daniel Katz)
Daniel Katz	1953	Field studies: exploratory versus hypothesis-testing
Irving Piliavin	1969	Subway field experiment on helping a collapsed passenger (with colleagues)
Burrhus Frederic Skinner	1904 to 1990	Experimental analysis of behaviour, the origin of single-subject designs
Murray Sidman	1960	Tactics of Scientific Research: group averages can mislead; generality comes from replication
Alan E. Kazdin	1982	Single-Case Research Designs: systematised single-case designs for clinical work
Jacob Moreno	1934	Sociometry: mapping choices and relations within a group
Sigmund Freud	1909	Little Hans, a classic case study
Émile Durkheim	1897	Suicide: archival research built from official statistics
K. Warner Schaie	1965	Sequential developmental designs (several cohorts tested on several occasions, the commonest being cross-sequential) to tell age changes from cohort differences
Lewis Madison Terman	1877 to 1956	Genetic Studies of Genius (from 1921), the classic longitudinal study of gifted children
William S. Robinson	1950	Demonstrated the ecological fallacy with US state-level census data
Richard Doll	1912 to 2005	British Doctors Study of smoking (prospective cohort, 1951 to 2001), with Bradford Hill
Austin Bradford Hill	1897 to 1991	British Doctors Study of smoking (from 1951), with Richard Doll
Paul Everett Meehl	1920 to 2003	With Albert Rosen (1955): base rates limit the predictive value of test signs and cutting scores
Albert Rosen	1955	With Meehl: base rates (antecedent probability) and the efficiency of psychometric signs and cutting scores
Hans Eysenck	1916 to 1997	1952 claim that neurotic patients improve as often without psychotherapy; critic of meta-analysis
Mary Lee Smith	1977	With Glass, pooled 375 controlled psychotherapy studies: average effect size 0.68
James P. Spradley	1933 to 1982	The Ethnographic Interview (1979): the informant explains their own cultural world; phases of participant observation (1980): descriptive, focused, selective
Bernard Berelson	1912 to 1979	Content analysis (1952): objective, systematic, quantitative description of manifest content; method of universe, unit of analysis, quantification (1954 handbook chapter)
Jacob Levy Moreno	1889 to 1974	Sociometry and the sociogram (Who Shall Survive?, 1934): stars, isolates, mutual choices; founder of psychodrama
Wilhelm Dilthey	1833 to 1911	Verstehen (understanding) versus explanation in the human sciences, 1883
Max Weber	1864 to 1920	Made Verstehen the method of interpretive sociology
William Stephenson	1935	Q-methodology: factor analysis of persons through Q-sorts
Rensis Likert	1903 to 1981	Summated rating attitude scale (taught in Chapter 2)
Louis Leon Thurstone	1887 to 1955	Equal-appearing interval attitude scale (taught in Chapter 2)
Louis Guttman	1916 to 1987	Cumulative attitude scale (taught in Chapter 2)
Egon Guba	1924 to 2008	Research paradigms with Lincoln (1994); trustworthiness criteria (Naturalistic Inquiry, 1985)
Yvonna Lincoln	1985 and 1994	Paradigms and trustworthiness with Guba; historical moments of qualitative research with Denzin (2005)

Name	When	Examined for
Herbert Blumer	1900 to 1987	Coined symbolic interactionism (1937); Symbolic Interactionism (1969)
George Herbert Mead	1863 to 1931	Ideas on meaning and social interaction behind symbolic interactionism
Norman Denzin	1970	Four types of triangulation (The Research Act): data, investigator, theory, methodological; historical moments with Lincoln (2005)
Michael Quinn Patton	2002	Twelve themes of qualitative inquiry under design, fieldwork and analysis
Kenneth Pike	1912 to 2000	Coined emic and etic (1954)
Clifford Geertz	1926 to 2006	Thick description (The Interpretation of Cultures, 1973)
Bronislaw Malinowski	1884 to 1942	Trobriand Islands fieldwork (1915 to 1918) and Argonauts of the Western Pacific (1922): participant observation, the native's point of view
Barney Glaser	1930 to 2022	Grounded theory with Strauss (The Discovery of Grounded Theory, 1967)
Anselm Strauss	1916 to 1996	Grounded theory (1967); open, axial and selective coding with Corbin (1990)
Juliet Corbin	1990	Basics of Qualitative Research with Strauss: open, axial, selective coding
Edmund Husserl	1859 to 1938	Founder of phenomenology; bracketing (epoche); descriptive phenomenology, the detached approach (called direct in one source)
Martin Heidegger	1889 to 1976	Hermeneutic (interpretive) phenomenology (Being and Time, 1927); the involved approach (called indirect in one source)
Jonathan Smith	1996	Interpretative phenomenological analysis (IPA)
Robert Yin	1984	Case study research
Harold Garfinkel	1917 to 2011	Ethnomethodology (Studies in Ethnomethodology, 1967); breaching experiments
Virginia Braun	2006	Thematic analysis in six phases, with Clarke
Victoria Clarke	2006	Thematic analysis in six phases, with Braun
Jane Ritchie	1994	Framework analysis, with Spencer
Liz Spencer	1994	Framework analysis, with Ritchie
John Heron	Late 20th century	Co-operative inquiry: participants as co-researchers and co-subjects
Matthew Miles	1994	Qualitative analysis with Huberman: data reduction, data display, conclusion drawing and verification
A. Michael Huberman	1994	Qualitative analysis with Miles: data reduction, data display, conclusion drawing and verification
John Creswell	2018	Three core mixed-methods designs (with Vicki Plano Clark): convergent, explanatory sequential, exploratory sequential
John B. Watson	1878 to 1958	Little Albert study (1920) with Rosalie Rayner: conditioned fear in an infant, an ethically debated case
Rosalie Rayner	1898 to 1935	Co-author of the Little Albert study (1920)
Philip Zimbardo	1933 to 2024	Stanford prison study (1971), stopped after six days
Stanley Milgram	1933 to 1984	Obedience study (1963): deception and stress; 26 of 40 (65 per cent) went to 450 volts
Diana Baumrind	1927 to 2018	Ethical critique of Milgram's obedience study (1964)
David S. Holmes	1976	Named the two parts of debriefing: dehoaxing and desensitising
William Russell	1959	The 3Rs of animal research (replacement, reduction, refinement), with Rex Burch
Rex Burch	1959	The 3Rs of animal research, with William Russell
Frederic Lord	1912 to 2000	On the statistical treatment of football numbers (1953), a critique of Stevens's permissible statistics
John Wilder Tukey	1915 to 2000	Exploratory Data Analysis (1977): stem-and-leaf display and box plot
Friedrich Wilhelm Bessel	1784 to 1846	Bessel's correction: dividing by N − 1 for the unbiased sample variance
Karl Pearson	1857 to 1936	Coefficient of skewness, 3(Mean − Median)/SD
Abraham de Moivre	1667 to 1754	First derived the normal curve (1733)
Pierre-Simon Laplace	1749 to 1827	Developed the normal curve
Carl Friedrich Gauss	1777 to 1855	Normal curve of error (1809), hence Gaussian; the keyed propounder of the normal curve
Adolphe Quetelet	1796 to 1874	Applied the normal curve to human characteristics (the average man)

Name	When	Examined for
Francis Galton	1822 to 1911	Applied the normal curve to human ability
Siméon Denis Poisson	1781 to 1840	Poisson distribution of rare events; distractor for the normal curve
Jacob (James) Bernoulli	Ars Conjectandi published 1713	Law of large numbers and Bernoulli trials; distractor for the normal curve
Wilhelm Reich	1897 to 1957	Psychoanalyst; distractor in a normal-curve question
David Wechsler	1896 to 1981	Deviation IQ with mean 100 and SD 15
Muriel Bristol	1920s	The lady tasting tea at Rothamsted; Fisher's opening example of a significance test
Jacob Bernoulli	1655 to 1705	Law of large numbers (published 1713)
Egon Pearson	1895 to 1980	With Jerzy Neyman: Type I and Type II errors, beta and power
Jacob Cohen	1923 to 1998	Power analysis and effect-size benchmarks (d = 0.2, 0.5, 0.8; r = .10, .30, .50); Statistical Power Analysis for the Behavioral Sciences (2nd edition, 1988)
Brian Nosek	2015	Coordinated the Open Science Collaboration's replication of 100 psychology studies (36 percent replicated)
William Sealy Gosset	1876 to 1937	The t-test, published as 'Student' in 1908 while at the Guinness brewery
George Waddel Snedecor	1881 to 1974	Named the F ratio in Fisher's honour
Sidney Siegel	1916 to 1961	Nonparametric Statistics for the Behavioral Sciences (1956); power-efficiency
Howard Levene	1914 to 2003	Levene's test of homogeneity of variance (1960)
Henry Scheffé	1907 to 1977	Scheffé's test, the most conservative post hoc test
Carlo Emilio Bonferroni	1892 to 1960	Bonferroni correction: alpha divided by the number of comparisons
Frank Yates	1902 to 1994	Correction for continuity in 2 x 2 chi-square tables (1934)
Quinn McNemar	1900 to 1986	McNemar test for paired nominal data (1947)
Henry Berthold Mann	1905 to 2000	Mann-Whitney U test for two independent samples (1947)
Donald Ransom Whitney	1915 to 2007	Mann-Whitney U test for two independent samples (1947)
Frank Wilcoxon	1892 to 1965	Wilcoxon matched-pairs signed-rank test (1945); rank-sum test
William Kruskal	1919 to 2005	Kruskal-Wallis H test, one-way ANOVA by ranks (1952)
W. Allen Wallis	1912 to 1998	Kruskal-Wallis H test, one-way ANOVA by ranks (1952)
Milton Friedman	1912 to 2006	Friedman test for three or more related samples, two-way ANOVA by ranks (1937)
Sir Francis Galton	1822 to 1911	Described co-relations (1888); regression towards mediocrity, now regression to the mean (1886)
Michelle Iaffaldano	1985	With Muchinsky, meta-analysis: job satisfaction and performance correlate only about .17
Paul Muchinsky	1985	With Iaffaldano, meta-analysis of job satisfaction and performance
Timothy Judge	2001	With colleagues, put the satisfaction and performance correlation at about .30
Sir Ronald Aylmer Fisher	1890 to 1962	r-to-z transformation; Fisher's z-test for the difference between two correlations
Charles Spearman	1863 to 1945	Rank-difference correlation (rho) and factor analysis, both 1904
Maurice Kendall	1938	Kendall's tau, a rank correlation based on agreeing and disagreeing pairs
Adrien-Marie Legendre	1752 to 1833	Published the method of least squares (1805)
Reuben Baron	1986	With Kenny, regression tests of mediation and moderation
David Kenny	1986	With Baron, regression tests of mediation and moderation
Michael Sobel	1982	Sobel test of an indirect (mediated) effect
Sewall Wright	1889 to 1988	Path analysis (1921)
Harold Hotelling	1933	Principal components analysis
Henry Kaiser	1958	Varimax rotation; the eigenvalue-above-1 rule; KMO measure
Raymond Cattell	1905 to 1998	Scree test for the number of factors (1966)
John Horn	1965	Parallel analysis for the number of factors
Karl Jöreskog	1969	Confirmatory factor analysis

Key terms

method of tenacity, method of authority, a priori method, elenchus, Empiricism, Replicability, Parsimony, Falsifiability, paradigm, Induction, Deduction, Abduction, hypothetico-deductive method, Research methods, Research methodology, Basic research, applied research, Action research, Exploratory research, Descriptive research, Explanatory research, nomothetic approach, idiographic approach, research problem, unsolvable problem, literature, review of literature, grey literature, hypothesis, null hypothesis, alternative hypothesis, directional hypothesis, non-directional hypothesis, universal hypothesis, existential hypothesis, variable, construct, independent variable, dependent variable, active variables, attribute variables, extraneous variable, confounding variable, moderator, mediator, intervening variable, operational definition, floor effect, ceiling effect, Control, MAXMINCON principle, order effects, Counterbalancing, randomisation, Statistical control, Bias, Hawthorne effect, Demand characteristics, evaluation apprehension, experimenter expectancy effect, Pygmalion effect, halo effect, Placebo effect, double-blind, Statistical conclusion validity, Internal validity, Construct validity, External validity, ecological validity, threat to internal validity, population, target population, accessible population, element, sampling unit, sampling frame, sample, census, parameter, statistic, Probability sampling, non-probability sampling, Mixed sampling, Simple random sampling, sampling with replacement, Systematic sampling, periodicity, Stratified random sampling, proportionate stratified sampling, disproportionate stratified sampling, Optimum allocation, Cluster sampling, multistage sampling, multiphase sampling, panel sampling, Convenience sampling, purposive sampling, Quota sampling, snowball sampling, saturation sampling, dense sampling, Sampling error, Non-sampling error, sampling bias, experiment, random assignment, control group, method of difference, replication, local control, pretest sensitisation, Solomon four-group design, quasi-experiment, between-subjects design, randomised block design, repeated-measures design, Latin square, crossover design, factorial design, main effect, interaction effect, field experiment, field study, single-subject design, reversal design, multiple-baseline design, correlational research, third-variable problem, survey research, ex post facto research, external criticism, internal criticism, longitudinal design, cross-sectional design, cohort effect, cross-sequential design, epidemiology, prevalence, point prevalence, incidence, morbidity rate, mortality rate, ecological fallacy, case-control study, cohort study, prospective cohort, retrospective cohort, relative risk, odds ratio, attributable risk, absolute risk reduction, number needed to treat, sensitivity, specificity, positive predictive value, base rate, receiver operating characteristic (ROC) curve, likelihood ratios, randomised controlled trial, allocation concealment, intention-to-treat analysis, per-protocol analysis, analogue research, systematic review, meta-analysis, publication bias, file-drawer problem, hierarchy of evidence, CONSORT, PRISMA, Primary data, Secondary data, participant observation, reactivity, observer bias, structured interview, focus group, open-ended question, ethnographic interview, questionnaire, schedule, Double-barrelled question, funnel sequence, Content analysis, Manifest content, Sociometry, sociogram, Verstehen, ontology, epistemology, Positivism, Postpositivism, Constructivism, Qualitative research, symbolic interactionism, Ethnography, emic, thick description, Grounded theory, axial coding, Phenomenology, bracketing, trustworthiness, Triangulation, Mixed-methods research, Nuremberg Code, Declaration of Helsinki, Belmont Report, APA Ethics Code, ICMR National Ethical Guidelines, institutional ethics committee, institutional review board, Informed consent, consent form, assent, Voluntary participation, right to withdraw, Protection from harm, risk-benefit analysis, confidentiality, Anonymity, Vulnerable populations, Deception, debriefing, Dehoaxing, desensitising, 3Rs, Research misconduct, fabrication, falsification, plagiarism, citation, Referencing, research report, References, bibliography, research proposal, Assumptions, Limitations, Delimitations, levels of measurement, nominal scale, ordinal scale,

interval scale, ratio scale, categorical variable, discrete variable, continuous variable, Editing, Coding, Classification, Tabulation, frequency distribution, exact limits, Cumulative frequency, Relative frequencies, bar graph, histogram, frequency polygon, ogive, scatter plot, measure of central tendency, mean, median, mode, quartile deviation, variance, standard deviation, coefficient of variation, Skewness, Kurtosis, normal probability curve, z-score, percentile rank, standard scores, Descriptive statistics, Inferential statistics, biostatistics, Classical probability, Empirical probability, addition rule, multiplication rule, conditional probability, binomial distribution, sampling distribution of the mean, standard error of the mean, central limit theorem, law of large numbers, point estimate, confidence interval, significance level, critical value, two-tailed test, one-tailed test, degrees of freedom, p-value, q-value, Type I error, Type II error, Power, Power analysis, effect size, Cohen's d, Eta squared, partial eta squared, clinical significance, p-hacking, replication crisis, parametric test, non-parametric test, power-efficiency, Shapiro-Wilk test, homoscedasticity, Levene's test, sphericity, t distribution, one-sample t-test, independent-samples t-test, Welch's t-test, paired-samples t-test, family-wise error rate, Analysis of variance, one-way ANOVA, mean square, F ratio, F distribution, Post hoc tests, Planned contrasts, Bonferroni correction, factorial ANOVA, Repeated-measures ANOVA, Analysis of covariance, homogeneity of regression slopes, Multivariate analysis of variance, chi-square test, test of independence, McNemar test, Mann-Whitney U test, Wilcoxon matched-pairs signed-rank test, Kruskal-Wallis H test, Friedman test, correlation, Pearson product-moment correlation coefficient, scatterplot, coefficient of determination, heteroscedasticity, autocorrelation, restriction of range, Fisher's z-test, Spearman's rho, partial correlation, semi-partial correlation, multiple correlation, regression to the mean, regression coefficient, method of least squares, standard error of estimate, beta weights, multicollinearity, variance inflation factor, logistic regression, factor analysis, simple structure, factor loading, communality, eigenvalue, Bartlett's test of sphericity, principal components analysis, Kaiser-Guttman rule, scree test, parallel analysis, orthogonal rotation, oblique rotation, second-order factor, exploratory factor analysis, confirmatory factor analysis

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From the past papers

1. 'Reinforcement will have positive effect on performance,' which type of hypothesis is this? (RML 2024)

   (a) Null and directional (b) Alternate and directional (c) Null and non-directional (d) Alternate and non-directional

   Answer: (b) Alternate and directional. The statement predicts an effect, so it is the alternative hypothesis (the null would predict no effect), and it says which way the effect goes (positive), so it is directional and would be tested one-tailed.

2. Nomothetic approach of study is involved in (JEMAS 2025)

   (a) Qualitative method (b) Experimental method (c) Quasi-experimental method (d) Ethnography

   Answer: (b) Experimental method. The nomothetic approach seeks general laws across many people, and the experiment is its purest method. Quasi-experiments are nomothetic too but are the weaker choice; qualitative methods and ethnography study particular cases in depth and are idiographic.

3. In which method does the researcher start with a theory and then test it with observations? (RML 2023)

   (a) Inductive reasoning (b) Deductive reasoning (c) Abductive reasoning (d) Hypothetico-deductive method

   Answer: (b) Deductive reasoning. Deduction runs top-down from theory to hypothesis to observation. Induction runs the other way (observations to theory), abduction infers the best explanation for a surprising fact, and the hypothetico-deductive method is the whole cycle of conjecture and test, not the direction of reasoning asked for.

4. Maximization of ____ variance, minimization of ____ variance, control of ____ variance. Select the appropriate option. (JEMAS 2025)

   (a) secondary, true, error (b) true, primary, secondary (c) primary, error, secondary (d) primary, true, error

   Answer: (c) primary, error, secondary. This is Kerlinger's MAXMINCON principle: maximise primary (experimental) variance due to the IV, minimise error variance, control secondary (extraneous) variance. True variance belongs to test theory, not to this principle.

5. Which of the following can help control experimenter's bias? (RML 2022)

   (a) Double-blinding (b) Randomly assigning subjects to groups (c) Using control groups (d) Demand characteristics

   Answer: (a) Double-blinding. When the experimenters who treat and assess participants do not know who is in which condition, their expectations cannot shape the results. Random assignment equalises groups and a control group gives a comparison, but the experimenter still knows who got what; demand characteristics are a source of bias, not a control.

6. A researcher is interested in the impact of anxiety on performance. To manipulate anxiety, some subjects were given plain chips to eat while others were given barbecue flavoured chips to eat. The most obvious problem with this experiment is a lack of: (RML 2022)

   (a) Statistical Validity (b) Empirical Validity (c) Internal Validity (d) Construct Validity

   Answer: (d) Construct Validity. Chip flavour does not manipulate anxiety, so the operation does not represent the construct named. No rival variable explains the result (internal validity), and statistical or empirical validity (an old name for a test's criterion-related validity) do not name this flaw.

7. A researcher wants to study the prevalence of test anxiety among all school students in India. Due to feasibility, they first randomly select 5 districts from each state, then randomly select 10 schools from each district, and finally randomly select 30 students from each school. This sampling technique is known as: (Real PYQ 2024)

   (a) Stratified Random Sampling (b) Quota Sampling (c) Cluster Sampling (d) Purposive Sampling

   Answer: (c) Cluster Sampling. Intact groups (districts, then schools) are chosen at random at successive levels, which is multistage cluster sampling. Stratified sampling would draw individuals from every stratum, and quota and purposive sampling involve no random selection at all.

8. Which of the following sampling methods relies on the researcher's judgment to select participants who are "typical" or "representative" of the population, and does NOT allow for the calculation of sampling error? (Real PYQ 2024)

   (a) Simple Random Sampling (b) Stratified Random Sampling (c) Quota Sampling (d) Cluster Sampling

   Answer: (c) Quota Sampling. It is the only non-probability method offered, so its sampling error cannot be calculated; the other three are probability methods. Strictly, a judgement-based choice of typical cases describes purposive sampling, which would be the better answer if it were offered.

9. A statistical measure based upon a population is known as: (IHBAS 2020)

   (a) Statistic (b) Parameter (c) Sample (d) None of the above

   Answer: (b) Parameter. A parameter (Greek letters such as mu and sigma) describes a population; a statistic (Roman letters) is computed from a sample, and a sample is a group of people, not a measure.

10. In experimental design, the principle that ensures every participant has an equal chance of being assigned to any group is: (Real PYQ 2025)

    (a) Replication (b) Local control (c) Randomization (d) Homogeneity

    Answer: (c) Randomization. Randomisation, one of Ronald Fisher's three principles (*The Design of Experiments*, 1935), gives every participant an equal chance of assignment to any group. Replication and local control (blocking) are the other two principles; homogeneity is not one of them.

11. Researchers studying meditation contacted over 10000 randomly selected adults in 10 nations. The results showed that those who meditated on a regular basis were less likely to suffer from stress than adults who meditate less regularly or not at all. Can we conclude that meditation lowers stress? Why? (RML 2021)

    (a) Yes, because the sample size was large (b) Yes, because the adults were randomly selected (c) Yes, because the adults were randomly assigned (d) Not necessarily, because this wasn't an experiment

    Answer: (d) Not necessarily, because this wasn't an experiment. The researchers only measured who already meditated; nobody was randomly assigned, so meditators may differ in other ways that lower stress. A large or randomly selected sample helps generalisation, not causal inference, and option (c) describes something the study did not do.

12. Longitudinal and cross-sectional designs can be used to study how behaviour changes as a function of age. A disadvantage of cross-sectional studies is that the result can be influenced by : (IHBAS 2024)

    (a) Random assignment (b) Socio-economic factors (c) Cohort difference (d) Participant attrition

    Answer: (c) Cohort difference. A cross-sectional study compares different age groups at one time, so generational (cohort) differences can masquerade as age changes. Attrition is the problem of longitudinal studies, random assignment plays no part in age comparisons, and socio-economic differences are a general confound rather than the design's characteristic flaw.

13. A famous study examined the smoking habits of British doctors and whether or not the doctors development various types of cancers. Participants responded to questionnaires about their smoking habits and health in 1951, 1957, 1966 and many more times all the way up to 2001. What type of observation study is this? (RML 2021)

    (a) A retrospective observational study (b) A prospective observational study (c) A sample survey (d) None of these. It's an experimental study.

    Answer: (b) A prospective observational study. This is Doll and Hill's British Doctors Study. Nobody was assigned to smoke, so it is observational rather than experimental, and the doctors were followed forward from exposure (smoking in 1951) to outcome, so it is a prospective cohort, not a retrospective one. A sample survey describes a population at one point in time.

14. What is the main purpose of using meta-analysis in clinical psychology? (IHBAS 2024)

    (a) To summarize findings from multiple studies (b) To conduct a single large experiment (c) To analyze individual case studies (d) To collect primary data

    Answer: (a) To summarize findings from multiple studies. The paper's key gives (b), which is wrong: a single experiment, however large, is primary research. Meta-analysis statistically pools the effect sizes of many existing studies; it neither collects primary data nor analyses individual case studies.

15. In a cohort study investigating the association between a new psychological intervention and relapse rates in schizophrenia, the risk ratio is calculated as 1.0. This finding indicates that: (Real PYQ 2024)

    (a) The incidence of relapse in the exposed group is double that of the unexposed group. (b) The intervention has a significant protective effect against relapse. (c) There is no difference in the risk of relapse between the intervention and control groups. (d) The odds of exposure are equal in both the relapse and non-relapse groups.

    Answer: (c) There is no difference in the risk of relapse between the groups. A risk ratio divides the risk in one group by the risk in the other, so 1.0 means equal risks. Double the incidence would be 2.0 and a protective effect below 1.0; (d) is how a case-control study's odds ratio is phrased, not what a cohort's risk ratio measures.

16. 'Can you tell me why this band is so popular among your age group?' is an example of: (RML 2020)

    (a) psychodynamic inquiry (b) a task-focused inquiry (c) an ethnographic inquiry (d) a strength-based inquiry

    Answer: (c) An ethnographic inquiry. The question treats the teenager as an informant and asks them to explain the culture of their own peer group, as in Spradley's ethnographic interview. Psychodynamic inquiry probes feelings and hidden motives, task-focused inquiry a defined task, and strength-based inquiry the person's resources.

17. Uniting various qualitative methods with quantitative methods can be called as: (IHBAS 2020)

    (a) Coalition (b) Triangulation (c) Multidimensional approach (d) Co-operative inquiry

    Answer: (b) Triangulation. Combining methods so that each one's errors are offset by the others is Denzin's (1970) methodological triangulation. Coalition is not a research term, a multidimensional approach merely studies several aspects of a problem, and co-operative inquiry is Heron's participatory method.

18. Which one of the following does not apply to qualitative research? (IHBAS 2020)
    (a) Recognition of the role of language and discourse (b) Emphasis on objective truth (c) Understanding of processes (d) Concern for human individuality

    Answer: (b) Emphasis on objective truth. Qualitative research rests on constructivism, which sees realities as constructed and multiple; language and discourse, processes and individuality are its hallmarks, while objective truth belongs to positivist, quantitative work.

19. 'The ethical principle that research participants should be told enough about a piece of research to be able to make a decision about whether to participate in it.' Which research principle is this a glossary definition of? (RML 2021)
    (a) Informed consent (b) Debriefing (c) Expectation management (d) De-concealment

    Answer: (a) Informed consent. Being told enough to decide, before taking part, is informed consent. Debriefing comes after participation; expectation management and de-concealment are not research-ethics principles (revealing a deception is called dehoaxing).

20. The principle of ethical research under which the subjects should be given full information about all aspects of the research study after it is done is known as: (IHBAS 2020)
    (a) Debriefing (b) Description (c) Explanation (d) Interpretation

    Answer: (a) Debriefing. Debriefing gives full information after participation, including any deception (dehoaxing) and help with distress (desensitising). Description and explanation are goals of science, and interpretation belongs to a report's Discussion.

21. Which of the following is NOT a form of plagiarism? (RML 2023)
    (a) Collusion (b) Citation (c) Patchwriting (d) Ghost-writing

    Answer: (b) Citation. Citing openly acknowledges a source and gives credit to its authors, so it is the cure for plagiarism. Collusion (unauthorised joint work), patchwriting (lightly reworded source text) and ghost-writing (work written by someone else) are all plagiarism.

22. Arbitrary zero is the characteristic of which scale? (RML 2024)
    (a) Ratio scale (b) Ordinal scale (c) Interval scale (d) Nominal scale

    Answer: (c) Interval scale. The source keys (a), which is wrong: an interval scale such as Celsius has equal intervals but an arbitrary zero, so ratios are meaningless, while a ratio scale is defined by a true zero. Ordinal and nominal scales have no meaningful zero or equal intervals at all.

23. A person wants to determine the representative income of the people living in a community. He would be best advised to use the (JEMAS 2025)
    (a) Mean (b) Median (c) Mode (d) Standard deviation

    Answer: (b) Median. Income is positively skewed: a few very rich households pull the mean upward, while the median, depending only on position, stays with the typical family. The mode is unstable, and the standard deviation measures spread, not central tendency.

24. In a negatively skewed distribution, the position of the mean with reference to the median is: (RML 2022)
    (a) Left of the median (b) Right of the median (c) Same as median (d) Same as a mode

    Answer: (a) Left of the median. In negative skew the long tail points to low scores and drags the mean furthest into it, so from low to high the order is mean, median, mode. The mean lies right of the median in positive skew, and the three coincide only in a symmetrical distribution.

25. Type II error occurs : (RML 2022)

    (a) whenever a true null hypothesis is accepted (b) whenever a false null hypothesis is rejected (c) whenever a false null hypothesis is accepted (d) whenever a true null hypothesis is rejected

    Answer: (c) whenever a false null hypothesis is accepted. A Type II error (β) misses a real effect: a false negative. Rejecting a true null (d) is a Type I error; (a) and (b) are the two correct decisions.

26. The statistical theorem states that given a sufficiently large sample size, the sampling distribution of the mean for a variable will be approximately normally distributed, regardless of that variable's distribution in the Population. What is this definition of? (RML 2021)

    (a) The normal distribution theorem (b) The central limit theorem (c) The sampling distribution theorem (d) The law of large numbers

    Answer: (b) The central limit theorem. With N of about 30 or more, the distribution of sample means is near normal whatever the population's shape. The law of large numbers says only that a sample mean approaches the population mean as N grows; (a) and (c) are not standard theorems.

27. What is the p-value in statistical testing? (IHBAS 2024)

    (a) The probability of the null hypothesis being true (b) The probability of obtaining the observed data if the null hypothesis is true (c) The effect size of the experiment (d) The power of the study

    Answer: (b). p is the probability of data at least this extreme *assuming* H0 is true. Option (a) reverses the conditional; effect size and power (1 − β) are separate quantities.

28. Which non-parametric test is equivalent to the independent samples t-test? (RML 2023)

    (a) Mann-Whitney U test (b) Kruskal-Wallis test (c) Wilcoxon signed-rank test (d) Friedman test

    Answer: (a) Mann-Whitney U test. It ranks the pooled scores of two independent groups and is the rank partner of the independent t-test. Wilcoxon signed-rank is for two related samples, Kruskal-Wallis for three or more independent groups, and Friedman for three or more related samples.

29. Which of the following is a pooled variance estimate that constitutes the denominator of an F-ratio? (RML 2021)

    (a) The between-cells mean square (MSB). (b) The mean of the sampling distribution. (c) The sum of the deviations from the grand mean. (d) The within-cells mean square (MSw).

    Answer: (d) The within-cells mean square (MSw). F = MS between / MS within, and MS within pools the error variance inside the groups. MS between is the numerator; deviations from the grand mean always sum to zero, which is why they are squared; the mean of the sampling distribution plays no part in F.

30. A researcher wants to calculate the degrees of freedom (df) for a 3 x 4 contingency table in a Chi-square test. The df is: (Real PYQ 2025)

    (a) 12 (b) 7 (c) 6 (d) 11

    Answer: (c) 6. For a test of independence df = (rows - 1)(columns - 1) = 2 x 3 = 6. Twelve multiplies the raw dimensions, 7 adds them, and 11 is the number of cells minus 1.

31. A researcher uses a Scree plot to determine the number of factors to retain in an EFA. The plot shows a sharp drop from eigenvalue 3.5 to 1.8, followed by a gradual plateau where eigenvalues are 0.9, 0.7, and 0.5. According to the Kaiser-Guttman rule and the Scree plot, how many factors should ideally be retained? (Real PYQ 2024)

    (a) 5 (b) 3 (c) 2 (d) 1

    Answer: (c) 2. Only 3.5 and 1.8 exceed the Kaiser-Guttman cut-off of 1.0, and both lie above the elbow at the third factor, so the two rules agree. Three wrongly keeps the elbow factor, five counts every eigenvalue and one keeps only the largest.

32. When two variables that are being used as predictors in a statistical analysis have a high correlation with each other, this statistical problem is known as: (Real PYQ 2025)

    (a) Heteroscedasticity (b) Multicollinearity (c) Autocorrelation (d) Homogeneity of variance

    Answer: (b) Multicollinearity. Highly intercorrelated predictors make regression weights unstable and inflate their standard errors (flagged by a VIF above 10). Heteroscedasticity is unequal residual spread, autocorrelation is residuals correlated over time, and homogeneity of variance is equal variance across groups.

33. Which of the following correlation methods is used when one variable is continuous and the other is dichotomous? (RML 2023)

    (a) Pearson correlation coefficient (b) Spearman's rank correlation coefficient (c) Point-biserial correlation coefficient (d) Biserial correlation coefficient

    Answer: (c) Point-biserial correlation coefficient. A continuous variable paired with a true (natural) dichotomy calls for point-biserial. Biserial is for an artificial dichotomy cut from a continuous variable, Pearson for two continuous variables and Spearman for ranks.

High-yield sheet

The facts this chapter's questions ask most, most-asked first. Last read before the exam.

1 A **z-score**, (X − M)/SD, states a score's distance from the mean in SD units (mean 0, SD 1) so scores from different distributions can be compared; on the normal curve z = 0, +1, +1.5 and +2 lie at about the 50th, 84th, 93rd and 98th percentiles, so deviation IQs of 85, 100 and 130 sit at the 16th, 50th and 98th. ASKED 12 TIMES

2 A **Type I error** (α) is rejecting a true null hypothesis (false positive); a **Type II error** (β) is retaining, or accepting, a false null hypothesis (false negative). The significance level α is the probability of a Type I error, in ANOVA as in every test. ASKED 11 TIMES

3 **Stevens (1946)**: nominal = unordered categories (marital status, car colour); ordinal = rank order (preference or agreement); interval = equal intervals with an **arbitrary zero** (Celsius); ratio = equal intervals with a **true zero** (height, number of books). ASKED 9 TIMES

4 **One-way ANOVA** compares the means of three or more independent groups on one IV with a continuous DV (it also works for two, where F = t squared); repeated t-tests would inflate the family-wise Type I error. ASKED 9 TIMES

5 The **Mann-Whitney U test** (Mann and Whitney, 1947; the same as the Wilcoxon rank-sum test) is the non-parametric equivalent of the independent-samples t-test; Wilcoxon signed-rank partners the paired t, Kruskal-Wallis the one-way ANOVA and Friedman the repeated-measures ANOVA. ASKED 9 TIMES

6 The **p-value** is the probability of obtaining the observed result, or one more extreme, if H0 is true; it is not the probability that H0 is true. If $p \leq \alpha$ ($p = .03$ with $\alpha = .05$), reject H0. ASKED 8 TIMES

7 **Chi-square** (Karl Pearson, 1900) is a non-parametric test for nominal frequency data; the test of independence asks whether two categorical variables are associated, with df = (rows - 1)(columns - 1): 3 x 4 gives 6, 4 x 5 gives 12. ASKED 8 TIMES

8 The **standard deviation** is the square root of the variance and measures variability; it is affected by extreme scores (the IQR and quartile deviation are not), and multiplying every score by c multiplies the variance by c^2 while adding a constant changes nothing (variance 4, scores × 3 + 10: variance 36). ASKED 7 TIMES

9 **Cluster sampling** randomly selects whole natural groups (districts, schools) and studies all or some units inside them; districts, then schools, then students is multistage cluster sampling, whereas **stratified random sampling** samples randomly within every stratum. ASKED 6 TIMES

10 In **non-probability sampling** (convenience, purposive, quota, snowball) the chance of selection is unknown, randomisation is impossible and sampling error cannot be calculated; quota sampling, the commonest option-list answer, fills proportional quotas non-randomly. ASKED 6 TIMES

11 The **normal curve** is symmetrical and unimodal with mean = median = mode, and about 68, 95 and 99.7 per cent of cases lie within ±1, ±2 and ±3 SD of the mean; a bimodal curve is not normal. ASKED 6 TIMES

12 The **median** is the average for skewed data (community income, anxiety scores), the **mode** the only one for nominal data; range, SD and mean deviation are measures of variability, not central tendency. ASKED 6 TIMES

13 **Cohen's d**, the standardised difference between two means, is small at 0.2, medium at 0.5 and large at 0.8; for r the benchmarks are .10, .30 and .50, so r = −.30 is a medium negative relationship. ASKED 6 TIMES

14 **F = MS between / MS within**; MS within, the pooled error variance, is the denominator. F is about 1 when H0 is true, so an F below 1 (0.99) is never significant; F rises as chance variability falls, can never be negative, and the test is one-tailed. ASKED 6 TIMES

15 **Post hoc tests** (Tukey's HSD, Scheffé the most conservative, Newman-Keuls, Duncan, Fisher's LSD the least conservative) are used only after a significant F; k levels give k(k - 1)/2 pairs (4 levels give 6); the **Bonferroni correction** divides alpha by the number of comparisons to control family-wise Type I error. ASKED 6 TIMES

16 **Induction** moves from specific observations to broad generalisations and builds theory (the theory-development stage; typical of qualitative research); **deduction** starts with a theory and tests it against observations (typical of quantitative, experimental research). ASKED 5 TIMES

17 **Degrees of freedom**: one-sample t = N - 1 (N 20 gives 19); independent t = N1 + N2 - 2 (25 per group gives 48); one-way ANOVA within = N - k (3 groups of 10 give 27); randomised block error = (treatments - 1)(blocks - 1) (4 and 5 give 12). ASKED 5 TIMES

18 **r = -0.85** is a strong negative relationship: the sign of *r* gives the direction and its size gives the strength. ASKED 5 TIMES

19 A **double-blind** study keeps both the participants and the experimenters who treat and assess them unaware of the conditions, so it controls both experimenter bias and participant expectancy effects. ASKED 4 TIMES

20 In **probability sampling** every element has a known, non-zero chance of selection, so sampling error can be calculated and the sample is likely, not guaranteed, to be representative. ASKED 4 TIMES

21 Only a true **experiment** (manipulation, control, random assignment) licenses cause-and-effect conclusions; correlational and observational studies cannot, however large or randomly selected the sample, because of the directionality and third-variable problems. ASKED 4 TIMES

22 **NNT = 1/ARR**, where the absolute risk reduction is the control event rate minus the treated event rate; an NNT of 8 means treating 8 patients for one additional patient to benefit. ASKED 4 TIMES

23 **Sensitivity** = TP/(TP + FN), the true positive rate; **specificity** = TN/(TN + FP). Sensitivity 90 percent and specificity 70 percent means the test correctly identifies 90 percent of ill and 70 percent of healthy people. ASKED 4 TIMES

24 **Qualitative research** is naturalistic, context-specific and inductive (bottom-up evidence), seeks meaning, process and the insider's view, and claims neither objective truth nor the researcher's authority; the **survey** is quantitative, whereas case study, ethnography and grounded theory are qualitative. ASKED 4 TIMES

25 **Random assignment** (randomisation) gives every participant an equal chance of being placed in any group, making groups equivalent on known and unknown variables so the main difference is the treatment; it controls selection bias and confounding and is not random sampling. ASKED 3 TIMES

26 The **independent variable** is the presumed cause, manipulated or selected (having heart disease in 'residents with heart disease die earlier'); the **dependent variable** is the measured outcome expected to change. ASKED 3 TIMES

27 **Stratified random sampling** divides the population into mutually exclusive strata and samples randomly within each; it is preferred when important subgroups must be represented and is more precise than simple random sampling when strata are internally homogeneous. ASKED 3 TIMES

28 A **longitudinal design** studies the same people repeatedly over time and is the best design for examining change over time; its problems are attrition, practice effects, cost and time. ASKED 3 TIMES

29 A **relative risk or odds ratio of 1.0** means no association between exposure and outcome; above 1 marks a risk factor, below 1 a protective one. ASKED 3 TIMES

30 **Intention-to-treat analysis** analyses every participant in the group to which they were randomised, regardless of adherence, drop-out or crossover, and so preserves randomisation.
ASKED 3 TIMES

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